

Geometry of Black Holes

PIOTR CHRUSCIEL



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Piotr T. Chruściel

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Preface

Black holes are regions of spacetime which are causally disconnected from our region in the following sense: no objects, radiation, or other non-quantum form of information can reach us from the black-hole region; on the other hand, we can fall into, or send spaceships inside. Although they have been studied for years, they still attract tremendous attention in the physics and astrophysics literature. They present one of the most fascinating predictions of Einstein's general relativity. There is strong evidence of their existence from observations of active galactic nuclei, including the centre of our galaxy, from the analysis of gravitational wave signals, and others.

There exists a large scientific literature on black holes, including many excellent textbooks at various levels. However, most texts steer clear from the mathematical niceties needed to make the theory of black holes a mathematically well-posed theory. Those which maintain a high standard of mathematical rigour are either focused on specific topics or skip over many details. The object of this book is to fill this gap, and present a detailed, mathematically oriented, extended introduction to the subject.

More specifically, the material presented here was written to serve as background for the theory of uniqueness of stationary black holes, as presented in detail in [113, 114]. Yet another motivation was to present in a coherent way some of my contributions to the understanding of the geometry of black holes, with suitable background to make the presentation self-contained.

At the core of this book lies a mix of lectures that I held in various summer schools, and suitably reworked papers that I authored or coauthored. Emphasis is on those aspects of the theory of black holes relevant to classical physics. We do not address directly the quantum aspects of the theory of black holes. The reader is referred to [66, 170, 259, 276, 393, 455] and references therein for these.

The book should be of interest to those mathematically minded advanced students and researchers who want to find something more than, or different from, the material found in other books on the subject.

In any case, the reader is assumed to be familiar with elementary pseudo-Riemannian geometry, as presented e.g. in [110, Chapter 1].

INCIDENTALLY: Throughout this work we use indented text, typeset in smaller font, for material which plays secondary—informative or auxiliary—role, and may be skipped without affecting the understanding of the main line of development of the subject. $\square$

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Part I

Global Lorentzian geometry

Causality theory is a prerequisite to understanding the nature of black holes, and the core of Part I is an introduction to this theory. For this, in Chapter 1, we review some basic facts of Lorentzian geometry. This provides an opportunity to present our notations and conventions. Chapter 2 provides the promised introduction to the causality theory; it is an expanded, revised, and corrected version of notes posted on the arXiv [106]. In Chapter 3 we review some key applications of the formalism developed in the previous chapter. Section 3.3 has a somewhat different character from the rest of the chapter, as we prove in detail there a few versions of *topological censorship theorem*, which are essential for understanding the topology of black holes.

1

Basic notions

The aim of this chapter is to lay the groundwork for further studies. We present our conventions and notations in Section 1.1. We review some basic facts about the topology of manifolds in Section 1.2. Lorentzian manifolds and spacetimes are introduced in Section 1.3. Elementary facts concerning the Levi-Civita connection and its curvature are reviewed in Section 1.4, with proofs of the usual properties of the curvature tensor deferred to Appendix C, p. 341. Some properties of geodesics, as needed in the remainder of this book, are presented in Section 1.5. The formalism of moving frames is outlined in Section 1.6, where it is used to calculate the curvature of some metrics of interest.

1.1 Conventions

All manifolds are Hausdorff and paracompact. The letter n usually denotes the space dimension of the manifold under consideration; thus, to emphasize the distinct character of the space and time variables, spacetimes will always have dimension $n + 1$. Greek indices α, β , etc., correspond to spacetime coordinates x^α and take values $0, 1, \dots, n$, while Latin indices i, j , etc., take values $1, \dots, n$ and correspond to space coordinates x^i . We shall use the summation convention throughout, which means that every pair of indices with one index up and one index down has to be summed over, e.g.

$$A_\alpha{}^\beta{}_\gamma{}_\delta X^\gamma Y_{\beta\sigma} := \sum_{\gamma, \beta=0}^n A_\alpha{}^\beta{}_\gamma{}_\delta X^\gamma Y_{\beta\sigma}, \quad A_i X^i := \sum_{i=1}^n A_i X^i.$$

If f is a function, then we will freely switch between the following notations to denote its derivatives:

$$\frac{\partial f}{\partial x^\mu} = \partial_\mu f = f_{;\mu} = f_{;\mu} = \nabla_\mu f = \nabla_{\partial_\mu} f.$$

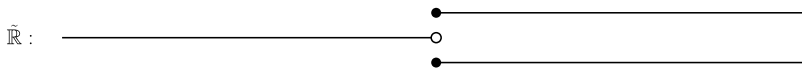
The last three notations will also be interchangeably used for the covariant derivatives of tensor fields. We will further use the symbol D for the covariant derivative, mainly—but not exclusively—in a Riemannian context, e.g. on Riemannian submanifolds of Lorentzian ones.

1.2 Topology and manifolds

Perhaps the main revolution brought by general relativity is that physics takes place on manifolds rather than on $\mathbb{R}^4$. Now, manifolds are topological spaces to start with, and a short discussion of the topology of manifolds is in order.

We will assume that the reader is familiar with elementary topology, as usually taught to physicists, and start by reviewing a few notions which are sometimes omitted from such courses. A very useful reference to ‘advanced-but-not-too-much-so’ topology is the textbook by Munkres [369].

It is natural to raise the question, what kind of topologies are considered acceptable in physics? Anyone who has sat in on a course on mechanics will automatically

Fig. 1.2.1 The bifurcating real axis $\tilde{\mathbb{R}}$.

think of physics in terms of the standard topology on $\mathbb{R}^3$, without even considering alternatives. In the context of n -dimensional manifolds, part of this standard topology is reflected in the local coordinate systems, modelled on $\mathbb{R}^n$, near every point. However, the mere requirement of the existence of local coordinate systems is compatible with extremely pathological behaviour, because local coordinates can be patched together in various unexpected ways. Some terminology will be needed to describe those pathologies.

Recall that a topological space $(X, \mathcal{T})$ is said to be *metrizable* if there exists a distance function d so that $\mathcal{T}$ coincides with the topology defined by d . Readers of this work are likely to be familiar with metric topologies, and metrizable topologies possess of course all the properties of the metric ones, without having to invoke a preferred metric. This makes sense from a physical point of view, because even in special relativity there is no preferred metric on spacetime: there exists a preferred Lorentzian metric tensor, but there is no natural distance function there. Every inertial observer in Minkowski spacetime $\mathbb{R}^{3,1} := (\mathbb{R}^4, -dt^2 + dx^2 + dy^2 + dz^2)$ can define a Riemannian metric $h = dt^2 + dx^2 + dy^2 + dz^2$, and use the associated topology. This metric is *not* invariant under Lorentz transformations, but all the Riemannian metrics obtained by Lorentz-transforming h lead to equivalent distance functions, resulting in the same topology. So the standard metric topology of $\mathbb{R}^{3,1}$ is Lorentz-invariant indeed.

Metric topologies have nice *separation properties*. Those have to do with the possibility of separating closed objects using open ones. Readers used to metric concepts will find the separation properties trivial, being obviously satisfied in metric topologies. However, one needs to keep in mind that such separation properties do not have to hold for general topological spaces.

In this context, we have the following terminology: a topological space is said to be *Hausdorff* if for each pair x, y of distinct points of X there exist disjoint open sets $\mathcal{U}, \mathcal{V}$, with $x \in \mathcal{U}$ and $y \in \mathcal{V}$. An example of a topological space, with usual local coordinates near every point but with non-Hausdorff topology, can be found in Figure 1.2.1. The space $\tilde{\mathbb{R}}$ illustrated there can be defined as two copies of $\mathbb{R}$ in which the negative reals have been identified. One of the main features of non-Hausdorff spaces is non-uniqueness of limits: the sequence of reals $-1/n$, $n \in \mathbb{N}$, accumulates both to the ‘upper zero’ and to the ‘lower zero’ of Figure 1.2.1. An observer travelling on $\tilde{\mathbb{R}}$ from the negatives to the positives could, when hitting the origin, continue either on the upper sheet or on the lower sheet, or duplicate herself to continue on both. None of this appears to be compatible with our everyday experience of physics, so one is tempted to postulate that *acceptable physical models should be Hausdorff*.

INCIDENTALLY: There exist two somewhat stronger notions of separability: a topological space $(X, \mathcal{T})$ is said to be *regular* if for each $x \in X$ and for each closed set $B \subset X$ not containing x there exist disjoint open sets $\mathcal{U}$ and $\mathcal{V}$ such that $x \in \mathcal{U}$ and $B \subset \mathcal{V}$. Finally, *normality* is defined by the requirement that for every disjoint closed sets A and B there exist disjoint open sets $\mathcal{U}$ and $\mathcal{V}$ such that $A \subset \mathcal{U}$ and $B \subset \mathcal{V}$. As already pointed out, every metric topology is normal, hence regular, hence Hausdorff. $\square$

Yet another class of pathologies that can occur has to do with the fact that topological spaces can be ‘very big’. Two instructive examples of very unpleasant,

not-metrizable, manifold-like objects are the one-dimensional *long line*, and the two-dimensional *Prüfer manifold*, both discussed in detail in [434, Vol. I, Appendix A]; the reader is strongly encouraged to study those examples. The behaviour exhibited there can be gotten rid of by invoking *countability properties* of the topology, as made clear by the following theorem of Urysohn:

THEOREM 1.2.1 (Urysohn's metrization theorem [369, Section 34]) *Every regular topological space with a countable basis is metrizable.*

An apparently unrelated notion is that of *paracompactness*. To define this, recall that a collection $\mathcal{B}$ of subsets of a set X is called a *refinement* of a collection $\mathcal{A}$ of subsets of X if for each element $B \in \mathcal{B}$ there is an element $A \in \mathcal{A}$ such that $B \subset A$. A refinement is called *open* if all its sets are open. Next, a collection $\mathcal{A}$ of subsets of X is said to be *locally finite* if every point of X has a neighbourhood that intersects only finitely many elements of $\mathcal{A}$. Then, a topological space $(X, \mathcal{T})$ is called *paracompact* if every open covering of X has a locally finite open refinement that covers X .

The reader should think of this as a version of local compactness: using the above terminology, the usual definition of compactness is equivalent to the property that every open covering of X has a finite open refinement.

A theorem of Stone's shows that *every metrizable space is paracompact* [385].

A key property of paracompact spaces is the existence of *partitions of unity*. Those are used in many constructions, and are defined as follows: consider a collection $\{\mathcal{U}_i\}$ of sets covering X . Then a family of continuous functions $\varphi_i : X \rightarrow [0, 1]$ is said to form a *partition of unity dominated by $\{\mathcal{U}_i\}$* if the supports $\text{supp } \varphi_i$ satisfy $\text{supp } \varphi_i \subset \mathcal{U}_i$, if for every $x \in X$ only a finite number of the functions φ_i are non-zero at x , and if $\sum_i \varphi_i = 1$. Typically the functions φ_i will be smooth in the applications at hand.

An obvious property of topological spaces with local charts modelled on $\mathbb{R}^n$ is that of *local metrizability*: this is defined by requiring that every point has a neighbourhood that is metrizable in the subspace topology. The following theorem of Smirnov's ties all the above properties together:

THEOREM 1.2.2 (Smirnov metrization theorem [369, Section 42]) *A topological space $(X, \mathcal{T})$ is metrizable if and only if it is a paracompact Hausdorff space that is locally metrizable.*

Yes, another topological notion, which this time invokes *both* countability *and* a local version of compactness, is that of σ -compactness: a topological space $(X, \mathcal{T})$ is said to be σ -compact if X can be covered by a countable family of open sets, each of which is contained in a compact one. The following should be clear from what has been said already (see [434, Vol. I, Appendix A] for a detailed proof):

THEOREM 1.2.3 *Let $(X, \mathcal{T})$ be a Hausdorff topological space with the property that every point $x \in X$ has an open neighbourhood $\mathcal{U}_x$ homeomorphic to $\mathbb{R}^n$ for some $n = n(x)$. Then $n(x)$ is constant on each connected component of X , and the following are equivalent:*

1. *Each component of X is σ -compact.*
2. *Each component of M has a countable base for its topology.*
3. *$(X, \mathcal{T})$ is paracompact.*
4. *$(X, \mathcal{T})$ is metrizable.*

Throughout this book we will be interested in manifolds equipped with a Lorentzian metric, hence a connection. It is then hard to see how metrizability can be avoided in view of the following observation of Geroch [218, Appendix] (compare [378] for the Riemannian case):

THEOREM 1.2.4 *Every connected smooth Hausdorff manifold equipped with a C^1 connection is metrizable.*

INCIDENTALLY: As pointed out by Geroch in [218], this does not have to be the case for all natural geometric structures: An example of a non-metrizable two-dimensional Hausdorff manifold with a conformal Lorentzian metric is provided by the Cartesian product of the long line by itself, with the two null directions along the factors. This example is tied to the infinite dimensionality of the group of conformal transformations in dimension 2, and such behaviour does not arise in higher dimensional conformal geometry—this follows from the existence of a natural connection on one of the bundles relevant in this context. $\square$

Summarizing, topological spaces with local coordinate charts can have unpleasant properties unless metrizability is imposed. Assuming that the spacetime is Hausdorff, metrizability is then equivalent to paracompactness or to σ -compactness or to second-countability. Metrizability will be assumed throughout this book, and only metrizable manifolds will be considered.

1.3 Lorentzian manifolds

A couple $(\mathcal{M}, g)$ will be called a Lorentzian manifold if $\mathcal{M}$ is an $(n+1)$ -dimensional, differentiable, paracompact, Hausdorff, connected manifold, and g is a non-degenerate symmetric twice-covariant continuous tensor field on $\mathcal{M}$ of signature $(- + \dots +)$. By an abuse of terminology g is often called a metric. We will only consider differentiable manifolds with continuous metrics. By a theorem of Whitney [459] there is no loss of generality in assuming that the manifold is smooth.¹ Unless explicitly indicated otherwise we assume that the metric is smooth; however, we will often indicate weaker differentiability conditions which are sufficient for the specific problems at hand.

A vector $X \in T_p\mathcal{M}$ is said to be *timelike* if $g(X, X) < 0$; *null* or *lightlike* if $g(X, X) = 0$ and $X \neq 0$; *causal* if $g(X, X) \leq 0$ and $X \neq 0$; and *spacelike* if $g(X, X) > 0$. Given any basis of T_pM with the first vector timelike, we can use the standard Gram–Schmidt procedure to construct an *ON-basis* of $T_p\mathcal{M}$, that is, vectors $e_a \in T_p\mathcal{M}$ such that

$$g(e_a, e_b) = \eta_{ab},$$

where η_{ab} stands for the Minkowski metric $\text{diag}(-1, +1, \dots, +1)$; indices a, b run from 0 to n . If $X = X^a e_a$, $Y = Y^a e_a$ (recall that we use the summation convention; i.e., repeated indices in different positions must be summed over), then

$$g(X, Y) = -X^0 Y^0 + X^1 Y^1 + \dots + X^n Y^n =: -X^0 Y^0 + \vec{X} \cdot \vec{Y}, \quad (1.3.1)$$

$$g(X, X) = -(X^0)^2 + (X^1)^2 + \dots + (X^n)^2 =: -(X^0)^2 + |\vec{X}|^2. \quad (1.3.2)$$

It follows that X is lightlike if and only if $X \neq 0$ and

$$X^0 = \pm \sqrt{\sum_i (X^i)^2}.$$

And X is timelike if and only if

$$|X^0| > \sqrt{\sum_i (X^i)^2}.$$

(Recall again that lower Latin indices i, j , etc. run from 1 to n .) Thus, in every tangent space the sets of timelike, lightlike, etc., vectors are linearly isomorphic to

¹However, such a restriction might be inconvenient for some purposes. For example, the initial value problem for general relativity leads naturally to manifolds with a Sobolev-type differentiable structure, discussed in detail in [31, Appendix A].

those of Minkowski spacetime. In particular the set of timelike vectors is the union of two disjoint open convex cones, the closures of which meet at the origin.

We will say that a causal vector X is *future pointing* if $X^0 > 0$ (compare Section 2.1).

We will often use the following elementary facts:

PROPOSITION 1.3.1 *Let X be timelike future pointing; then*

$$g(X, Y) < 0$$

for all future-pointing causal Y 's.

Proof. Suppose that we can choose an ON frame with

$$e_0 = X/\sqrt{-g(X, X)} \iff X = X^0 e_0;$$

the result is then obvious from (1.3.1). We will see shortly that such a basis exists, but it is not completely obvious at this stage that this can be done, so one must argue directly: since X is timelike future pointing, we have, in an ON basis, $X^0 > |\vec{X}|$. Similarly $0 \neq Y^0 \geq |\vec{Y}|$. Then, using Cauchy–Schwarz,

$$g(X, Y) = -X^0 Y^0 + \vec{X} \cdot \vec{Y} \leq -X^0 Y^0 + \underbrace{|\vec{X}|}_{< X^0} \underbrace{|\vec{Y}|}_{\leq Y^0} < -X^0 Y^0 + X^0 Y^0 = 0. \quad \square$$

It follows from Proposition 1.3.1 that the scalar product of a timelike vector with a causal vector never vanishes. We conclude the following:

COROLLARY 1.3.2 *Let X be timelike. Then the space*

$$X^\perp := \{Y : g(X, Y) = 0\}$$

is spacelike, in the sense that all non-zero vectors in $X^\perp$ are spacelike.

This allows us to prove the following:

PROPOSITION 1.3.3 *Let X be timelike; then there exists an ON basis $\{e_0, e_1, \dots, e_n\}$ such that X is proportional to e_0 .*

Proof. Complete $\{X\}$ to any basis $\{X, X_1, \dots, X_n\}$. Define $e_0 = X/\sqrt{-g(X, X)}$. For $i = 1, \dots, n$ set

$$Y_i = X_i + g(N, X_i)N.$$

Then the Y_i 's form a basis of $X^\perp$, which is spacelike by Corollary 1.3.2. We can now apply the usual Gram–Schmidt procedure to $\{Y_1, \dots, Y_n\}$ to obtain an ON basis $\{e_1, \dots, e_n\}$ of $X^\perp$. This provides the desired ON basis. $\square$

The scalar product of causal vectors satisfies the *inverse Cauchy–Schwarz inequality*:

PROPOSITION 1.3.4 *Let X, Y be two causal vectors; then*

$$|g(X, Y)| \geq \sqrt{-g(X, X)}\sqrt{-g(Y, Y)}, \quad (1.3.3)$$

with equality holding if and only if X is proportional to Y .

Proof. For null X 's (1.3.3) is trivial, so assume that X is timelike. By performing a Gram–Schmidt orthonormalization we can construct an ON basis in which $e^0 = X/\sqrt{-g(X, X)}$, hence $X = \sqrt{-g(X, X)}e^0$. Since Y is causal we have $|Y^0| \geq \sqrt{\sum_i (Y^i)^2}$, hence

$$\begin{aligned} |g(X, Y)| &= |-X^0 Y^0| \\ &= \sqrt{-g(X, X)}|Y^0| \end{aligned}$$

$$\begin{aligned}
&\geq \sqrt{-g(X, X)} \sqrt{(Y^0)^2 - \sum_i (Y^i)^2} \\
&= \sqrt{-g(X, X)} \sqrt{-g(Y, Y)},
\end{aligned}$$

with equality holding if and only if $\sum_i (Y^i)^2 = 0$, in which case Y is proportional to X .

If X is null, and equality holds in (1.3.3), then $g(X, Y) = 0$. In this case a Gram–Schmidt orthonormalization starting from any timelike vector e^0 as a first vector, and X as a second vector, leads to a basis in which $X = X^0(e_0 + e_1)$, so that

$$0 = g(X, Y) = -X^0(Y^0 - Y^1).$$

Since X^0 is non-zero we conclude that $Y^0 = Y^1$. Causality of Y gives

$$|Y^0| \geq \sum_i (Y^i)^2 \implies Y = Y^0(e_0 + e_1) \sim X. \quad \square$$

As an immediate corollary of Proposition 1.3.1 we obtain the following:

COROLLARY 1.3.5 *Let X, Y be two null vectors satisfying*

$$g(X, Y) = 0.$$

Then X is proportional to Y . $\square$

1.4 The Levi-Civita connection, curvature

In this section we give a short overview of the properties of the Levi-Civita connection for Lorentzian manifolds, and its curvature. The main point here is to codify our notations and conventions for those readers who are already familiar with those notions. Newcomers will find an extended exposition of the results reviewed here in e.g. [110, Chapter 1].

Consider a Lorentzian manifold $(\mathcal{M}, g)$; in a way completely analogous to the Riemannian case, to any Lorentzian metric one can associate a connection ∇ which is uniquely defined by the requirements that (a) ∇ is g compatible, that is, for all differentiable vector fields X, Y, Z on $\mathcal{M}$ we have

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z);$$

and (b) ∇ is torsion free:

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

∇ is called the *Levi-Civita* connection associated with the metric g . Given any connection ∇ , its Riemann tensor is defined as

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z. \quad (1.4.1)$$

Let $\partial_\alpha \equiv \partial/\partial x^\alpha$ be a basis of $T\mathcal{M}$ associated with a coordinate system x^α , and let dx^α be the associated dual basis

$$\langle dx^\alpha, \partial_\beta \rangle = \delta^\alpha_\beta, \quad (1.4.2)$$

where $\langle \cdot, \cdot \rangle$ is the duality pairing (if ϕ is a covector and X is a vector, then $\langle \phi, X \rangle := \phi(X)$) and δ^α_β is the Kronecker delta, equal to 1 when $\alpha = \beta$, and 0 otherwise. One defines

$$\begin{aligned}
g_{\alpha\beta} &:= g(\partial_\alpha, \partial_\beta) \iff g = g_{\alpha\beta} dx^\alpha \otimes dx^\beta, \\
R^\alpha_{\beta\gamma\delta} &:= \langle dx^\alpha, R(\partial_\gamma, \partial_\delta)\partial_\beta \rangle.
\end{aligned} \quad (1.4.3)$$

It would be logical to use the symbol R for the Riemann tensor; however, that symbol is often used to denote the Ricci scalar defined in (1.4.12). For this reason

we will write ‘Riem’ instead for the Riemann tensor, so that (1.4.3) can be rewritten as

$$\text{Riem} = R^\alpha{}_{\beta\gamma\delta} \partial_\alpha \otimes dx^\beta \otimes dx^\gamma \otimes dx^\delta. \quad (1.4.4)$$

There are actually two ways of understanding objects with indices, such as $R^\alpha{}_{\beta\gamma\delta}$, or $g_{\beta\gamma}$. The first refers to a given coordinate system $\{x^\mu\}$, and then (1.4.3) defines a collection of functions $\{R^\alpha{}_{\beta\gamma\delta}\}$, associated with the tensor Riem. This provides a practical way of computing things, and then (1.4.4) tells us how to recover Riem from the $\{R^\alpha{}_{\beta\gamma\delta}\}$ ’s. The second way, used throughout this work, is to understand the indices in $R^\alpha{}_{\beta\gamma\delta}$ or in $g_{\beta\gamma}$ as markers which indicate the tensor type, independently of any coordinate system. Those markers are very useful when operations on tensors are performed, such as taking contractions (e.g. $B^\alpha{}_\beta \rightarrow B^\alpha{}_\alpha$) or traces (e.g. $A^{\alpha\beta} \rightarrow A^\alpha{}_\alpha := A^{\alpha\beta} g_{\alpha\beta}$). This is, essentially, *Penrose’s abstract index notation* [399].

To summarize, a notation such as $W^\alpha{}_{\beta\gamma}$, when referring to a tensor field W , provides a shorthand for saying that W is a section of the bundle $TM \otimes T^*M \otimes TM$ (we will write $W \in \Gamma(TM \otimes T^*M \otimes TM)$, or $W \in \Gamma_{C^k}(TM \otimes T^*M \otimes TM)$ to indicate the C^k differentiability class, etc.). This shorthand is consistent with the local equations

$$W^\alpha{}_{\beta\gamma} = W(dx^\alpha, \partial_\beta, dx^\gamma) \iff W = W^\alpha{}_{\beta\gamma} \partial_\alpha \otimes dx^\beta \otimes \partial_\gamma,$$

when a local coordinate system $\{x^\mu\}$ has been given, but the reader *should not* assume that some coordinate system has been singled out.

Returning to the Riemann tensor, it follows from (1.4.1) that $R^\alpha{}_{\beta\gamma\delta}$ is antisymmetric in its last two indices,

$$R^\alpha{}_{\beta\gamma\delta} = -R^\alpha{}_{\beta\delta\gamma}.$$

As in the Riemannian case we have

$$R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta};$$

here and throughout we use the metric to raise and lower the indices,

$$R_{\alpha\beta\gamma\delta} := g_{\alpha\lambda} R^\lambda{}_{\beta\gamma\delta}.$$

This, and some other identities, are proved in Appendix C. We further have the *first Bianchi identity*,

$$R^\alpha{}_{\beta\gamma\delta} + R^\alpha{}_{\gamma\delta\beta} + R^\alpha{}_{\delta\beta\gamma} = 0. \quad (1.4.5)$$

We use the convention that adding a subscript ‘ ${}_{;\alpha}$ ’ to a tensor field denotes covariant differentiation in the direction ∂_α , e.g.

$$R^\alpha{}_{\beta\gamma\delta;\sigma} := \langle dx^\alpha, (\nabla_\sigma R)(\partial_\gamma, \partial_\delta) \partial_\beta \rangle,$$

where

$$\nabla_\sigma := \nabla_{\partial_\sigma}.$$

In this notation the *second Bianchi identity* reads

$$R^\alpha{}_{\beta\gamma\delta;\sigma} + R^\alpha{}_{\beta\delta\sigma;\gamma} + R^\alpha{}_{\beta\sigma\gamma;\delta} = 0. \quad (1.4.6)$$

A metric—whether Lorentzian or Riemannian—defines an isomorphism $\flat : T\mathcal{M} \rightarrow T^*\mathcal{M}$ by the formula

$$X_{\flat}(Y) := g(X, Y).$$

If we write X in the coordinate basis ∂_{α} as $X^{\alpha}\partial_{\alpha}$ (hence $X^{\alpha} = \langle dx^{\alpha}, X \rangle$), we then have

$$X_{\flat} = g_{\alpha\beta}X^{\alpha}dx^{\beta} =: X_{\beta}dx^{\beta}.$$

The operation which takes X^{α} to X_{α} is sometimes called ‘the lowering of indices’ in the physics literature, while its inverse is called ‘the raising of indices’.

We can use the map $\flat$ and its inverse, denoted by $\sharp$, to define a metric $g^{\sharp}$ on $T^*\mathcal{M}$, by transporting the metric g from $T\mathcal{M}$ to $T^*\mathcal{M}$. If we let $g^{\alpha\beta}$ denote the matrix inverse to $g_{\alpha\beta}$, one easily finds that

$$g^{\sharp}(dx^{\alpha}, dx^{\beta}) = g^{\alpha\beta}. \quad (1.4.7)$$

In the physics literature it is customary to use the same symbol g for g , $g^{\sharp}$, and extensions of g to tensor fields of any order, and we shall often do this.

The *Christoffel symbols* $\Gamma^{\alpha}_{\beta\gamma}$ are defined as

$$\Gamma^{\alpha}_{\beta\gamma} := \langle dx^{\alpha}, \nabla_{\partial_{\gamma}}\partial_{\beta} \rangle. \quad (1.4.8)$$

It follows that

$$\nabla_{\alpha}X = (\partial_{\alpha}X^{\beta} + \Gamma^{\beta}_{\sigma\alpha}X^{\sigma})\partial_{\beta}.$$

We have the classical formulae for the Christoffel symbols of the Levi-Civita connection

$$\Gamma^{\alpha}_{\beta\gamma} = \frac{1}{2}g^{\alpha\sigma}(\partial_{\beta}g_{\sigma\gamma} + \partial_{\gamma}g_{\sigma\beta} - \partial_{\sigma}g_{\beta\gamma}), \quad (1.4.9)$$

$$R^{\alpha}_{\beta\gamma\delta} = \partial_{\gamma}\Gamma^{\alpha}_{\beta\delta} - \partial_{\delta}\Gamma^{\alpha}_{\beta\gamma} + \Gamma^{\alpha}_{\sigma\gamma}\Gamma^{\sigma}_{\beta\delta} - \Gamma^{\alpha}_{\sigma\delta}\Gamma^{\sigma}_{\beta\gamma}. \quad (1.4.10)$$

We will interchangeably write $\Gamma^{\alpha}_{\beta\gamma}$ and $\Gamma^{\alpha}_{\beta\gamma}$ for the Christoffel symbols.

The *Ricci tensor* $\text{Ric} = R_{\alpha\beta}dx^{\alpha} \otimes dx^{\beta}$ is defined as

$$R_{\alpha\beta} = R^{\gamma}_{\alpha\gamma\beta}. \quad (1.4.11)$$

The scalar curvature R , defined as

$$R = g^{\alpha\beta}R_{\alpha\beta}, \quad (1.4.12)$$

is also called *the Ricci scalar* in the physics literature. The *Einstein tensor* $G = G_{\alpha\beta}dx^{\alpha} \otimes dx^{\beta}$ is defined as

$$G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}Rg_{\alpha\beta}. \quad (1.4.13)$$

The interest of this tensor field stems from the *divergence identity*,

$$G^{\beta}_{\alpha}{}^{;\beta} = 0, \quad (1.4.14)$$

which is obtained as follows: contracting the second Bianchi identity (1.4.6) over α and σ gives

$$R^{\alpha}_{\beta\gamma\delta;\alpha} - R_{\beta\delta;\gamma} + R_{\beta\gamma;\delta} = 0. \quad (1.4.15)$$

Contracting this equation with $g^{\beta\gamma}$ yields

$$-R^{\alpha}_{\delta;\alpha} - R^{\gamma}_{\delta;\gamma} + R_{;\delta} = 0, \quad (1.4.16)$$

which is, up to some renaming of indices, Eq. (1.4.14).

1.5 Geodesics

An *affinely parameterized geodesic* γ is a maximally extended solution of the equation

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0. \quad (1.5.1)$$

The parameter s defining the tangent $\dot{\gamma} \equiv d\gamma/ds$ is called an *affine parameter*.

Note that $\dot{\gamma} \equiv 0$ trivially solves the equation, to avoid this we will also require that $\dot{\gamma}$ does not vanish. While equation (1.5.1) is a convenient abuse of notation, with a coordinate-independent flavour, what is really meant is, in local coordinates,

$$\frac{d^2 \gamma^\mu}{ds^2} + \Gamma^\mu_{\alpha\beta} \frac{d\gamma^\alpha}{ds} \frac{d\gamma^\beta}{ds} = 0. \quad (1.5.2)$$

It is a fundamental postulate of general relativity that freely falling physical observers move on timelike geodesics. This motivates the following definition: a *freely falling observer* is a maximally extended future-directed timelike geodesics.

INCIDENTALLY: It is sometimes convenient to consider geodesics which are not affinely parameterized. Those are solutions of

$$\nabla_{\frac{d\gamma}{d\lambda}} \frac{d\gamma}{d\lambda} = \chi \frac{d\gamma}{d\lambda}, \quad (1.5.3)$$

for some function χ .

Let us show that a change of parameter obtained by solving the equation

$$\frac{d^2 \lambda}{ds^2} + \chi \left(\frac{d\lambda}{ds} \right)^2 = 0 \quad (1.5.4)$$

brings (1.5.4) to the form (1.5.1): under a reparameterization $\lambda = \lambda(s)$ we have

$$\frac{d\gamma^\mu}{ds} = \frac{d\lambda}{ds} \frac{d\gamma^\mu}{d\lambda},$$

and

$$\begin{aligned} \frac{D}{ds} \frac{d\gamma^\nu}{ds} &= \frac{D}{ds} \left(\frac{d\lambda}{ds} \frac{d\gamma^\nu}{d\lambda} \right) \\ &= \frac{d^2 \lambda}{ds^2} \frac{d\gamma^\nu}{d\lambda} + \frac{d\lambda}{ds} \frac{D}{ds} \frac{d\gamma^\nu}{d\lambda} \\ &= \frac{d^2 \lambda}{ds^2} \frac{d\gamma^\nu}{d\lambda} + \left(\frac{d\lambda}{ds} \right)^2 \frac{D}{d\lambda} \frac{d\gamma^\nu}{d\lambda} \\ &= \frac{d^2 \lambda}{ds^2} \frac{d\gamma^\nu}{d\lambda} + \left(\frac{d\lambda}{ds} \right)^2 \chi \frac{d\gamma^\nu}{d\lambda}, \end{aligned}$$

and the choice indicated above gives zero, as desired. $\square$

Let f be a smooth function and let $\lambda \mapsto \gamma(\lambda)$ be any integral curve of ∇f ; by definition, this means that $d\gamma^\mu/d\lambda = \nabla^\mu f$. The following provides a convenient tool for finding geodesics:

PROPOSITION 1.5.1 (Integral curves of gradients) *Let f be a function satisfying*

$$g(\nabla f, \nabla f) = \psi(f),$$

for some function ψ . Then the integral curves of ∇f are geodesics, affinely parameterized if $\psi' = 0$.

Proof. We have

$$\dot{\gamma}^\alpha \nabla_\alpha \dot{\gamma}^\beta = \nabla^\alpha f \nabla_\alpha \nabla^\beta f = \nabla^\alpha f \nabla^\beta \nabla_\alpha f = \frac{1}{2} \nabla^\beta (\nabla^\alpha f \nabla_\alpha f) = \frac{1}{2} \nabla^\beta \psi(f) = \frac{1}{2} \psi' \nabla^\beta f. \quad (1.5.5)$$

Let λ be the natural parameter on the integral curves of ∇f ,

$$\frac{d\gamma^\mu}{d\lambda} = \nabla^\mu f;$$

then (1.5.5) can be rewritten as

$$\frac{D}{d\lambda} \frac{d\gamma^\mu}{d\lambda} = \frac{1}{2} \psi' \frac{d\gamma^\mu}{d\lambda}. \quad \square$$

A significant special case is that of a coordinate function $f = x^i$. Then

$$g(\nabla f, \nabla f) = g(\nabla x^i, \nabla x^i) = g^{ii} \text{ (no summation)}.$$

For example, in Minkowski spacetime, all $g^{\mu\nu}$'s are constant, which shows that the integral curves of the gradient of any such coordinate, and hence also of any linear combination of any such coordinates, are affinely parameterized geodesics. Another example is provided by the coordinate r in Schwarzschild spacetime, where $g^{rr} = 1 - 2m/r$; this is indeed a function of r , so the integral curves of $\nabla r = (1 - 2m/r)\partial_r$ are (non-affinely parameterized) geodesics.

A vector field X is called a *Killing vector* if the local flow of X generates isometries. Equivalently,

$$\mathcal{L}_X g = 0 \iff \nabla_\alpha X_\beta + \nabla_\beta X_\alpha = 0. \quad (1.5.6)$$

Similarly to the proof of Proposition 1.5.1 one shows the following:

PROPOSITION 1.5.2 *Suppose that $d(g(X, X)) = 0$ along an orbit γ of a Killing vector field X . Then γ is a geodesic.*

1.6 Moving frames

A formalism which is very convenient for practical calculations is that of *moving frames*; it also plays a key role when considering spinors. By definition, a moving frame is a (locally defined) field of bases $\{e_a\}$ of TM such that the scalar products

$$g_{ab} := g(e_a, e_b) \quad (1.6.1)$$

are point independent. In most standard applications one assumes that the e_a 's form an orthonormal basis, so that g_{ab} is a diagonal matrix with plus and minus ones on the diagonal. However, it is sometimes convenient to allow other such frames, e.g. with isotropic vectors being members of the frame.

It is customary to denote by ω^a_{bc} the associated connection coefficients,

$$\omega^a_{bc} := \theta^a(\nabla_{e_c} e_b) \iff \nabla_X e_b = \omega^a_{bc} X^c e_a, \quad (1.6.2)$$

where, as elsewhere, $\{\theta^a(p)\}$ is a basis of T_p^*M dual to $\{e_a(p)\} \subset T_pM$; we will refer to θ^a as a *coframe*. The *connection one forms* ω^a_b are defined as

$$\omega^a_b(X) := \theta^a(\nabla_X e_b) \iff \nabla_X e_b = \omega^a_b(X) e_a. \quad (1.6.3)$$

As always we use the metric to raise and lower indices, even though the ω^a_{bc} 's do not form a tensor, so that

$$\omega_{abc} := g_{ad} \omega^d_{bc}, \quad \omega_{ab} := g_{ad} \omega^d_b. \quad (1.6.4)$$

When ∇ is metric compatible, the ω_{ab} 's are anti-antisymmetric: indeed, as the g_{ab} 's are point independent, for any vector field X we have

$$\begin{aligned}
0 &= X(g_{ab}) = X(g(e_a, e_b)) = g(\nabla_X e_a, e_b) + g(e_a, \nabla_X e_b) \\
&= g(\omega^c{}_a(X)e_c, e_b) + g(e_a, \omega^d{}_b(X)e_d) \\
&= g_{cb}\omega^c{}_a(X) + g_{ad}\omega^d{}_b(X) \\
&= \omega_{ba}(X) + \omega_{ab}(X).
\end{aligned}$$

Hence

$$\boxed{\omega_{ab} = -\omega_{ba} \iff \omega_{abc} = -\omega_{bac}}. \quad (1.6.5)$$

One can obtain a formula for the ω_{ab} 's in terms of Christoffels, the frame vectors and their derivatives: In order to see this, we note that

$$g(e_a, \nabla_{e_c} e_b) = g(e_a, \omega^d{}_{bc} e_d) = g_{ad} \omega^d{}_{bc} = \omega_{abc}. \quad (1.6.6)$$

Rewritten the other way round this gives an alternative equation for the ω 's with all indices down:

$$\omega_{abc} = g(e_a, \nabla_{e_c} e_b) \iff \omega_{ab}(X) = g(e_a, \nabla_X e_b). \quad (1.6.7)$$

Then, writing

$$e_a = e_a{}^\mu \partial_\mu,$$

we find that

$$\begin{aligned}
\omega_{abc} &= g(e_a{}^\mu \partial_\mu, e_c{}^\lambda \nabla_\lambda e_b) \\
&= g_{\mu\sigma} e_a{}^\mu e_c{}^\lambda (\partial_\lambda e_b{}^\sigma + \Gamma_{\lambda\nu}^\sigma e_b{}^\nu).
\end{aligned} \quad (1.6.8)$$

Next, it turns out that we can calculate the ω_{ab} 's in terms of the Lie brackets of the vector fields e_a , without having to calculate the Christoffel symbols. This shouldn't be too surprising, since an ON frame defines the metric uniquely. If ∇ has no torsion, from (1.6.7) we find that

$$\omega_{abc} - \omega_{acb} = g(e_a, \nabla_{e_c} e_b - \nabla_{e_b} e_c) = g(e_a, [e_c, e_b]).$$

We can now carry out the usual cyclic-permutation calculation to obtain

$$\begin{aligned}
\omega_{abc} - \omega_{acb} &= g(e_a, [e_c, e_b]), \\
-(\omega_{bca} - \omega_{bac}) &= -g(e_b, [e_a, e_c]), \\
-(\omega_{cab} - \omega_{cba}) &= -g(e_c, [e_b, e_a]).
\end{aligned}$$

So, if the connection is the Levi-Civita connection, summing the three equations and using (1.6.5) leads to

$$\boxed{\omega_{cba} = \frac{1}{2} \left(g(e_a, [e_c, e_b]) - g(e_b, [e_a, e_c]) - g(e_c, [e_b, e_a]) \right)}. \quad (1.6.9)$$

Equations (1.6.8)–(1.6.9) provide explicit expressions for the ω 's; yet another formula can be found in (1.6.11) below. While it is useful to know that there are such expressions, and while those expressions are useful for estimating things for PDE purposes, they are rarely used for practical calculations; see Example 1.6.3 for more comments about this.

It turns out that one can obtain a simple expression for the torsion tensor T of ω using exterior differentiation. Recall that if α is a one-form, then its exterior derivative $d\alpha$ can be calculated using the formula

$$d\alpha(X, Y) = X(\alpha(Y)) - Y(\alpha(X)) - \alpha([X, Y]). \quad (1.6.10)$$

14 Basic notions

EXERCISE 1.6.1 Use (1.6.9) and (1.6.10) to show that

$$\omega_{cba} = \frac{1}{2} \left(-g_{ad} d\theta^d(e_c, e_b) + g_{bd} d\theta^d(e_a, e_c) + g_{cd} d\theta^d(e_b, e_a) \right). \quad (1.6.11)$$

□

We set

$$T^a(X, Y) := \theta^a(T(X, Y)),$$

and using (1.6.10) together with the definition (B.7) of the torsion tensor T we calculate as follows:

$$\begin{aligned} T^a(X, Y) &= \theta^a(\nabla_X Y - \nabla_Y X - [X, Y]) \\ &= X(Y^a) + \omega^a_b(X)Y^b - Y(X^a) - \omega^a_b(Y)X^b - \theta^a([X, Y]) \\ &= X(\theta^a(Y)) - Y(\theta^a(X)) - \theta^a([X, Y]) + \omega^a_b(X)\theta^b(Y) - \omega^a_b(Y)\theta^b(X) \\ &= d\theta^a(X, Y) + (\omega^a_b \wedge \theta^b)(X, Y). \end{aligned}$$

It follows that

$$T^a = d\theta^a + \omega^a_b \wedge \theta^b. \quad (1.6.12)$$

In particular when the torsion vanishes we obtain the so-called *Cartan's first structure equation*

$$\boxed{d\theta^a + \omega^a_b \wedge \theta^b = 0}. \quad (1.6.13)$$

EXAMPLE 1.6.2 As a simple example, we consider a two-dimensional metric of the form

$$g = dx^2 + e^{2f} dy^2, \quad (1.6.14)$$

where f could possibly depend upon x and y . A natural frame is given by

$$\theta^1 = dx, \quad \theta^2 = e^f dy.$$

The first Cartan structure equations read

$$0 = \underbrace{d\theta^1}_0 + \omega^1_b \wedge \theta^b = \omega^1_2 \wedge \theta^2,$$

since $\omega^1_1 = \omega_{11} = 0$ by antisymmetry, and

$$0 = \underbrace{d\theta^2}_{e^f \partial_x f dx \wedge dy} + \omega^2_b \wedge \theta^b = \partial_x f \theta^1 \wedge \theta^2 + \omega^2_1 \wedge \theta^1.$$

It should then be clear that both equations can be solved by choosing ω_{12} proportional to θ^2 , and such an ansatz leads to

$$\omega_{12} = -\omega_{21} = -\partial_x f \theta^2 = -\partial_x(e^f) dy. \quad (1.6.15)$$

We continue this example in Example 1.6.5, p. 16. □

EXAMPLE 1.6.3 As another example of the moving frame technique we consider (the most general) three-dimensional spherically symmetric metric

$$g = e^{2\gamma(r)} d\tilde{r}^2 + e^{2\gamma(r)} d\theta^2 + e^{2\gamma(r)} \sin^2 \theta d\varphi^2. \quad (1.6.16)$$

There is an obvious choice of ON coframe for g given by

$$\theta^1 = e^{\beta(r)} dr, \quad \theta^2 = e^{\gamma(r)} d\theta, \quad \theta^3 = e^{\gamma(r)} \sin \theta d\varphi, \quad (1.6.17)$$

leading to

$$g = \theta^1 \otimes \theta^1 + \theta^2 \otimes \theta^2 + \theta^3 \otimes \theta^3,$$

so that the frame e_a dual to the θ^a 's will be ON, as desired:

$$g_{ab} = g(e_a, e_b) = \text{diag}(1, 1, 1).$$

The idea of the calculation which we are about to do is the following: there is only one connection which is compatible with the metric, and which is torsion free. If we find a set of one-forms ω_{ab} which exhibit the properties just mentioned, then they must be the connection forms of the Levi-Civita connection. As shown in the calculation leading to (1.6.5), the compatibility with the metric will be ensured if we require

$$\omega_{11} = \omega_{22} = \omega_{33} = 0,$$

$$\omega_{12} = -\omega_{21}, \quad \omega_{13} = -\omega_{31}, \quad \omega_{23} = -\omega_{32}.$$

Next, we have the equations for the torsion vanishing:

$$\begin{aligned} 0 = d\theta^1 &= -\underbrace{\omega^1_1}_{=0} \wedge \theta^1 - \omega^1_2 \wedge \theta^2 - \omega^1_3 \wedge \theta^3 \\ &= -\omega^1_2 \wedge \theta^2 - \omega^1_3 \wedge \theta^3, \\ d\theta^2 &= \gamma' e^\gamma dr \wedge d\theta = \gamma' e^{-\beta} \theta^1 \wedge \theta^2 \\ &= -\underbrace{\omega^2_1}_{=-\omega^1_2} \wedge \theta^1 - \underbrace{\omega^2_2}_{=0} \wedge \theta^2 - \omega^2_3 \wedge \theta^3 \\ &= \omega^1_2 \wedge \theta^1 - \omega^2_3 \wedge \theta^3, \\ d\theta^3 &= \gamma' e^\gamma \sin \theta dr \wedge d\varphi + e^\gamma \cos \theta d\theta \wedge d\varphi = \gamma' e^{-\beta} \theta^1 \wedge \theta^3 + e^{-\gamma} \cot \theta \theta^2 \wedge \theta^3 \\ &= -\underbrace{\omega^3_1}_{=-\omega^1_3} \wedge \theta^1 - \underbrace{\omega^3_2}_{=-\omega^2_3} \wedge \theta^2 - \underbrace{\omega^3_3}_{=0} \wedge \theta^3 \\ &= \omega^1_3 \wedge \theta^1 + \omega^2_3 \wedge \theta^2. \end{aligned}$$

Summarizing,

$$\begin{aligned} -\omega^1_2 \wedge \theta^2 - \omega^1_3 \wedge \theta^3 &= 0, \\ \omega^1_2 \wedge \theta^1 &- \omega^2_3 \wedge \theta^3 = \gamma' e^{-\beta} \theta^1 \wedge \theta^2, \\ \omega^1_3 \wedge \theta^1 + \omega^2_3 \wedge \theta^2 &= \gamma' e^{-\beta} \theta^1 \wedge \theta^3 + e^{-\gamma} \cot \theta \theta^2 \wedge \theta^3. \end{aligned}$$

It should be clear from the first and second lines that an ω^1_2 proportional to θ^2 should do the job; similarly from the first and third lines one sees that an ω^1_3 proportional to θ^3 should work. It is then easy to find the relevant coefficient, as well as to find ω^2_3 :

$$\omega^1_2 = -\gamma' e^{-\beta} \theta^2 = -\gamma' e^{-\beta+\gamma} d\theta, \quad (1.6.18a)$$

$$\omega^1_3 = -\gamma' e^{-\beta} \theta^3 = -\gamma' e^{-\beta+\gamma} \sin \theta d\varphi, \quad (1.6.18b)$$

$$\omega^2_3 = -e^{-\gamma} \cot \theta \theta^3 = -\cos \theta d\varphi. \quad (1.6.18c)$$

We continue this example in Example 1.6.6, p. 16. □

It is convenient to define *curvature two-forms*:

$$\Omega^a_b = R^a_{bcd} \theta^c \otimes \theta^d = \frac{1}{2} R^a_{bcd} \theta^c \wedge \theta^d. \quad (1.6.19)$$

The *second Cartan structure equation* reads

$$\boxed{\Omega^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b}. \quad (1.6.20)$$

This identity is easily verified using (1.6.10):

$$\begin{aligned}
\Omega^a{}_b(X, Y) &= \frac{1}{2} R^a{}_{bcd} \underbrace{\theta^c \wedge \theta^d}_{=X^c Y^d - X^d Y^c}(X, Y) \\
&= R^a{}_{bcd} X^c Y^d \\
&= \theta^a (\nabla_X \nabla_Y e_b - \nabla_Y \nabla_X e_b - \nabla_{[X, Y]} e_b) \\
&= \theta^a (\nabla_X (\omega^c{}_b(Y) e_c) - \nabla_Y (\omega^c{}_b(X) e_c) - \omega^c{}_b([X, Y]) e_c) \\
&= \theta^a \left(X(\omega^c{}_b(Y)) e_c + \omega^c{}_b(Y) \nabla_X e_c \right. \\
&\quad \left. - Y(\omega^c{}_b(X)) e_c - \omega^c{}_b(X) \nabla_Y e_c - \omega^c{}_b([X, Y]) e_c \right) \\
&= X(\omega^a{}_b(Y)) + \omega^c{}_b(Y) \omega^a{}_c(X) \\
&\quad - Y(\omega^a{}_b(X)) - \omega^c{}_b(X) \omega^a{}_c(Y) - \omega^a{}_b([X, Y]) \\
&= \underbrace{X(\omega^a{}_b(Y)) - Y(\omega^a{}_b(X)) - \omega^a{}_b([X, Y])}_{=d\omega^a{}_b(X, Y)} \\
&\quad + \omega^a{}_c(X) \omega^c{}_b(Y) - \omega^a{}_c(Y) \omega^c{}_b(X) \\
&= (d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b)(X, Y).
\end{aligned}$$

Equation (1.6.20) provides an efficient way of calculating the curvature tensor of any metric.

EXAMPLE 1.6.4 In dimension two the only non-vanishing components of $\omega^a{}_b$ are $\omega^1{}_2 = -\omega^2{}_1$, and it follows from (1.6.20) that

$$\Omega^1{}_2 = d\omega^1{}_2 + \omega^1{}_a \wedge \omega^a{}_2 = d\omega^1{}_2. \quad (1.6.21)$$

In particular, denoting by $d\mu_g$ the volume form, and assuming that θ^2 is dual to a spacelike vector, whatever the signature of the metric,

$$\begin{aligned}
R d\mu_g &= R \theta^1 \wedge \theta^2 = 2R^{12}{}_{12} \theta^1 \wedge \theta^2 = R^1{}_{2ab} \theta^a \wedge \theta^b = 2\Omega^1{}_2 \\
&= 2d\omega^1{}_2.
\end{aligned} \quad (1.6.22)$$

□

EXAMPLE 1.6.5 (Example 1.6.2 continued) We have seen that the connection one-forms for the metric

$$g = dx^2 + e^{2f} dy^2 \quad (1.6.23)$$

read

$$\omega_{12} = -\omega_{21} = -\partial_x f \theta^2 = -\partial_x(e^f) dy.$$

By symmetry the only non-vanishing curvature two-forms are $\Omega_{12} = -\Omega_{21}$. From (1.6.20) we find that

$$\begin{aligned}
\Omega_{12} &= d\omega_{12} + \underbrace{\omega_{1b} \wedge \omega^b{}_2}_{= \omega_{12} \wedge \omega^2{}_2 = 0} = -\partial_x^2(e^f) dx \wedge dy = -e^{-f} \partial_x^2(e^f) \theta^1 \wedge \theta^2.
\end{aligned}$$

We conclude that

$$R_{1212} = -e^{-f} \partial_x^2(e^f). \quad (1.6.24)$$

For instance, if g is the unit round metric on the two-sphere, then $e^f = \sin x$, and $R_{1212} = 1$. If $e^f = \sinh x$, then g is the canonical metric on hyperbolic space, and $R_{1212} = -1$. Finally, the function $e^f = \cosh x$ defines a *hyperbolic wormhole*, with again $R_{1212} = -1$. □

EXAMPLE 1.6.6 (Example 1.6.3 continued) From (1.6.18) we find that

$$\begin{aligned}
\Omega^1{}_2 &= d\omega^1{}_2 + \underbrace{\omega^1{}_1 \wedge \omega^1{}_2}_{=0} + \omega^1{}_2 \wedge \underbrace{\omega^2{}_2}_{=0} + \underbrace{\omega^1{}_3 \wedge \omega^3{}_2}_{\sim \theta^3 \wedge \theta^3 = 0} \\
&= -d(\gamma' e^{-\beta+\gamma} d\theta) \\
&= -(\gamma' e^{-\beta+\gamma})' dr \wedge d\theta
\end{aligned}$$

$$\begin{aligned}
&= -(\gamma' e^{-\beta+\gamma})' e^{-\beta-\gamma} \theta^1 \wedge \theta^2 \\
&= \sum_{a < b} R^1{}_{2ab} \theta^a \wedge \theta^b,
\end{aligned}$$

which shows that the only non-trivial coefficient (up to permutations) with the pair 12 in the first two slots is

$$R^1{}_{212} = -(\gamma' e^{-\beta+\gamma})' e^{-\beta-\gamma}. \quad (1.6.25)$$

A similar calculation, or arguing by symmetry, leads to

$$R^1{}_{313} = -(\gamma' e^{-\beta+\gamma})' e^{-\beta-\gamma}. \quad (1.6.26)$$

Finally,

$$\begin{aligned}
\Omega^2{}_3 &= d\omega^2{}_3 + \omega^2{}_1 \wedge \omega^1{}_3 + \underbrace{\omega^2{}_2}_{=0} \wedge \omega^2{}_3 + \omega^2{}_3 \wedge \underbrace{\omega^3{}_3}_{=0} \\
&= -d(\cos \theta d\varphi) + (\gamma' e^{-\beta} \theta^2) \wedge (-\gamma' e^{-\beta} \theta^3) \\
&= (e^{-2\gamma} - (\gamma')^2 e^{-2\beta}) \theta^2 \wedge \theta^3,
\end{aligned}$$

yielding

$$R^2{}_{323} = e^{-2\gamma} - (\gamma')^2 e^{-2\beta}. \quad (1.6.27)$$

The curvature scalar can easily be calculated now to be

$$\begin{aligned}
R &= R^{ij}{}_{ij} = 2(R^{12}{}_{12} + R^{13}{}_{13} + R^{23}{}_{23}) \\
&= -4(\gamma' e^{-\beta+\gamma})' e^{-\beta-\gamma} + 2(e^{-2\gamma} - (\gamma')^2 e^{-2\beta}).
\end{aligned} \quad (1.6.28)$$

□

EXAMPLE 1.6.7 Consider an n -dimensional Riemannian metric of the form

$$g = e^{2h(r)} d\tilde{r}^2 + e^{2f(r)} \underbrace{\mathring{h}_{AB}(x^C) dx^A dx^B}_{=: \mathring{h}}. \quad (1.6.29)$$

Let $\hat{\theta}^A$ be an ON -frame for $\mathring{h}$, with corresponding connection coefficients $\mathring{\omega}^A{}_B$:

$$d\hat{\theta}^A + \mathring{\omega}^A{}_B \wedge \hat{\theta}^B = 0.$$

Set

$$\theta^1 = e^h dr, \quad \theta^A = e^f \hat{\theta}^A.$$

Then the first structure equations,

$$\begin{aligned}
&\underbrace{d\theta^1}_0 + \omega^1{}_B \wedge \theta^B = 0, \\
d(e^f \hat{\theta}^A) + e^h \omega^A{}_1 \wedge dr + e^f \omega^A{}_B \wedge \hat{\theta}^B &= 0,
\end{aligned}$$

are solved by

$$\omega^A{}_1 = e^{-h} (e^f)' \hat{\theta}^A, \quad \omega^A{}_B = \mathring{\omega}^A{}_B. \quad (1.6.30)$$

This leads to

$$\begin{aligned}
\Omega^A{}_1 &= d\omega^A{}_1 + \omega^A{}_B \wedge \omega^B{}_1 \\
&= e^{-h-f} (e^{-h} (e^f)')' \theta^1 \wedge \theta^A,
\end{aligned} \quad (1.6.31)$$

$$\Omega^A{}_B = \mathring{\Omega}^A{}_B - e^{-2h-2f} ((e^f)')^2 \theta^A \wedge \theta_B, \quad (1.6.32)$$

where $\mathring{\Omega}^A{}_B$ are the curvature two-forms of the metric $\mathring{h}$,

$$\mathring{\Omega}^A{}_B = \frac{1}{2} \mathring{R}^A{}_{BCD} \hat{\theta}^A \wedge \hat{\theta}^B = \frac{e^{-2f}}{2} \mathring{R}^A{}_{BCD} \theta^A \wedge \theta^B. \quad (1.6.33)$$

Hence

$$R^A{}_{1B1} = -e^{-h-f} (e^{-h} (e^f)')' \delta^A{}_B, \quad (1.6.34)$$

$$R^A{}_{1BC} = 0, \quad (1.6.35)$$

$$R^A{}_{BCD} = e^{-2f} \dot{R}^A{}_{BCD} - e^{-2h-2f} ((e^f)')^2 \delta_{[C}^A g_{D]B}, \quad (1.6.36)$$

$$R^A{}_C = -e^{-h-f} \left((e^{-h}(e^f)')' + (n-2)e^{-h-f} ((e^f)')^2 \right) \delta_C^A \\ + e^{-2f} \dot{R}^A{}_C, \quad (1.6.37)$$

$$R^1{}_1 = -(n-1)e^{-h-f} (e^{-h}(e^f)')', \quad (1.6.38)$$

$$R^1{}_B = 0, \quad (1.6.39)$$

$$R = -(n-1)e^{-h-f} \left(2(e^{-h}(e^f)')' + (n-2)e^{-h-f} ((e^f)')^2 \right) \\ + e^{-2f} \dot{R}. \quad (1.6.40)$$

Let g be the space-part of the Birmingham metrics of Section 5.5, p. 250, thus g takes the form (1.6.29) with

$$e^f = r, \quad e^{-2h} = \beta - \frac{2m}{r^{n-2}} - \epsilon \frac{\tilde{r}^2}{\ell^2}, \quad \epsilon \in \{0, \pm 1\}, \quad (1.6.41)$$

where β , m , and ℓ are real constants. Then

$$R^A{}_{1B1} = \left(\frac{\epsilon}{\ell^2} - m(n-2)r^{-n} \right) \delta_B^A, \quad (1.6.42)$$

$$R^A{}_{1BC} = 0 = R^1{}_B, \quad (1.6.43)$$

$$R^A{}_{BCD} = \dot{R}^A{}_{BCD} - \left(\frac{\beta}{\tilde{r}^2} - \frac{\epsilon}{\ell^2} + 2mr^{-n} \right) \delta_{[C}^A g_{D]B}, \quad (1.6.44)$$

$$R^A{}_B = \frac{\ell^2(n-2)(m\tilde{r}^2 - \beta r^n) + \epsilon(n-1)r^{n+2}}{\ell^2 r^{n+2}} \delta_B^A \\ + r^{-2} \dot{R}^A{}_B, \quad (1.6.45)$$

$$R^1{}_1 = (n-1) \left(\frac{\epsilon}{\ell^2} - m(n-2)r^{-n} \right), \quad (1.6.46)$$

$$R = \frac{(n-1)(\epsilon n \tilde{r}^2 - \beta \ell^2(n-2))}{\ell^2 \tilde{r}^2} + r^{-2} \dot{R}. \quad (1.6.47)$$

If $\dot{h}$ is Einstein, with

$$\dot{R}_{AB} = (n-2)\beta \dot{h}_{AB}, \quad (1.6.48)$$

the last formulae above simplify to

$$R^A{}_B = \frac{\ell^2(n-2)r^{-n-2}(\beta(n-2)r^n + m\tilde{r}^2) + \epsilon(n-1)}{\ell^2} \delta_B^A, \quad (1.6.49)$$

$$R = \frac{\epsilon(n-1)n}{\ell^2}. \quad (1.6.50)$$

We see that the scalar curvature R is constant, consistently with the general relativistic scalar constraint equation (G.1), p. 363, in Appendix G. $\square$

EXAMPLE 1.6.8 We can use (1.6.11),

$$\omega_{cba} = \frac{1}{2} \left(-g_{ad} d\theta^d(e_c, e_b) + g_{bd} d\theta^d(e_a, e_c) + g_{cd} d\theta^d(e_b, e_a) \right), \quad (1.6.51)$$

to determine how the curvature tensor transforms under conformal rescalings. For this let $g = \eta_{ab} \theta^a \theta^b$ with $d\eta_{ab} = 0$, and let

$$\bar{g} = e^{2f} g = \eta_{ab} \underbrace{e^f \theta^a}_{=: \bar{\theta}^a} \otimes e^f \theta^b \equiv \eta_{ab} \bar{\theta}^a \bar{\theta}^b. \quad (1.6.52)$$

If the vector fields $\{e_a\}$ form a basis dual to the basis $\{\theta^a\}$, then the vector fields $\bar{e}_a = e^{-f} e_a$ provide a basis dual to $\{\bar{\theta}^b\}$,

$$\begin{aligned}
\bar{\omega}_{cba} &= \frac{1}{2} \left(-\eta_{ad} d(e^f \theta^d)(e^{-f} e_c, e^{-f} e_b) \right. \\
&\quad \left. + \eta_{bd} d(e^f \theta^d)(e^{-f} e_a, e^{-f} e_c) + \eta_{cd} d(e^f \theta^d)(e^{-f} e_b, e^{-f} e_a) \right) \\
&= e^{-f} \left(\omega_{cba} + \frac{1}{2} \left(-\eta_{ad} (df \wedge \theta^d)(e_c, e_b) + \eta_{bd} (df \wedge \theta^d)(e_a, e_c) \right. \right. \\
&\quad \left. \left. + \eta_{cd} (df \wedge \theta^d)(e_b, e_a) \right) \right) \\
&= e^{-f} (\omega_{cba} - \eta_{a[b} e_{c]}(f) + \eta_{b[c} e_{a]}(f) + \eta_{c[a} e_{b]}(f)) \\
&= e^{-f} (\omega_{cba} - \eta_{ab} e_c(f) + \eta_{ac} e_b(f)) .
\end{aligned} \tag{1.6.53}$$

Equivalently,

$$\bar{\omega}_{cb} = \bar{\omega}_{cba} \bar{\theta}^a = e^f \bar{\omega}_{cba} \theta^a = \omega_{cb} + (\eta_{ac} \nabla_b f - \eta_{ab} \nabla_c f) \theta^a . \tag{1.6.54}$$

Taking the exterior derivative one finds that

$$\begin{aligned}
\bar{\Omega}_{cb} &= \Omega_{cb} + \left(\eta_{ab} \nabla_d \nabla_c f - \eta_{ac} \nabla_d \nabla_b f \right. \\
&\quad \left. + \eta_{ac} \nabla_d f \nabla_b f + \eta_{db} \nabla_c f \nabla_a f - \eta_{ac} \eta_{db} |df|_g^2 \right) \theta^a \wedge \theta^d .
\end{aligned} \tag{1.6.55}$$

Re-expressed in terms of the Riemann tensor, this reads as

$$\begin{aligned}
e^{2f} \bar{R}_{cbad} &= R_{cbad} + 2 \left(\eta_{b[a} \nabla_{d]} \nabla_c f - \eta_{c[a} \nabla_{d]} \nabla_b f \right. \\
&\quad \left. + \eta_{c[a} \nabla_{d]} f \nabla_b f + \eta_{b[d} \nabla_{a]} f \nabla_c f - \eta_{c[a} \eta_{d]b} |df|_g^2 \right) ,
\end{aligned} \tag{1.6.56}$$

where the components $\bar{R}_{cbad}$ are taken with respect to a $\bar{g}$ -ON frame, and all components of the right-hand side are taken with respect to a g -ON frame. Taking traces we obtain, in dimension d ,

$$e^{2f} \bar{R}_{ac} = R_{ac} + (2-d) \left(\nabla_a \nabla_c f - \nabla_a f \nabla_c f + |df|_g^2 \eta_{ac} \right) - \Delta_g f \eta_{ac} , \tag{1.6.57}$$

$$e^{2f} \bar{R} = R + (1-d) \left(2\Delta_g f + (d-2) |df|_g^2 \right) . \tag{1.6.58}$$

□

The Bianchi identities have a particularly simple proof in the moving frame formalism. For this, let ψ^a be any vector-valued differential form, and define

$$D\psi^a = d\psi^a + \omega^a_b \wedge \psi^b . \tag{1.6.59}$$

Thus, in this notation the vanishing of torsion reads as

$$D\theta^a = 0 . \tag{1.6.60}$$

Whether or not the torsion vanishes, we find that

$$\begin{aligned}
D\tau^a &= d\tau^a + \omega^a_b \wedge \tau^b = d(d\theta^a + \omega^a_b \wedge \theta^b) + \omega^a_c \wedge (d\theta^c + \omega^c_b \wedge \theta^b) \\
&= d\omega^a_b \wedge \theta^b - \omega^a_b \wedge d\theta^b + \omega^a_c \wedge (d\theta^c + \omega^c_b \wedge \theta^b) \\
&= \Omega^a_b \wedge \theta^b .
\end{aligned}$$

If the torsion vanishes the left-hand side is zero, and we find that

$$\Omega^a_b \wedge \theta^b = 0 . \tag{1.6.61}$$

This is equivalent to the first Bianchi identity:

$$0 = \Omega^a_b \wedge \theta^b = \frac{1}{2} R^a_{bcd} \theta^c \wedge \theta^d \wedge \theta^b = R^a_{[bcd]} \theta^c \wedge \theta^d \wedge \theta^b \iff R^a_{[bcd]} = 0. \quad (1.6.62)$$

Next, for any differential form α_b with two-frame indices, such as the curvature two-form, we define

$$D\alpha^a_b := d\alpha^a_b + \omega^a_c \wedge \alpha^c_b - \omega^c_b \wedge \alpha^a_c. \quad (1.6.63)$$

(The reader will easily workout the obvious generalization of this definition to differential forms with any number of frame indices.) For the curvature two-form we find that

$$\begin{aligned} D\Omega^a_b &= d(d\omega^a_b + \omega^a_c \wedge \omega^c_b) + \omega^a_c \wedge \Omega^c_b - \omega^c_b \wedge \Omega^a_c \\ &= d\omega^a_c \wedge \omega^c_b - \omega^a_c \wedge d\omega^c_b + \omega^a_c \wedge \Omega^c_b - \omega^c_b \wedge \Omega^a_c \\ &= (\Omega^a_c - \omega^a_e \wedge \omega^e_c) \wedge \omega^c_b - \omega^a_c \wedge (\Omega^c_b - \omega^c_e \wedge \omega^e_b) \\ &\quad + \omega^a_c \wedge \Omega^c_b - \omega^c_b \wedge \Omega^a_c = 0. \end{aligned}$$

Thus

$$D\Omega^a_b = 0. \quad (1.6.64)$$

Let us show that this is equivalent to the second Bianchi identity:

$$\begin{aligned} 0 &= D\Omega^a_b = \frac{1}{2} R^a_{bc;d} \theta^d \wedge \theta^b \wedge \theta^c = \frac{1}{2} R^a_{[bc;d]} \theta^d \wedge \theta^b \wedge \theta^c \\ &\iff R^a_{[bc;d]} = 0. \end{aligned} \quad (1.6.65)$$

Here the only not-obviously-apparent fact is, if any, the second equality in the first line of (1.6.65):

$$\begin{aligned} D\Omega^a_b &= \frac{1}{2} \left(d(R^a_{bef} \theta^e \wedge \theta^f) + \omega^a_c \wedge R^c_{bef} \theta^e \wedge \theta^f - \omega^c_b \wedge R^a_{cef} \theta^e \wedge \theta^f \right) \\ &= \frac{1}{2} \left(\underbrace{dR^a_{bef}}_{e_k(R^a_{bef}) \theta^k} \wedge \theta^e \wedge \theta^f + R^a_{bef} \underbrace{d\theta^e}_{-\omega^e_k \wedge \theta^k} \wedge \theta^f + R^a_{bef} \theta^e \wedge \underbrace{d\theta^f}_{-\omega^f_k \wedge \theta^k} \right. \\ &\quad \left. + R^c_{bef} \omega^a_c \wedge \theta^e \wedge \theta^f - R^a_{cef} \omega^c_b \wedge \theta^e \wedge \theta^f \right) \\ &= \frac{1}{2} \nabla_{e_k} R^a_{bef} \theta^k \wedge \theta^e \wedge \theta^f, \end{aligned} \quad (1.6.66)$$

as desired.

2

Elements of causality

A standard part of studies of black holes, and in fact of mathematical general relativity, is *causality theory*, which is the study of causal relations on Lorentzian manifolds. An essential issue here is the understanding of the influence of energy conditions on those relations. The highlights of such studies include the incompleteness theorems, known also as singularity theorems, of Penrose, Hawking, and Geroch, the area theorem of Hawking, and the topology theorems of Hawking and others; see Chapter 3 for references and details.

The aim of this chapter is to provide an introduction to the subject with a complete exposition of those topics which are needed for the global treatment of the general relativistic Cauchy problem, and for the uniqueness theory of black holes.

INCIDENTALLY: Alternative, and sometimes more complete expositions of the subject for smooth metrics can be found in [38, 244, 300, 355, 356, 383, 398]. A presentation of some aspects of the theory for C^2 metrics, in a spirit somewhat similar to ours, appeared in [214]. An alternative approach to the theory based on subsets of $\mathcal{M} \times \mathcal{M}$ rather than subsets of $\mathcal{M}$ can be found in [349, 350]. $\square$

Unless explicitly indicated otherwise, or clear from the context, we consider manifolds equipped with a C^1 atlas and a continuous metric. Now, some of the constructions in what follows, such as approximations by smooth fields, might require a smooth atlas. For this, recall that a C^1 atlas can always be replaced by an equivalent smooth one, and we might make this replacement without further comments when needed.

The above notwithstanding, our aim here is to provide a coherent introduction to causality theory for metrics which are C^2 and manifolds which are C^3 . This threshold can be traced back to the use of normal coordinates as a tool for understanding the local aspects of causality. However, for many considerations a metric of C^0 differentiability class suffices, and we will sometimes indicate explicitly the differentiability of the metric needed. Readers interested in causality theory for continuous metrics are invited to consult [125, 188, 233, 353].

2.1 Time orientation

Recall that at each point $p \in \mathcal{M}$ the set of timelike vectors in $T_p\mathcal{M}$ has precisely two components. A *time-orientation* of $T_p\mathcal{M}$ is the assignment of the name ‘future-pointing vectors’ to one of those components; vectors in the remaining component are then called ‘past pointing’. The set of future-pointing timelike, or causal, vectors is stable under addition and multiplication by positive numbers; similarly for past-pointing ones. (In particular this implies convexity.) In order to see this, suppose that $X = (X^0, \vec{X})$ and $Y = (Y^0, \vec{Y})$ are timelike future pointing, in an ON-frame this is equivalent to

$$|\vec{X}| < X^0, \quad |\vec{Y}| < Y^0,$$

and the inequality

$$|\vec{X} + \vec{Y}| \leq |\vec{X}| + |\vec{Y}| < X^0 + Y^0$$

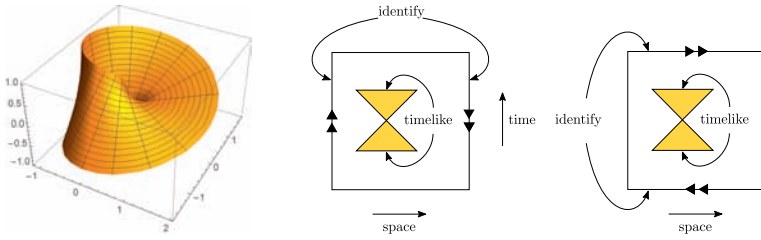


Fig. 2.1.1 The Möbius strip: embedded in $\mathbb{R}^3$ (left); with the flat metric $g = -dt^2 + dx^2$ (so that the light cones are at 45°), providing an example of a two-dimensional Lorentzian manifold which is not time-orientable (middle); with the flat metric $-g$ (right), which makes it time-orientable (but not space-orientable). I am grateful to Maciek Maliborski for providing the figures.

follows. Two timelike vectors X and Y have the same time orientation if and only if

$$g(X, Y) < 0; \quad (2.1.1)$$

this is immediate in an ON frame in which X is proportional to e_0 , so that $g(X, Y) = -X^0 Y^0$.

A time orientation of $T_p \mathcal{M}$ can always be propagated to a neighbourhood of p by choosing any continuous vector field X defined around p which is timelike and future pointing at p . By continuity of the metric and of X , the vector field X will be timelike in a sufficiently small neighbourhood $\mathcal{O}_p$ of p , and for $q \in \mathcal{O}_p$ one can define future-pointing vectors at q as those lying in the same component of the set of timelike vectors as $X(q)$: for $q \in \mathcal{O}_p$ the vector $Y \in T_q \mathcal{M}$ will be said to be timelike future pointing if and only if $g(Y, X(q)) < 0$. A Lorentzian manifold is said to be *time-orientable* if such locally defined time orientations can be defined globally in a consistent way; that is, we can cover $\mathcal{M}$ by coordinate neighbourhoods $\mathcal{O}_p$, each equipped with a vector field $X_{\mathcal{O}_p}$, such that $g(X_{\mathcal{O}_p}, X_{\mathcal{O}_q}) < 0$ on $\mathcal{O}_p \cap \mathcal{O}_q$.

Some Lorentzian manifolds will not be time-orientable, as is shown by the flat metric on the Möbius strip. Indeed, we can view the Möbius strip as the quotient of the strip

$$(t, x) \in (-1, 1) \times \mathbb{R}$$

equipped with the flat metric $g := -dt^2 + dx^2$, where the quotient map identifies (t, x) with $(-t, x + 1)$. On the resulting manifold there is no way of continuously distinguishing the futures and the pasts of points lying on, e.g., the loop

$$[0, 1] \ni s \mapsto [(t = 0, x = s)].$$

In two dimensions the tensor field $-g$ is a Lorentzian metric whenever g is: in this case the operation $g \rightarrow -g$ has the effect of interchanging the role of space and of time. The reader will notice that while the Möbius strip with the flat metric g of the middle figure in Fig. 2.1.1 is not time-orientable, it becomes time-orientable when equipped with $-g$.

On a time-orientable manifold there are precisely two choices of time orientation possible, and $(\mathcal{M}, g)$ will be said *time-oriented* when such a choice has been made. This leads us to the fundamental definition:

DEFINITION 2.1.1 A pair $(\mathcal{M}, g)$ will be called a *spacetime* if $(\mathcal{M}, g)$ is a time-oriented Lorentzian manifold. We will write $(\mathcal{M}, g)_{C^k}$ to denote a spacetime with a metric of C^k -differentiability class.

REMARK 2.1.2 A Lorentzian manifold $(\mathcal{M}, g)$ which is *not* time-orientable has a double cover which is [217]. The proof proceeds as follows: Choose any $p_0 \in \mathcal{M}$ and set

$$\widehat{\mathcal{M}} := \{(p, \gamma) : p \text{ is a point of } \mathcal{M} \text{ and } \gamma \text{ is a continuous curve from } p \text{ to } p_0\} / \sim,$$

where $\sim$ is the following equivalence relation: $(p, \gamma) \sim (p', \gamma')$ if $p = p'$ and if there exists a continuous timelike vector field defined along the curve obtained by first following γ from p to p_0 and then γ' from p_0 to $p' = p$. The usual arguments from the theory of covering spaces show that $\widehat{\mathcal{M}}$ can be equipped with a manifold structure, and covers $\mathcal{M}$ twice. $\widehat{\mathcal{M}}$ is then equipped with the pull-back $\hat{g}$ of g using the covering map; time-orientability of $(\widehat{\mathcal{M}}, \hat{g})$ should be clear. Furthermore, any time-orientable cover of $\mathcal{M}$ also covers $\widehat{\mathcal{M}}$, so $\widehat{\mathcal{M}}$ can be thought of as the smallest time-oriented covering of $\mathcal{M}$. $\square$

On any spacetime there exists a globally defined future-directed timelike vector field. To show this, consider the locally defined timelike vector fields $X_{\mathcal{O}_p}$ defined on neighbourhoods $\mathcal{O}_p$ as described above. One can choose a locally finite covering of $\mathcal{M}$ by such neighbourhoods $\mathcal{O}_{p_i}$, $i \in \mathbb{N}$, and construct a globally defined vector field X on $\mathcal{M}$ by setting

$$X = \sum_i \phi_i X_{\mathcal{O}_{p_i}},$$

where the functions ϕ_i form a partition of unity dominated by the covering $\{\mathcal{O}_{p_i}\}_{i \in \mathbb{N}}$. The resulting vector field will be timelike future pointing everywhere, as follows from the fact that the sum of an arbitrary number of future-pointing timelike vectors is a future-pointing timelike vector.

Now, non-compact manifolds always admit a nowhere-vanishing vector field. For instance, [252, Theorem 4.8] shows that a non-compact manifold always admits a function f without critical points. Then ∇f provides the desired vector field.

Next, compact manifolds possess a nowhere-vanishing vector field if and only if they have vanishing Euler characteristic χ . Indeed, if M is a compact manifold, then the Poincaré–Hopf theorem (see, e.g., [236])¹ implies that the index of any smooth vector field, say X , on M satisfies

$$\text{index}(X) = \chi(M).$$

For a nowhere-vanishing vector field X one has $\text{index}(X) = 0$ and, hence, $\chi(M) = 0$. Conversely, if M has $\chi(M) = 0$, then any smooth vector field X on M is of index zero. A theorem of Hopf [257] then implies that there exists a non-vanishing vector field on M homotopic to X .

We conclude that the vanishing of the Euler characteristic is a necessary condition of topological nature for a compact manifold to admit a time-orientable Lorentzian metric.

Regardless of any topological issues, the following holds:

PROPOSITION 2.1.3 *A manifold $\mathcal{M}$ admits a spacetime structure if and only if there exists a nowhere-vanishing, continuous vector field on $\mathcal{M}$.*

Proof. The necessity of the existence of a nowhere-vanishing vector field on $\mathcal{M}$ has already been established. Conversely, suppose that such a vector field X exists, and let h be any Riemannian metric on $\mathcal{M}$. Then the formula

$$g(Y, Z) = h(Y, Z) - 2 \frac{h(Y, X)h(Z, X)}{h(X, X)} \quad (2.1.2)$$

¹In [236] the orientability of M is assumed. However, one can use a diffeomorphism to move all the zeros of a vector field to a small ball, in which case the question of orientability becomes clearly irrelevant. I am grateful to S. Jackowski for pointing out this argument.

defines a Lorentzian metric on $\mathcal{M}$. Finally, the existence of a globally defined time-like vector field X on a Lorentzian manifold $(\mathcal{M}, g)$ implies the time-orientability of $\mathcal{M}$ in the obvious way: choose $\mathcal{O}_p = \mathcal{M}$ and $X_{\mathcal{O}_p} = X$. $\square$

Summarizing, non-compact $\mathcal{M}$'s always admit both a Lorentzian metric and a spacetime structure. Now, because the Euler characteristic of a double-cover of $\mathcal{M}$ is zero if and only if that of $\mathcal{M}$ is, it follows from Remark 2.1.2 and Proposition 2.1.3 that compact $\mathcal{M}$'s admit a Lorentzian metric if and only if they have a vanishing Euler characteristic. For example, no Lorentzian metrics exist on S^2 .

2.2 Normal coordinates

Given a C^2 metric, for $p \in \mathcal{M}$ the exponential map

$$\exp_p : T_p \mathcal{M} \rightarrow \mathcal{M}$$

is defined as follows; if X is a vector in the tangent space $T_p \mathcal{M}$, then $\exp_p(X) \in \mathcal{M}$ is the point reached by following a geodesic with initial point p and initial tangent vector $X \in T_p \mathcal{M}$ for an affine distance one, provided that the geodesic in question can be continued that far. Now an affinely parameterized geodesic solves the equation

$$\nabla_{\dot{x}} \dot{x} = 0 \quad \Longleftrightarrow \quad \frac{d^2 x^\mu}{ds^2} = -\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds}, \quad (2.2.1)$$

where the $\Gamma_{\alpha\beta}^\mu$'s are the Christoffel symbols of the metric g , defined as

$$\Gamma_{\alpha\beta}^\mu := \frac{1}{2} g^{\mu\sigma} \left(\frac{\partial g_{\sigma\alpha}}{\partial x^\beta} + \frac{\partial g_{\sigma\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \right), \quad (2.2.2)$$

$$g^{\mu\sigma} := g^\#(dx^\mu, dx^\sigma), \quad g_{\alpha\beta} := g(\partial_\alpha, \partial_\beta). \quad (2.2.3)$$

Here, as elsewhere, we sometimes use the symbol $g^\#$ to denote the ‘contravariant metric’, that is, the metric on $T^* \mathcal{M}$ constructed out from g in the canonical way (the matrix $g^{\alpha\beta}$ is thus the matrix inverse to $g_{\alpha\beta}$). However, as already mentioned, it is usual in the literature to use the same symbol g for the metric $g^\#$ and for all other metrics induced by g on tensor bundles over $\mathcal{M}$, and we will often do so.

Equations (2.2.2)–(2.2.3) show that when the metric is of the $C^{1,1}$ differentiability class, then the Christoffel symbols are Lipschitz continuous, which guarantees local existence and uniqueness of the solutions of (2.2.1).² (Here $C^{k,\alpha}$ is the space of k times differentiable functions (or maps, or sections—whichever is the case should be clear from the context), the k th derivatives of which satisfy, locally, a Hölder condition of order α .) This leads to a mismatch in differentiability between the Cauchy problem and causality theory which has not been satisfactorily clarified so far.³

The domain $\mathcal{U}_p$ of $\exp_p$ is always the largest subset of $T_p \mathcal{M}$ on which the exponential map is defined. By construction, and by homogeneity properties of solutions of (2.2.1) under a linear change of parameterization (see (2.2.17)), the set $\mathcal{U}_p$ is star-shaped with respect to the origin (this means that if $X \in \mathcal{U}_p$, then we also have $\lambda X \in \mathcal{U}_p$ for all $\lambda \in [0, 1]$). When the metric is $C^{1,1}$, the continuity of solutions of ODEs upon initial values shows that $\mathcal{U}_p$ is an open neighbourhood of the origin of $T_p \mathcal{M}$.

²Examples of $C^{1,\alpha}$ metrics with non-unique null geodesics for $0 < \alpha < 1$ can be found in [96, Appendix F] and [125]; compare [241] for spacelike geodesics in a Riemannian context; see also [420].

³One can construct large classes of solutions to the Cauchy problem for the vacuum Einstein equations which are *not* of the $C^{1,1}$ differentiability class [22, 290, 291, 294, 431].

The exponential map is neither surjective nor injective in general. For example, on $\mathbb{R} \times S^1$ with the flat metric $-dt^2 + dx^2$, the ‘left-directed’ null geodesics $\Gamma_-(s) = (s, -s \bmod 2\pi)$ and the ‘right-directed’ null geodesics $\Gamma_+(s) = (s, s \bmod 2\pi)$ meet again after going each ‘half of the way around S^1 ’, and injectivity fails. An example of non-surjectivity is provided by the spacetime $(\mathcal{M}, g)$ obtained by removing a point q from Minkowski spacetime $\mathbb{R}^{1,n}$. Indeed, let γ be a timelike geodesic through q in $\mathbb{R}^{1,n}$, and let p be any point lying on γ to the past of q . Then all the points on γ lying to the future of q in Minkowski spacetime belong to the future of p in $(\mathcal{M}, g)$ but do not lie on any geodesic starting at q .

EXAMPLE 2.2.1 A striking example of non-surjectivity of the exponential map is obtained by letting $(\mathcal{M}, g)$ be the anti-de Sitter spacetime. We choose coordinates in which the metric takes the form

$$g = -W dt^2 + \frac{dr^2}{W} + r^2 d\Omega^2, \quad (2.2.4)$$

where $W = 1 + r^2$ and where $d\Omega^2$ is the unit round metric on S^{n-1} . Let

$$p \in \{t = 0\} \subset \mathcal{M},$$

and let γ be a future-directed timelike geodesic starting at p and parameterized by proper time. The isometry group of $(\mathcal{M}, g)$ acts transitively on the tangent bundle of $\{t = 0\}$, which implies that to understand causal geodesics in $(\mathcal{M}, g)$ it suffices to consider geodesics of the form $\gamma(s) = (t(s), r(s), v^A)$ with $t(0) = r(0) = 0$ and $dv^A/ds = 0$, where the v^A ’s are local coordinates on S^{n-1} . Since ∂_t is a Killing vector the scalar product

$$E := -g(\partial_t, \dot{\gamma}) = W \frac{dt}{ds}$$

is a constant of motion. We have $E \geq W$ by the inverse Cauchy–Schwarz inequality. From $g(\dot{\gamma}, \dot{\gamma}) = -1$ one has

$$\begin{aligned} -1 &= -W \left(\frac{dt}{ds} \right)^2 + \frac{1}{W} \left(\frac{dr}{ds} \right)^2 \iff \\ \left(\frac{dr}{ds} \right)^2 &= E^2 - W = -r^2 + E^2 - 1. \end{aligned} \quad (2.2.5)$$

The associated second-order equation for a geodesic can be found by differentiating (2.2.5),

$$\frac{d^2 r}{ds^2} = -r, \quad (2.2.6)$$

leading to

$$r(s) = \sqrt{E^2 - 1} \sin(s). \quad (2.2.7)$$

Hence all timelike geodesics return to their space-departure-point after a proper time $s = \pi$. When reaching the return point, the coordinate t equals

$$\begin{aligned} t(\pi) &= \int_0^\pi \frac{dt}{ds} ds = E \int_0^\pi \frac{ds}{W(r(s))} = E \int_0^\pi \frac{ds}{(E^2 - 1) \sin^2(s) + 1} \\ &= E \int_0^\pi \frac{ds}{E^2 \sin^2(s) + \cos^2(s)} = \pi. \end{aligned} \quad (2.2.8)$$

We conclude that all timelike geodesics through $p = (\tau, r, v^A)$ meet again at $p' := (\tau + \pi, r, v^A)$.

Now, it will be seen later that, quite generally, the timelike future of p is open for smooth metrics; in particular, it contains a neighbourhood of p' within the level set $\{t = \tau + \pi\}$. And we have just seen that the only point within this level set which lies on a timelike geodesic emanating from p is p' . Hence the exponential map is not surjective from timelike vectors at p to the timelike future of p .

(In fact, for future-directed *null* radial geodesics one gets directly

$$0 = -W \left(\frac{dt}{ds} \right)^2 + \frac{1}{W} \left(\frac{dr}{ds} \right)^2 \iff \frac{dr}{dt} = W, \quad (2.2.9)$$

leading to

$$t(r) = \tan(r).$$

This implies that every point $(t, \hat{r}, \hat{v}^A) =: (t, \hat{q})$ with

$$t \geq \tau + \pi/2$$

lies on a causal curve obtained by following a relevant null geodesic from $p = (\tau, r, v^A)$ to $(t(\hat{s}), \hat{q})$, with a suitable value of $\hat{s}$, and then the timelike geodesic $\lambda \mapsto (\lambda, \hat{q})$ from $(t(\hat{s}), \hat{q})$ to $(t, \hat{q})$. So in fact every point with time coordinate larger than $\tau + \pi/2$ lies to the timelike future of p . $\square$

A Lorentzian manifold is said to be *geodesically complete* if all geodesics can be defined for all real values of affine parameter; this is equivalent to the requirement that for all $p \in \mathcal{M}$ the domain of the exponential map is $T_p\mathcal{M}$. One also talks about *timelike geodesically complete* spacetimes, *future timelike geodesically complete* spacetimes, etc., with those notions defined in an obvious way.

EXAMPLE 2.2.2 It follows from the *Hopf–Rinow* theorem [258, 348] that **compact Riemannian manifolds are geodesically complete**. There is no Lorentzian analogue of this; the standard counter-example proceeds as follows.

Consider the following symmetric tensor field on $\mathbb{R}^2$:

$$g = \frac{2dxdy}{x^2 + y^2}. \quad (2.2.10)$$

We have

$$g_{\mu\nu} = \frac{1}{x^2 + y^2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \implies \det g_{\mu\nu} = -\frac{1}{(x^2 + y^2)^2}, \quad (2.2.11)$$

which shows that g is indeed a Lorentzian metric. Note that for all $\lambda \in \mathbb{R}^*$ the maps

$$\mathbb{R}^2 \ni (x, y) \rightarrow \phi_\lambda(x, y) := (\lambda x, \lambda y)$$

are isometries of g :

$$\phi_\lambda^* g = \frac{2d(\lambda x)d(\lambda y)}{(\lambda x)^2 + (\lambda y)^2} = \frac{2dxdy}{x^2 + y^2} = g.$$

It follows that for any $1 \neq \lambda > 0$ the metric g passes to the quotient space

$$\{\mathbb{R}^2 \setminus \{0\}\} / \phi_\lambda = \{(x, y) \sim (\lambda x, \lambda y)\} \approx S^1 \times S^1 = \mathbb{T}^2.$$

(Clearly the quotient spaces with λ and $1/\lambda$ are the same, so without loss of generality one can assume $\lambda > 1$.) In order to show geodesic incompleteness of g we will use the following result:

PROPOSITION 2.2.3 Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let f be a function such that $g(\nabla f, \nabla f)$ is a constant. Then the integral curves of ∇f are affinely parameterized geodesics.

Proof. Let $X := \nabla f$; we have

$$\begin{aligned} (\nabla_X X)^j &= \nabla^i f \nabla_i \nabla^j f = \nabla^i f \nabla^j \nabla_i f \\ &= \frac{1}{2} \nabla^j (\nabla^i f \nabla_i f) = \frac{1}{2} \nabla^j (g(\nabla f, \nabla f)) = 0. \end{aligned} \quad \square$$

Returning to the metric (2.2.10), let $f = x$; by (2.2.11) we have

$$g^{\mu\nu} = (g_{\mu\nu})^{-1} = (x^2 + y^2) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

so that

$$\nabla f = (x^2 + y^2)\partial_y \implies g(\nabla f, \nabla f) = 0.$$

Proposition 2.2.3 shows that the integral curves of ∇f are null affinely parameterized geodesics. Let $\gamma(s) = (x^\mu(s))$ be any such integral curve; thus,

$$\frac{dx^\mu}{ds} = \nabla^\mu f \implies \frac{dx}{ds} = 0, \quad \frac{dy}{ds} = (x^2 + y^2).$$

It follows that $x(s) = x(0)$ for all s . The equation for y is easily integrated; for our purposes it is sufficient to consider that integral curve which passes through $(0, y_0) \in \mathbb{R}^2 \setminus \{0\}$, $y_0 > 0$ —we then have $x(s) = 0$ for all s and

$$\frac{dy}{ds} = y^2 \implies y(s) = \frac{y_0}{1 - y_0 s}. \quad (2.2.12)$$

This shows that $y(s)$ runs away to infinity as s approaches

$$s_\infty := \frac{1}{y_0}.$$

It follows that γ is indeed incomplete on $\mathbb{R}^2 \setminus \{0\}$. To see that it is also incomplete on the quotient torus $\{\mathbb{R}^2 \setminus \{0\}\} / \phi_\lambda$, $\lambda > 1$, note that the image of $\gamma(s) = (0, y(s))$ under the equivalence relation $\sim$ is a circle, and there exists a sequence $s_j \rightarrow s_\infty$ such that $\gamma(s_j)$ passes again and again through its starting point:

$$y(s_j) = \lambda^j y_0 \implies (0, y(s_j)) \sim (0, y_0) \text{ in } \{\mathbb{R}^2 \setminus \{0\}\} / \phi_\lambda.$$

By (2.2.12) we have

$$\frac{dy}{ds}(s_j) = (y(s_j))^2 = (\lambda_j y_0)^2 \longrightarrow_{s_j \rightarrow s_\infty} \infty,$$

which shows that the sequence of tangents $(dy/ds)(s_j)$ at $(0, y_0)$ blows up as j tends to infinity. This clearly implies that γ cannot be extended beyond s_∞ as a C^1 curve. $\square$

When the metric is C^2 , the inverse function theorem⁴ shows that there exists a neighbourhood $\mathcal{V}_p \subset \mathcal{U}_p$ of the origin in $\mathbb{R}^{\dim \mathcal{M}}$ on which the exponential map is a diffeomorphism between $\mathcal{V}_p$ and its image

$$\mathcal{O}_p := \exp_p(\mathcal{V}_p) \subset \mathcal{M}.$$

This allows one to define *normal coordinates* centred at p .

PROPOSITION 2.2.4 *Let $(\mathcal{M}, g)$ be a C^3 Lorentzian manifold with C^2 metric g . For every $p \in \mathcal{M}$ there exists an open coordinate neighbourhood $\mathcal{O}_p$ of p , in which p is mapped to the origin of $\mathbb{R}^{n+1}$, such that the coordinate rays $s \rightarrow sx^\mu$ are affinely parameterized geodesics. If the metric g is expressed in the resulting coordinates $(x^\mu) = (x^0, \vec{x}) \in \mathcal{V}_p$, then*

$$g_{\mu\nu}(0) = \eta_{\mu\nu}. \quad (2.2.13)$$

⁴It has been shown in [308] that one can construct normal coordinates for metrics which are only $C^{1,1}$. However, the resulting coordinates are only Lipschitz-continuous for such general metrics, which is a problem for some arguments later; note that one cannot even calculate the metric functions everywhere in such coordinates. We have not investigated this line of thought any further, as causality for poorly differentiable metrics g can be efficiently studied by approximating g by well-chosen smooth metrics, cf. e.g. [125].

If g is C^3 then we also have

$$\partial_\sigma g_{\mu\nu}(0) = 0. \quad (2.2.14)$$

Further, if the function $\sigma_p : \mathcal{O}_p \rightarrow \mathbb{R}$ is defined by the formula

$$\sigma_p(\exp_p(x^\mu)) := \eta_{\mu\nu} x^\mu x^\nu \equiv -(x^0)^2 + (\vec{x})^2, \quad (2.2.15)$$

then

$$\nabla \sigma_p \text{ is } \begin{cases} \text{timelike} & \begin{cases} \text{past directed} & \text{on } \{q \mid \sigma_p(q) < 0, x^0(q) < 0\}, \\ \text{future directed} & \text{on } \{q \mid \sigma_p(q) < 0, x^0(q) > 0\}, \end{cases} \\ \text{null} & \begin{cases} \text{past directed} & \text{on } \{q \mid \sigma_p(q) = 0, x^0(q) < 0\}, \\ \text{future directed} & \text{on } \{q \mid \sigma_p(q) = 0, x^0(q) > 0\}, \end{cases} \\ \text{spacelike} & \text{on } \{q \mid \sigma_p(q) > 0\}. \end{cases} \quad (2.2.16)$$

REMARK 2.2.5 The definition of normal coordinates leads to C^{k-1} coordinate functions if the metric is C^k . Hence the metric, when expressed in normal coordinates, will be of C^{k-2} differentiability class. This implies that there is a C^2 threshold for the introduction of normal coordinates, and that two derivatives are lost when expressing the metric in those coordinates. This is irrelevant for our purposes, as the main point here is that for C^2 metrics there exists a function σ_p satisfying (2.4.7) below, together with the following three facts:

1. $q \mapsto \sigma_p(q)$ is differentiable;
2. $(q, p) \mapsto \sigma_p(q)$ is continuous; and
3. If g_n converges to g in C^2 , then the corresponding functions σ_p converge in C^0 .

These are standard properties of solutions of ODEs (cf., e.g., [444]).

REMARK 2.2.6 The coefficients of a Taylor expansion of $g_{\mu\nu}$ in normal coordinates can be expressed in terms of the Riemann tensor and its covariant derivatives (compare [363, 368]).

Proof. Let us start by justifying that the implicit function theorem can indeed be applied: let x^μ be any coordinate system around p , and let $e_a = e_a^\mu \partial_\mu$ be any ON frame at p . Let

$$X = X^a e_a = X^a e_a^\mu \partial_\mu \in T_p \mathcal{M}$$

and let $x^\mu(s, X)$ denote the affinely parameterized geodesic passing by p at $s = 0$, with tangent vector

$$\dot{x}^\mu(0, X) := \left. \frac{dx^\mu(s, X)}{ds} \right|_{s=0} = X^a e_a^\mu.$$

Homogeneity properties of the ODE (2.2.1) under the change of parameter $s \rightarrow \lambda s$ together with uniqueness of solutions of ODEs show that for any constant $a \neq 0$ we have

$$x^\mu(as, X/a) = x^\mu(s, X).$$

This, in turn, implies that there exist functions γ^μ such that

$$x^\mu(s, X) = \gamma^\mu(sX). \quad (2.2.17)$$

From (2.2.1) and (2.2.17) we have

$$x^\mu(s, X) = x_0^\mu + sX^a e_a^\mu + O((s|X|)^2).$$

Here x_0^μ are the coordinates of p , $|X|$ denotes the norm of X with respect to some auxiliary Riemannian metric on M , while the $O((s|X|)^2)$ term is justified

by (2.2.17). The usual considerations of the proof that solutions of ODEs are differentiable functions of their initial conditions show that

$$\begin{aligned}\frac{\partial x^\mu(s, X)}{\partial X^a} &= \frac{\partial(x_0^\mu + sX^a e_a^\mu)}{\partial X^a} + O(s^2)|X| \\ &= s e_a^\mu + O(s^2)|X|.\end{aligned}$$

At $s = 1$ one thus obtains

$$\frac{\partial x^\mu(1, X)}{\partial X^a} = e_a^\mu + O(|X|). \quad (2.2.18)$$

This shows that $\partial x^\mu / \partial X^a$ will be bijective at $X = 0$ provided that $\det e_a^\mu \neq 0$. But this last inequality can be obtained by taking the determinant of the equation

$$g(e_a, e_b) = g_{\mu\nu} e_a^\mu e_b^\nu \implies -1 = (\det g_{\mu\nu})(\det e_a^\mu)^2. \quad (2.2.19)$$

This justifies the use of the implicit function theorem to obtain existence of the neighbourhood $\mathcal{O}_p$ announced in the statement of the proposition.

Clearly $\mathcal{O}_p$ can be chosen to be star-shaped with respect to p . Equation (2.2.18) and the fact that e_a^μ is an ON-frame show that

$$g(\partial_a, \partial_b) \Big|_{X^a=0} = g_{\mu\nu} e_a^\mu e_b^\nu \Big|_{X^a=0} = \eta_{ab},$$

which establishes in (2.2.13).

By construction the rays

$$s \rightarrow \gamma^a(s) := sX^a$$

are affinely parameterized geodesics with tangent $\dot{\gamma} = X^a \partial_a$, which gives

$$\begin{aligned}0 = (\nabla_{\dot{\gamma}} \dot{\gamma})^a &= \underbrace{\frac{d^2(sX^a)}{ds^2}}_{=0} + \Gamma^a_{bc}(sX^d) X^b X^c \\ &= \Gamma^a_{bc}(sX^d) X^b X^c.\end{aligned}$$

Setting $s = 0$ and differentiating this equation twice with respect to X^d and X^e one obtains

$$\Gamma^a_{de}(0) = 0.$$

The equation

$$0 = \nabla_a g_{bc} = \partial_a g_{bc} - \Gamma^d_{ba} g_{dc} - \Gamma^d_{ca} g_{bd}$$

evaluated at $X = 0$ gives (2.2.14).

Let us pass now to the proof of the main point here, namely (2.2.16). From now on we will denote by x^μ the normal coordinates obtained so far, and which were denoted by X^a in the arguments just done. For $x \in \mathcal{O}_p$ define

$$f(x) := \eta_{\mu\nu} x^\mu x^\nu, \quad (2.2.20)$$

and let $\mathcal{H}_\tau \subset \mathcal{O}_p \setminus \{p\}$ be the level sets of f :

$$\mathcal{H}_\tau := \{x : f(x) = \tau, x \neq 0\}. \quad (2.2.21)$$

We will show that

$$\text{the vector field } x^\mu \partial_\mu \text{ is normal to the } \mathcal{H}_\tau \text{'s.} \quad (2.2.22)$$

Now, $x^\mu \partial_\mu$ is tangent to the geodesic rays $s \rightarrow \gamma^\mu(s) := sx^\mu$. As the causal character of the field of tangents to a geodesic⁵ is point-independent along the geodesic, we have

$$\begin{aligned} x^\mu \partial_\mu \text{ is timelike at } \gamma(s) &\iff f(x) < 0, \\ x^\mu \partial_\mu \text{ is null at } \gamma(s) &\iff f(x) = 0, \quad x \neq 0, \\ x^\mu \partial_\mu \text{ is spacelike at } \gamma(s) &\iff f(x) > 0. \end{aligned} \quad (2.2.23)$$

This follows from the fact that the right-hand side is precisely the condition that the geodesic be timelike, spacelike, or null, at $\gamma(0)$. Since ∇f is always normal to the level sets of f , when (2.2.22) holds we will have

$$x^\mu \partial_\mu \text{ is proportional to } \nabla^\mu f. \quad (2.2.24)$$

This shows that (2.2.16) will follow from (2.2.23) when (2.2.22) holds.

It remains to establish (2.2.22). In order to do that, consider any differentiable curve $\lambda \mapsto x^\mu(\lambda)$ lying on $\mathcal{H}_\tau$:

$$\eta_{\mu\nu} x^\mu(\lambda) x^\nu(\lambda) = \tau \implies \eta_{\mu\nu} x^\mu(\lambda) \partial_\lambda x^\nu(\lambda) = 0. \quad (2.2.25)$$

Let $\gamma^\mu(\lambda, s)$ be the one-parameter family of geodesic rays

$$\gamma^\mu(\lambda, s) := sx^\mu(\lambda).$$

For any function f set

$$T(f) = \partial_s(f \circ \gamma(s, \lambda)), \quad X(f) = \partial_\lambda(f \circ \gamma(s, \lambda)),$$

so that

$$T(\lambda, s) := \left(\partial_s \gamma^\mu(\lambda, s) \right) \partial_\mu = x^\mu(\lambda, s) \partial_\mu, \quad X(\lambda, s) := \left(\partial_\lambda \gamma^\mu(\lambda, s) \right) \partial_\mu.$$

For any fixed value of λ the curves $s \rightarrow \gamma^\mu(\lambda, s)$ are geodesics, which shows that

$$\nabla_T T = 0.$$

This gives

$$\frac{d(g(T, T))}{ds} = 2g(\nabla_T T, T) = 0,$$

hence

$$g(T, T)(s) = g(T, T)(0) = \eta_{\mu\nu} x^\mu(\lambda) x^\nu(\lambda) = \tau$$

by (2.2.25); in particular, $g(T, T)$ is λ -independent.

Next, for any twice-differentiable function ψ we have

$$[T, X](\psi) := T(X(\psi)) - X(T(\psi)) = \partial_s \partial_\lambda (\psi(\gamma^\mu(s, \lambda))) - \partial_\lambda \partial_s (\psi(\gamma^\mu(s, \lambda))) = 0,$$

because of the symmetry of the matrix of second partial derivatives. It follows that

$$[T, X] = \nabla_T X - \nabla_X T = 0.$$

Finally,

$$\begin{aligned} \frac{d(g(T, X))}{ds} &= g(\underbrace{\nabla_T T}_0, X) + g(T, \underbrace{\nabla_T X}_{=\nabla_X T}) \\ &= g(T, \nabla_X T) = \frac{1}{2} X(g(T, T)) = \frac{1}{2} \partial_\lambda (g(T, T)) = 0. \end{aligned}$$

This yields

$$g(T, X)(s, \lambda) = g(T, X)(0, \lambda) = \eta_{\mu\nu} x^\mu(\lambda) \partial_\lambda x^\nu(\lambda) = 0$$

again by (2.2.25). Thus T is normal to the level sets of f , which had to be established. $\square$

⁵Without loss of generality an affine parameterization of a geodesic γ can be chosen; the result follows then from the calculation $d(g(\dot{\gamma}, \dot{\gamma}))/ds = 2g(\nabla_{\dot{\gamma}} \dot{\gamma}, \dot{\gamma}) = 0$.

As already pointed out, some regularity of the metric is lost when going to normal coordinates; this can be avoided using coordinates which are only approximately normal up to a required order, which is often sufficient for several purposes.

INCIDENTALLY: It is sometimes useful to have a *geodesic convexity property* at our disposal. This is made precise by the following proposition.

PROPOSITION 2.2.7 Let $\mathcal{O}$ be the domain of the definition of a coordinate system $\{x^\mu\}$. Let $p \in \mathcal{O}$ and let $B_p(r) \subset \mathcal{O}$ denote an open coordinate ball of radius r centred at p . There exists $r_0 > 0$ such that every geodesic segment $\gamma : [a, b] \rightarrow \overline{B_p(r_0)} \subset \mathcal{O}$, satisfying

$$\gamma(a), \gamma(b) \in B_p(r), \quad r < r_0$$

is entirely contained in $B_p(r)$.

Proof. Let $x^\mu(s)$ be the coordinate representation of γ ; set

$$f(s) := \sum_{\mu} (x^\mu - x_0^\mu)^2,$$

where x_0^μ is the coordinate representation of p . We have

$$\begin{aligned} \frac{df}{ds} &= 2 \sum_{\mu} (x^\mu - x_0^\mu) \frac{dx^\mu}{ds}, \\ \frac{d^2 f}{ds^2} &= 2 \sum_{\mu} \left(\frac{dx^\mu}{ds} \right)^2 + 2 \sum_{\mu} (x^\mu - x_0^\mu) \frac{d^2 x^\mu}{ds^2} \\ &= 2 \sum_{\mu} \left(\frac{dx^\mu}{ds} \right)^2 - 2 \sum_{\mu} (x^\mu - x_0^\mu) \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds}. \end{aligned}$$

Compactness of $\overline{B_p(r_0)}$ implies that there exists a constant C such that we have

$$\left| \sum_{\mu} (x^\mu - x_0^\mu) \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} \right| \leq C r_0 \sum_{\mu} \left(\frac{dx^\mu}{ds} \right)^2.$$

It follows that $d^2 f / ds^2 \geq 0$ for r_0 small enough. This shows that f has no interior maximum if r_0 is small enough, whence the result. $\square$

It is convenient to introduce the following notion.

DEFINITION 2.2.8 An elementary region is an open coordinate ball $\mathcal{O}$ within the domain of a normal coordinate neighbourhood $\mathcal{U}_p$ such that

1. $\mathcal{O}$ has compact closure $\overline{\mathcal{O}}$ in $\mathcal{U}_p$, and
2. ∇t and ∂_t are timelike on $\overline{\mathcal{O}}$.

Note that ∂_t is timelike if and only if

$$g_{tt} \equiv g(\partial_t, \partial_t) < 0,$$

while ∇t is timelike if and only if

$$g^{tt} \equiv g^\#(dt, dt) < 0.$$

Existence of elementary regions containing some point $p \in M$ follows immediately from Proposition 2.2.4: in normal coordinates centred at p one chooses $\mathcal{O}$ be an open coordinate ball of sufficiently small radius.

2.3 Causal paths

Let $(\mathcal{M}, g)$ be a spacetime. The basic objects in causality theory are paths. We shall always use *parameterized paths*: by definition, these are *continuous* maps from some

interval to $\mathcal{M}$. We will use interchangeably the terms ‘path’, ‘parameterized path’, or ‘curve’, but we note that some authors make the distinction. (For example, in [86] a path is a map and a curve is the image of a path, oriented by parameterization.)

Let $\gamma : I \rightarrow \mathcal{M}$ and let $\mathcal{U} \subset \mathcal{M}$, we will write

$$\gamma \subset \mathcal{U}$$

whenever the image $\gamma(I)$ of I by γ is a subset of $\mathcal{U}$. We will sometimes write

$$\gamma \cap \mathcal{U}$$

for the path obtained by removing from I those parameters s for which $\gamma(s) \notin \mathcal{U}$. Strictly speaking, our definition of a path requires γ to be defined on a connected interval, so if the last construction gives several intervals I_i , then $\gamma \cap \mathcal{U}$ will actually describe the collection of paths $\gamma|_{I_i}$.

INCIDENTALLY: Some authors define a path in $\mathcal{M}$ as what would be in our terminology the *image of a parameterized path*. In this approach one forgets about the parameterization of γ , and identifies two paths which have the same image and differ only by a reparameterization. This leads to various difficulties when considering end points of causal paths (cf. Section 2.5), or limits of sequences of paths (cf. Section 2.6), and therefore we do *not* adopt this approach. $\square$

If $\gamma : I \rightarrow \mathcal{M}$ where $I = [a, b)$ or $I = [a, b]$, then $\gamma(a)$ is called the *starting point* of the path γ , or of its image $\gamma(I)$. If $I = (a, b]$ or $I = [a, b]$, then $\gamma(b)$ is called the *end point*. (This definition will be extended in Section 2.5, but it is sufficient for our purposes here.) We shall say that $\gamma : [a, b] \rightarrow \mathcal{M}$ is a path from p to q if $\gamma(a) = p$ and $\gamma(b) = q$.

It is convenient to choose once and for all some auxiliary Riemannian metric h on $\mathcal{M}$, such that $(\mathcal{M}, h)$ is complete; such a metric always exists [378]. We denote by dist_h the associated distance function. A parameterized path $\gamma : I \rightarrow \mathcal{M}$ from an interval $I \subset \mathbb{R}$ to $\mathcal{M}$ is called *locally Lipschitzian*, or *locally Lipschitz*, if for every compact subset K of I there exists a constant $C(K)$ such that

$$\forall s_1, s_2 \in K \quad \text{dist}_h(\gamma(s_1), \gamma(s_2)) \leq C(K)|s_1 - s_2|.$$

INCIDENTALLY: It is natural to enquire whether the class of paths so defined depends upon the background metric h :

PROPOSITION 2.3.1 Let h_1 and h_2 be two complete Riemannian metrics on $\mathcal{M}$. Then a path $\gamma : I \rightarrow \mathcal{M}$ is locally Lipschitzian with respect to h_1 if and only if it is locally Lipschitzian with respect to h_2 .

Proof. Let $K \subset I$ be a compact set, then $\gamma(K)$ is compact. Let L_a , $a = 1, 2$ denote the h_a -length of γ ; set

$$\mathcal{K}_a := \cup_{s \in K} B_{h_a}(\gamma(s), L_a),$$

where $B_{h_a}(p, r)$ denotes a closed geodesic ball, with respect to the metric h_a , centred at p , of radius r . Then the $\mathcal{K}_a$'s are compact. Likewise the sets

$$\widehat{\mathcal{K}}_a \subset T\mathcal{M},$$

defined as the sets of h_a -unit vectors over $\mathcal{K}_a$, are compact. This implies that there exists a constant C_K such that for all $X \in T_p\mathcal{M}$, $p \in \mathcal{K}_a$, we have

$$C_K^{-1}h_1(X, X) \leq h_2(X, X) \leq C_K h_1(X, X).$$

Let γ_{a,s_1,s_2} denote any minimizing h_a -geodesic between $\gamma(s_1)$ and $\gamma(s_2)$, then

$$\forall s_1, s_2 \in K \quad \gamma_{a,s_1,s_2} \subset \mathcal{K}_a.$$

This implies that

$$\text{dist}_{h_2}(\gamma(s_1), \gamma(s_2)) = \int_{\sigma=\gamma_{2,s_1,s_2}} \sqrt{h_2(\dot{\sigma}, \dot{\sigma})}$$

$$\begin{aligned}
&\geq C_K^{-1} \int_{\sigma=\gamma_2, s_1, s_2} \sqrt{h_1(\dot{\sigma}, \dot{\sigma})} \\
&\geq C_K^{-1} \inf_{\sigma} \int_{\sigma} \sqrt{h_1(\dot{\sigma}, \dot{\sigma})} \\
&= C_K^{-1} \int_{\sigma=\gamma_1, s_1, s_2} \sqrt{h_1(\dot{\sigma}, \dot{\sigma})} \\
&= C_K^{-1} \text{dist}_{h_1}(\gamma(s_1), \gamma(s_2)).
\end{aligned}$$

From symmetry with respect to the interchange of h_1 with h_2 we conclude that for all $s_1, s_2 \in K$

$$C_K^{-1} \text{dist}_{h_1}(\gamma(s_1), \gamma(s_2)) \leq \text{dist}_{h_2}(\gamma(s_1), \gamma(s_2)) \leq C_K \text{dist}_{h_1}(\gamma(s_1), \gamma(s_2)),$$

and the result easily follows. $\square$

More generally, if (N, k) and (M, h) are Riemannian manifolds, then a map $\phi : N \rightarrow M$ is called *locally Lipschitzian*, or *locally Lipschitz*, if for every compact subset K of N there exists a constant $C(K)$ such that

$$\forall p, q \in K \quad \text{dist}_h(\phi(p), \phi(q)) \leq C(K) \text{dist}_k(p, q).$$

A map is called *Lipschitzian* if the constant $C(K)$ above can be chosen independently of K .

The following important theorem of Rademacher will play a key role in our considerations.

THEOREM 2.3.2 (Rademacher) *Let $\phi : M \rightarrow N$ be a locally Lipschitz map from a manifold M to a manifold N . Then we have the following.*

1. ϕ is classically differentiable almost everywhere, with ‘almost everywhere’ understood in the sense of the Lebesgue measure in local coordinates on M .
2. The distributional derivatives of ϕ are in L_{loc}^{∞} and coincide with the classical ones almost everywhere.
3. Suppose that M is an open subset of $\mathbb{R}$ and N is an open subset of $\mathbb{R}^n$. Then ϕ is the integral of its distributional derivative,

$$\phi(x) - \phi(y) = \int_y^x \frac{d\phi}{dt} dt. \quad (2.3.1)$$

Proof. Point 1 is the classical statement of Rademacher, the proof of which can be found in [187, Theorem 2, p. 235]. Point 2 is Theorem 5 in [187, p. 131] and Theorem 1 of [187, p. 235]. (In that last theorem the classical differentiability a.e. is actually established for all $W_{\text{loc}}^{1,p}$ functions with $p > n$). Point 3 can be established by approximating ϕ by C^1 functions as in [187, Theorem 1, p. 251], and passing to the limit. $\square$

Point 2 shows that the usual properties of the derivatives of continuously differentiable functions—such as the Leibniz rule, or the chain rule—hold almost everywhere for the derivatives of locally Lipschitzian functions. By point 3 those properties can be used freely whenever integration is involved.

We will use the symbol

$$\dot{\gamma}$$

to denote the *classical* derivative of a path γ , wherever defined. A parameterized path γ will be called *causal future directed* if γ is locally Lipschitzian, with $\dot{\gamma}$ —

causal and future directed almost everywhere.⁶ Thus, $\dot{\gamma}$ is defined almost everywhere; and it is causal future directed almost everywhere on the set on which it is defined. A parameterized path γ will be called *timelike future directed* if γ is locally Lipschitzian, with $\dot{\gamma}$ —timelike future directed almost everywhere. *Past-directed* parameterized paths are defined by changing ‘future’ to ‘past’ in the definitions above.

INCIDENTALLY: In many treatments of causality theory [38, 219, 244, 383, 398, 454] one defines *future-directed timelike* paths as those paths γ which are *piecewise differentiable*, with $\dot{\gamma}$ timelike and future directed wherever defined; at break points one further assumes that both the left-sided and right-sided derivatives are timelike. This definition turns out to be quite inconvenient for several purposes. For instance, when studying the global causal structure of spacetimes one needs to take limits of timelike curves, obtaining thus—by definition—*causal future-directed* paths. Such limits will not be piecewise differentiable most of the time, which leads one to the necessity of considering paths with poorer differentiability properties. One then faces the unhandy situation in which timelike and causal paths have completely different properties. In several theorems separate proofs have then to be given. The approach we present avoids this, leading—we believe—to a considerable simplification of the conceptual structure of the theory. $\square$

A useful property of locally Lipschitzian paths is that they can be parameterized by h -length. (Recall that h is an auxiliary Riemannian metric such that $(\mathcal{M}, h)$ is complete.) Let $\gamma : [a, b) \rightarrow \mathcal{M}$ be a path, and suppose that $\dot{\gamma}$ is non-zero almost everywhere—this is certainly the case for causal paths. By Rademacher’s theorem the integral

$$s(t) = \int_a^t |\dot{\gamma}|_h(u) du$$

is well defined. Clearly $s(t)$ is a continuous strictly increasing function of t , so that the map $t \rightarrow s(t)$ is a bijection from $[a, b)$ to its image. The new path $\hat{\gamma} := \gamma \circ s^{-1}$ differs from γ only by a reparameterization, so it has the same image in $\mathcal{M}$. The reader will easily check that $|\dot{\hat{\gamma}}|_h = 1$ almost everywhere. Further, $\hat{\gamma}$ is Lipschitz continuous with Lipschitz constant smaller than or equal to 1. In order to see that, let dist_h be the associated distance function; we claim that

$$\text{dist}_h(\hat{\gamma}(s), \hat{\gamma}(s')) \leq |s - s'|. \quad (2.3.2)$$

For $s > s'$ we have

$$\begin{aligned} s - s' &= \int_{s'}^s dt \\ &= \int_{s'}^s \underbrace{\sqrt{h(\dot{\hat{\gamma}}, \dot{\hat{\gamma}})}(t)}_{=1 \text{ a.e.}} dt \\ &\geq \inf_{\tilde{\gamma}} \int_{\tilde{\gamma}} \sqrt{h(\dot{\tilde{\gamma}}, \dot{\tilde{\gamma}})}(t) dt = \text{dist}_h(\hat{\gamma}(s), \hat{\gamma}(s')), \end{aligned} \quad (2.3.3)$$

where the infimum is taken over $\tilde{\gamma}$ ’s which start at $\gamma(s')$ and finish at $\gamma(s)$. This is precisely (2.3.2).

Summarizing, we have the following.

PROPOSITION 2.3.3 *A locally Lipschitzian path γ can be reparameterized so that it is Lipschitzian, with Lipschitz constant equal to 1.*

⁶Some authors allow constant paths to be causal, in which case the sets $J^\pm(\mathcal{U}; \mathcal{O})$ defined in Section 2.4 automatically contain $\mathcal{U}$. This leads to unnecessary discussions when concatenating causal paths, such that we find it convenient not to allow such paths in our definition.

One could be tempted to impose the h -length-parameterization on all causal curves. This is, however, not a good idea. First, it is often convenient to parameterize timelike curves by proper-time or by a local time coordinate, and geodesics by an affine parameter. More significantly, the h -length-parameterization does not necessarily pass to the limit when sequences of curves are considered, cf. Example 2.5.1, p. 45 and Example 2.6.1, p. 48.

2.4 Futures, pasts

Let $\mathcal{U} \subset \mathcal{O} \subset \mathcal{M}$. One sets

$$\begin{aligned} I^+(\mathcal{U}; \mathcal{O}) &:= \{q \in \mathcal{O} : \text{there exists a timelike future-directed path} \\ &\quad \text{from } \mathcal{U} \text{ to } q \text{ contained in } \mathcal{O}\}, \\ J^+(\mathcal{U}; \mathcal{O}) &:= \{q \in \mathcal{O} : \text{there exists a causal future-directed path} \\ &\quad \text{from } \mathcal{U} \text{ to } q \text{ contained in } \mathcal{O} \cup \mathcal{U}\}. \end{aligned}$$

$I^-(\mathcal{U}; \mathcal{O})$ and $J^-(\mathcal{U}; \mathcal{O})$ are defined by replacing ‘future’ with ‘past’ in the definitions above. The set $I^+(\mathcal{U}; \mathcal{O})$ is called the *timelike future of $\mathcal{U}$ in $\mathcal{O}$* , while $J^+(\mathcal{U}; \mathcal{O})$ is called the *causal future of $\mathcal{U}$ in $\mathcal{O}$* , with similar terminology for the timelike past and the causal past. We will write $I^\pm(\mathcal{U})$ for $I^\pm(\mathcal{U}; \mathcal{M})$, similarly for $J^\pm(\mathcal{U})$, and we then omit the qualification ‘in $\mathcal{M}$ ’ when talking about the causal or timelike futures and pasts of $\mathcal{U}$. We will write $I^\pm(p; \mathcal{O})$ for $I^\pm(\{p\}; \mathcal{O})$, $I^\pm(p)$ for $I^\pm(\{p\}; \mathcal{M})$, etc.

INCIDENTALLY: Although our definitions of timelike and causal curves do not coincide with the usual ones [38, 244, 398, 454], the resulting sets $J^\pm$ and $I^\pm$ are identical to the standard ones (compare Proposition 2.4.11, p. 41). $\square$

It is legitimate to raise the question, why is it interesting to consider sets such as $J^+(\mathcal{O})$? The answer is twofold: From a mathematical point of view, those sets appear naturally when describing the *finite speed of the propagation* property of wave-type equations, such as Einstein’s equations. From a physical point of view, such constructs are related to the fundamental postulate of general relativity, that *no signal can travel faster than the speed of light*. This is equivalent to the statement that the only events of spacetimes influenced by an event $p \in \mathcal{M}$ are those which belong to $J^+(\mathcal{M})$.

EXAMPLE 2.4.1 Let $\mathcal{M} = S^1 \times S^1$ with the flat metric $g = -dt^2 + d\varphi^2$. Geodesics of g through $(0, 0)$ are of the form

$$\gamma(s) = (\alpha s \bmod 2\pi, \beta s \bmod 2\pi), \quad (2.4.1)$$

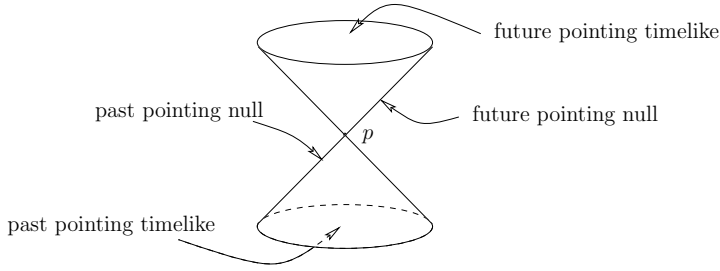
where α and β are constants, not simultaneously zero; the remaining geodesics are obtained by a rigid translation of (2.4.1). Clearly any two points of $\mathcal{M}$ can be joined by a timelike geodesic, which shows that for all $p \in \mathcal{M}$ we have

$$I^+(p) = J^+(p) = \mathcal{M}.$$

It is of some interest to point out that for irrational β/α in (2.4.1) the corresponding geodesic is dense in $\mathcal{M}$.

There is an obvious *meta-rule* in the theory of causality that whenever a property involving I^+ or J^+ holds, then an identical property will be true with I^+ replaced by I^- , and with J^+ replaced by J^- , or both. This is proved by changing the time orientation of the manifold. Thus we will only make formal statements for the futures.

Example 2.4.1 shows that in causally pathological spacetimes the notions of futures and pasts need not carry interesting information. On the other hand, those

Fig. 2.4.1 The light cone at p .

objects are useful tools for studying the global structure of those spacetimes which possess reasonable causal properties.

We start with some elementary properties of futures and pasts.

PROPOSITION 2.4.2 *We have:*

1. $I^+(\mathcal{U}) \subset J^+(\mathcal{U})$.
2. $p \in I^+(q) \iff q \in I^-(p)$.
3. $\mathcal{V} \subset I^+(\mathcal{U}) \implies I^+(\mathcal{V}) \subset I^+(\mathcal{U})$.

Similar properties hold with I^+ replaced by J^+ .

Proof. 1. A timelike curve is a causal curve.

2. If $[0, 1] \ni s \rightarrow \gamma(s)$ is a future-directed causal curve from q to p , then $[0, 1] \ni s \rightarrow \gamma(1-s)$ is a past-directed causal curve from p to q .

3. Let us start by introducing some notation: consider $\gamma_a : [0, 1] \rightarrow \mathcal{M}$, $a = 1, 2$, two curves such that $\gamma_1(1) = \gamma_2(0)$. We define the *concatenation operation* $\gamma_1 \cup \gamma_2$ as follows:

$$(\gamma_1 \cup \gamma_2)(s) = \begin{cases} \gamma_1(s), & s \in [0, 1], \\ \gamma_2(s-1), & s \in [1, 2]. \end{cases} \quad (2.4.2)$$

There is an obvious extension of this definition when the ranges of parameters of the γ_a 's are not $[0, 1]$, or when a finite number $i \geq 3$ of paths is considered; we leave the formal definition to the reader.

Let, now, $r \in I^+(\mathcal{V})$; then there exists $q \in \mathcal{V}$ and a future-directed timelike curve γ_2 from q to r . Since $\mathcal{V} \subset I^+(\mathcal{U})$ there exists a future-directed timelike curve γ_1 from some point $p \in \mathcal{U}$ to q . Then the curve $\gamma_1 \cup \gamma_2$ is a future-directed timelike curve from $\mathcal{U}$ to r . $\square$

We have the following, intuitively obvious, description of futures and pasts of points in Minkowski spacetime (see Fig. 2.4.1); in Proposition 2.4.5 we will shortly prove a similar local result in general spacetimes, with a considerably more complicated proof.

PROPOSITION 2.4.3 *Let $(\mathcal{M}, g)$ be the $(n+1)$ -dimensional Minkowski spacetime $\mathbb{R}^{1,n} := (\mathbb{R}^{1+n}, \eta)$, with Minkowskian coordinates $(x^\mu) = (x^0, \vec{x})$ so that*

$$\eta(\partial_\mu, \partial_\nu) = \text{diag}(-1, +1, \dots, +1).$$

Then

1. $I^+(0) = \{x^\mu : \eta_{\mu\nu} x^\mu x^\nu < 0, x^0 > 0\}$,
2. $J^+(0) = \{x^\mu : \eta_{\mu\nu} x^\mu x^\nu \leq 0, x^0 \geq 0\}$,
3. *in particular the boundary, denoted as $\dot{J}^+(0)$, of $J^+(0)$ is the union of $\{0\}$ together with all null future-directed geodesics with initial point at the origin.*

Proof. Let $\gamma(s) = (x^\mu(s))$ be a parameterized causal path in $\mathbb{R}^{1,n}$ with $\gamma(0) = 0$. At points at which γ is differentiable we have

$$\eta(\dot{\gamma}, \dot{\gamma}) = - \left(\frac{dx^0}{ds} \right)^2 + \left| \frac{d\vec{x}}{ds} \right|_{\delta}^2 \leq 0, \quad \frac{dx^0}{ds} \geq \left| \frac{d\vec{x}}{ds} \right|_{\delta} \geq 0.$$

Now, similarly to a differentiable function, a locally Lipschitzian function is the integral of its distributional derivative (see Theorem 2.3.2); hence,

$$x^0(s) = \int_0^s \frac{dx^0}{ds}(u) du \quad (2.4.3a)$$

$$\geq \int_0^s \left| \frac{d\vec{x}}{ds} \right|_{\delta}(u) du =: \ell(\gamma_s). \quad (2.4.3b)$$

Here $\ell(\gamma_s)$ is the length, with respect to the flat Riemannian metric δ , of the path γ_s , defined as

$$[0, s] \ni u \rightarrow \vec{x}(u) \in \mathbb{R}^n.$$

Let dist_{δ} denote the distance function of the metric δ ; thus

$$\text{dist}_{\delta}(\vec{x}, \vec{y}) = |\vec{x} - \vec{y}|_{\delta}.$$

It is well known that

$$\ell_s \geq \text{dist}_{\delta}(\vec{x}(s), \vec{x}(0)) = |\vec{x}(s) - \vec{x}(0)|_{\delta} = |\vec{x}(s)|_{\delta}.$$

Therefore

$$x^0(s) \geq |\vec{x}(s)|_{\delta},$$

and point 2 follows. For timelike curves the same proof applies, with all inequalities becoming strict, establishing point 1. Point 3 is a straightforward consequence of point 2. $\square$

INCIDENTALLY: There is a natural generalization of Proposition 2.4.3 to the following class of metrics on $\mathbb{R} \times \mathcal{S}$,

$$g = -\varphi dt^2 + h, \quad \partial_t \varphi = \partial_t h = 0, \quad (2.4.4)$$

where h is a Riemannian metric on $\mathcal{S}$, and φ is a strictly positive function. (Such metrics are called *warped-products*, with *warping function* φ .) This proceeds as follows.

PROPOSITION 2.4.4 Let $\mathcal{M} = \mathbb{R} \times \mathcal{S}$ with the metric (2.4.4), and let $p \in \mathcal{S}$. Then $J^+((0, p))$ is the graph over $\mathcal{S}$ of the distance function $\text{dist}_{\hat{h}}(p, \cdot)$ of the *optical metric*

$$\hat{h} := \varphi^{-1} h,$$

while $I^+((0, p))$ is the epigraph of $\text{dist}_{\hat{h}}(p, \cdot)$,

$$I^+((0, p)) = \{(t, q) : t > \text{dist}_{\hat{h}}(p, q)\}.$$

Proof. Since the causal character of a curve is invariant under conformal transformations, the causal and timelike futures with respect to the metric g coincide with those with respect to the metric

$$\varphi^{-1} g = -dt^2 + \hat{h}.$$

Arguing as in the proof of Proposition 2.4.3, (2.4.3) becomes

$$x^0(s) = \int_0^s \frac{dx^0}{ds}(u) du \quad (2.4.5a)$$

$$\geq \int_0^s \left| \frac{d\vec{x}}{ds} \right|_{\hat{h}}(u) du =: \ell_{\hat{h}}(\gamma_s), \quad (2.4.5b)$$

where $\ell_{\hat{h}}(\gamma_s)$ denotes the length of γ_s with respect to $\hat{h}$, and one concludes as before. $\square$

The next result shows that, locally, causal behaviour is identical to that of Minkowski spacetime. The proof of this ‘obvious’ fact turns out to be surprisingly involved.

PROPOSITION 2.4.5 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let $\mathcal{O}_p$ be a domain of normal coordinates x^μ centred at $p \in \mathcal{M}$ as in Proposition 2.2.4. Let*

$$\mathcal{O} \subset \mathcal{O}_p$$

be any normal-coordinate ball such that ∇x^0 is timelike on $\mathcal{O}$. Recall (compare (2.2.15)) that the function $\sigma_p : \mathcal{O}_p \rightarrow \mathbb{R}$ has been defined by the formula

$$\sigma_p(\exp_p(x^\mu)) := \eta_{\mu\nu} x^\mu x^\nu. \quad (2.4.6)$$

Then

$$\mathcal{O} \ni q = \exp_p(x^\mu) \in \begin{cases} I^+(p; \mathcal{O}) & \iff \sigma_p(q) < 0, \ x^0 > 0, \\ J^+(p; \mathcal{O}) & \iff \sigma_p(q) \leq 0, \ x^0 \geq 0, \\ \dot{J}^+(p; \mathcal{O}) & \iff \sigma_p(q) = 0, \ x^0 \geq 0, \end{cases} \quad (2.4.7)$$

with the obvious analogues for pasts. In particular, a point $q = \exp_p(x^\mu) \in \dot{J}^+(p; \mathcal{O}_p)$ if and only if q lies on the null geodesic segment $[0, 1] \ni s \rightarrow \gamma(s) = \exp_p(sx^\mu) \in \dot{J}^+(p; \mathcal{O}_p)$.

REMARK 2.4.6 Example 2.4.1 shows that $I^\pm(p; \mathcal{O})$, etc., cannot be replaced by $I^\pm(p)$, because causal paths through p can exit $\mathcal{O}_p$ and re-enter it; this can actually happen again and again.

Before proving Proposition 2.4.5, we note the following straightforward implication thereof.

PROPOSITION 2.4.7 *Let $\mathcal{O}$ be as in Proposition 2.4.5; then $I^+(p; \mathcal{O})$ is open.* $\square$

Proof of Proposition 2.4.5. As the coordinate rays are geodesics, the implications ‘ $\Leftarrow$ ’ in (2.4.7) are obvious. It remains to prove ‘ $\Rightarrow$ ’. We start with a lemma.

LEMMA 2.4.8 *Let τ be a time function, i.e., a differentiable function with timelike past-pointing gradient. For any τ_0 , a future-directed causal path γ cannot leave the set $\{q : \tau(q) > \tau_0\}$; the same holds for sets of the form $\{q : \tau(q) \geq \tau_0\}$. In fact, τ is non-decreasing along γ , strictly increasing if γ is timelike.*

Proof. Let $\gamma : I \rightarrow \mathcal{M}$ be a future-directed parameterized causal path; then $\tau \circ \gamma$ is a locally Lipschitzian function. Hence, it equals the integral of its derivative on any compact subset of its domain of definition, such that

$$\begin{aligned} \tau(\gamma(s_2)) - \tau(\gamma(s_1)) &= \int_{s_1}^{s_2} \frac{d(\tau \circ \gamma)}{du}(u) du \\ &= \int_{s_1}^{s_2} \langle d\tau, \dot{\gamma} \rangle(u) du \\ &= \int_{s_1}^{s_2} g(\nabla \tau, \dot{\gamma})(u) du \geq 0, \end{aligned} \quad (2.4.8)$$

since $\nabla \tau$ is timelike past directed, while $\dot{\gamma}$ is causal future directed or zero wherever defined. The function $s \rightarrow \tau(\gamma(s))$ is strictly increasing when γ is timelike, since then the integrand in (2.4.8) is strictly positive almost everywhere. $\square$

Applying Lemma 2.4.8 to the time function x^0 we obtain the claim about x^0 in (2.4.7). To justify the remaining claims of Proposition 2.4.5, we recall Eq. (2.2.16)

$$\nabla \sigma_p \text{ is } \begin{cases} \text{timelike future directed} & \text{on } \{q : \sigma_p(q) < 0, \ x^0(q) > 0\}, \\ \text{null future directed} & \text{on } \{q : \sigma_p(q) = 0, \ x^0(q) > 0\}. \end{cases} \quad (2.4.9)$$

Let $\gamma = (\gamma^\mu) : I \rightarrow \mathcal{O}$ be a parameterized future-directed causal path with $\gamma(0) = p$; then $\sigma_p \circ \gamma$ is a locally Lipschitzian function. Hence

$$\begin{aligned}\sigma_p \circ \gamma(t) &= \int_0^t \frac{d(\sigma_p \circ \gamma)(s)}{ds} ds \\ &= \int_0^t g(\nabla \sigma_p, \dot{\gamma})(s) ds.\end{aligned}\tag{2.4.10}$$

We note the following.

LEMMA 2.4.9 *A future-directed causal path $\gamma \subset \mathcal{O}_p$ cannot leave the set $\{q : x^0(q) > 0, \sigma_p(q) < 0\}$.*

Proof. The time function x^0 remains positive along γ by Lemma 2.4.8. If $-\sigma_p$ were also a time function we would be done by the same argument. The problem is that $-\sigma_p$ is a time function only on the set where σ_p is negative, so some care is needed; we proceed as follows: the vector field $\nabla \sigma_p$ is causal future directed on $\{x^0 > 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}$, while $\dot{\gamma}$ is causal future directed or zero wherever defined. Hence $g(\nabla \sigma_p, \dot{\gamma}) \leq 0$ as long as γ stays in $\{x^0 > 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}$. By Eq. (2.4.9) the function σ_p is non-increasing along γ as long as γ stays in $\{x^0 > 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}$. Suppose that $\sigma_p(\gamma(s_1)) < 0$ and let

$$s_* = \sup\{u \in I : \sigma_p(\gamma(s)) < 0 \text{ on } [s_1, u]\}.$$

If $s_* \in I$, then $\sigma_p \circ \gamma(s_*) = 0$ and $\sigma_p \circ \gamma$ is *not* non-increasing on $[s_1, s_*)$, which is not possible since $\gamma(s) \in \{x^0 > 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}$ for $s \in [s_1, s_*)$. It follows that $\sigma_p \circ \gamma < 0$, as desired. $\square$

Proposition 2.4.5 immediately follows for those future direct causal paths through p which *do enter* the set $\{\eta_{\mu\nu} x^\mu x^\nu < 0\}$. This is the case for γ 's such that $\dot{\gamma}(0) = (\dot{\gamma}^\mu(0))$ exists and is timelike: We then have

$$\gamma^\mu(s) = s\dot{\gamma}^\mu(0) + o(s);$$

hence

$$\eta_{\mu\nu} \gamma^\mu(s) \gamma^\nu(s) = s^2 \eta_{\mu\nu} \dot{\gamma}^\mu(0) \dot{\gamma}^\nu(0) + o(s^2) < 0$$

for s small enough. It follows that γ enters the set $\{\eta_{\mu\nu} x^\mu x^\nu < 0\} \equiv \{q : \sigma_p(q) < 0\}$, and remains there for $|s|$ small enough. We conclude using Lemma 2.4.9.

We continue with arbitrary parameterized future-directed *timelike* paths $\gamma : [0, b) \rightarrow \mathcal{M}$, with $\gamma(0) = p$; thus $\dot{\gamma}$ exists and is timelike future directed for almost all $s \in [0, b)$. In particular there exists a sequence $s_i \rightarrow_{i \rightarrow \infty} 0$ such that $\dot{\gamma}(s_i)$ exists and is timelike.

Standard properties of solutions of ODEs show that for each $q \in \mathcal{O}_p$ there exists a neighbourhood $\mathcal{W}_{p,q}$ of p such that the function

$$\mathcal{W}_{p,q} \ni r \rightarrow \sigma_r(q)$$

is defined, continuous in r . For i large enough we will have $\gamma(s_i) \in \mathcal{W}_{p,\gamma(s)}$; for such i 's we have just shown that

$$\sigma_{\gamma(s_i)}(\gamma(s)) < 0.$$

Passing to the limit $i \rightarrow \infty$, by continuity one obtains

$$\sigma_p(\gamma(s)) \leq 0;\tag{2.4.11}$$

thus

$$\gamma \subset \{x^0 \geq 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}.\tag{2.4.12}$$

Since $\dot{\gamma}$ is timelike future directed wherever defined, and $\nabla \sigma_p$ is causal future directed on $\{x^0 > 0, \eta_{\mu\nu} x^\mu x^\nu \leq 0\}$, Eqs. (2.4.10) and (2.4.12) show that the inequality in (2.4.11) must be strict.

To finish the proof, we reduce the general case to the last one by considering perturbed metrics, as follows: let e_0 be any unit timelike vector field on $\mathcal{O}$ (e_0 can, e.g., be chosen as $\nabla x^0 / \sqrt{-g(\nabla x^0, \nabla x^0)}$); for $\epsilon > 0$ define a family of Lorentzian metrics g_ϵ on $\mathcal{O}$ by the formula

$$g_\epsilon(X, Y) = g(X, Y) - \epsilon g(e_0, X)g(e_0, Y).$$

Consider any vector X which is causal for g ; then

$$\begin{aligned} g_\epsilon(X, X) &= g(X, X) - \epsilon(g(e_0, X))^2 \\ &\leq -\epsilon(g(e_0, X))^2 < 0, \end{aligned}$$

so that X is timelike for g_ϵ . Let $\sigma(g_\epsilon)_p$ be the associated functions defined as in (2.4.6), where the exponential map there is the one associated to the metric g_ϵ . Standard properties of solutions of ODEs (see, e.g., [444]) imply that for any compact subset K of $\mathcal{O}_p$ there exists an $\epsilon_K > 0$ and a neighbourhood $\mathcal{O}_{p,K}$ of K such that for all $\epsilon \in [0, \epsilon_K]$ the functions

$$\mathcal{O}_{p,K} \ni q \rightarrow \sigma(g_\epsilon)_p(q)$$

are defined, and depend continuously upon ϵ . We take K to be $\gamma([0, s])$, where s is such that $[0, s] \subset I$, and consider any ϵ in $(0, \epsilon_{\gamma([0, s])})$. Since γ is timelike for g_ϵ , the results already established show that we have

$$\sigma(g_\epsilon)_p(\gamma(s)) < 0.$$

Continuity in ϵ implies that

$$\sigma_p(\gamma(s)) \leq 0.$$

Since s is arbitrary in I , Proposition 2.4.5 is established. $\square$

INCIDENTALLY: For certain considerations it is useful to have the following.

COROLLARY 2.4.10 Let $\mathcal{O}_p$ be a domain of normal coordinates x^μ centred at $p \in \mathcal{M}$ as in Proposition 2.2.4, and let $\mathcal{O} \subset \mathcal{O}_p$ be any normal-coordinates ball such that ∇x^0 is timelike on $\mathcal{O}$. If $\gamma \subset \mathcal{O}$ is a causal curve from p to

$$q = \exp_p(x^\mu) \in J^+(p; \mathcal{O}),$$

then γ lies entirely in $J^+(p; \mathcal{O})$, and there exists a reparameterization $s \rightarrow r(s)$ of γ so that γ is a null-geodesic segment through p :

$$[0, 1] \ni s \rightarrow \gamma(r(s)) = \exp_p(sx^\mu).$$

Proof. Proposition 2.4.5 shows that $\sigma_p(q) = 0$. It follows that

$$0 = \sigma_p \circ \gamma(t) = \int_0^t g(\nabla \sigma_p, \dot{\gamma})(s) ds. \quad (2.4.13)$$

Since $\nabla \sigma_p$ and $\dot{\gamma}$ are causal oppositely directed we have $g(\nabla \sigma_p, \dot{\gamma}) \geq 0$ almost everywhere. It thus follows from (2.4.13) that

$$g(\nabla \sigma_p, \dot{\gamma}) = 0$$

almost everywhere. Recall, now, that the scalar product of two causal consistently directed vectors vanishes only if they are proportional. Thus

$$\nabla \sigma_p \sim \dot{\gamma} \quad (2.4.14)$$

almost everywhere. Thus $\nabla \sigma_p$ is null a.e. along γ . But $\nabla \sigma_p$ is null only on $J^+(p; \mathcal{O})$, thus $\gamma(s) \in J^+(p; \mathcal{O})$ a.e.; by continuity this is true for all s , and we have shown that γ lies entirely on $J^+(p; \mathcal{O})$.

To continue, in normal coordinates (2.4.14) reads (see point 4 of Proposition B.3, p. 340)

$$\frac{dx^\mu(s)}{ds} = f(s) x^\mu(s)$$

a.e., for some strictly positive function $f \in L^\infty$. Define $r(s)$ by

$$r(s) = \int_0^s f(t) dt;$$

then r is strictly increasing, hence a bijection from the interval of definition of γ to some interval $[0, r_0]$. Further r is Lipschitz, differentiable on a set of full measure, and on that set it holds that

$$\frac{dr}{ds} = f.$$

Now, this equation shows that the set where the map $r \mapsto s(r)$ might fail to be differentiable is the image by r of the set Ω_1 where r fails to be differentiable, together with the image by r of the set of points Ω_2 where f vanishes. Both Ω_1 and Ω_2 have zero measure, and the image of a negligible set by a Lipschitz map is a negligible set. We thus obtain, almost everywhere,

$$\frac{dx^\mu(s(r))}{dr} = \frac{dx^\mu(s(r))}{ds} \frac{ds}{dr} = x^\mu(s(r)), \quad (2.4.15)$$

so dx^μ/dr can be extended by continuity to a continuous function. It easily follows that $x^\mu(s(\lambda))$ is C^1 , and the result is obtained by integration of (2.4.15). $\square$

INCIDENTALLY: Penrose's approach [398] to the theory of causality is based on the notion of *timelike or causal trips*: by definition, a causal trip is a piecewise broken causal geodesic. The following result can be used to show equivalence of the definitions of I^+ , etc., given here, to those of Penrose.

COROLLARY 2.4.11 Consider a spacetime $(\mathcal{M}, g)_{C^2}$. If $q \in I^+(p)$, then there exists a future-directed piecewise broken future-directed timelike geodesic from p to q . Similarly, if $q \in J^+(p)$, then there exists a future-directed piecewise broken future-directed causal geodesic from p to q .

Proof. Let $\gamma : [0, 1] \rightarrow \mathcal{M}$ be a parameterized future-directed causal path with $\gamma(0) = p$ and $\gamma(1) = q$. Continuity of γ implies that for every $s \in [0, 1]$ there exists $\epsilon_s > 0$ such that

$$\gamma(u) \in \mathcal{O}_{\gamma(s)}$$

for all

$$u \in (s - 2\epsilon_s, s + 2\epsilon_s) \cap [0, 1] = [\max(0, s - 2\epsilon_s), \min(1, s + 2\epsilon_s)],$$

where $\mathcal{O}_r$ is a normal-coordinates ball centred at r , and satisfying the requirements of Proposition 2.4.5. Compactness of $[0, 1]$ implies that from the covering

$$\{(s - \epsilon_s, s + \epsilon_s)\}_{s \in [0, 1]},$$

a finite covering $\{(s_i - \epsilon_{s_i}, s_i + \epsilon_{s_i})\}_{i=0, \dots, N}$ can be extracted, with $s_0 = 0$, $s_N = 1$. Reordering the s_i 's if necessary we may assume that $s_i < s_{i+1}$. By definition we have

$$\gamma|_{[s_i, s_{i+1}]} \subset \mathcal{O}_{\gamma(s_i)},$$

and by Proposition 2.4.5 there exists a causal future-directed geodesic segment from $\gamma(s_i)$ to $\gamma(s_{i+1})$: if $\gamma(s_{i+1}) = \exp_{\gamma(s_i)}(x^\mu)$, then the required geodesic segment is given by

$$[0, 1] \ni s \rightarrow \exp_{\gamma(s_i)}(sx^\mu).$$

If γ is timelike, then all the segments are timelike. Concatenating the segments together provides the claimed piecewise broken geodesic. $\square$

Proposition 2.4.3 shows that the sets $I^\pm(p)$ are open in Minkowski spacetime. Similarly it follows from Proposition 2.4.5 that the sets $I^\pm(p; \mathcal{O}_p)$ are open. This turns out to be true in general.

PROPOSITION 2.4.12 Consider a spacetime $(\mathcal{M}, g)_{C^2}$. For all $\mathcal{U} \subset \mathcal{M}$ the sets $I^\pm(\mathcal{U})$ are open.

Proof. Let $q \in I^+(\mathcal{U})$, and let, as in the proof of Corollary 2.4.11, s_{N-1} be such that $q \in \mathcal{O}_{\gamma(s_{N-1})}$. Then

$$\mathcal{O}_{\gamma(s_{N-1})} \cap I^+(\gamma(s_{N-1}); \mathcal{O})$$

is an open neighbourhood of q by Corollary 2.4.7. Clearly

$$I^+(\gamma(s_{N-1}); \mathcal{O}) \subset I^+(\gamma(s_{N-1})).$$

Since $\gamma(s_{N-1}) \in I^+(\mathcal{U})$ we have

$$I^+(\gamma(s_{N-1})) \subset I^+(\mathcal{U})$$

(see point 3 of Proposition 2.4.2). It follows that

$$\mathcal{O}_{\gamma(s_{N-1})} \cap I^+(\gamma(s_{N-1}); \mathcal{O}) \subset I^+(\mathcal{U}),$$

which implies our claim. $\square$

In Minkowski spacetime the sets $J^\pm(p)$ are closed, with

$$\overline{I^\pm(p)} = J^\pm(p). \quad (2.4.16)$$

We will show in Corollary 2.4.18 that we always have

$$\overline{I^\pm(p)} \supset J^\pm(p), \quad (2.4.17)$$

but this requires some work. Before proving (2.4.17), let us point out that (2.4.16) does not need to be true in general.

EXAMPLE 2.4.13 Let $(\mathcal{M}, g)$ be the two-dimensional Minkowski spacetime $\mathbb{R}^{1,1}$ from which the set $\{x^0 = 1, x^1 \leq -1\}$ has been removed. Then

$$J^+(0; \mathcal{M}) = J^+(0, \mathbb{R}^{1,1}) \setminus \{x^0 = -x^1, x^1 \in (-\infty, 1]\},$$

cf. Figure 2.4.2. Hence $J^+(0; \mathcal{M})$ is neither open nor closed, and Eq. (2.4.16) does

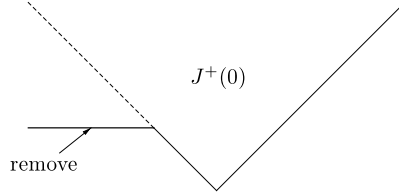


Fig. 2.4.2 $J^+(p)$ is not closed unless some causal regularity conditions are imposed on $(\mathcal{M}, g)$.

not hold.

We have the following.

LEMMA 2.4.14 ('Push-up Lemma 1') *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. For any $\Omega \subset \mathcal{M}$ we have*

$$I^+(J^+(\Omega)) = I^+(\Omega). \quad (2.4.18)$$

Proof. The obvious property

$$\mathcal{U} \subset \mathcal{V} \implies I^+(\mathcal{U}) \subset I^+(\mathcal{V})$$

provides inclusion of the right-hand side of (2.4.18) into the left-hand side. It remains to prove that

$$I^+(J^+(\Omega)) \subset I^+(\Omega).$$

Let $r \in I^+(J^+(\Omega))$; thus there exists a past-directed timelike curve γ_0 from r to a point $q \in J^+(\Omega)$. Since $q \in J^+(\Omega)$, then either $q \in \Omega$, and there is nothing to prove,

or there exists a past-directed causal curve $\gamma : I \rightarrow \mathcal{M}$ from q to some point $p \in \Omega$. We want to show that there exists a past-directed timelike curve $\hat{\gamma}$ starting at r and ending at p . The curve $\hat{\gamma}$ can be obtained by ‘pushing-up’ γ slightly to make it timelike, the construction proceeds as follows: using compactness, we cover γ by a finite collection $\mathcal{U}_i$, $i = 0, \dots, N$, of elementary regions $\mathcal{U}_i$ centred at $p_i \in \gamma(I)$, with

$$p_0 = q, \quad p_i \in \mathcal{U}_i \cap \mathcal{U}_{i+1}, \quad p_{i+1} \subset J^-(p_i), \quad p_N = p.$$

Let $\gamma_0 : [0, s_0] \rightarrow \mathcal{M}$ be the already-mentioned causal curve from r to $q \in \mathcal{U}_1$; let $s_1 \neq s_0$ be close enough to s_0 such that $\gamma_0(s_1) \in \mathcal{U}_1$. By Proposition 2.2.4 together with the definition of elementary regions there exists a past-directed timelike curve $\gamma_1 : [0, 1] \rightarrow \mathcal{U}_1$ from $\gamma_0(s_1)$ to $p_1 \in \mathcal{U}_1 \cap \mathcal{U}_2$. For s close enough to 1 the curve γ_1 enters $\mathcal{U}_2$; choose an $s_2 \neq 1$ such that $\gamma_1(s_2) \in \mathcal{U}_2$. Again by Proposition 2.2.4 there exists a past-directed timelike curve $\gamma_2 : [0, 1] \rightarrow \mathcal{U}_2$ from $\gamma_1(s_2)$ to p_2 . One repeats that construction, iteratively obtaining a (finite) sequence of past-directed timelike curves $\gamma_i \subset I^+(\gamma) \cap \mathcal{O}$ such that the end point $\gamma_i(s_{i+1})$ of $\gamma_i|_{[0, s_{i+1}]}$ coincides with the starting point of γ_{i+1} . Concatenating those curves together gives the desired path $\hat{\gamma}$. $\square$

We have the following, slightly stronger, version of Lemma 2.4.14, which gives a sufficient condition to be able to deform a *causal* curve to a *timelike* one, keeping the deformation as small as desired.

COROLLARY 2.4.15 *Let the metric be twice differentiable. Consider a causal future-directed curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ from p to q . If there exist $s_1 < s_2 \in [0, 1]$ such that $\gamma|_{[s_1, s_2]}$ is timelike, then in any neighbourhood $\mathcal{O}$ of γ there exists a timelike future-directed curve $\hat{\gamma}$ from p to q .*

REMARK 2.4.16 The so-called *maximizing* null geodesics can *not* be deformed as above to timelike curves, whether locally or globally. We note that all null geodesics in Minkowski spacetime are maximizing.

Proof. If $s_2 = 1$, then Corollary 2.4.15 is essentially a special case of Lemma 2.4.14: the only difference is the statement about the neighbourhood $\mathcal{O}$. This last requirement can be satisfied by choosing the sets $\mathcal{U}_i$ in the proof of Lemma 2.4.14 so that $\mathcal{U}_i \subset \mathcal{O}$. If $s_1 = 0$ (and regardless of the value of s_2) the result is obtained by changing time orientation, applying the result already established to the path $\gamma'(s) = \gamma(1 - s)$, and changing time orientation again. The general case is reduced to those already covered by first deforming the curve $\gamma|_{[0, s_2]}$ to a new timelike curve $\tilde{\gamma}$ from p to $\gamma(s_2)$, and then applying the result again to the curve $\tilde{\gamma} \cup \gamma|_{[s_2, 1]}$. $\square$

Another result in the same spirit is provided by the following.

PROPOSITION 2.4.17 *Let γ be a causal curve from p to q in $(\mathcal{M}, g)_{C^2}$ which is not a null geodesic. Then there exists a timelike curve from p to q .*

Proof. By Corollary 2.4.11 we can without loss of generality assume that γ is a piecewise broken geodesic. If one of the geodesics forming γ is timelike, the result follows from Corollary 2.4.15. It remains to consider curves which are piecewise broken null geodesics with at least one break point, say p . Let $q \in J^-(p)$ be close enough to p so that p belongs to a domain of normal coordinates $\mathcal{O}_q$ centred at q . Corollary 2.4.10 shows that points on γ lying to the causal future of p are not in $J^+(q, \mathcal{O}_q)$; hence, they are in $I^+(q, \mathcal{O}_q)$, and so γ can be deformed within $\mathcal{O}_q$ to a timelike curve. The result follows now again from Corollary 2.4.15. $\square$

As another straightforward corollary of Lemma 2.4.14 one obtains a property of J , which is wrong in general for metrics which are not of $C^{0,1}$ differentiability class [125].

COROLLARY 2.4.18 Consider a spacetime $(\mathcal{M}, g)_{C^2}$. For any $p \in \mathcal{M}$ we have

$$J^+(p) \subset \overline{I^+(p)}.$$

Proof. Let $q \in J^+(p)$, and let $r_i \in I^+(q)$ be any sequence of points accumulating at q ; then $r_i \in I^+(p)$ by Lemma 2.4.14, hence $q \in \overline{I^+(p)}$. $\square$

2.5 Extendible and inextendible paths

To avoid ambiguities, recall that we only assume continuity of the metric unless explicitly indicated otherwise.

A useful concept, when studying causality, is that of a causal path which cannot be extended any further. Recall that, from a physical point of view, the image in spacetime of a timelike path is supposed to represent the history of some observer, and it is sometimes useful to have at hand idealized observers which never cease to exist. Here it is important to have the geometrical picture in mind, where all that matters is the image in spacetime of the path, independently of any parameterization: if that image ‘stops’, then one can sometimes continue the path by concatenating with a further one; continuing in this way one hopes to be able to obtain paths which are inextendible.

In order to make things precise, let $\gamma : [a, b) \rightarrow \mathcal{M}$ be a parameterized, causal, future-directed path. A point p is called a *future end point* of γ if $\lim_{s \rightarrow b} \gamma(s) = p$. Past end points are defined in the obvious analogous way. An *end point* is a point which is either a past end point or a future end point.

Given γ as above, together with an end point p , one is tempted to extend γ to a new path $\hat{\gamma} : [a, b] \rightarrow \mathcal{M}$ defined as

$$\hat{\gamma}(s) = \begin{cases} \gamma(s), & s \in [a, b), \\ p, & s = b. \end{cases} \quad (2.5.1)$$

The first problem with this procedure is that the resulting curve might fail to be locally Lipschitz in general. An example is given by the timelike future-directed path

$$[0, 1) \ni s \rightarrow \gamma_1(s) = (-(1-s)^{1/2}, 0) \in \mathbb{R}^{1,1},$$

which is locally Lipschitzian on $[0, 1)$, but is *not* on $[0, 1]$ when extended using (2.5.1). (This follows from the fact that the difference quotient $(f(s) - f(s'))/(s - s')$ blows up as s and s' tend to 1 when $f(s) = (1-s)^{1/2}$.) Recall that in our definition of a causal curve γ , a prerequisite condition is the locally Lipschitz character, so that the extension $\hat{\gamma}_1$ fails to be causal even though γ_1 is.

The problem is even worse if $b = \infty$: consider the timelike future-directed curve

$$[1, \infty) \ni s \rightarrow \gamma_2(s) = (-1/s, 0) \in \mathbb{R}^{1,1}.$$

Here there is no way to extend the curve to the future as an application from a subset of $\mathbb{R}$ to $\mathcal{M}$, because the range of parameters already covers all $s \geq 1$. Now, the image of both γ_1 and γ_2 is simply the interval $[-1, 0) \times \{0\}$, which can be extended to a longer causal curve in $\mathbb{R}^{1,1}$ in many ways if one thinks in terms of images rather than of maps.

Both problems above can be taken care of by requiring that the parameter s be the length parameter of an auxiliary Riemannian metric h . (At this stage h is not required to be complete.) This might require reparameterizing the path, as in Proposition 2.3.3. From the point of view of our definition this means that we are passing to a different path, but the image in spacetime of the new path coincides with the previous one. If one thinks of timelike paths as describing observers, the

new observer will thus have experienced identical events, even though he or she will be experiencing those events at different times on his or her time-measuring device. We note, moreover, that (locally Lipschitz) reparameterizations do not change the timelike or causal character of paths.

EXAMPLE 2.5.1 Here some care is needed when passing to a limit with a sequence of curves: consider a sequence of null geodesics in $\mathbb{R}^{1,1} = (\mathbb{R}^2, g = -dt^2 + dx^2)$, with $h = dt^2 + dx^2$ as the Riemannian background metric, threading back and forth up to a space-distance $1/n$ around the $\{x = 0\}$ axis. The limit curve is $\gamma(s) = (s/\sqrt{2}, 0)$ which is *not* h -length-parameterized. $\square$

We have already shown in Section 2.3 that a locally Lipschitzian path can always be reparameterized by h -length, leading to a uniformly Lipschitzian path, with Lipschitz constant 1. It should be clear from the examples given earlier, as well as from the examples to be discussed at the beginning of Section 2.6, that it is sensible to use such a parameterization, and it is tempting to build this requirement into the definition of a causal path. One reason for not doing that is the existence of affine parameterization for geodesics, which is geometrically significant, and which is convenient for several purposes. Another reason is the arbitrariness related to the choice of h . Last but not least, a limit curve for a sequence of h -length-parameterized curves does not have to be h -length-parameterized, as seen in Example 2.5.1. Therefore we will not assume a priori an h -length parameterization, but such a reparameterization will often be used in the proofs.

Returning to (2.5.1), we want to show that $\hat{\gamma}$ will be uniformly Lipschitz if h -length-parameterization is used for γ . More generally, suppose that γ is uniformly Lipschitz with Lipschitz constant L ,

$$\text{dist}_h(\gamma(s), \gamma(s')) \leq L|s - s'|. \quad (2.5.2)$$

Passing with s' to b in that equation we obtain

$$\text{dist}_h(\gamma(s), p) \leq L|s - b|,$$

and the Lipschitzian character of $\hat{\gamma}$ easily follows. We have therefore proved the following.

LEMMA 2.5.2 *Let $\gamma : [a, b) \rightarrow \mathcal{M}$, $b < \infty$, be a uniformly Lipschitzian path with an end point p . Then γ can be extended to a uniformly Lipschitzian path $\hat{\gamma} : [a, b] \rightarrow \mathcal{M}$, with $\hat{\gamma}(b) = p$.*

Let $\gamma : [a, b) \rightarrow \mathcal{M}$, $b \in \mathbb{R} \cup \{\infty\}$ be a path; then p is said to be an ω -limit point of γ if there exists a sequence $s_k \rightarrow b$ such that $\gamma(s_k) \rightarrow p$. An end point is always an ω -limit point, but the inverse does not need to be true in general (consider $\gamma(s) = \exp(is) \in \mathbb{C}$; then every point $\exp(ix) \in S^1 \subset \mathbb{C}^1$ is seen to be a ω -limit point of γ by setting $s_k = x + 2\pi k$). For $b < \infty$ and for uniformly Lipschitz paths the notions of end point and of ω -limit point coincide.

LEMMA 2.5.3 *Let $\gamma : [a, b) \rightarrow \mathcal{M}$, $b < \infty$, be a uniformly Lipschitzian path. Then every ω -limit point of γ is an end point of γ . In particular, γ has at most one ω -limit point.*

Proof. By (2.5.2) we have

$$\text{dist}_h(\gamma(s_i), \gamma(s)) \leq L|s_i - s|,$$

and since dist_h is a continuous function of its arguments we obtain, passing to the limit $i \rightarrow \infty$,

$$\text{dist}_h(p, \gamma(s)) \leq L|b - s|.$$

Thus p is an end point of γ . Since there can be at most one end point, the result follows. $\square$

A future-directed causal curve $\gamma : [a, b) \rightarrow \mathcal{M}$ will be said to be *future extendible* if there exists $b < c \in \mathbb{R} \cup \{\infty\}$ and a causal curve $\tilde{\gamma} : [a, c) \rightarrow \mathcal{M}$ such that

$$\tilde{\gamma}|_{[a,b)} = \gamma. \quad (2.5.3)$$

The path $\tilde{\gamma}$ is then said to be an *extension* of γ . The curve γ will be said to be *future inextendible* if it is not future extendible. The notions of *past extendibility* and of *extendibility* are defined in the obvious way.

Extendibility in the class of causal paths forces a causal $\gamma : [a, b) \rightarrow \mathcal{M}$ to be uniformly Lipschitzian: This follows from the fact that $[a, b]$ is a compact subset of the domain of definition of any extension $\tilde{\gamma}$, so that $\tilde{\gamma}|_{[a,b]}$ is uniformly Lipschitzian there. But then $\tilde{\gamma}|_{[a,b)}$ is also uniformly Lipschitzian, and the result follows from (2.5.3).

Whenever a uniformly Lipschitzian path can be extended by adding an end point, it can also be extended as a strictly longer path.

LEMMA 2.5.4 *A uniformly Lipschitzian causal path $\gamma : [a, b) \rightarrow \mathcal{M}$, $b < \infty$ is extendible if and only if it has an end point.*

Proof. Let $\hat{\gamma}$ be given by Proposition 2.5.2, and let $\tilde{\gamma} : [0, d)$ be any maximally extended to the future, future-directed causal geodesic starting at p , for an appropriate $d \in (0, \infty)$. Then $\hat{\gamma} \cup \tilde{\gamma}$ is an extension of γ . $\square$

It turns out that the paths considered in Lemma 2.5.4 are always extendible:

THEOREM 2.5.5 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let $\gamma : [a, b) \rightarrow \mathcal{M}$, $b \in \mathbb{R} \cup \{\infty\}$, be a future-directed causal path parameterized by h -distance, where h is any complete auxiliary Riemannian metric. Then γ is future inextendible if and only if $b = \infty$.*

Proof. Suppose that $b < \infty$. Let $B_h(p, r)$ denote the open h -distance ball, with respect to the metric h , of radius r , centred at p . Since γ is parameterized by h -length we have, by (2.3.2),

$$\gamma([a, b)) \subset B_h(\gamma(a), b - a).$$

The Hopf–Rinow theorem [258, 348] asserts that $\overline{B_h(\gamma(a), b - a)}$ is compact; therefore there exists $p \in \overline{B_h(\gamma(a), b - a)}$ and a sequence s_i such that

$$[a, b) \ni s_i \rightarrow_{i \rightarrow \infty} b \quad \text{and} \quad \gamma(s_i) \rightarrow p.$$

Thus p is an ω -limit point of γ . Clearly γ is uniformly Lipschitzian (with Lipschitz modulus 1), and Lemma 2.5.3 shows that p is an end point of γ . The result follows now from Lemma 2.5.4. $\square$

2.5.1 Maximally extended geodesics

Consider the Cauchy problem for an affinely parameterized geodesic γ :

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0, \quad \gamma(0) = p, \quad \dot{\gamma}(0) = X. \quad (2.5.4)$$

This is a second-order ODE which, by the standard theory [241] for $C^{1,1}$ metrics, has unique solutions defined on a maximal interval $I = I(p, X) \ni 0$. The interval I is maximal in the sense that if I' is another interval containing 0 on which a solution of (2.5.4) is defined, then $I' \subset I$. When I is maximal the geodesic will be called *maximally extended*. Now, it is not immediately obvious that a *maximally extended* geodesic is *inextendible* in the sense just defined: to start with, the notion of inextendibility involves only the pointwise properties of a path, while the notion of maximally extended geodesic involves the ODE (2.5.4), which involves both the first and second derivatives of γ . Next, the inextendibility criteria given above have

been formulated in terms of uniformly Lipschitzian parameterizations. While an affinely parameterized geodesic is certainly locally Lipschitzian, there is no a priori reason why it should be uniformly so, when maximally extended. All these issues turn out to be irrelevant, and we have the following.

PROPOSITION 2.5.6 *Consider a spacetime $(\mathcal{M}, g)_{C^{1,1}}$. A geodesic $\gamma : I \rightarrow \mathcal{M}$ is maximally extended as a geodesic if and only if γ is inextendible as a causal path.*

Proof. Suppose, for contradiction, that γ is a maximally extended geodesic which is extendible as a path; thus γ can be extended to a path $\hat{\gamma}$ by adding its end point p as in (2.5.1). Working in a normal coordinate neighbourhood $\mathcal{O}_p$ around p , $\hat{\gamma} \cap \mathcal{O}_p$ has a last component which is a geodesic segment which ends at p . By construction of normal coordinates the component of $\hat{\gamma}$ in question is simply a half-ray through the origin, which can be clearly continued through p as a geodesic. This contradicts maximality of γ as a geodesic. It follows that a maximally extended geodesic is inextendible. Now, if γ is inextendible as a path, then γ can clearly not be extended as a geodesic, which establishes the reverse implication. $\square$

A result often used in causality theory is the following.

THEOREM 2.5.7 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let γ be a future-directed causal, respectively timelike, path. Then there exists an inextendible causal, respectively timelike, extension of γ .*

Proof. If $\gamma : [a, b) \rightarrow \mathcal{M}$ is inextendible there is nothing to prove; otherwise the path $\hat{\gamma} \cup \tilde{\gamma}$, where $\hat{\gamma}$ is given by Proposition 2.5.2, and $\tilde{\gamma}$ is any maximally extended future-directed causal geodesic as in the proof of Lemma 2.5.4, provides an extension. This extension is inextendible by Proposition 2.5.6. $\square$

INCIDENTALLY: Theorem 2.5.7 remains true for metrics which are merely assumed to be continuous; this proceeds as follows: suppose that γ is extendible; in particular, γ has an end point p . Let Ω_p denote the collection of all future-directed, parameterized by h -proper distance, timelike paths starting at p . Obviously Ω_p is non-empty. Ω_p can be directed using the property of ‘being an extension’: we write $\gamma_1 < \gamma_2$ if γ_2 is an extension of γ_1 . The existence of inextendible paths in Ω_p follows now from the Kuratowski–Zorn lemma.

If γ_1 is any maximal element of Ω_p , then $\hat{\gamma} \cup \gamma_1$, with $\hat{\gamma}$ given by Lemma 2.5.2, is an inextendible future-directed extension of γ . $\square$

2.6 Accumulation curves

A key tool in the analysis of global properties of spacetimes is the analysis of sequences of curves. One typically wants to obtain a limiting curve, and study its properties. The object of this section is to establish the existence of such limiting curves.

We wish, first, to find the ingredients needed for a useful notion of a limit of curves. It is enlightening to start with several examples.

The first question that arises is whether to consider a sequence of curves γ_n defined on a common interval I , or whether one should allow different domains I_n for each γ_n . To illustrate that this last option is very unpractical, consider the family of timelike curves

$$(-1/n, 1/n) \ni s \rightarrow \gamma_n(s) = (s, 0) \in \mathbb{R}^{1,1}. \quad (2.6.1)$$

The only sensible geometric object to which the $\gamma_n(s)$ converge is the constant map

$$\{0\} \ni s \rightarrow \gamma_\infty(s) = 0 \in \mathbb{R}^{1,1}, \quad (2.6.2)$$

which is quite reasonable, except that it takes us away from the class of causal curves. To avoid such behaviour we will therefore assume that all the curves γ_n have a common domain of definition I .

Next, there are various reasons why a sequence of curves might fail to have an ‘accumulation curve’. First, the whole sequence might simply run to infinity. (Consider, for example, the sequence

$$\mathbb{R} \ni s \rightarrow \gamma_n(s) = (s, n) \in \mathbb{R}^{1,1}.)$$

This is avoided when one considers curves such that $\gamma_n(0)$ converges to some point $p \in \mathcal{M}$.

Further, there might be a problem with the way the curves are parameterized. As an example, let γ_n be defined as

$$(-1, 1) \ni s \rightarrow \gamma_n(s) = (s/n, 0) \in \mathbb{R}^{1,1}.$$

As in (2.6.1), the $\gamma_n(s)$ converge to the constant map

$$(-1, 1) \ni s \rightarrow \gamma_\infty(s) = 0 \in \mathbb{R}^{1,1}, \quad (2.6.3)$$

again not a causal curve. This kind of behaviour can be avoided by requiring a *uniform inverse-Lipschitz bound*,

$$\text{dist}_h(\gamma_n(s), \gamma_n(s')) \geq c|s - s'|, \quad (2.6.4)$$

for some $c > 0$. A curve satisfying (2.6.4) will be called *inverse-Lipschitz*. It will be called *locally inverse-Lipschitz* if for every s in the parameter-domain I there exists a neighbourhood $I_s \subset I$ of s and a constant $c = c(s)$ such that (2.6.4) holds for $s' \in I_s$.

We show below (cf. (2.6.11)) that causal curves are locally inverse-Lipschitz.

EXAMPLE 2.6.1 Requiring (2.6.4) globally is too restrictive in some situations even on a single causal curve. Consider, for instance, $S^1 \times \mathbb{R}$ with the metric $-d\varphi^2 + dx^2$, where the φ factor runs over S^1 with periodicity 2π , and let the auxiliary Riemannian metric be $h = d\varphi^2 + dx^2$. The curve

$$\mathbb{R} \ni s \mapsto \gamma(s) = (\varphi = s, x = 0)$$

is timelike, h -length-parameterized, and (2.6.4) holds on γ for $|s - s'| \leq \pi$ with $c = 1$. However, it clearly cannot hold globally since, e.g., $\gamma(2\pi) = \gamma(0)$. $\square$

Another example of pathological parameterizations is given by the family of curves

$$\mathbb{R} \ni s \rightarrow \gamma_n(s) = (ns, 0) \in \mathbb{R}^{1,1}.$$

In this case one is tempted to say that the γ_n ’s accumulate at the path, say γ_1 , if parameterization is not taken into account. However, such a convergence is extremely awkward to deal with when attempting to prove something. This last behaviour can be avoided by assuming that all the curves are uniformly Lipschitz continuous, with the same Lipschitz constant. One way of ensuring this is to parameterize all the curves by a length parameter with respect to our auxiliary complete Riemannian metric h .

Yet another problem arises when considering the family of Euclidean-distance-parameterized causal curves

$$\mathbb{R} \ni s \rightarrow \gamma_n(s) = (s + n, 0) \in \mathbb{R}^{1,1}.$$

This can be gotten rid of by shifting the distance parameter so that the sequence $\gamma_n(s_0)$ converges, or stays in a compact set, for some s_0 in the common domain I .

The above discussion motivates the hypotheses of the following result.

PROPOSITION 2.6.2 *Let $(\mathcal{M}, g)$ be a C^3 Lorentzian manifold with a C^2 metric. Let $\gamma_n : I \rightarrow \mathcal{M}$ be a sequence of future-directed causal curves and suppose that there exist a covering of I by closed intervals $K_i \subset I$, $i \in \mathbb{N}$, and constants $L_i \equiv L(K_i)$ (and thus independent of n) such that for $s, s' \in K_i$ we have*

$$L_i^{-1}|s - s'| \leq \text{dist}_h(\gamma_n(s), \gamma_n(s')) \leq L_i|s - s'|. \quad (2.6.5)$$

If there exists $p \in \mathcal{M}$ such that

$$\gamma_n(0) \rightarrow p, \quad (2.6.6)$$

then there exists a future-directed causal curve $\gamma : I \rightarrow \mathcal{M}$ and a subsequence γ_{n_i} converging to γ in the topology of uniform convergence on compact subsets of I .

REMARK 2.6.3 The hypothesis that g is C^2 is made to guarantee that the function $(p, q) \mapsto \sigma_q(p)$ is continuous, and depends continuously upon the metric. $\square$

Proposition 2.6.2 provides the justification for the following definition.

DEFINITION 2.6.4 *Let $\gamma_n : I \rightarrow \mathcal{M}$ be a sequence of paths in $(\mathcal{M}, g)_{C^0}$. We shall say that $\gamma : I \rightarrow \mathcal{M}$ is an accumulation curve of the γ_n 's, or that the γ_n 's accumulate at γ , if there exists a subsequence γ_{n_i} that converges to γ uniformly on compact subsets of I .*

INCIDENTALLY: In their treatment of causal theory, Hawking and Ellis [244] introduce a notion of *limit curve* for paths, regardless of parameterization, which we find very awkward to work with. A related but slightly more convenient notion of *cluster curve* is considered in [300], where the name of 'limit curve' is used for yet another notion of convergence. As discussed in [38, 300], those definitions lead to pathological behaviour in some situations. We have found the above notion of 'accumulation curve' the most convenient to work with.

A sensible terminology, in the context of Definition 2.6.4, could be ' C^0_{loc} -limits of curves', but we prefer not to use the term 'limit' in this context, as limits are usually unique, while Definition 2.6.4 allows for sequences that have more than one accumulation curve. $\square$

Proof of Proposition 2.6.2. Equation (2.6.7) shows that the family $\{\gamma_n\}$ is equicontinuous on every compact subset of I , and (2.6.6) together with the Arzela–Ascoli theorem implies that for every compact set $K \subset I$ there exists a curve $\gamma_K : K \rightarrow \mathcal{M}$ and a subsequence γ_{n_i} which converges uniformly to γ_K on K . One can obtain a K -independent curve γ by the so-called *diagonalization* procedure.

INCIDENTALLY: The diagonalization procedure goes as follows: first, if I is compact, there is nothing to do. Next, for ease of notation we consider $I = \mathbb{R}$; the same argument applies on any other non-closed interval with obvious modifications. Let $\gamma_{n(i,1)}$ be the sequence which converges to $\gamma_{[-1,1]}$; applying Arzela–Ascoli to this sequence one can extract a subsequence $\gamma_{n(i,2)}$ of $\gamma_{n(i,1)}$ which converges uniformly to some curve $\gamma_{[-2,2]}$ on $[-2,2]$. Since $\gamma_{n(i,2)}$ is a subsequence of $\gamma_{n(i,1)}$, and since $\gamma_{n(i,1)}$ converges to $\gamma_{[-1,1]}$ on $[-1,1]$, one finds that $\gamma_{[-2,2]}$ restricted to $[-1,1]$ equals $\gamma_{[-1,1]}$. One continues iteratively: suppose that $\{\gamma_{n(i,k)}\}_{i \in \mathbb{N}}$ has been defined for some k , and converges to a curve $\gamma_{[-k,k]}$ on $[-k,k]$; then the sequence $\{\gamma_{n(i,k+1)}\}_{i \in \mathbb{N}}$ is defined as a subsequence of $\{\gamma_{n(i,k)}\}_{i \in \mathbb{N}}$ which converges to some curve $\gamma_{[-(k+1),k+1]}$ on $[-(k+1),k+1]$. The curve γ is finally defined as

$$\gamma(s) = \gamma_{[-k,k]}(s),$$

where k is any number such that $s \leq k$. The construction guarantees that $\gamma_{[-k,k]}(s)$ does not depend upon k as long as $s \leq k$. $\square$

It remains to show that γ is causal. Passing to the limit $n \rightarrow \infty$ in (2.6.5) one finds on K_i

$$L_i^{-1}|s - s'| \leq \text{dist}_h(\gamma(s), \gamma(s')) \leq L_i|s - s'|. \quad (2.6.7)$$

For $q \in \mathcal{M}$ let $\mathcal{O}_q$ be an elementary neighbourhood of q as in Proposition 2.4.5, and let σ_q be the associated function defined by (2.4.6). Let $s \in \mathbb{R}$ and consider any point $\gamma(s) \in \mathcal{M}$. Now, the size of the sets $\mathcal{O}_q$ can be controlled uniformly when q varies

over compact subsets of $\mathcal{M}$. It follows that for all s' close enough to s and for all n large enough we have $\gamma_n(s') \in \mathcal{O}_{\gamma_n(s)}$. Since the γ_n 's are causal, Proposition 2.4.5 shows that we have

$$\sigma_{\gamma_n(s)}(\gamma_n(s')) \leq 0. \quad (2.6.8)$$

Passing to the limit in (2.6.8) gives

$$\sigma_{\gamma(s)}(\gamma(s')) \leq 0. \quad (2.6.9)$$

This is only possible if γ is causal, which can be seen as follows: suppose that γ is differentiable at s . In normal coordinates on $\mathcal{O}_{\gamma(s)}$ we have, by definition of the derivative,

$$\gamma^\mu(s') = \underbrace{\gamma^\mu(s)}_{=0} + \dot{\gamma}^\mu(s)(s' - s) + o(s' - s),$$

hence

$$0 \geq \sigma_{\gamma(s)}(\gamma(s')) \equiv \eta_{\mu\nu} \gamma^\mu(s') \gamma^\nu(s') = \eta_{\mu\nu} \dot{\gamma}^\mu(s) \dot{\gamma}^\nu(s) (s' - s)^2 + o((s' - s)^2).$$

For $s' - s$ small enough this is only possible if

$$\eta_{\mu\nu} \dot{\gamma}^\mu(s) \dot{\gamma}^\nu(s) \leq 0.$$

The local inverse-Lipschitz bound (2.6.7) shows that $\dot{\gamma}$ does not vanish. We conclude that $\dot{\gamma}$ is causal, as we desired to show. $\square$

Let us address now the question of inextendibility of accumulation curves. We note the following.

LEMMA 2.6.5 *Let γ_n be a sequence of h -length-parameterized inextendible causal curves converging to γ uniformly on compact subsets of $\mathbb{R}$. Then γ is inextendible.*

Proof. Note that the parameter range of γ is $\mathbb{R}$, and the result would follow from Theorem 2.5.5 if γ were h -length-parameterized, but this might fail to be the case, as seen in Example 2.5.1.

So we need to show that both $\gamma|_{[0, \infty)}$ and $\gamma|_{(-\infty, 0]}$ are of infinite length. As usual it suffices to consider $\gamma|_{[0, \infty)}$; we retain the name γ for this last path. Suppose that this is not the case; then there exists $a < \infty$ so that γ is defined on $[0, a)$, when reparameterized by h -length. By Theorem 2.5.5 the curve γ can be extended to a causal curve defined on $[0, a]$, still denoted by γ .

Let $\mathcal{U}$ be an elementary neighbourhood centred at $\gamma(a)$, and let $0 < b < a$ be such that $\gamma(b) \in \mathcal{U}$. By definition of accumulation curve there exists a sequence $n_i \in \mathbb{N}$, a compact interval $[-k, k] \subset \mathbb{R}$, and a sequence $s_i \in [-k, k]$ such that $\gamma_{n_i}(s_i)$ converges to $\gamma(b)$. In particular we will have $\gamma_{n_i}(s_i) \in \mathcal{U}$ for i large enough. We note the following.

LEMMA 2.6.6 *Let $\mathcal{U}$ be an elementary neighbourhood, as defined in Definition 2.2.8. There exists a constant ℓ such that for any causal curve $\gamma : I \rightarrow \mathcal{U}$ the h -length $|\gamma|_h$ of γ is bounded by ℓ .*

To prove Lemma 2.6.6 we need the following variation of the inverse Cauchy-Schwarz inequality (compare Proposition 1.3.4, p. 7).

LEMMA 2.6.7 *Let K be a compact set and let X be a continuous timelike vector field defined there. There exists a strictly positive constant C such that for all $q \in K$ and for all causal vectors $Y \in T_q \mathcal{M}$ we have*

$$|g(X, Y)| \geq C|Y|_h. \quad (2.6.10)$$

Proof. By homogeneity it is sufficient to establish (2.6.10) for causal $Y \in T_q \mathcal{M}$ such that $|Y|_h = 1$; let us denote by $U_q(h)$ this last set. The result follows then by continuity of the strictly positive function

$$\cup_{q \in K} U_q(h) \ni Y \rightarrow |g(X, Y)|$$

on the compact set $\cup_{q \in K} U_q(h)$. $\square$

Returning to the proof of Lemma 2.6.6, let x^0 be the local time coordinate on $\mathcal{U}$; since $X := \nabla x^0$ is timelike we can use Lemma 2.6.7 with $K = \mathcal{U}$ to conclude that there exists a constant C such that for any causal curve $\gamma \subset \mathcal{U}$ we have

$$|g(X, \dot{\gamma})| \geq C > 0$$

at all points at which γ is differentiable. This implies, for $s_2 \geq s_1$ that

$$\begin{aligned} |x^0(s_2) - x^0(s_1)| &\geq \int_{s_1}^{s_2} |g(\nabla x^0, \dot{\gamma})| ds \\ &\geq C \int_{s_1}^{s_2} ds = C |s_2 - s_1|. \end{aligned} \quad (2.6.11)$$

It follows that

$$|\gamma|_h \leq \ell := \frac{2}{C} \sup_{\mathcal{U}} |x^0| < \infty, \quad (2.6.12)$$

as desired. $\square$

Returning to the proof of Lemma 2.6.5, it follows from Lemma 2.6.6 applied to γ_{n_i} that $\gamma_{n_i}|_{[s_i, s_i + \ell]}$ must exit $\mathcal{U}$. This implies that $\gamma_{n_i}|_{[-k, k + \ell]}$ cannot accumulate at a curve which has an end point $\gamma(b) \in \mathcal{U}$, and the result follows. $\square$

In view of Proposition 2.6.2, the following definition is useful.

DEFINITION 2.6.8 *We shall say that a sequence of causal curves $\gamma_n : I \rightarrow \mathcal{M}$ is well parameterized if for every compact subset K of I there exists constants $L(K)$, $\epsilon(K) > 0$ such that*

$$L(K)^{-1} |s - s'| \leq \text{dist}_h(\gamma_n(s), \gamma_n(s')) \leq L(K) |s - s'| \quad (2.6.13)$$

holds for $|s - s'| \leq \epsilon(K)$.

We have the following.

PROPOSITION 2.6.9 *Any sequence γ_n of inextendible causal curves such that $\gamma_n(0)$ converges can be well parameterized.*

Proof. We reparameterize the γ_n 's by h -length from $\gamma_n(0)$ where, as usual, h is a complete Riemannian metric. This shows that any $L(K) \geq 1$ provides an upper bound in (2.6.13) for all $\epsilon(K) > 0$.

In order to obtain the lower bound, set $p = \lim_{n \rightarrow \infty} \gamma_n(0)$. Let $R > 0$; there exists $r \geq 0$ so that all the image-curves $\gamma_n([-R, R])$ are contained in the open dist_h -ball $B_p(R + r)$. We can cover $\overline{B_p(R + r)}$ by a finite number N of balls $B_{p_i}(\epsilon_i)$ such that $B_{p_i}(2\epsilon_i)$ is contained in an elementary region $\mathcal{U}_i$. Set

$$\epsilon(R) = \min_{i=1, \dots, N} \epsilon_i.$$

Suppose that $\gamma_n(s_j) \in \mathcal{O}_i$ for some $|s_j| \leq R$; let $I_{n,i,j}$ be the connected component containing s_j of the set

$$\{s \in \mathbb{R} : \gamma_n(s) \in B_{p_i}(2\epsilon_i)\}.$$

It follows from (2.6.11) that for $s, s' \in I_{n,i,j}$ with

$$|s - s'| \leq \epsilon(R)$$

there exists a constant $c_i > 0$, independent of j and n , such that

$$\text{dist}_h(\gamma_n(s), \gamma_n(s')) \geq c_i |x^0(s) - x^0(s')| \geq c_i C_i |s - s'|. \quad (2.6.14)$$

For $K \subset [-R, R]$ the result is obtained by setting

$$L(K) := \frac{1}{\min_{i=1,\dots,N} c_i C_i} + 1, \quad \epsilon(K) := \epsilon(R). \quad \square$$

Summarizing, from Lemma 2.5.4 and 2.6.5 together with Proposition 2.6.2 we have the following.

THEOREM 2.6.10 *Let $(\mathcal{M}, g)$ be a C^3 Lorentzian manifold with a C^2 metric. Consider a sequence of future-directed, inextendible, causal curves which accumulates at a point $p \in \mathcal{M}$. Then we have the following.*

1. *The curves in the sequence can be reparameterized if necessary so that the sequence becomes well parameterized.*
2. *Any such well-parameterized sequence accumulates at some future-directed, inextendible, causal curve through p .* $\square$

One is sometimes interested in sequences of maximally extended geodesics.

PROPOSITION 2.6.11 *Let γ_n be a sequence of maximally extended geodesics accumulating at γ in $(\mathcal{M}, g)_{C^{1,1}}$. Then γ is a maximally extended geodesic.*

Proof. If we use a h -length-parameterization of the γ_n 's and of γ such that $\gamma_n(0) \rightarrow \gamma(0)$, then by the Arzela–Ascoli Theorem (passing to a subsequence if necessary) the γ_n 's converge to γ , uniformly on compact subsets of $\mathbb{R}$. Let $\mathcal{K}$ be a compact neighbourhood of $\gamma(0)$; compactness of $\cup_{p \in \mathcal{K}} U_p \mathcal{M}$, where $U_p \mathcal{M} \subset T_p \mathcal{M}$ is the set of h -unit vectors tangent to $\mathcal{M}$, implies that there exists a subsequence such that $\dot{\gamma}_n(0)$ converges to some vector $X \in U_{\gamma(0)} \mathcal{M} \subset T_{\gamma(0)} \mathcal{M}$. Let $\sigma : (a, b) \rightarrow \mathcal{M}$, $a \in \mathbb{R} \cup \{-\infty\}$, $b \in \mathbb{R} \cup \{\infty\}$, be an affinely parameterized maximally extended geodesic through $\gamma(0)$ with initial tangent vector X . By continuous dependence of ODEs upon initial values it follows that (1) for any $a < \alpha < \beta < b$ all the γ_n 's, except perhaps for a finite number, are defined on $[\alpha, \beta]$ when affinely parameterized, and (2) they converge to $\sigma|_{[\alpha, \beta]}$ in the (uniform) $C^1([\alpha, \beta], \mathcal{M})$ topology. Thus $\dot{\gamma}_n(s) \rightarrow \dot{\sigma}(s)$ uniformly on compact subsets of (a, b) , which implies that a h -length-parameterization is preserved under taking limits. Hence the γ_n 's, when h -length-parameterized, converge uniformly to a h -length-reparameterization of σ on compact subsets of $\mathbb{R}$; call it μ . It follows that $\gamma = \mu$, and γ is a maximally extended geodesic. $\square$

2.6.1 Achronal causal curves

A curve $\gamma : I \rightarrow \mathcal{M}$ is called *achronal* if

$$\forall s, s' \in I \quad \gamma(s) \notin I^+(\gamma(s')).$$

Any spacelike geodesic in Minkowski spacetime is achronal. More interestingly, it follows from Proposition 2.4.3 that this is also true for null Minkowskian geodesics. However, null geodesics do not have to be achronal in general: consider, e.g., the two-dimensional spacetime $\mathbb{R} \times S^1$ with the flat metric $-dt^2 + dx^2$, where x is an angle-type coordinate along S^1 with periodicity, say, 2π . Then the points $(0, 0)$ and $(2\pi, 0)$ both lie on the null geodesic

$$s \rightarrow (s, s \bmod 2\pi),$$

and are clearly timelike related to each other.

In this section we will be interested in causal curves that are achronal. We start with the following.

PROPOSITION 2.6.12 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. If γ is an achronal causal curve, then γ is a null geodesic.*

Proof. Let $\mathcal{O}$ be any elementary neighbourhood; then any connected component of $\gamma \cap \mathcal{O}$ is a null geodesic by Corollary 2.4.10. $\square$

THEOREM 2.6.13 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let $\gamma_n : I \rightarrow \mathcal{M}$ be a sequence of achronal causal curves accumulating at γ . Then γ is achronal.*

REMARKS 2.6.14 1. Propositions 2.6.12 and 2.6.11 show that γ is inextendible if the γ_n 's are.

2. The proof uses the fact that timelike futures and pasts are open, which is not necessarily true if the metric is not C^2 [233].

Proof. Suppose γ is not achronal; then there exist $s_1, s_2 \in I$ such that $\gamma(s_2) \in I^+(\gamma(s_1))$. Thus there exists a *timelike* curve $\hat{\gamma} : [s_1, s_2] \rightarrow \mathcal{M}$ from $\gamma(s_1)$ to $\gamma(s_2)$. Choose some $\hat{s} \in (s_1, s_2)$. We have $\gamma(s_2) \in I^+(\hat{\gamma}(\hat{s}))$, and since $I^+(\hat{\gamma}(\hat{s}))$ is open there exists an open neighbourhood $\mathcal{O}_2$ of $\gamma(s_2)$ such that $\mathcal{O}_2 \subset I^+(\hat{\gamma}(\hat{s}))$. Similarly there exists an open neighbourhood $\mathcal{O}_1$ of $\gamma(s_1)$ such that $\mathcal{O}_1 \subset I^-(\hat{\gamma}(\hat{s}))$. This shows that any point $p_2 \in \mathcal{O}_2$ lies in the timelike future of any point $p_1 \in \mathcal{O}_1$: indeed, one can go from p_1 along some timelike path to $\hat{\gamma}(\hat{s})$, and continue along another timelike path from $\hat{\gamma}(\hat{s})$ to p_2 .

Passing to a subsequence if necessary, there exist sequences $s_{1,n}$ and $s_{2,n}$ such that $\gamma_n(s_{1,n})$ converges to $\gamma(s_1)$ and $\gamma_n(s_{2,n})$ converges to $\gamma(s_2)$. Then $\gamma_n(s_{1,n}) \in \mathcal{O}_1$ and $\gamma_n(s_{2,n}) \in \mathcal{O}_2$ for n large enough, leading to $\gamma_n(s_{2,n}) \in I^+(\gamma_n(s_{1,n}))$, contradicting the achronality of γ_n . $\square$

2.7 Causality conditions

Spacetimes can exhibit various causal pathologies, most of which are undesirable from a physical point of view. The simplest example of unwanted causal behaviour is the existence of closed timelike curves. A spacetime is said to be *chronological* if no such curves exist. An example of a spacetime which is not chronological is provided by $S^1 \times \mathbb{R}$ with the flat metric $-dt^2 + dx^2$, where t is a local coordinate defined modulo 2π on S^1 . Then every circle $x = \text{const}$ is a closed timelike curve.

INCIDENTALLY: The class of compact manifolds is a very convenient one from the point of view of Riemannian geometry. The following result of Geroch shows that such manifolds are always pathological from a Lorentzian perspective.

PROPOSITION 2.7.1 (Geroch [217]) Every compact spacetime $(\mathcal{M}, g)_{C^2}$ contains a closed timelike curve.

Proof. Consider the covering of $\mathcal{M}$ by the collection of open sets $\{I^-(p)\}_{p \in \mathcal{M}}$; by compactness a finite covering $\{I^-(p_i)\}_{i=1, \dots, I}$ can be chosen. The possibility $p_1 \in I^-(p_1)$ yields immediately a closed timelike curve through p_1 ; otherwise, there exists $p_{i(1)}$ such that $p_1 \in I^-(p_{i(1)})$. Again if $p_{i(1)} \in I^-(p_{i(1)})$ we are done; otherwise, there exists $p_{i(2)}$ such that $p_{i(1)} \in I^-(p_{i(2)})$. Continuing in this way we obtain a—finite or infinite—sequence of points $p_{i(j)}$ such that

$$p_{i(j)} \in I^-(p_{i(j+1)}). \quad (2.7.1)$$

If the sequence is finite we are done. Now, we have only a finite number of p_i 's at our disposal; therefore if the sequence is infinite it has to contain repetitions,

$$p_{i(j+\ell)} = p_{i(j)},$$

for some j , and some $\ell > 0$. It should be clear from (2.7.1) that there exists a closed timelike curve through $p_{i(j)}$. $\square$

REMARK 2.7.2 Galloway [201] has shown that in compact spacetimes $(\mathcal{M}, g)$ there exist closed timelike curves through any two points p and q , under the supplementary condition that the Ricci tensor Ric satisfies the energy condition

$$\text{Ric}(X, X) > 0 \quad \text{for all causal vectors } X. \quad (2.7.2)$$

□

The chronology condition excludes *closed timelike* curves, but it just fails to exclude the possibility of occurrence of *closed causal* curves. A spacetime is said to be *causal* if no such curves can be found. The existence of spacetimes which are *chronological* but not *causal* requires a little work.

EXAMPLE 2.7.3 Let $\mathcal{M} = \mathbb{R} \times S^1$ with the metric

$$g = 2dt\,dx + f(t)dx^2,$$

where f is any function satisfying

$$f \geq 0, \quad \text{with } f(t) = 0 \text{ iff } t = 0.$$

(The function $f(t) = t^2$ will do.) Here t runs over the $\mathbb{R}$ factor of $\mathcal{M}$, while x is a coordinate defined modulo 2π on S^1 . In matrix notation we have

$$[g_{\mu\nu}] = \begin{bmatrix} 0 & 1 \\ 1 & f \end{bmatrix},$$

which leads to the inverse metric

$$[g^{\mu\nu}] = \begin{bmatrix} -f & 1 \\ 1 & 0 \end{bmatrix}.$$

It follows that

$$g(\nabla t, \nabla t) = -f \leq 0, \text{ with } g(\nabla t, \nabla t) = 0 \text{ iff } t = 0. \quad (2.7.3)$$

Recall, now, that a function τ is called a *time function* if $\nabla\tau$ is timelike, past-pointing. Equation (2.7.3) shows that t is a time function on the set $\{t \neq 0\}$. Since a time function is strictly increasing on any causal curve (see Lemma 2.4.8), one easily concludes that no closed causal curve in $\mathcal{M}$ can intersect the set $\{t \neq 0\}$. In other words, closed causal curves—if they do exist—must be entirely contained in the set $\{t = 0\}$. Now, any curve γ contained in this last set is of the form

$$\gamma(s) = (0, x(s)),$$

with tangent vector

$$\dot{\gamma} = \dot{x}\partial_x \implies g(\dot{\gamma}, \dot{\gamma}) = (\dot{x})^2 g(\partial_x, \partial_x) = (\dot{x})^2 g_{xx} = 0.$$

This shows in particular that

- $\mathcal{M}$ does contain closed causal curves: an example is given by $x(s) = s \bmod 2\pi$; and
- All closed causal curves are null.

It follows that $(\mathcal{M}, g)$ is indeed chronological, but not causal, as claimed. □

It is desirable to have a condition of causality which is stable under small changes of the metric. By way of example, consider a spacetime which contains a family of causal curves γ_n with both $\gamma_n(0)$ and $\gamma_n(1)$ converging to p . Such curves can be thought of as being ‘almost closed’. Further, it is clear that one can produce an

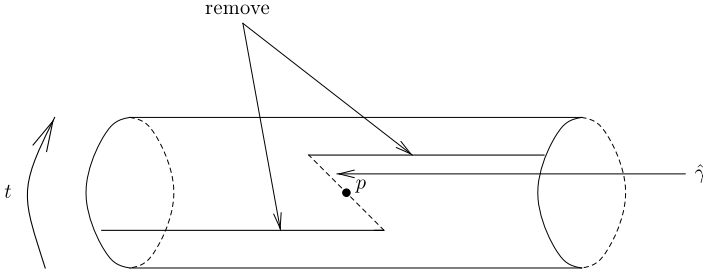


Fig. 2.7.1 A causal spacetime which is not strongly causal. Here the metric is the flat one $-dt^2 + dx^2$, with t a parameter along S^1 , so that the light cones are at 45° ; in particular $\hat{\gamma}$ is a null geodesic. It should be clear that no matter how small the neighbourhood $\mathcal{U}$ of p is, there will exist a causal curve which will intersect this neighbourhood twice. In order to show that $(\mathcal{M}, g)$ is causal one can proceed as follows: suppose that γ is a closed causal curve in $\mathcal{M}$; then γ has to intersect the hypersurfaces $\{t = \pm 1\}$ at some points $x_\pm$, with $x_- > 1$ and $x_+ < -1$. If we parameterize γ so that $\gamma(s) = (s, x(s))$ we obtain $-2 > x_+ - x_- = \int_{-1}^1 \frac{dx}{ds} ds$; hence there must exist $s_* \in [-1, 1]$ such that $dx/ds < -1$, contradicting causality of γ .

arbitrarily small deformation of the metric which will allow one to obtain a closed causal curve in the deformed spacetime. The object of our next causality condition is to exclude this behaviour. A spacetime will be said to be *strongly causal* if every neighbourhood $\mathcal{O}$ of a point $p \in \mathcal{M}$ contains a neighbourhood $\mathcal{U}$ such that for every causal curve $\gamma : I \rightarrow \mathcal{M}$ the set

$$\{s \in I : \gamma(s) \in \mathcal{U}\} \subset I$$

is a connected subset of I . In other words, γ does not re-enter $\mathcal{U}$ once it has left it.

Clearly, a strongly causal spacetime is necessarily causal. However, the inverse does not always hold. An example is given in Fig. 2.7.1.

The definition of strong causality appears, at first sight, somewhat unwieldy to verify, so simpler conditions are desirable. The following provides a useful criterion: a spacetime $(\mathcal{M}, g)$ is said to be *stably causal* if there exists a time function globally defined on $\mathcal{M}$. Recall—see Lemma 2.4.8—that time functions are strictly increasing on causal curves. It then easily follows that *stable causality* implies *strong causality*.

PROPOSITION 2.7.4 *If $(\mathcal{M}, g)_{C^0}$ is stably causal, then it is strongly causal.*

Proof. Let $\mathcal{O}$ be a connected open neighbourhood of $p \in \mathcal{M}$, and let φ be a non-negative smooth function such that $\varphi(p) \neq 0$ and such that the support $\text{supp}\varphi$ of φ is a compact set contained in $\mathcal{O}$. Let τ be a time function on $\mathcal{M}$, for $a \in \mathbb{R}$ set

$$\tau_a := \tau + a\varphi.$$

As $\nabla\tau$ is timelike, the function $g(\nabla\tau, \nabla\tau)$ is bounded away from 0 on the compact set $\text{supp}\varphi$, which implies that there exists $\epsilon > 0$ small enough so that $\tau_{\pm\epsilon}$ are time functions on $\text{supp}\varphi$. Now, $\tau_{\pm\epsilon}$ coincides with τ away from $\text{supp}\varphi$, so that the $\tau_{\pm\epsilon}$'s are actually time functions on $\mathcal{M}$ as well. We set

$$\mathcal{U} := \{q : \tau_{-\epsilon}(q) < \tau(p) < \tau_{+\epsilon}(q)\}.$$

We have that:

- $p \in \mathcal{U}$, therefore $\mathcal{U}$ is not empty;
- $\mathcal{U}$ is open because the τ_a 's are continuous; and

- $\mathcal{U} \subset \mathcal{O}$ because φ vanishes outside of $\mathcal{O}$.

Consider any causal curve γ the image of which intersects $\mathcal{U}$; γ can enter or leave $\mathcal{U}$ only through

$$\partial\mathcal{U} \subset \{q : \tau_{-\epsilon}(q) = \tau(p)\} \cup \{q : \tau(p) = \tau_{+\epsilon}(q)\}. \quad (2.7.4)$$

At a point s_- at which $\gamma(s_-) \in \{q : \tau_{-\epsilon}(q) = \tau(p)\}$ we have

$$\tau(p) = \tau_{-\epsilon}(\gamma(s_-)) = \tau(\gamma(s_-)) - \epsilon\varphi(\gamma(s_-)) \implies \tau(\gamma(s_-)) > \tau(p).$$

Similarly at a point s_+ at which $\gamma(s_+) \in \{q : \tau_{+\epsilon}(q) = \tau(p)\}$ we have

$$\tau(p) = \tau_{+\epsilon}(\gamma(s_+)) = \tau(\gamma(s_+)) + \epsilon\varphi(\gamma(s_+)) \implies \tau(\gamma(s_+)) < \tau(p).$$

As τ is increasing along γ , we conclude that γ can enter $\mathcal{U}$ only through $\{q : \tau_{+\epsilon}(q) = \tau(p)\}$, and leave $\mathcal{U}$ only through $\{q : \tau_{-\epsilon}(q) = \tau(p)\}$. Lemma 2.4.8 shows that γ can intersect each of the two sets at the right-hand side of (2.7.4) at most once. Those facts obviously imply connectedness of the intersection of (the image of) γ with $\mathcal{U}$. $\square$

INCIDENTALLY: There exist various alternative definitions of stable causality which are equivalent for C^2 metrics, but note that equivalence is not obvious and its proof requires work. For example, Yvonne Choquet-Bruhat [86] defines stable causality as the requirement of existence of a timelike vector field v such that $g - v \otimes v$ is chronological. Hawking and Ellis [244] define stable causality by requiring that C^0 -small perturbations of the metric preserve causality. Compare [351, 352]. $\square$

The strongest causality condition is that of *global hyperbolicity*, considered in the next section.

2.8 Global hyperbolicity

DEFINITION 2.8.1 *A spacetime $(\mathcal{M}, g)$ is said to be globally hyperbolic if it is strongly causal, and if for every $p, q \in \mathcal{M}$ the sets $J^+(p) \cap J^-(q)$ are compact.*

The sets $J^+(p) \cap J^-(q)$ will be referred to as *causal diamonds*.

INCIDENTALLY: The definition above is the one most widely used. It is, however, often convenient to use the equivalent requirement of stable causality together with compactness of the sets $J^+(p) \cap J^-(q)$; compare Theorem 2.11.1, p. 79. The landmark result of Hounnonkpe and Minguzzi, Theorem 2.8.17, p. 60 shows that, in non-compact spacetimes of dimension larger than or equal to 3, causality conditions can be dropped altogether in the definition of global hyperbolicity. One can only deplore that it has only been discovered in 2019, as it can be used to simplify many proofs. The Hounnonkpe–Minguzzi theorem builds on a theorem of Bernal and Sánchez [51], Theorem 2.8.9 here, which shows that the requirement of strong causality can be replaced by that of causality. From a Cauchy-problem-for-wave-equations point of view, a natural definition is by requiring the existence of a Cauchy surface; compare Section 2.9. Theorem 2.11.1, p. 79, shows that the last definition is again equivalent for C^2 metrics. $\square$

It is not too difficult to show that Minkowski spacetime $\mathbb{R}^{1,n}$ is globally hyperbolic: first, the Minkowski time x^0 provides a time function on $\mathbb{R}^{n,1}$; this implies strong causality. Compactness of $J^+(p) \cap J^-(q)$ for all p 's and q 's is easily checked by drawing pictures; it is also easy to write a formal proof using Proposition 2.4.3; this is left as an exercise to the reader.

The notion of global hyperbolicity provides excellent control over causal properties of $(\mathcal{M}, g)$. This will be made clear at several other places in this work. Anticipating, let us list a few of those:

1. Let $(\mathcal{M}, g)$ be globally hyperbolic. If $J^+(p) \cap J^-(q) \neq \emptyset$, then there exists a causal geodesic from p to q . Similarly if $I^+(p) \cap I^-(q) \neq \emptyset$, then there exists a timelike geodesic from p to q .

2. The Cauchy problem for linear wave equations is globally solvable on globally hyperbolic spacetimes.
3. A key theorem of Choquet-Bruhat and Geroch asserts that *maximal globally hyperbolic* solutions of the Cauchy problem for Einstein's equations are unique up to diffeomorphism.

We start our study of globally hyperbolic spacetimes with the following property.

PROPOSITION 2.8.2 *Let $(\mathcal{M}, g)_{C^2}$ be globally hyperbolic, and let γ_n be a family of causal curves accumulating at both p and q . Then there exists a causal curve γ , accumulation curve of the (perhaps reparameterized) γ_n 's which passes both through p and q .*

REMARK 2.8.3 If the γ_n 's are uniformly Lipschitz with an n -independent bound on the Lipschitz constant, then the result holds without the need for reparameterizing the γ_n 's.

EXAMPLE 2.8.4 The result is wrong if stable causality is assumed only. Indeed, let $(\mathcal{M}, g)$ be the two-dimensional Minkowski spacetime with the origin removed. Let γ_n be obtained by following a timelike geodesic from $p = (-1, 0)$ to $(0, 1/n)$ and then another timelike geodesic to $q = (1, 0)$. Then γ_n has exactly two accumulation curves $s \rightarrow (s, 0)$, with $s \in [-1, 0)$ for the first one and $s \in (0, 1]$ for the second, none of which passes through both p and q .

EXAMPLE 2.8.5 One wonders whether the result remains true when all curves lie within a compact set, regardless of causality conditions. This is not the case, as seen by the following example: consider the flat torus $S^1 \times S^1$ with the product Lorentzian metric g . Points on $S^1 \times S^1$ can be viewed as pairs of complex numbers $(e^{i\alpha}, e^{i\beta})$, $\alpha, \beta \in [0, 2\pi)$, with $g = -d\alpha^2 + d\beta^2$. For $n \in \mathbb{N}$ let γ_n , $n \geq 1$, be the sequence of causal curves from $p := (1, 1)$ to $q := (1, -1)$ given by $\gamma_n(s) = (e^{2is}, e^{is/n})$, $s \in [0, n\pi]$. Then the inextendible accumulation curves of the family $\{\gamma_n\}$ are the curves $s \mapsto (e^{2is}, e^{i\beta})$, with $\beta \in [0, \pi]$ fixed. None of them pass simultaneously through p and q .

Proof of Proposition 2.8.2. Extending the γ_n 's to inextendible curves, and reparameterizing if necessary, we can assume that the γ_n 's are h -length-parameterized, with the common domain of definition $I = \mathbb{R}$, and with $\gamma_n(0)$ converging to p . If $p = q$ the result has already been established in Proposition 2.6.2, so we assume that $p \neq q$. Consider the compact set

$$\mathcal{K} := (J^+(p) \cap J^-(q)) \cup (J^+(q) \cap J^-(p)) \quad (2.8.1)$$

(since a globally hyperbolic spacetime is causal, one of those sets is, of course, necessarily empty). $\mathcal{K}$ can be covered by a finite number of elementary domains $\mathcal{U}_i$, $i = 1, \dots, N$. Strong causality allows us to choose the $\mathcal{U}_i$'s small enough so that for every n the image of γ_n is a connected subset in $\mathcal{U}_i$. We can choose a parameterization of the γ_n 's by h -length so that, passing to a subsequence of the γ_n 's if necessary, we have $\gamma_n(0) \rightarrow p$. Extending the γ_n 's if necessary we can assume that all the γ_n 's are defined on $\mathbb{R}$. Now, Lemma 2.6.6, p. 50, shows that there exists a constant L_i —independent of n —such that the h -length $|\gamma_n \cap \mathcal{U}_i|_h$ is bounded by L_i . Consequently the h -length $|\gamma_n \cap \mathcal{K}|_h$, with $\mathcal{K}$ as in (2.8.1), is bounded by

$$|\gamma_n \cap \mathcal{K}|_h \leq L := L_1 + L_2 + \dots + L_I. \quad (2.8.2)$$

By hypothesis the γ_n 's accumulate at q ; therefore there exists a sequence s_n (passing again to a subsequence if necessary) such that

$$\gamma_n(s_n) \rightarrow q.$$

Equation (2.8.2) shows that the sequence s_n is bounded, hence—perhaps passing to a subsequence—we have $s_n \rightarrow s_*$ for some $s_* \in \mathbb{R}$.

At this stage we could use Proposition 2.6.2, but one might as well argue directly: by our choice of parametrization we have

$$\text{dist}_h(\gamma_n(s), \gamma_n(s')) \leq |s - s'| \quad (2.8.3)$$

(see (2.3.2)–(2.3.3)). This shows that the family $\{\gamma_n\}$ is equicontinuous, and (2.8.3) together with the Arzela–Ascoli theorem (on the compact set $[-L, L]$) implies existence of a curve $\gamma : [-L, L] \rightarrow \mathcal{M}$ and a subsequence γ_{n_i} which converges uniformly to γ on $[-L, L]$. As $\gamma_{n_i}(s_{n_i})$ converges both to $\gamma(s_*)$ and to q we have

$$\gamma(s_*) = q.$$

This shows that γ is the desired causal curve joining p with q . $\square$

REMARK 2.8.6 It should be clear from the proof above that, as emphasized in [86], a spacetime is globally hyperbolic if and only if the length of causal paths between two points, as measured with respect to a smooth complete Riemannian metric, is bounded by a number independent of the path. $\square$

As a straightforward corollary of Proposition 2.8.2 we obtain the following.

COROLLARY 2.8.7 *Let $(\mathcal{M}, g)_{C^2}$ be globally hyperbolic, then*

$$\overline{I^\pm(p)} = J^\pm(p);$$

in particular causal futures and pasts are closed.

Proof. Let $q_n \in I^+(p)$ be a sequence of points accumulating at q ; thus there exists a sequence γ_n of causal curves from p to q , then $q \in J^+(p)$ by Proposition 2.8.2. Hence

$$\overline{I^\pm(p)} \subset J^\pm(p).$$

The reverse inclusion is provided by Corollary 2.4.18, p. 44. $\square$

REMARK 2.8.8 We say that $(\mathcal{M}, g)$ is *causally simple* if $(\mathcal{M}, g)$ is causal and if all the sets $J^\pm(p)$ are closed [51]. It follows from Corollary 2.8.7 that global hyperbolicity of $(\mathcal{M}, g)_{C^2}$ implies causal simplicity. An example of causally simple spacetime which is not globally hyperbolic is provided by the strip $\{t \in \mathbb{R}, -1 < x < 1\}$, in two-dimensional Minkowski spacetime. $\square$

Bernal and Sánchez have shown that, in the definition of global hyperbolicity, the condition of *strong causality* can be replaced by that of *causality*.

THEOREM 2.8.9 (Bernal and Sánchez [51]) *Assume that $(\mathcal{M}, g)_{C^2}$ has the property that*

$$\begin{aligned} &\text{for all } p, q \in \mathcal{M} \text{ the causal diamond } J^+(p) \cap J^-(q) \\ &\text{is compact when non-empty.} \end{aligned} \quad (2.8.4)$$

Then the following two conditions are equivalent:

1. $(\mathcal{M}, g)$ is causal.
2. $(\mathcal{M}, g)$ is strongly causal.

Proof. The implication $2 \Rightarrow 1$ is trivial. As a first step to prove the converse, let us show that (2.8.4) suffices to obtain the conclusions of Corollary 2.8.7.

LEMMA 2.8.10 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$ in which (2.8.4) holds. Then the sets $J^\pm(p)$ are closed for all $p \in \mathcal{M}$.*

Proof. Suppose not, then (changing time orientation if necessary) there exists $r \in J^+(p) \setminus J^+(p)$ and a sequence $r_n \rightarrow r$ with $r_n \in J^+(p)$. Let $q \in I^+(r)$; then $I^-(q)$ forms an open neighbourhood of r and so we have

$$r_n \in J^+(p) \cap I^-(q) \subset J^+(p) \cap J^-(q)$$

for all n large enough. Thus, $\{r_n\}_{n \in \mathbb{N}} \subset J^+(p) \cap J^-(q)$, which is compact, but converges to the point r which does not lie in $J^+(p) \cap J^-(q)$, a contradiction. $\square$

Theorem 2.8.9 follows immediately from Lemma 2.8.10 together with Proposition 2.8.11. $\square$

PROPOSITION 2.8.11 *Causal spacetimes $(\mathcal{M}, g)_{C^2}$ in which all the sets $J^\pm(p)$ are closed are strongly causal.*

Proof. Suppose not, then there exists a point p at which strong causality is violated. Let $\mathcal{U}$ be an elementary domain containing p . There exists a sequence of future-directed, causal curves γ_i through p , leaving $\mathcal{U}$ at a point $q_i \ni \partial\mathcal{U}$, entering $\mathcal{U}$ again and ending at a point $p_i \in \mathcal{U}$, with p_i converging to p . Then $q_i \in J^+(p)$, and passing to a subsequence if necessary, we can assume that the q_i 's converge to some point $q \in \partial\mathcal{U}$. Clearly $q \neq p$, since $q \in \partial\mathcal{U}$. Using standard properties of elementary domains, or invoking the fact that $J^+(p)$ is closed, we have $q \in J^+(p)$.

Let $r_n \in I^-(q)$ be any sequence of points converging to q . We can construct a future-directed causal curve from r_n to p_i by first following a timelike curve from r_n to q_i , at least for all i large enough, and then following γ_i from q_i to p_i . Hence $p_i \in J^+(r_n)$. Since $J^+(r_n)$ is closed and $p_i \rightarrow p$ we have $p \in J^+(r_n)$.

Equivalently, $r_n \in J^-(p)$. Since $J^-(p)$ is closed and $r_n \rightarrow q$ we have $q \in J^-(p)$.

Hence q is distinct from p and belongs to both $J^+(p)$ and $J^-(p)$, contradicting causality of $(\mathcal{M}, g)$. $\square$

INCIDENTALLY: For future use we note the following strengthening of Proposition 2.8.11, also due to Bernal and Sánchez [51].

PROPOSITION 2.8.12 Consider a spacetime $(\mathcal{M}, g)_{C^2}$ in which all the sets $J^\pm(p)$ are closed. The following conditions are equivalent:

1. $(\mathcal{M}, g)$ is causal.
2. $(\mathcal{M}, g)$ is distinguishing, i.e., if $p \neq q$ then $I^+(p) \neq I^+(q)$ and $I^-(p) \neq I^-(q)$.
3. $(\mathcal{M}, g)$ is strongly causal.

REMARK 2.8.13 Recall that a spacetime $(\mathcal{M}, g)$ is said to be *causally simple* if $(\mathcal{M}, g)$ is causal with all the sets $J^\pm(p)$ closed. Proposition 2.8.12 shows that, for twice-differentiable metrics, the condition of causality in the definition of causal simplicity can be equivalently replaced by the distinguishing condition, or by the requirement of strong causality. $\square$

Proof. The implication ' $3 \Rightarrow 1$ ' is trivial. The implication ' $1 \Rightarrow 3$ ' is the main content of Proposition 2.8.11. Next:

$1 \Rightarrow 2$: Otherwise there exist two points $p \neq q$ with, say $I^+(p) = I^+(q)$. We always have $q \in \overline{I^+(q)}$, which in the current context coincides with $\overline{I^+(p)}$. Since $\overline{I^+(p)} = \overline{J^+(p)}$ in any spacetime $(\mathcal{M}, g)_{C^2}$, we obtain $q \in \overline{J^+(p)}$. The hypothesis that causal futures are closed implies that $q \in J^+(p)$.

Analogously, $p \in J^+(q)$, and since $p \neq q$ the spacetime is not causal, a contradiction.

The implication ' $2 \Rightarrow 1$ ' is the contents of Proposition 2.8.14, which completes the proof. $\square$

PROPOSITION 2.8.14 Every distinguishing spacetime $(\mathcal{M}, g)_{C^2}$ is causal.

Proof. Suppose that $(\mathcal{M}, g)$ is not causal; let p and q be two distinct points lying on the image of a closed causal future-directed curve γ . If $r \in I^+(p)$, then one can construct a causal future-directed curve γ_{qr} by following γ from q to p , and then a timelike curve from p to r . The curve γ_{qr} can be deformed to a timelike curve from q to r ; hence $I^+(p) \subset I^+(q)$. The reverse inclusion follows by interchanging p with q in the argument just given. Hence $I^+(p) = I^+(q)$, a contradiction. $\square$

For further reference, we note the following corollary of Remark 2.8.13 and Proposition 2.8.12.

COROLLARY 2.8.15 *Let $(\mathcal{M}, g)_{C^2}$ be globally hyperbolic; then $I^\pm(p) \neq I^\pm(q)$ for $p \neq q$, similarly for $J^\pm$.*

As already mentioned, global hyperbolicity gives us control over causal geodesics; no proof will be given.

THEOREM 2.8.16 *Let $(\mathcal{M}, g)_{C^2}$ be globally hyperbolic, if $q \in I^+(p)$, respectively $q \in J^+(p)$; then there exists a timelike, respectively causal, future-directed geodesic from p to q . $\square$*

It is rather surprising that the following fundamental result has only been discovered in 2019.

THEOREM 2.8.17 (Hounnonkpe and Minguzzi [262]) *Assume that $(\mathcal{M}, g)_{C^2}$ has the property that*

$$\begin{aligned} &\text{for all } p, q \in \mathcal{M} \text{ the causal diamond } J^+(p) \cap J^-(q) \\ &\text{is compact when non-empty.} \end{aligned} \tag{2.8.5}$$

If $\dim \mathcal{M} \geq 3$ and $\mathcal{M}$ is not compact, then $(\mathcal{M}, g)$ is globally hyperbolic.

Proof. We need to show that a spacetime satisfying (2.8.5) which is *not* globally hyperbolic is either compact or two-dimensional.

Suppose, thus, that $(\mathcal{M}, g)$ is not globally hyperbolic. By Theorem 2.8.9 there exists a closed causal curve, say γ from p to p . Hence the causal diamond $J^+(p) \cap J^-(p)$ is non-trivial, and compact by hypothesis.

Case 1. Suppose that γ is not achronal; then γ can be deformed to a timelike curve, still denoted by γ . It follows that $p \in I^+(p) \cap I^-(p)$, which forms an open neighbourhood of p contained in $J^+(p) \cap J^-(p)$. Let $q \in J^+(p) \cap J^-(p)$, and let $\hat{\gamma}$ be a causal curve from q to p ; then the concatenation of $\hat{\gamma}$ and γ provides a causal curve from q to p which can be deformed to a timelike one. It follows that $J^+(p) \cap J^-(p)$ is open; it is closed by hypothesis, whence $\mathcal{M} = J^+(p) \cap J^-(p)$ and $\mathcal{M}$ is compact, as claimed.

Case 2. Otherwise γ is achronal, and thus an achronal null geodesic by Proposition 2.6.12, p. 53.

Suppose that $\partial(J^+(p) \cap J^-(p))$ is not empty. If $p \notin \partial(J^+(p) \cap J^-(p))$ then $J^+(p) \cap J^-(p)$ contains a neighbourhood $\mathcal{O}$ of p . One can then construct a timelike curve from p to p by first going from p to a nearby point $p' \in \mathcal{O} \setminus \{p\}$ and then from p' to p . We conclude as in case 1.

Otherwise $p \in \partial(J^+(p) \cap J^-(p))$ and there is an achronal null geodesic γ from p to p .

Suppose that there exists a point q lying on $(I^+(p) \cap I^-(p)) \cap \gamma$. Let γ_1 be a timelike curve from p to q . Then the concatenation of γ with γ_1 provides a causal curve from p to q which can be deformed to a timelike one, contradicting the fact that $q \in \partial(J^+(p) \cap J^-(p))$. Thus $\gamma \subset \partial(J^+(p) \cap J^-(p))$.

Summarizing: either $\partial(J^+(p) \cap J^-(p))$ is empty, thus $\mathcal{M} = J^+(p) \cap J^-(p)$ which is compact, or there exists an achronal null geodesic γ lying on $\partial(J^+(p) \cap J^-(p))$.

It remains to show that the latter case cannot occur in dimensions larger than two. For this, consider the Lipschitz topological hypersurface $\dot{I}^+(\gamma)$. If $\dim \mathcal{M} \geq 3$ there exists $q \in \dot{I}^+(\gamma) \setminus \gamma$, and we note that this is not necessarily the case in dimension two. Since $J^-(q) \cap J^+(p)$ is compact there exists a causal curve, say γ_2 , from p to q . The curve γ_2 is not a reparameterization of γ since $q \notin \gamma$. One can now go from p to q by concatenating γ with γ_2 . The resulting curve can be deformed to a timelike curve from p to q , contradicting the fact that $q \in \dot{I}^+(\gamma)$. So γ cannot lie on the boundary of $J^+(p) \cap J^-(p)$, and the proof is completed. $\square$

REMARK 2.8.18 As already pointed out, the argument fails in dimension two because then $\dot{I}^+(p) \setminus \gamma$ can be empty. The following example, pointed out to us by M. Sánchez, shows that this happens in the spacetime $S^1 \times \mathbb{R}$ with coordinates (u, v) and metric $du dv$, which is not globally hyperbolic. If $p_a = (u_a, v_a)$, $a \in \{1, 2\}$, then $J^+(p_1) \cap J^-(p_2)$ is the compact set $S^1 \times [v_1, v_2]$ if $v_1 \leq v_2$, with $J^+(p_1) \cap J^-(p_2) = \emptyset$ if $v_1 > v_2$. $\square$

2.9 $\mathcal{D}_I$ -Domains of dependence

A set $\mathcal{U} \subset \mathcal{M}$ is said to be *achronal* if

$$I^+(\mathcal{U}) \cap I^-(\mathcal{U}) = \emptyset.$$

There is an obvious analogous definition of an *acausal* set

$$J^+(\mathcal{U}) \cap J^-(\mathcal{U}) = \emptyset.$$

Let $\mathcal{S}$ be an achronal topological hypersurface in a spacetime $(\mathcal{M}, g)$. (By a hypersurface we mean an embedded submanifold of codimension one, i.e., a submanifold of dimension $\dim \mathcal{M} - 1$.) Unless explicitly indicated otherwise we will assume that $\mathcal{S}$ has no boundary. The *future domain of dependence* $\mathcal{D}_I^+(\mathcal{S})$ of $\mathcal{S}$ is defined as the set of points $p \in \mathcal{M}$ with the property that *every past-directed past-inextendible timelike curve starting at p meets $\mathcal{S}$ precisely once*. The *past domain of dependence* $\mathcal{D}_I^-(\mathcal{S})$ is defined by changing *past-directed past-inextendible* to *future-directed future-inextendible* above. Finally one sets

$$\mathcal{D}_I(\mathcal{S}) := \mathcal{D}_I^+(\mathcal{S}) \cup \mathcal{D}_I^-(\mathcal{S}). \quad (2.9.1)$$

The ‘precisely’ in ‘precisely once’ above follows of course already from achronality of $\mathcal{S}$; the repetitiveness in our definition is deliberate, to emphasize the property. We always have

$$\mathcal{S} \subset \mathcal{D}_I^\pm(\mathcal{S}).$$

INCIDENTALLY: We have found it useful to build in the fact that $\mathcal{S}$ is a topological hypersurface in the definition of $\mathcal{D}_I^\pm(\mathcal{S})$. Some authors do not impose this restriction [219], which can lead to various pathologies. From the point of view of differential equations the only interesting case is that of a hypersurface anyway. $\square$

The domain of dependence is usually denoted by $\mathcal{D}(\mathcal{S})$ in the literature, and we will sometimes do so. We have added the subscript I to emphasize that the definition is based on timelike curves. Hawking and Ellis [244] define the domain of dependence using causal curves instead of timelike ones; we will denote the resulting domains of dependence by

$$\mathcal{D}_J^\pm(\mathcal{S}), \text{ together with } \mathcal{D}_J(\mathcal{S}) := \mathcal{D}_J^+(\mathcal{S}) \cup \mathcal{D}_J^-(\mathcal{S}). \quad (2.9.2)$$

On the other hand, timelike curves are used by Geroch [219] and by Penrose [398]. For C^3 metrics and spacelike acausal hypersurfaces $\mathcal{S}$, the resulting sets differ by a boundary. The definition with causal curves has the advantage that the resulting set $\mathcal{D}_J(\mathcal{S})$ is open when $\mathcal{S}$ is an acausal topological hypersurface. However, this excludes piecewise null hypersurfaces as Cauchy surfaces, and this is why we use the definition based on timelike curves in the current treatment. It appears that the definition using causal curves is easier to handle when continuous metrics are considered [125].

The following examples are instructive, and are left as exercises to the reader; note that some of the results proved later in this section might be helpful in verifying our claims.

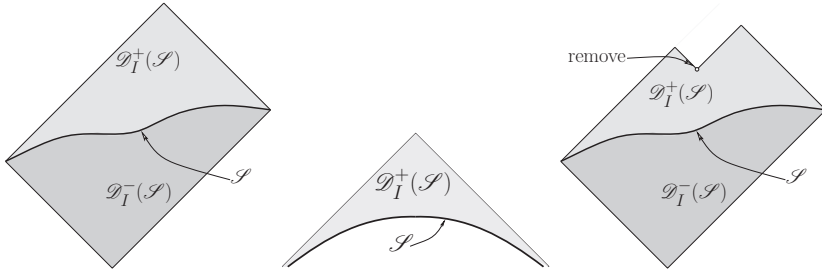


Fig. 2.9.1 Examples of domains of dependence.

EXAMPLE 2.9.1 Let $\mathcal{S} = \{x^0 = 0\}$ in Minkowski spacetime $\mathbb{R}^{1,n}$, where x^0 is the usual time coordinate on $\mathbb{R}^{1,n}$. Then $\mathcal{D}_I(\mathcal{S}) = \mathbb{R}^{1,n}$. Thus both $\mathcal{D}_I^+(\mathcal{S})$ and $\mathcal{D}_I^-(\mathcal{S})$ are non-trivial, and their union covers the whole spacetime. $\square$

EXAMPLE 2.9.2 Let $\mathcal{S} = \{\text{the set of points in } \mathbb{R}^n \text{ with rational coordinates}\} \subset \{x^0 = 0\}$ in Minkowski spacetime $\mathbb{R}^{1,n}$, where x^0 is the usual time coordinate on $\mathbb{R}^{1,n}$. Then ' $\mathcal{D}_I^+(\mathcal{S}) = \mathcal{S}$ ', in the sense that 'the set of points $p \in \mathcal{M}$ with the property that every past-directed past-inextendible timelike curve starting at p meets $\mathcal{S}$ precisely once' coincides with $\mathcal{S}$. Such examples are why we assumed that $\mathcal{S}$ is a hypersurface in the definition of $\mathcal{D}_I(\mathcal{S})$. $\square$

EXAMPLE 2.9.3 Let $\mathcal{S} = \{x^0 - x^1 = 0\}$ in Minkowski spacetime, where the x^μ 's are the usual Minkowskian coordinates on $\mathbb{R}^{1,n}$. Then $\mathcal{D}_I^+(\mathcal{S}) = \mathcal{D}_I^-(\mathcal{S}) = \mathcal{D}_I(\mathcal{S}) = \mathcal{S}$ (this can be proved using, e.g., Lemma 2.9.10). $\square$

EXAMPLE 2.9.4 Let $\mathcal{S} = \{x^0 = |x^1|\}$ in Minkowski spacetime $\mathbb{R}^{1,n}$. Then $\mathcal{D}_I^-(\mathcal{S}) = \mathcal{S}$, $\mathcal{D}_I^+(\mathcal{S}) = \{x^0 \geq |x^1|\}$. The fact that $\mathcal{D}_I^-(\mathcal{S}) = \mathcal{S}$ makes $\mathcal{D}_I^-(\mathcal{S})$ rather uninteresting. On the other hand, $\mathcal{D}_I^+(\mathcal{S})$ coincides with the causal future of $\mathcal{S}$. $\square$

EXAMPLE 2.9.5 Let $\mathcal{S} = \dot{J}^+(0)$, the forward light cone of the origin in Minkowski spacetime $\mathbb{R}^{1,n}$. Then $\mathcal{D}_I^+(\mathcal{S}) = J^+(0)$ is the forward causal cone of the origin, while $\mathcal{D}_I^-(\mathcal{S}) = \mathcal{S}$. On the other hand, if one removes the origin from $\dot{J}^+(0)$, so that $\mathcal{S} = \dot{J}^+(0) \setminus \{0\}$, then $\mathcal{D}_I^+(\mathcal{S}) = \mathcal{D}_I^-(\mathcal{S}) = \mathcal{S}$. $\square$

EXAMPLE 2.9.6 Let $\mathcal{S} = \{\eta_{\mu\nu}x^\mu x^\nu = -1, x^0 > 0\}$ be the upper component of the unit spacelike hyperboloid in Minkowski spacetime. Then $\mathcal{D}_I(\mathcal{S}) = J^+(0)$. Thus both $\mathcal{D}_I^-(\mathcal{S})$ and $\mathcal{D}_I^+(\mathcal{S})$ are non-trivial; however, $\mathcal{D}_I^-(\mathcal{S})$ does not cover the whole past of $\mathcal{S}$. $\square$

As a warm-up, let us prove the following elementary property of domains of dependence.

PROPOSITION 2.9.7 Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let $p \in \mathcal{D}_I^+(\mathcal{S})$; then

$$I^-(p) \cap J^+(\mathcal{S}) \subset \mathcal{D}_I^+(\mathcal{S}).$$

Proof. Let $q \in I^-(p) \cap J^+(\mathcal{S})$; thus, there exists a past-directed timelike curve γ_0 from p to q . Let γ_1 be a past-inextendible timelike curve γ_1 starting at q . The curve $\gamma := \gamma_0 \cup \gamma_1$ is a past-inextendible past-directed timelike curve starting at p ; thus, it meets $\mathcal{S}$ precisely once at some point $r \in \mathcal{S}$. Suppose that γ passes through r before passing through q ; as $q \in J^+(\mathcal{S})$ from Lemma 2.4.14 we find that $r \in I^+(\mathcal{S})$, contradicting achronality of $\mathcal{S}$. This shows that γ must meet $\mathcal{S}$ after passing through q ; hence γ_1 meets $\mathcal{S}$ precisely once. $\square$

Let $\mathcal{S}$ be achronal; we shall say that a set $\mathcal{O}$ forms a one-sided future neighbourhood of $p \in \mathcal{S}$ if there exists an open set $\mathcal{U} \subset \mathcal{M}$ such that $\mathcal{U}$ contains p

and

$$\mathcal{U} \cap J^+(\mathcal{S}) \subset \mathcal{O}.$$

As $I^-(p)$ is open, Proposition 2.9.7 immediately implies the following.

COROLLARY 2.9.8 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Suppose that $\mathcal{D}_I^+(\mathcal{S}) \neq \mathcal{S}$; consider any point $p \in \mathcal{D}_I^+(\mathcal{S}) \setminus \mathcal{S}$. For any $q \in \mathcal{S} \cap I^-(p)$ the set $\mathcal{D}_I^+(\mathcal{S})$ forms a one-sided future neighbourhood of q . $\square$*

Transversality considerations near $\mathcal{S}$ should make it clear that the hypothesis of Corollary 2.9.8 is satisfied for achronal, C^1 , spacelike hypersurfaces without boundary, and therefore for such $\mathcal{S}$ the set $\mathcal{D}_I(\mathcal{S})$ forms a neighbourhood of $\mathcal{S}$. Example 2.9.3 shows that this will not be the case for general $\mathcal{S}$'s.

The next theorem shows that achronal topological hypersurfaces can be used to produce globally hyperbolic spacetimes.

THEOREM 2.9.9 *Let $\mathcal{S}$ be an achronal hypersurface in $(\mathcal{M}, g)_{C^2}$, and suppose that the interior $\mathring{\mathcal{D}}_I(\mathcal{S})$ of the domain of dependence $\mathcal{D}_I(\mathcal{S})$ of $\mathcal{S}$ is not empty. Then $\mathring{\mathcal{D}}_I(\mathcal{S})$ equipped with the metric obtained by restriction from g is globally hyperbolic.*

Proof. We need first to show that a causal curve can be pushed up by an amount as small as desired to yield a timelike curve.

LEMMA 2.9.10 ('Push-up Lemma II') *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let*

$$\gamma : \mathbb{R}^+ \rightarrow \mathcal{M}$$

be a past-inextendible past-directed causal curve starting at p , and let $\mathcal{O}$ be a neighbourhood of the image $\gamma(\mathbb{R}^+)$ of γ . Then for every $r \in I^+(q) \cap \mathcal{O}$ there exists a past-inextendible past-directed timelike curve $\hat{\gamma}$ starting at r such that

$$\hat{\gamma} \subset I^+(\gamma) \cap \mathcal{O}, \quad (2.9.3)$$

$$\forall s \in [0, \infty) \quad I^-(\hat{\gamma}(s)) \cap \gamma(\mathbb{R}^+) \neq \emptyset. \quad (2.9.4)$$

Proof. The construction is essentially identical to that of the proof of Lemma 2.4.14, p. 42, except that we will have to deal with a countable collection of curves, rather than a finite number. We also need to make sure that the final curve is inextendible. As usual, we parameterize γ by h -distance as measured from p . Using an exhaustion of $[0, \infty)$ by compact intervals $[m, m+1]$ we cover γ by a countable collection $\mathcal{U}_i \subset \mathcal{O}$, $i \in \mathbb{N}$ of elementary regions $\mathcal{U}_i$ centred at

$$p_i = \gamma(r_i)$$

with

$$p_1 = p, \quad p_i \in \mathcal{U}_i \cap \mathcal{U}_{i+1}, \quad p_{i+1} \subset J^-(p_i).$$

We further impose the following condition on the $\mathcal{U}_i$'s: if $r_i \in [j, j+1)$, then the corresponding $\mathcal{U}_i$ is contained in a h -distance ball $B_h(p_i, 1/(j+1))$.

Let $\gamma_0 : [0, s_0] \rightarrow \mathcal{M}$ be a past-directed causal curve from r to $p \in \mathcal{U}_1 \cap \mathcal{U}_2$; let s_1 be close enough to s_0 so that

$$\gamma_0(s_1) \in \mathcal{U}_2.$$

Proposition 2.2.4, p. 27, together with the definition of elementary regions, shows that there exists a past-directed timelike curve $\gamma_1 : [0, 1] \rightarrow \mathcal{U}_1 \subset \mathcal{O}$ from q to p_2 . (In particular $\gamma_1 \setminus \{p\} \subset I^+(p) \subset I^+(\gamma)$.) Similarly, for any $s \in [0, 1]$ there exists a past-directed timelike curve $\gamma_{2,s} : [0, 1] \rightarrow \mathcal{U}_2 \subset \mathcal{O}$ from $\gamma_1(s)$ to p_2 . We choose $s =: s_2$ small enough so that

$$\gamma_1(s_2) \in \mathcal{U}_3.$$

One repeats that construction iteratively, obtaining a sequence of past-directed timelike curves $\gamma_i \subset I^+(\gamma) \cap \mathcal{U}_i \subset I^+(\gamma) \cap \mathcal{O}$ such that the end point of γ_i lies

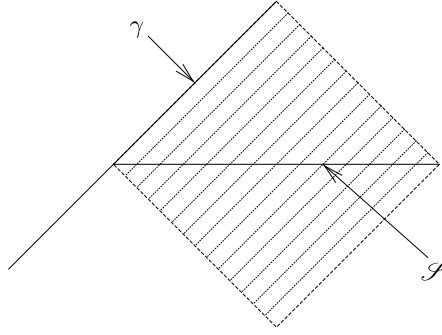


Fig. 2.9.2 Let $\mathcal{S} = \{t = 0, x \in (-1, 1)\} \subset \mathbb{R}^{1,1}$; then $\mathcal{D}_I(\mathcal{S})$ is the closed dotted diamond region *without* the two rightmost and leftmost points that lie on the closure of $\mathcal{S}$. The past-directed null geodesic γ starting at $(1, 1) \in \mathcal{D}_I^+(\mathcal{S})$ does not intersect $\mathcal{S}$.

in $\mathcal{U}_{i+1}$ and coincides with the starting point of γ_{i+1} . Concatenating those curves together gives the desired path $\hat{\gamma}$. Since every path γ_i lies in $I^+(\gamma) \cap \mathcal{O}$, so does their union.

Since $\gamma_i \subset \mathcal{U}_i \subset B_h(p_i, 1/(j+1))$ when $r_i \in [j, j+1)$ we obtain, for $r \in [j, j+1)$,

$$\text{dist}_h(\gamma(r), \hat{\gamma}) \leq \text{dist}_h(\gamma(r), \gamma(r_i)) + \text{dist}_h(\gamma(r_i), \gamma_i) \leq \frac{2}{j+1},$$

where we have ensured that $\text{dist}_h(\gamma(r), \gamma(r_i)) < 1/(j+1)$ by choosing r_i appropriately. It follows that

$$\text{dist}_h(\gamma(r), \hat{\gamma}) \leq \frac{2}{r}. \quad (2.9.5)$$

To finish the proof, suppose that $\hat{\gamma} : [0, s_*) \rightarrow \mathcal{M}$ is extendible; call $\hat{p}$ the end point of $\hat{\gamma}$. By (2.9.5)

$$\lim_{r \rightarrow \infty} \text{dist}_h(\gamma(r), \hat{p}) = 0.$$

Thus $\hat{p}$ is an end point of γ , which together with Theorem 2.5.5 contradicts the inextendibility of γ . $\square$

By the definition of domains of dependence, inextendible *timelike* curves through $p \in \mathcal{D}_I^+(\mathcal{S})$ intersect all the sets $\mathcal{S}$, $I^+(\mathcal{S})$, and $I^-(\mathcal{S})$. This is wrong in general for inextendible *causal* curves through points in $\mathcal{D}_I^+(\mathcal{S}) \setminus \mathring{\mathcal{D}}_I^+(\mathcal{S})$, as shown in Fig. 2.9.2. Nevertheless we have the following.

LEMMA 2.9.11 *If $p \in \mathring{\mathcal{D}}_I(\mathcal{S})$, then every inextendible causal curve γ through p intersects $\mathcal{S}$, $I^-(\mathcal{S})$ and $I^+(\mathcal{S})$.*

REMARK 2.9.12 In contradistinction with timelike curves, for causal curves the intersection of γ with $\mathcal{S}$ does not have to be a point. An example is given by the hypersurface $\mathcal{S}$ of Fig. 2.9.3. $\square$

Proof. Changing time orientation if necessary we may suppose that $p \in \mathcal{D}_I^+(\mathcal{S})$. Let $\gamma : I \rightarrow \mathcal{M}$ be any past-directed inextendible causal curve through p . Since p is an interior point of $\mathcal{D}_I^+(\mathcal{S})$ there exists $q \in I^+(p) \cap \mathcal{D}_I^+(\mathcal{S})$. By the Push-up Lemma 2.9.10 with $\mathcal{O} = \mathcal{M}$ there exists a past-inextendible past-directed timelike curve $\hat{\gamma}$ starting at q which lies to the future of γ . The inextendible timelike curve $\hat{\gamma}$ enters $I^-(\mathcal{S})$, and so does γ by (2.9.4).

If $p \in \mathcal{S}$, we can repeat the argument above with the time orientation changed, showing that γ enters $I^+(\mathcal{S})$ as well, and we are done.

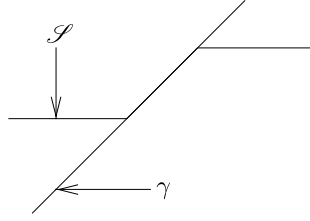


Fig. 2.9.3 A null geodesic γ intersecting an achronal topological hypersurface $\mathcal{S}$ at more than one point.

Otherwise $p \notin \mathcal{S}$, then p is necessarily in $I^+(\mathcal{S})$; hence γ meets $I^+(\mathcal{S})$ as well. Now, each of the two disjoint sets

$$I_{\pm} := \{s \in I : \gamma(s) \in I^{\pm}(\mathcal{S})\} \subset \mathbb{R}$$

is open in the connected interval I . They cover I if γ does not meet $\mathcal{S}$, which implies that either I_+ or I_- must be empty when $\gamma \cap \mathcal{S} = \emptyset$. But we have shown that both I_+ and I_- are not empty, and so γ meets $\mathcal{S}$, as desired. $\square$

Returning to the proof of Theorem 2.9.9, suppose that $\mathring{\mathcal{D}}_I(\mathcal{S})$ is not strongly causal. (In view of Theorem 2.8.17, this part of the proof can be skipped if $\dim \mathcal{M} \geq 3$; note that $\mathcal{M}$ cannot be compact under the current hypotheses.) Then there exists $p \in \mathring{\mathcal{D}}_I(\mathcal{S})$ and a sequence $\gamma_n : \mathbb{R} \rightarrow \mathring{\mathcal{D}}_I(\mathcal{S})$ of inextendible causal curves which exit the h -distance geodesic ball $B_h(p, 1/n)$ (centred at p and of radius $1/n$) and re-enter $B_h(p, 1/n)$ again. Changing the time orientation of $\mathcal{M}$ if necessary, without loss of generality we may assume that $p \in I^-(\mathcal{S}) \cup \mathcal{S}$. Note that the property ‘leaves and re-enters’ is invariant under the change $\gamma_n(s) \rightarrow \gamma_n(-s)$, so that by changing the orientation of (some of) the γ_n ’s if necessary, there is no loss of generality in assuming the γ_n ’s to be future-directed. Finally, we reparameterize the γ_n ’s by h -distance, with $\gamma_n(0) \in B_h(p, 1/n)$. Then, there exists a sequence $s_n > 0$ such that $\gamma_n(s_n) \in B_h(p, 1/n)$, with $\gamma_n(0)$ and $\gamma_n(s_n)$ lying on different connected components of $\gamma \cap B_h(p, 1/n)$.

Let $\mathcal{O}$ be an elementary neighbourhood of p , as in Definition 2.2.8, p. 31, and let n_0 be large enough so that $B_h(p, 1/n_0) \subset \mathcal{O}$. Note that the local coordinate x^0 on $\mathcal{O}$ is monotonic along every connected component of $\gamma_n \cap \mathcal{O}$, which implies, for $n \geq n_0$, that any causal curve which exits and re-enters $B_h(p, 1/n)$ also must exit and re-enter $\mathcal{O}$. This in turn guarantees the existence of an $\epsilon > 0$ such that $s_n > \epsilon$ for all $n \geq n_0$.

Let γ be an accumulation curve through p of the γ_n ’s; passing to a subsequence if necessary, the γ_n ’s converge uniformly to γ on compact subsets of $\mathbb{R}$. The curve γ is causal and p is, by hypothesis, in the interior of $\mathring{\mathcal{D}}_I(\mathcal{S})$; we can therefore invoke Lemma 2.9.11 to conclude that there exist $s_{\pm} \in \mathbb{R}$ such that $\gamma(s_-) \in I^-(\mathcal{S})$ and $\gamma(s_+) \in I^+(\mathcal{S})$. Since $\mathcal{S}$ is achronal and γ is future directed we must have $s_- < s_+$. Since the $I^{\pm}(\mathcal{S})$ ’s are open, and since (passing to a subsequence if necessary) $\gamma_n(s_{\pm}) \rightarrow \gamma(s_{\pm})$, we have $\gamma_n(s_{\pm}) \in I^{\pm}(\mathcal{S})$ for n large enough.

The situation which is simplest to exclude is the one where the sequence $\{s_n\}$ is bounded. Then there exists $s_* \in \mathbb{R}$ such that, passing again to a subsequence if necessary, we have $s_n \rightarrow s_*$. Note that $\gamma_n(s_*) \rightarrow p$ and that $s_* \geq \epsilon$. Since $\gamma_n|_{[0, s_*]}$ converges uniformly to $\gamma|_{[0, s_*]}$, we obtain an inextendible periodic causal curve γ' through p by repetitively circling from p to p along $\gamma|_{[0, s_*]}$. By Lemma 2.9.11 γ' meets all of $\mathcal{S}$, $I^+(\mathcal{S})$ and $I^-(\mathcal{S})$, which is clearly incompatible with periodicity of γ' and achronality of $\mathcal{S}$. (In detail, there exist points $q_{\pm} \in \gamma' \cap I^{\pm}(\mathcal{S})$. Following backwards γ' from p to q_+ we obtain $q_+ \in J^-(p)$. But $I^-(q_+) \cap \mathcal{S} \neq \emptyset$, and

Lemma 2.4.14 implies that $I^-(p) \cap \mathcal{S} \neq \emptyset$. Since $p \in I^-(\mathcal{S}) \cup \mathcal{S}$, this contradicts achronality of $\mathcal{S}$.)

Note that if $p \in I^-(\mathcal{S})$ we would need to have $s_n \leq s_+$ for n large enough. Otherwise, for n large, we could follow γ_n in the future direction from $\gamma_n(s_+) \in I^+(\mathcal{S})$ to $\gamma_n(s_n) \in I^-(\mathcal{S})$, which is not possible if $\mathcal{S}$ is achronal. But then the sequence s_n would be bounded, which has already been excluded. So $p \in I^-(\mathcal{S})$ cannot occur.

There remains the possibility that $p \in \mathcal{S}$. We then must have $\gamma_n|_{[0,\infty)} \cap I^-(\mathcal{S}) = \emptyset$; otherwise, we would obtain a contradiction with achronality of $\mathcal{S}$ by following γ_n to the future from $p = \gamma_n(0) \in \mathcal{S}$ to a point where γ_n intersects $I^-(\mathcal{S})$. Set

$$\hat{\gamma}_n(s) = \gamma_n(s + s_n);$$

then $\hat{\gamma}_n$ accumulates at p since $\hat{\gamma}_n(0) \rightarrow p$. Therefore, there exists an inextendible accumulation curve $\hat{\gamma} : \mathbb{R} \rightarrow \mathcal{M}$ of the $\hat{\gamma}_n$'s passing through p . As

$$\hat{\gamma}_n([-s_n, \infty)) \cap I^-(p) = \gamma_n([0, \infty)) \cap I^-(\mathcal{S}) = \emptyset$$

we have $\hat{\gamma}(\mathbb{R}) \cap I^-(p) = \emptyset$ as well. This is, however, not possible if $p \in \mathring{\mathcal{D}}_I(\mathcal{S})$ by Lemma 2.9.11. We see that the possibility that $p \in \mathcal{S}$ cannot occur either. We conclude that $\mathring{\mathcal{D}}_I(\mathcal{S})$ is strictly causal, as desired.

To finish the proof, we need to prove compactness of the sets of the form

$$J^+(p) \cap J^-(q), \quad p, q \in \mathring{\mathcal{D}}_I(\mathcal{S}).$$

If p and q are such that this set is empty or equals $\{p\}$ there is nothing to prove. Otherwise, consider a sequence $r_n \in J^+(p) \cap J^-(q)$. One of the following is true:

1. We have $r_n \in I^-(\mathcal{S}) \cup \mathcal{S}$ for all $n \geq n_0$, or
2. There exists a subsequence, still denoted by r_n , such that $r_n \in I^+(\mathcal{S})$.

In the second case we change the time orientation, pass to a subsequence, and rename p and q , reducing the analysis to the first case. Note that this leads to $p \in I^-(\mathcal{S}) \cup \mathcal{S}$.

By definition, there exists a future-directed causal curve $\hat{\gamma}_n$ from p to q which passes through r_n ,

$$\hat{\gamma}_n(s_n) = r_n. \quad (2.9.6)$$

Let γ_n be any h -length-parameterized, inextendible future-directed causal curve extending $\hat{\gamma}_n$, with $\gamma_n(0) = p$. Let γ be an inextendible accumulation curve of the γ_n 's; then γ is a future-inextendible causal curve through

$$p \in (\mathcal{D}_I^-(\mathcal{S}) \cup \mathcal{S}) \cap \mathring{\mathcal{D}}_I(\mathcal{S}).$$

By Lemma 2.9.11 there exists s_+ such that $\gamma(s_+) \in I^+(\mathcal{S})$. Passing to a subsequence, the γ_n 's converge uniformly to γ on $[0, s_+]$, which implies that for n large enough the $\gamma_n|_{[0, s_+]}$'s enter $I^+(\mathcal{S})$. This, together with achronality of $\mathcal{S}$, shows that the sequence s_n defined by (2.9.6) is bounded; in fact, we must have $0 \leq s_n \leq s_+$. Eventually passing to another subsequence we thus have $s_n \rightarrow s_\infty$ for some $s_\infty \in \mathbb{R}$. This implies that

$$r_n \rightarrow \gamma(s_\infty) \in J^+(p) \cap J^-(q),$$

which had to be established. $\square$

We have the following characterization of *interiors* of domains of dependence.

THEOREM 2.9.13 *Consider a spacetime $(\mathcal{M}, g)_{C^2}$. Let $\mathcal{S}$ be a differentiable acausal spacelike hypersurface. A point $p \in \mathcal{M}$ is in $\mathring{\mathcal{D}}_I^+(\mathcal{S})$ if and only if*

$$\text{the set } J^-(p) \cap \mathcal{S} \text{ is non-empty, and compact as a subset of } \mathring{\mathcal{S}}. \quad (2.9.7)$$

REMARK 2.9.14 The set $\mathcal{S} = \{t = 0, x \in [-1, 1]\} \subset \mathbb{R}^{1,1}$ (compare Fig. 2.9.2, but note that a different $\mathcal{S}$ was meant there) shows that the condition (2.9.7) cannot be replaced by the requirement that the set $\overline{I^-(p)} \cap \mathcal{S}$ is non-empty, and *compact as a subset of $\mathcal{M}$* .

Proof. For $p \in \mathring{\mathcal{D}}_I^+(\mathcal{S})$ the compactness of $\overline{I^-(p)} \cap \mathcal{S}$ can be established by an argument very similar to that given in the last part of the proof of Theorem 2.9.9, the details are left to the reader.

In order to prove the reverse implication, assume that (2.9.7) holds; then there exists a future-directed causal curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ from some point $q \in \mathcal{S}$ to p . Set

$$I := \{t \in (0, 1] : \gamma(s) \in \mathring{\mathcal{D}}^+(\mathcal{S}) \text{ for all } 0 < s \leq t\} \subset (0, 1].$$

Now, elementary considerations show that for C^1 , spacelike, acausal hypersurfaces we have $\gamma(t) \in \mathring{\mathcal{D}}_I^+(\mathcal{S})$ for $t > 0$ small enough; hence, I is not empty. Clearly I is open in $(0, 1]$. In order to show that it equals $(0, 1]$ set

$$t_* := \sup I.$$

Consider any past-inextendible past-directed causal curve $\hat{\gamma}$ starting at $\gamma(t_*)$. For $t < t_*$ let $\hat{\gamma}_t$ be a family of past-inextendible causal push-downs of $\hat{\gamma}$ which start at $\gamma(t)$, and which have the property that

$$\text{dist}_h(\gamma_t(s), \gamma(s)) \leq |t - t_*| \text{ for } 0 \leq s \leq 1/|t - t_*|.$$

Then $\hat{\gamma}_t$ intersects $\mathcal{S}$ at some point $q_t \in J^-(p)$. The compactness of $J^-(p) \cap \mathcal{S}$ implies that the curve $t \rightarrow q_t \in \mathcal{S}$ accumulates at some point $q_* \in \mathcal{S}$, which clearly is the point of intersection of γ with $\mathcal{S}$. This shows that every causal curve γ through $\gamma(t_*)$ meets $\mathcal{S}$, in particular $\gamma(t_*) \in \mathring{\mathcal{D}}_I^+(\mathcal{S})$. So I is both open and closed in $(0, 1]$, hence $I = (0, 1]$, and the result is proved. $\square$

2.10 Cauchy horizons

Geometrically, *Cauchy horizons* are the boundaries of domains of dependence. They play a key role from a PDE perspective, being hypersurfaces where uniqueness and existence of solutions of wave-type equations are expected to break down, and are known to fail in some examples. In this section we define the notion, provide some examples, prove the existence of *generators*, and establish *semi-convexity* of a class of horizons.

DEFINITION 2.10.1 Let $\mathcal{S}$ be an achronal topological hypersurface. We define the future Cauchy horizon $\mathcal{H}_I^+(\mathcal{S})$ of $\mathcal{S}$ as

$$\mathcal{H}_I^+(\mathcal{S}) = \mathring{\mathcal{D}}_I^+(\mathcal{S}) \setminus I^-(\mathring{\mathcal{D}}_I^+(\mathcal{S})),$$

with an obvious corresponding definition for the past Cauchy horizon $\mathcal{H}_I^-(\mathcal{S})$. One defines the Cauchy horizon as

$$\mathcal{H}_I(\mathcal{S}) = \mathcal{H}_I^-(\mathcal{S}) \cup \mathcal{H}_I^+(\mathcal{S}).$$

Our definition follows that of Penrose [398]. Similarly to the domains of dependence, the usual notation for Cauchy horizons is $\mathcal{H}$ and not $\mathcal{H}_I$, and we will sometimes write so. The analogous definition of future Cauchy horizon with $\mathring{\mathcal{D}}_I^+(\mathcal{S})$ replaced by $\mathring{\mathcal{D}}_J^+(\mathcal{S})$ leads to identical Cauchy horizons for Lipschitz-continuous metrics, but one obtains sometimes essentially different sets in general for metrics which are merely assumed to be continuous.

It is instructive to consider a few examples.

EXAMPLE 2.10.2 Let $\mathcal{S} = \{x^0 = 0\}$ in Minkowski spacetime $\mathbb{R}^{1,n}$, where x^0 is the usual time coordinate on $\mathbb{R}^{1,n}$. Then $\mathcal{H}_I(\mathcal{S}) = \emptyset$. $\square$

EXAMPLE 2.10.3 Let $\mathcal{S}$ be the open unit ball in $\mathbb{R}^n$, viewed as a subset of $\{x^0 = 0\}$ in Minkowski spacetime $\mathbb{R}^{1,n}$, where x^0 is the usual time coordinate on $\mathbb{R}^{1,n}$. Then $\mathcal{H}_I^+(\mathcal{S})$ is the intersection of the past light cone of the point $(x^0 = 1, \vec{x} = 0)$ with $\{x^0 \geq 0\}$. $\square$

EXAMPLE 2.10.4 Let $\mathcal{S} = \{\eta_{\mu\nu}x^\mu x^\nu = -1, x^0 > 0\}$ be the upper component of the unit spacelike hyperboloid in Minkowski spacetime. Then $\mathcal{H}_I^+(\mathcal{S}) = \emptyset$, while $\mathcal{H}_I^-(\mathcal{S})$ coincides with the future light cone of the origin. $\square$

EXAMPLE 2.10.5 The Taub-NUT spacetimes [375, 443] provide examples of spacetimes with an achronal spacelike S^3 and with two corresponding past and future Cauchy horizons, each diffeomorphic to S^3 [359]. $\square$

For any open set Ω one has $\Omega \setminus I^-(\Omega) = \emptyset$, which shows that

$$\mathring{\mathcal{D}}_I(\mathcal{S}) \cap \mathcal{H}_I^+(\mathcal{S}) = \emptyset. \quad (2.10.1)$$

It follows that $\mathcal{H}_I^+(\mathcal{S})$ is a subset of the topological boundary $\partial\mathcal{D}_I^+(\mathcal{S})$ of $\mathcal{D}_I^+(\mathcal{S})$:

$$\mathcal{H}_I^+(\mathcal{S}) \subset \partial\mathcal{D}_I^+(\mathcal{S}) := \overline{\mathcal{D}_I^+(\mathcal{S})} \setminus \mathring{\mathcal{D}}_I^+(\mathcal{S}). \quad (2.10.2)$$

The important notion of *generators* of horizons stems from the following result in where we assume that $\mathcal{S}$ is differentiable and spacelike, which is sufficient for many purposes.

PROPOSITION 2.10.6 *Let $\mathcal{S}$ be a spacelike achronal C^1 hypersurface in $(\mathcal{M}, g)_{C^2}$. For any $p \in \mathcal{H}_I^+(\mathcal{S})$ there exists a past-directed null geodesic $\gamma_p \subset \mathcal{H}_I^+(\mathcal{S})$ starting at p which either does not have an end point in $\mathcal{M}$ or has an end point on $\overline{\mathcal{S}} \setminus \mathcal{S}$.*

REMARK 2.10.7 There might be more than one such geodesic for some points on the horizon. $\square$

Proof. Let $p \in \mathcal{H}_I^+(\mathcal{S})$; then there exists a sequence of points $p_n \notin \mathcal{D}_I^+(\mathcal{S})$ which converge to p , and past-inextendible timelike curves γ_n through p_n that do not meet $\mathcal{S}$. Let γ be an accumulation curve of the γ_n through p . Then γ does not meet $\mathcal{S}$: indeed, if it did, then the γ_n 's would be meeting $\mathcal{S}$ as well for all n large enough. If γ meets $\mathcal{S}$, we let γ_p be the segment of γ from p to the intersection point with $\mathcal{S}$; otherwise we let $\gamma_p = \gamma$.

The curve γ_p is achronal: otherwise γ_p would enter the interior of $\mathcal{D}_I^+(\mathcal{S})$, but then it would have to intersect $\mathcal{S}$ by Lemma 2.9.11. We can thus invoke Proposition 2.6.12, p. 53, to conclude that γ_p is a null geodesic.

It remains to show that $\gamma_p \subset \mathcal{H}_I^+(\mathcal{S})$. Let γ be an inextendible past-directed timelike curve through a point q on γ_p , with $q \notin \mathcal{S}$. Let $\mathcal{O}$ be a neighbourhood of q that does not meet $\mathcal{S}$, and let $r \in \mathcal{O}$ be a point on γ lying to the timelike past of q . By Corollary 2.4.15 there exists a timelike curve γ_1 from p to r . Consider the past-inextendible timelike curve, say γ_2 , obtained by following γ_1 from p to r , and then following γ to the past. Since $p \in \mathcal{H}_I^+(\mathcal{S})$ the curve γ_2 has to meet $\mathcal{S}$. As γ_1 does not meet $\mathcal{S}$, it must be the case that γ meets $\mathcal{S}$, and so $q \in \mathcal{H}_I^+(\mathcal{S})$. $\square$

For any $p \in \mathcal{H}_I^+(\mathcal{S})$ let $\hat{\gamma}_p$ denote a maximal future extension of the geodesic segment γ_p of Proposition 2.10.6, and set $\tilde{\gamma}_p = \hat{\gamma}_p \cap \mathcal{H}_I^+(\mathcal{S})$. (Note that $\hat{\gamma}_p$ might exit $\mathcal{H}_I^+(\mathcal{S})$ when followed to the future; an example of this can be seen in Fig. 2.9.2.) Then $\tilde{\gamma}_p$ is called a *generator* of $\mathcal{H}_I^+(\mathcal{S})$.

Using this terminology, Proposition 2.10.6 can be reworded as the property that *every $p \in \mathcal{H}_I^+(\mathcal{S})$ is either an interior point or a future end point of a generator of $\mathcal{H}_I^+(\mathcal{S})$* . If $\overline{\mathcal{S}} = \mathcal{S}$, then generators of $\mathcal{H}_I^+(\mathcal{S})$ do not have past end points, remaining forever on $\mathcal{H}_I^+(\mathcal{S})$ to the past.

2.10.1 Semi-convexity

A hypersurface $\mathcal{H} \subset \mathcal{M}$ will be said to be *future null geodesically ruled* if every point $p \in \mathcal{H}$ belongs to a future-inextendible null geodesic $\Gamma \subset \mathcal{H}$; those geodesics are called *the generators* of $\mathcal{H}$. We emphasize that the generators are allowed to have past end points on $\mathcal{H}$, but no future end points. *Past null geodesically ruled* hypersurfaces are defined by changing the time orientation. Examples of future geodesically ruled hypersurfaces include past Cauchy horizons $\mathcal{D}^-(\mathcal{S})$ of achronal sets $\mathcal{S}$ of Proposition 2.10.6 (compare [398, Theorem 5.12]) and black-hole event horizons $\dot{J}^-(\mathcal{S}^+)$ of Section 4.3.8, p. 166.

Note that our definition involves explicitly geodesics, and therefore throughout this section we assume that the metric is twice-continuously differentiable. It should be kept in mind that the notion of the generator of a horizon in spacetimes with merely continuous metrics is not completely clear, and allowing metrics of poor differentiability appears to require a reformulation of the problem.

As elsewhere, the spacetime dimension $\dim \mathcal{M}$ is $n + 1$.

Suppose that $\mathcal{O}$ is a domain in $\mathbb{R}^n$. Recall that a continuous function $f : \mathcal{O} \rightarrow \mathbb{R}$ is called semi-convex if there exists a C^2 function $\phi : \mathcal{O} \rightarrow \mathbb{R}$ such that $f + \phi$ is convex. We shall say that the graph of f is a semi-convex hypersurface if f is semi-convex. A hypersurface $\mathcal{H}$ in a manifold $\mathcal{M}$ will be said semi-convex if $\mathcal{H}$ can be covered by coordinate patches $\mathcal{U}_\alpha$ such that $\mathcal{H} \cap \mathcal{U}_\alpha$ is a semi-convex graph for each α .

Consider an achronal hypersurface $\mathcal{H} \neq \emptyset$ in a globally hyperbolic spacetime $(\mathcal{M}, g)$. Let t be a time function on $\mathcal{M}$ which induces a diffeomorphism of $\mathcal{M}$ with $\mathbb{R} \times \mathcal{S}$ in the standard way [219, 427], with the level sets

$$\mathcal{S}_\tau \equiv \{p \mid t(p) = \tau\}$$

of t being Cauchy surfaces. As usual we identify $\mathcal{S}_0$ with $\mathcal{S}$ and, in the identification above, the curves $\mathbb{R} \times \{q\}$, $q \in \mathcal{S}$, are integral curves of ∇t . Define

$$\mathcal{S}_\mathcal{H} = \{q \in \mathcal{S} \mid \mathbb{R} \times \{q\} \text{ intersects } \mathcal{H}\}. \quad (2.10.3)$$

For $q \in \mathcal{S}_\mathcal{H}$ the set $(I \times \{q\}) \cap \mathcal{H}$ is a point by achronality of $\mathcal{H}$, which will be denoted by $(f(q), q)$. Thus an achronal hypersurface $\mathcal{H}$ in a globally hyperbolic spacetime is a graph over $\mathcal{S}_\mathcal{H}$ of a function f . The invariance-of-the-domain theorem shows that $\mathcal{S}_\mathcal{H}$ is an open subset of $\mathcal{S}$. We have the following.

THEOREM 2.10.8 *Let $\mathcal{H} \neq \emptyset$ be an achronal null geodesically ruled hypersurface in $(\mathcal{M}, g)_{C^2}$. Then $\mathcal{H}$ is locally Lipschitz, and semi-convex.*

Proof. Let $p \in \mathcal{H}$; changing the time orientation we can assume that $\mathcal{H}$ is future null geodesically ruled. When $\mathcal{M}$ is replaced with a globally hyperbolic neighbourhood of p , Theorem 2.10.8 follows from the fact that semi-convex functions are locally Lipschitz (compare Corollary 2.10.11), together with the following theorem. $\square$

THEOREM 2.10.9 *Let $\mathcal{H} \neq \emptyset$ be an achronal future null geodesically ruled hypersurface in a globally hyperbolic spacetime $(\mathcal{M} = \mathbb{R} \times \mathcal{S}, g)_{C^2}$. Then $\mathcal{H}$ is the graph of a semi-convex function f defined on an open subset $\mathcal{S}_\mathcal{H}$ of $\mathcal{S}$; in particular $\mathcal{H}$ is semi-convex.*

Proof. As discussed earlier, $\mathcal{H}$ is the graph of a function f . The idea of the proof is to show that f satisfies a variational principle; the semi-concavity of f follows then by a standard argument. Let $p \in \mathcal{H}$ and let $\mathcal{O}$ be a coordinate patch in a neighbourhood of p such that $x^0 = t$, with $\mathcal{O}$ of the form $I \times B(3R)$, where $B(R)$ denotes a coordinate ball centred at 0 of radius R in $\mathbb{R}^3$, with $p = (t(p), 0)$. Here I is the range of the coordinate x^0 ; we require it to be a bounded interval the size

of which will be determined later on. We further assume that the curves $I \times \{\vec{x}\}$, $\vec{x} \in B(3R)$, are integral curves of ∇t . Define

$$\mathcal{U}_0 = \{\vec{x} \in B(3R) \mid \text{the causal path } I \ni t \rightarrow (t, \vec{x}) \text{ intersects } \mathcal{H}\} .$$

We note that $\mathcal{U}_0$ is non-empty, since $0 \in \mathcal{U}_0$. Set

$$\mathcal{H}_\sigma = \mathcal{H} \cap \mathcal{S}_\sigma , \quad (2.10.4)$$

and choose σ large enough so that $\mathcal{O} \subset I^-(\mathcal{S}_\sigma)$. Now p lies on a future-inextendible generator Γ of $\mathcal{H}$, and global hyperbolicity of $(\mathcal{M}, g)$ implies that $\Gamma \cap \mathcal{S}_\sigma$ is non-empty; hence $\mathcal{H}_\sigma$ is non-empty.

For $\vec{x} \in B(3R)$ let $\mathcal{P}(x)$ denote the collection of piecewise differentiable future-directed null curves $\Gamma : [a, b] \rightarrow \mathcal{M}$ with $\Gamma(a) \in \mathbb{R} \times \{\vec{x}\}$ and $\Gamma(b) \in \mathcal{H}_\sigma$. We define

$$\tau(\vec{x}) = \sup_{\Gamma \in \mathcal{P}(x)} t(\Gamma(a)) . \quad (2.10.5)$$

We emphasize that we allow the domain of definition $[a, b]$ to depend upon Γ , and that the ‘ a ’ occurring in $t(\Gamma(a))$ in (2.10.5) is the lower bound for the domain of definition of the curve Γ under consideration.

We have the following result (compare [20, 223, 400]).

PROPOSITION 2.10.10 (Fermat principle) *For $\vec{x} \in \mathcal{U}_0$ we have*

$$\tau(\vec{x}) = f(\vec{x}) .$$

Proof. Let Γ be any generator of $\mathcal{H}$ such that $\Gamma(0) = (f(\vec{x}), \vec{x})$; clearly $\Gamma \in \mathcal{P}(x)$ so that $\tau(\vec{x}) \geq f(\vec{x})$. To show that this inequality must be an equality, suppose for contradiction that $\tau(\vec{x}) > f(\vec{x})$; thus there exists a null future-directed curve $\tilde{\Gamma}$ such that $t(\tilde{\Gamma}(0)) > f(\vec{x})$ and $\tilde{\Gamma}(1) \in \mathcal{H}_\sigma \subset \mathcal{H}$. Then the curve $\tilde{\Gamma}$ is obtained by following $\mathbb{R} \times \{\vec{x}\}$ from $(f(\vec{x}), \vec{x})$ to $(t(\tilde{\Gamma}(0)), \vec{x})$ and following Γ from there on is a causal curve with end points on $\mathcal{H}$ which is not a null geodesic. By Proposition 2.4.17 the curve $\tilde{\Gamma}$ can be deformed to a timelike curve with the same end points, which is impossible by achronality of $\mathcal{H}$. $\square$

The Fermat principle, Proposition 2.10.10, shows that f is a solution of the variational principle (2.10.5). Now this variational principle can be rewritten in a somewhat more convenient form as follows: the identification of $\mathcal{M}$ with $\mathbb{R} \times \mathcal{S}$ by flowing from $\mathcal{S}_0 \equiv \mathcal{S}$ along the gradient of t leads to a global decomposition of the metric of the form

$$g = \alpha(-dt^2 + h_t) ,$$

where h_t denotes a t -dependent family of Riemannian metrics on $\mathcal{S}$. Any future-directed differentiable null curve $\Gamma(s) = (t(s), \gamma(s))$ satisfies

$$\frac{dt(s)}{ds} = \sqrt{h_{t(s)}(\dot{\gamma}, \dot{\gamma})} ,$$

where $\dot{\gamma}$ is a shorthand for $d\gamma(s)/ds$. It follows that for any $\Gamma \in \mathcal{P}(x)$ it holds that

$$\begin{aligned} t(\Gamma(a)) &= t(\Gamma(b)) - \int_a^b \frac{dt}{ds} ds \\ &= \sigma - \int_a^b \sqrt{h_{\tilde{\sigma}}(\dot{\gamma}, \dot{\gamma})} ds . \end{aligned}$$

This allows us to rewrite (2.10.5) as

$$\tau(\vec{x}) = \sigma - \mu(\vec{x}) , \quad \mu(\vec{x}) \equiv \inf_{\Gamma \in \mathcal{P}(x)} \int_a^b \sqrt{h_{t(s)}(\dot{\gamma}, \dot{\gamma})} ds . \quad (2.10.6)$$

We note that in static spacetimes $\mu(\vec{x})$ is the Riemannian distance from $\vec{x}$ to $\mathcal{H}_\sigma$. In particular (2.10.6) implies the well-known fact that in globally hyperbolic static

spacetimes Cauchy horizons of open subsets of level sets of t are graphs of the distance function from the boundary of those sets.

Let $\gamma : [a, b] \rightarrow \mathcal{S}$ be a piecewise differentiable path; for any $p \in \mathbb{R} \times \{\gamma(b)\}$ we can find a null future-directed curve $\hat{\gamma} : [a, b] \rightarrow \mathcal{M}$ of the form $\hat{\gamma}(s) = (\phi(s), \gamma(s))$ with future end point p by solving the problem

$$\begin{cases} \phi(b) = t(p) , \\ \frac{d\phi(s)}{ds} = \sqrt{h_{\phi(s)}(\dot{\gamma}(s), \dot{\gamma}(s))} . \end{cases} \quad (2.10.7)$$

The path $\hat{\gamma}$ will be called the *null lift of γ with end point p* .

As an example application of Proposition 2.10.10 we recover the following well-known result [398].

COROLLARY 2.10.11 *f is Lipschitz continuous on any compact subset of its domain of definition.*

Proof. For $\vec{y}, \vec{z} \in B(2R)$ let $K \subset \mathbb{R} \times B(2R)$ be a compact set which contains all the null lifts $\Gamma_{\vec{y}, \vec{z}}$ of the coordinate segments $[\vec{y}, \vec{z}] := \{\lambda \vec{y} + (1 - \lambda) \vec{z} , \lambda \in [0, 1]\}$ with end points $(\tau(\vec{z}), \vec{z})$. Define

$$\hat{C} = \sup \{ \sqrt{h_p(n, n)} \mid p \in K, |n|_\delta = 1 \} , \quad (2.10.8)$$

where the supremum is taken over all points $p \in K$ and over all vectors $n \in T_p \mathcal{M}$ such that the coordinate-components vector (n^i) has Euclidean length $|n|_\delta$ equal to 1. Choose I to be a bounded interval large enough so that $K \subset I \times B(2R)$ and, as before, choose σ large enough so that $I \times B(2R)$ lies to the past of $\mathcal{S}_\sigma$. Let $\vec{y}, \vec{z} \in B(2R)$ and consider the causal curve $\Gamma = (t(s), \gamma(s))$ obtained by following the null lift $\Gamma_{\vec{y}, \vec{z}}$ in the parameter interval $s \in [0, 1]$, and then a generator of $\mathcal{H}$ from $(\tau(\vec{z}), \vec{z})$ until $\mathcal{H}_\sigma$ in the parameter interval $s \in [1, 2]$. Then we have

$$\mu(\vec{z}) = \int_1^2 \sqrt{h_{\vec{\sigma}}(\dot{\gamma}, \dot{\gamma})} ds .$$

Further $\Gamma \in \mathcal{P}(x)$ so that

$$\begin{aligned} \mu(\vec{y}) &\leq \int_0^2 \sqrt{h_{\vec{\sigma}}(\dot{\gamma}, \dot{\gamma})} ds \\ &= \int_0^1 \sqrt{h_{\vec{\sigma}}(\dot{\gamma}, \dot{\gamma})} ds + \int_1^2 \sqrt{h_{\vec{\sigma}}(\dot{\gamma}, \dot{\gamma})} ds \\ &\leq \hat{C} |\vec{y} - \vec{z}|_\delta + \mu(\vec{z}) , \end{aligned} \quad (2.10.9)$$

where $|\cdot|_\delta$ denotes the Euclidean norm of a vector, and with $\hat{C}$ defined in (2.10.8). Setting first $\vec{y} = \vec{x}$, $\vec{z} = \vec{x} + \vec{h}$ in (2.10.9) and then $\vec{z} = \vec{x}$, $\vec{y} = \vec{x} + \vec{h}$, the Lipschitz continuity of f on $B(2R)$ follows. The general result is obtained now by a standard covering argument. $\square$

Returning to the proof of Theorem 2.10.9, for $\vec{x} \in B(R)$ let $\Gamma_{\vec{x}}$ be a generator of $\mathcal{H}$ such that $\Gamma_{\vec{x}}(0) = (\tau(\vec{x}), \vec{x})$, and, if we write $\Gamma_{\vec{x}}(s) = (\phi_{\vec{x}}(s), \gamma_{\vec{x}}(s))$, then we require that $\gamma_{\vec{x}}(s) \in B(2R)$ for $s \in [0, 1]$. For $s \in [0, 1]$ and $\vec{h} \in B(R)$ let $\gamma_{\vec{x}, \pm}(s) \in \mathcal{S}$ be defined by

$$\gamma_{\vec{x}, \pm}(s) = \gamma_{\vec{x}}(s) \pm (1 - s) \vec{h} = s \gamma_{\vec{x}}(s) + (1 - s)(\gamma_{\vec{x}}(s) \pm \vec{h}) \in B(2R) .$$

We note that

$$\gamma_{\vec{x}, \pm}(0) = \vec{x} \pm \vec{h} , \quad \gamma_{\vec{x}, \pm}(1) = \gamma_{\vec{x}}(1) , \quad \dot{\gamma}_{\vec{x}, \pm} - \dot{\gamma}_{\vec{x}} = \mp \vec{h} .$$

Let $\Gamma_{\vec{x}, \pm} = (\phi_{\pm}(s), \gamma_{\vec{x}, \pm})$ be the null lifts of the paths $\gamma_{\vec{x}, \pm}$ with end points $\Gamma_{\vec{x}}(1)$. Let K be a compact set containing all the $\Gamma_{\vec{x}, \pm}$'s, where $\vec{x}$ and $\vec{h}$ run through $B(R)$.

Let I be any bounded interval such that $I \times B(2R)$ contains K . As before, choose σ so that $I \times B(2R)$ lies to the past of $\mathcal{S}_\sigma$, and let b be such that $\Gamma_{\vec{x}}(b) \in \mathcal{H}_\sigma$. (The value of the parameter b will of course depend upon $\vec{x}$.) Let $\Gamma_\pm$ be the null curve obtained by following $\Gamma_{\vec{x},\pm}$ for parameter values $s \in [0, 1]$, and then $\Gamma_{\vec{x}}$ for parameter values $s \in [1, b]$. Then $\Gamma_\pm \in \mathcal{P}(\vec{x} \pm \vec{h})$ so that we have

$$\mu(\vec{x} \pm \vec{h}) \leq \int_0^1 \sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x},\pm}, \dot{\gamma}_{\vec{x},\pm})} ds + \int_1^b \sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x}}, \dot{\gamma}_{\vec{x}})} ds .$$

Further

$$\mu(\vec{x}) = \int_0^b \sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x}}, \dot{\gamma}_{\vec{x}})} ds ;$$

hence

$$\begin{aligned} & \frac{\mu(\vec{x} + \vec{h}) + \mu(\vec{x} - \vec{h})}{2} - \mu(\vec{x}) \\ & \leq \int_0^1 \left(\frac{\sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x},+}, \dot{\gamma}_{\vec{x},+})} + \sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x},-}, \dot{\gamma}_{\vec{x},-})}}{2} - \sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x}}, \dot{\gamma}_{\vec{x}})} \right) ds . \end{aligned} \quad (2.10.10)$$

Since solutions of ODEs with parameters are differentiable functions of those, we can write

$$\phi_\pm(s) = \phi_{\vec{x}}(s) + \psi_i(s)h^i + r(s, h), \quad |r(s, h)| \leq C|h|_\delta^2, \quad (2.10.11)$$

for some functions ψ_i , with a constant C which is independent of $\vec{x}, \vec{h} \in B(R)$ and $s \in [0, 1]$. Inserting (2.10.11) in (2.10.10), second-order Taylor expanding the function $\sqrt{h_{\phi_\pm(s)}(\dot{\gamma}_{\vec{x},\pm}, \dot{\gamma}_{\vec{x},\pm})}(s)$ in all its arguments around $(\phi_{\vec{x}}(s), \gamma_{\vec{x}}(s), \dot{\gamma}_{\vec{x}}(s))$, and using the compactness of K we obtain

$$\frac{\mu(\vec{x} + \vec{h}) + \mu(\vec{x} - \vec{h})}{2} - \mu(\vec{x}) \leq C|\vec{h}|_\delta^2, \quad (2.10.12)$$

for some constant C . Set

$$\psi(\vec{x}) = \mu(\vec{x}) - C|\vec{x}|_\delta^2 .$$

Equation (2.10.12) shows that

$$\forall \vec{x}, \vec{h} \in B(R) \quad \psi(\vec{x}) \geq \frac{\psi(\vec{x} + \vec{h}) + \psi(\vec{x} - \vec{h})}{2} .$$

A standard argument implies that ψ is concave. It follows that

$$f(\vec{x}) + C|\vec{x}|_\delta^2 = \tau(\vec{x}) + C|\vec{x}|_\delta^2 = \sigma - \mu(\vec{x}) + C|\vec{x}|_\delta^2 = \sigma - \psi(\vec{x})$$

is convex, which is what had to be established. $\square$

2.10.2 Points of differentiability

Theorem 2.10.8 establishes the Lipschitz character of null geodesically ruled hypersurfaces. Since Lipschitz functions are differentiable almost everywhere, we conclude that such hypersurfaces have tangent planes almost everywhere. In this section we will give a sharp geometric description of points at which tangent planes exist, following [37, 119].

To put things in perspective, we note that in some works it is (implicitly or explicitly) assumed that Cauchy or event horizons have a fair amount of differentiability, except perhaps for a well-behaved lower dimensional subset thereof. For instance, in the classical proof of the area theorem for black holes [244, Proposition 9.2.7] (compare Section 3.5) something close to C^2 differentiability ‘almost everywhere’ (in a sense *not* made precise there) of the event horizon seems to have been assumed. In the classical proofs of the rigidity theorem for black holes (see [113, 114] and references therein) it is assumed that the event horizon is an analytic submanifold of spacetime. The closely related rigidity theorem of Cauchy horizons of Isenberg and Moncrief [364, 365] assumes compactness and analyticity. This should be contrasted with the following example constructed in [119].

THEOREM 2.10.12 *There exists a connected set $K \subset \mathbb{R}^2 \approx \{t = 0\} \subset \mathbb{R}^{1,2}$ with the following properties:*

1. *The boundary $\partial K = \overline{K} \setminus \text{int} K$ of K is a connected, compact, Lipschitz topological submanifold of $\mathbb{R}^2$. K is the complement of a compact set of $\mathbb{R}^2$.*
2. *There exists no open set $\Omega \subset \mathbb{R}^{1,2}$ such that $\Omega \cap \mathcal{D}_J^+(K) \cap \{0 < t < 1\}$ is a differentiable submanifold of $\mathbb{R}^{1,2}$.*

The idea of the proof of Theorem 2.10.12 is to keep adding recursively ‘small ripples’ to a circle in a way which ensures the lack of differentiability of the resulting Cauchy horizon. We will not give any details here, but we note that a key element of the analysis is the understanding of the points where the horizon is, or is not, differentiable, and this is the question addressed in this section.

It is convenient to start with some definitions: let $\mathcal{H}$ be a null geodesically ruled hypersurfaces. Following [119], a vector tangent to a generator of $\mathcal{H}$ at p will be called a *semi-tangent* to $\mathcal{H}$. Here p is possibly, but not necessarily, an end point of such a generator. We assume that the semi-tangents are oriented to the past for future horizons, and to the future for past horizons. We also normalize all semi-tangents to length one with respect to some fixed auxiliary Riemannian metric.

DEFINITION 2.10.13 *We denote by $\mathcal{N}_p(\mathcal{H})$ the set of null semi-tangents to $\mathcal{H}$ at p . The multiplicity $\#\mathcal{N}_p(\mathcal{H})$ of a point $p \in \mathcal{H}$ is the number (perhaps infinite) of generators which pass through p , or end at p .*

EXAMPLE 2.10.14 Let $\Omega \subset \mathbb{R}^2$ be an open disc of radius R centred at $\vec{0}$, and view $\mathbb{R}^2$ as the hypersurface $\{t = 0\}$ in $\mathbb{R}^{1,2}$. Then $\mathcal{H} := \partial\mathcal{D}_J^+(\Omega) = \partial\mathcal{D}_I^+(\Omega)$ is an inverted cone with vertex at $p := (R, \vec{0})$. This is a smooth manifold except at the vertex, with $\#\mathcal{N}_p(\mathcal{H}) = \infty$. $\square$

EXAMPLE 2.10.15 The following example is due to Beem and Królak [37]. Let

$$\Omega := \{y > x^2\} \subset \mathbb{R}^2 \quad (2.10.13)$$

be the open epigraph of a parabola, and view again $\mathbb{R}^2$ as the hypersurface $\{t = 0\}$ in $\mathbb{R}^{1,2}$. Then $\mathcal{H} := \partial\mathcal{D}_J^+(\Omega) = \partial\mathcal{D}_I^+(\Omega)$ is the graph $t = f(\vec{x})$ of a function f over Ω . The generators of $\mathcal{H}$ are segments of future null geodesics through $\partial\Omega$, orthogonal to $\partial\Omega$, and directed towards Ω . Symmetry considerations show that a generator, say Γ_x , issued from $(0, x, x^2) \in \partial\Omega$, with $x \neq 0$, remains on $\mathcal{H}$ until it meets the plane $\{x = 0\}$. Since the direction orthogonal to $\partial\Omega$ at (x, x^2) is spanned by

$$\vec{n}_x := \frac{1}{\sqrt{1+4x^2}}(-2x, 1),$$

we can write

$$\Gamma_x(s) = \left(s, x - \frac{2xs}{\sqrt{1+4x^2}}, x^2 + \frac{s}{\sqrt{1+4x^2}}\right). \quad (2.10.14)$$

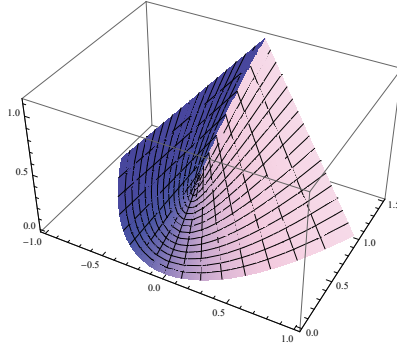


Fig. 2.10.1 The future Cauchy horizon $\partial\mathcal{D}^+(\Omega)$, with Ω given by (2.10.13).

It follows that Γ_x and Γ_{-x} both meet the plane $\{x = 0\}$ at $s = \sqrt{1 + 4x^2}/2$, at which value they leave $\mathcal{H}$. It should then be clear that the generator with initial point $(0, 0, 0)$ leaves the horizon at $(1/2, 0, 1/2)$; see Fig. 2.10.1.

We leave it as an exercise to the reader to explicitly calculate the graphing function f of $\mathcal{H}$ using (2.10.14).

Either from general considerations based on what has been said so far, or from the explicit formula, or from Fig. 2.10.1, one sees that f is smooth away from the line $\{x = 0, y \geq 1/2\}$. The end point $(1/2, 0, 1/2)$ of the generator Γ_0 of $\mathcal{H}$ has multiplicity one, while all the remaining end points $(t, 0, t^2 + 1/4)$, with $t > 1/2$, have multiplicity two. In particular the collection of end points with multiplicity two is a smooth one-dimensional submanifold of $\mathbb{R}^{1,2}$; compare Theorem 2.10.23. $\square$

We have the following.

THEOREM 2.10.16 (Beem and Królak [37]) *Let $p \in \mathcal{H}$ be such that $\#\mathcal{N}_p(\mathcal{H}) = 1$. Then $\mathcal{H}$ is differentiable at p .*

Theorem 2.10.16 follows immediately from the following more general statement.

THEOREM 2.10.17 *Let $\mathcal{H}$ be a topological hypersurface in $(\mathcal{M}, g)_{C^2}$ satisfying the following:*

1. $\mathcal{H}$ is locally achronal; i.e., for any $p \in \mathcal{H}$ there exists a neighbourhood $\mathcal{O}$ of p such that $\mathcal{H} \cap \mathcal{O}$ is achronal in the spacetime $(\mathcal{O}, g|_{\mathcal{O}})$.
2. Every point p of $\mathcal{H}$ is either an interior point of a null geodesic $\Gamma \subset \mathcal{H}$ or a future end point thereof. Such Γ 's will be called generators of $\mathcal{H}$.

Then $\mathcal{H}$ is differentiable at every point p which belongs to only one generator of $\mathcal{H}$.

Our proof of Theorem 2.10.17 follows [101], and invokes the following function-theoretic result.

LEMMA 2.10.18 *Let $\mathcal{U} \subset \mathbb{R}^n$ be an open set, suppose that $f \in C^{0,1}(\mathcal{U})$, and let $\mathcal{H}$ be the graph of f :*

$$\mathcal{H} = \{t = f(\vec{x}), \vec{x} \in \mathcal{U}\}.$$

Then $\mathcal{H}$ is differentiable at $\vec{x}_0 \in \mathcal{U}$ if and only if there exists a hypersurface $T \subset \mathbb{R} \times \mathbb{R}^n$ such that for every sequence $(f(\vec{x}_0) + \epsilon_i w_i, \vec{x}_0 + \epsilon_i \vec{v}_i) \in \mathcal{H}$ for which $\epsilon_i \rightarrow 0$, $w_i \rightarrow w$, and $\vec{v}_i \rightarrow \vec{v}$, we have $(w, \vec{v}) \in T$.

Proof. $\Rightarrow$ By a slight abuse of notation consider f to be a function on $\mathbb{R} \times \mathcal{U}$ satisfying $\partial f / \partial t = 0$, where t is the variable running along the $\mathbb{R}$ factor. Let dt be the differential of t at $(f(\vec{x}_0), \vec{x}_0)$, and let df be the differential of f at $(f(\vec{x}_0), \vec{x}_0)$; then $T = \ker(dt - df)$.

$\Leftarrow$ Let (e_0, e_i) denote the standard basis of $\mathbb{R} \times \mathbb{R}^n$ and let (f^0, f^i) be the corresponding dual basis. Consider any $\alpha \in (\mathbb{R} \times \mathbb{R}^n)^*$ such that $T = \ker \alpha$. On this basis α can be written as $\alpha_\mu f^\mu$; note that $\alpha \neq 0$ since $\text{codim } T = 1$. Let $\vec{v} \in \mathbb{R}^n$ be such that $\sum (v^i)^2 = 1$ and let ϵ_i be any sequence converging to 0; consider the sequence $(f(\vec{x}_0 + \epsilon_i \vec{v}), \vec{x}_0 + \epsilon_i \vec{v}) \rightarrow (f(\vec{x}_0), \vec{x}_0)$. Since f is Lipschitz continuous we have $|f(\vec{x}_0 + \epsilon_i \vec{v}) - f(\vec{x}_0)| \leq L\epsilon_i$ for some constant L . Compactness of $[-L, L]$ implies that there exists a subsequence ϵ_{i_j} such that

$$\left(\frac{f(\vec{x}_0 + \epsilon_{i_j} \vec{v}) - f(\vec{x}_0)}{\epsilon_{i_j}}, \vec{v} \right)$$

converges to $(v^0, \vec{v})$. By hypothesis $(v^0, \vec{v}) \in T$; thus $\alpha_0 v^0 + \alpha_i v^i = 0$. Note that $\alpha_0 v^0 = 0$ implies that $\alpha_i v^i = 0$ and, hence, $\alpha_i = 0$ by arbitrariness of v^i . It follows that we can always normalize α so that $\alpha_0 = 1$ and we obtain $v^0 = -\alpha_i v^i$. We thus have

$$\lim_{j \rightarrow \infty} \frac{f(\vec{x}_0 + \epsilon_{i_j} \vec{v}) - f(\vec{x}_0)}{\epsilon_{i_j}} = -\alpha_i v^i. \quad (2.10.15)$$

As the right-hand side of (2.10.15) does not depend upon the sequence ϵ_i , we must actually have

$$\lim_{\epsilon \rightarrow 0} \frac{f(\vec{x}_0 + \epsilon \vec{v}) - f(\vec{x}_0)}{\epsilon} = -\alpha_i v^i.$$

This can be rewritten as

$$f(\vec{x}_0 + \epsilon \vec{v}) = f(\vec{x}_0) - \epsilon \alpha_i v^i + o(\epsilon),$$

which is what had to be established. $\square$

We will also need the following.

PROPOSITION 2.10.19 *Let $\mathcal{H}$ be as in Theorem 2.10.17. If p is an interior point of a null geodesic Γ such that the image of Γ lies in $\mathcal{H}$ near p , then Γ is a generator of $\mathcal{H}$, and there is then only one generator of $\mathcal{H}$ at p .*

Proof. Let Γ_1 be a generator of $\mathcal{H}$ starting at, or passing through p . We orient Γ consistently with Γ_1 (thus to the past). We write $\Gamma : I \rightarrow \mathcal{H}$, with the interval I containing $(-\epsilon, \epsilon)$ for some $\epsilon > 0$, and with $\Gamma(0) = p$. Suppose that Γ does not coincide with Γ_1 near p . Then the curve obtained by following Γ from $\Gamma(-\epsilon)$ to p , and then following Γ_1 , is not a null geodesic, and therefore can be deformed to a timelike curve by Proposition 2.4.17, p. 43. This is not possible for achronal $\mathcal{H}$'s. $\square$

REMARK 2.10.20 Proposition 2.10.19 shows that $\#\mathcal{N}_p(\mathcal{H}) = 1$ for interior points of generators. This, together with Theorem 2.10.17, establishes the differentiability of $\mathcal{H}$ at interior points of generators. $\square$

Proof of Theorem 2.10.17. Let $p_i \in \mathcal{H}$ be any sequence such that $p_i \rightarrow p$, and let $\gamma_i \in \mathcal{N}_{p_i}(\mathcal{H})$ converge to $\gamma \in T_p \mathcal{M}$. Then $\gamma \in \mathcal{N}_p(\mathcal{H})$, as the following shows.

LEMMA 2.10.21 $\cup_{p \in \mathcal{H}} \mathcal{N}_p(\mathcal{H})$ is a closed subset of $\cup_{p \in \mathcal{H}} T_p \mathcal{M}$.

Proof. By definition of a topological hypersurface there exists an open set $\mathcal{U} \subset \mathcal{M}$ such that $\mathcal{H} \cap \mathcal{U}$ is closed. Now, there exists an open subset $\mathcal{V}$ of $\mathcal{U}$ and $\epsilon > 0$ such that the exponential map is defined for all vectors $\cup_{p \in \mathcal{V}} T_p \mathcal{M}$ which have h -length smaller than ϵ , and for such vectors has an image in $\mathcal{U}$. Let $\gamma_i \in \mathcal{N}_{p_i}(\mathcal{H})$, $p_i \in \mathcal{V}$, converge to $\gamma \in T_p \mathcal{M}$. Let $s \mapsto \Gamma_i(s)$ be the corresponding generators of $\mathcal{H}$, with $\dot{\Gamma}_i(0) = \gamma_i$; then the points $\Gamma_i(\epsilon) \in \mathcal{U} \cap \mathcal{H}$ converge to $\Gamma(\epsilon) \in \mathcal{U}$, where Γ is the null geodesic segment through p , maximally extended to the past, such that $\dot{\Gamma}(0) = \gamma$.

Since $\mathcal{H} \cap \mathcal{U}$ is closed, $\Gamma(\epsilon) \in \mathcal{H}$. Hence $\Gamma(s)$ remains on $\mathcal{H}$ at least for $s > 0$ small enough. By Proposition 2.10.19 the geodesic segment Γ coincides near $\Gamma(s)$ with the generator of $\mathcal{H}$ through $\Gamma(s)$. But then the image of Γ is included in $\mathcal{H}$ by definition of $\mathcal{H}$. Thus Γ is also a generator of $\mathcal{H}$ at p . $\square$

Returning to the proof of Theorem 2.10.17, in normal coordinates centred at p we can write

$$p_i = p + d_i v_i, \quad 0 \leq d_i \rightarrow 0,$$

where the length of the vectors v_i has been normalized to 1 using some auxiliary Riemannian metric h . For $\epsilon, \epsilon_i \geq 0$ let $p_i(\epsilon_i) \in \mathcal{H}$, respectively $p(\epsilon) \in \mathcal{H}$, denote the point lying an affine distance ϵ_i , respectively ϵ , on the null generator of $\mathcal{H}$ with semi-tangent γ_i , respectively γ . From $\gamma_i \rightarrow \gamma$ we have $\gamma_i - \gamma = o(1)$, and from the fact that, in normal coordinates, null geodesics through $p + d_i v_i$ at affine distance ϵ_i differ from straight lines by terms which are $o(d_i + \epsilon_i)$ we obtain

$$\begin{aligned} p_i(\epsilon_i) &= p + d_i v_i + \epsilon_i \gamma_i + o(d_i + \epsilon_i), \\ p(\epsilon) &= p + \epsilon \gamma. \end{aligned}$$

Let η be the Minkowski metric; consider the quantity

$$\begin{aligned} A &= \eta(p_i(\epsilon_i) - p, p_i(\epsilon_i) - p) \\ &= d_i^2 \eta(v_i, v_i) + 2d_i \epsilon_i \eta(v_i, \gamma) + o((d_i + \epsilon_i)^2). \end{aligned} \quad (2.10.16)$$

Suppose that $v_i \rightarrow v$, and suppose, first, that $\eta(v, v) < 0$. Equation (2.10.16) with $\epsilon_i = 0$ gives $A < 0$ for i large enough. It follows that the coordinate line through p and $p_i(0) = p_i$ is timelike, which contradicts the achronality of $\mathcal{H}$. Hence

$$\eta(v, v) \geq 0. \quad (2.10.17)$$

Suppose, next, that $\eta(v, v) > 0$ and $\eta(v, \gamma) < 0$. In that case Eq. (2.10.16) with $\epsilon_i = \frac{\eta(v, v)}{|\eta(v, \gamma)|} d_i$ gives $A < 0$. It follows that the coordinate line through p and $p_i(\epsilon_i)$ is timelike which is again impossible, so that

$$\eta(v, \gamma) \geq 0. \quad (2.10.18)$$

If $\eta(v, v) = 0$, Eq. (2.10.16) with $\epsilon_i = d_i$ leads similarly to (2.10.18).

To show that the inequality (2.10.18) must be an equality, consider the coordinate lines starting at $p + v_i$ and ending at $p + \epsilon_i \gamma$:

$$[0, 1] \ni s \rightarrow \Gamma_i(s) = p + (1 - s)d_i v_i + s\epsilon_i \gamma.$$

On $\Gamma_i(s)$ we have

$$\begin{aligned} g(v_i, v_i) &= g(v, v) + o(1) = \eta(v, v) + o(1), \\ g(\gamma, v_i) &= g(\gamma, v) + o(1) = \eta(\gamma, v) + o(1), \\ g(\gamma, \gamma) &= g(\gamma, \gamma) + o(1) = o(1), \end{aligned}$$

which implies that

$$g\left(\frac{d\Gamma_i}{ds}, \frac{d\Gamma_i}{ds}\right) = d_i^2 \eta(v, v) - 2\epsilon_i d_i \eta(v, \gamma) + o((d_i + \epsilon_i)^2). \quad (2.10.19)$$

Note that Eq. (2.10.19) differs from Eq. (2.10.16) only by the sign of the $\eta(v_i, \gamma)$ terms, so that a similar analysis shows that Γ_i will be timelike for i large enough unless

$$\eta(v, \gamma) = 0.$$

It follows that $v \in T := \gamma^\perp$. The local achronality of $\mathcal{H}$ implies that $\mathcal{H}$ is Lipschitz, and differentiability of $\mathcal{H}$ at p follows now from Lemma 2.10.18. $\square$

Let us denote by $\mathcal{H}_{\text{end}}$ the set of end points of generators of $\mathcal{H}$; for simplicity we will refer to those points as *end points*. In some applications it is useful to know how ‘large’ this set can be. It follows from Proposition 2.10.19 that if $\#\mathcal{N}_p(\mathcal{H}) > 1$, then p is necessarily the end point of all relevant generators. It further follows from Theorem 2.10.17 that the set of points with multiplicity $\#\mathcal{N}_p(\mathcal{H}) > 1$ coincides with the set where the horizon fails to be differentiable. Now, because achronal hypersurfaces are Lipschitz, the set of points at which a horizon is non-differentiable has vanishing n -dimensional Hausdorff measure in spacetime dimension $n + 1$, which gives one some control over the size of the set of end points with multiplicity $\#\mathcal{N}_p(\mathcal{H}) > 1$. Thus, in order to control the dimension of $\mathcal{H}_{\text{end}}$ it remains to estimate that of the set of end points with multiplicity $\#\mathcal{N}_p(\mathcal{H}) = 1$. Let us denote by $\mathcal{H}_{\text{diff}}$ the set of points of $\mathcal{H}$ at which $\mathcal{H}$ is differentiable; what has been said shows that the set of end points with multiplicity $\#\mathcal{N}_p(\mathcal{H}) = 1$ coincides with the set $\mathcal{H}_{\text{end}} \cap \mathcal{H}_{\text{diff}}$. (An example where the set $\mathcal{H}_{\text{end}} \cap \mathcal{H}_{\text{diff}}$ is not empty is given in Example 2.10.15.) In [118] the reader will find a proof of the following.

THEOREM 2.10.22 *Let $\mathcal{H}$ be an achronal, closed, future null geodesically ruled topological hypersurface in an $(n + 1)$ -dimensional spacetime $(\mathcal{M}, g)$. Then the set $\mathcal{H}_{\text{end}} \cap \mathcal{H}_{\text{diff}}$ has a vanishing n -dimensional Hausdorff measure. Moreover, for any C^2 spacelike hypersurface $\mathcal{S}$ the set $\mathcal{H}_{\text{end}} \cap \mathcal{H}_{\text{diff}} \cap \mathcal{S}$ has a vanishing $(n - 1)$ -dimensional Hausdorff measure.*

The set of points where the multiplicity $\#\mathcal{N}_p(\mathcal{H})$ is large has a more precise structure. Define

$$\mathcal{C}_p := \text{convex cone generated by } \mathcal{N}_p(\mathcal{H});$$

thus $\mathcal{C}_p \subset T_p\mathcal{M}$. We can measure the size of the set of generators through p by $\dim(\mathcal{C}_p)$, defined as the dimension of the linear span of $\mathcal{C}_p$ in T_pM . This is a different measure than is $\#\mathcal{N}_p(\mathcal{H})$; in particular this gives finer information when $\#\mathcal{N}_p(\mathcal{H}) = \infty$. We also set

$$\mathcal{H}[k] := \{p : \dim(\mathcal{C}_p) \geq k\}. \quad (2.10.20)$$

We have $\mathcal{H}[1] = \mathcal{H}$, as every point is on at least one generator.

For $k = 2$, $\mathcal{H}[k]$ is the set of points of $\mathcal{H}$ that are on more than one generator. As $\dim(\mathcal{C}_p)$ is the dimension of the span of $\mathcal{N}_p(\mathcal{H})$ and $\mathcal{N}_p(\mathcal{H})$ contains $\#\mathcal{N}_p(\mathcal{H})$ vectors, $\dim(\mathcal{C}_p) \leq \#\mathcal{N}_p(\mathcal{H})$. This implies that

$$\mathcal{H}[k] \subseteq \{p \in \mathcal{H} : \#\mathcal{N}_p(\mathcal{H}) \geq k\}.$$

Note that for $1 \leq k \leq 3$, any k distinct elements of $\mathcal{N}_p(\mathcal{H})$ are linearly independent. Indeed, the linear span of two future-pointing h -unit null vectors is a two-dimensional timelike subspace and there are only two future-pointing null rays in this subspace. So a third future-pointing h -unit vector cannot be in the span of the first two.

Therefore if $1 \leq k \leq 3$ and $\#\mathcal{N}_p(\mathcal{H}) \geq k$, then choosing k distinct, and thus linearly independent, elements of $\mathcal{N}_p(\mathcal{H})$ shows that $\dim(\mathcal{C}_p) \geq k$. Whence

$$\mathcal{H}[k] = \{p \in \mathcal{H} : \#\mathcal{N}_p(\mathcal{H}) \geq k\} \quad \text{for } 1 \leq k \leq 3.$$

One of the main results of [118], based on the deep results in [9], is that the sets $\mathcal{H}[k]$ are ‘almost C^2 submanifolds of dimension $n + 1 - k$, up to singular sets of lower dimension’. To make this statement precise, let $\mathfrak{H}^m$ be the m -dimensional Hausdorff measure on $\mathcal{M}$ defined using some auxiliary Riemannian metric h . Recall [9, Def. 1.1 p.19] that a Borel set $\Sigma \subset M$ is said to be ‘ C^2 rectifiable of dimension m ’ iff Σ can be covered, up to a set of a vanishing $\mathfrak{H}^m$ measure, by a countable collection of m -dimensional C^2 submanifolds of $\mathcal{M}$. This definition is independent of the choice of the Riemannian metric h . We then have the following.

THEOREM 2.10.23 ([116]) For $1 \leq k \leq n+1$ the set $\mathcal{H}[k]$ is a C^2 rectifiable set of dimension $n+1-k$. Therefore $\mathcal{H}[k]$ has Hausdorff dimension $\leq n+1-k$.

Using $k=1$, and that $\mathcal{H}[1] = \mathcal{H}$, this implies that horizons are C^2 rectifiable of dimension n . As they are also locally Lipschitz graphs they have the further property that $\mathfrak{H}^n(\mathcal{H} \cap K) < \infty$ for all compact sets $K \subseteq M$. When $k=n+1$ this implies that $\mathcal{H}[n+1]$ is a countable set.

Both in Example 2.10.15 and in an example analysed in [268], the set $\{\#\mathcal{N}_p(\mathcal{H}) = 2\}$ is (up to a lower dimensional set in [268]) a smooth submanifold of codimension two.

2.10.3 Alexandrov differentiability

A key object associated with a black-hole region $\mathcal{B}$ (whether defined as in Section 3.1 or Section 4.3.8) is that of an *event horizon*,

$$\mathcal{H}^+ := \partial\mathcal{B}. \quad (2.10.21)$$

(Actually, when $\mathcal{H}^+$ is not connected some care is required for a useful definition, but this does not affect the discussion that follows.) Event horizons are a special case of a family of objects called *future horizons*: by definition, these are closed topological hypersurfaces $\mathcal{H}$ threaded by null geodesics, called *generators*, with no *future end* points, and possibly with *past end* points. At the latter, differentiability of $\mathcal{H}$ breaks down in general; a necessary and sufficient condition for this breakdown has been given in Theorem 2.10.16.

Many authors have taken for granted that horizons are nice, piecewise smooth hypersurfaces. This is, however, not the case: it is simple to construct *nowhere* C^1 black-hole horizons using the set K of Theorem 2.10.12, p. 73. In fact, nowhere C^1 horizons are generic in the class of convex horizons in Minkowski spacetime [70]. This leads to various difficulties when trying to study their structure, e.g., attempting to prove results such as the area theorem discussed in Section 3.5.

Corollary 2.10.11 shows that various horizons, including event horizons, are *locally Lipschitz-continuous* topological hypersurfaces; they are locally graphs of functions satisfying

$$|\phi(x) - \phi(y)| \leq C|x - y|.$$

One of the key objects characterizing a twice-differentiable future horizon $\mathcal{H}$ is its expansion θ , cf. (F.8), p. 357. The function θ plays a key role in many considerations, including the area theorem and the incompleteness theorems. It turns out that, despite the potential low differentiability of $\mathcal{H}$, a useful equivalent of θ can be defined almost everywhere in an *Alexandrov sense*. For this, recall that we have seen in Theorem 2.10.9 that future horizons are semi-convex. A deep theorem of Alexandrov allows one to define second derivatives almost everywhere.

THEOREM 2.10.24 (Alexandrov [193, Appendix E]) *Semi-convex functions* $f : B \rightarrow \mathbb{R}$ are ‘twice differentiable’ almost everywhere in the following sense: there exists a set $B_{\mathcal{A}l}$ with full measure in B such that

$$\begin{aligned} \forall x \in B_{\mathcal{A}l} \exists Q \in (\mathbb{R}^p)^* \otimes (\mathbb{R}^p)^* \text{ such that } \forall y \in B \\ f(y) - f(x) - df(x)(y-x) = Q(x-y, x-y) + r_2(x, y), \end{aligned}$$

with $r_2(x, y) = o(|x-y|^2)$. The symmetric quadratic form Q above will be denoted by $\frac{1}{2}D^2f(x)$, and will be called the second Alexandrov derivative of f at x .

Points in $B_{\mathcal{A}l}$ will be referred to as the *Alexandrov points* of B .

One can now use Theorem 2.10.24 to define the equivalent to the usual divergence θ of the horizon, by writing $\mathcal{H}$ locally as the graph of a function f , and using

the second Alexandrov derivatives of f to define (almost everywhere on $\mathcal{H}$) the divergence θ_{Al} of the horizon. More precisely, let $p = (t = f(q), q)$ be an Alexandrov point of $\mathcal{H}$, one sets

$$\begin{aligned} L_\mu(p)dx^\mu &:= -dt + df(q), \quad \partial_\mu L_t(p) = \partial_t L_\mu(p) := 0, \quad \partial_i L_j(p) := D_{ij}^2 f(q), \\ \nabla_\mu L_\nu(p) &:= \partial_\mu L_\nu(p) - \Gamma^\alpha_{\mu\nu} L_\alpha(p), \end{aligned} \quad (2.10.22)$$

where the Hessian matrix $D_{ij}^2 f$ above is calculated using the second Alexandrov derivatives of f . One can now insert (2.10.22) in the usual definition (F.9), p. 358 of θ .

Clearly, the definition coincides with the standard definition of θ when $f \in C^2$. Several results of causality theory go through with θ replaced by θ_{Al} , though the standard arguments sometimes have to be replaced by completely different ones; the reader is referred to [117] for details. In particular the Raychaudhuri equation (F.20), p. 360, as well as the remaining optical equations, hold on almost all generators of $\mathcal{H}$.

2.11 Cauchy surfaces, time functions

A topological hypersurface $\mathcal{S}$ is said to be a *Cauchy surface* if

$$\mathcal{D}_J(\mathcal{S}) = \mathcal{M}.$$

(Note that it does not matter whether $\mathcal{D}_J(\mathcal{S})$ or $\mathcal{D}_I(\mathcal{S})$ is chosen in the definition when the metric is C^2 .) Theorem 2.9.9, p. 63, shows that a necessary condition for this equality is that $\mathcal{M}$ be globally hyperbolic. A celebrated theorem, with independent proofs due to Geroch and Seifert, shows that this condition is also sufficient.

THEOREM 2.11.1 (Geroch [219], Seifert [427]) *A spacetime $(\mathcal{M}, g)_{C^2}$ is globally hyperbolic if and only if there exists on $\mathcal{M}$ a continuous time function τ ranging over $\mathbb{R}^+$ with the property that all its level sets are Cauchy surfaces.*

Proof. The proof uses volume functions, defined as follows: let φ_i , $i \in \mathbb{N}$, be any partition of unity on $\mathcal{M}$, set

$$V_i := \int_{\mathcal{M}} \varphi_i d\mu,$$

where $d\mu$ is, say, the Riemannian measure associated to the auxiliary Riemannian metric h on $\mathcal{M}$. Define

$$\nu := \sum_{i \in \mathbb{N}} \frac{1}{2^i V_i} \varphi_i.$$

Then ν is smooth, positive, nowhere vanishing, with

$$\int_{\mathcal{M}} \nu d\mu = 1.$$

Following Geroch, we define

$$V_\pm(p) := \int_{J^\pm(p)} \nu d\mu.$$

We clearly have

$$\forall p \in \mathcal{M} \quad 0 < V_\pm(p) < 1.$$

The functions $V_\pm$ may fail to be continuous in general, an example is given in Figure 2.11.1. It turns out that such behaviour cannot occur under the current conditions.

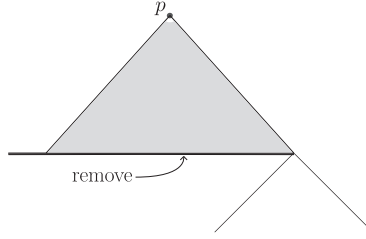


Fig. 2.11.1 The volume function V_- is discontinuous at p .

LEMMA 2.11.2 *On C^2 globally hyperbolic spacetimes the functions $V_{\pm}$ are continuous.*

Proof. Lemma 2.11.2 is an immediate consequence of Proposition 2.8.2, p. 57, which guarantees that in a globally hyperbolic spacetime the sets $J^{\pm}(p)$ are closed for all p , and from the following general fact. $\square$

LEMMA 2.11.3 *Let $(\mathcal{M}, g)_{C^2}$ have the property that the sets $J^{\pm}(p)$ are closed for all p . Then the functions $V_{\pm}$ are continuous.*

Proof. Let p_i be any sequence converging to p , and let the symbol φ_{Ω} denote the characteristic function of a set Ω . Let q be any point such that $q \in I^-(p) \Leftrightarrow p \in I^+(q)$. Since $I^+(q)$ forms a neighbourhood of p we have $p_i \in I^+(q) \Leftrightarrow q \in I^-(p_i)$ for i large enough. Equivalently,

$$\forall i \geq i_0 \quad \varphi_{I^-(p_i)}(q) = 1 = \varphi_{I^-(p)}(q). \quad (2.11.1)$$

Since the right-hand side of (2.11.1) is zero for $q \notin I^-(p)$ we obtain

$$\forall q \quad \liminf_{i \rightarrow \infty} \varphi_{I^-(p_i)}(q) \geq \varphi_{I^-(p)}(q). \quad (2.11.2)$$

By Corollary 2.8.7, p. 58, $J^-(p)$ differs from $I^-(p)$ by a topological hypersurface so that

$$\liminf_{i \rightarrow \infty} \varphi_{J^-(p_i)} \geq \varphi_{J^-(p)} \quad \text{a.e.} \quad (2.11.3)$$

To obtain the inverse inequality for $\limsup$, let q be such that

$$\limsup_{i \rightarrow \infty} \varphi_{J^-(p_i)}(q) = 1;$$

hence there exists a subsequence p_{i_j} such that $p_{i_j} \in J^+(q)$. Since $J^+(q)$ is closed by hypothesis, $p \in J^+(q)$, then also $q \in J^-(p)$ and so $\varphi_{J^-(p)}(q) = 1$.

This establishes the implication that

$$\limsup_{i \rightarrow \infty} \varphi_{J^-(p_i)}(q) = 1 \implies \varphi_{J^-(p)}(q) = 1.$$

Since the $\limsup$ at the left-hand side of the implication above can only take values zero or one, it follows that

$$\limsup_{i \rightarrow \infty} \varphi_{J^-(p_i)} \leq \varphi_{J^-(p)}. \quad (2.11.4)$$

Equations (2.11.3)–(2.11.4) show that

$$\limsup_{i \rightarrow \infty} \varphi_{J^-(p_i)} \leq \varphi_{J^-(p)} \leq \liminf_{i \rightarrow \infty} \varphi_{J^-(p_i)} \quad \text{a.e.} \quad (2.11.5)$$

Hence

$\lim_{i \rightarrow \infty} \varphi_{J^-(p_i)}$ exists a.e., and equals $\varphi_{J^-(p)}$ a.e.

Since

$$0 \leq \varphi_{J^-(p)} \leq 1 \in \mathcal{L}^1(\nu d\mu),$$

the Lebesgue-dominated convergence theorem gives

$$V_-(p) = \int_{\mathcal{M}} \varphi_{J^-(p)} \nu d\mu = \lim_{i \rightarrow \infty} \int_{\mathcal{M}} \varphi_{J^-(p_i)} \nu d\mu = \lim_{i \rightarrow \infty} V_-(p_i).$$

Changing time orientation one also obtains the continuity of V_+ . $\square$

We continue with the following observation.

LEMMA 2.11.4 *V_- tends to zero along any past-inextendible causal curve $\gamma : [a, b) \rightarrow \mathcal{M}$.*

Proof. Let X_i be any partition of $\mathcal{M}$ by sets with compact closure, the dominated convergence theorem shows that

$$\lim_{k \rightarrow \infty} \sum_{i \geq k} \int_{X_i} \nu d\mu = 0. \quad (2.11.6)$$

Suppose that there exists $k < \infty$ such that

$$\forall s \quad J^-(\gamma(s)) \cap \left(\bigcup_{i=1}^k X_i \right) \neq \emptyset.$$

Equivalently, there exists a sequence $s_i \rightarrow b$ such that

$$\gamma(s_i) \in K := \overline{\bigcup_{i=1}^k X_i}.$$

Compactness of K implies that there exists (passing to a subsequence if necessary) a point $q_\infty \in K$ such that $\gamma(s_i) \rightarrow q_\infty$. Strong causality of $\mathcal{M}$ implies that there exists an elementary neighbourhood $\mathcal{O}$ of q_∞ such that $\gamma \cap \mathcal{O}$ is connected, and Lemma 2.6.6 shows that $\gamma \cap \mathcal{O}$ has finite h -length, which contradicts inextendibility of γ (compare Theorem 2.5.5). This implies that for any k we have

$$J^-(\gamma(s)) \cap \left(\bigcup_{i=1}^k X_i \right) = \emptyset$$

for s large enough, say $s \geq s_k$. In particular

$$s \geq s_k \implies \int_{J^-(\gamma(s)) \cap \left(\bigcup_{i=1}^k X_i \right)} \nu d\mu = 0.$$

This implies that

$$\forall s \geq s_k \quad V_-(\gamma(s)) = \int_{J^-(\gamma(s)) \cap \left(\bigcup_{i=k+1}^\infty X_i \right)} \nu d\mu \leq \sum_{i \geq k+1} \int_{X_i} \nu d\mu,$$

which, in view of (2.11.6), can be made as small as desired by choosing k sufficiently large. $\square$

We are ready now to pass to the proof of Theorem 2.11.1. Set

$$\tau := \frac{V_-}{V_+}.$$

Then τ is continuous by Lemma 2.11.2. Let $\gamma : (a, b) \rightarrow \mathcal{M}$ be any inextendible future-directed causal curve. By Lemma 2.11.4

$$\lim_{s \rightarrow b} \tau(\gamma(s)) = \infty, \quad \lim_{s \rightarrow a} \tau(\gamma(s)) = 0.$$

Thus τ runs from 0 to ∞ on any such curves; in particular γ intersects every level set of τ at least once. From the definition of the measure $\nu d\mu$ and from Corollary 2.8.15,

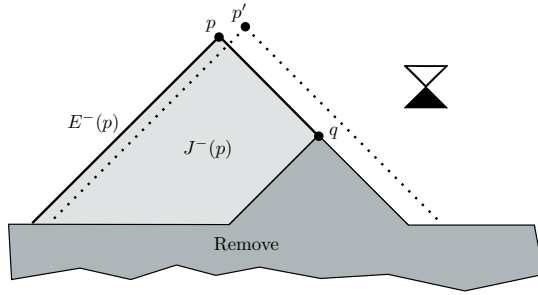


Fig. 2.11.2 The volume of past light cones is Lipschitz-continuous but not differentiable at $J^+(q)$. Reprinted from [126] with permission.

p. 60, the function τ is actually strictly increasing on any causal curve; hence, the level sets of τ are met by causal curves precisely once. $\square$

The question arises whether the volume time functions are differentiable. In [150] the following has been proved.

THEOREM 2.11.5 *Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime with a $C^{2,1}$ metric. There exists a class of smooth functions $\varphi > 0$ such that the functions*

$$\tau_\varphi^\pm(p) = \int_{J^\pm(p)} \varphi d\mu_g, \quad (2.11.7)$$

where $d\mu_g$ is the volume element of g , are continuously differentiable with timelike gradient.

We will only provide a sketch of the proof. One starts by writing a light cone integral formula for a candidate derivative of $\tau_\varphi^\pm$. The integrand involves Jacobi fields which might be blowing up as one approaches the end of the interval of existence of the geodesic generators of

$$E^\pm(p) := J^\pm(p) \setminus I^\pm(p).$$

So the weighting function φ must compensate for this, which provides one of the constraints on the set of admissible functions φ .

Next, the domain of integration that arises in the derivative formula is generated by lightlike geodesics that may be either complete or incomplete. The difficulties associate with this are avoided by conformally rescaling g so as to make all the null geodesics complete [35].

Finally, there might exist generators that do not meet the cut-locus and which span an area region on $E^\pm(p)$ that does *not* vary continuously with p . An illustration of this behaviour can be found in Example 2.11.6. The discontinuous behaviour illustrated by the example happens ‘close to infinity’ in a complete auxiliary Riemannian metric, where a fast fall-off of φ amends the problem.

EXAMPLE 2.11.6 Let (N, g) be any bounded globally hyperbolic subset in 2-dimensional Minkowski spacetime, with the weighting function φ in (2.11.7) equal to 1. Let $q \in N$ and set $\mathcal{M} = N \setminus J^-(q)$, with the induced metric. Then τ_φ^- is not differentiable at the boundary of the future of q ; see Figure 2.11.2. $\square$

Theorem 2.11.5 is not the end of the story, because one would like to have smooth time functions. Now, both the function τ of Theorem 2.11.1 and the functions $\tau_\varphi^\pm$ of Theorem 2.11.5 have the following property: let us parameterize all curves by h -length, where h is an auxiliary smooth complete Riemannian metric on $\mathcal{M}$. By inspection of light cones in local coordinates it is easily seen that on every compact subset K of $\mathcal{M}$ there exists a constant c_K such that we have, for any differentiable

future-directed causal curve γ , with $\gamma([s_1, s_2]) \subset K$, a locally uniform anti-Lipschitz bound:

$$\tau(\gamma(s_2)) - \tau(\gamma(s_1)) \geq c_K(s_2 - s_1). \quad (2.11.8)$$

The point is that such functions have good smoothability properties. Indeed, [126] shows the following.

THEOREM 2.11.7 *Let t be a continuous function on $(\mathcal{M}, g)_{C^2}$ satisfying, on every compact set K , an anti-Lipschitz condition along h -length-parameterized, future-directed, causal curves γ :*

$$t(\gamma(s_2)) - t(\gamma(s_1)) \geq C(K)(s_2 - s_1), \quad (2.11.9)$$

for $s_2 > s_1$, with some constant $C(K) > 0$. Then there exists on $\mathcal{M}$ a sequence of smooth time functions t_i which, as i tends to infinity, converge to t , uniformly on every compact set. $\square$

INCIDENTALLY: An alternative proof of the existence of smooth time functions has been provided in [48]; see also [49, 50]. $\square$

We are ready to prove now the following.

THEOREM 2.11.8 *A globally hyperbolic spacetime $(\mathcal{M}, g)_{C^2}$ is diffeomorphic to $\mathbb{R} \times \mathcal{S}$, with the coordinate along the $\mathbb{R}$ factor having timelike gradient.*

Proof. Choose φ so that the functions $\tau_\varphi^\pm$ of Theorem 2.11.5 are C^1 with timelike gradient. Set

$$t := \ln \left(\frac{\tau_\varphi^+}{\tau_\varphi^-} \right).$$

Let us denote by τ any of the smoothings of t provided by Theorem 2.11.7. Let X by any smooth timelike vector field on X , e.g. $X = \nabla \tau$, but any other timelike vector field will do.

Choose any number $\tau_0 > 0$. For $p \in \mathcal{M}$ let $q(p)$ be the point on the level set $\mathcal{S}_0 = \{r \in \mathcal{M} : \tau(r) = \tau_0\}$ which lies on the integral curve of X through p . Such a point exists because any inextendible timelike curve in $\mathcal{M}$ meets $\mathcal{S}_0$; it is unique by the achronality of $\mathcal{S}_0$. Set

$$\phi(p) := (t(p), q(p)).$$

This is clearly a bijection $\phi : \mathcal{M} \rightarrow \mathbb{R} \times \mathcal{S}$. The map ϕ is continuous by continuous dependence of ODEs upon initial values. Since τ is differentiable, its level sets are differentiable, X meets those level sets transversely, and the fact that φ is a diffeomorphism follows from the implicit function theorem. $\square$

INCIDENTALLY: Remark that if the function τ in the last proof were merely continuous, one could invoke the invariance of domain theorem⁷ to conclude that the map ϕ is a homeomorphism.

We close this section by a collection of remarks on related issues.

It is not easy to decide whether a hypersurface $\mathcal{S}$ is a Cauchy hypersurface, except in *spatially compact* spacetimes.

THEOREM 2.11.9 (Budić et al. [69], Galloway [202]) *Let $(\mathcal{M}, g)$ be a smooth globally hyperbolic spacetime and suppose that $\mathcal{M}$ contains a smooth, compact, connected spacelike hypersurface $\mathcal{S}$. Then $\mathcal{S}$ is a Cauchy surface for $\mathcal{M}$.*

REMARK 2.11.10 Some further results concerning Cauchy surface criteria can be found in [202, 239]. $\square$

⁷See, e.g., http://www.encyclopediaofmath.org/index.php?title=Domain_invariance

We also note the following ‘sandwich result’ of Galloway, which does not assume compactness or fall-off conditions.

PROPOSITION 2.11.11 (Galloway [202]) *Let $\mathcal{S}_a$, $a = 1, 2$, be two Cauchy surfaces in $(\mathcal{M}, g)_{C^2}$. Let $\mathcal{S}$ be a connected closed spacelike hypersurface (without boundary) in $\mathcal{M}$ and suppose that*

$$\mathcal{S} \subset I^+(\mathcal{S}_1), \quad \mathcal{S}_2 \subset I^+(\mathcal{S}). \quad (2.11.10)$$

Then $\mathcal{S}$ is a Cauchy surface. □

Bernal and Sánchez have established the following *extension result*.

THEOREM 2.11.12 (Bernal and Sánchez [50]) *Let H be a m -dimensional, acausal, compact submanifold with boundary in a globally hyperbolic spacetime $(\mathcal{M}, g)_{C^2}$. Then there exists a Cauchy hypersurface $\mathcal{S}$ in $\mathcal{M}$ such that*

$$H \subset \mathcal{S}. \quad \square$$

Let g and $\hat{g}$ be two Lorentzian metrics. We will say that the light-cones of $\hat{g}$ are *strictly larger* than those of g , or that those of g are *strictly smaller* than those of $\hat{g}$, and we shall write

$$\hat{g} \succ g, \quad (2.11.11)$$

if it is true that

$$g(T, T) \leq 0, \quad T \neq 0 \implies \hat{g}(T, T) < 0. \quad (2.11.12)$$

We note, again without proof, the following result.

THEOREM 2.11.13 (Navarro and Minguzzi [45]) *Let $(\mathcal{M}, g)_{C^2}$ be globally hyperbolic. Then there exists a smooth metric $\hat{g} \succ g$ so that $(\mathcal{M}, \hat{g})$ is globally hyperbolic.* □

3

Some applications

Any formalism is only useful if it leads to interesting applications. The aim of this chapter is to present some of these, as relevant to black-hole spacetimes. For this we need to introduce the concept of conformal completions, which is done in Section 3.1. We continue, in Section 3.2, with a review of the null splitting theorems of Galloway. Section 3.3 has a somewhat different character than the remaining sections in this chapter: we give there complete proofs of a few versions of the *topological censorship theorems*, which are otherwise scattered across the literature, and which play a basic role in the understanding of the topology of black holes. In Section 3.4 we review the basic *incompleteness theorems*, also known under the misleading name of *singularity theorems*. Section 3.5 is devoted to the presentation of a few versions of the *area theorem*, which is a cornerstone of ‘black-hole thermodynamics’. We close this chapter with a short discussion of the role played by causality theory in the analysis of wave equations (Section 3.6).

3.1 Conformal completions

The starting point of many studies of global properties of spacetimes are the conformal completions at infinity introduced by Penrose [395].

DEFINITION 3.1.1 *A pair $(\tilde{\mathcal{M}}, \tilde{g})$ will be called a conformal completion at infinity of $(\mathcal{M}, g)$ if $\tilde{\mathcal{M}}$ is a manifold with boundary such that:*

1. $\mathcal{M}$ is the interior of $\tilde{\mathcal{M}}$,
2. on $\tilde{\mathcal{M}}$ there exists a function Ω with the property that the metric $\tilde{g}$, defined as $\Omega^2 g$ on $\mathcal{M}$, extends by continuity to the boundary of $\tilde{\mathcal{M}}$, with the extended metric maintaining its signature on the boundary,
3. Ω is (strictly) positive on $\mathcal{M}$, differentiable on $\tilde{\mathcal{M}}$, vanishes precisely on

$$\mathcal{I} := \partial \tilde{\mathcal{M}},$$

with $d\Omega$ nowhere vanishing on $\mathcal{I}$.

One sets

$$\mathcal{I}^+ := \mathcal{I} \cap J^+(\mathcal{M}), \quad \mathcal{I}^- := \mathcal{I} \cap J^-(\mathcal{M}). \quad (3.1.1)$$

In the context of Definition 3.1.1, the *domain of outer communications* $\langle\langle \mathcal{I} \rangle\rangle$ is defined as

$$\langle\langle \mathcal{I} \rangle\rangle := I^+(\mathcal{I}^-) \cap I^-(\mathcal{I}^+). \quad (3.1.2)$$

(An alternative definition of domain of outer communications, more convenient to work with but requiring stationarity, and equivalent to this one under natural conditions, will be introduced in Section 4.3.8, p. 166.)

EXAMPLE 3.1.2 Minkowski spacetime provides the simplest example of this construction. For this, on $\mathbb{R}^{n+1} \setminus \{r = 0\}$ introduce coordinates $u = t - r$, $x = 1/r$, x^A , with (r, x^A) being the usual spherical coordinates on $\mathbb{R}^n$. In these coordinates the Minkowski metric η takes the form

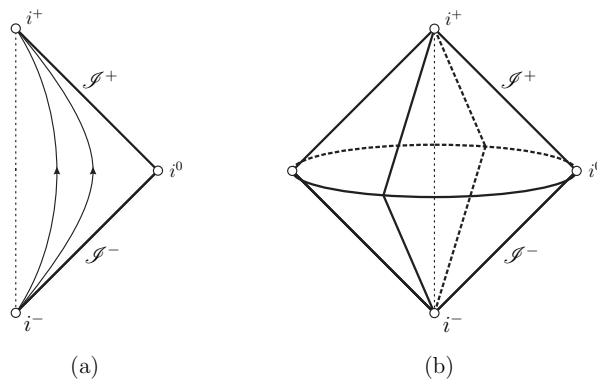


Fig. 3.1.1 Conformal structure of $(n+1)$ -dimensional Minkowski spacetime, $n \geq 2$. The dotted line is the space-origin $r = 0$. Every point in (a) represents a sphere S^{n-1} except for the centre of rotations $r = 0$, which is a point in space. (b) is obtained by rotating (a) around $r = 0$. The points $i^\pm$ and i^0 are defined in Remark 3.1.3.

$$\begin{aligned}\eta &\equiv -dt^2 + dr^2 + r^2 d\Omega^2 = -du^2 - 2dudr + r^2 d\Omega^2 \\ &= x^{-2} (x^2 du^2 + 2dudx + d\Omega^2) .\end{aligned}\quad (3.1.3)$$

One can then use $\Omega := x$ as the conformal factor to rescale the metric, obtaining

$$\tilde{\eta} := \Omega^2 \eta = x^2 du^2 + 2dudx + d\Omega^2 .$$

The metric $\tilde{\eta}$ clearly extends smoothly (in fact, analytically) to and beyond the boundary

$$\mathcal{J}^+ := \{x = 0\} .$$

In dimensions $n \geq 2$ one has $\mathcal{J}^+ \approx \mathbb{R} \times S^{n-1}$. See Section 6.2.1, p. 263, for the case $n = 1$.

We will refer to this construction as the *standard conformal completion of the Minkowski spacetime $\mathbb{R}^{1,n}$ at future null infinity*, and denote $\mathbb{R}^4$ with the hypersurface $\{x = 0\}$ attached by $\tilde{\mathcal{M}}^+$.

One can similarly obtain a *standard conformal completion of the Minkowski spacetime at past null infinity*, which will be denoted by $\tilde{\mathcal{M}}^-$, by replacing t with its negative in all formulae above. The added null hypersurface $\{x = 0\}$ becomes $\mathcal{J}^- \approx \mathbb{R} \times S^2$ in the extended spacetime.

Finally, the standard completion is obtained by adding simultaneously $\mathcal{J}^+$ and $\mathcal{J}^-$; see Fig. 3.1.1.

An alternative construction is presented in Section 6.2.2, p. 264, via a conformal embedding of the Minkowski spacetime in the static Einstein cylinder. $\square$

REMARK 3.1.3 It turns out that there is a way of adding further boundary points to the conformal completion so far, namely a point i^0 , called *spatial infinity*, and a point i^+ , called *future timelike infinity*, as well as a point i^- , called *past timelike infinity*. The construction remains conformal but the reader is warned that this does *not* lead to a manifold with boundary, as the resulting topological space is not a manifold at these points unless some identifications are made. The construction proceeds as follows.

Let x^α denote the usual coordinates on Minkowski spacetime, in which the metric is diagonal with constant entries. In the region where $\eta_{\mu\nu} x^\mu x^\nu < 0$ and $x^0 < 0$ define new coordinates y^μ by the formula

$$y^\alpha = \frac{x^\alpha}{\eta_{\mu\nu} x^\mu x^\nu} .\quad (3.1.4)$$

One readily checks that

$$\eta_{\mu\nu}y^\mu y^\nu = \frac{1}{\eta_{\mu\nu}x^\mu x^\nu} \quad (3.1.5)$$

and $y^0 > 0$, which easily implies that the map $x \mapsto y$ is

a diffeomorphism from $\{\eta_{\mu\nu}x^\mu x^\nu < 0, x^0 < 0\}$ to $\{\eta_{\mu\nu}y^\mu y^\nu < 0, y^0 > 0\}$.

Equivalently, the Minkowskian past light cone of the origin of the x -coordinates becomes the Minkowskian future light cone of the origin of the y -coordinates. The point i^- is defined as the origin of the y -coordinates.

The point i^+ is defined by reversing the time orientation in the construction above.

To obtain i^0 we use the transformation (3.1.4) on the set $\{\eta_{\mu\nu}x^\mu x^\nu > 0\}$, which is then diffeomorphically mapped to the set $\{\eta_{\mu\nu}y^\mu y^\nu > 0\}$. The point i^0 is again defined as the origin of the coordinates y . We prefer to think of the y -coordinates used here as having nothing to do with those used to define i^+ , as well as those used to define i^- , but identifications are possible.

We note that (3.1.4) transforms the Minkowski metric to a conformal image of itself. Indeed, we have

$$dy^\alpha = \frac{\eta_{\beta\gamma}x^\beta x^\gamma dx^\alpha - 2\eta_{\mu\nu}x^\mu dx^\nu x^\alpha}{(\eta_{\beta\gamma}x^\beta x^\gamma)^2}, \quad (3.1.6)$$

which leads to

$$\eta_{\mu\nu}dy^\mu dy^\nu = \frac{\eta_{\mu\nu}dx^\mu dx^\nu}{(\eta_{\beta\gamma}x^\beta x^\gamma)^2}. \quad (3.1.7)$$

Note that with this construction, Minkowski spacetime becomes the timelike future of i^- , as well as the past future of i^+ ; $\mathcal{I}^+$ becomes the boundary of the future of i^0 , and it is natural to make it coincide with the boundary of the past of i^+ , as can be seen by repeating the construction after having changed the origin of the Minkowskian coordinates x . Finally, $\mathcal{I}^-$ becomes the boundary of the past of i^0 , which can be identified with the boundary of the future of i^- . See Fig. 3.1.1.b; the identifications become evident when the construction of Section 6.2.2, p. 264 is carried out instead. $\square$

Further examples of spacetimes with a conformal boundary at infinity are provided by the Schwarzschild spacetime (see Remark 4.2.6, p. 131), or by the Robinson–Trautman spacetimes (see (8.1.8), p. 314).

We show in Section 5.3.10 (see the discussion around (5.3.92), p. 240) how to construct conformal completions of $(n+1)$ -dimensional vacuum stationary spacetimes with $n \geq 5$.

While we mainly have Lorentzian metrics in mind in this work, note that the definition applies regardless of the signature of the metric. Examples of Riemannian metrics with a conformal boundary at infinity are constructed in Section 5.5.3, cf. Eq. (5.5.50), p. 257.

The following property of $\mathcal{I}$ follows immediately from the transformation properties of the Ricci tensor under conformal transformations (cf. Appendix H.6, p. 366).

PROPOSITION 3.1.4 *Consider a conformal completion at infinity with a C^2 conformal metric. If the physical metric is vacuum near $\mathcal{I}$, or if the matter fields decay sufficiently fast, then $\mathcal{I}$ is*

$$\begin{cases} \text{timelike,} & \text{when } \Lambda < 0; \\ \text{null, with vanishing shear,} & \text{when } \Lambda = 0; \\ \text{spacelike,} & \text{when } \Lambda > 0. \end{cases} \quad (3.1.8)$$

Here the *shear* of a null hypersurface is defined as the trace-free part of its null second fundamental form defined in (8.4.5), p. 329.

The conformal boundary $\mathcal{I}$ is called *null infinity* when it consists of a null hypersurface. In this case Geroch and Horowitz [221] propose to add the conditions that (1) one can find a gauge such that the Hessian of the conformal factor Ω vanishes on $\mathcal{I}$, and that (2) the integral curves of $\tilde{\nabla}\Omega$ on $\mathcal{I}$ are complete in this gauge. This is useful for the purpose of a meaningful definition of a black hole (compare Example 3.5.2, p. 110), but does not seem to play any obvious role in our treatment of topological censorship later.

The standard definition of a black-hole region in a spacetime admitting a conformal completion at future null infinity $\mathcal{I}^+$ is

$$\mathcal{B} := \mathcal{M} \setminus J^-(\mathcal{I}^+). \quad (3.1.9)$$

White-hole regions are defined by changing the time orientation:

$$\mathcal{W} := \mathcal{M} \setminus J^+(\mathcal{I}^-). \quad (3.1.10)$$

Similarly to the notion of the domain of outer communication definitions, of black-hole/white-hole regions for stationary spacetimes, which are more convenient for many purposes, and which become equivalent to (3.1.9)–(3.1.10) in situations of main interest after $\mathcal{I}^+$ has been suitably constructed, are given by (4.3.72)–(4.3.73), p. 166.

3.2 Null splitting theorems

A *null line* is an inextendible null geodesic such that no points thereof are timelike related to each other.

A spacetime is said to be *asymptotically simple* if it has a conformal completion at infinity in the sense of Definition 3.1.1, p. 85, and if every future-directed inextendible geodesic in $\mathcal{M}$ has an end point on $\mathcal{I}^-$ and an end point on $\mathcal{I}^+$.

In a beautiful study of null hypersurfaces associated with null lines, Galloway established the following result.

THEOREM 3.2.1 (Galloway [205, 206]) *Suppose that $\mathcal{M}$ is an asymptotically simple vacuum spacetime which contains a null line. Then $\mathcal{M}$ is isometric to Minkowski space.* $\square$

The key to the proof is a careful study of null hypersurfaces which, a priori, might be only continuous.

The fundamental notions of the geometry of smooth null hypersurfaces are outlined in Appendix F; in particular the null mean curvature is defined in (F.8), p. 357.

The main tool for the analysis of *smooth* null hypersurfaces is the following comparison principle.

THEOREM 3.2.2 (Galloway [205]) *Let S_1 and S_2 be smooth null hypersurfaces in a spacetime $\mathcal{M}$. Suppose:*

- (1) *S_1 and S_2 meet at $p \in \mathcal{M}$ and S_2 lies to the future side of S_1 near p , and*
- (2) *the null mean curvature scalars θ_1 of S_1 , and θ_2 of S_2 , satisfy,*

$$\theta_2 \leq 0 \leq \theta_1.$$

Then S_1 and S_2 coincide near p and this common null hypersurface has null mean curvature $\theta = 0$.

However, real-life null hypersurfaces might not be differentiable; an example of a horizon which is non-differentiable on a dense set has been described in Theorem 2.10.12. To give a precise definition of what it means for a topological hypersurface S to be null, recall that a set $A \subset \mathcal{M}$ is said to be achronal if no two

points of A can be joined by a timelike curve. $A \subset \mathcal{M}$ is locally achronal if for each $p \in A$ there is a neighbourhood U of p such that $A \cap U$ is achronal in U . One then introduces the following.

DEFINITION 3.2.3 (Galloway [205]) 1. A C^0 non-timelike hypersurface in M is a topological hypersurface S in M which is locally achronal.

2. A C^0 future null hypersurface in $\mathcal{M}$ is a non-timelike hypersurface S in $\mathcal{M}$ such that for each $p \in S$ and any neighbourhood U of p in which S is achronal, there exists a point $q \in S$, $q \neq p$, such that $q \in J^+(p, U)$.

For such null hypersurfaces one can define the following notion of null mean curvature, where all null tangents have been normalized to unit norm with respect to some auxiliary Riemannian metric.

DEFINITION 3.2.4 (Galloway [205]) Let S be a C^0 future null hypersurface in $\mathcal{M}$. We say that S has null mean curvature $\theta_{\text{sh}} \geq 0$ in the sense of support hypersurfaces provided for each $p \in S$ and for each $\epsilon > 0$ there exists a C^2 null hypersurface $S_{p,\epsilon}$ such that:

1. $S_{p,\epsilon}$ is a past support hypersurface for S at p , i.e., $S_{p,\epsilon}$ passes through p and lies to the past side of S near p , and
2. the (standard) null mean curvature, say $\theta_{p,\epsilon}$, of $S_{p,\epsilon}$ at p satisfies $\theta_{p,\epsilon} \geq -\epsilon$.

A spacetime is said to satisfy the null energy condition, or the null convergence condition, if

$$\text{Ric}(X, X) \geq 0 \quad (3.2.1)$$

for all null vectors X . A useful result is the following.

LEMMA 3.2.5 (Galloway [205]) Let $\mathcal{M}$ be a spacetime which satisfies the null energy condition. Suppose S is an achronal C^0 future null hypersurface whose null generators are future geodesically complete. Then S has null mean curvature $\theta_{\text{sh}} \geq 0$ in the sense of support hypersurfaces, with null second fundamental forms of the supporting hypersurfaces locally bounded from below.

One then has the following comparison principle.

THEOREM 3.2.6 (Galloway [205]) Let S_1 be a C^0 future null hypersurface and let S_2 be a C^0 past null hypersurface in a spacetime $\mathcal{M}$. Suppose that:

1. S_1 and S_2 meet at $p \in \mathcal{M}$ and S_2 lies to the future side of S_1 near p ,
2. S_1 has null mean curvature $\theta_{\text{sh}} \geq 0$ in the sense of support hypersurfaces, with null second fundamental forms $\{B_{p,\epsilon} : p \in S_1, \epsilon > 0\}$ of the supporting hypersurfaces locally bounded from below, and
3. S_2 has null mean curvature $\theta_{\text{sh}} \leq 0$ in the sense of support hypersurfaces.

Then S_1 and S_2 coincide near p ; i.e., there is a neighbourhood $\mathcal{O}$ of p such that $S_1 \cap \mathcal{O} = S_2 \cap \mathcal{O}$. Moreover, $S_1 \cap \mathcal{O} = S_2 \cap \mathcal{O}$ is a smooth null hypersurface with null mean curvature $\theta = 0$.

As an application one has the following result, which is the main ingredient in the proof of Theorem 3.2.1.

THEOREM 3.2.7 (Galloway [205]) Let $\mathcal{M}$ be a null geodesically complete spacetime which obeys the null energy condition, $\text{Ric}(X, X) \geq 0$ for all null vectors X , and contains a null line η . Then η is contained in a smooth closed achronal totally geodesic null hypersurface S .

3.3 Topological censorship

The aim of this section is to present various shades of the *topological censorship principle* of Friedman, Schleich, and Witt [194]. In its original form, this is the statement that causal curves starting and ending in a simply connected asymptotic

region do not experience any non-trivial topology, in that they can be deformed to a curve entirely contained within the asymptotic region. This result imposes restrictions on the topology of black-hole spacetimes, providing a stepping stone for the black-holes uniqueness theorems.

Our presentation, including figures, draws extensively on [124], with permission, and on unpublished joint work with Greg Galloway available as [121]. This author (PTC) assumes responsibility for all errors and omissions which might have been introduced when adapting the work to the current setting.

INCIDENTALLY: Various topological censorship results have been established in the literature under diverse conditions [67, 122, 124, 149, 203, 204, 208, 209, 277]. Here we follow the approach of [204], where topological censorship is reduced to a *null convexity* property of timelike boundaries. The reader will find related results formulated in a Cauchy data setting in [23, 181]. $\square$

3.3.1 Horizon topology

In what follows we will present a series of theorems that guarantee, under suitable conditions, that distinct asymptotic regions cannot communicate with each other, which will then imply that *domains of outer communication* (d.o.c.'s) are simply connected. Now, we provided a precise definition of a d.o.c. in (3.1.2) for spacetimes with conformal completions, and we will provide a more convenient one in (3.3.18), p. 98, for stationary spacetimes. But at this stage the reader can simply think of the d.o.c. as *the region exterior to all black holes and all white holes*. In well-behaved spacetimes the d.o.c.'s are, by definition of 'well behaved', globally hyperbolic. Let $\mathcal{S}$ be a Cauchy surface for the d.o.c. Assuming asymptotic flatness, in (again well-behaved) four-dimensional black-hole spacetimes the hypersurfaces $\mathcal{S}$ are assumed to have a finite number of asymptotically flat ends diffeomorphic to $\mathbb{R}^+ \times S^2$, and a finite number of compact boundaries lying on event horizons.

One of the interests of the theorems that we are about to establish in the forthcoming sections stems from the following observation in [149], generalizing a previous result proved by Hawking using different methods and where toroidal black holes were not excluded.

THEOREM 3.3.1 (Horizon topology theorem) *In the setting above, $\mathcal{S}$ has only one asymptotically flat region, and all the horizons have spherical topology.*

Proof. Global hyperbolicity implies that the d.o.c. is diffeomorphic to $\mathbb{R} \times \mathcal{S}$, which in turn implies that $\mathcal{S}$ is simply connected when the d.o.c. is. It follows immediately from [246, Lemma 4.9] and simple connectedness that all components of the boundary of $\mathcal{S}$ are spheres. The existence of only one asymptotic region in a d.o.c. follows from the fact, to be established shortly, that distinct asymptotic regions cannot communicate with each other. $\square$

A variation of Theorem 3.3.1 can be found in Theorem 8.4.5, p. 331. Theorem 8.4.1, p. 329, provides a higher-dimensional generalization of the above.

3.3.2 Trapped surfaces

One of the key notions in the current context is that of 'trapped surfaces'. In fact, we will see shortly in Theorem 3.3.18 that the occurrence of 'trapped', or 'marginally trapped' surfaces provides a tool for detecting black-hole regions. We therefore start with a discussion of those.

Let $(\mathcal{M}, g)$ be a spacetime of dimension $n + 1 \geq 3$, and consider a spacelike manifold $S \subset \mathcal{M}$ of codimension two. We assume that there exists a smooth unit spacelike vector field N normal to S . If S is a two-sided boundary of a set contained within a spacelike hypersurface $\mathcal{S}$, we shall choose N to be the outward-directed normal tangent to $\mathcal{S}$; this justifies the name of *outward normal* for N . If $S \subset$

$\{r = R\}$ in a KK -asymptotically flat or AdS spacetime, as defined in Section 3.3.5, then the outward normal is defined to be the one for which $N(r) > 0$.

At every point $p \in S$ there exists then a unique future-directed null vector field N^+ normal to S such that $g(N, N^+) = 1$, which we shall call the *outward future null normal* to S . The *inward future null normal* N^- is defined by the requirement that N^- is null, future directed, with $g(N, N^-) = -1$.

We define the *null future inward and outward mean curvatures* $\theta^\pm$ of S by the formula

$$\theta^\pm := \text{tr}_\gamma(\nabla N^\pm), \quad (3.3.1)$$

where tr_γ denotes the trace with respect to the metric γ induced on S . In (3.3.1) the symbol $N^\pm$ should be understood as representing any extension of the null normals $N^\pm$ to a neighbourhood of S . The definition is independent of the extension chosen.

One way to make (3.3.1) explicit is to let e_i , $i = 1, \dots, n-1$ be a local ON frame on S , then

$$\theta^\pm = \sum_{i=1}^{n-1} g(\nabla_{e_i} N^\pm, e_i). \quad (3.3.2)$$

We shall say that a compact S is *weakly outer future trapped* if $\theta^+ \leq 0$. The notion of *weakly inner future trapped* is defined by requiring $\theta^- \leq 0$. A similar notion of *weakly outer or inner past trapped* is defined by changing $\leq$ to $\geq$ in the defining inequalities above. We will say *outer future trapped* if $\theta^+ < 0$, etc. Finally, S is called *marginally future outer trapped* if $\theta^+ = 0$.

One can think of conditions on the sign of $\theta^\pm$ as *mean null convexity*.

Clearly, the hypothesis of compactness is not needed to formulate the definition, but essentially all applications of the notion assume this, and therefore it is convenient to impose compactness.

INCIDENTALLY: It is sometimes convenient to talk about trapped surfaces in terms of geometry of spacelike hypersurfaces. Let thus $\mathcal{S}$ be a spacelike hypersurface in $(\mathcal{M}, g)$, and consider a submanifold $S \subset \mathcal{S}$ of codimension one. Assume that S is two-sided in $\mathcal{S}$, which means that there exists a globally defined field, say m , of unit normals to S within $\mathcal{S}$. There are actually two such fields, m and $-m$; we arbitrarily choose one and call it *outer pointing*. In situations where S bounds a compact region, the outer-pointing normal is chosen to point away from the compact region. We let H denote the mean extrinsic curvature of S within $\mathcal{S}$,

$$H := D_i m^i, \quad (3.3.3)$$

where D is the covariant extrinsic of the metric h induced on $\mathcal{S}$ by the spacetime metric g . The requirement that S is *outer-future-trapped* translates into the requirement that

$$\theta^+ := H + K_{ij}(h^{ij} - m^i m^j) > 0, \quad (3.3.4)$$

where K is the extrinsic curvature tensor of $\mathcal{S}$ in $\mathcal{M}$, while S is *inner-future-trapped* if

$$\theta^- := -H + K_{ij}(h^{ij} - m^i m^j) > 0. \quad (3.3.5)$$

The *past* version of these inequalities is obtained by changing the sign in front of K .

A marginally trapped surface lying within a spacelike hypersurface is sometimes referred to as an *apparent horizon*. We return to this in Section 8.4, p. 328. $\square$

Let $\mathcal{T}$ be a smooth timelike hypersurface in $\mathcal{M}$ with a globally defined smooth field N of unit normals to $\mathcal{T}$. We shall say that $\mathcal{T}$ is *weakly outer-past-trapped* with respect to a time function t if the level sets of t on $\mathcal{T}$ are weakly outer-past-trapped. A similar definition is used for the notion of *weakly outer-future-trapped* timelike hypersurfaces, etc.

EXAMPLE 3.3.2 As an example, let t be the usual time function in the $n+1$ -dimensional exterior Schwarzschild metric with mass parameter M ,

$$g = -Vdt^2 + \underbrace{\frac{dr^2}{V} + r^2 d\Omega^2}_{=h},$$

with $V = 1 - 2Mr^{2-n}$. We have $K_{ij} = 0$ since the metric is static. Let $\mathcal{T}$ be any level set of the radial coordinate r . The field of outer-pointing spacetime unit normals N to the level sets of r is $N = \sqrt{V}\partial_r$, which happens to be tangent to the level sets of t , hence the field, say m , of normals within the level sets of t coincides with N : $m = \sqrt{V}\partial_r$. This gives

$$\begin{aligned} H &= D_i m^i = \frac{1}{\sqrt{\det h}} \partial_i \left(\sqrt{\det h} m^i \right) = \sqrt{V} r^{-(n-1)} \partial_r \left(r^{(n-1)} \right) \\ &= \frac{(n-1)\sqrt{V}}{r}. \end{aligned} \quad (3.3.6)$$

We conclude that all level sets of V in the exterior region are both inner future and inner past trapped. $\square$

INCIDENTALLY: It is shown in [124, Appendix] that the hypersurfaces of large constant radius in electro-vacuum spacetimes constructed in [325] by evolving nearly Minkowskian asymptotically flat initial data, as well as their higher-dimensional counterparts constructed by Loizelet [328, 329] (compare [87]) are past-inward-trapped timelike hypersurfaces. One expects that the asymptotic behaviour of the metric derived in those papers will be typical for solutions of the Einstein–Maxwell equations constructed by evolution of asymptotically flat initial data, and that therefore the result will be true without the near-Minkowskian condition, and for general topologies of the initial data surface. $\square$

3.3.3 Causality in spacetimes with boundary

All the definitions of the causality theory introduced so far will be carried over to spacetimes with boundaries, regardless of their causal character. The reader is warned that some results proved for boundaryless spacetimes might fail to carry over to the new setup unless further conditions are imposed.

In particular the definition of global hyperbolicity for spacetimes with timelike boundary, which will be of main interest here, is identical to that of spacetimes without boundary. To avoid ambiguities, a spacetime $(\mathcal{M}, g)$ with *timelike* boundary $\mathcal{T}$ will be said *globally hyperbolic* if $(\mathcal{M}, g)$ is strongly causal and if for all $p, q \in \mathcal{M}$ the sets $J^+(p) \cap J^-(q)$ are empty or compact. In this case a hypersurface $\mathcal{S}$ is said to be a *Cauchy surface* if $\mathcal{S}$ is met by every inextendible causal curve precisely once. A smooth function t is said to be a *Cauchy time function* if it ranges over $\mathbb{R}$, if ∇t is timelike past directed, and if all its level sets are Cauchy surfaces.

The causal theory of smooth spacetimes with timelike boundary has been studied in detail in [432]. It is explicitly checked there that most of the causality theory developed in the previous chapters remains valid, with essentially identical proofs. For instance, chronological future and past sets are open, and global hyperbolicity as just defined implies causal simplicity. The following basic property of Cauchy surfaces holds as well, and is stated for future reference.

PROPOSITION 3.3.3 *Let $\mathcal{S}$ be a Cauchy surface and let K be a compact subset of a spacetime with timelike boundary. Then $J^+(K) \cap J^-(\mathcal{S})$ and $J^+(K) \cap \mathcal{S}$ are compact.* $\square$

3.3.4 Spacetimes with timelike boundary

We will prove the following key fact [124] (see also Galloway [204, Theorem 1]).

THEOREM 3.3.4 *Let t be a Cauchy time function on a spacetime $(\mathcal{M}, g)$ with timelike boundary $\mathcal{T} = \cup_{\alpha \in \Omega} \mathcal{T}_\alpha$. Assume that $(\mathcal{M}, g)$ satisfies the null energy condition (NEC)*

$$R_{\mu\nu}X^\mu X^\nu \geq 0 \quad \text{for all null vectors } X^\mu. \quad (3.3.7)$$

Suppose that there exists a component $\mathcal{T}_1$ of $\mathcal{T}$ with compact level sets $t|_{\mathcal{T}_1}$ such that

$\mathcal{T}_1$ is weakly inner-future-trapped

with respect to t . If

all connected components $\mathcal{T}_\alpha$, $\alpha \neq 1$, of $\mathcal{T}$ are inner-past-trapped

with respect to t , then

$$J^+(\mathcal{T}_1) \cap J^-(\mathcal{T}_\alpha) = \emptyset \quad \text{for } \mathcal{T}_\alpha \neq \mathcal{T}_1.$$

REMARKS 3.3.5 1. Nothing is assumed about the nature of the index set Ω .

2. The condition that at least one of the defining inequalities is strict is necessary. Indeed, let

$$\mathcal{M}' = \mathbb{R} \times \underbrace{S^1 \times \dots \times S^1}_{n \text{ factors}}$$

be equipped with a flat product metric. Let t be a standard coordinate on the $\mathbb{R}$ factor, and let $\varphi \in [0, 2\pi]$ be a standard angular coordinate on the first S^1 factor. Then $\mathcal{M} = \{0 \leq \varphi \leq \pi\} \subset \mathcal{M}'$ is a spacetime with boundaries $\mathcal{T}_1 := \{\varphi = 0\}$ and $\mathcal{T}_2 := \{\varphi = \pi\}$, which are weakly inner-past- and future-trapped, $\theta^\pm \equiv 0$, thus satisfies the hypotheses above except for the strictness condition, and does not satisfy the conclusion. $\square$

Let t be a time function on $\mathcal{M}$, and let $\gamma : [a, b] \rightarrow \mathcal{M}$ be a future-directed causal curve. The *time of flight* t_γ of γ is defined as

$$t_\gamma = t(\gamma(b)) - t(\gamma(a)).$$

As a step in the proof of Theorem 3.3.4, we will need the following result.

PROPOSITION 3.3.6 *Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime satisfying the null energy condition and containing a past-inward-trapped hypersurface $\mathcal{T}$. Let $S \subset \mathcal{M}$ be weakly future-inward-trapped. Then there are no future-directed causal curves, starting inwardly at S , meeting $\mathcal{T}$ inwardly, and minimizing, amongst nearby causal curves, the time of flight between S and $\mathcal{T}$.*

Proof. Suppose that the result is wrong; thus, there exists a future-directed causal curve $\gamma : [a, b] \rightarrow \mathcal{M}$ with $\gamma(a) \in S$, $\gamma(b) \in \mathcal{T}$, locally minimizing the time of flight. It is not too difficult to show that γ is a null geodesic emanating orthogonally from S without S -conjugate points on $[a, b]$. In particular $\dot{J}^+(S)$ is a smooth null hypersurface near $\gamma([a, b))$. Let $t_+ = t(\gamma(b))$, set $S_+ = \{t = t_+\} \cap \mathcal{T}$; then $\dot{J}^-(S_+)$ is a smooth null hypersurface near S_+ that contains a segment $\gamma([b - \varepsilon, b])$ for some $\varepsilon > 0$. Moreover, $\dot{J}^-(S_+)$ lies to the causal past of $\dot{J}^+(S)$ close to γ since otherwise we could construct a timelike curve from S to S_+ close to γ , thus violating the minimization character of γ . Since $\dot{\gamma}(a)$ is inward pointing and $\dot{\gamma}(b)$ outward pointing, the Raychaudhuri equation (F.20), p. 360, shows that the null divergence of $\dot{J}^+(S)$ along γ is non-positive, whereas the null divergence of $\dot{J}^-(S_+)$ is positive near S_+ . For points $\gamma(s)$, with $s \neq b$ but close to b , this contradicts the maximum principle for null hypersurfaces. Theorem 3.2.2, p. 88, establishes the result. $\square$

EXAMPLE 3.3.7 An example to keep in mind is the following: let $p, q \in \mathbb{R}^{n+1}$ be two spatially separated points in Minkowski spacetime $\mathbb{R}^{1,n}$. Let $S = \dot{J}^-(p) \cap \dot{J}^-(q)$. The null generators of $\dot{J}^-(p)$ and $\dot{J}^-(q)$ are converging, when followed to the future,

and they meet S normally, which shows that S is an outward- and inward-future-trapped (non-compact) submanifold of $\mathbb{R}^{1,n}$. Of course, in this case the choice of ‘inward’ and ‘outward’ is a pure matter of convention.

Let $n = 3$, choose $p = (2, 2, 0, 0)$, $q = (2, -2, 0, 0)$; let $\mathcal{T}$ be the timelike cylinder $\mathcal{T} = \{r = 1\}$, which is both future- and past-inward-trapped. The null achronal geodesic segments $\gamma_{\pm}(t) = (t, \pm t, 0, 0)$, $0 \leq t < 2$, are in $J^+(S)$ and, by symmetry considerations, *maximize* the time of flight between S and $\mathcal{T}$. Since $J^+(S)$ lies below $J^-(\{t = 0\} \cap \mathcal{T})$, the argument in the proof below, when applied to $\gamma_{\pm}$, does not lead to a contradiction. But the example shows that the existence of an achronal null geodesic segment between S and $\mathcal{T}$ is compatible with the hypotheses above. In particular ‘locally minimizing’ cannot be replaced by ‘extremizing’. $\square$

Proof of Theorem 3.3.4. Let S be a weakly inner-trapped compact Cauchy surface of $\mathcal{T}_1$. Suppose there exists a causal curve c from S to a point p in a different component $\mathcal{T}_2$. Let $\mathcal{S}$ be the Cauchy surface $\{t = t(p)\}$ for $\mathcal{M}$. We want to construct the fastest null geodesic from S to $\mathcal{T} \setminus \mathcal{T}_1$; for this we need to show that only finitely many components, $\mathcal{T}_a$, of $\mathcal{T} \setminus \mathcal{T}_1$ meet the set $A = J^+(S) \cap \mathcal{S}$, which is compact by Proposition 3.3.3. Suppose, to the contrary, that there are infinitely many of these components that meet A . Then we obtain an infinite sequence of points $\{x_n\}$ in A , each point in a different component. Since A is compact we can pass to a convergent subsequence, still called $\{x_n\}$, such that $x_n \rightarrow x \in A$. Since $\mathcal{T} \cap \mathcal{S}$ is closed, x is in $\mathcal{T}$. But this contradicts the half-neighbourhood property of manifolds with boundary.

The time function t on $\mathcal{M}$ restricts to a time function on $\mathcal{T}$. By the observation in the preceding paragraph, the set $(\mathcal{T} \setminus \mathcal{T}_1) \cap J^+(S) \cap J^-(\mathcal{S})$ is compact and thus we can now minimize t on causal curves from S to $\mathcal{T} \setminus \mathcal{T}_1$ contained in the aforementioned compact set to obtain a fastest causal curve γ from S to $\cup_{a \neq 1} \mathcal{T}_a$. Since t has been minimized, γ meets $\mathcal{T}$ only at its end points, and hence must be a null geodesic. This contradicts Proposition 3.3.6, and establishes the result. $\square$

We now proceed to establish a general topological censorship result for globally hyperbolic spacetimes with timelike boundary.

THEOREM 3.3.8 *Let $(\mathcal{M}, g)$ be a spacetime with a connected timelike boundary $\mathcal{T}$ satisfying the null energy condition (3.2.1). Let*

$$\langle\langle \mathcal{T} \rangle\rangle := I^+(\mathcal{T}) \cap I^-(\mathcal{T})$$

be the domain of communications of $\mathcal{T}$. Further assume $\langle\langle \mathcal{T} \rangle\rangle$ has a Cauchy time function such that the level sets $t|_{\mathcal{T}}$ are compact. If the null energy condition holds on $\langle\langle \mathcal{T} \rangle\rangle$, and if $\mathcal{T}$ is inner-past-trapped and weakly inner-future-trapped with respect to t , then topological censorship holds; i.e., any causal curve included within $\langle\langle \mathcal{T} \rangle\rangle$ with end points on $\mathcal{T}$ can be deformed, keeping end points fixed, to a curve included in $\mathcal{T}$.

Proof. First notice that the inclusion $j: \mathcal{T} \hookrightarrow \langle\langle \mathcal{T} \rangle\rangle$ induces a homomorphism of fundamental groups $j_*: \pi_1(\mathcal{T}) \rightarrow \pi_1(\langle\langle \mathcal{T} \rangle\rangle)$. Thus there exists a covering $\pi: \mathcal{M} \rightarrow \langle\langle \mathcal{T} \rangle\rangle$ associated to the subgroup $j_*(\pi_1(\mathcal{T}))$ of $\pi_1(\langle\langle \mathcal{T} \rangle\rangle)$. This covering is characterized as the largest covering of $\langle\langle \mathcal{T} \rangle\rangle$ containing a homeomorphic copy $\mathcal{T}_0$ of $\mathcal{T}$; that is, $\pi|_{\mathcal{T}_0}$ is a homeomorphism onto $\mathcal{T}$ [238]. Furthermore, this covering has the property that the map $i_*: \pi_1(\mathcal{T}_0) \rightarrow \pi_1(\mathcal{M})$ induced by the inclusion $i: \mathcal{T}_0 \rightarrow \mathcal{M}$ is surjective. Endowing $\mathcal{M}$ with the pullback metric $\pi^*(g)$ we obtain a globally hyperbolic spacetime with timelike boundary $\pi^{-1}(\mathcal{T})$. Now, let $\gamma: [a, b] \rightarrow \langle\langle \mathcal{T} \rangle\rangle$ be a causal curve with end points in $\mathcal{T}$. Lift γ to $\gamma_0: [a, b] \rightarrow \mathcal{M}$ with $\gamma_0(a) \in \mathcal{T}_0$. By Theorem 3.3.4 we know that $\mathcal{T}_0$ cannot communicate with any other component of $\pi^{-1}(\mathcal{T})$, hence $\gamma_0(b) \in \mathcal{T}_0$. As a consequence, γ_0 is homotopic to a curve in $\mathcal{T}_0$ and the result follows. $\square$

As noted in [208], topological censorship can be viewed as the statement that *any* curve in $\langle\langle\mathcal{T}\rangle\rangle$ with end points in $\mathcal{T}$ is homotopic to a curve in $\mathcal{T}$, or equivalently that the map $j_*: \pi_1(\mathcal{T}) \rightarrow \pi_1(\langle\langle\mathcal{T}\rangle\rangle)$ is surjective.

THEOREM 3.3.9 *Under the hypotheses of Theorem 3.3.8, the map $j_*: \pi_1(\mathcal{T}) \rightarrow \pi_1(\langle\langle\mathcal{T}\rangle\rangle)$ induced by the inclusion $j: \mathcal{T} \rightarrow \langle\langle\mathcal{T}\rangle\rangle$ is surjective.*

Proof. Let $\pi: M \rightarrow \langle\langle\mathcal{T}\rangle\rangle$ be the universal cover of $\langle\langle\mathcal{T}\rangle\rangle$ and let $\{\mathcal{I}_\alpha\}$, $\alpha \in A$, be the collection of connected components of the timelike boundary $\pi^{-1}(\mathcal{T})$. Let us define $\langle\langle I \rangle\rangle_{\alpha,\beta} := I^+(\mathcal{I}_\alpha) \cap I^-(\mathcal{I}_\beta)$. We claim that the collection of sets $\langle\langle I \rangle\rangle_{\alpha,\beta}$ forms an open cover of $\mathcal{M}$. Indeed, let $p \in \mathcal{M}$; since $\pi(p) \in \langle\langle\mathcal{T}\rangle\rangle$ there exists a causal curve through $\pi(p)$ which starts and ends in $\mathcal{T}$. Then γ lifts to a causal curve through p which starts in some $\mathcal{I}_\alpha$ and ends in some $\mathcal{I}_\beta$, hence the result. Now, by Theorem 3.3.8 the sets $I^+(\mathcal{I}_\alpha) \cap I^-(\mathcal{I}_\beta)$ are empty if $\alpha \neq \beta$. It follows that the sets $\langle\langle I \rangle\rangle_{\alpha,\alpha}$ are pairwise disjoint, cover $\mathcal{M}$, and since $\mathcal{M}$ is connected we conclude that $|A| = 1$ and hence $\pi^{-1}(\mathcal{T})$ is connected. The result now follows from the following topological result [208, Lemma 3.2]. $\square$

PROPOSITION 3.3.10 *Let $\mathcal{M}$ and S be topological manifolds, $\iota: S \hookrightarrow \mathcal{M}$ an embedding and $\pi: \mathcal{M}^* \rightarrow \mathcal{M}$ the universal cover of $\mathcal{M}$. If $\pi^{-1}(S)$ is connected, then the induced group homomorphism $\iota_*: \pi_1(S) \rightarrow \pi_1(\mathcal{M})$ is surjective.* $\square$

3.3.5 Kaluza–Klein asymptotics

In the sections that follow we shall apply Theorem 3.3.8 to obtain topological information about spacetimes with *Kaluza–Klein* asymptotics: we shall say that $\mathcal{S}_{\text{ext}}$ is a *Kaluza–Klein asymptotic end*, or *asymptotic end* for short, if $\mathcal{S}_{\text{ext}}$ is diffeomorphic to $\mathbb{R}_r \times \check{N} \times Q$, where $\check{N}$ and Q are compact manifolds. The notation $\mathbb{R}_r$ is meant to convey the information that we denote by r the coordinate running along an $\mathbb{R}$ factor. Let $\check{h}_r$ be a family of Riemannian metrics parameterized by r , let $\check{k}$ be a fixed Riemannian metric on Q , and finally let $\check{\lambda}$ and $\check{\nu}$ be two functions on $\mathbb{R}$. The *reference metric* $\check{g}$ on $\mathbb{R}_t \times \mathcal{S}_{\text{ext}}$ is defined as

$$\check{g} = -e^{2\check{\lambda}(r)} dt^2 + \underbrace{e^{-2\check{\nu}(r)} dr^2 + \check{h}_r + \check{k}}_{=: \check{h}}. \quad (3.3.8)$$

The reason for treating $\check{N}$ and Q separately is that the metrics $\check{h}_r$ are allowed to depend on r (in examples to follow we will actually have $\check{h}_r = r^2 \check{h}$, for a fixed metric $\check{h}$), while $\check{k}$ is not. The manifold $\mathbb{R}_t \times \mathbb{R}_r \times \check{N}$ will be referred to as the *base manifold*, while Q can be thought of as the internal space of Kaluza–Klein theory (see, e.g., [153]).

To apply our previous results, we will need the hypothesis that the hypersurfaces

$$\mathcal{T}_R := \{r = R\}$$

are inner-future- and past-trapped for the reference metric $\check{g}$. We define the *outward-pointing* $\check{g}$ -normal to $\mathcal{T}_R$ to be

$$\check{N} := e^{\check{\nu}} \partial_r,$$

and the two null future normals $N^\pm$ to $\{t = \text{const}, r = \text{const}'\}$ are given by $N^\pm = e^{-\check{\lambda}} \partial_t \pm \check{N}$. The requirement of ‘mean outward null $\check{g}$ -convexity’ of $\mathcal{T}_R$ reads

$$\pm \check{\theta}^\pm = \frac{e^{\check{\nu}-\check{\lambda}}}{\sqrt{\det \check{h}_r}} \partial_r (\sqrt{\det \check{h}_r} e^{\check{\lambda}}) > 0. \quad (3.3.9)$$

We will be interested in metrics g which are asymptotic, as r goes to infinity, to metrics of the above form. The convergence of g to $\check{g}$ should be such that the

positivity of $\pm\theta^\pm$ holds, for R large enough, uniformly over compact sets of the t variable. Two special cases seem to be of particular interest, with asymptotically flat, or asymptotically anti-de Sitter base metrics.

Asymptotically flat base manifolds. A special case of the above arises when $\mathcal{S}_{\text{ext}}$ is diffeomorphic to $(\mathbb{R}^n \setminus \bar{B}(R)) \times Q$, where $\bar{B}(R)$ is a closed coordinate ball of radius R ; thus the manifold $\check{N}$ is an $(n-1)$ -dimensional sphere. In dimension $n \geq 3$ we take $\check{g} = -dt^2 \oplus \check{h}$, where $\check{h} = \delta \oplus \check{k}$, and where δ is the Euclidean metric on $\mathbb{R}^n$. If $n = 2$, in (3.3.8) we take $\check{\lambda} = \check{\nu} = 0$ and $\check{h}_r = r^2 d\varphi^2$, where φ is a coordinate on S^1 which ranges perhaps, but *not necessarily*, over $[0, 2\pi]$. Thus, for all $n \geq 2$ we have $\check{\lambda} = \check{\nu} = 0$ and $\check{h}_r = r^2 d\Omega^2$, where $d\Omega^2$ is the round metric on S^{n-1} ; strictly speaking, $\mathbb{R}_r$ is then (R, ∞) , a set diffeomorphic to $\mathbb{R}$. Metrics g which asymptote to this $\check{g}$ as r tends to infinity will be said to have an *asymptotically flat base manifold*. Equation (3.3.9) gives

$$\pm\theta^\pm = \frac{n-1}{r}, \quad (3.3.10)$$

which is positive, as required.

We shall say that a Riemannian metric h on $\mathcal{S}_{\text{ext}}$ is *Kaluza–Klein asymptotically flat*, or *KK-asymptotically flat* for short, if there exists $\alpha > 0$ and $k \geq 1$ such that for $0 \leq \ell \leq k$ we have

$$\check{D}_{i_1} \cdots \check{D}_{i_\ell} (h - \check{h}) = O(r^{-\alpha-\ell}), \quad (3.3.11)$$

where $\check{D}$ denotes the Levi-Civita connection of $\check{h}$, and r is the radius function in $\mathbb{R}^n$, $r := \sqrt{(x^1)^2 + \cdots + (x^n)^2}$, with the x^i 's being any Euclidean coordinates of $(\mathbb{R}^n, \delta)$.

We shall say that a general relativistic initial data set $(\mathcal{S}_{\text{ext}}, h, K)$ (cf. Appendix G, p. 363) is *Kaluza–Klein asymptotically flat*, or *KK-asymptotically flat*, if $(\mathcal{S}_{\text{ext}}, h)$ is *KK-asymptotically flat* and if for $0 \leq \ell \leq k-1$ we have

$$\check{D}_{i_1} \cdots \check{D}_{i_\ell} K = O(r^{-\alpha-1-\ell}). \quad (3.3.12)$$

A spacetime $(\mathcal{M}, g)$ will be said to contain a *Kaluza–Klein asymptotically flat region* if there exists a subset of $\mathcal{M}$, denoted by $\mathcal{M}_{\text{ext}}$, and a time function t on $\mathcal{M}_{\text{ext}}$, such that the initial data (g, K) induced by g on the level sets of t are *KK-asymptotically flat*.

REMARK 3.3.11 All this reduces to the usual notion of asymptotic flatness when Q is the manifold consisting of a single point. $\square$

Let $\mathcal{T}_R = \{r = R\}$ be a level set of r in $\mathcal{M}_{\text{ext}}$. Then the unit outward-pointing conormal $n^\flat$ to $\mathcal{T}_R$ is $n^\flat = (1 + O(r^{-\alpha}))dr$. This implies that the future-directed null vector fields normal to the foliation of $\mathcal{T}_R$ by the level sets of t take the form $N^p m = \partial_t \pm \frac{x^i}{r} \partial_i + O(r^{-\alpha})$, leading to (compare (3.3.10))

$$\pm\theta^\pm = \frac{n-1}{r} + O(r^{-\alpha-1}) > 0 \text{ for } r \text{ large enough.} \quad (3.3.13)$$

Asymptotically anti-de Sitter base manifolds. We consider, now, manifolds with asymptotically anti-de Sitter base metrics. The base reference metric is taken of the Birmingham form (see Section 5.5, p. 250),

$$-e^{2\check{\lambda}(r)} dt^2 + e^{-2\check{\nu}(r)} dr^2 + \check{h}_r, \quad \check{h}_r = r^2 \check{h}, \quad (3.3.14)$$

where $\check{h}$ is an Einstein metric on the compact $(n-1)$ -dimensional manifold $\check{N}$, $n \geq 2$. Furthermore,

$$e^{2\check{\lambda}(r)} = e^{2\check{\nu}(r)} = \check{\alpha} r^2 + \check{\beta},$$

for some suitable constants $\check{\alpha} > 0$ and $\check{\beta} \in \mathbb{R}$, which can be chosen so that (3.3.14) is an Einstein metric: indeed, if Q has dimension k , then $\check{g}$ will satisfy the vacuum

Einstein equations with cosmological constant Λ if $\check{k}$ is an Einstein metric with scalar curvature $2k\Lambda/(n+k-1)$, while $\check{\alpha} = -2\Lambda/n(n+k-1)$, and $\check{\beta} = R(\check{h})/(n-1)(n-2)$ for $n > 2$, while $\check{\beta}$ is arbitrary if $n = 2$, where $R(\check{h})$ is the scalar curvature of the metric $\check{h}$.

In a manner somewhat analogous to the previous section, with decay requirements adapted to the problem at hand, we shall say that a Riemannian metric h on $\mathcal{S}_{\text{ext}}$ is *KK-asymptotically AdS* if there exist a real number $\alpha > 1$ and an integer $k \geq 1$ such that for $0 \leq \ell \leq k$ we have

$$|\check{D}_{i_1} \dots \check{D}_{i_\ell}(h - \check{h})|_{\check{h}} = O(r^{-\alpha}), \quad (3.3.15)$$

where $|\cdot|_{\check{h}}$ is the norm of a tensor with respect to $\check{h}$, and r is a ‘radial coordinate’ as in (3.3.14). We shall say that a general relativistic initial data set $(\mathcal{S}_{\text{ext}}, h, K)$ is *KK-asymptotically AdS*, if $(\mathcal{S}_{\text{ext}}, h)$ is *KK-asymptotically AdS* and if for $0 \leq \ell \leq k-1$ we have

$$|\check{D}_{i_1} \dots \check{D}_{i_\ell} K|_{\check{h}} = O(r^{-\alpha}). \quad (3.3.16)$$

Finally, a spacetime $(\mathcal{M}, g)$ will be said to contain a *Kaluza–Klein asymptotically AdS region* if there exists a subset of $\mathcal{M}$, denoted by $\mathcal{M}_{\text{ext}}$, and a time function t on $\mathcal{M}_{\text{ext}}$, such that the initial data (g, K) induced by g on the level sets of t are *KK-asymptotically AdS*.

The fact that *KK-asymptotically AdS* metrics have the right null convexity properties follows from the calculation in (3.3.27), using (3.3.15)–(3.3.16) with $\alpha > 1$.

Uniform KK-asymptotic ends. We shall say that a *KK-asymptotically flat region*, or a *KK-asymptotically AdS region*, is *uniform of order k* if there exists a time function t such that the estimates (3.3.11)–(3.3.12), or (3.3.15)–(3.3.16) hold with constants independent of t .

Metrics which are t -independent in the asymptotic region provide the main examples.

3.3.6 Spacetimes with uniform Kaluza–Klein ends

In this section we consider manifolds with *KK-asymptotically flat regions* or with *KK-asymptotically AdS regions*. This includes in particular asymptotically flat or asymptotically AdS manifolds.

Now, the approach to topological censorship here requires uniformity in time of the mean null extrinsic curvatures of the spheres $\{t = \text{const}, r = \text{const}'\}$. This might conceivably hold for a wide class of dynamical metrics, but how large the corresponding class of metrics is remains to be seen. As such, the main applications of our results concern stationary metrics [104], in which case the uniformity is easy to guarantee by an obvious choice of time functions.

Consider, first, a spacetime with a Killing vector field X , with complete orbits, containing a *KK-asymptotic end* $\mathcal{S}_{\text{ext}}$. Then X will be called *stationary* if X is timelike on $\mathcal{S}_{\text{ext}}$ and approaches, as r goes to infinity, ∂_t in the coordinate system of (3.3.8)¹; $(\mathcal{M}, g)$ will then be called *stationary*. Similarly to the standard asymptotically flat case, we set

$$\mathcal{M}_{\text{ext}} := \cup_{t \in \mathbb{R}} \phi_t[X](\mathcal{S}_{\text{ext}}), \quad (3.3.17)$$

¹For metrics which are asymptotically flat in the usual (rather than *KK*) sense, the existence of such coordinates can be established for Killing vectors which are timelike on $\mathcal{S}_{\text{ext}}$, whenever the initial data set satisfies the conditions of the positive energy theorem (cf., e.g., [133, 265, 313, 327, 426]). It is likely that a similar result holds for *KK-asymptotically flat* or *AdS* metrics, but we have not investigated this issue any further.

where $\phi_t[X]$ denotes the flow of X . Assuming stationarity, the *domain of outer communications* is defined as

$$\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle := I^-(\mathcal{M}_{\text{ext}}) \cap I^+(\mathcal{M}_{\text{ext}}). \quad (3.3.18)$$

More generally, let $(\mathcal{M}, g)$ admit a time function t ranging over an open interval I (not necessarily equal $\mathbb{R}$), and a radius function r as in (3.3.8), with KK -asymptotic level sets which are *uniform of order zero*. We then set

$$\mathcal{M}_{\text{ext}} := \{p \in \mathcal{M} : r(p) \geq R_0\}$$

for some R_0 chosen large enough so that for any $R \geq R_0$ we have

$$J^\pm(\mathcal{M}_{\text{ext}}) = J^\pm(\{r = R\}). \quad (3.3.19)$$

INCIDENTALLY: To see that such an R_0 exists, note that the inclusion $J^\pm(\mathcal{M}_{\text{ext}}) \supset J^\pm(\{r = R\})$ is obvious whenever $\{r = R\} \subset \mathcal{M}_{\text{ext}}$. To justify the opposite inclusion let, say, $x \in J^-(\mathcal{M}_{\text{ext}})$, so there exists a future-directed causal curve from x to some point $(t, p) \in \mathcal{M}_{\text{ext}}$; thus $p \in \mathcal{S}_{\text{ext}}$. We need to show that there exists a future-directed causal curve from (t, p) to a point $(t', q) \in \{r = R\}$. This follows from the somewhat more general fact; that for any t and for any two points $p, q \in \mathcal{S}_{\text{ext}}$ such that $r(p) \geq R_0$ and $r(q) \geq R_0$, there exists a causal curve $\gamma(s) = (t + \alpha s, \sigma(s))$ such that $\sigma(0) = p$ and $\sigma(1) = q$, with α and σ independent of t . Now, the existence of σ follows from connectedness of $\mathcal{S}_{\text{ext}}$. Next, the existence of a t -independent (large) constant α so that γ is causal for $\hat{g}$ follows immediately from the form of the metric $\hat{g}$. Finally, it should be clear from uniformity in time of the error terms that, increasing α if necessary, γ will also be causal for g , independently of t , provided R_0 is chosen large enough. $\square$

If the asymptotic estimates are moreover uniform to order one, we choose R_0 large enough so that all level sets of $\{r = R\}$, with R sufficiently large are *future- and past-inner-trapped*.

Let us consider a spacetime $(\mathcal{M}, g)$ with several KK -asymptotic regions $\mathcal{M}_{\text{ext}}^\lambda$, $\lambda \in \Lambda$ for some index set Λ , each generating its own domain of outer communications. We assume that all regions are uniform to order one with respect to a Cauchy time function t . Let $\mathcal{T}_\lambda \subset \mathcal{M}_{\text{ext}}^\lambda$ be defined as $\{r = \hat{R}_\lambda\}$ for an appropriately large $\hat{R}_\lambda$. Consider the manifold obtained by removing from the original spacetime the asymptotic regions $\{r > \hat{R}_\lambda\} \subset \mathcal{M}_{\text{ext}}^\lambda$; this is a manifold with boundary $\mathcal{T} = \cup_\lambda \mathcal{T}_\lambda$, each connected component $\mathcal{T}_\lambda$ being both future- and past-inward-trapped. From (3.3.19) we have $J^\pm(\mathcal{M}_{\text{ext}}^\lambda) = J^\pm(\mathcal{T}_\lambda)$. Then the following result is a straightforward consequence of Theorem 3.3.4.

THEOREM 3.3.12 *If $(\mathcal{M}, g)$ is a globally hyperbolic spacetime with KK -asymptotic ends, uniform to order one, satisfying the null energy condition (3.2.1), then*

$$J^+(\mathcal{M}_{\text{ext}}^{\lambda_1}) \cap J^-(\mathcal{M}_{\text{ext}}^{\lambda_2}) = \emptyset \quad \text{whenever} \quad \mathcal{M}_{\text{ext}}^{\lambda_1} \cap \mathcal{M}_{\text{ext}}^{\lambda_2} = \emptyset. \quad (3.3.20)$$

Next, Theorems 3.3.8 and 3.3.9 yield the following result on topological censorship for stationary KK -asymptotically flat spacetimes.

THEOREM 3.3.13 *Let $(\mathcal{M}, g)$ be a spacetime satisfying the null energy condition, and containing a KK -asymptotic end $\mathcal{M}_{\text{ext}}$, uniform to order one. Suppose further that $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ is globally hyperbolic. Then every causal curve in $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ with end points in $\mathcal{M}_{\text{ext}}$ is homotopic to a curve in $\mathcal{M}_{\text{ext}}$. Moreover the map $j_*: \pi_1(\mathcal{M}_{\text{ext}}) \rightarrow \pi_1(\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle)$ is surjective.*

Proof. It suffices to prove the second statement. Let $\hat{R} > R$ and $\mathcal{T} = \{r = \hat{R}\}$ be defined as in the previous result. Let α be a loop in $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ based at p_0 , and let c be the radial curve from p_0 to $p \in \mathcal{T}$. Then since $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle = \langle\langle \mathcal{T} \rangle\rangle$, by Theorem 3.3.9 the loop $c * \alpha * c^-$, where c^- denotes c followed backwards, is homotopic to a loop

β in $\mathcal{T}$ based at p . Thus α is in turn homotopic to $c^- * \beta * c$, which is a loop that lies entirely in $\mathcal{M}_{\text{ext}}$, hence establishing the result. $\square$

For future reference, we point out the following special case of Proposition 3.3.6, which follows immediately from the fact that large level sets of r are inner trapped.

PROPOSITION 3.3.14 *Let $(\mathcal{M}, g)$ be a stationary, KK -asymptotically flat or asymptotically flat globally hyperbolic spacetime satisfying the null energy condition. Let $S \subset \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ be future inward marginally trapped. There exists a large constant R_1 such that for all $R_2 \geq R_1$ there are no future-directed null geodesics starting inwardly at S , ending inwardly at $\{r = R_2\} \subset \mathcal{M}_{\text{ext}}$, and locally minimizing the time of flight.* $\square$

We proceed now to prove the main theorem for quotients of KK -asymptotically flat spacetimes.

THEOREM 3.3.15 *Let $(\mathcal{M}, g)$ be a spacetime satisfying the null energy condition, and containing a KK -asymptotically flat region, or a KK -asymptotically AdS region, with the asymptotic estimates uniform to order one. Suppose that $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ is globally hyperbolic, and that there exists an action of a group G_s on $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ by isometries which, on $\mathcal{M}_{\text{ext}} \approx \mathbb{R} \times \mathcal{S}_{\text{ext}}$, takes the form*

$$g \cdot (t, p) = (t, g \cdot p).$$

If $\mathcal{S}_{\text{ext}}/G_s$ simply connected, then so is $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G_s$.

REMARK 3.3.16 A variation on the proof below, using an exhaustion argument, shows that the result remains valid if the asymptotic decay estimates are uniform in t to order zero, and uniform over compact sets in t to order one. In this case the hypersurfaces $\{r = R\}$ are not necessarily trapped, but there exists a sequence R_k such that the hypersurfaces $\{r = R_k, |t| < k\}$ are. $\square$

Proof. If the action of G_s is such that the projection $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \rightarrow \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G_s$ has the homotopy lifting property (see, e.g., [242]), then the following argument applies: consider the commutative diagram

$$\begin{array}{ccc} \mathcal{M}_{\text{ext}} & \xrightarrow{i} & \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \\ q \downarrow & & \downarrow p \\ \mathcal{M}_{\text{ext}}/G_s & \xrightarrow{j} & \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G_s, \end{array}$$

where p and q are the standard projections, i is the standard inclusion, and j is the map induced by i . Thus, we have the corresponding commutative diagram

$$\begin{array}{ccc} \pi_1(\mathcal{M}_{\text{ext}}) & \xrightarrow{i_*} & \pi_1(\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle) \\ q_* \downarrow & & \downarrow p_* \\ \pi_1(\mathcal{M}_{\text{ext}}/G_s) & \xrightarrow{j_*} & \pi_1(\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G_s) \end{array}$$

of fundamental groups. By Theorem 3.3.12, i_* is onto. Finally notice that p_* and q_* are onto since p and q have the homotopy lifting property. Hence j_* is onto and as a consequence $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G_s$ is simply connected if $\mathcal{M}_{\text{ext}}/G_s = \mathbb{R} \times (\mathcal{S}_{\text{ext}}/G_s)$ is.

The homotopy lifting property of the action is known to hold in many significant cases (e.g., when the action is free), but it is not clear whether it holds in sufficient generality. However, one can proceed as follows: Let π denote the projection map

$$\pi : \mathcal{M} \rightarrow \mathcal{M}/G_s.$$

We start by constructing a covering space, $\widehat{\mathcal{M}}$, of $\mathcal{M}$: choose $p \in \mathcal{M}$ and let Ω be the set of continuous paths in $\mathcal{M}$ starting at p . We shall say that the paths $\gamma_a \in \Omega$,

$a = 1, 2$, are equivalent, writing $\gamma_1 \sim \gamma_2$, if they share their end point, and if the projection $\pi(\gamma_1 * \gamma_2^-)$ of the path $\gamma_1 * \gamma_2^-$, obtained by concatenating γ_1 with γ_2 followed backwards, is homotopically trivial in $\mathcal{M}/G_s$. We set

$$\widehat{\mathcal{M}} := \Omega / \sim .$$

By the usual arguments (see, e.g., the proof of [315, Theorem 12.8]) $\widehat{\mathcal{M}}$ is a topological covering of $\mathcal{M}$, while [316, Proposition 2.12] shows that $\widehat{\mathcal{M}}$ is a smooth manifold. (In fact, $\widehat{\mathcal{M}}$ is the covering space of $\mathcal{M}$ associated with the subgroup $\text{Ker } \pi_* \subset \pi_1(\mathcal{M})$.) The covering is trivial if and only if $\mathcal{M}/G_s$ is simply connected.

Since $\mathcal{S}_{\text{ext}}/G_s$ is simply connected, the quotient $\mathcal{M}_{\text{ext}}/G_s = \mathbb{R} \times (\mathcal{S}_{\text{ext}}/G_s)$ also is, which implies that $\pi^{-1}(\mathcal{M}_{\text{ext}}) \subset \widehat{\mathcal{M}}$ is the union of pairwise disjoint diffeomorphic copies $\mathcal{M}_{\text{ext}}^\lambda$, $\lambda \in \Lambda$, of $\mathcal{M}_{\text{ext}}$, for some index set Λ . Each $\mathcal{M}_{\text{ext}}^\lambda$ comes with an associated open domain of outer communications $\langle\langle \mathcal{M}_{\text{ext}}^\lambda \rangle\rangle \subset \mathcal{M}$. As in the proof of Theorem 3.3.9, the $\langle\langle \mathcal{M}_{\text{ext}}^\lambda \rangle\rangle$'s form an open cover of $\widehat{\mathcal{M}}$. Moreover, by Theorem 3.3.12 they are pairwise disjoint. Connectedness of $\widehat{\mathcal{M}}$ implies that Λ is a singleton $\{\lambda_*\}$, with $\widehat{\mathcal{M}} = \langle\langle \mathcal{M}_{\text{ext}}^{\lambda_*} \rangle\rangle$, hence $\widehat{\mathcal{M}} = \mathcal{M}$, and the result follows. $\square$

Existence of twist potentials. We turn now our attention to the question of the existence of twist potentials. The problem is the following: suppose that ω is a closed one form on a domain of outer communications $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$. For $i = 1, \dots, r$ let X_i be the basis of a Lie algebra of Killing vectors generating a connected group G of isometries and suppose that

$$\forall i \quad \mathcal{L}_{X_i} \omega = 0 = \omega(X_i). \quad (3.3.21)$$

INCIDENTALLY: The application of main interest here is to the following: in spacetime dimension $r + 3$, the twist forms associated to the Killing vector fields X_i are defined as

$$\omega_i = *(dX_i^b \wedge X_1^b \wedge \dots \wedge X_r^b), \quad (3.3.22)$$

in which case the functions ν_i such that $\omega_i = d\nu_i$ are called the *twist potentials*. $\square$

If $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ is simply connected and (3.3.21) holds, then there exists a G -invariant function v such that

$$\omega = dv.$$

More generally, if $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$ is a simply connected *manifold*, then ω descends to a closed one form on $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$, and again existence of the potential v follows.

Let us show that the condition that $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$ is a manifold can be replaced by the weaker condition, that the projection map $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \rightarrow \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$ has the *path homotopy lifting property*, namely, every homotopy of paths in $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$ can be lifted to a continuous family of paths in $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$.

PROPOSITION 3.3.17 *If $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle/G$ is simply connected, and if the path homotopy lifting property holds, then there exists a G -invariant function v on $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ so that $\omega = dv$.*

Proof. To simplify notations, let $\mathcal{M} = \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ with the induced metric. Choose a point $p \in \mathcal{M}$, let $\gamma : [0, 1] \rightarrow \mathcal{M}$ be any path with $\gamma(0) = p$, and set

$$v_\gamma = \int_\gamma \omega.$$

We need to show that $v_\gamma = 0$ whenever $\gamma(0) = \gamma(1)$. Let $\tilde{\gamma}^b$ be the projection to $\mathcal{M}/G$ of a loop $\tilde{\gamma}$ through p ; since $\mathcal{M}/G$ is simply connected, there exists a continuous one-parameter family of paths γ_t^b , $t \in [0, 1]$, so that $\gamma_0^b = \tilde{\gamma}^b$, $\gamma_t^b(0) =$

$\gamma_t^b(1) = \hat{\gamma}^b(0)$, $\gamma_1^b(s) = \hat{\gamma}^b(0)$. Let γ_t be any continuous lift of γ_t^b to $\mathcal{M}$ which is also continuous in t , such that $\gamma_t(1) = p$. Then $\gamma_t(0) = g_t p$ for some continuous $g_t \in G$. We can thus obtain a closed path through p , denoted by $\hat{\gamma}_t$, by following γ_t from p to $\gamma_t(0)$, and then following the path

$$[0, t] \ni s \mapsto g_{t-s} p.$$

Since γ_1 is trivial, so is $\hat{\gamma}_1 = \gamma_1$, so that $v_{\hat{\gamma}_1} = 0$. The family $\hat{\gamma}_t$ provides thus a homotopy of $\hat{\gamma}_0$ with $\hat{\gamma}_1$, and by homotopy invariance

$$0 = v_{\gamma_1} = v_{\hat{\gamma}_1} = v_{\hat{\gamma}_0} = v_{\gamma_0}.$$

Next, using the fact that both γ_0 and $\hat{\gamma}$ project to $\hat{\gamma}^b$, we will show that

$$v_{\gamma_0} = v_{\hat{\gamma}}, \quad (3.3.23)$$

which will establish the result.

Let $s \in [0, 1]$, set $r := \hat{\gamma}(s)$, let $\mathcal{O}_r \subset \mathcal{O}$ denote any sufficiently small simply connected neighbourhood of r , and let v_r denote the solution on $\mathcal{O}_r$ of

$$dv_r = \omega, \quad v_r(r) = 0. \quad (3.3.24)$$

Let $\mathcal{U}_r = G\mathcal{O}_r$ be the orbit of G through $\mathcal{O}_r$; for $p' \in \mathcal{U}_p$ there exists $\hat{p} \in \mathcal{O}_r$ and $g \in G$ such that $p' = g\hat{p}$. Set $v_r(p') := v_r(\hat{p})$; this is well defined as the right-hand side is independent of the choice of g and q by (3.3.21). Then v_r is a solution of (3.3.24) on $\mathcal{U}_r$, and for all s such that $\hat{\gamma}(s) \in \mathcal{U}_r$ we have

$$v_r(\hat{\gamma}(s)) = v_r(\gamma_0(s)).$$

It follows that for any interval $[s_1, s_2]$ such that $\hat{\gamma}([s_1, s_2]) \subset \mathcal{O}_r$ we have

$$\int_{\hat{\gamma}([s_1, s_2])} \omega = v_r(\hat{\gamma}(s_2)) - v_r(\hat{\gamma}(s_1)) = v_r(\gamma_0(s_2)) - v_r(\gamma_0(s_1)) = \int_{\gamma_0([s_1, s_2])} \omega.$$

A covering argument finishes the proof. $\square$

3.3.7 Weakly future trapped surfaces are invisible

One of the key applications of the ideas above is the following result.

THEOREM 3.3.18 (Invisibility theorem) *Let $(\mathcal{M}, g)$ admit a future conformal completion*

$$\tilde{\mathcal{M}} = \mathcal{M} \cup \mathcal{I}^+,$$

where $\mathcal{I}^+$ is a connected null hypersurface, and suppose that

$$\tilde{\mathcal{D}} := \mathcal{D} \cup \mathcal{I}^+, \quad \text{where } \mathcal{D} := I^-(\mathcal{I}^+) \cap \mathcal{M},$$

is globally hyperbolic. If the null energy condition holds on $\mathcal{D}$, then there are no compact weakly future-trapped spacelike submanifolds of codimension two within $\mathcal{D}$.

INCIDENTALLY: Theorem 3.3.18 is [124, Theorem 6.1] with the (not-made-clear there) notion of ‘regular $\mathcal{I}$ ’ being captured here by Definition 3.1.1, together with the observation in Lemma 3.3.20 that the ‘ i^0 -avoidance condition’ of [124, Theorem 6.1] already follows² from hypothesis 1 of [124, Theorem 6.1]. $\square$

REMARKS 3.3.19 1. If $\mathcal{M} \cup \mathcal{I}^+$ is globally hyperbolic, then $\tilde{\mathcal{D}}$ also is.

²Note that for spacelike $\mathcal{I}$ s the property needed for topological censorship, that the future of a trapped surface does *not* include the whole of $\mathcal{I}^+$, cannot be inferred from causality conditions; see [122].

2. Theorem 3.3.18 is often invoked for surfaces which are inner-trapped rather than weakly inner-trapped. In such a case its proof can be simplified by invoking [36, Propositions 12.32 and 12.33], or [383, Proposition 48, p. 296] (compare [204, Section 2]), rather than using the weak comparison principle for null hypersurfaces to exclude the borderline cases.
3. Compare Section 3.5.1 for a discussion of issues that arise in a related context.

Proof of Theorem 3.3.18. We start with a lemma, the conclusion of which is known as the ‘ i^0 -avoidance condition’.

LEMMA 3.3.20 *Under the hypotheses of Theorem 3.3.18, for any compact set $K \subset \mathcal{D}$, $J^+(K, \tilde{\mathcal{D}})$ does not contain all of $\mathcal{I}^+$.*

Proof. Let $p \in J^+(K, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$. By global hyperbolicity $J^-(p, \tilde{\mathcal{D}}) \cap J^+(K, \tilde{\mathcal{D}})$ is compact. Suppose that $J^+(K, \tilde{\mathcal{D}})$ contains all of $\mathcal{I}^+$; then the generator of $\mathcal{I}^+$ through p is contained in the compact set $J^-(p, \tilde{\mathcal{D}}) \cap J^+(K, \tilde{\mathcal{D}})$, which is impossible in a globally hyperbolic, hence strongly causal, spacetime. $\square$

Returning to the proof of Theorem 3.3.18, we note that the global hyperbolicity of $\tilde{\mathcal{D}}$ implies that $\tilde{\mathcal{D}}$ is causally simple, i.e., that there are no closed causal curves and the sets of the form $J^+(K, \tilde{\mathcal{D}})$ are closed in $\tilde{\mathcal{D}}$ for all compact sets K . Suppose S is a compact future weakly trapped submanifold in $\mathcal{D}$. Let q be a point on

$$\partial(J^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+) = \dot{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+,$$

which is non-empty by Lemma 3.3.20. Since $\dot{J}^+(S, \tilde{\mathcal{D}}) = J^+(S, \tilde{\mathcal{D}}) \setminus I^+(S, \tilde{\mathcal{D}})$, there exists an achronal null geodesic $\gamma : [a, b] \rightarrow \tilde{\mathcal{D}}$, with $\gamma(a) \in S$ and $\gamma(b) = q$, emanating orthogonally from S , without S -conjugate points on $[a, b]$. In particular, $\dot{J}^+(S)$ is a smooth null hypersurface near $\gamma([a, b])$. Below we show that for a suitably chosen point $q \in \dot{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$, there exists a spacelike hypersurface S_+ in $\mathcal{I}^+$ that passes through q and does not meet $I^+(S, \tilde{\mathcal{D}})$. Given this, the proof may now be completed along the lines of the proof of Proposition 3.3.6. Since S_+ does not meet $I^+(S, \tilde{\mathcal{D}})$, one easily argues that $\dot{J}^-(S_+)$ is a smooth null hypersurface near S_+ that contains a segment $\gamma([b - \varepsilon, b])$ and lies to the causal past of $\dot{J}^+(S)$.

Let $\tilde{K}$ be a future-directed outward pointing null vector at q orthogonal to S_+ in the unphysical metric $\tilde{g} = \Omega^2 g$. Since Ω decreases to the future along γ near q , we can choose $\tilde{K}$ so that

$$\tilde{K}(\Omega) = \tilde{g}(\tilde{K}, \tilde{\nabla} \Omega) = -1.$$

Now extend $\tilde{K}$ to a null vector field tangent to $\dot{J}^-(S_+)$ near q , and let $K = \Omega \tilde{K}$. Let $e_a = e_a^\mu \partial_\mu$, $a = 0, \dots, n$ be a g -orthonormal frame with e_a , $a = 1, \dots, n-1$ spacelike and tangent to S_+ . Set $\tilde{e}_a := \Omega^{-1} e_a$; then $\tilde{e}_a$, $a = 0, \dots, n$ is $\tilde{g}$ -orthonormal. The divergence θ of S_+ with respect to K is defined as

$$\theta = \sum_1^{n-1} g(\nabla_{e_a} K, e_a). \quad (3.3.25)$$

The rescaling $\tilde{g} = \Omega^2 g$ implies that

$$\Gamma_{\nu\rho}^\mu = \tilde{\Gamma}_{\nu\rho}^\mu - \Omega^{-1}(\delta_\nu^\mu \partial_\rho \Omega + \delta_\rho^\mu \partial_\nu \Omega - \tilde{\nabla}^\mu \Omega \tilde{g}_{\nu\rho}), \quad (3.3.26)$$

which gives

$$\theta = \sum_1^{n-1} g(e_a^\rho (\partial_\rho K^\mu + \Gamma_{\nu\rho}^\mu K^\nu) \partial_\mu, e_a)$$

$$\begin{aligned}
&= \sum_1^{n-1} g(\tilde{\nabla}_{e_a} K - \Omega^{-1} e_a^\rho (\delta_\rho^\mu \partial_\rho \Omega + \delta_\rho^\mu \partial_\nu \Omega - \tilde{\nabla}^\mu \Omega \tilde{g}_{\nu\rho}) K^\nu) \partial_\mu, e_a) \\
&= \sum_1^{n-1} g(\tilde{\nabla}_{e_a} K - \Omega^{-1} (e_a(\Omega) K + K(\Omega) e_a - \underbrace{\tilde{g}(e_a, K)}_0) \tilde{\nabla} \Omega, e_a) \\
&= \sum_1^{n-1} g(\tilde{\nabla}_{e_a} (\Omega \tilde{K}) - \Omega^{-1} K(\Omega) e_a, e_a) = \sum_1^{n-1} \Omega^{-1} \tilde{g}(\tilde{\nabla}_{e_a} \tilde{K}, e_a) - (n-1) \tilde{K}(\Omega) \\
&= \Omega \sum_1^{n-1} \tilde{g}(\tilde{\nabla}_{\tilde{e}_a} \tilde{K}, \tilde{e}_a) - (n-1) \tilde{K}(\Omega). \tag{3.3.27}
\end{aligned}$$

Denoting by $\tilde{\theta}$ the divergence of $\tilde{J}^-(S_+)$ with respect to $\tilde{K}$ in the unphysical metric $\tilde{g}$, this can be rewritten as

$$\theta = -(n-1) \tilde{K}(\Omega) + \Omega \tilde{\theta} = (n-1) + \Omega \tilde{\theta}. \tag{3.3.28}$$

It follows that $\tilde{J}^-(S_+)$ will have positive null divergence at points of γ close to q . On the other hand, as in the proof of Proposition 3.3.6, $\tilde{J}^+(S)$, has nonpositive null divergence along γ , and we are again led to a contradiction with the maximum principle for null hypersurfaces.

We conclude the proof by explaining how to choose q and S_+ . For this purpose we introduce a Riemannian metric on $\mathcal{I}^+$, with respect to which the following constructions are carried out. Fix $q_0 \in \tilde{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$, and let $U \subset \mathcal{I}^+$ be a convex normal neighbourhood of q_0 . By choosing a point $p \in U$, $p \notin J^+(S, \tilde{\mathcal{D}})$, sufficiently close to q_0 , we obtain a point $q \in \tilde{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$, such that the geodesic segment $\overline{pq}$ in U realizes the distance from p to $\tilde{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$. Now let S_+ be the distance sphere in U centred at p and passing through q . S_+ is a smooth hypersurface in $\mathcal{I}^+$ that does not meet $I^+(S, \tilde{\mathcal{D}})$. It follows that S_+ intersects the generator of $\mathcal{I}^+$ through q transversely, and hence is spacelike near q . To see this, let γ be the null geodesic from S to q as in the preceding paragraph. For $x \in \gamma$ sufficiently close to q , $S' = \tilde{J}^+(x, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$ will be, in the vicinity of q , a smooth hypersurface in $\mathcal{I}^+$ transverse to the null generator of $\mathcal{I}^+$ through q . But, since $S' \subset J^+(S, \tilde{\mathcal{D}})$, S_+ must meet S' tangentially at q . Hence q is the desired point in $\tilde{J}^+(S, \tilde{\mathcal{D}}) \cap \mathcal{I}^+$ and S_+ , suitably restricted, is the desired spacelike hypersurface in $\mathcal{I}^+$. $\square$

Essentially identical proofs provide the following variations of Theorem 3.3.18 for spacetimes $(\mathcal{M}, g)$ which are asymptotically anti-de Sitter (in which case $\mathcal{I}$ is timelike) or which contain KK -asymptotic ends.

THEOREM 3.3.21 *Let $(\mathcal{M}, g)$ be an asymptotically anti-de Sitter spacetime with the asymptotic estimates uniform to order one. If $(\mathcal{M}, g)$ contains a globally hyperbolic domain of outer communications $\langle\langle \mathcal{I} \rangle\rangle$ on which the null energy condition holds, then there are no compact future weakly trapped submanifolds within $\langle\langle \mathcal{I} \rangle\rangle$.* $\square$

THEOREM 3.3.22 *Let $(\mathcal{M}, g)$ be a KK -asymptotically flat or KK -asymptotically anti-de Sitter spacetime with the asymptotic estimates uniform to order one. If $(\mathcal{M}, g)$ contains a globally hyperbolic domain of outer communications $\langle\langle \mathcal{I} \rangle\rangle$ on which the null energy condition holds, then there are no compact future weakly trapped submanifolds within $\langle\langle \mathcal{I} \rangle\rangle$.* $\square$

REMARK 3.3.23 Theorems 3.3.18 and 3.3.22 may be adapted to rule out the visibility from an infinity of submanifolds S bounding compact acausal hypersurfaces $\mathcal{S}$ with weakly outer-future-trapped boundary. In fact $\mathcal{S}$ is allowed to have non-weakly outer-trapped components of the boundary as long as those lie in a black-hole region. Here the outer direction at S is defined as pointing away from $\mathcal{S}$. $\square$

For further purposes we will need yet another variation of Theorem 3.3.18. Consider a compact codimension-two spacelike submanifold of $\mathcal{M}$, say S , and let us suppose that S is two-sided, in the sense of existence of a well-defined field of spacelike normals. Let us choose to denote one of those sides ‘inner’. Instead of supposing, as in Theorem 3.3.18, that both families of null future-directed geodesics are weakly trapped at S , we will only assume that the inner family is weakly trapped. The proof of Theorem 3.3.18 applies as is to this setting to give the following.

THEOREM 3.3.24 *Under the hypotheses of Theorem 3.3.18, let S be a future weakly inner-trapped submanifold contained in $\mathcal{D}$. Then there are no null geodesics on $j^+(S)$ connecting $\mathcal{I}^+$ with the inner side of S .* $\square$

3.3.8 Spacetimes with a conformal completion at null infinity

The setting for the original topological censorship argument of Friedman, Schleich, and Witt [194] is that of asymptotically flat spacetimes in the sense of Definition 3.1.1. We therefore turn our attention now to such spacetimes.

Uniform Scris. The simplest version of topological censorship in the current context is obtained by taking the Kaluza–Klein group G in Theorem 3.3.15 to be trivial, and rephrasing its hypotheses in terms of Penrose’s conformal framework. For this some terminology is useful.

We will say that $\mathcal{S}$ satisfies *asymptotic estimates uniform to order one* if there exists:

1. a ‘radial function’ r and a constant R_0 such that the set $\{r \geq R_0\}$ forms a neighbourhood of $\mathcal{S}$, with the level sets of r *timelike* there, and
2. a time function t on $\mathcal{M}$ such that for all $r \geq R_0$ and for all $\tau \in \mathbb{R}$ the sets

$$\mathcal{S}_{r,\tau} := \{r = R, t = \tau\}$$

are *smooth past and future-inner-trapped compact spacelike submanifolds* of $\mathcal{M}$.

The terminology is related to the formula (3.3.28) for the conformal transformation of the expansion scalar of null hypersurfaces, which guarantees the above in spacetimes which are asymptotically flat in a coordinate sense as in Remark 3.3.11, *provided that* the metric and its first derivatives satisfy bounds which are uniform in time in the asymptotic region $\{r \geq R_0\}$.

We have the following rewording of Theorem 3.3.15.

THEOREM 3.3.25 *Let $(\mathcal{M}, g)$ be a spacetime satisfying the null energy condition, i.e.*

$$R_{\mu\nu}X^\mu X^\nu \geq 0 \text{ for all null vector fields } X, \quad (3.3.29)$$

and admitting a conformal completion with null boundary $\mathcal{S}$ satisfying asymptotic estimates uniform to order one.

If the domain of outer communications $\langle\langle \mathcal{S} \rangle\rangle$ is globally hyperbolic and if there exists $R_1 \geq R_0$ such that the set $\{r \geq R_1\}$ is simply connected, then $\langle\langle \mathcal{S} \rangle\rangle$ is simply connected. $\square$

Scris which are not necessarily uniform. We have the following alternative to Theorem 3.3.15, essentially due to Friedman, Schleich, and Witt.

THEOREM 3.3.26 *Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime satisfying the null energy condition. Assume that $(\mathcal{M}, g)$ admits a conformal completion at infinity $(\tilde{\mathcal{M}}, \tilde{g})$ such that both $\mathcal{I}^+$ and $\mathcal{I}^-$ are connected null hypersurfaces, and that*

$$\tilde{\mathcal{D}} := \mathcal{D} \cup \mathcal{I}^+, \text{ where } \mathcal{D} := I^-(\mathcal{I}^+) \cap \mathcal{M} \quad (3.3.30)$$

is globally hyperbolic. Suppose that there exists a simply connected neighbourhood $\mathcal{U}$ of $\mathcal{S}$ and a foliation of a (perhaps smaller) neighbourhood of $\mathcal{I}^-$ with spacelike

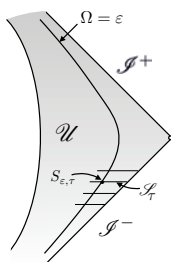


Fig. 3.3.1 The sets $\mathcal{U}$, $\mathcal{S}_\tau$, and $S_{\tau, \varepsilon}$.

up-to-boundary and smooth up-to-boundary acausal hypersurfaces $\mathcal{S}_\tau \subset \tilde{\mathcal{M}}$, $\tau \in R$, such that for each τ the intersections, say $S_{\tau, \varepsilon}$, of $\mathcal{S}_\tau$ with the ε -level sets of the conformal factor Ω

are non-empty, smooth, connected, and compact for all $\varepsilon \geq 0$ small enough. (3.3.31)

(See Figure 3.3.1.) If, again for each τ and for all ε small enough,

the null geodesics normal to $S_{\tau, \varepsilon}$ with $\dot{\Omega}|_{S_{\tau, \varepsilon}} < 0$ never leave $\mathcal{U}$, (3.3.32)

where $\dot{\Omega}$ denotes the derivative of Ω along the geodesic, then the domain of outer communications $I^+(\mathcal{J}^-) \cap I^-(\mathcal{J}^+)$ is simply connected.

REMARK 3.3.27 The key property of the collection of smooth compact manifolds $S_{\tau, \varepsilon}$ is that they are future-inner trapped, cover a neighbourhood of $\mathcal{J}^-$, and satisfy (3.3.32). There are many ways of obtaining such a family.

First, one could, e.g., assume that there exists a foliation of $\mathcal{J}^-$ by smooth compact submanifolds, each intersecting the generators of $\mathcal{J}^-$ exactly once, and extend this foliation to a foliation of a neighbourhood of $\mathcal{J}^-$ in $\mathcal{M}$ by moving along a field of spacelike directions, leading to the family $\mathcal{S}_\tau$ of the theorem.

Next, the manifolds $S_{\tau, \varepsilon}$ can then, e.g., be obtained by moving a $\tilde{g}$ -distance ε away from $\mathcal{J}^-$ along $\mathcal{S}_\tau$, rather than intersecting with the level sets of Ω .

Finally, one could define an alternative τ -foliation, as needed in the statement of the theorem, by extending the just-mentioned foliation of $\mathcal{J}^-$ into $\mathcal{M}$ along null rather than spacelike hypersurfaces $\mathcal{S}_\tau$, with the manifolds $S_{\tau, \varepsilon}$ obtained by moving cuts of $\mathcal{J}^-$ an affine-parameter-distance ε along the generators of $\mathcal{S}_\tau$.

In each case the resulting $S_{\tau, \varepsilon}$'s will be future-inner-trapped for ε small enough. But note that in each case the additional condition (3.3.32) needs to be imposed. $\square$

REMARK 3.3.28 The proof of Theorem 3.3.26 actually requires the weaker, but somewhat less transparent condition that for each τ and for all ε small enough

those generators of $\mathcal{J}^+(S_{\tau, \varepsilon})$ on which $\frac{d\Omega}{ds}|_{S_{\tau, \varepsilon}} < 0$ never leave $\mathcal{U}$. (3.3.33)

We note that (3.3.32) clearly implies (3.3.33). $\square$

REMARK 3.3.29 For simplicity we assume that both metrics $\tilde{g}$ and g are smooth up to boundary. One can check, using e.g. [125] and the conformal transformation formula (3.3.28) for the divergence of null hypersurfaces, that the proof applies if g is C^2 on $\mathcal{M}$ and $\tilde{g}$ is C^1 up-to-boundary. $\square$

REMARK 3.3.30 The theorem remains true, with an essentially identical proof, when the null energy condition is replaced by the following integrated null energy condition: for all half-geodesics γ that start near $\mathcal{J}^-$ it holds that

$$\int_0^\infty \text{Ric}(\gamma', \gamma') ds \geq 0;$$

compare [208]. $\square$

Proof of Theorem 3.3.26. Suppose that the domain of outer communications $\langle\langle \mathcal{I} \rangle\rangle$ is non-empty and not simply connected; otherwise, there is nothing to prove.

Let $\mathcal{M}_1$ be the universal cover of $\langle\langle \mathcal{I} \rangle\rangle \cup \mathcal{I}$ equipped with the covering metrics which we continue to denote by g and $\tilde{g}$. The covering manifold $\mathcal{M}_1$ contains several copies, say $\mathcal{U}_\alpha$, with $\alpha \in A$ for some index set A , of the simply connected neighbourhood $\mathcal{U}$ of the original $\mathcal{I}$. Each $\mathcal{U}_\alpha$ comes with its adjacent conformal boundaries $\mathcal{I}_\alpha^+$, $\mathcal{I}_\alpha^-$ and $\mathcal{I}_\alpha := \mathcal{I}_\alpha^+ \cup \mathcal{I}_\alpha^-$.

Let us choose two distinct lifts of $\mathcal{U}$, call them $\mathcal{U}_1$ and $\mathcal{U}_2$, with conformal boundaries $\mathcal{I}_1$ and $\mathcal{I}_2$, and suppose that there exists a causal curve, say $\tilde{\gamma}$, connecting $\mathcal{I}_1^-$ with $\mathcal{I}_2^+$. Let $\tilde{\tau}$ be such that $\tilde{\gamma}$ meets $\mathcal{I}_1^-$ at $\mathcal{S}_{\tilde{\tau}}$. We can choose a small parameter $\epsilon > 0$ and a value of τ_0 near $\tilde{\tau}$ so that

1. $\tilde{\gamma}$ intersects $\mathcal{S}_{\tau_0} \cap \{\Omega \leq \epsilon\} \cap \mathcal{M}$,
2. $S := \mathcal{S}_{\tau_0} \cap \{\Omega = \epsilon\}$ is compact,
3. (3.3.32) holds, and
4. S is inner-future-trapped, in the sense that it has negative divergence on those null geodesics which initially move normally away from S to the future and in the direction of increasing Ω . This follows from (3.3.28), which provides the following transformation formula³ for the future and past divergences $\theta^\pm$, in spacetime dimension $n + 1$,

$$\theta^\pm = (n - 1)(\tilde{T} \pm \tilde{N})(\Omega) + \Omega \tilde{\theta}^\pm, \quad (3.3.34)$$

where $\tilde{\theta}^\pm$ are the divergences calculated using the metric $\tilde{g}$, $\tilde{T}$ is the field of unit future-directed normals to $\mathcal{S}_{\tau_0}$, and $\tilde{N}$ is the field of outer-pointing normals to the level sets of Ω within $\mathcal{S}_{\tau_0}$. By hypothesis $\nabla\Omega$ is null and past-directed on $\mathcal{I}^-$, tangent to $\mathcal{I}^-$, while $(\tilde{T} \pm \tilde{N})$ is null, future directed, and transverse to $\mathcal{I}^-$. This shows that $\theta^\pm$ is positive, bounded away from 0 on $\mathcal{S}_{\tau_0} \cap \mathcal{I}^-$, hence positive for all ϵ small enough.

Let S_1 denote the lift of S to $\mathcal{U}_1$. We claim that S_1 is a future-outer-trapped surface as seen from $\mathcal{I}_2^+$. For this, note that the achronal boundary $\dot{J}^+(S_1)$ forms near S_1 a union of two smooth null hypersurfaces ruled by null geodesic segments normal to S_1 . Let us denote by $\mathcal{N}_1^L$ the subset of $\dot{J}^+(S_1)$ on which Ω initially increases when moving away from S_1 along the generators, and $\mathcal{N}_1^R$ the subset on which Ω initially decreases. As in Theorem 3.3.18 there exists a null geodesic γ in $\dot{J}^+(S_1)$ connecting S_1 to $\mathcal{I}_2^+$. The geodesic γ intersects $\partial\mathcal{U}_1$, and therefore cannot lie on $\mathcal{N}_1^R$ which is entirely included in $\mathcal{U}_1$ by (3.3.32).

Hence γ lies on $\mathcal{N}_1^L$, which has negative divergence on S_1 , as claimed. But this contradicts Theorem 3.3.24. We conclude that

$$I^-(\mathcal{I}_\alpha^+) \cap I^+(\mathcal{I}_\beta^-) = \emptyset \quad \text{for } \alpha \neq \beta. \quad (3.3.35)$$

To finish the proof, note that the collection of open sets

$$I^-(\mathcal{I}_\alpha^+) \cap I^+(\mathcal{I}_\alpha^-) \quad \alpha \in \Omega$$

covers $\tilde{\mathcal{M}}_1$. The sets are pairwise disjoint by (3.3.35). Since $\tilde{\mathcal{M}}_1$ is connected there is only one such set, contradicting the assumption of non-simple connectivity, and Theorem 3.3.26 is established. $\square$

³We take this opportunity to correct Eq. (4.10) in [124], which should be replaced by (3.3.34). This change has no impact on the remaining arguments there.

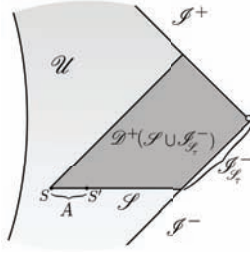


Fig. 3.3.2 The sets $\mathcal{U}$, $S \equiv S_{\tau,\varepsilon}$, $\mathcal{S} \equiv \mathcal{S}_{\tau,\varepsilon} := \mathcal{S}_{\tau} \cap \{\Omega \leq \varepsilon\}$, $\mathcal{J}_{\tau}^{-}$ and $\mathcal{D}^{+}(\mathcal{S} \cup \mathcal{J}_{\tau}^{-})$.

The least satisfactory condition in Theorem 3.3.26 appears to be (3.3.32), and the question arises whether it can be replaced by other natural conditions. We list here some alternatives; we have not been able to verify that (3.3.32) can be gotten rid of altogether.

PROPOSITION 3.3.31 *Under the remaining hypotheses of Theorem 3.3.26, suppose that instead of (3.3.32) we have*

1. *either for each τ and for some ε small enough the connected component of the level set $\{\Omega = \varepsilon\}$ containing $S_{\tau,\varepsilon}$ (cf. Fig. 3.3.1) is a timelike hypersurface entirely contained in $\mathcal{U}$ and separating $\mathcal{U}$ (compare Lemma 1 in [194]);*
2. *or the set*

$$\tilde{\mathcal{D}}^{-} := \mathcal{D}^{-} \cup \mathcal{J}^{-}, \quad \text{where } \mathcal{D}^{-} = I^{+}(\mathcal{J}^{-}) \cap \mathcal{M} \quad (3.3.36)$$

is also globally hyperbolic, and there exists a foliation $\mathcal{S}_{\tau}$ as in Theorem 3.3.26 such that for all τ and for some ε small enough we have a

$$\overline{\mathcal{D}^{+}((\mathcal{S}_{\tau} \cap \{\Omega \leq \varepsilon\}) \cup \mathcal{J}_{\tau}^{-})} \subset \mathcal{U}, \quad (3.3.37)$$

where $\mathcal{J}_{\tau}^{-} := \mathcal{J}^{-} \cap J^{+}(\mathcal{S}_{\tau})$ (cf. Fig. 3.3.2).

Then the conclusions of Theorem 3.3.26 hold.

Proof. 1. We wish to show that (3.3.33) holds. For this, let us denote by

$$\{\Omega = \varepsilon\}_0$$

that component of $\{\Omega = \varepsilon\}$ which contains $S_{\tau,\varepsilon}$. It is, by hypothesis, a timelike hypersurface. It contains a compact achronal spacelike surface $S_{\tau,\varepsilon}$ and therefore constitutes a globally hyperbolic spacetime of its own, with Cauchy surface $S_{\tau,\varepsilon}$, by [69, 202]. Since $\{\Omega = \varepsilon\}_0$ separates $\mathcal{U}$ by hypothesis, every generator of $J^{+}(S_{\tau,\varepsilon})$ which starts with $\dot{\Omega} < 0$ at $S_{\tau,\varepsilon}$ and which leaves $\mathcal{U}$ must cross $\{\Omega = \varepsilon\}_0$ again. But global hyperbolicity of $\{\Omega = \varepsilon\}_0$ implies that through every point of $\{\Omega = \varepsilon\}_0$ there exists a timelike curve meeting $S_{\tau,\varepsilon}$, which is not possible since the generators of $J^{+}(S_{\tau,\varepsilon})$ do not intersect $I^{+}(S_{\tau,\varepsilon})$.

2. Let τ_0 , as well as the remaining notation, be as in the proof of Theorem 3.3.26.

Let $\mathcal{S}_{\tau_0,\varepsilon} = \mathcal{S}_{\tau_0} \cap \{\Omega \leq \varepsilon\}$, and set $\widehat{\mathcal{S}} = \mathcal{S}_{\tau_0,\varepsilon} \cup \mathcal{J}_{\tau_0}^{-}$. From (3.5) we have

$$\overline{\mathcal{D}^{+}(\widehat{\mathcal{S}})} \subset \mathcal{U}. \quad (3.3.38)$$

Note that the manifold $S \equiv \mathcal{S}_{\tau_0} \cap \{\Omega = \varepsilon\}$ of the proof of Theorem 3.3.26 satisfies $S = \text{edge}(\widehat{\mathcal{S}})$. Let $A \subset \mathcal{S}_{\tau_0,\varepsilon} \cap \mathcal{M}$ be a smooth compact manifold diffeomorphic to $[0, 1] \times S$ with boundary $\partial A = S \cup S'$; see Fig. 3.3.2.

Suppose $\langle\langle \mathcal{S} \rangle\rangle$ is not simply connected. Let $\mathcal{U}_1$ and $\mathcal{U}_2$ be as in the proof of Theorem 3.3.26. Let $S_1, S'_1, \mathcal{A}_1, \widehat{\mathcal{S}}_1$, and A_1 be the lifts of $S, S', \mathcal{S}_{\tau_0,\varepsilon}, \widehat{\mathcal{S}}$, and A ,

respectively, to $\mathcal{U}_1$. We may assume that $J^+(A_1)$ meets $\mathcal{S}_2^+$ but, by Lemma 2.4, does not contain all of it. Then there exists a future complete null geodesic γ on $\widehat{J}(A_1)$ that extends from $\mathcal{S}_2^+$ to either S_1 or S'_1 when followed to the past.

Suppose, first, that γ extends to S'_1 . Then it will have to cross $D^+(\widehat{\mathcal{S}}_1) \subset \mathcal{U}_1$ at some point, say $p \in H^+(\widehat{\mathcal{S}}_1) \cap \mathcal{M}$ (not on S_1). Let us denote by $\hat{\gamma}$ a generator of $H^+(\widehat{\mathcal{S}}_1)$ passing through p .

Now, generators of $H^+(\widehat{\mathcal{S}}_1)$ either are past inextendible or have an end point on the edge of $\widehat{\mathcal{S}}_1$. We wish to show that $\hat{\gamma}$ extends to S_1 , and therefore the former case cannot occur. In order to see this, we first note that any point q lying on a generator of $H^+(\widehat{\mathcal{S}}_1)$ is in $J^+(\mathcal{S}_1, \widehat{\mathcal{D}}_1^-)$, where

$$\widehat{\mathcal{D}}_1^- := \mathcal{D}_1^- \cup \mathcal{S}_1^-, \quad \text{with } \mathcal{D}_1^- := I^+(\mathcal{S}_1^-) \cap \mathcal{M}.$$

Indeed, since $q \in H^+(\widehat{\mathcal{S}}_1)$ there exists a causal curve connecting q with $\widehat{\mathcal{S}}_1 = \mathcal{S}_1 \cup \mathcal{S}_{\widehat{\mathcal{S}}}$. If the curve connects q with $\mathcal{S}_1$ there is nothing to prove; otherwise, it connects to $\mathcal{S}_{\widehat{\mathcal{S}}}$, but then we can concatenate it with a segment of generator of $\mathcal{S}_1^-$ which slides down from the intersection point to $\mathcal{S}_{\widehat{\mathcal{S}}}$. We conclude that the part of the generator of $H^+(\widehat{\mathcal{S}}_1)$ which lies to the past of p is included in

$$\mathcal{K} := J^-(p, \widehat{\mathcal{D}}_1^-) \cap J^+(\mathcal{S}_1, \widehat{\mathcal{D}}_1^-).$$

The set $\mathcal{K}$ is compact by global hyperbolicity of $\widehat{\mathcal{D}}_1^-$. Since $\hat{\gamma}$ is contained in the compact set $\mathcal{K}$, it has to end on the edge of $\widehat{\mathcal{S}}_1$ to avoid a violation of strong causality, as claimed.

Now, since the edge of $\widehat{\mathcal{S}}_1$ is S_1 , $\hat{\gamma}$ clearly enters the timelike future of A_1 . But this implies that γ enters the timelike future of A_1 , which is not possible for a curve lying on $\widehat{J}^+(A_1)$, a contradiction. Thus γ extends to S_1 , and necessarily to the inside. Hence γ lies on $\mathcal{N}_1^L$, and the proof continues as in the proof of Theorem 3.3.26. $\square$

3.4 Incompleteness theorems

One of the first successes of causality theory were the so-called *singularity theorems*, which are, in fact, *incompleteness theorems*. We will review two such theorems here, especially relevant for black-hole spacetimes. The reader is referred to [68, 231, 309, 354, 355, 448] and references therein for further results in this vein.

We say that $(\mathcal{M}, g)$ satisfies the *timelike focusing condition*, or *timelike convergence condition*, if the Ricci tensor satisfies

$$R_{\mu\nu}n^\mu n^\nu \geq 0 \quad \text{for all timelike vectors } n^\mu. \quad (3.4.1)$$

By continuity, the inequality in (3.4.1) will also hold for causal vectors. Condition (3.4.1) can of course be rewritten as a condition on the matter fields using the Einstein equation, and is satisfied in many cases of interest, including vacuum general relativity, or the Einstein–Maxwell theory, or the Einstein–Yang–Mills theory. These last two examples actually have the property that the corresponding energy–momentum tensor is trace-free; whenever this happens, (3.4.1) is simply the requirement that the energy density of the matter fields is non-negative for all observers:

$$8\pi T_{\mu\nu}n^\mu n^\nu = (R_{\mu\nu} - \underbrace{\frac{1}{2}R}_{=0 \text{ if } g^{\alpha\beta}T_{\alpha\beta}=0} g_{\mu\nu})n^\mu n^\nu = R_{\mu\nu}n^\mu n^\nu. \quad (3.4.2)$$

Recall that $(\mathcal{M}, g)$ satisfies the *null energy condition* if

$$R_{\mu\nu}n^\mu n^\nu \geq 0 \quad \text{for all null vectors } n^\mu. \quad (3.4.3)$$

Clearly, the timelike focusing condition implies the null energy condition. Because $g_{\mu\nu}n^\mu n^\nu = 0$ for all null vectors, the R term in the calculation (3.4.2) drops out regardless of whether $T_{\mu\nu}$ is traceless, so the null energy condition is equivalent to the positivity of the energy density of matter fields without any provisos.

The simplest *geodesic incompleteness theorem* is the following.

THEOREM 3.4.1 (Hawking's geodesic incompleteness theorem [243]) *Let $(\mathcal{M}, g)$ be a smooth globally hyperbolic spacetime satisfying the timelike focusing condition, and suppose that $\mathcal{M}$ contains a compact Cauchy surface $\mathcal{S}$ with strictly negative mean curvature,*

$$\mathrm{tr}_h K < 0,$$

where (h, K) are the usual Cauchy data induced on $\mathcal{S}$ by g . Then $(\mathcal{M}, g)$ is future timelike geodesically incomplete.

Yet another incompleteness theorem involves *trapped surfaces*, as defined in Section 3.3.2, p. 90. The celebrated theorem of Penrose asserts the following.

THEOREM 3.4.2 (Penrose's geodesic incompleteness theorem [396]) *Let $(\mathcal{M}, g)$ be a smooth globally hyperbolic spacetime satisfying the null energy condition, and suppose that $\mathcal{M}$ contains a non-compact Cauchy surface $\mathcal{S}$. If there exists a compact trapped surface within $\mathcal{S}$, then $(\mathcal{M}, g)$ is geodesically incomplete.*

The significance of this theorem stems from the fact, that the Schwarzschild solution, as well as the non-degenerate Kerr black holes, possess trapped surfaces under the horizon. (See Example 3.3.2, p. 91.) A small perturbation of the metric will preserve this. It follows that the geodesic incompleteness of these black holes is not an accident of the large isometry group involved, but is stable under perturbations of the metric.

3.5 Area theorem

The thermodynamics of black holes rests upon *Hawking's area theorem* [244], which asserts that in appropriate spacetimes the area of cross-sections of black-hole horizons is non-decreasing towards the future. The simplest version of this theorem reads as follows.

THEOREM 3.5.1 ([115]) *Consider a spacetime $(\mathcal{M}, g)_{C^\infty}$. Let $\mathcal{E}$ be a future geodesically complete acausal null hypersurface, and let $\mathcal{S}_1, \mathcal{S}_2$ be two spacelike acausal hypersurfaces. If the null energy condition (3.4.3) holds on $\mathcal{E}$ and if*

$$\mathcal{E} \cap \mathcal{S}_1 \subset J^-(\mathcal{E} \cap \mathcal{S}_2),$$

then

$$\mathrm{Area}(\mathcal{E} \cap \mathcal{S}_1) \leq \mathrm{Area}(\mathcal{E} \cap \mathcal{S}_2).$$

The simplicity of the above relies on the very strong hypothesis of geodesic completeness. This hypothesis does not seem to be natural in view of the incompleteness theorems of Section 3.4, whence alternative hypotheses are desirable. We will present some such sets.

In what follows we will consider spacetimes with conformal completions $\mathcal{I}$ as defined in Section 3.1. The event horizon $\mathcal{H}$ can then be defined as

$$\mathcal{H} := \dot{J}^-(\mathcal{I}^+). \quad (3.5.1)$$

EXAMPLE 3.5.2 It turns out that the definition (3.5.1) is not very useful without some *completeness conditions* on $\mathcal{I}^+$. As an example, consider the standard conformal completion of the Minkowski spacetime at future null infinity $\tilde{\mathcal{M}}^+$ of Example 3.1.2. One clearly has $\mathcal{B} = \emptyset$ for this completion.

One can obtain a different completion by replacing $\mathcal{I}^+$ by an open subset thereof, e.g. restricting the range of the coordinate u in Example 3.1.2 to $(-\infty, 0)$. This will give a non-empty black-hole region $\mathcal{B}$ in Minkowski spacetime,

$$\mathcal{B} \equiv \mathcal{M} \setminus J^-(\mathcal{I}^+) = J^+(0), \quad (3.5.2)$$

which is clearly a misunderstanding. $\square$

A way out of the last problem (as well as of some related ones, discussed in [221]) has been proposed by Geroch and Horowitz in [221], for spacetimes which are vacuum near the conformal boundary, assuming that the cosmological constant is zero.⁴ As already mentioned in Section 3.1, Geroch and Horowitz introduce a preferred family of conformal factors Ω such that the matrix Hess Ω of second covariant derivatives of Ω vanishes at $\mathcal{I}$. It is then required that both $\mathcal{I}^+$ and $\mathcal{I}^-$ be diffeomorphic to $\mathbb{R} \times S^2$, with the $\mathbb{R}$ factor corresponding to integral curves of $\tilde{\nabla}\Omega$, which are further *required* to be complete. One readily checks that this enforces the range of u on the conformal boundary of Example 3.5.2 to range over $\mathbb{R}$.

In order to establish a precise version of the area theorem, the conformal completion of the spacetime under consideration will have to satisfy some further regularity conditions. To define these, some terminology is useful.

- DEFINITION 3.5.3 1. A point q in a set $A \subset B$ is said to be a *past point* of A with respect to B if $J^-(q; B) \cap A = \{q\}$.
2. We shall say that $\mathcal{I}^+$ is $\mathcal{H}$ -regular if there exists a neighbourhood $\mathcal{O}$ of $\mathcal{H}$ such that for every compact set $C \subset \mathcal{O}$ satisfying $I^+(C; \tilde{\mathcal{M}}) \cap \mathcal{I}^+ \neq \emptyset$ there exists a past point with respect to $\mathcal{I}^+$ in $\partial I^+(C; \tilde{\mathcal{M}}) \cap \mathcal{I}^+$.

When $\mathcal{I}^+$ is null or timelike, the purpose of the condition is to exclude pathological situations in which the closure of the domain of influence of a compact set C in $\mathcal{M}$ contains points which are arbitrarily far in the past on $\mathcal{I}^+$ in an uncontrollable way.

The notion of $\mathcal{H}$ -regularity provides a specific version of ‘the i^0 avoidance condition’, already mentioned in the proof of Theorem 3.3.18, p. 101.

REMARK 3.5.4 When $\mathcal{I}^+$ is *null throughout*, the condition of $\mathcal{H}$ -regularity is equivalent to the existence of a neighbourhood $\mathcal{O}$ of $\mathcal{H}$ such that for every compact set $C \subset \mathcal{O}$ satisfying $\overline{I^+(C; \tilde{\mathcal{M}})} \cap \mathcal{I}^+ \neq \emptyset$ there exists at least one generator of $\mathcal{I}^+$ which intersects $\overline{I^+(C; \tilde{\mathcal{M}})}$ and leaves it when followed to the past. Those points at which the generators of $\mathcal{I}^+$ exit $\overline{I^+(C; \tilde{\mathcal{M}})}$ are past points of $\partial I^+(C; \tilde{\mathcal{M}}) \cap \mathcal{I}^+$. $\square$

REMARK 3.5.5 When $\mathcal{I}^+$ is *timelike throughout* the condition of $\mathcal{H}$ -regularity is equivalent to the existence of a neighbourhood $\mathcal{O}$ of $\mathcal{H}$ such that for every compact set $C \subset \mathcal{O}$ satisfying $\overline{I^+(C; \tilde{\mathcal{M}})} \cap \mathcal{I}^+ \neq \emptyset$, there is a point x in $\overline{I^+(C; \tilde{\mathcal{M}})} \cap \mathcal{I}^+$ such that every past-inextendible causal curve in $\mathcal{I}^+$ from x leaves $\overline{I^+(C; \tilde{\mathcal{M}})}$. $\square$

REMARK 3.5.6 When $\mathcal{I}^+$ is *spacelike throughout* the condition of $\mathcal{H}$ -regularity is equivalent to the existence of a neighbourhood $\mathcal{O}$ of $\mathcal{H}$ such that for every compact set $C \subset \mathcal{O}$ satisfying $\overline{I^+(C; \tilde{\mathcal{M}})} \cap \mathcal{I}^+ \neq \emptyset$, the set $\overline{I^+(C; \tilde{\mathcal{M}})}$ does not contain all of $\mathcal{I}^+$. $\square$

⁴The ‘WASE setup’ of [244], reviewed in Section 3.5.1, does not solve this particular problem, cf. [221].

The simplest way to guarantee the $\mathcal{H}$ -regularity condition is to assume that the conformal completion $(\mathcal{M}, \tilde{g})$ is globally hyperbolic as a manifold with boundary. For example, Minkowski spacetime with the *standard* $\mathcal{I}^+$ attached (cf. Example 3.1.2, p. 85) is a globally hyperbolic manifold with boundary. (However, if both $\mathcal{I}^+$ and $\mathcal{I}^-$ are attached, then it is not. Further, if $\mathcal{I}^+$ and $\mathcal{I}^-$ and i^0 are attached to Minkowski spacetime, then it is *not* a manifold with boundary any more. Likewise, the conformal completions considered in [454] are not manifolds with boundary.) Similarly the Schwarzschild spacetime with the usual $\mathcal{I}^+$, as constructed in Remark 4.2.6, p. 131, attached to it is a globally hyperbolic manifold with boundary. Further, the usual conformal completions of de Sitter, or anti-de Sitter spacetime as well as those of the Kottler spacetimes (sometimes called Schwarzschild–de Sitter and Schwarzschild–anti-de Sitter spacetimes, and discussed in Section 5.5, p. 250) are globally hyperbolic manifolds with boundary.

In [117] the following version of the area theorem has been proved.

THEOREM 3.5.7 (The area theorem) *Let $\mathcal{H}$ be a black-hole event horizon in a smooth spacetime $(\mathcal{M}, g)$. Suppose that either*

- (a) *$(\mathcal{M}, g)$ is globally hyperbolic, and there exists a conformal completion $(\tilde{\mathcal{M}}, \tilde{g}) = (\mathcal{M} \cup \mathcal{I}^+, \Omega^2 g)$ of $(\mathcal{M}, g)$ with a $\mathcal{H}$ -regular $\mathcal{I}^+$, and the null energy condition holds on the past $I^-(\mathcal{I}^+; \tilde{\mathcal{M}}) \cap \mathcal{M}$ of $\mathcal{I}^+$ in $\mathcal{M}$; or*
- (b) *There exists a globally hyperbolic conformal completion $(\tilde{\mathcal{M}}, \tilde{g}) = (\mathcal{M} \cup \mathcal{I}^+, \Omega^2 g)$ of $(\mathcal{M}, g)$, with the null energy condition holding on $I^-(\mathcal{I}^+; \tilde{\mathcal{M}}) \cap \mathcal{M}$.*

Let Σ_a , $a = 1, 2$ be two achronal spacelike embedded hypersurfaces of C^2 differentiability class, set $S_a = \Sigma_a \cap \mathcal{H}$. Then:

1. *The area $\text{Area}(S_a)$ of S_a is well defined.*
2. *If*

$$S_1 \subset J^-(S_2) ,$$

then the area of S_2 is larger than or equal that of S_1 . (Moreover, this is true even if the area of S_1 is counted with multiplicity of generators (cf. Definition 2.10.13) provided that $S_1 \cap S_2 = \emptyset$.)

We stress that it is *not* assumed that $\mathcal{I}^+$ is null, as is the case for spacetimes which are asymptotically flat and asymptotically vacuum in null directions with vanishing cosmological constant; in fact it could be even changing causal type from point to point. In particular Theorem 3.5.7 also applies when the cosmological constant does not vanish. In point (b) global hyperbolicity of $(\tilde{\mathcal{M}}, \tilde{g})$ should of course be understood as that of a manifold with boundary.

The fact that the potential lack of differentiability of the horizons is an important issue in the current context was made apparent in [119], where a *nowhere continuously differentiable* event horizon was constructed (this is, essentially, the example presented in Theorem 2.10.12, p. 73). Given the existence of such examples, it is not immediately clear that the area of cross-sections of general horizons can even be defined, hence point 1 above. The reading of the proofs of the area theorem in [244, 454] suggests that the horizons have been assumed to be piecewise C^2 there. Such a hypothesis is certainly incompatible with the examples of Theorem 2.10.12, which are *nowhere* C^2 . The main part of the work in [117] was to show that *no hypotheses concerning the differentiability of the horizon* are needed for a proof of the area theorem.

A key notion used in the proof is that of *Alexandrov divergence* of the generators of the horizon, discussed in Section 2.10.3.

REMARK 3.5.8 Królak has previously extended the definition of a black hole to settings more general, in various ways, than the standard setting considered above. In each of

the papers [302–304] Królak establishes an area theorem under the implicit assumption of piecewise smoothness. As explained in detail in [117, Appendix C], the area theorems of Królak remain true without the supplementary hypothesis of piecewise smoothness.

Yet another approach to the area theorem has been presented in [454], where one considers globally hyperbolic completions in a setup that includes i^0 . This introduces some unnecessary difficulties related to the low differentiability of the conformally rescaled metric at i^0 . $\square$

It is of interest to consider what happens when the area does not change. This question is relevant to the classification of stationary black holes, as well as to the understanding of compact Cauchy horizons. One has the following [117].

THEOREM 3.5.9 *Under the hypotheses of Theorem 3.5.7, suppose that the area of S_1 equals that of S_2 . Then*

$$(J^+(S_1) \setminus S_1) \cap (J^-(S_2) \setminus S_2)$$

is a smooth (analytic if the metric is analytic) null hypersurface with vanishing null second fundamental form. Moreover, if γ is a null generator of $\mathcal{H}$ with $\gamma(0) \in S_1$ and $\gamma(1) \in S_2$, then the curvature tensor of $(\mathcal{M}, g)$ satisfies $R(X, \gamma'(t))\gamma'(t) = 0$ for all $t \in [0, 1]$ and $X \in T_{\gamma(t)}\mathcal{H}$.

Theorem 3.5.9 can be used to show that, e.g. in electrovacuum stationary spacetimes, the event horizons are as differentiable as the metric allows: thus smooth if the metric is smooth, real-analytic if the metric is.

3.5.1 Hawking and Ellis’s area theorem

The original statement of the area theorem is due to Hawking and Ellis [244], and it appears appropriate to present the causal regularity conditions which are imposed in Hawking and Ellis’s approach to that problem; we follow the presentation of [117, Appendix B]. One of the conditions of the Hawking–Ellis area theorem [244, Proposition 9.2.7, p. 318] is that the spacetime $(\mathcal{M}, g)$ is *weakly asymptotically simple and empty* (‘WASE’, [244, p. 225]). This is defined through the requirement of existence of an open set $\mathcal{U} \subset \mathcal{M}$ which is isometric to $\mathcal{U}' \cap \mathcal{M}'$, where $\mathcal{U}'$ is a neighbourhood of null infinity in an asymptotically simple and empty (ASE) spacetime $(\mathcal{M}', g')$. Here a spacetime is said to be ASE if it has a conformal completion as in Section 3.1, with the metric being vacuum near the conformal boundary at infinity $\mathcal{I}$, and if every null geodesic in $\mathcal{M}$ has two end points on $\mathcal{I}$ [244, p. 222]. Next, it is assumed that $\mathcal{M}$ admits a partial Cauchy surface $\mathcal{S}$ with respect to which $\mathcal{M}$ is *future asymptotically predictable* ([244], p. 310). This last notion is defined by the requirement that $\mathcal{S}^+$ is contained in the closure of the future domain of dependence $\mathcal{D}^+(\mathcal{S}; \mathcal{M})$ of $\mathcal{S}$, where the closure is taken in the conformally completed manifold $\tilde{\mathcal{M}} = \mathcal{M} \cup \mathcal{S}^+ \cup \mathcal{S}^-$, with both $\mathcal{S}^+$ and $\mathcal{S}^-$ being null hypersurfaces. Next, one says that $(\mathcal{M}, g)$ is *strongly future asymptotically predictable* [244, p. 313] if it is future asymptotically predictable and if $J^+(\mathcal{S}) \cap J^-(\mathcal{S}^+; \tilde{\mathcal{M}})$ is contained in $\mathcal{D}^+(\mathcal{S}; \mathcal{M})$. Finally ([244], p. 318), $(\mathcal{M}, g)$ is said to be a *regular predictable spacetime* if $(\mathcal{M}, g)$ is strongly future asymptotically predictable and if the following three conditions hold:

- (α) $\mathcal{S} \cap J^-(\mathcal{S}^+; \tilde{\mathcal{M}})$ is homeomorphic to $\mathbb{R}^3 \setminus \{\text{point}\}$ (an open set with compact closure).
- (β) $\mathcal{S}$ is simply connected.
- (γ) The family of hypersurfaces $\mathcal{S}(\tau)$ constructed in [244, Proposition 9.2.3, p. 313] has the property that for sufficiently large τ the sets $\mathcal{S}(\tau) \cap \overline{J^-(\mathcal{S}^+; \tilde{\mathcal{M}})}$ are contained in $J^+(\mathcal{S}^-; \tilde{\mathcal{M}})$.

As such, it is not clear whether this set of conditions suffices for a proof of the area theorem, as asserted in [244, Proposition 9.2.7, p. 318]: indeed, in the

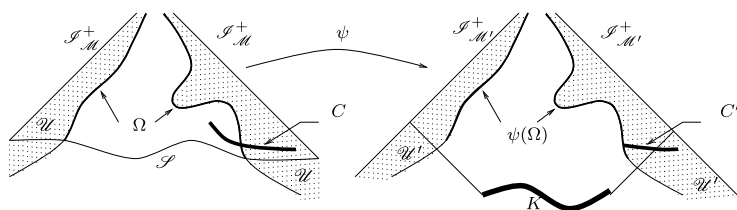


Fig. 3.5.1 The set $\Omega \equiv J^+(\mathcal{S}; \mathcal{M}) \cap \partial\mathcal{U}$ and its image under ψ . From [117], reprinted with permission.

proof of [244, Proposition 9.2.1, p. 311] (which is one of the results used in the proof of [244, Proposition 9.2.7, p. 318]) Hawking and Ellis write: ‘This shows that if $\mathcal{W}$ is any compact set of $\mathcal{S}$, every generator of $\mathcal{S}^+$ leaves $J^+(\mathcal{W}; \tilde{\mathcal{M}})$.’ The justification of this given in [244] is wrong. If one is willing to impose the sentence in quotation marks as *yet one more regularity hypothesis* on $\mathcal{S}^+$, then the arguments given in [244] apply to prove the area theorem for black holes with a piecewise smooth event horizon. The results in [117] show that this remains true with no supplementary conditions on the differentiability of the horizon.

An alternative way to guarantee that the area theorem will hold in the ‘future asymptotically predictable WASE’ setup is to impose the following supplementary conditions [117, Appendix B]: let $\psi : \mathcal{U} \rightarrow \mathcal{U}' \cap \mathcal{M}'$ denote the isometry arising in the definition of the WASE spacetime $\mathcal{M}$. First, one requires that ψ can be extended by continuity to a continuous map, still denoted by ψ , defined on $\overline{\mathcal{U}}$. Next, one demands that there exists a compact set $K \subset \mathcal{M}'$ such that

$$\psi(J^+(\mathcal{S}; \mathcal{M}) \cap \partial\mathcal{U}) \subset J^+(K; \mathcal{M}'), \quad (3.5.3)$$

see Fig. 3.5.1. Under those conditions the area theorem does again hold.

Some other proposals on how to modify the WASE conditions of [244] to obtain sufficient control of the spacetimes at hand for the validity of the area theorem have been set forth in [151, 305, 376].

3.6 Causality and wave equations

The following shows the key role of the notion of global hyperbolicity when studying the wave equation (cf., e.g., [26, Section 3.2]).

THEOREM 3.6.1 *Let $\mathcal{S}$ be a smooth achronal spacelike hypersurface in a smooth spacetime $(\mathcal{M}, g)$. Then the Cauchy problem for the wave equation has a unique globally defined solution on $\mathcal{D}_I(\mathcal{S})$ for all smooth initial data on $\mathcal{S}$.*

The theorem begs the question, what happens with solutions of the wave equation in spacetimes which are not globally hyperbolic, i.e. when Cauchy horizons occur.

First, examples are known where solutions of wave equations blow up at an event horizon, cf., e.g., [128, 287].

Next, a simple example where solutions extend smoothly to solutions of the wave equation, but uniqueness fails, proceeds as follows: let $\mathcal{S}$ be the unit ball within the hypersurface $\{t = 0\}$ in Minkowski spacetime. Let $p = (1, \vec{0})$ and let $q = (-1, \vec{0})$; then the Cauchy horizon is the union of two inverted cones with tips at p and q :

$$\mathcal{H}_I(\mathcal{S}) = (J^-(p) \cap \{t > 0\}) \cup J^+(q) \cap \{t < 0\}.$$

Any two distinct solutions of the wave equation on Minkowski spacetime which have the same Cauchy data on $\mathcal{S}$ coincide on $\mathcal{D}_I(\mathcal{S})$, and provide examples of solutions which differ beyond the event horizon.

To conclude, we have both global existence and uniqueness of solutions of the Cauchy problem for the wave equation in globally hyperbolic spacetimes. On the other hand, uniqueness or existence are problematic when the spacetime is not globally hyperbolic.

The counterpart of Theorem 3.6.1 for the Einstein equations is the celebrated Choquet-Bruhat–Geroch theorem (compare [107, 416, 421]). The proof invokes many elements of the causality theory, as developed in the previous chapter.

THEOREM 3.6.2 (Choquet-Bruhat and Geroch [88]) *Consider a triple $(\mathcal{S}, \gamma, K)$, where $\mathcal{S}$ is an n -dimensional manifold, γ is a Riemannian metric on $\mathcal{S}$, and K is a symmetric two-tensor on $\mathcal{S}$, satisfying the general relativistic vacuum constraint equations (cf. Appendix G, p. 363). Then there exists a unique up to isometries vacuum spacetime $(\mathcal{M}, g)$, called the maximal globally hyperbolic development of $(\mathcal{S}, \gamma, K)$, with an embedding $i : \mathcal{S} \rightarrow \mathcal{M}$ such that $i^*g = \gamma$, and such that K corresponds to the extrinsic curvature tensor (‘second fundamental form’) of $i(\mathcal{S})$ in $\mathcal{M}$. The spacetime $(\mathcal{M}, g)$ is inextendible in the class of globally hyperbolic spacetimes with a vacuum metric.*

This theorem is the starting point of many studies in mathematical general relativity. Similarly to the wave equation, examples show that uniqueness fails beyond horizons.

Further information concerning wave equations in the presence of Cauchy horizons can be found in [362, 401, 402] and references therein.

Part II

Black holes

Having the introductory material and the basic notions of causality theory out of the way, we are ready now to pass to the main topic of this book, namely an introduction to the geometry of black holes. In Chapter 4 we present the key notions of the theory, and illustrate them with a collection of black-hole metrics which should be familiar to any relativist. In Chapter 5 we present some further illustrative and significant solutions. Chapter 6 is devoted to the construction of extensions across horizons and their visualizations for block-diagonal metrics. In Chapter 7 the construction of diagrams is extended to cover more general metrics, using projection diagrams. Chapter 8 reviews some facts about dynamical black holes.

4

An introduction to black holes

In this chapter the basics of the geometry of stationary black-hole spacetimes are presented. We start in Section 4.1 with a brief review of astrophysical black holes. We continue in Section 4.2 with the presentation of the flagship black hole—the Schwarzschild solution: we construct there its various extensions, and analyse some of its properties. The general notions arising in the context of black-hole geometries are presented in Section 4.3. A systematic discussion of extensions of spacetimes is carried out in Section 4.4. The charged counterparts of the Schwarzschild metric, namely the Reissner–Nordström metrics, are analysed in Section 4.5. The Kerr metric, expected to describe the most general vacuum, stationary, and rotating black holes, is presented in Section 4.6. The electrovacuum Majumdar–Papapetrou spacetimes, containing two or more disconnected black-hole regions, are described in Section 4.7.

INCIDENTALLY: Before continuing, it might be of interest to point out that several field theories are known to possess solutions which exhibit black-hole properties:

- Einstein’s theory of gravitation predicts existence of ‘standard’, gravitational, black-hole spacetimes which, according to our current understanding, contain event horizons shielding away perturbations of all classical fields.
- The ‘dumb holes’ are the sonic counterparts of black holes, first discussed by Unruh [450].
- The ‘optical black holes’ arise in the theory of moving dielectric media, or in non-linear electrodynamics [319, 379].
- The ‘numerical black holes’ are objects constructed by numerical general relativists. They belong to general relativity but, in view of the complexity and the methods involved, can be viewed as objects of their own. An example can be seen in Fig. 4.0.1.

An even longer list of models and submodels can be found in [28]. □

INCIDENTALLY: The reader will find a concise review of the history of the concept of a black hole in the introduction to [83]. More extensive such reviews can be found in [82, 274]. □

4.1 Black holes as astrophysical objects

We start with a short review of the observational status of black holes in astrophysics.

When a star runs out of nuclear fuel, it must find ways to fight gravity. Current physics predicts that dead stars with masses up to the Chandrasekhar limit $1.4M_{\odot}$, become white dwarfs, where electron degeneracy supplies the necessary pressure. Here and elsewhere $M_{\odot}$ denotes the mass of our Sun. Above the Chandrasekhar limit $1.4M_{\odot}$, and up to a second mass limit, $M_{\text{NS,max}} \sim 2 - 3M_{\odot}$, dead stars are expected to become neutron stars, where neutron degeneracy pressure holds them up. If a dead star has a mass $M > M_{\text{NS,max}}$, there is no known force that can prevent the star from collapsing. What we have then is a black hole.

While there is growing evidence that black holes do indeed exist in astrophysical objects, and that alternative explanations for the observations discussed in what follows seem less convincing, it should be borne in mind that no undisputed evidence of occurrence of black holes has been presented so far. The flagship black-hole

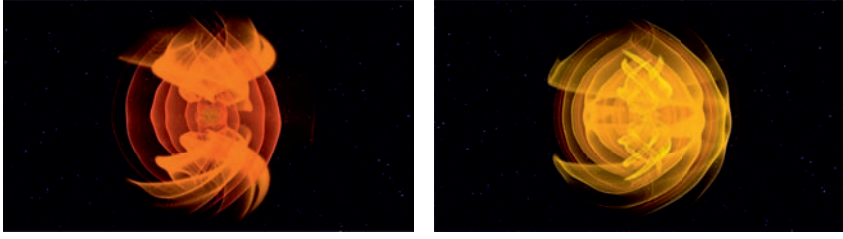


Fig. 4.0.1 Numerical relativity simulation of two inspiralling black holes that merge to form a new black hole. Shown are the black-hole horizon, the strong gravitational field surrounding the black holes, and the gravitational waves produced. © S. Ossokine, A. Buonanno (Max Planck Institute for Gravitational Physics), Simulating eXtreme Space-times project, W. Bengel (Airborne Hydro Mapping GmbH), reproduced with permission from URL <https://www.aei.mpg.de/1824987/Detection?page=2>.

Table 4.1.1 Stellar mass black-hole candidates (from [331])

Type	Binary system	$M_c/M_\odot$	$M_*/M_\odot$
HMXB:	Cygnus X-1	11–21	24–42
	LMC X-3	5.6–7.8	20
	LMC X-1	≥ 4	4–8
LMXB:	V 404 Cyg	10–15	≈ 0.6
	A 0620-00	5–17	0.2–0.7
	GS 1124-68 (Nova Musc)	4.2–6.5	0.5–0.8
	GS 2000+25 (Nova Vul 88)	6–14	≈ 0.7
	GRO J 1655-40	4.5–6.5	≈ 1.2
	H 1705-25 (Nova Oph 77)	5–9	≈ 0.4
	J 04224+32	6–14	≈ 0.3 –0.6

candidate used to be Cygnus X-1, known and studied for years (cf., e.g., [83, 386]), and it still remains a strong one. Table 4.1.1¹ lists a series of further strong black-hole candidates in *X*-ray binary systems; M_c is the mass of the compact object and M_* is that of its optical companion. Further candidates, as well as references, can be found in [76, 346, 367, 372]. The binaries have been divided into two families: the high mass *X*-ray binaries (HMXB), where the companion star is of (relatively) high mass, and the low mass *X*-ray binaries (LMXB), where the companion is typically below a solar mass. The LMXB's include the ‘*X*-ray transients’, so called because of flaring-up behaviour. This particularity makes it possible to make detailed studies of their optical properties during the quiescent periods, which would be impossible during the periods of intense *X*-ray activity. The stellar systems listed have *X*-ray spectra which are neither periodic (that would correspond to a rotating neutron star) nor recurrent (which is interpreted as thermonuclear explosions on a neutron star's hard surface). The final selection criterion is that of the mass M_c exceeding in a clear way the Chandrasekhar limit, say $M_c \gtrsim M_C \approx 3$ solar masses.² According to the authors of [83], the strongest stellar-mass black-hole candidate in 1999 was V404 Cygni, which belongs to the LMXB class. Table 4.1.1 can be put into perspective by realizing that, by some estimates [331], a typical galaxy—such as ours—can harbour 10^7 – 10^8 stellar black mass holes. We note an interesting proposal, put forward in [84], to carry out observations by gravitational microlensing of some

¹The review [344] lists 40 binaries containing a black-hole candidate.

²See [346] for a discussion and references concerning the value of M_C .

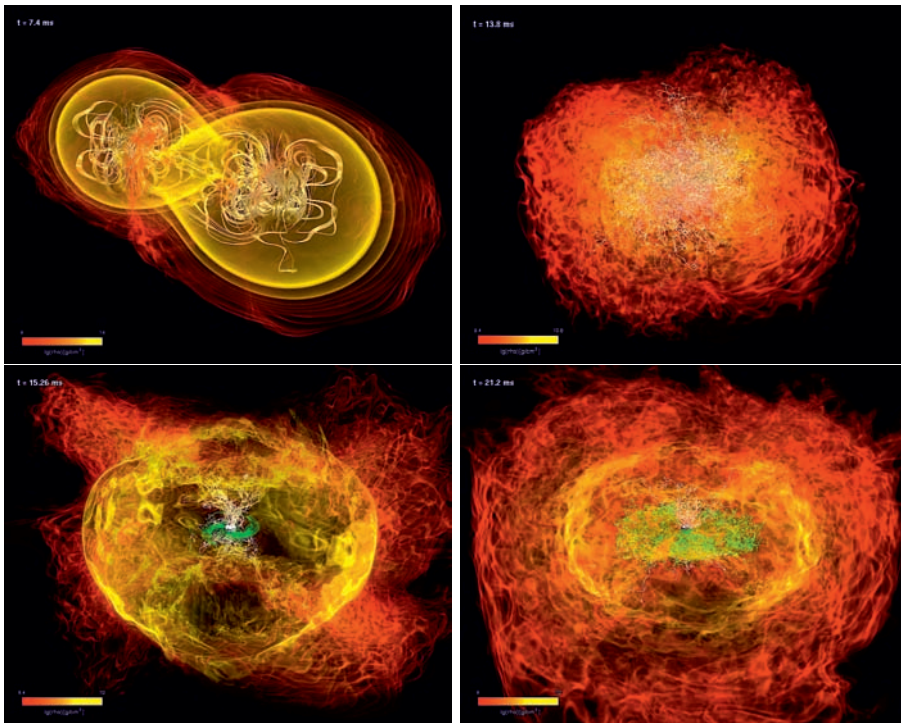


Fig. 4.1.1 Snapshots at representative times of the evolution of a neutron-star binary and of the formation of a large-scale ordered magnetic field, from [414]; see also URL <https://relastro.uni-frankfurt.de/relativistic-binary-neutron-star-inspirals/>. Shown with a colour-code map is the density over which the magnetic-field lines are superposed. The panels in the upper row refer to the binary during the merger and before the collapse to black hole (BH), while those in the lower row to the evolution after the formation of the BH. Green lines sample the magnetic field in the torus and on the equatorial plane, while white lines show the magnetic field outside the torus and near the BH spin axis. The inner/outer part of the torus has a size of about 90/170 km, while the horizon has a diameter on the order of 9 km. © AAS, reproduced by permission.

20,000 stellar-mass black holes predicted [357] to cluster within 0.7 pc of Sgr A*, the centre of our galaxy.

It is now widely accepted that quasars and active galactic nuclei are powered by accretion onto massive black holes [190, 333, 464]. Further, over the past few years there has been increasing evidence that massive dark objects may reside at the centres of most, if not all, galaxies [332, 411]. In several cases the best explanation for the nature of those objects is that they are ‘heavyweight’ black holes, with masses ranging from 10^6 to 10^{10} solar masses. Table 4.1.2³ lists some supermassive black-hole candidates; some other candidates, as well as precise references, can be found in [298, 346, 347, 410]. The main criterion for finding candidates for such black holes is the presence of a large mass within a small region; this is determined by maser line spectroscopy, gas spectroscopy, or by measuring the motion of stars orbiting around the galactic nucleus.

³The table lists those galaxies which are listed in both [347] and [298]; we note that some candidates from earlier lists [410] do not occur any more in [298, 347].

Table 4.1.2 Selected supermassive black-hole candidates (from [298, 347])

dynamics of	Host galaxy	$M_h/M_\odot$	Host galaxy	$M_h/M_\odot$
Water maser discs	NGC 4258	4×10^7		
Gas discs	IC 1459	2×10^8	M 87	3×10^9
	NGC 2787	4×10^7	NGC 3245	2×10^8
	NGC 4261	5×10^8	NGC 4374	4×10^8
	NGC 5128	2×10^8	NGC 6251	6×10^8
	NGC 7052	3×10^8		
Stars	NGC 821	4×10^7	NGC 1023	4×10^7
	NGC 2778	1×10^7	NGC 3115	1×10^9
	NGC 3377	1×10^8	NGC 3379	1×10^8
	NGC 3384	1×10^7	NGC 3608	1×10^8
	NGC 4291	2×10^8	NGC 4342	3×10^8
	NGC 4473	1×10^8	NGC 4486B	5×10^8
	NGC 4564	6×10^7	NGC 4649	2×10^9
	NGC 4697	2×10^8	NGC 4742	1×10^7
	NGC 5845	3×10^8	NGC 7457	4×10^6
	Milky Way	3.7×10^6		

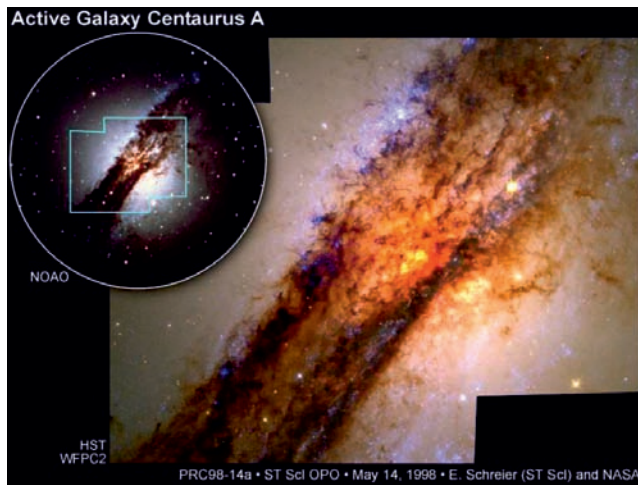


Fig. 4.1.2 A massive black hole hidden at the centre of Centaurus A galaxy (NGC 5128), feeding on a smaller galaxy. From the Hubble Space Telescope, <http://hubblesite.org/newscenter/newsdesk/archive/releases/1998/14/text>. Image credit: E. J. Schreier (STScI) and NASA, reproduced with permission.

There seems to be consensus [298, 347, 373, 411] that the two most convincing supermassive black-hole candidates are the galactic nuclei of NGC 4258 and of our own Milky Way [222, 227]. The determination of mass of the galactic nuclei via direct measurements of star motions has been made possible both by the unprecedentedly high angular resolution and sensitivity of the Hubble Space Telescope (HST), see also Fig. 4.1.3, and by the adaptive-optics Keck Telescopes [461].

The reader is referred to [366] for a discussion of the maser emission lines and their analysis for the supermassive black-hole candidate NGC 4258. An example of measurements via gas spectrography is given by the analysis of the HST obser-

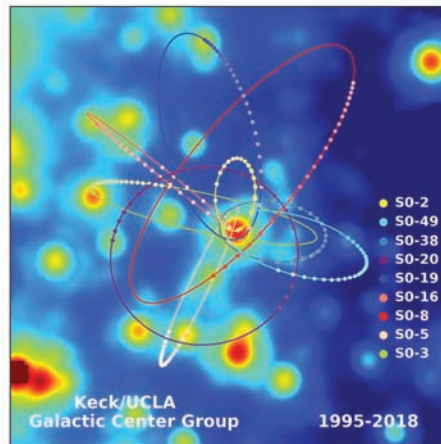


Fig. 4.1.3 The orbits of stars within the central 1.0×1.0 arcseconds of our Galaxy, providing evidence of a supermassive black hole with a mass of 3.7 million times the mass of the Sun. The image was created by Andrea Ghez and her research team at UCLA, from data sets obtained with the W. M. Keck Telescopes. See also <http://www.astro.ucla.edu/~ghezgroup/gc/blackhole.html>.

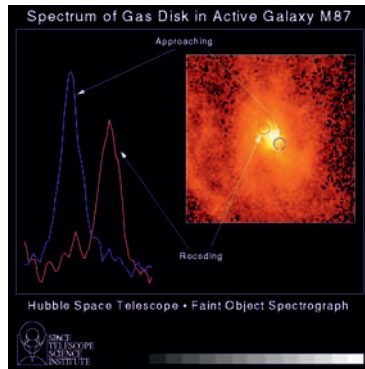


Fig. 4.1.4 Hubble Space Telescope observations of spectra of gas in the vicinity of the nucleus of the radio galaxy M 87. © NASA and H. Ford, Space Telescope Science Institute/Johns Hopkins University; R. Harms, Applied Research Corp.; Z. Tsvetanov, A. Davidsen, and G. Kriss at Johns Hopkins; R. Bohlin and G. Hartig at the Space Telescope Science Institute [433]; L. Dressel and A. K. Kochhar at Applied Research Corp. in Landover, MD; and B. Margon from the University of Washington in Seattle.

vations of the radio galaxy M 87 [449] (compare [333]): a spectral analysis shows the presence of a disk-like structure of ionized gas in the innermost few arc seconds in the vicinity of the nucleus of M 87. The velocity of the gas measured by spectroscopy (cf. Fig. 4.1.4) at a distance from the nucleus on the order of 6×10^{17} m shows that the gas recedes from us on one side, and approaches us on the other, with a velocity difference of about 920 km s^{-1} . This leads to a mass of the central object of $\sim 3 \times 10^9 M_{\odot}$, and no known form of matter with this mass is likely to occupy such a (relatively) small region except for a black hole. Figure 4.1.5 shows another image, reconstructed out of HST observations, of a recent candidate for a supermassive black hole—the (active) galactic nucleus of NGC 4438 [286].

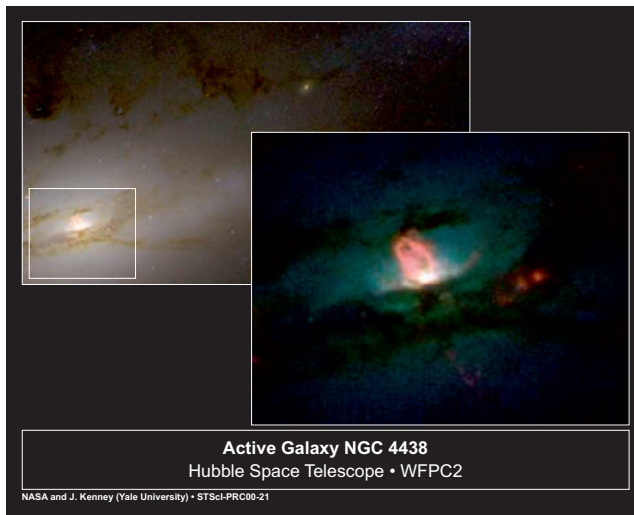


Fig. 4.1.5 Hubble Space Telescope observations [286] of the nucleus of the galaxy NGC 4438, from the STScI Public Archive [433]. Image credit: J.Kenney and NASA, reproduced with permission.

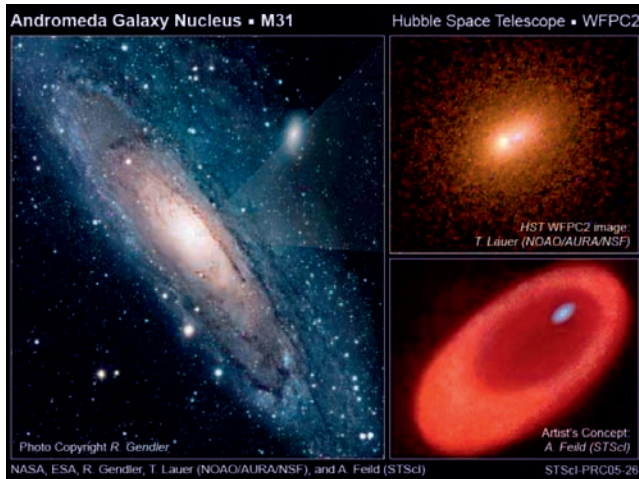


Fig. 4.1.6 A disk of young, blue stars around a 140-million- $M_{\odot}$ black hole in the centre of the Andromeda Galaxy's (M31). Image credit: R. Gendler, T. Lauer, A. Feild, NASA and ESA, reproduced with permission. From the Hubble Space Telescope, <http://hubblesite.org/newscenter/newsdesk/archive/releases/2005/26/image/a>

There have been suggestions of the existence of an intermediate-mass black hole orbiting three light-years from Sagittarius A*. This black hole of 1,300 solar masses is within a cluster of seven stars, possibly the remnant of a massive star cluster that has been stripped down by the Galactic Centre [334]. See [234] for a list of further intermediate-mass candidates.

New twists to the observations of black holes have been added by the first direct detection of a gravitational wave by the Laser Interferometric Gravitational Observatory (LIGO) in September 2015 [3], with a second one in December 2015 [2], a third one in January 2017 [5], and several others since [7]. A real-time list of candidate detections of gravitational wave can be accessed on <https://gracedb.ligo.org>.

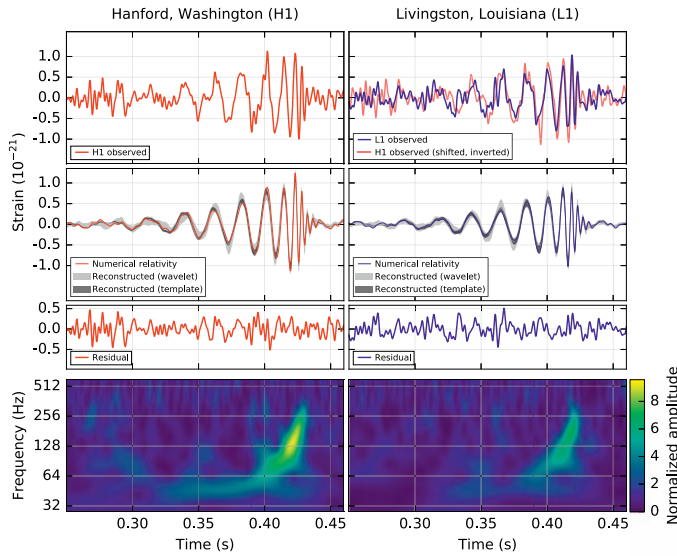


Fig. 4.1.7 GW150914 as observed in the Hanford and Livingston detectors, from [3]. The top row is the signal observed, after filtering out the low- and high-frequency noise. In the top-right corner, the Hanford signal has been inverted when superposing with the Livingston signal because of an opposite orientation of the detector arms. The bottom row shows the evolution of frequency of the signal in time.

While there is widespread consensus that the waves have been detected by now, some scientific scepticism is in order. The observation requires the extraction of an absurdly small signal from overwhelmingly noisy data using sophisticated data analysis techniques. Even though the scientists working on the problem have made many efforts to ensure the validity of the claim, there always remains the possibility of instrumental, interpretational, or data analysis errors. One needs also to keep in mind the possibility that the interpretation of the waves, as originating from black-hole mergers, might be flawed. In any case there is strong evidence for a direct observation of gravitational waves now, and we can only hope that this evidence will keep growing stronger.

Having said this, the first event, christened GW150914 (for ‘Gravitation Wave observed on 14 September, 2015’), is thought to have been created by two black holes with respective masses $36^{+5}_{-4}M_{\odot}$ and $29^{+4}_{-4}M_{\odot}$, merging into a final black hole with mass $62^{+4}_{-4}M_{\odot}$. An astounding $3^{+5}_{-5}M_{\odot}c^2$ amount of energy has been released within a fraction of a second into gravitational waves. The signal observed can be seen in Fig. 4.1.7.

The second event GW151226, illustrated in Fig. 4.1.8, is interpreted as representing the merger of two black holes of masses $14^{+8.3}_{-3.7}M_{\odot}$ and $7.5^{+2.3}_{-2.3}M_{\odot}$, leading to a final black hole of mass $21^{+5.9}_{-1.9}M_{\odot}$. Inspection of Figs. 4.1.7 and 4.1.8 reveals that the GW151226 signal is nowhere as striking as GW150914, with a maximal amplitude smaller than the residual noise. Nevertheless, the estimated probability of a false detection for GW151226 is smaller than the convincingly small number 10^{-7} .

The LIGO events provide thus the first evidence of the existence of black-hole binaries, and of black holes with masses in the range $100M_{\odot}$ – $10M_{\odot}$. The spectrum of lightweight-to-middleweight black holes, as known in early 2019, is illustrated in Fig. 4.1.9.

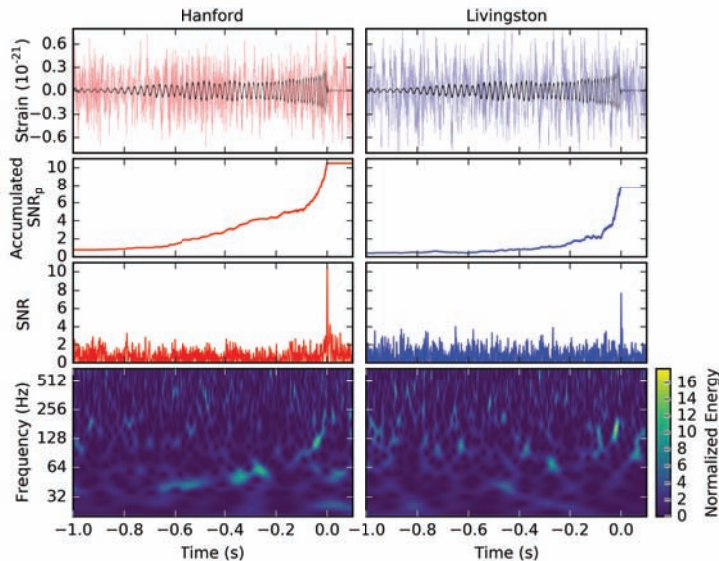


Fig. 4.1.8 GW151226 as observed in the Hanford and Livingston detectors, from [2]. The top row is the signal observed, after filtering out the low- and high-frequency noise, superposed with the black curves corresponding to the best-fit general-relativistic template. The second row shows the accumulated in-time signal-to-noise ratio. The third row shows the signal-to-noise ratio (SNR) time series produced by time shifting the best-match template waveform and computing the integrated SNR at each point in time. The bottom row shows the evolution of frequency of the signal in time.

A strong confirmation of the gravitational-wave nature of the signals seen in the instruments goes back to 17 August 2017. The waveform GW170817 observed corresponds to a catastrophic merger of two neutron stars, resulting in a kilonova. The event has been seen in several electromagnetic bands, see Fig. 4.1.10, opening the era of ‘multimessenger astronomy’. It turns out that no signal was seen in the VIRGO detector, which suggested a source located in VIRGO’s dead angle. This location was subsequently confirmed by optical observations; see Fig. 4.1.11.

A compilation of black-hole candidates can be found on Wikipedia, https://en.wikipedia.org/wiki/List_of_black_holes. The list there needs to be interpreted with the usual care.

We close this section by pointing out the review paper [77] which discusses both theoretical and experimental issues concerning *primordial* black holes.

4.2 The Schwarzschild solution and its extensions

In this section we describe various basic properties of the Schwarzschild metric and its extensions. Our presentation is an extended version of [110, Chapter 3], after ignoring some observation-directed properties of the geometry discussed there.

Stationary solutions are of interest for a variety of reasons. As models for compact objects at rest, or in steady rotation, they play a key role in astrophysics. They are easier to study than non-stationary systems because stationary solutions are governed by elliptic rather than hyperbolic equations. Further, like in any field theory, one expects that large classes of dynamical solutions approach a stationary state in the final stages of their evolution. Last but not least, explicit stationary solutions are easier to come by than dynamical ones.

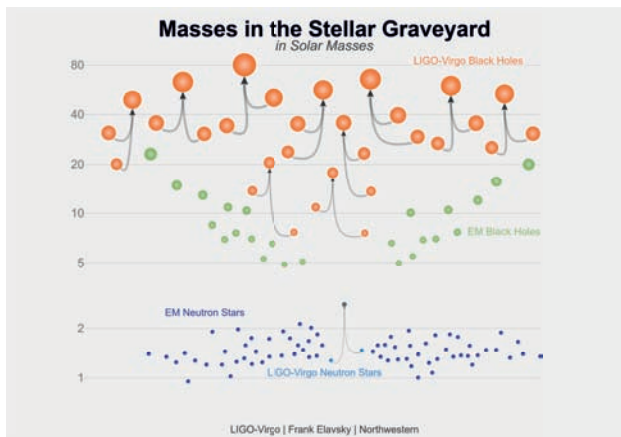


Fig. 4.1.9 Masses of ‘dead stars’, as known in early 2019, from <https://ciera.northwestern.edu/gallery>, © Frank Elavsky, Northwestern IT, LIGO-Virgo, reproduced with permission. I am grateful to T. Bulik for the colour-inversion of the original figure.

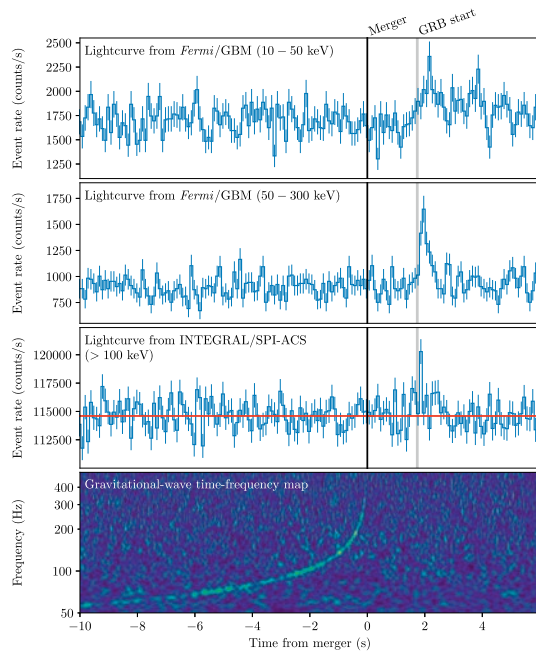


Fig. 4.1.10 A merger of two neutron stars, as observed in three γ -ray bands and as a gravitational wave in July 2017, from [4]. © AAS. Reproduced with permission.

The flagship example in general relativity is provided by the *Schwarzschild metric*:

$$g = -\left(1 - \frac{2m}{r}\right)dt^2 + \frac{dr^2}{1 - \frac{2m}{r}} + r^2 d\Omega^2, \quad (4.2.1)$$

$$t \in \mathbb{R}, \quad r \neq 2m, 0. \quad (4.2.2)$$

Here and elsewhere $d\Omega^2$ denotes the metric of the round unit 2-sphere,

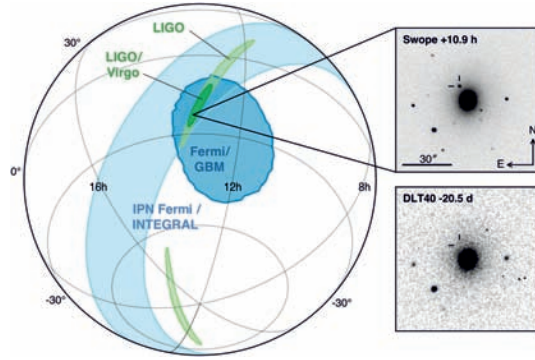


Fig. 4.1.11 The localization of the neutron-star merger at the origin of GW170817 using gravitational waves, gamma rays, and visible light, from [6]. © AAS. Reproduced with permission.

$$d\Omega^2 := d\theta^2 + \sin^2 \theta d\varphi^2.$$

In Section 5.5.2, p. 252, we verify that the metric (4.2.2) satisfies the vacuum Einstein equations; see (5.5.38)–(5.5.40) (compare Besse [52] for a very different calculation).

Generators of isometries are called *Killing vectors*; see Section 4.3.1, p. 149. A theorem due to Jebsen [278], but usually attributed to Birkhoff [54], shows the following.

THEOREM 4.2.1 *Any spherically symmetric vacuum metric has a further local Killing vector, say X , orthogonal to the orbits of spherical symmetry. Near any point at which X is not null the metric can be locally written in the Schwarzschild form (4.2.1), for some mass parameter m .*

REMARK 4.2.2 Note that a locally defined Killing vector does not necessarily extend to a global one. A simple example of this is provided by a flat torus: the collection of Killing vector fields on sufficiently small balls contains all the generators of rotations and translations, but only the translational Killing vectors extend to globally defined ones. Example 4.4.1, p. 168, is also instructive in this context. $\square$

INCIDENTALLY: One can find in the literature several results referred to as ‘Birkhoff theorems’; see [425] for an overview. Theorem 4.2.1 is a special case of the classification of ‘warped product spacetimes’ in [13], carried out for various Einstein-matter systems. In fact, in [13] one does not even need the full Einstein equations to be satisfied. Further, existence of isometries is not assumed there; instead, one considers metrics of a block-diagonal form which would follow in the presence of a suitable group of isometries.

Specializing [13, Theorem 3.2] to the case of vacuum spacetimes with a cosmological constant one has the following.

THEOREM 4.2.3 Consider a spacetime

$$(M = Q \times F, \bar{g} = g + r^2 h)$$

satisfying the vacuum Einstein equations with cosmological constant Λ , where (Q, g) is a 2-dimensional manifold, (F, h) an $n \geq 2$ dimensional one, and r a function on Q . Then

1. Either $\bar{g}$ takes the standard Eddington–Finkelstein form

$$\bar{g} = - \left(\frac{R^{[h]}}{n(n-1)} - \frac{2m}{r^{n-1}} - \frac{2\Lambda}{n(n+1)} r^2 \right) du^2 \pm 2dudr + r^2 h,$$

where $R^{[h]} = \text{const}$ is the scalar curvature of h , or

2. $\Lambda = 0$, the Ricci tensor of h vanishes, and

$$\bar{g} = -dt^2 + dr^2 + (t \pm r)^2 h,$$

or

3. r is constant, (Q, g) is maximally symmetric, (F, h) is Einstein, $R^{[h]} = 2r^2\Lambda$, and $R^{[g]} = 4\Lambda/n$. $\square$

When (F, h) is $\mathbb{S}^n$ with the round metric and $\Lambda = 0$ this reduces to the classic Birkhoff Theorem 4.2.1. In that case (2) does not apply, while (3) with $\Lambda \neq 0$ gives the Nariai metrics, cf. Example 6.3.5, p. 273 (see also [424], Section 4). $\square$

Theorem 4.2.1 implies that the hypothesis of spherical symmetry implies in vacuum, at least locally, the existence of two further symmetries: translations in t and t -reflections $t \rightarrow -t$. More precisely, we obtain *time translations* and *time reflections* in the region where $1 - 2m/r > 0$ (a metric with those two properties is called *static*). However, in the region where $r < 2m$ the notation ' t ' for the coordinate appearing in (4.2.1) is misleading, as t is then a space-coordinate, and r is a time one. So in this region t -translations are actually translations in space.

The above requires some comments and definitions, which will be useful for our further analysis. Recall that *time orientation* has been defined as a decision as to which timelike vectors are future pointing and which are past pointing. In asymptotically flat time-orientable spacetimes it is usual to say that a globally defined timelike vector field is future pointing if $X^0 > 0$ in the manifestly asymptotically flat coordinates in the asymptotic region, and we will use this convention.

Recall, next, that a function f is called a *time function* if ∇f is everywhere timelike past pointing. A coordinate, say y^0 , will be said to be a *time coordinate* if y^0 is a time function.

So, for example, $f = t$ on Minkowski spacetime is a time function: indeed, in canonical coordinates in which $\eta = \text{diag}(-1, 1, \dots, 1)$ we have

$$\nabla t = \eta^{\mu\nu} \partial_\mu t \partial_\nu = \eta^{0\nu} \partial_\nu = -\partial_t, \quad (4.2.3)$$

and so

$$\eta(\nabla t, \nabla t) = \eta(\partial_t, \partial_t) = -1.$$

(The minus sign in (4.2.3) is at the origin of the requirement that ∇f be past pointing, rather than future pointing.)

On the other hand, consider $f = t$ in the Schwarzschild metric: the inverse metric now reads

$$g^{\mu\nu} \partial_\mu \partial_\nu = -\frac{1}{1 - \frac{2m}{r}} \partial_t^2 + \left(1 - \frac{2m}{r}\right) \partial_r^2 + r^{-2} (\partial_\theta^2 + \sin^{-2} \theta \partial_\varphi^2), \quad (4.2.4)$$

and so

$$\nabla t = g^{\mu\nu} \partial_\mu t \partial_\nu = g^{0\nu} \partial_\nu = -\frac{1}{1 - \frac{2m}{r}} \partial_t.$$

The length-squared of ∇t is thus

$$g(\nabla t, \nabla t) = \frac{g(\partial_t, \partial_t)}{\left(1 - \frac{2m}{r}\right)^2} = -\frac{1}{1 - \frac{2m}{r}} = \frac{r}{2m - r} \begin{cases} < 0, & r > 2m > 0; \\ > 0, & 0 < r < 2m. \end{cases}$$

When $m > 0$, we conclude that t is a time function in the region $\{r > 2m\}$ when the usual time orientation is chosen there, but is *not* on the manifold $\{r < 2m\}$.

A similar calculation for ∇r gives

$$\begin{aligned} \nabla r &= \left(1 - \frac{2m}{r}\right) \partial_r, \\ g(\nabla r, \nabla r) &= \left(1 - \frac{2m}{r}\right)^2 g(\partial_r, \partial_r) = \left(1 - \frac{2m}{r}\right) = \frac{r-2m}{r} \begin{cases} > 0, & r > 2m > 0; \\ < 0, & 0 < r < 2m. \end{cases} \end{aligned}$$

So, assuming again that $m > 0$, r is a *time function* in the region $\{0 < r < 2m\}$ if the time orientation is chosen so that ∂_r is future pointing. On the other hand,

the alternative choice of time orientation implies that *minus* r is a time function in this region. We will return to the implications of this shortly.

4.2.1 The singularity $r = 0$

Some of the metric functions in (4.2.1) grow without bound as the sets $\{r = 0\}$ and $\{r = 2m\}$ are approached. In order to understand what happens there, we start with an analysis of the geometry near $\{r = 0\}$. A useful tool for this is provided by the formula (see, e.g., <http://grtensor.phy.queensu.ca/NewDemo>)

$$R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = \frac{48m^2}{r^6}, \quad (4.2.5)$$

which shows that the scalar $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$, called the *Kretschmann scalar*, tends to infinity as r tends to 0.

This is true regardless of the sign of $m \neq 0$, but the sign of m makes a difference: namely, if $m > 0$, any continuous curve starting in the region $\{r < 2m\}$ must cross the coordinate hypersurface $\{r = 2m\}$ before reaching ‘the exterior world’, where r is allowed to grow without bound; but the value $r = 2m$ is not allowed in (4.2.1). (What happens at $r = 2m > 0$ will be addressed shortly.) But when $m < 0$, nothing prevents a curve starting near $r = 0$ to reach any value of r . One says that $m < 0$ leads to metrics which contain ‘naked singularities’, in the following sense: *any point in the asymptotically flat region lies on causal curves for which the curvature scalar $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ grows without bound in finite proper time when followed to the past.*

In fact:

- Far-away observers can reach the set $r = 0$ in finite proper time by travelling on a timelike curve, and
- Timelike curves starting arbitrarily close to $r = 0$ can reach arbitrarily large values of r .

In view of time-reversal symmetry, $t \mapsto -t$, of the metric (4.2.1), and of the time-translation invariance, $t \mapsto t - t_0$ for any $t_0 \in \mathbb{R}$, it suffices to exhibit one single timelike curve which starts at $t = 0$ with an arbitrarily initial value r_0 of the coordinate r and reaches the set $\{r = 0\}$ in finite proper time. An example is provided by the curve

$$[0, r_0] \ni t \mapsto \gamma(t) = (t, r = r_0 - t, \theta = \theta_0, \varphi = \varphi_0), \quad (4.2.6)$$

where θ_0 and φ_0 are constants. Let us check that γ is timelike. For negative m we have

$$\begin{aligned} g(\dot{\gamma}, \dot{\gamma}) &= - \left(1 + \frac{2|m|}{r(t)}\right) \left(\frac{dt}{dt}\right)^2 + \frac{1}{1 + \frac{2|m|}{r(t)}} \left(\frac{dr}{dt}\right)^2 \\ &= - \left(1 + \frac{2|m|}{r(t)}\right) + \frac{1}{1 + \frac{2|m|}{r(t)}} \\ &= - \frac{4|m|}{r(t)} \times \frac{1 + \frac{|m|}{r(t)}}{1 + \frac{2|m|}{r(t)}} \end{aligned} \quad (4.2.7)$$

$$< 0, \quad (4.2.8)$$

as needed for a timelike curve. Next, recall that the proper time s along a timelike curve parameterized by a parameter λ is defined as

$$s(\lambda_2) - s(\lambda_1) = \int_{\lambda_1}^{\lambda_2} \sqrt{g\left(\frac{d\gamma}{d\lambda}, \frac{d\gamma}{d\lambda}\right)} d\lambda. \quad (4.2.9)$$

On the curve (4.2.6) we have the inequality

$$\frac{1 + \frac{|m|}{r(t)}}{1 + \frac{2|m|}{r(t)}} < 1$$

which, together with (4.2.7), leads to

$$\begin{aligned} s(t) &= \int_0^t \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt < 2 \int_0^t \sqrt{\frac{|m|}{r(t)}} dt \\ &= 2 \int_0^t \sqrt{\frac{|m|}{r_0 - t}} dt = 4\sqrt{|m|}(\sqrt{r_0} - \sqrt{r_0 - t}) \\ &\leq 4\sqrt{|m|r_0}. \end{aligned} \quad (4.2.10)$$

We see that the curve reaches the singularity at $r = 0$ both in finite coordinate time and in finite proper time.

INCIDENTALLY: A similar (and simpler) calculation shows that, for $m < 0$, on each space-like surface $\{t = \text{const}\}$ the set $\{r = 0\}$ can be reached along (spacelike) curves of finite length. $\square$

REMARK 4.2.4 When $m < 0$, any spacetime event $(t_0, r_0, \theta_0, \varphi_0)$ lies on the null radial geodesic

$$(0, \infty) \ni r \mapsto \gamma(r) = (t(r), r, \theta_0, \varphi_0), \quad (4.2.11)$$

where $t(r)$ is given by

$$t(r) = t_0 - r_0 + r - 2|m| \ln \left(\frac{r + 2|m|}{r_0 + 2|m|} \right). \quad (4.2.12)$$

Keeping in mind that photons travel on null geodesics, we see that, when $m < 0$, photons ‘starting’ at the singularity $r = 0$ can reach any event in spacetime. Equivalently, the singularity ‘can be seen’ from any point in spacetime.

Unless explicitly indicated otherwise, in the remainder of this section we will assume

$$m > 0.$$

4.2.2 Eddington–Finkelstein extension

The metric (4.2.1) is likewise singular as $r = 2m$ is approached. In the physics literature this is presented as a poor choice of coordinates: one talks about ‘a coordinate singularity’. This is justified if one already has a larger manifold to work with. But at this stage we only have the region $\{r > 2m\}$ at our disposal, in which case the correct point of view stems from the observation that the manifold $\{t \in \mathbb{R}, r > 2m\} \times S^2$ equipped with the metric (4.2.1) can be *extended* to a larger manifold, with the metric extending smoothly across the surface $\{r = 2m\}$.

The simplest such extension is due to Eddington and Finkelstein, and will be referred to as the *Eddington–Finkelstein extension*. It proceeds via a replacement of t by a new coordinate v , chosen to cancel the singularity in g_{rr} : if we set

$$v = t + f(r),$$

we find that $dv = dt + f' dr$, so that

$$\begin{aligned} \left(1 - \frac{2m}{r}\right) dt^2 &= \left(1 - \frac{2m}{r}\right) (dv - f' dr)^2 \\ &= \left(1 - \frac{2m}{r}\right) (dv^2 - 2f' dv dr + (f')^2 dr^2). \end{aligned}$$

Substituting in (4.2.1), the offending g_{rr} terms will go away if we choose f to satisfy

$$\left(1 - \frac{2m}{r}\right)(f')^2 = \frac{1}{1 - \frac{2m}{r}}.$$

There are two possibilities for the sign; we choose

$$f' = \frac{1}{1 - \frac{2m}{r}} = \frac{r}{r - 2m} = \frac{r - 2m + 2m}{r - 2m} = 1 + \frac{2m}{r - 2m}, \quad (4.2.13)$$

leading to

$$v = t + r + 2m \ln \left(\frac{r - 2m}{2m} \right). \quad (4.2.14)$$

The alternative choice amounts to introducing another coordinate

$$u = t - f(r), \quad (4.2.15)$$

with f still as in (4.2.13).

The choice (4.2.14) brings g to the form

$$g = - \left(1 - \frac{2m}{r} \right) dv^2 + 2dv dr + r^2 d\Omega^2, \quad (4.2.16)$$

and note that the choice (4.2.15) would lead to a non-diagonal term $-2du dr$ instead in the metric above. Now, all coefficients of g in the new coordinate system are smooth. Further,

$$\det g = -r^4 \sin^2 \theta,$$

which is non-zero for $r > 0$ except at the north and south poles, where we have the usual spherical-coordinates singularity. Since g obviously has signature $(-, +, +, +)$ for $r > 2m$, the signature cannot change across $r = 2m$, as for this the determinant would have had to vanish there. We conclude that g is a well-defined smooth Lorentzian metric on the set

$$\{v \in \mathbb{R}, r \in (0, \infty)\} \times S^2. \quad (4.2.17)$$

More precisely, (4.2.16)–(4.2.17) defines an analytic extension of the original space-time (4.2.1).

The coordinates (v, r, θ, φ) will be referred to as ‘*advanced Eddington–Finkelstein coordinates*’.

We claim the following.

THEOREM 4.2.5 *The region $\{r \leq 2m\}$ for the metric (4.2.16) is a black-hole region, in the sense that*

$$\textit{observers, or signals, can enter this region, but can never leave it.} \quad (4.2.18)$$

Proof. Let $\gamma(s) = (v(s), r(s), \theta(s), \varphi(s))$ be a future-directed timelike curve; for the metric (4.2.16) the condition $g(\dot{\gamma}, \dot{\gamma}) < 0$ reads

$$-\left(1 - \frac{2m}{r}\right)\dot{v}^2 + 2\dot{v}\dot{r} + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2) < 0.$$

This implies that

$$\dot{v} \left(- \left(1 - \frac{2m}{r} \right) \dot{v} + 2\dot{r} \right) < 0. \quad (4.2.19)$$

It follows that $\dot{v}$ does not change sign on a timelike curve. As already pointed out, the standard choice of time orientation in the exterior region corresponds to $\dot{v} > 0$ on future-directed curves, so $\dot{v}$ must be positive everywhere, which leads to

$$-(1 - \frac{2m}{r})\dot{v} + 2\dot{r} < 0.$$

For $r \leq 2m$ the first summand is non-negative, which enforces $\dot{r} < 0$ on all future-directed timelike curves in that region. Thus, r is a strictly decreasing function along such curves, which implies that future-directed timelike curves can cross the hypersurface $\{r = 2m\}$ only if coming from the region $\{r > 2m\}$. The same conclusion applies for future-directed causal curves: it suffices to approximate a causal curve by a sequence of future-directed timelike ones. $\square$

The last theorem motivates the name *black-hole event horizon* for $\{r = 2m, v \in \mathbb{R}\} \times S^2$.

INCIDENTALLY: The analogous construction using the *advanced* coordinate u instead of v leads to a *white-hole* spacetime, with $\{r = 2m\}$ being a *white-hole event horizon*. The latter can only be crossed by those future-directed causal curves which originate in the region $\{r < 2m\}$. In either case, $\{r = 2m\}$ is a causal membrane which prevents future-directed causal curves to go *back and forth*. This will become clearer in Section 4.2.3. $\square$

From (4.2.16) one easily finds the inverse metric

$$g^{\mu\nu} \partial_\mu \partial_\nu = 2\partial_v \partial_r + (1 - \frac{2m}{r})\partial_r^2 + r^{-2}\partial_\theta^2 + r^{-2}\sin^2\theta\partial_\varphi^2. \quad (4.2.20)$$

In particular,

$$0 = g^{vv} = g(\nabla v, \nabla v),$$

which, in view of Proposition 1.5.1, p. 11, shows that the integral curves of

$$\nabla v = \partial_r$$

are null, affinely parameterized geodesics. They are called *generators* of the event horizon.

We also have

$$g(\nabla r, \nabla r) = g^{rr} = \left(1 - \frac{2m}{r}\right), \quad (4.2.21)$$

and since this vanishes at $r = 2m$ we say that the hypersurface $\{r = 2m\}$ is *null*. It is reached by all the radial null geodesics $v = \text{const}$, $\theta = \text{const}'$, $\varphi = \text{const}''$, in finite affine time.

We further see from (4.2.21) and Proposition 1.5.1 that the integral curves of ∇r are geodesics as well. Now, in the (v, r, θ, φ) coordinates one finds from (4.2.20) that

$$\nabla r = \partial_v + \left(1 - \frac{2m}{r}\right) \partial_r,$$

which equals ∂_v at $r = 2m$. This provides an alternative proof that the curves $(v = s, r = 2m, \theta = \theta_0, \varphi = \varphi_0)$ are null geodesics, though the affine parameterization thereof requires further work.

REMARK 4.2.6 The Eddington–Finkelstein advanced and retarded coordinates u and v are well suited for the construction of a conformal completion, in the sense of Section 3.1, p. 85, of the exterior Schwarzschild metric. For this, setting $x = 1/r$ brings (4.2.16) to the form

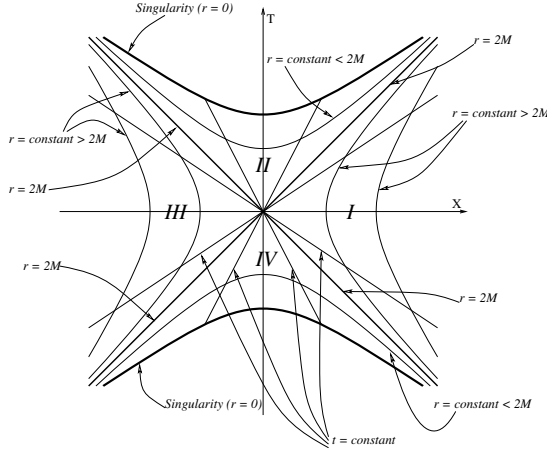


Fig. 4.2.1 The Kruskal–Szekeres extension of the Schwarzschild solution, reprinted with permission from [377].

$$\begin{aligned}
 g &= -\left(1 - \frac{2m}{r}\right) dv^2 + 2dv dr + r^2 d\Omega^2 \\
 &= x^{-2} \left(-(1 - 2mx) dv^2 - 2dv dx + d\Omega^2 \right) \\
 &=: x^{-2} \tilde{g}.
 \end{aligned} \tag{4.2.22}$$

Letting $\Omega := x$, we see that the metric $\tilde{g} \equiv \Omega^2 g$ extends analytically through the conformal boundary $\mathcal{I}^- := \{x = 0\}$.

An identical construction using the retarded coordinate u provides an extension through $\mathcal{I}^+$. $\square$

4.2.3 The Kruskal–Szekeres extension

The transition from (4.2.1) to (4.2.16) is not the end of the story, as further extensions are possible, which will be clear from the calculations that we will do shortly. For the metric (4.2.1) the standard maximal analytic extension, which we are about to present, has been found independently by Kruskal [306], Szekeres [438], and Fronsdal [200]; for some obscure reason Fronsdal is almost never mentioned in this context. This extension is visualized in Fig. 4.2.1. Region *I* there corresponds to the spacetime (4.2.1), while the Eddington–Finkelstein extension just constructed using the advanced coordinate v corresponds to regions *I* and *II*.

The general construction for spherically symmetric metrics proceeds as follows: let us write the metric in the form

$$g = -V^2 dt^2 + V^{-2} dr^2 + r^2 d\Omega^2, \tag{4.2.23}$$

where V^2 is a smooth function which depends only upon r and which we allow to be negative. (The notation V^2 , which we use because of its convenience for analysing properties of general static metrics, can be viewed as merely a notation for a smooth function, ignoring the question of existence and properties of an auxiliary function V .) We invoke the coordinate u , defined by changing a sign in (4.2.13),

$$u = t - f(r), \quad f' = \frac{1}{V^2}. \tag{4.2.24}$$

In the Schwarzschild case one obtains

$$u = t - r - 2m \ln \left(\frac{r - 2m}{2m} \right).$$

We could now replace (t, r) with (u, r) , obtaining an extension of the exterior region I of Fig. 4.2.1 into the ‘white hole’ region IV . We leave the details of this extension as an exercise for the reader, and we pass to the complete extension. The associated calculations are standard; for the convenience of the reader we reproduce here the presentation from [110]. First, we replace (t, r) with (u, v) . We note that

$$V du = V dt - \frac{1}{V} dr, \quad V dv = V dt + \frac{1}{V} dr,$$

which gives

$$V dt = \frac{V}{2}(du + dv), \quad \frac{1}{V} dr = \frac{V}{2}(dv - du).$$

Inserting this into (4.2.1) brings g to the form

$$\begin{aligned} g &= -V^2 dt^2 + V^{-2} dr^2 + r^2 d\Omega^2 \\ &= -V^2 du dv + r^2 d\Omega^2. \end{aligned} \quad (4.2.25)$$

The metric so obtained is still degenerate at $\{V = 0\}$. The desingularization is now obtained by setting

$$\hat{u} = -\exp(-cu), \quad \hat{v} = \exp(cv), \quad (4.2.26)$$

with

$$4mc = 1.$$

Indeed, using

$$d\hat{u} = c \exp(-cu) du, \quad d\hat{v} = c \exp(cv) dv,$$

we obtain

$$\begin{aligned} V^2 du dv &= \frac{V^2}{c^2} \exp(-c(-u + v)) d\hat{u} d\hat{v} \\ &= \frac{V^2}{c^2} \exp(-2cf(r)) d\hat{u} d\hat{v}. \end{aligned}$$

In the Schwarzschild case this reads

$$\begin{aligned} \frac{V^2}{c^2} \exp(-2cf(r)) &= \frac{r-2m}{c^2 r} \exp\left(-2c\left(r + 2m \ln\left(\frac{r-2m}{2m}\right)\right)\right) \\ &= \frac{\exp(-2cr)}{c^2 r} (r-2m) \exp\left(-4mc \ln\left(\frac{r-2m}{2m}\right)\right), \end{aligned}$$

and, with the above choice of c , the dangerous term $r - 2m$ cancels out. Thus, the desired coordinate transformation is

$$\hat{u} = -\exp(-cu) = -\exp\left(\frac{r-t}{4m}\right) \sqrt{\frac{r-2m}{2m}}, \quad (4.2.27)$$

$$\hat{v} = \exp(cv) = \exp\left(\frac{r+t}{4m}\right) \sqrt{\frac{r-2m}{2m}}, \quad (4.2.28)$$

with g taking the form

$$\begin{aligned} g &= -V^2 du dv + r^2 d\Omega^2 \\ &= -\frac{32m^3 \exp(-\frac{r}{2m})}{r} d\hat{u} d\hat{v} + r^2 d\Omega^2. \end{aligned} \quad (4.2.29)$$

Here r is a function of $\hat{u}$ and $\hat{v}$ defined implicitly by the equation

$$-\hat{u}\hat{v} = \underbrace{\exp\left(\frac{r}{2m}\right) \frac{(r-2m)}{2m}}_{=:G(r)}. \quad (4.2.30)$$

A comment about this definition is in order. We have

$$\left(\exp\left(\frac{r}{2m}\right)(r-2m)\right)' = \frac{r}{2m} \exp\left(\frac{r}{2m}\right) > 0,$$

which shows that the function G defined at the right-hand side of (4.2.30) is a smooth strictly increasing function of $r > 0$. We have $G(0) = -1$, and G tends to infinity as r does, so G defines a bijection of $(0, \infty)$ with $(-1, \infty)$. The real-analytic implicit-function theorem guarantees analyticity of the inverse G^{-1} , and hence the existence of a real-analytic function $r = G^{-1}(-\hat{u}\hat{v})$ solving (4.2.30) on the set $\hat{u}\hat{v} \in (-\infty, 1)$.

Note that so far we had $r > 2m$, but there are a priori no reasons for the function $r(u, v)$ defined above to satisfy this constraint. In fact, we already know from our experience with the (v, r, θ, φ) coordinate system that a restriction $r > 2m$ would lead to a spacetime with poor global properties.

We have

$$\det g = -(32m^3)^2 \exp\left(-\frac{r}{m}\right) r^2 \sin^2 \theta,$$

with all coefficients of g smooth, which shows that (4.2.29) defines a smooth Lorentzian metric on the set

$$\{\hat{u}, \hat{v} \in \mathbb{R} \text{ such that } r > 0\} \times S^2 = \{\hat{u}, \hat{v} \in \mathbb{R} \text{ such that } \hat{u}\hat{v} < 1\} \times S^2. \quad (4.2.31)$$

This is the *Kruskal–Szekeres* extension of the original spacetime (4.2.1). Figure 4.2.1 gives a representation of the extended spacetime in coordinates

$$X = (\hat{v} - \hat{u})/2, \quad T = (\hat{v} + \hat{u})/2. \quad (4.2.32)$$

Recall that (4.2.5) shows that the so-called *Kretschmann scalar* $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ diverges as r^{-6} when r approaches zero; we conclude that the metric cannot be extended across the set $r = 0$, at least in the class of C^2 metrics.

Let us discuss some features of Fig. 4.2.1:

1. The sets $r = \text{const}$ coincide with hyperboloids $X^2 - T^2 = \text{const}'$, which are timelike in regions *I* and *III* (since $X^2 - T^2 < 0$ there), and which are spacelike in regions *II* and *IV*. In particular the singular set $r = 0$ corresponds to the spacelike hyperboloids

$$(T^2 - X^2)|_{r=0} = \hat{u}\hat{v}|_{r=0} = 1.$$

2. We have

$$g(\nabla T, \nabla T) = g^\sharp(dT, dT) = \frac{1}{4}g^\sharp(d\hat{u} + d\hat{v}, d\hat{u} + d\hat{v}) = \frac{1}{2}g^\sharp(d\hat{u}, d\hat{v}) < 0,$$

which shows that T is a time coordinate. A similar calculation shows that X is a space coordinate, so that in Fig. 4.2.1 time flows along the vertical axis, while displacements along the horizontal one are spacelike.

3. The map

$$(\hat{u}, \hat{v}) \rightarrow (-\hat{u}, -\hat{v})$$

leaves the metric invariant, hence is an isometry. This shows that region *I* is isometric to region *III*, and region *II* is isometric to region *IV*. In particular the extended manifold has two asymptotically flat regions, the original region *I* and region *III*.

4. The hypersurface $t = 0$ from region I corresponds to $\hat{v} = -\hat{u} > 0$, which is the same as the subset $X > 0$ of the hypersurface $T = 0$. We see that this hypersurface can be smoothly continued to negative X , which corresponds to a second, isometric copy of the original hypersurface. The hypersurface resulting from the union is called the *Einstein–Rosen bridge*.

It is instructive to do the continuation directly using the Riemannian metric γ induced by g on $t = 0$:

$$\gamma = \frac{dr^2}{1 - \frac{2m}{r}} + r^2 d\Omega^2, \quad r > 2m.$$

A convenient coordinate ρ is given by

$$\rho = \sqrt{r^2 - 4m^2} \iff r = \sqrt{\rho^2 + 4m^2}.$$

This brings γ to the form

$$\gamma = \left(1 + \frac{2m}{\sqrt{\rho^2 + 4m^2}}\right) d\rho^2 + (\rho^2 + 4m^2) d\Omega^2, \quad (4.2.33)$$

which can be smoothly continued from the original range $\rho > 0$ to $\rho \in \mathbb{R}$.

Equation (4.2.33) exhibits explicitly asymptotic flatness of both asymptotic regions $\rho \rightarrow \infty$ and $\rho \rightarrow -\infty$. Indeed,

$$\gamma \sim d\rho^2 + \rho^2 d\Omega^2$$

to leading order, for large $|\rho|$, which is the flat metric in radial coordinates with radius $|\rho|$.

5. In order to understand how the Eddington–Finkelstein extension using the v coordinate fits into Fig. 4.2.1, we need to express $\hat{u}$ in terms of v and r . For this we have

$$u = t - f(r) = v - 2f(r) = v - 2r - 4m \ln\left(\frac{r}{2m} - 1\right),$$

hence

$$\hat{u} = -e^{-\frac{u}{4m}} = -e^{-\frac{v-2r}{4m}} \left(\frac{r}{2m} - 1\right), \quad \hat{v} = e^{\frac{v}{4m}}.$$

So $\hat{v}$ remains positive but $\hat{u}$ is allowed to become negative as r crosses $r = 2m$ from above. This corresponds to the region above the diagonal $T = -X$ in the coordinates (X, T) of Fig. 4.2.1.

A similar calculation shows that the Eddington–Finkelstein extension using the coordinate u corresponds to the region $\hat{u} < 0$ within the Kruskal–Szekeres extension, which is the region below the diagonal $T = X$ in the coordinates of Fig. 4.2.1.

6. As already mentioned, vector fields generating isometries are called *Killing vector fields*. Since time-translations are isometries in our case, the vector field $K := \partial_t$ is a Killing vector field. In the Kruskal–Szekeres coordinate system the Killing vector field K takes the form

$$\begin{aligned} K &= \partial_t = \frac{\partial \hat{u}}{\partial t} \partial_{\hat{u}} + \frac{\partial \hat{v}}{\partial t} \partial_{\hat{v}} \\ &= \frac{1}{4m} (-\hat{u} \partial_{\hat{u}} + \hat{v} \partial_{\hat{v}}). \end{aligned} \quad (4.2.34)$$

More precisely, the Killing vector field ∂_t defined on the original Schwarzschild region extends to a Killing vector field X defined throughout the Kruskal–Szekeres manifold by the second line of (4.2.34).

We note that K is tangent to the level sets of $\hat{u}$ or $\hat{v}$ at $\hat{u}\hat{v} = 0$, and therefore is null there. Moreover, it vanishes at the sphere $\hat{u} = \hat{v} = 0$, which is called *the bifurcation surface of a bifurcate Killing horizon*. The justification of this last terminology should be clear from Fig. 4.2.1.

A hypersurface $\mathcal{H}$ is called *null* if the pull-back of the spacetime metric to $\mathcal{H}$ is degenerate; see Appendix F, p. 356. Quite generally, an embedded null hypersurface to which a Killing vector is tangent, and null there, is called a *Killing horizon*.⁴ Therefore the union $\{\hat{u}\hat{v} = 0\}$ of the black-hole horizon $\{\hat{u} = 0\}$ and the white-hole event horizon $\{\hat{v} = 0\}$ can be written as the union of four Killing horizons and of their bifurcation surface.

The bifurcate horizon structure and the formula (4.2.34) are rather reminiscent of what happens when considering the Killing vector $t\partial_x + x\partial_t$ in Minkowski spacetime; this is left as an exercise to the reader.

The Kruskal–Szekeres extension is *inextendible* within the class of C^2 -extensions (compare Theorem 4.2.9), which can be proved as follows: first, (4.2.5) shows that the *Kretschmann scalar* $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ diverges as r approaches zero. As already pointed out, this implies that no C^2 extension of the metric is possible across the set $\{r = 0\}$. Next, an analysis of the geodesics of the Kruskal–Szekeres metric shows that all (maximally extended) geodesics which *do not* approach $\{r = 0\}$ are complete. This, fact just explained, together with Theorem 4.4.2 and Proposition 4.4.3, both p. 169, imply inextendibility. Together with Corollary 4.4.7 we obtain the following.

THEOREM 4.2.7 *The Kruskal–Szekeres spacetime is the unique extension, within the class of simply connected analytic extensions of the Schwarzschild region $r > 2m$, with the property that all maximally extended causal geodesics on which $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ is bounded are complete.* $\square$

REMARK 4.2.8 Nevertheless, it should be realized that the exterior Schwarzschild spacetime (4.2.1) admits many non-isometric vacuum extensions, *even in the class of maximal, analytic, simply connected ones*: indeed, let S be any two-dimensional closed submanifold entirely included in, say, the black-hole region of the Kruskal–Szekeres manifold $(\mathcal{M}, g)$, such that $\mathcal{M} \setminus S$ is *not* simply connected. (A natural example is obtained by removing the ‘bifurcation sphere’ $\{\hat{u} = \hat{v} = 0\}$.) Then, for any such S the universal covering manifold $(\mathcal{M}_S, \hat{g})$ of $(\mathcal{M} \setminus S, g|_{\mathcal{M} \setminus S})$ has the claimed properties. While maximal, these extensions will contain inextendible geodesics on which the geometry is bounded, consistently with Theorem 4.2.7. We return to such issues in Section 4.4.

Yet another particularly interesting extension of the region $\{r > 2m\}$ is provided by the ‘ $\mathbb{RP}^3$ geon’ of [194]; see Example 4.4.1. $\square$

A beautiful theorem of Sbierski’s [422] asserts the following.

THEOREM 4.2.9 *The Kruskal–Szekeres spacetime is inextendible within the class of Lorentzian spacetimes with continuous metrics.* $\square$

4.2.4 Other coordinate systems, higher dimensions

A convenient coordinate system for the Schwarzschild metric is given by the so-called *isotropic coordinates*: introducing a new radial coordinate $\tilde{r}$, implicitly defined by the formula

$$r = \tilde{r} \left(1 + \frac{m}{2\tilde{r}} \right)^2, \quad (4.2.35)$$

with a little work one obtains

⁴More precisely, let X be a Killing vector field. A *Killing horizon* is a connected component of the set $\{g(X, X) = 0, X \neq 0\}$ which forms an embedded hypersurface.

$$g_m = \left(1 + \frac{m}{2|x|}\right)^4 \left(\sum_{i=1}^3 (dx^i)^2\right) - \left(\frac{1 - m/(2|x|)}{1 + m/(2|x|)}\right)^2 dt^2, \quad (4.2.36)$$

where x^i are coordinates on $\mathbb{R}^3$ with $|x| = \tilde{r}$. Those coordinates show explicitly that the space part of the metric is conformally flat (as follows from spherical symmetry in any case).

The Schwarzschild spacetime has the curious property of possessing *flat* spacelike hypersurfaces. They appear miraculously when introducing the *Painlevé–Gullstrand coordinates* [237, 317, 388]: starting from the standard coordinate system of (4.2.1) one introduces a new time τ via the equation

$$t = \tau - 2r \sqrt{\frac{2m}{r}} + 4m \operatorname{arctanh} \left(\sqrt{\frac{2m}{r}} \right), \quad (4.2.37)$$

so that

$$dt = d\tau - \frac{\sqrt{2m/r}}{1 - 2m/r} dr.$$

This leads to

$$g = - \left(1 - \frac{2m}{r}\right) d\tau^2 + 2\sqrt{\frac{2m}{r}} dr d\tau + dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2),$$

or, passing from spherical to standard coordinates,

$$g = - \left(1 - \frac{2m}{r}\right) d\tau^2 + 2\sqrt{\frac{2m}{r}} dr d\tau + dx^2 + dy^2 + dz^2. \quad (4.2.38)$$

(Note that each such slice has zero ADM mass.)

A useful tool for the PDE analysis of spacetimes is provided by *wave coordinates*. In spherical coordinates associated to wave coordinates $(t, \hat{x}, \hat{y}, \hat{z})$, with radius function $\hat{r} = \sqrt{\hat{x}^2 + \hat{y}^2 + \hat{z}^2}$, the Schwarzschild metric takes the form [326, 436]

$$g = -\frac{\hat{r} - m}{\hat{r} + m} dt^2 + \frac{\hat{r} + m}{\hat{r} - m} d\hat{r}^2 + (\hat{r} + m)^2 d\Omega^2. \quad (4.2.39)$$

This is clearly obtained by replacing r with $\hat{r} = r - m$ in (4.2.1).

INCIDENTALLY: In order to verify the harmonic character of the coordinates associated with (4.2.39), consider a general spherically symmetric static metric of the form

$$\begin{aligned} g &= -e^{2\alpha} dt^2 + e^{2\beta} dr^2 + e^{2\gamma} r^2 d\Omega^2 \\ &= -e^{2\alpha} dt^2 + e^{2\beta} dr^2 + e^{2\gamma} (\delta_{ij} dx^i dx^j - dr^2) \\ &= -e^{2\alpha} dt^2 + \left(e^{2\gamma} \delta_{ij} + (e^{2\beta} - e^{2\gamma}) \frac{x^i x^j}{r^2} \right) dx^i dx^j, \end{aligned} \quad (4.2.40)$$

where α , β , and γ depend only upon r . We need to calculate

$$\square_g x^\alpha = \frac{1}{\sqrt{|\det g|}} \partial_\mu (\sqrt{|\det g|} g^{\mu\nu} \partial_\nu x^\alpha) = \frac{1}{\sqrt{|\det g|}} \partial_\mu (\sqrt{|\det g|} g^{\mu\alpha}).$$

Clearly $g^{0i} = 0$, which makes the calculation for $x^0 = t$ straightforward:

$$\square_g t = \frac{1}{\sqrt{|\det g|}} \partial_\mu (\sqrt{|\det g|} g^{\mu 0}) = \frac{1}{\sqrt{|\det g|}} \partial_t (\sqrt{|\det g|} g^{00}) = 0,$$

as nothing depends upon t . For $\square_g x^i$ we have to calculate $\sqrt{|\det g|}$ and $g^{\mu\nu}$. For the latter, it is clear that $g^{00} = -e^{-2\alpha}$, while by symmetry considerations we must have

$$g^{ij} = e^{-2\gamma} \left(\delta^{ij} + \chi \frac{x^i x^j}{r^2} \right),$$

for a function χ to be determined. The equation

$$\begin{aligned} \delta_i^j &= g^{j\mu} g_{\mu i} = g^{jk} g_{ki} = e^{-2\gamma} \left(\delta^{jk} + \chi \frac{x^j x^k}{r^2} \right) \left(e^{2\gamma} \delta_{ki} + (e^{2\beta} - e^{2\gamma}) \frac{x^k x^i}{r^2} \right) \\ &= \delta_i^j + e^{-2\gamma} \left(\chi e^{2\gamma} + e^{2\beta} - e^{2\gamma} + \chi (e^{2\beta} - e^{2\gamma}) \right) \frac{x^i x^j}{r^2} \\ &= \delta_i^j + e^{-2\gamma} \left(e^{2\beta} - e^{2\gamma} + \chi e^{2\beta} \right) \frac{x^i x^j}{r^2} \end{aligned}$$

gives $\chi = e^{2(\gamma-\beta)} - 1$, and finally

$$g^{ij} = e^{-2\gamma} \delta^{ij} + (e^{-2\beta} - e^{-2\gamma}) \frac{x^i x^j}{r^2}.$$

Next, $\sqrt{|\det g|}$ is best calculated in a coordinate system in which the vector (x, y, z) is aligned along the x axis, $(x, y, z) = (r, 0, 0)$. Then (4.2.40) reads, in spacetime dimension $n+1$,

$$g = \begin{pmatrix} -e^{2\alpha} & 0 & 0 & \cdots & 0 \\ 0 & e^{2\beta} & 0 & \cdots & 0 \\ 0 & 0 & e^{2\gamma} & \cdots & 0 \\ 0 & 0 & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & e^{2\gamma} \end{pmatrix}$$

which implies that

$$\det g = -e^{2(\alpha+\beta)+2(n-1)\gamma},$$

still at $(x, y, z) = (r, 0, 0)$. Spherical symmetry implies that this equality holds everywhere.

In order to continue, it is convenient to set

$$\phi = e^{\alpha+\beta+(n-3)\gamma}, \quad \psi = e^{\alpha+\beta+(n-1)\gamma} (e^{-2\beta} - e^{-2\gamma}).$$

We then have

$$\begin{aligned} \sqrt{|\det g|} \square_g x^i &= \partial_\mu (\sqrt{|\det g|} g^{\mu i}) = \partial_j (\sqrt{|\det g|} g^{ji}) \\ &= \partial_j \left(\underbrace{e^{\alpha+\beta+(n-3)\gamma}}_{\phi} \delta^{ij} + \underbrace{e^{\alpha+\beta+(n-1)\gamma} (e^{-2\beta} - e^{-2\gamma})}_{\psi} \frac{x^i x^j}{r^2} \right) \\ &= (\phi' + \psi') \frac{x^i}{r} + \psi \partial_j \left(\frac{x^i x^j}{r^2} \right) = \left(\phi' + \psi' + \frac{(n-1)}{r} \psi \right) \frac{x^i}{r}. \end{aligned} \tag{4.2.41}$$

For the metric (4.2.39) we have

$$e^{2\alpha} = \frac{\hat{r} - m}{\hat{r} + m}, \quad \beta = -\alpha, \quad e^{2\gamma} \hat{r}^2 = (\hat{r} + m)^2,$$

so that

$$\phi = 1, \quad \psi = e^{2\gamma} \times e^{2\alpha} - 1 = \frac{(\hat{r} + m)^2}{\hat{r}^2} \times \frac{\hat{r} - m}{\hat{r} + m} - 1 = -\frac{m^2}{\hat{r}^2},$$

and if $n = 3$ we obtain $\square_g x^\mu = 0$, as desired.

More generally, consider the Schwarzschild metric in any dimension $n \geq 3$,

$$g_m = - \left(1 - \frac{2m}{r^{n-2}} \right) dt^2 + \frac{dr^2}{1 - \frac{2m}{r^{n-2}}} + r^2 d\Omega^2, \tag{4.2.42}$$

where, as usual, $d\Omega^2$ is the round unit metric on S^{n-1} . In order to avoid confusion we keep the symbol r for the coordinate appearing in (4.2.42), and rewrite (4.2.40) as

$$g = -e^{2\alpha} dt^2 + e^{2\beta} d\hat{r}^2 + e^{2\gamma} \hat{r}^2 d\Omega^2. \quad (4.2.43)$$

It follows from (4.2.41) that the harmonicity condition reads

$$0 = \frac{d(\phi + \psi)}{d\hat{r}} + \frac{(n-1)}{\hat{r}} \psi = \frac{d(\phi + \psi)}{d\hat{r}} + \frac{(n-1)}{\hat{r}} (\psi + \phi) - \frac{(n-1)}{\hat{r}} \phi. \quad (4.2.44)$$

Equivalently,

$$\frac{d[\hat{r}^{n-1}(\phi + \psi)]}{d\hat{r}} = (n-1)\hat{r}^{n-2}\phi. \quad (4.2.45)$$

Transforming r to $\hat{r}$ in (4.2.70) and comparing with (4.2.43) we find that

$$e^\alpha = \sqrt{1 - \frac{2m}{r^{n-2}}}, \quad e^\beta = e^{-\alpha} \frac{dr}{d\hat{r}}, \quad e^\gamma = \frac{r}{\hat{r}}.$$

Note that $\phi + \psi = e^{\alpha-\beta+(n-1)\gamma}$; chasing through the definitions one obtains $\phi = \frac{dr}{d\hat{r}}(r/\hat{r})^{n-3}$, leading eventually to the following form of (4.2.45):

$$\frac{d}{dr} \left[r^{n-1} \left(1 - \frac{2m}{r^{n-2}} \right) \frac{d\hat{r}}{dr} \right] = (n-1)r^{n-3}\hat{r}.$$

Introducing $x = 1/r$, one obtains an equation with a Fuchsian singularity at $x = 0$:

$$\frac{d}{dx} \left[x^{3-n} \left(1 - 2mx^{n-2} \right) \frac{d\hat{r}}{dx} \right] = (n-1)x^{1-n}\hat{r}.$$

The characteristic exponents are -1 and $n-1$ so that, after matching a few leading coefficients, the standard theory of such equations provides solutions with the behaviour

$$\hat{r} = r - \frac{m}{(n-2)r^{n-3}} + \begin{cases} \frac{m^2}{4}r^{-3} \ln r + O(r^{-5} \ln r), & n = 4; \\ O(r^{5-2n}), & n \geq 5 \end{cases}$$

Somewhat surprisingly, we find logarithms of r in an asymptotic expansion of $\hat{r}$ in dimension $n = 4$. However, for $n \geq 5$ there is a complete expansion of $\hat{r}$ in terms of inverse powers of r , without any logarithmic terms in those dimensions. $\square$

As already made clear in (4.2.42), there exist higher dimensional counterparts of metrics (4.2.1), which have been found by Tangherlini [440]. In spacetime dimension $n+1$, the metrics take the form (4.2.1) with

$$V^2 = 1 - \frac{2m}{r^{n-2}}, \quad (4.2.46)$$

and with $d\Omega^2$ —the unit round metric on S^{n-1} . The parameter m is the *Arnowitt–Deser–Misner (ADM) mass* in spacetime dimension four, and is proportional to that mass in higher dimensions. Assuming again $m > 0$, a maximal analytic extension can be constructed by a simple modification of the calculations above, leading to a spacetime with global structure identical to that of Fig. 4.2.7, p. 148, except for the replacement $2M \rightarrow (2M)^{1/(n-2)}$ there.

REMARK 4.2.10 For further reference we present a general construction of Walker [456]. We summarize the calculations already done: the starting point is a metric of the form

$$g = -F dt^2 + F^{-1} dr^2 + \underbrace{h_{AB} dx^A dx^B}_{=: h}, \quad (4.2.47)$$

with $F = F(r)$, where $h := h_{AB}(t, r, x^C) dx^A dx^B$ is a family of Riemannian metrics on an $(n-2)$ -dimensional manifold which possibly depends on t and r . It is convenient

to write F for V^2 , as the sign of F did not play any role; similarly the metric h was irrelevant for the calculations we did earlier. We assume that F is defined for r in a neighbourhood of $r = r_0$, at which F vanishes, with a simple zero there. Equivalently,

$$F(r_0) = 0, \quad F'(r_0) \neq 0.$$

Defining

$$u = t - f(r), \quad v = t + f(r), \quad f' = \frac{1}{F}, \quad (4.2.48)$$

$$\hat{u} = -\exp(-cu), \quad \hat{v} = \exp(cv), \quad (4.2.49)$$

one is led to the following form of the metric

$$g = -\frac{F}{c^2} \exp(-2cf(r)) d\hat{u} d\hat{v} + h. \quad (4.2.50)$$

Since F has a simple zero, it factorizes as

$$F(r) = (r - r_0)H(r), \quad H(r_0) = F'(r_0),$$

for a function H which has no zeros in a neighbourhood of r_0 . This follows immediately from the formula

$$\begin{aligned} F(r) - F(r_0) &= \int_0^1 \frac{dF(t(r - r_0) + r_0)}{dt} dt \\ &= (r - r_0) \int_0^1 F'(t(r - r_0) + r_0) dt. \end{aligned} \quad (4.2.51)$$

Now,

$$\begin{aligned} \frac{1}{F(r)} &= \frac{1}{H(r_0)(r - r_0)} + \frac{1}{F(r)} - \frac{1}{H(r_0)(r - r_0)} \\ &= \frac{1}{H(r_0)(r - r_0)} + \frac{H(r_0) - H(r)}{H(r)H(r_0)(r - r_0)}. \end{aligned}$$

An analysis of $H(r) - H(r_0)$ as in (4.2.51) allows us to integrate the equation $f' = 1/F$ in the form

$$f(r) = \frac{1}{F'(r_0)} \ln |r - r_0| + \hat{f}(r),$$

for some function $\hat{f}$ which is smooth near r_0 . Inserting all this into (4.2.50) with

$$c = \frac{F'(r_0)}{2} \quad (4.2.52)$$

gives

$$g = \mp \frac{4H(r)}{(F'(r_0))^2} \exp(-\hat{f}(r)F'(r_0)) d\hat{u} d\hat{v} + h, \quad (4.2.53)$$

with a negative sign if we started in the region $r > r_0$, and positive otherwise.

The function r is again implicitly defined by the equation

$$\hat{u}\hat{v} = \mp(r - r_0) \exp(\hat{f}(r)F'(r_0)).$$

The right-hand side has a derivative which equals $\mp \exp(\hat{f}(r_0)/F'(r_0)) \neq 0$ at r_0 , and therefore this equation defines a smooth function $r = r(\hat{u}\hat{v})$ for r near r_0 by the implicit function theorem.

The above discussion applies to F which are of the C^k differentiability class, with some losses of differentiability. Indeed, (4.2.53) provides an extension of the C^{k-2} differentiability class, which leads to the restriction $k \geq 2$. However, the implicit function argument just given requires h to be differentiable, so we need in fact $k \geq 3$ for a coherent analysis. Note that for real-analytic F s the extension so constructed is real analytic; this follows from the analytic version of the implicit function theorem.

Supposing we start with a region where $r > r_0$, with F positive there. Then we are in a situation reminiscent of that we encountered with the Schwarzschild metric, where a single region of the type *I* in Fig. 4.2.1 leads to the attachment of *three* new regions to the initial manifold, through ‘a lower left horizon, and an upper left horizon, meeting at a corner’. On the other hand, if we start with $r < r_0$ and F is negative there, we are in the situation of Fig. 4.2.1 where a region of type *II* is extended through ‘an upper left horizon, and an upper right horizon, meeting at a corner’. The reader should have no difficulties examining all remaining possibilities. We return to this in Chapter 6. $\square$

The function f of (4.2.48) for a $(4 + 1)$ -dimensional Schwarzschild–Tangherlini solution can be calculated to be

$$f = r + \sqrt{2m} \ln \left(\frac{r - \sqrt{2m}}{r + \sqrt{2m}} \right).$$

A direct calculation leads to

$$g = -\frac{32m(r + \sqrt{2m})^2}{r^2} \exp(-r/2m) d\hat{u} d\hat{v} + d\Omega^2. \quad (4.2.54)$$

One can similarly obtain (non-very-enlightening) explicit formulae in dimension $(5 + 1)$.

The isotropic coordinates in higher dimensions lead to the following form of the Schwarzschild–Tangherlini metric [392]:

$$g_m = \left(1 + \frac{m}{2|x|^{n-2}} \right)^{\frac{4}{n-2}} \left(\sum_{i=1}^n (dx^i)^2 \right) - \left(\frac{1 - m/2|x|^{n-2}}{1 + m/2|x|^{n-2}} \right)^2 dt^2. \quad (4.2.55)$$

The radial coordinate $|x|$ in (4.2.55) is related to the radial coordinate r of (4.2.46) by the formula

$$r = \left(1 + \frac{m}{2|x|^{n-2}} \right)^{\frac{2}{n-2}} |x|.$$

INCIDENTALLY: It may be considered unsatisfactory that the function r appearing in the globally regular form of the metric (4.2.29) is not given by an explicit elementary function of the coordinates. Here is an explicit form of the extended Schwarzschild metric due to Israel [273]:⁵

$$g = -8m \left[dx dy + \frac{y^2}{xy + 2m} dx^2 \right] - (xy + 2m)^2 d\Omega^2. \quad (4.2.56)$$

The coordinates (x, y) are related to the standard Schwarzschild coordinates (t, r) as follows:

$$r = xy + 2m, \quad (4.2.57)$$

$$t = xy + 2m(1 + \ln |y/x|), \quad (4.2.58)$$

$$|x| = \sqrt{|r - 2m|} \exp \left(\frac{r - t}{4m} \right), \quad (4.2.59)$$

$$|y| = \sqrt{|r - 2m|} \exp \left(\frac{t - r}{4m} \right). \quad (4.2.60)$$

In higher dimensions one also has an explicit, though again not very enlightening, manifestly globally regular form of the metric [312], in spacetime dimension $n + 1$:

⁵The Israel coordinates have been found independently in [389]; see also [296].

$$ds^2 = -2 \frac{w^2(-r)^{-n+2} 2^{n+1} m^{n+1} + 4m^2((n+1)(2m-r) + 3r - 4m)}{m(2m-r)^2} dU^2 + 8mdU dw + r^2 d\Omega_{n-1}^2, \quad (4.2.61)$$

where $r \geq 0$ is the function

$$r(U, w) \equiv 2m + (n-2)Uw, \quad (4.2.62)$$

while $d\Omega_{n-1}^2$ is the metric of a unit round $n-1$ sphere. $\square$

4.2.5 Some geodesics

The geodesics in the Schwarzschild metric have been studied extensively in the literature (cf., e.g., [85]), so we will only make a few general comments about those.

First, we have already encountered a family of outgoing and incoming radial null geodesics $t \mp (r + 2m \ln(r - 2m)) = \text{const.}$

Next, we have seen that the horizon $\{r = 2m\}$ is threaded by a family of null geodesics, its generators.

We continue by noting that each Killing vector X produces a constant of motion $g(X, \dot{\gamma})$ along an affinely parameterized geodesic. So we have a conserved energy-per-unit-mass

$$\mathcal{E} := g(\partial_t, \dot{\gamma}) = -\left(1 - \frac{2m}{r}\right)\dot{t},$$

and a conserved angular-momentum-per-unit-mass J

$$J := g(\partial_\varphi, \dot{\gamma}) = r^2 \dot{\varphi}.$$

Yet another constant of motion arises from the length of $\dot{\gamma}$,

$$g(\dot{\gamma}, \dot{\gamma}) = -\left(1 - \frac{2m}{r}\right)\dot{t}^2 + \frac{\dot{r}^2}{1 - \frac{2m}{r}} + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2) = \varepsilon \in \{-1, 0, 1\}. \quad (4.2.63)$$

INCIDENTALLY: To simplify things somewhat, let us show that all motions are *planar*. One way of doing this is to write the equations explicitly. The Lagrangian for geodesics reads

$$\mathcal{L} = \frac{1}{2} \left(-V^2 \left(\frac{dt}{ds} \right)^2 + V^{-2} \left(\frac{dr}{ds} \right)^2 + r^2 \left(\frac{d\theta}{ds} \right)^2 + r^2 \sin^2 \theta \left(\frac{d\varphi}{ds} \right)^2 \right).$$

Those Euler–Lagrange equations which are not already covered by the conservation laws read

$$\begin{aligned} \frac{d}{ds} \left(V^{-2} \frac{dr}{ds} \right) &= -\partial_r V \left(V \left(\frac{dt}{ds} \right)^2 + V^{-3} \left(\frac{dr}{ds} \right)^2 \right) \\ &\quad + r \left[\left(\frac{d\theta}{ds} \right)^2 + \sin^2 \theta \left(\frac{d\varphi}{ds} \right)^2 \right], \end{aligned} \quad (4.2.64)$$

$$\frac{d}{ds} \left(r^2 \frac{d\theta}{ds} \right) = r^2 \sin \theta \cos \theta \left(\frac{d\varphi}{ds} \right)^2. \quad (4.2.65)$$

Consider any geodesic, and think of the coordinates (r, θ, φ) as spherical coordinates on $\mathbb{R}^3$. Then the initial position vector (which is, for obvious reasons, assumed *not* to be the origin) and the initial velocity vector, which is assumed *not* to be radial (otherwise the geodesic will be radial, and the claim follows), define a unique plane in $\mathbb{R}^3$. We can then choose the spherical coordinates so that this plane is the plane $\theta = \pi/2$. We then have $\theta(0) = \pi/2$ and $\dot{\theta}(0) = 0$, and then $\theta(s) \equiv \pi/2$ is a solution of (4.2.65) satisfying the initial values. By uniqueness this is *the* solution. $\square$

So, without loss of generality we can assume that $\sin \theta = 1$ throughout the motion; from (4.2.63) we then obtain the following ODE for $r(s)$:

$$\dot{r}^2 = \mathcal{E}^2 + \left(1 - \frac{2m}{r}\right)\left(\varepsilon - \frac{J^2}{r^2}\right). \quad (4.2.66)$$

The radial part of the geodesic equation can be obtained by calculating directly the Christoffel symbols of the metric. A more efficient way is to use the variational principle for geodesics, with the Lagrangian $\mathcal{L} = g(\dot{\gamma}, \dot{\gamma})$ —this can be read off from the middle term in (4.2.63). But the reader should easily convince herself that, at this stage, the desired equation can be obtained by differentiating (4.2.66) with respect to s , obtaining

$$2 \frac{d^2 r}{ds^2} = \frac{d}{dr} \left(\mathcal{E}^2 + \left(1 - \frac{2m}{r}\right)\left(\varepsilon - \frac{J^2}{r^2}\right) \right). \quad (4.2.67)$$

We wish to point out the existence of a striking class of *null* geodesics for which $r(s) = \text{const}$. It follows from (4.2.67), and from the uniqueness of solutions of the Cauchy problem for ODEs, that such a curve will be a null geodesic provided that the right-hand sides of (4.2.66) and of (4.2.67) (with $\varepsilon = 0$) vanish:

$$\mathcal{E}^2 - \left(1 - \frac{2m}{r}\right)\frac{J^2}{r^2} = 0 = \frac{2J^2}{r^3}(-r + 3m). \quad (4.2.68)$$

Simple algebra shows now that the curves

$$s \mapsto \gamma_{\pm}(s) = (t = s, r = 3m, \theta = \pi/2, \varphi = \pm 3^{3/2}m^{-1}s) \quad (4.2.69)$$

are null geodesics spiraling on the timelike cylinder $\{r = 3m\}$.

4.2.6 The Flamm paraboloid

We write again the Schwarzschild metric in dimension $n + 1$,

$$g_m = - \left(1 - \frac{2m}{r^{n-2}}\right) dt^2 + \frac{dr^2}{1 - \frac{2m}{r^{n-2}}} + r^2 d\Omega^2, \quad (4.2.70)$$

where, as usual, $d\Omega^2$ is the round unit metric on S^{n-1} . Because of spherical symmetry, the geometry of the $t = \text{const}$ slices can be realized by an embedding into $(n + 1)$ -dimensional Euclidean space. If we set

$$\dot{g} = dz^2 + (dx^1)^2 + \dots + (dx^n)^2 = dz^2 + dr^2 + r^2 d\Omega^2,$$

the metric h induced by $\dot{g}$ on the surface $z = z(r)$ reads

$$h = \left(\left(\frac{dz}{dr} \right)^2 + 1 \right) dr^2 + r^2 d\Omega^2. \quad (4.2.71)$$

This will coincide with the space part of (4.2.70) if we require that

$$\frac{dz}{dr} = \pm \sqrt{\frac{2m}{r^{n-2} - 2m}}.$$

The equation can be explicitly integrated in dimensions $n = 3$ and 4 in terms of elementary functions, leading to

$$z = z_0 \pm \sqrt{2m} \times \begin{cases} 2\sqrt{r - 2m}, & r > 2m, n = 3, \\ \ln(r + \sqrt{r^2 - 2m}), & r > \sqrt{2m}, n = 4. \end{cases}$$

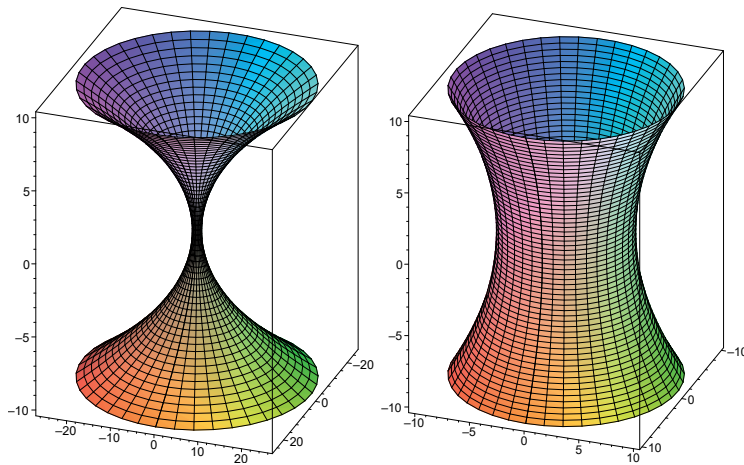


Fig. 4.2.2 Isometric embedding of the space-geometry of a three-dimensional Schwarzschild black hole into four-dimensional Euclidean space, near the throat of the Einstein–Rosen bridge $r = 2m$, with $2m = 1$ (left) and $2m = 6$ (right). Reproduced from [110].

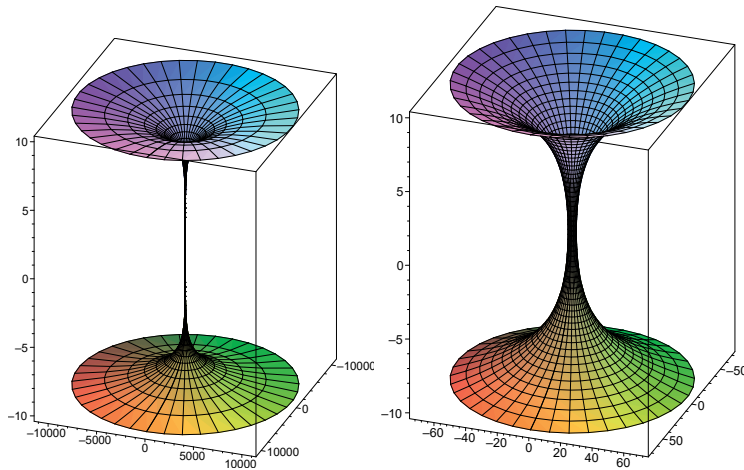


Fig. 4.2.3 Isometric embedding of the space-geometry of an $n = 4$ -dimensional Schwarzschild black hole into five-dimensional Euclidean space, near the throat of the Einstein–Rosen bridge $r = (2m)^{1/2}$, with $2m = 1$ (left) and $2m = 6$ (right). The extents of the vertical axes are the same as those in Fig. 4.2.2.

The positive sign corresponds to the usual black-hole exterior, while the negative sign corresponds to the second asymptotically flat region, on the ‘other side’ of the *Einstein–Rosen bridge*. Solving for $r(z)$, a convenient choice of z_0 leads to

$$r = \begin{cases} 2m + z^2/8m, & n = 3, \\ \sqrt{2m} \cosh(z/\sqrt{2m}), & n = 4. \end{cases}$$

In dimension $n = 3$ one obtains a paraboloid, as first noted by Flamm. The embeddings are visualized in Figs. 4.2.2 and 4.2.3.

The qualitative behaviour in dimensions $n \geq 5$ is somewhat different, as then $z(r)$ asymptotes to a finite value as r tends to infinity. The embeddings in $n = 5$ are

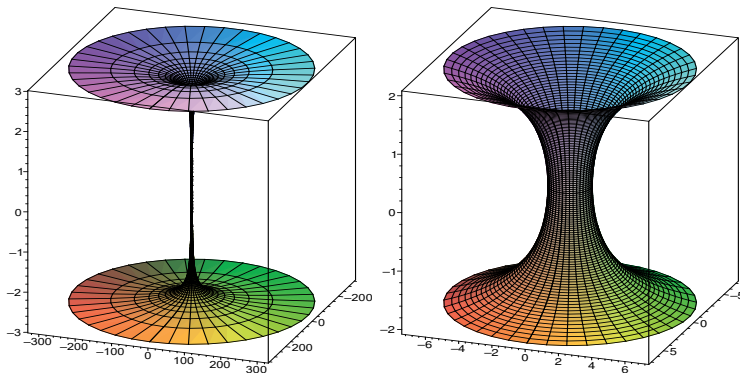


Fig. 4.2.4 Isometric embedding of the *space-geometry* of a $(5 + 1)$ -dimensional Schwarzschild black hole into six-dimensional Euclidean space, near the throat of the Einstein–Rosen bridge $r = (2m)^{1/3}$, with $2m = 2$. The variable along the vertical axis asymptotes to $\approx \pm 3.06$ as r tends to infinity. The right picture is a zoom to the centre of the throat.

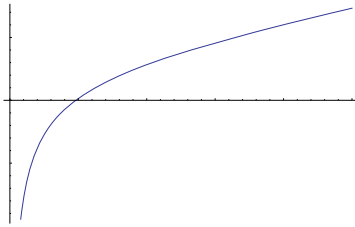


Fig. 4.2.5 The function z^2/m of (4.2.72) in terms of r/m .

visualized in Fig. 4.2.4; in that dimension $z(r)$ can be expressed in terms of elliptic functions, but the final formula is not very illuminating.

4.2.7 Fronsdal's embedding

An embedding of the full Schwarzschild geometry into six-dimensional Minkowski spacetime has been constructed by Fronsdal [200] (compare [174–176]). For this, let us write the flat metric η on $\mathbb{R}^6$ as

$$\eta = -(dz^0)^2 + (dz^1)^2 + (dz^2)^2 + (dz^3)^2 + (dz^4)^2 + (dz^5)^2.$$

For $r > 2m$ the required embedding is obtained by setting

$$\begin{aligned} z^0 &= 4m\sqrt{1 - 2m/r} \sinh(t/4m), & z^1 &= 4m\sqrt{1 - 2m/r} \cosh(t/4m), \\ z^2 &= \int \sqrt{2m(r^2 + 2mr + 4m^2)/r^3} dr, & (4.2.72) \\ z^3 &= r \sin \theta \sin \phi, & z^4 &= r \sin \theta \cos \phi, & z^5 &= r \cos \phi. \end{aligned}$$

(The function z^2 , plotted in Fig. 4.2.5, can be found explicitly in terms of elliptic integrals, but the final formula is not very enlightening.) The embedding is visualized in Fig. 4.2.6. Note that z^2 is defined and analytic for all $r > 0$, which allows one to extend the map (4.2.72) analytically to the whole Kruskal–Szekeres manifold. This led Fronsdal to his own discovery of the Kruskal–Szekeres extension of the Schwarzschild metric, but somehow his name is rarely mentioned in this context.

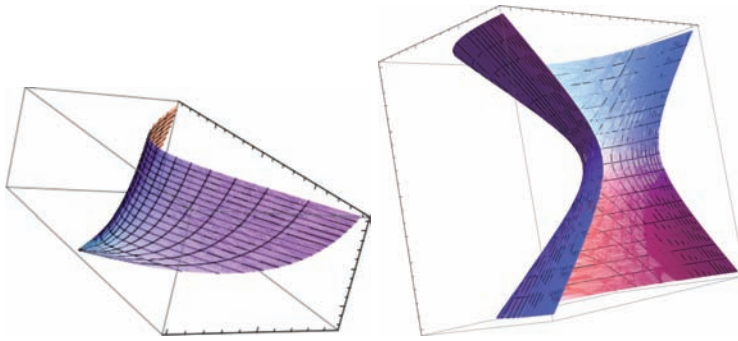


Fig. 4.2.6 Fronsda's embedding $(t, r) \mapsto (x = z^1, \tau = z^0, y = z^2)$, with target metric $-d\tau^2 + dx^2 + dy^2$, of the region $r > 2m$ (left) and of the whole Kruskal-Szekeres manifold (right) with $m = 1$.

EXERCISE 4.2.11 Show that the formulae

$$\begin{aligned} z^0 &= 4m\sqrt{1 - 2m/r^{n-2}} \sinh(t/4m), & z^1 &= 4m\sqrt{1 - 2m/r^{n-2}} \cosh(t/4m), \\ z^2 &= \sqrt{2} \sqrt{\frac{mr^{2-n}(r^n - 8m^3(n-2)^2)}{r^n - 2mr^2}} \end{aligned} \quad (4.2.73)$$

can be used to construct an embedding of $(n+1)$ -dimensional Schwarzschild metric with $n \geq 3$ into $\mathbb{R}^{1,n+2}$. $\square$

EXERCISE 4.2.12 Prove that no embedding of $(n+1)$ -dimensional Schwarzschild metric with $n \geq 3$ into $\mathbb{R}^{1,n+1}$ exists. $\square$

EXERCISE 4.2.13 Find an embedding of the Schwarzschild metric into $\mathbb{R}^6$ with metric $\eta = -(dz^0)^2 - (dz^1)^2 + (dz^2)^2 + (dz^3)^2 + (dz^4)^2 + (dz^5)^2$. You may wish to assume $z^0 = f(r) \cos(t/4m)$, $z^1 = f(r) \sin(t/4m)$, $z^2 = h(r)$, with the remaining functions as in (4.2.72). $\square$

4.2.8 Conformal Carter-Penrose diagrams

Consider an $(n+1)$ -dimensional metric with the product structure

$$g = \underbrace{g_{rr}(t, r)dr^2 + 2g_{rt}(t, r)dtdr + g_{tt}(t, r)dt^2}_{=: {}^2g} + \underbrace{h_{AB}(t, r, x^A)dx^A dx^B}_{=: h}, \quad (4.2.74)$$

where $h(t, r, \cdot)$ is a *Riemannian* metric in dimension $n-1$ at fixed (t, r) . Then any causal vector for g is also a causal vector for 2g , and drawing light cones for 2g gives a good idea of the causal structure of $(\mathcal{M}, g)$. We have already done that in Fig. 4.2.1 to depict the black-hole character of the Kruskal-Szekeres spacetime.

Now, it is not too difficult to prove that any two-dimensional Lorentzian metric can be locally written in the form

$${}^2g = 2g_{uv}(u, v)dudv = 2g_{uv}(-dt^2 + dr^2) \quad (4.2.75)$$

in which the light cones have slopes of ± 1 , just as in Minkowski spacetime. When using such coordinates, it is sufficient to draw their domain of definition to visualize the global causal structure of the spacetime.

EXERCISE 4.2.14 Prove (4.2.75). (*Hint: introduce coordinates associated with right-going and left-going null geodesics.*) $\square$

The above are the first two ingredients behind the idea of conformal Carter-Penrose diagrams. The last ingredient is to bring any infinite domain of definition of the (u, v) coordinates to a finite one. We will discuss how to do this quite generally in

Chapter 6, but it is of interest to do it explicitly for the Kruskal–Szekeres spacetime. For this, let $\bar{u}$ and $\bar{v}$ be defined by the equations

$$\tan \bar{u} = \hat{u}, \quad \tan \bar{v} = \hat{v},$$

where $\hat{v}$ and $\hat{u}$ have been defined in (4.2.27)–(4.2.28). Using

$$d\hat{u} = \frac{1}{\cos^2 \bar{u}} d\bar{u}, \quad d\hat{v} = \frac{1}{\cos^2 \bar{v}} d\bar{v},$$

the Schwarzschild metric (4.2.29) takes the form

$$\begin{aligned} g &= -\frac{32m \exp(-\frac{r}{2m})}{r} d\hat{u} d\hat{v} + r^2 d\Omega^2 \\ &= -\frac{32m \exp(-\frac{r}{2m})}{r \cos^2 \bar{u} \cos^2 \bar{v}} d\bar{u} d\bar{v} + r^2 d\Omega^2. \end{aligned} \quad (4.2.76)$$

Introducing new time and space coordinates $\bar{t} = (\bar{u} + \bar{v})/2$, $\bar{x} = (\bar{v} - \bar{u})/2$, so that

$$\bar{u} = \bar{t} - \bar{x}, \quad \bar{v} = \bar{t} + \bar{x},$$

one obtains a more familiar-looking form,

$$g = \frac{32m \exp(-\frac{r}{2m})}{r \cos^2 \bar{u} \cos^2 \bar{v}} (-d\bar{t}^2 + d\bar{x}^2) + r^2 d\Omega^2.$$

This is regular except at $\cos \bar{u} = 0$, or $\cos \bar{v} = 0$, or $r = 0$. The first set corresponds to the straight lines $\bar{u} = \bar{t} - \bar{x} \in \{\pm\pi/2\}$, while the second is the union of the lines $\bar{v} = \bar{t} + \bar{x} \in \{\pm\pi/2\}$.

The analysis of $\{r = 0\}$ requires some work: recall that $r \rightarrow 0$ corresponds to $\hat{u}\hat{v} \rightarrow 1$, which is equivalent to

$$\tan(\bar{u}) \tan(\bar{v}) \rightarrow 1.$$

Using the formula

$$\tan(\bar{u} + \bar{v}) = \frac{\tan \bar{u} + \tan \bar{v}}{1 - \tan \bar{u} \tan \bar{v}}$$

we obtain $\tan(\bar{u} + \bar{v}) \rightarrow_{r \rightarrow 0} \pm\infty$ unless perhaps the numerator tends to 0. Except for the last borderline cases, this is equivalent to

$$\bar{u} + \bar{v} = 2\bar{t} \rightarrow \pm\pi/2.$$

So the Kruskal–Szekeres metric is conformal to a smooth Lorentzian metric on $C \times S^2$, where C is the set of Fig. 4.2.7.

REMARK 4.2.15 The ‘ $r = \infty$ ’ open intervals in Fig. 4.2.7 correspond to the conformal boundaries $\mathscr{J}^\pm$ constructed in Remark 4.2.6. We see that the conformal boundary of the Kruskal–Szekeres extension of the exterior region of the Schwarzschild spacetime has four connected components, with two components for $\mathscr{J}^+$ and two for $\mathscr{J}^-$.

4.2.9 Weyl coordinates

A set of coordinates well suited to study static axisymmetric metrics has been introduced by Weyl. In those coordinates the Schwarzschild metric takes the form (cf., e.g., [436, Eq. (20.12)])

$$g = -e^{2U_{\text{Schw}}} dt^2 + e^{-2U_{\text{Schw}}} \rho^2 d\varphi^2 + e^{2\lambda_{\text{Schw}}} (d\rho^2 + dz^2), \quad (4.2.77)$$

where

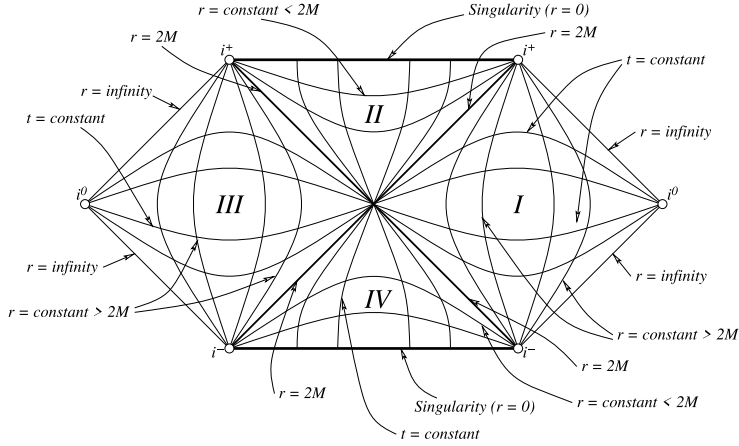


Fig. 4.2.7 The Carter–Penrose diagram for the Kruskal–Szekeres spacetime with mass M . There are actually two asymptotically flat regions, with corresponding event horizons defined with respect to the second region. Each point in this diagram represents a two-dimensional sphere, and coordinates are chosen so that light cones have slopes of ± 1 . Regions are numbered as in Fig. 4.2.1. Fig. by J. P. Nicolas and M. Maliborski, reprinted with permission.

$$U_{\text{Schw}} = \ln \rho - \ln \left(m \sin \tilde{\theta} + \sqrt{\rho^2 + m^2 \sin^2 \tilde{\theta}} \right) \quad (4.2.78)$$

$$= \frac{1}{2} \ln \left[\frac{\sqrt{(z-m)^2 + \rho^2} + \sqrt{(z+m)^2 + \rho^2} - 2m}{\sqrt{(z-m)^2 + \rho^2} + \sqrt{(z+m)^2 + \rho^2} + 2m} \right], \quad (4.2.79)$$

$$\lambda_{\text{Schw}} = -\frac{1}{2} \ln \left[\frac{(r_{\text{Schw}} - m)^2 - m^2 \cos^2 \tilde{\theta}}{r_{\text{Schw}}^2} \right] \quad (4.2.80)$$

$$= -\frac{1}{2} \ln \left[\frac{4\sqrt{(z-m)^2 + \rho^2}\sqrt{(z+m)^2 + \rho^2}}{\left[2m + \sqrt{(z-m)^2 + \rho^2} + \sqrt{(z+m)^2 + \rho^2} \right]^2} \right]. \quad (4.2.81)$$

In (4.2.78) the angle $\tilde{\theta}$ is a Schwarzschild angular variable, with the relations

$$\begin{aligned} 2m \cos \tilde{\theta} &= \sqrt{(z+m)^2 + \rho^2} - \sqrt{(z-m)^2 + \rho^2}, \\ 2(r_{\text{Schw}} - m) &= \sqrt{(z+m)^2 + \rho^2} + \sqrt{(z-m)^2 + \rho^2}, \\ \rho^2 &= r_{\text{Schw}}(r_{\text{Schw}} - 2m) \sin^2 \tilde{\theta}, \quad z = (r_{\text{Schw}} - m) \cos \tilde{\theta}, \end{aligned}$$

where r_{Schw} is the usual Schwarzschild radial variable such that $e^{2U_{\text{Schw}}} = 1 - 2m/r_{\text{Schw}}$. It is easily seen that the function U_{Schw} is smooth on $\mathbb{R}^3$ except on the set $\{\rho = 0, -m \leq z \leq m\}$. From (4.2.78) we find, at fixed z in the interval $-m < z < m$ and for small ρ ,

$$U_{\text{Schw}}(\rho, z) = \ln \rho - \ln(2\sqrt{(m+z)(m-z)}) + O(\rho^2) \quad (4.2.82)$$

(with the error term *not* uniform in z).

The value $\dot{\lambda}$ of λ on the rod equals

$$\dot{\lambda}(z) = -\frac{1}{2} \ln \left[\frac{(m-z)(z+m)}{(2m)^2} \right].$$

4.3 Some general notions

Before continuing, some general notions are in order.

4.3.1 Isometries

A *Killing field* X , by definition, is a vector field, the local flow of which preserves the metric. Equivalently, X satisfies the *Killing equation*

$$0 = \mathcal{L}_X g_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu. \quad (4.3.1)$$

The set of solutions of this equation forms a Lie algebra, where the bracket operation is the bracket of vector fields (see Appendix A, p. 337).

One of the features of the Schwarzschild metric (4.2.1) is its *stationarity*, with Killing vector field $X = \partial_t$: a spacetime is called *stationary* if there exists a Killing vector field X which approaches ∂_t in the asymptotically flat region (where r goes to ∞ ; see Section 4.3.7 for precise definitions) and generates one-parameter groups of isometries. A spacetime is called *static* if it is stationary and if the stationary Killing vector X is hypersurface-orthogonal, i.e.

$$X^\flat \wedge dX^\flat = 0, \quad (4.3.2)$$

where

$$X^\flat = X_\mu dx^\mu = g_{\mu\nu} X^\nu dx^\mu.$$

EXERCISE 4.3.1 Show that the Schwarzschild metric, as well as the Reissner–Nordström metrics of Section 4.5, is static but the Kerr metrics with $a \neq 0$, presented in Section 4.6, are not. $\square$

Any metric with a Killing vector field X can be locally written, away from the zeros of X , in the form

$$g = -F(dt + \underbrace{\theta_i dx^i}_{=: \theta})^2 + h_{ij} dx^i dx^j, \quad (4.3.3)$$

with $X = \partial_t$ and hence $\partial_t g_{\mu\nu} = 0$. Then

$$X^\flat = -F(dt + \theta), \quad X^\flat = -dF \wedge (dt + \theta) - Fd\theta, \quad X^\flat \wedge dX^\flat = F^2(dt + \theta) \wedge d\theta. \quad (4.3.4)$$

It follows that g is static if and only if $d\theta = 0$. Therefore, for static metrics, on any simply connected subset of $\mathcal{M}$ there exists a function $f = f(x^i)$ such that $\theta = df$. Introducing a new time coordinate $\tau := t + f$, we conclude that any static metric g can locally be written as

$$g = -F d\tau^2 + h_{ij} dx^i dx^j. \quad (4.3.5)$$

We also see that staticity leads to, and is equivalent to, the existence of a supplementary local discrete isometry of g obtained by mapping τ to its negative, $\tau \mapsto -\tau$.

On a simply connected spacetime $\mathcal{M}$ the representation (4.3.5) is global, with a function V without zeros, provided that there exists in $\mathcal{M}$ a hypersurface $\mathcal{S}$ which is transverse to a globally timelike Killing vector X , with every orbit of X meeting $\mathcal{S}$ precisely once.

A spacetime is called *axisymmetric* if there exists a Killing vector field Y , which generates a one-parameter group of isometries, and which behaves like a *rotation*: this last property is captured by requiring that all orbits are 2π periodic, and that the set $\{Y = 0\}$, called the *axis of rotation*, is non-empty. Killing vector fields which are a non-trivial linear combination of a time translation and of a rotation in the asymptotically flat region are called *stationary-rotating*, or *helical*. Note that those definitions require completeness of orbits of all Killing vector fields (this means that

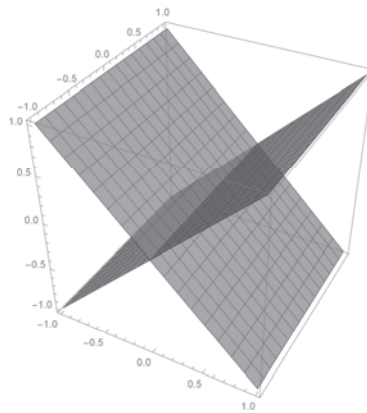


Fig. 4.3.1 The four branches of a bifurcate horizon and the bifurcation surface for the boost Killing vector $x\partial_t + t\partial_x$ in three-dimensional Minkowski spacetime.

the equation $\dot{x} = X$ has a global solution for all initial values); see [99] and [215] for some results concerning this question.

In the extended Schwarzschild spacetime the set $\{r = 2m\}$ is a null hypersurface $\mathcal{E}$, the Schwarzschild event horizon. The stationary Killing vector $X = \partial_t$ extends to a Killing vector $\hat{X}$ in the extended spacetime which becomes tangent to and null on $\mathcal{E}$, except at the ‘bifurcation sphere’ right in the middle of Fig. 4.2.7, where $\hat{X}$ vanishes.

4.3.2 Killing horizons

A *null* hypersurface which *coincides* with a connected component of the set

$$\mathcal{N}_X := \{g(X, X) = 0, X \neq 0\}, \quad (4.3.6)$$

where X is a Killing vector, with X *tangent* to $\mathcal{N}$, is called a *Killing horizon* associated to X .

Here it is implicitly assumed that the hypersurface is *embedded*.

We will sometimes write $\mathcal{N}(X)$ instead of $\mathcal{N}_X$.

EXAMPLE 4.3.2 The simplest example is provided by the ‘boost Killing vector field’

$$X = z\partial_t + t\partial_z \quad (4.3.7)$$

in Minkowski spacetime: the Killing horizon $\mathcal{N}_X$ of X has four connected components

$$\mathcal{N}(X)_{\epsilon\delta} := \{t = \epsilon z, \delta t > 0\}, \quad \epsilon, \delta \in \{\pm 1\}. \quad (4.3.8)$$

Indeed, we have

$$\{g(X, X) = 0\} = \{t = \pm z\}, \quad (4.3.9)$$

but from this we need to remove the set of points $\{z = t = 0\}$, where X vanishes. The closure $\overline{\mathcal{N}_X}$ of $\mathcal{N}_X$ is the set $\{|t| = |z|\}$, as in (4.3.9), which is *not* a manifold, because of the crossing of the null hyperplanes $\{t = \pm z\}$ at $t = z = 0$; see Fig. 4.3.1. As already mentioned, horizons of this type are referred to as *bifurcate Killing horizons*. More precisely, a set will be called a *bifurcate Killing horizon* if it is the union of a smooth submanifold S of codimension two, on which the Killing vector vanishes, called the *bifurcation surface*, and of four Killing horizons obtained by shooting null geodesics in the four distinct null directions orthogonal to S . So, the Killing vector $z\partial_t + t\partial_z$ in Minkowski spacetime has a bifurcate Killing horizon, with the bifurcation surface $\{t = z = 0\}$.

EXAMPLE 4.3.3 Figure 4.2.1, p. 132, makes it clear that the set $\{r = 2m\}$ in the Kruskal–Szekeres spacetime is the union of four Killing horizons and of the bifurcation surface, with respect to the Killing vector field which equals ∂_t in the asymptotically flat region. $\square$

It turns out that the above examples are typical. Indeed, consider a spacelike submanifold S of codimension two in a spacetime $(\mathcal{M}, g)$, and suppose that there exists a (non-trivial) Killing vector field X which vanishes on S . Then the one-parameter group of isometries $\phi_t[X]$ generated by X leaves S invariant and, at $p \in S$, the tangent maps $\phi_t[X]_*$ induce isometries of $T_p\mathcal{M}$ to itself. At every $p \in S$ there exist precisely two null directions $\text{Vect}\{n_\pm\} \subset T_p\mathcal{M}$, where $n_\pm$ are two distinct null future-directed vectors normal to S . Since every geodesic is uniquely determined by its initial point and its initial direction, we conclude that the null geodesics through p are mapped to themselves by the flow of X . Thus X is tangent to those geodesics. There exist two null hypersurfaces $\mathcal{N}_\pm$ threaded by those null geodesics, intersecting at S . We define $\mathcal{N}_{\pm+}$ to be the connected components of $\mathcal{N}_\pm \setminus \{X = 0\}$ lying to the *future* of S and accumulating at S . Similarly we define $\mathcal{N}_{\pm-}$ to be the connected components of $\mathcal{N}_\pm \setminus \{X = 0\}$ lying to the *past* of S and accumulating at S . Then the $\mathcal{N}_{\pm\pm}$ are Killing horizons which, together with S , form a bifurcate Killing horizon with bifurcation surface S .

EXAMPLE 4.3.4 One more noteworthy example, in Minkowski spacetime, is provided by the Killing vector

$$X = y\partial_t + t\partial_y + x\partial_y - y\partial_x = y\partial_t + (t+x)\partial_y - y\partial_x. \quad (4.3.10)$$

Thus, X is the sum of a boost $y\partial_t + t\partial_y$ and a rotation $x\partial_y - y\partial_x$. Note that X vanishes if and only if

$$y = t + x = 0,$$

which is a two-dimensional isotropic (null) submanifold of Minkowski spacetime $\mathbb{R}^{1,3}$. Further,

$$g(X, X) = (t+x)^2 = 0,$$

which is an isotropic hyperplane in $\mathbb{R}^{1,3}$.

The example shows that the Killing horizon $\{t = x, y > 0\}$ admits non-colinear Killing vectors tangent to its generators. Properties of such horizons have been investigated in [337, 338].

REMARK 4.3.5 When attempting to prove uniqueness of black holes, one is naturally led to the following notion: let X be a Killing vector, then every connected, not necessarily embedded, null hypersurface $\mathcal{N}_0 \subset \mathcal{N}_X$, with $\mathcal{N}_X$ as in (4.3.6), with the property that X is tangent to $\mathcal{N}_0$, is called a *Killing prehorizon*.

One of the fundamental differences between prehorizons and horizons is that the latter are necessarily embedded, while the former are allowed not to be. Thus, a Killing horizon is also a Killing prehorizon, but the reverse implication is not true in general. As an example, consider $\mathbb{R} \times \mathbb{T}^2$ with the flat product metric, and let Y be any covariantly constant unit vector on $\mathbb{T}^2$, the orbits of which are dense on $\mathbb{T}^2$. Let $\Gamma \subset \mathbb{T}^2$ be such an orbit; then $\mathbb{R} \times \Gamma$ provides an example of non-embedded prehorizon associated with the null Killing vector $X := \partial_t + Y$.

Prehorizons are a major headache to handle in the uniqueness theory of stationary black holes, and one of the key steps there is to prove that they do not exist within the domain of outer communications of well-behaved black-hole spacetimes [113, 120, 339]. $\square$

4.3.3 Surface gravity

The *surface gravity* κ of a Killing horizon $\mathcal{N}(X)$ is defined by the formula

$$(X^\alpha X_\alpha)_{,\mu} \Big|_{\mathcal{N}(X)} = -2\kappa X_\mu. \quad (4.3.11)$$

A word of justification is in order here: since $g(X, X) = 0$ on $\mathcal{N}(X)$, the differential of $g(X, X)$ is conormal to $\mathcal{N}(X)$. (A form α is said to be *conormal* to S if for every vector $Y \in TS$ we have $\alpha(Y) = 0$.) Recalling (cf. Appendix F, p. 356) that on a null hypersurface the conormal is proportional to $g(\ell, \cdot)$, where ℓ is any null vector tangent to $\mathcal{N}$ (those are defined uniquely up to a proportionality factor), we obtain that $d(g(X, X))$ is proportional to $X^\flat = X_\mu dx^\mu$; whence (4.3.11). We will show shortly that κ is a constant under fairly general conditions.

The surface gravity of black holes plays an important role in *black-hole thermodynamics*, cf. e.g. [66] and references therein.

REMARK 4.3.6 The surface gravity measures the acceleration a of the integral curves of the Killing vector X , in the following sense: let $\tau \mapsto x^\mu(\tau)$ be an integral curve of X ; thus, $x^\mu(\tau)$ solves the equation

$$\dot{x}^\mu(\tau) \equiv \frac{dx^\mu}{d\tau}(\tau) = X^\mu(x^\alpha(\tau)).$$

The acceleration $a = a^\mu \partial_\mu$ of the curve $x^\mu(\tau)$ is defined as $D\dot{x}^\mu/d\tau$. On the Killing horizon $\mathcal{N}_X$ we have

$$\begin{aligned} a^\mu &= \frac{D\dot{x}^\mu}{d\tau} = \dot{x}^\sigma \nabla_\sigma X^\mu = X^\sigma \nabla_\sigma X^\mu = -X^\sigma \nabla^\mu X_\sigma = -\frac{1}{2} \nabla^\mu (X^\sigma X_\sigma) \\ &= \kappa \dot{x}^\mu = \kappa X^\mu. \end{aligned}$$

Thus

$$a = \kappa X \tag{4.3.12}$$

on the Killing horizon. $\square$

As an example of calculation of surface gravity, consider the Killing vector X of (4.3.7). We have

$$d(g(X, X)) = d(-z^2 + t^2) = 2(-zdz + tdt).$$

On $\mathcal{N}(X)_{\epsilon\delta}$ we have $t = \epsilon z$, and

$$X^\flat = -zdt + tdz = z(-dt + \epsilon dz) = -\frac{1}{2} \epsilon d(g(X, X))|_{\mathcal{N}(X)_{\epsilon\delta}},$$

and so

$$\kappa = \epsilon \in \{\pm 1\}. \tag{4.3.13}$$

As another example, for the Killing vector X of (4.3.10) we have

$$d(g(X, X)) = 2(t+x)(dt+dx),$$

which vanishes on each of the Killing horizons $\{t = -x, y \neq 0\}$. We conclude that $\kappa = 0$ on both horizons.

A Killing horizon $\mathcal{N}_X$ is said to be *degenerate*, or *extreme*, if κ vanishes throughout $\mathcal{N}_X$; it is called *non-degenerate* if κ has no zeros on $\mathcal{N}_X$. Thus, the Killing horizons $\mathcal{N}(X)_{\epsilon\delta}$ of (4.3.8) are non-degenerate, while both Killing horizons of X given by (4.3.10) are degenerate.

EXAMPLE 4.3.7 Consider the Schwarzschild metric, as extended in (4.2.16),

$$g = -\left(1 - \frac{2m}{r}\right) dv^2 + 2dvdr + r^2 d\Omega^2. \tag{4.3.14}$$

We have

$$d(g(X, X)) = d(g(\partial_v, \partial_v)) = -\frac{2m}{r^2} dr.$$

Now, $X^\flat = g(\partial_v, \cdot) = -(1 - \frac{2m}{r})dv + dr$, which equals dr at the hypersurface $r = 2m$. Comparing with (4.3.11) gives

$$\kappa \equiv \kappa_m := \frac{1}{4m}.$$

We see that the black-hole event horizon in the above extension of the Schwarzschild metric is a non-degenerate Killing horizon, with surface gravity $(4m)^{-1}$.

The same calculation for the extension based on the retarded time coordinate u of (4.2.15) proceeds identically except for various sign changes, resulting in $\kappa = -1/4m$ for the white-hole event horizon.

Note that there are no black holes with degenerate Killing horizons within the Schwarzschild family. In fact [142], there are no suitably regular, degenerate, static vacuum black holes at all. $\square$

In Kerr spacetimes (see Section 4.6, p. 174) we have $\kappa = 0$ if and only if $m = a$. On the other hand, all horizons in the multi-black-hole Majumdar–Papapetrou solutions of Section 4.7 are degenerate.

4.3.4 Zeroth law

In what follows we will prove that κ , or its square, is constant on Killing horizons under various circumstances. This is a fundamental property of Killing horizons in spacetimes of physical interest. It plays a key role in many results, including the classification theory of stationary black holes. It is used when assigning a temperature to Killing horizons in black-hole thermodynamics; in that context it is referred to as the *zeroth law of black-hole thermodynamics*.

In the proofs we will need to differentiate the equivalent defining equation for surface gravity

$$X^\gamma \nabla_\alpha X_\gamma \big|_{\mathcal{N}} = -\kappa X_\alpha. \quad (4.3.15)$$

For this some preliminary work is needed.

Let $t_{\alpha_1 \dots \alpha_\ell}$ be any tensor field vanishing on $\mathcal{N}$. Then

$$k^\beta \nabla_\beta t_{\alpha_1 \dots \alpha_\ell} \big|_{\mathcal{N}} = 0 \quad (4.3.16)$$

for any vector field k^β tangent to $\mathcal{N}$. Since X_β spans the space of covectors annihilating $\mathcal{N}$, (4.3.16) holds if and only if $\nabla_\beta t_{\alpha_1 \dots \alpha_\ell} \big|_{\mathcal{N}}$ equals $X_\beta s_{\alpha_1 \dots \alpha_\ell}$ for some tensor field $s_{\alpha_1 \dots \alpha_\ell}$. Equivalently,

$$X_{[\gamma} \nabla_{\beta]} t_{\alpha_1 \dots \alpha_\ell} \big|_{\mathcal{N}} = 0. \quad (4.3.17)$$

This is our desired differential consequence of the vanishing of $t_{\alpha_1 \dots \alpha_\ell} \big|_{\mathcal{N}}$.

We have the following.

THEOREM 4.3.8 κ^2 is a non-zero constant on bifurcate Killing horizons.

REMARK 4.3.9 Both (4.3.13) and Example 4.3.7 show that κ , as defined in (4.3.11), is *not* constant on a bifurcate Killing horizon, because its sign changes when passing from one Killing horizon component to another. One can fiddle with the sign in (4.3.11) when passing from one branch to another to obtain constant value of κ throughout a bifurcate Killing horizon, but we will not proceed in this manner. $\square$

Proof. We follow the argument in [285, p. 59]. Consider, quite generally, a smooth hypersurface $\mathcal{N}$ with defining function f ; by definition, this means that f vanishes precisely on $\mathcal{N}$, with df different from 0 on $\mathcal{N}$. Thus, on each connected component $\mathcal{N}$ of our bifurcate Killing horizon we have such a function f . Next, we claim that there exists a function h such that on $\mathcal{N}$ we have

$$X^\flat := g_{\mu\nu} X^\mu dx^\nu = h df. \quad (4.3.18)$$

(This property is called *hypersurface orthogonality*.) Indeed, if Y is any vector tangent to $\mathcal{N}$, then $Y^\mu \partial_\mu f = 0$ on $\mathcal{N}$, since f is constant on $\mathcal{N}$. On the other hand,

$Y^\mu X_\mu = 0$, because a null vector tangent to a null hypersurface $\mathcal{N}$ is orthogonal to all vectors tangent to $\mathcal{N}$. It follows that X^b is proportional to df , which justifies the existence of the function h in (4.3.18). We emphasize that we do *not* assume (4.3.18) everywhere, but only on $\mathcal{N}$.

We start by showing that (4.3.18) implies the identity

$$X_{[\mu} \nabla_\nu X_{\rho]} \big|_{\mathcal{N}} = 0. \quad (4.3.19)$$

Indeed, differentiating (4.3.18) we find that

$$\nabla_\nu X_\rho \big|_{\mathcal{N}} = \nabla_\nu h \nabla_\rho f + h \nabla_\nu \nabla_\rho f + X_\nu Z_\rho, \quad (4.3.20)$$

for some vector field Z . Here, as already explained earlier, Z accounts for the fact that the equality (4.3.18) only holds on $\mathcal{N}$, and therefore differentiation might introduce non-zero terms in directions transverse to $\mathcal{N}$. The first term in (4.3.20) drops out under antisymmetrization as in (4.3.19) since X is proportional to ∇f ; similarly for the last one. The second term is symmetric in ρ and ν , and also gives zero under antisymmetrization. This establishes (4.3.19). (In fact, (4.3.19) is a special case of the *Frobenius theorem*, keeping in mind that X_α is hypersurface-orthogonal.)

We continue with the identity

$$\nabla^\nu X^\rho \nabla_\nu X_\rho \big|_{\mathcal{N}} = -2\kappa^2. \quad (4.3.21)$$

To see this, we multiply (4.3.19) by $\nabla^\nu X^\rho$ and expand: using the symbol ‘ $=_{\mathcal{N}}$ ’ to denote equality on $\mathcal{N}$, we find that

$$\begin{aligned} 0 &=_{\mathcal{N}} \nabla^\nu X^\rho X_\mu \nabla_\nu X_\rho + \underbrace{\nabla^\nu X^\rho X_\nu}_{\kappa X^\rho} \nabla_\rho X_\mu + \underbrace{\nabla^\nu X^\rho X_\rho}_{-\kappa X^\nu} \nabla_\mu X_\nu \\ &=_{\mathcal{N}} \nabla^\nu X^\rho X_\mu \nabla_\nu X_\rho + \underbrace{\kappa X^\rho \nabla_\rho X_\mu}_{\kappa X_\mu} - \underbrace{\kappa X^\nu \nabla_\mu X_\nu}_{-\kappa X_\mu} \\ &=_{\mathcal{N}} (\nabla^\nu X^\rho \nabla_\nu X_\rho + 2\kappa^2) X_\mu. \end{aligned}$$

This proves (4.3.21) away from the set where X vanishes.

To continue, recall that a bifurcate horizon is the union of four Killing horizons, which are smooth hypersurfaces on which X has no zeros, and of the bifurcation surface S , where X vanishes. So the set $\{X \neq 0\}$ is dense on a bifurcate horizon. Recall also that κ has *not* been defined on S so far, as the definition needs the condition $X \neq 0$. But we can view instead (4.3.21) as the definition of κ , up to sign, including points at points at which X vanishes. Then the calculation just given shows that the function κ^2 , so extended to S , is a smooth function on a bifurcate Killing horizon $\mathcal{N}$, with (4.3.21) holding throughout $\mathcal{N}$.

Now, a Killing vector field satisfies the set of equations

$$\nabla_\alpha \nabla_\beta X_\gamma = R_{\sigma\alpha\beta\gamma} X^\sigma. \quad (4.3.22)$$

Indeed, the Killing equation

$$\nabla_\alpha X_\beta + \nabla_\beta X_\alpha = 0 \quad (4.3.23)$$

leads to a second-order system of equations, as follows: taking cyclic permutations of the equation obtained by differentiating (4.3.23) one has

$$\begin{aligned} -\nabla_\gamma \nabla_\alpha X_\beta - \nabla_\gamma \nabla_\beta X_\alpha &= 0, \\ \nabla_\alpha \nabla_\beta X_\gamma + \nabla_\alpha \nabla_\gamma X_\beta &= 0, \end{aligned}$$

$$\nabla_\beta \nabla_\gamma X_\alpha + \nabla_\beta \nabla_\alpha X_\gamma = 0.$$

Adding, and expressing commutators of derivatives in terms of the Riemann tensor, one obtains

$$\begin{aligned} 2\nabla_\alpha \nabla_\beta X_\gamma &= (R_{\sigma\gamma\beta\alpha} + R_{\sigma\alpha\beta\gamma} + \underbrace{R_{\sigma\beta\alpha\gamma}}_{=-R_{\sigma\alpha\gamma\beta}-R_{\sigma\gamma\beta\alpha}}) X^\sigma \\ &= 2R_{\sigma\alpha\beta\gamma} X^\sigma, \end{aligned}$$

as desired.

Differentiating (4.3.21) we obtain, using (4.3.22),

$$\nabla^\nu X_\rho \underbrace{\nabla_\sigma \nabla_\nu X_\rho}_{R_{\alpha\sigma\nu\rho} X^\alpha} \Big|_{\mathcal{N}} = -\nabla_\sigma(\kappa^2). \quad (4.3.24)$$

So the left-hand side vanishes on S . It follows that $\nabla\kappa^2$ vanishes on S . We conclude that κ^2 is constant on any connected component of S .

Contracting (4.3.24) with X^σ we further find that

$$-\mathcal{L}_X \kappa^2 \Big|_{\mathcal{N}} = -X^\sigma \nabla_\sigma(\kappa^2) \Big|_{\mathcal{N}} = \nabla^\nu X_\rho \underbrace{R_{\alpha\sigma\nu\rho} X^\alpha X^\sigma}_{=0} = 0. \quad (4.3.25)$$

Hence κ^2 is constant along the null Killing orbits threading each of the Killing horizons issued from S . Continuity implies that on each orbit the surface gravity κ^2 takes the same value as at its accumulation point at S . But we have already seen that κ^2 is constant on S . We conclude that κ^2 is constant throughout the bifurcate Killing horizon emanating from S .

It remains to show that κ^2 cannot vanish on S . For this, consider any point p on S , and let x^μ be a coordinate system at p such that the vectors $\{\partial_\mu\}_{\mu=0,\dots,n}$ form an orthonormal frame at p with the vectors ∂_A , $A = 2, \dots, n$ tangent to S . Then the components

$$\nabla_A X^\mu = \partial_A X^\mu + \Gamma^\mu_{\nu A} X^\nu$$

of ∇X vanish at p , since X vanishes on S and since the derivatives $\partial_A X^\mu$ are in directions tangent to S . So there exists a number a such that at p the antisymmetric matrix $\nabla_\mu X_\nu$ takes the form

$$(\nabla_\mu X_\nu) = \begin{pmatrix} 0 & a & 0 & \dots & 0 \\ -a & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}.$$

This gives $\nabla_a X_b \nabla^a X^b = -2a^2$, hence $a = \pm\kappa$.

So if κ vanishes at S , then both X and ∇X vanish at S . But we have the following.

PROPOSITION 4.3.10 *Let $\mathcal{M}$ be connected and let $p \in \mathcal{M}$. A Killing vector is uniquely defined by its value $X(p)$ and the value at p of the anti-symmetric tensor $\nabla X(p)$.*

Proof. Consider two Killing vectors X and Y such that $X(p) = Y(p)$ and $\nabla X(p) = \nabla Y(p)$. Let $q \in \mathcal{M}$ and let γ be any curve from p to q . Set

$$Z^\beta := X^\beta - Y^\beta, \quad A_{\alpha\beta} = \nabla_\alpha(X_\beta - Y_\beta).$$

Along the curve γ we have, using (4.3.22),

$$\frac{DZ_\alpha}{ds} = \dot{\gamma}^\mu \nabla_\mu Z_\alpha = \dot{\gamma}^\mu A_{\mu\alpha}$$

$$\frac{DA_{\alpha\beta}}{ds} = \dot{\gamma}^\mu \nabla_\mu \nabla_\alpha Z_\beta = R_{\gamma\mu\alpha\beta} \dot{\gamma}^\mu Z^\gamma. \quad (4.3.26)$$

This is a linear first-order system of ODEs along γ with vanishing Cauchy data at p . Hence the solution vanishes along γ , and thus $X^\mu(q) = Y^\mu(q)$. $\square$

Returning to the proof of Theorem 4.3.8, Proposition 4.3.10 gives thus $X \equiv 0$, contradicting the definition of a bifurcate Killing horizon. $\square$

Yet another class of spacetimes with constant κ (see [250], Theorem 7.1 or [454], Section 12.5) is provided by spacetimes satisfying the *dominant energy condition*: this means that

$$T_{\mu\nu} X^\mu Y^\nu \geq 0 \text{ for all causal future-directed vector fields } X \text{ and } Y. \quad (4.3.27)$$

Our aim now is to prove this.

Since X_α is hypersurface-orthogonal on $\mathcal{N}$, from the ‘Frobenius identity’ (4.3.19) we have

$$\begin{aligned} 0 &=_{\mathcal{N}} 3X_{[\beta} \nabla_\sigma X_{\gamma]} =_{\mathcal{N}} X_\beta \nabla_\sigma X_\gamma + X_\sigma \nabla_\gamma X_\beta + X_\gamma \nabla_\beta X_\sigma \\ &=_{\mathcal{N}} 2X_{[\beta} \nabla_\sigma] X_\gamma + X_\gamma \nabla_\beta X_\sigma. \end{aligned} \quad (4.3.28)$$

Equivalently,

$$X_{[\beta} \nabla_\sigma] X_\gamma \big|_{\mathcal{N}} = \frac{1}{2} X_\gamma \nabla_\sigma X_\beta. \quad (4.3.29)$$

Thus, applying the differential operator $X_{[\beta} \nabla_\sigma]$ to the left-hand side of (4.3.15), we find that

$$\begin{aligned} X_{[\beta} \nabla_\sigma] (X^\gamma \nabla_\alpha X_\gamma) &= \underbrace{(X_{[\beta} \nabla_\sigma] X^\gamma)}_{\frac{1}{2} X^\gamma \nabla_\sigma X_\beta} \nabla_\alpha X_\gamma + X^\gamma \underbrace{X_{[\beta} \nabla_\sigma] \nabla_\alpha X_\gamma}_{-R_{\sigma]\mu\alpha\gamma} X^\mu} \\ &= -\frac{\kappa}{2} X_\alpha \nabla_\sigma X_\beta - X_{[\beta} R_{\sigma]\mu\alpha\gamma} X^\gamma X^\mu \\ &= \kappa X_{[\sigma} \nabla_\beta] X_\alpha - X_{[\beta} R_{\sigma]\mu\alpha\gamma} X^\gamma X^\mu. \end{aligned} \quad (4.3.30)$$

Comparing with the corresponding derivatives of the right-hand side of (4.3.15), we conclude that

$$X_{[\beta} R_{\sigma]\mu\alpha\gamma} X^\gamma X^\mu \big|_{\mathcal{N}} = X_\alpha X_{[\sigma} \nabla_\beta] \kappa. \quad (4.3.31)$$

The next step is to show that

$$X_{[\beta} R_{\sigma]\mu\alpha\gamma} X^\gamma X^\mu \big|_{\mathcal{N}} = X_\alpha X_{[\beta} R_{\sigma]\gamma} X^\gamma. \quad (4.3.32)$$

For this, we apply $X_{[\mu} \nabla_\nu]$ to (4.3.19): letting $S_{\alpha\beta\gamma}$ denote a cyclic sum over $\alpha\beta\gamma$, and using (4.3.29) we obtain

$$\begin{aligned} 0 &=_{\mathcal{N}} X_{[\mu} \nabla_\nu] (S_{\alpha\beta\gamma} X_\alpha \nabla_\beta X_\gamma) \\ &=_{\mathcal{N}} S_{\alpha\beta\gamma} \left(\underbrace{(X_{[\mu} \nabla_\nu] X_\alpha)}_{\frac{1}{2} X_\alpha \nabla_\nu X_\mu} \nabla_\beta X_\gamma + X_\alpha X_{[\mu} \nabla_\nu] \underbrace{\nabla_\beta X_\gamma}_{R_{\nu]\sigma\gamma\beta} X^\sigma} \right) \\ &=_{\mathcal{N}} S_{\alpha\beta\gamma} X_\alpha X^\sigma X_{[\mu} R_{\nu]\sigma\gamma\beta}. \end{aligned} \quad (4.3.33)$$

Writing out this sum, and contracting with $g^{\alpha\mu}$, after a renaming of indices one obtains (4.3.32).

Comparing (4.3.31) and (4.3.32), since X does not vanish anywhere on a Killing horizon,

$$X_{[\alpha} \nabla_\beta] \kappa = -X_{[\alpha} R_{\beta]\gamma} X^\gamma. \quad (4.3.34)$$

We have therefore proved the following (a similar result is valid in the more general context of isolated horizons [321, Equation (3.29)]).

PROPOSITION 4.3.11 *Let $\mathcal{N}$ be a Killing horizon associated with a Killing vector X . The surface gravity κ is constant on $\mathcal{N}$ if and only if*

$$X_{[\alpha}R_{\mu]\nu}X^\nu = 0 \text{ on } \mathcal{N}. \quad (4.3.35)$$

□

Let us relate (4.3.35) to the dominant energy condition, alluded to earlier. In vacuum the Ricci tensor vanishes, so clearly (4.3.35) is satisfied. More generally, using the Einstein equation, (4.3.35) is equivalent to

$$X_{[\alpha}T_{\mu]\nu}X^\nu = 0 \text{ on } \mathcal{N}. \quad (4.3.36)$$

Now, multiplying (4.3.34) by X^α and using $X^\alpha \nabla_\alpha \kappa = 0$ one finds that

$$R_{\mu\nu}X^\mu X^\nu \big|_{\mathcal{N}} = 0, \quad (4.3.37)$$

and therefore also

$$T_{\mu\nu}X^\mu X^\nu \big|_{\mathcal{N}} = 0. \quad (4.3.38)$$

Assuming the dominant energy condition, this is possible if and only if (4.3.36) holds. Indeed, the condition that $T_{\mu\nu}X^\mu X^\nu$ is positive for all causal future-directed vectors implies that $-T^\mu{}_\nu X^\nu$ is causal future directed. But then $T_{\mu\nu}X^\mu X^\nu$ vanishes on $\mathcal{N}$ if and only if $T^\mu{}_\nu X^\nu$ is proportional to X^μ , which implies (4.3.36). We conclude that Proposition 4.3.11 applies, leading to the following.

THEOREM 4.3.12 *Let $\mathcal{N}$ be a Killing horizon and suppose that the energy-momentum tensor satisfies the dominant energy condition*

$$T_{\mu\nu}Z^\nu Y^\mu \geq 0 \text{ for all causal future-directed } Z \text{ and } Y. \quad (4.3.39)$$

Then κ is constant on $\mathcal{N}$.

We conclude this section with the following result, originally proved by Rácz and Wald [408] in spacetime dimension $n+1=4$.

THEOREM 4.3.13 *Let $\mathcal{N}$ be a Killing horizon associated with a Killing vector field X in an $(n+1)$ -dimensional spacetime. Then the surface gravity κ is constant on $\mathcal{N}$ if and only if the exterior derivative of the twist form field ω is zero on the horizon, i.e.*

$$\nabla_{[\alpha_0}\omega_{\alpha_1\dots\alpha_{n-2}]} \big|_{\mathcal{N}} = 0,$$

where ω is defined as

$$\omega_{\alpha_1\dots\alpha_{n-2}} = \epsilon_{\alpha_1\dots\alpha_{n-2}\beta\gamma\delta} X^\beta \nabla^\gamma X^\delta. \quad (4.3.40)$$

Proof. Recall (4.3.34):

$$X_{[\alpha}\nabla_{\beta]}\kappa \big|_{\mathcal{N}} = -X_{[\alpha}R_{\beta]\gamma}X^\gamma. \quad (4.3.41)$$

On the other hand, letting $\delta_{\gamma\delta}^{\alpha\beta} = \delta_{[\gamma}^{\alpha}\delta_{\delta]}^{\beta}$ we have

$$\begin{aligned} & \epsilon^{\alpha\beta\gamma\delta_1\dots\delta_{n-2}} \nabla_\gamma \omega_{\delta_1\dots\delta_{n-2}} \\ &= \epsilon^{\alpha\beta\gamma\delta_1\dots\delta_{n-2}} \nabla_\gamma (\epsilon_{\delta_1\dots\delta_{n-2}\mu\nu\rho} X^\mu \nabla^\nu X^\rho) \\ &= (-1)^{n-2} \epsilon^{\delta_1\dots\delta_{n-2}\alpha\beta\gamma} \epsilon_{\delta_1\dots\delta_{n-2}\mu\nu\rho} \nabla_\gamma (X^\mu \nabla^\nu X^\rho) \\ &= \underbrace{2(-1)^{n-1}(n-2)!}_{=:c_n} (\delta_\mu^\alpha \delta_{\nu\rho}^{\beta\gamma} + \delta_\mu^\beta \delta_{\nu\rho}^{\gamma\alpha} + \delta_\mu^\gamma \delta_{\nu\rho}^{\alpha\beta}) \nabla_\gamma (X^\mu \nabla^\nu X^\rho) \end{aligned}$$

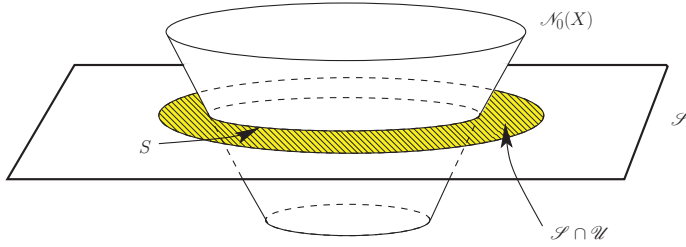


Fig. 4.3.2 A spacelike hypersurface $\mathcal{S}$ intersecting a Killing horizon $\mathcal{N}_0(X)$ in a compact cross-section S .

$$\begin{aligned}
 &= c_n \left(\nabla_\gamma (X^\alpha \nabla^\beta X^\gamma) + \nabla_\gamma (X^\beta \nabla^\gamma X^\alpha) + \nabla_\gamma (X^\gamma \nabla^\alpha X^\beta) \right) \\
 &= c_n \left(\underbrace{\nabla_\gamma X^\alpha \nabla^\beta X^\gamma}_{=:(*)} + X^\alpha \underbrace{\nabla_\gamma \nabla^\beta X^\gamma}_{=R_{\sigma\gamma}{}^{\beta\gamma} X^\sigma} + \underbrace{\nabla_\gamma X^\beta \nabla^\gamma X^\alpha}_{=-\nabla_\gamma X^\alpha \nabla^\beta X^\gamma, \text{ cancels out } (*)} \right. \\
 &\quad \left. + X^\beta \underbrace{\nabla_\gamma \nabla^\gamma X^\alpha}_{=R_{\sigma\gamma}{}^{\gamma\alpha} X^\sigma} + \underbrace{\nabla_\gamma X^\gamma \nabla^\alpha X^\beta}_{=0} + \underbrace{X^\gamma \nabla_\gamma \nabla^\alpha X^\beta}_{=X^\gamma R_{\sigma\gamma}{}^{\alpha\beta} X^\sigma=0} \right) \\
 &= 2c_n X^{[\alpha} R^{\beta]}_{\sigma} X^\sigma. \tag{4.3.42}
 \end{aligned}$$

Comparing with (4.3.41), we find that

$$X_{[\alpha} \nabla_{\beta]} \kappa|_{\mathcal{N}} = -\frac{1}{2c_n} \epsilon_{\alpha\beta}{}^{\gamma\delta_1 \dots \delta_{n-2}} \nabla_\gamma \omega_{\delta_1 \dots \delta_{n-2}}, \tag{4.3.43}$$

from which the theorem follows. $\square$

A vector field X is said to be *hypersurface orthogonal* if the twist form ω of (4.3.40) vanishes; compare (4.3.2). Recall that X is static if it is timelike (at least at large distances) and hypersurface orthogonal. It thus follows from Theorem 4.3.13 that the surface gravity is always constant (up to sign) on Killing horizons associated with *static* Killing vectors, regardless of field equations.

It is known that the twist form vanishes for stationary and axisymmetric electrovacuum spacetimes [307, 391]; we again infer that κ is constant for such spacetimes; of course this follows also from Theorem 4.3.12.

4.3.5 The orbit-space geometry near Killing horizons

Consider a spacetime $(\mathcal{M}, g)$ with a Killing vector field X . Let $\mathcal{S}$ be a spacelike hypersurface in $\mathcal{M}$. Near $\mathcal{S}$ we can introduce a function t so that $\mathcal{S} = \{t = 0\}$, with dt nowhere vanishing on $\mathcal{S}$. Let $\mathcal{U}$ be an open set intersecting $\mathcal{S}$ on which X is timelike. We can propagate local coordinates x^i from $\mathcal{S} \cap \mathcal{U}$ to a neighbourhood of $\mathcal{S}$ so that $X = \partial_t$, and write the metric as

$$g = -V^2(dt + \theta_i dx^i)^2 + h_{ij} dx^i dx^j, \quad \partial_t V = \partial_t \theta_i = \partial_t h_{ij} = 0, \tag{4.3.44}$$

where $h := h_{ij} dx^i dx^j$ has Riemannian signature. Indeed, the tensor field $g_{ij} dx^i dx^j$ induced by g on $\mathcal{S}$ is positive definite by hypothesis. We have

$$h_{ij} = g_{ij} + V^2 \theta_i \theta_j,$$

which shows that h is also positive definite. So h is a Riemannian metric on $\mathcal{S} \cap \mathcal{U}$ which will be called the *orbit-space metric*.

Note that h will *not* coincide with the metric induced by g on $\mathcal{S}$ in general.

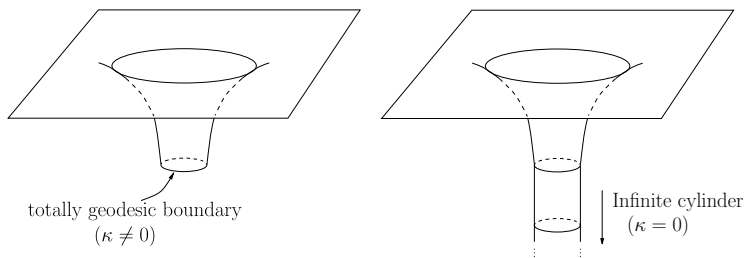


Fig. 4.3.3 The general features of the geometry of the orbit-space metric on a spacelike hypersurface intersecting a non-degenerate (left) and degenerate (right) Killing horizon, near the intersection, visualized by a codimension-one embedding in Euclidean space.

A coordinate-independent version of the above reads as follows: for $Y, Z \in T\mathcal{S}$

$$h(Y, Z) = g(Y, Z) - \frac{g(X, Y)g(X, Z)}{g(X, X)}. \quad (4.3.45)$$

To continue, we assume in addition that $\mathcal{S} \cap \mathcal{U}$ has a boundary component S which forms a compact cross-section of a Killing horizon $\mathcal{N}_0(X)$ as in Fig. 4.3.2. The vanishing, or not, of the surface gravity has a deep impact on the geometry of the orbit-space metric h near $\mathcal{N}_0(X)$ [102]:

1. Every differentiable such S , included in a C^2 *degenerate* Killing horizon $\mathcal{N}_0(X)$, corresponds to a *complete* asymptotic end of $(\mathcal{S} \cap \mathcal{U}, h)$. See Fig. 4.3.3.⁶ (A similar result holds for stationary and axisymmetric four-dimensional configurations without the hypothesis that X is timelike near the horizon [137].)
2. Every such S included in a smooth Killing horizon $\mathcal{N}_0(X)$ on which

$$\kappa > 0,$$

corresponds to a totally geodesic boundary of $(\mathcal{S} \cap \mathcal{U}, h)$, with h being smooth up to boundary at S . Moreover

- (a) A doubling of $(\mathcal{S} \cap \mathcal{U}, h)$ across S leads to a smooth metric on the doubled manifold,
- (b) With $\sqrt{-g(X, X)}$ extending smoothly to $-\sqrt{-g(X, X)}$ across S .

In the Majumdar–Papapetrou solutions of Section 4.7, the orbit-space metric h as in (4.3.44) asymptotes to the usual metric on a round cylinder as the event horizon is approached. One is therefore tempted to think of degenerate event horizons as corresponding to *asymptotically cylindrical ends* of $(\mathcal{S}, h)$.

4.3.6 Near-horizon geometry

A key feature of black-hole geometries is the existence of event horizons, which are null hypersurfaces. By Theorem 2.10.8 any connected null achronal hypersurface $\mathcal{H}$ is a Lipschitz topological hypersurface. Furthermore (cf. Appendix F, p. 356), through every point $p \in \mathcal{H}$ there is a future-inextendible null geodesic entirely contained in $\mathcal{H}$ (though it may leave $\mathcal{H}$ when followed to the past of p). Such geodesics are called *generators*.

A useful tool for studying geometry near *smooth* null hypersurfaces is provided by the *null Gaussian coordinates* of Isenberg and Moncrief [364]. (It should be kept in mind that general null hypersurfaces are not smooth; as already pointed out, Theorem 2.10.12 provides examples of horizons which are nowhere C^1 .)

⁶I am grateful to C. Williams for providing the figure.

PROPOSITION 4.3.14 ([364]) *Near a point on a smooth null hypersurface $\mathcal{H}$ one can introduce null Gaussian coordinates, in which the spacetime metric g takes the form*

$$g = x\varphi dv^2 + 2dvdx + 2xh_adx^a dv + h_{ab}dx^a dx^b, \quad (4.3.46)$$

with $\mathcal{H}$ given by the equation $\{x = 0\}$, and where v is an affine parameter along the generators of $\mathcal{H}$.

REMARK 4.3.15 The construction that follows is done so that v is an affine parameter. In situations where a field of null tangents ℓ to $\mathcal{H}$ is given a priori, after ignoring the equation $\nabla_\ell = \ell$ in the first paragraph of the proof one can instead construct a coordinate system as in (4.3.46) with $\ell = \partial_v$ on $\mathcal{H}$, in which case v becomes the natural flow parameter along the flow of ℓ . $\square$

REMARK 4.3.16 Suppose that $\mathcal{H}$ has a global cross-section, i.e. a smooth submanifold S of $\mathcal{H}$ such that each generator of $\mathcal{H}$ intersects S precisely once, and that $\mathcal{H}$ is two-sided. Then the coordinates (x, v) are defined globally in a neighbourhood of $\mathcal{H}$. $\square$

Proof of Proposition 4.3.14. Let $S \subset \mathcal{H}$ be any $(n-1)$ -dimensional submanifold of $\mathcal{H}$ which is transverse to the null generators of $\mathcal{H}$. Let x^a be any local coordinate system on S , and let $\ell|_S$ be any field of null vectors, defined on S , tangent to the generators of $\mathcal{H}$. Solving the equation $\nabla_\ell \ell = 0$, with initial values $\ell|_S$ on S , one obtains a null vector field ℓ defined on a $\mathcal{H}$ -neighbourhood $\mathcal{V} \subset \mathcal{H}$ of S , tangent to the generators of $\mathcal{H}$.

One can extend x^a to $\mathcal{V}$ by solving the equation $\ell(x^a) = 0$. The function $v|_{\mathcal{H}}$ is defined by solving the equation $\ell(v) = 1$ with initial value $v|_S = 0$. Passing to a subset of $\mathcal{V}$ if necessary, this defines a global coordinate system (v, x^a) on $\mathcal{V}$. By construction we have

$$\ell = \partial_v \text{ on } \mathcal{V},$$

in particular $g_{vv} = 0$ on $\mathcal{V}$. Further, ℓ is normal to $\mathcal{H}$ because $\mathcal{H}$ is a null surface, which implies that $g_{va} = 0$ on $\mathcal{V}$.

Let, next, $\bar{\ell}|_{\mathcal{V}}$ be a field of null vectors on $\mathcal{V}$ defined uniquely by the conditions

$$g(\bar{\ell}|_{\mathcal{V}}, \ell) = 1, \quad g(\bar{\ell}|_{\mathcal{V}}, \partial_a) = 0. \quad (4.3.47)$$

The first equation implies that $\bar{\ell}|_{\mathcal{V}}$ is everywhere transverse to $\mathcal{V}$. Then we define $\bar{\ell}$ in a spacetime neighbourhood $\mathcal{U} \subset \mathcal{M}$ of $\mathcal{V}$ by solving the geodesic equation $\nabla_{\bar{\ell}} \bar{\ell} = 0$ with initial value $\bar{\ell}|_{\mathcal{V}}$ at $\mathcal{V}$. The coordinates (v, x^a) are extended to $\mathcal{U}$ by solving the equations $\bar{\ell}(v) = \bar{\ell}(x^a) = 0$, and the coordinate x is defined by solving the equation $\bar{\ell}(x) = 1$, with initial value $x = 0$ at $\mathcal{V}$. Passing to a subset of $\mathcal{U}$ if necessary, this defines a global coordinate system (v, x, x^a) on $\mathcal{U}$.

By construction we have

$$\bar{\ell} = \partial_x; \quad (4.3.48)$$

hence ∂_x is a null, geodesic, vector field on $\mathcal{U}$. In particular

$$g_{xx} \equiv g(\partial_x, \partial_x) = 0.$$

Let $(z^A) = (x, x^a)$, and note that

$$\begin{aligned} \bar{\ell}\left(g(\bar{\ell}, \partial_A)\right) &= g(\bar{\ell}, \nabla_{\bar{\ell}} \partial_A) = g(\bar{\ell}, \nabla_{\partial_x} \partial_A) = g(\bar{\ell}, \nabla_{\partial_A} \partial_x) \\ &= g(\bar{\ell}, \nabla_{\partial_A} \bar{\ell}) = \frac{1}{2} \partial_A \left(g(\bar{\ell}, \bar{\ell})\right) = 0. \end{aligned}$$

This shows that the components g_{xA} of the metric are x -independent. On S we have $g_{xv} = 1$ and $g_{xa} = 0$ by (4.3.47), which finishes the proof. $\square$

EXAMPLE 4.3.17 An example of the coordinate system above is obtained by taking $\mathcal{H}$ to be the light cone of the origin in $(n+1)$ -dimensional Minkowski spacetime, with $x = r - t$, $y = (t + r)/2$; then the Minkowski metric η takes the form

$$\eta = -dt^2 + dr^2 + r^2 d\Omega^2 = 2dx dy + \frac{(x+2y)^2}{4} d\Omega^2.$$

REMARK 4.3.18 The coordinate x is not unique; it depends upon the choice of the initial section S and the vector $\ell|_S$ from the proof of Proposition 4.3.14.

An instructive example of this non-uniqueness is provided by the null hypersurface $\mathcal{H} := \{T + X = 0\}$ in four-dimensional Minkowski spacetime with metric

$$\eta = -dT^2 + dX^2 + dY^2 + dZ^2.$$

The Killing vector field $\partial_T - \partial_X$ is tangent to the generators of $\mathcal{H}$, hence providing a natural choice for ℓ . This leads to

$$\bar{\ell} = -\frac{1}{2}(\partial_X + \partial_T), \quad v = \frac{1}{2}(T - X), \quad x = -(T + X), \quad \eta = 2du dv + dY^2 + dZ^2. \quad (4.3.49)$$

Now, another natural candidate for ℓ is provided by the Killing vector already seen in (4.3.10),

$$K := Y\partial_T + T\partial_Y + X\partial_Y - Y\partial_X = Y\partial_T + (T + X)\partial_Y - Y\partial_X, \quad (4.3.50)$$

tangent to the generators of $\mathcal{H}$ and non-vanishing for $Y \neq 0$. As pointed out to me by R. Wald (private communication), choosing $K = \ell$, for

$$T^2 > X^2 \text{ and } Y > \sqrt{T^2 - X^2}$$

the coordinate transformation

$$\begin{aligned} x &= -\frac{1}{2}(T - X)\sqrt{Y^2 - (T - X)(T + X)}, \\ v &= 2\frac{Y - \sqrt{Y^2 - (T - X)(T + X)}}{T - X}, \\ y &= \sqrt{Y^2 - (T - X)(T + X)}, \\ z &= Z, \end{aligned} \quad (4.3.51)$$

with inverse

$$T - X = -\frac{2x}{y}, \quad T + X = yv - \frac{v^2 x}{2y}, \quad Y = y - \frac{vx}{y}, \quad Z = z, \quad (4.3.52)$$

brings the metric to the desired form:

$$\eta = \frac{x^2}{y^2} dv^2 + 2dv dx - \frac{4x}{y} dv dy + dy^2 + dz^2. \quad (4.3.53)$$

In the new coordinates the Killing vector K of (4.3.50) equals ∂_v . $\square$

EXAMPLE 4.3.19 The Eddington–Finkelstein coordinates, which bring the Schwarzschild metric to the form (4.2.16),

$$g = -\left(1 - \frac{2m}{r}\right) dv^2 + 2dv dr + r^2 d\Omega^2, \quad (4.3.54)$$

provide directly an example of null Gauss coordinates around the null hypersurface $\{r = 2m\}$.

Quite generally, metrics of the form

$$g = -F(r)dt^2 + \frac{dr^2}{F(r)} + h_{AB}dx^A dx^B, \quad (4.3.55)$$

where F vanishes at $r = r_0$, can be extended across the null-hypersurface $\{r = r_0\}$ by introducing a new coordinate

$$v = t + f(r), \text{ where } f' = \frac{1}{F}. \quad (4.3.56)$$

This leads to

$$g = -Fdv^2 + 2dv dr + h_{AB}dx^A dx^B, \quad (4.3.57)$$

as required in the null Gaussian form (4.3.46). $\square$

Average surface gravity. A topological submanifold S of a null achronal hypersurface $\mathcal{H}$ will be called a *local section*, or simply *section*, if S meets the generators of $\mathcal{H}$ transversally; it will be called a *cross-section* if it meets all the generators precisely once.

Let S be any smooth compact cross-section of $\mathcal{H}$; the *average surface gravity* $\langle \kappa \rangle_S$ is defined as

$$\langle \kappa \rangle_S = -\frac{1}{2|S|} \int_S \varphi d\mu_h, \quad (4.3.58)$$

where φ is as in (4.3.46), p. 160, $d\mu_h$ is the measure associated with the metric $h_{ab}dx^a dx^b$ of (4.3.46) on S , and $|S|$ is the volume of S in that measure.

We emphasize that while the notion of *surface gravity* was defined for Killing horizons, that of *average surface gravity* is defined for any sufficiently differentiable null hypersurface with compact cross-sections. The interest of the notion stems from the uniqueness theory of black holes. Indeed, one of the steps there is the proof that *event horizons are Killing horizons*; this is known as the *Hawking rigidity theorem*. The theorem assumes existence of a black-hole event horizon, which is a null hypersurface where only the average surface gravity is defined a priori.

Note that the requirement of compactness is crucial to guarantee that the defining integral is well defined and finite.

In situations where $X := \partial_v$ is a Killing vector and $\mathcal{H}$ is an associated Killing horizon, in physically relevant spacetimes the notions coincide. Indeed, from (4.3.46) we have

$$g(X, X) \equiv g_{vv} = x\varphi, \quad d(g(X, X))|_{x=0} = \varphi dx, \quad \nabla(g(X, X))|_{x=0} = \varphi \partial_v,$$

and the Definition 4.3.11, p. 151, of surface gravity κ gives

$$\kappa = -\frac{1}{2}\varphi.$$

So if κ is constant on $\mathcal{H}$ which, as discussed in Section 4.3.3, holds in many situations of interest, we obtain

$$\langle \kappa \rangle_S = \kappa.$$

The near-horizon geometry equations. When $\mathcal{H}$ is a degenerate Killing horizon, the surface gravity vanishes by definition. This implies that the function φ in (4.3.46) can be written as xA , for some smooth function A . The vacuum Einstein equations imply that (see [364, Eq. (2.9)] in dimension four and [321, Eq. (5.9)] in higher dimensions)

$$\hat{R}_{ab} = \frac{1}{2}\hat{h}_a\hat{h}_b - \hat{D}_{(a}\hat{h}_{b)}, \quad (4.3.59)$$

where, in the notation of (4.3.46), $\mathring{R}_{ab}$ is the Ricci tensor of $\mathring{h}_{ab} := h_{ab}|_{x=0}$, and $\mathring{D}$ is the covariant derivative thereof, while $\mathring{h}_a := h_a|_{x=0}$. The Einstein equations also determine $\mathring{A} := A|_{x=0}$ uniquely in terms of $\mathring{h}_a$ and $\mathring{h}_{ab}$:

$$\mathring{A} = \frac{1}{2} \mathring{h}^{ab} \left(\mathring{h}_a \mathring{h}_b - \mathring{D}_a \mathring{h}_b \right) \quad (4.3.60)$$

(this equation follows again, e.g. from [364, Eq. (2.9)] in dimension four, and can be checked by a calculation in all higher dimensions).

The triple $(S, \mathring{h}_{ab}, \mathring{h}_a)$ is called the *near-horizon geometry*. In view of the Taylor expansions

$$h_a = \mathring{h}_a + O(x), \quad h_{ab} = \mathring{h}_{ab} + O(x),$$

the pair $(\mathring{h}_{ab}, \mathring{h}_a)$ together with (4.3.60) describes the leading-order behaviour of the metric near $\{x = 0\}$, which justifies the name.

Suppose that g satisfies the vacuum Einstein equations, possibly with a cosmological constant. If ∂_v is a Killing vector (equivalently, $\partial_v A = 0 = \partial_v h_a = \partial_v h_{ab}$), then the *near-horizon metric*

$$\mathring{g} = x^2 \mathring{A} dv^2 + 2dvdx + 2x \mathring{h}_a dx^a dv + \mathring{h}_{ab} dx^a dx^b \quad (4.3.61)$$

is also a solution of the vacuum Einstein equations. To see this, let $\epsilon > 0$ and in the metric (4.3.46) replace the coordinates (v, x) by $(\epsilon^{-1}v, \epsilon x)$:

$$\begin{aligned} g_\epsilon &:= g|_{(v,x) \rightarrow (\epsilon^{-1}v, \epsilon x)} \\ &= \epsilon^2 x^2 A(\epsilon x, x^a) d(\epsilon^{-1}v)^2 + 2d(\epsilon^{-1}v) d(\epsilon x) + 2\epsilon x h_a(\epsilon x, x^a) dx^a d(\epsilon^{-1}v) \\ &\quad + h_{ab}(\epsilon x, x^a) dx^a dx^b \\ &= x^2 A(\epsilon x, x^a) dv^2 + 2dvdx + 2x h_a(\epsilon x, x^a) dx^a dv + h_{ab}(\epsilon x, x^a) dx^a dx^b \\ &\rightarrow_{\epsilon \rightarrow 0} \mathring{g}. \end{aligned} \quad (4.3.62)$$

Now, for every $\epsilon > 0$ the metric g_ϵ is in fact g written in a different coordinate system. Hence the before-last line of (4.3.62) provides a family g_ϵ of solutions of the vacuum Einstein equations depending smoothly on a parameter ϵ . Passing to the limit $\epsilon \rightarrow 0$, the conclusion readily follows.

The classification of near-horizon geometries turns out to be a key step towards a classification of degenerate black holes. We have⁷ the following partial results, where either staticity is assumed without restriction on dimensions or axial symmetry is required in spacetime dimension four [321].

THEOREM 4.3.20 ([142]) *Let the spacetime dimension be $n + 1$, $n \geq 3$; suppose that a degenerate Killing horizon $\mathcal{N}$ has a compact cross-section, and that $\mathring{h}_a = \partial_a \lambda$ for some function λ (which is necessarily the case in vacuum static spacetimes). Then (4.3.59) implies $\mathring{h}_{ab} \equiv 0$, so that $\mathring{h}_{ab}$ is Ricci-flat.* $\square$

REMARK 4.3.21 The form (4.3.53) of the Minkowski metric provides an example of a static vacuum example with a flat metric $\mathring{h}_{ab}$ and with a non-vanishing form $\mathring{h}_a$. This example shows that the hypothesis of compactness of the cross-section of the horizon in Theorem 4.3.20 is necessary. $\square$

THEOREM 4.3.22 ([321]) *In spacetime dimension four and in vacuum, suppose that a degenerate Killing horizon $\mathcal{N}$ has a spherical cross-section, and that $(\mathcal{M}, g)$ admits a second Killing vector field with periodic orbits. For every connected component $\mathcal{N}_0$ of $\mathcal{N}$ there exists an embedding of $\mathcal{N}_0$ into a Kerr spacetime which preserves $\mathring{h}_a$, $\mathring{h}_{ab}$, and $\mathring{A}$.* $\square$

⁷Further partial results with a non-zero cosmological constant have been proved in [142].

In the four-dimensional static case, Theorem 4.3.20 enforces toroidal topology of cross-sections of $\mathcal{N}$, with a flat $\dot{h}_{ab}$. This, together with Theorem 3.3.1, p. 90, shows the non-existence of static, degenerate, asymptotically flat, suitably regular vacuum black holes.

On the other hand, in the four-dimensional axi-symmetric case, Theorem 4.3.22 guarantees that the geometry tends to a Kerr one, up to second-order errors, when the horizon is approached. This is one of the key ingredients of the proof of the uniqueness theorem for axi-symmetric axisymmetric, degenerate, connected, vacuum, asymptotically flat, suitably regular black holes [137].

It would be of significant interest to obtain more information about solutions of (4.3.59), in all dimensions, without any restrictive conditions. For instance, it is expected that the hypothesis of the existence of a second vector field is not necessary for Theorem 4.3.22, and it would be of interest to prove, or disprove, this.

INCIDENTALLY: A partial result towards the existence of a second Killing vector has been obtained in [146], where small perturbations of the Kerr near-horizon geometry are studied. For such perturbations the problem can be reduced to a study of the linearized equations. Using a formalism introduced by Jezierski and Kaminski [279] and spherical harmonic decompositions one reduces the problem to the proof that an ODE operator associated with each spherical harmonic mode has no kernel. This is done there analytically except for the seven lowest modes, for which numerical evidence is provided. A key step of the analysis, established in [146] without any smallness assumptions, is the proof that $\dot{h}$ always has precisely two zeros of index one.

Further partial results concerning the problem can be found in [279, 380]. $\square$

As just seen, in the degenerate case the vacuum equations impose strong restrictions on the near-horizon geometry. It turns out that no such restrictions exist for non-degenerate horizons, at least in the analytic setting: Indeed, for any triple $(N, \dot{h}_a, \dot{h}_{ab})$, where N is a two-dimensional analytic manifold (compact or not), $\dot{h}_a$ is an analytic one-form on N , and $\dot{h}_{ab}$ is an analytic Riemannian metric on N , there exists a vacuum spacetime $(\mathcal{M}, g)$ with a bifurcate (and thus non-degenerate) Killing horizon, so that the metric g takes the form (4.3.46) near each Killing horizon branching out of the bifurcation surface $S \approx N$, with $\dot{h}_{ab} = h_{ab}|_{r=0}$ and $\dot{h}_a = h_a|_{r=0}$; in fact $\dot{h}_{ab}$ is the metric induced by g on S . When N is the two-dimensional torus $\mathbb{T}^2$ this can be inferred from [361] as follows: using [361, Theorem (2)] with $(\phi, \beta_a, g_{ab})|_{t=0} = (0, 2\dot{h}_a, \dot{h}_{ab})$ one obtains a vacuum spacetime $(\mathcal{M}' = S^1 \times \mathbb{T}^2 \times (-\epsilon, \epsilon), g')$ with a compact Cauchy horizon $S^1 \times \mathbb{T}^2$ and Killing vector X tangent to the S^1 factor of $\mathcal{M}'$. One can then pass to a covering space where S^1 is replaced by $\mathbb{R}$, and use a construction of Rácz and Wald (cf. Theorem 4.3.23) to obtain the desired $\mathcal{M}$ containing the bifurcate horizon.

This argument generalizes to any analytic $(N, \dot{h}_a, \dot{h}_{ab})$ without difficulties.

4.3.7 Asymptotically flat stationary metrics

Isolated black holes are modelled by asymptotically flat metrics; it is therefore important to provide a careful definition of asymptotic flatness.

In fact, there exist several ways of defining asymptotic flatness, all of them roughly equivalent in vacuum. We will adapt a Cauchy data point of view (cf. Appendix G, p. 363), as it appears to be the least restrictive. (An often-used alternative proceeds through the conformal completions of Section 3.1, p. 85.) So, a spacetime $(\mathcal{M}, g)$ will be said to possess an *asymptotically flat end* if $\mathcal{M}$ contains a spacelike hypersurface $\mathcal{S}_{\text{ext}}$ diffeomorphic to $\mathbb{R}^n \setminus B(R)$, where $B(R)$ is a coordinate ball of radius R , with the following properties: there exists a constant $\alpha > 0$ such that, in local coordinates on $\mathcal{S}_{\text{ext}}$ obtained from $\mathbb{R}^n \setminus B(R)$, the metric h induced by g on

$\mathcal{S}_{\text{ext}}$ and the extrinsic curvature tensor K of $\mathcal{S}_{\text{ext}}$ satisfy the fall-off conditions, for some $k > 1$,

$$h_{ij} - \delta_{ij} = O_k(r^{-\alpha}), \quad K_{ij} = O_{k-1}(r^{-1-\alpha}), \quad (4.3.63)$$

where we write $f = O_k(r^\alpha)$ if f satisfies

$$\partial_{k_1} \dots \partial_{k_\ell} f = O(r^{\alpha-\ell}), \quad 0 \leq \ell \leq k. \quad (4.3.64)$$

For simplicity we assume that the spacetime is vacuum, though similar results hold in general under appropriate conditions on matter fields; see [40, 133] and references therein.

Along any spacelike hypersurface $\mathcal{S}$, a Killing vector field X of $(\mathcal{M}, g)$ can be decomposed as

$$X = Nn + Y,$$

where Y is tangent to $\mathcal{S}$, and n is the unit future-directed normal to $\mathcal{S}_{\text{ext}}$. The fields N and Y are called ‘Killing initial data’, or KID for short. The vacuum field equations, together with the Killing equations, imply the following set of equations on $\mathcal{S}$,

$$D_i Y_j + D_j Y_i = 2N K_{ij}, \quad (4.3.65)$$

$$R_{ij}(h) + K^k{}_k K_{ij} - 2K_{ik} K^k{}_j - N^{-1}(\mathcal{L}_Y K_{ij} + D_i D_j N) = 0, \quad (4.3.66)$$

where $R_{ij}(h)$ is the Ricci tensor of h . Equations (4.3.65)–(4.3.66) are sometimes referred to as the *vacuum KID equations*, cf. [109] and references therein.

Under the boundary conditions (4.3.63), an analysis of these equations provides detailed information about the asymptotic behaviour of (N, Y) . In particular one can prove that if the asymptotic region $\mathcal{S}_{\text{ext}}$ is part of initial data set $(\mathcal{S}, h, K)$ satisfying the requirements of the positive energy theorem, and if X is timelike along $\mathcal{S}_{\text{ext}}$, then $(N, Y^i) \rightarrow_{r \rightarrow \infty} (A^0, A^i)$, where the A^μ ’s are constants satisfying $(A^0)^2 > \sum_i (A^i)^2$ [39, 133]. One can then choose adapted coordinates so that the metric can be, locally, written as

$$g = -V^2(dt + \underbrace{\theta_i dx^i}_{=: \theta})^2 + \underbrace{h_{ij} dx^i dx^j}_{=: h}, \quad (4.3.67)$$

with

$$\partial_t V = \partial_t \theta = \partial_t h = 0 \quad (4.3.68)$$

$$h_{ij} - \delta_{ij} = O_k(r^{-\alpha}), \quad \theta_i = O_k(r^{-\alpha}), \quad V - 1 = O_k(r^{-\alpha}). \quad (4.3.69)$$

As discussed in more detail in [41], in h -harmonic coordinates, and in e.g. a maximal time-slicing, the vacuum equations for g form a quasi-linear elliptic system with a diagonal principal part, with its principal symbol identical to that of the scalar Laplace operator. Methods known in principle show that, in this ‘gauge’, all metric functions have a full asymptotic expansion in terms of powers of $\ln r$ and inverse powers of r . In the new coordinates we can in fact take

$$\alpha = n - 2, \quad (4.3.70)$$

where n equals the dimension of $\mathcal{S}$, thus 1 less than the dimension of $\mathcal{M}$. By inspection of the equations one can further infer that the leading order corrections in the metric can be written in the Schwarzschild form (4.2.55).

Solutions without $\ln r$ terms are of special interest, because the associated space-times admit a *smooth* conformal completion at infinity as in Section 3.1, p. 85. In

even spacetime dimension, initial data sets containing such asymptotic regions, when close enough to Minkowskian data, lead to asymptotically simple spacetimes [16, 87, 196]. It has been shown by Beig and Simon that logarithmic terms can always be gotten rid of by a change of coordinates in space dimension three when the mass is non-zero [44, 429]. This has been generalized in [41] to all stationary metrics in even space-dimension $n \geq 6$, and to static metrics with non-vanishing mass in $n = 5$.

4.3.8 Domains of outer communications, event horizons

A key notion in the theory of asymptotically flat black holes is that of the *domain of outer communications*. In the *stationary* case the most convenient definition proceeds as follows: for $t \in \mathbb{R}$ let $\phi_t[X] : \mathcal{M} \rightarrow \mathcal{M}$ denote the one-parameter group of diffeomorphisms generated by a Killing vector field X ; we will write ϕ_t for $\phi_t[X]$ whenever ambiguities are unlikely to occur. Let $\mathcal{S}_{\text{ext}}$ be an asymptotic end as in Section 4.3.7. The exterior region $\mathcal{M}_{\text{ext}}$ and the *domain of outer communications* $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ are then defined as in (3.3.17)–(3.3.18), p. 98:⁸

$$\mathcal{M}_{\text{ext}} := \cup_t \phi_t(\mathcal{S}_{\text{ext}}), \quad \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle = I^+(\mathcal{M}_{\text{ext}}) \cap I^-(\mathcal{M}_{\text{ext}}). \quad (4.3.71)$$

The *black-hole region* $\mathcal{B}$ and the *black-hole event horizon* $\mathcal{H}^+$ are defined as

$$\mathcal{B} = \mathcal{M} \setminus I^-(\mathcal{M}_{\text{ext}}), \quad \mathcal{H}^+ = \partial \mathcal{B}. \quad (4.3.72)$$

The *white-hole region* $\mathcal{W}$ and the *white-hole event horizon* $\mathcal{H}^-$ are defined as above after changing time orientation:

$$\mathcal{W} = \mathcal{M} \setminus I^+(\mathcal{M}_{\text{ext}}), \quad \mathcal{H}^- = \partial \mathcal{W}. \quad (4.3.73)$$

It follows that the boundaries of $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ are included in the event horizons. We set

$$\mathcal{E}^\pm = \partial \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \cap I^\pm(\mathcal{M}_{\text{ext}}), \quad \mathcal{E} = \mathcal{E}^+ \cup \mathcal{E}^-. \quad (4.3.74)$$

There is considerable freedom in choosing the asymptotic region $\mathcal{S}_{\text{ext}}$. However, the argument following (3.3.19), p. 98, shows that $I^\pm(\mathcal{M}_{\text{ext}})$, and hence $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$, $\mathcal{H}^\pm$ and $\mathcal{E}^\pm$, are independent of the choice of $\mathcal{S}_{\text{ext}}$ as long as the associated $\mathcal{M}_{\text{ext}}$'s overlap.

For definiteness, our definitions of domain of outer communications, black-hole region, etc., have been tailored to asymptotically flat stationary spacetimes. The definitions carry over *verbatim* to spacetimes with different asymptotics when a preferred region $\mathcal{M}_{\text{ext}}$ is present, as e.g. for asymptotically anti-de Sitter spacetimes which have a timelike conformal boundary at infinity.

4.3.9 Adding bifurcation surfaces

When trying to prove results about spacetimes containing non-degenerate Killing horizons, it is convenient to have a compact bifurcation surface at hand. Indeed, this hypothesis is made throughout the classification theory of static (non-degenerate) black holes (cf. [113, 114] and references therein). The problem is that while we have good control of the geometry of the domain of outer communications, various unpleasant things can happen at its boundary. In particular, in [408] it has been shown that there might be an obstruction for the extendability of a domain of outer communications in such a way that the extension comprises a compact bifurcation

⁸Recall (see Section 2.4, p. 35) that $I^-(\Omega)$, respectively $J^-(\Omega)$, is the set covered by past-directed timelike, respectively causal, curves originating from Ω , while $\dot{I}^-$ denotes the boundary of I^- , etc. The sets I^+ , etc., are defined as I^- , etc., after changing time orientation.

surface. Nevertheless, as far as applications are concerned, it suffices to have the following: given a spacetime $(\mathcal{M}, g)$ with a domain of outer communications $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ and a non-degenerate Killing horizon, there exists a spacetime $(\mathcal{M}', g')$, with a domain of outer communications $\langle\langle \mathcal{M}'_{\text{ext}} \rangle\rangle$ which is isometrically diffeomorphic to $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$, such that all non-degenerate Killing horizons in $(\mathcal{M}', g')$ contain their bifurcation surfaces. Rácz and Wald have shown [408], under appropriate conditions, that this is indeed the case.

THEOREM 4.3.23 (I. Rácz & R. Wald) *Let $(\mathcal{M}, g)$ be a stationary, or stationary-rotating spacetime with Killing vector field X and with an asymptotically flat region $\mathcal{M}_{\text{ext}}$. Suppose that $J^+(\mathcal{M}_{\text{ext}})$ is globally hyperbolic with an asymptotically flat Cauchy surface Σ which intersects the event horizon $\mathcal{E}^+ = \partial\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \cap I^+(\mathcal{M}_{\text{ext}})$ in a compact cross-section. Suppose that X is tangent to the generators of $\mathcal{E}^+$ and that the surface gravity of every connected component of $\mathcal{E}^+$ is a non-zero constant. Then there exists a spacetime $(\mathcal{M}', g')$ and an isometric embedding*

$$\Psi : \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \rightarrow \langle\langle \mathcal{M}'_{\text{ext}} \rangle\rangle \subset \mathcal{M}' ,$$

where $\langle\langle \mathcal{M}'_{\text{ext}} \rangle\rangle$ is a domain of outer communications in $\mathcal{M}'$, such that:

1. *There exists a one-parameter group of isometries of $(\mathcal{M}', g')$, such that the associated Killing vector field X' coincides with $\Psi_* X$ on $\langle\langle \mathcal{M}'_{\text{ext}} \rangle\rangle$.*
2. *Every connected component of $\langle\langle \mathcal{M}'_{\text{ext}} \rangle\rangle$ is a Killing horizon which comprises a compact bifurcation surface.*
3. *There exists a local ‘wedge-reflection’ isometry about every connected component of the bifurcation surface.*

It should be emphasized that field equations, energy inequalities, and analyticity have not been assumed above. However, the constancy of surface gravity has been imposed; compare Section 4.3.3.

4.3.10 Black strings and branes

Consider a vacuum black-hole solution $(\mathcal{M}, g)$ without cosmological constant and let (N, h) be a Riemannian manifold with a Ricci flat metric, $\text{Ric}(h) = 0$. Then the spacetime $(\mathcal{M} \times N, g \oplus h)$ is again a vacuum spacetime, containing a black-hole region in the sense used so far. (Similarly if $\text{Ric}(g) = \sigma g$ and $\text{Ric}(h) = \sigma h$, then $\text{Ric}(g \oplus h) = \sigma(g \oplus h)$.) Objects of this type are called *black strings* when $\dim N = 1$, and *black branes* in general. See [66, 170, 185, 254, 259, 276, 336, 393, 455] and references therein.

4.4 Extensions

In what follows we will construct black-hole spacetimes by extending metrics to larger manifolds. Therefore, before continuing our studies of various aspects of black-hole spacetimes, it is convenient to discuss systematically the notion of extensions of Lorentzian manifolds, and of their properties. Our presentation follows closely that of [113].

Let $k \in \mathbb{R} \cup \{\infty\} \cup \{\omega\}$; we use the symbol C^ω to denote the set of real-analytic functions or tensor fields. The $(n+1)$ -dimensional spacetime $(\mathcal{M}, \tilde{g})$ is said to be a C^k -extension of an $(n+1)$ -dimensional spacetime $(\mathcal{M}, g)$ if there exists a C^k -immersion $\psi : \mathcal{M} \rightarrow \mathcal{M}$ such that $\psi^* \tilde{g} = g$, and such that $\psi(\mathcal{M}) \neq \mathcal{M}$. A spacetime $(\mathcal{M}, g)$ is said to be C^k -maximal, or C^k -inextendible, if no C^k -extensions of $(\mathcal{M}, g)$ exist.

4.4.1 Distinct extensions

We start by noting that maximal analytic extensions of manifolds are not unique. Simple examples of nonuniqueness have already been discussed in Remark 4.2.8: remove a subset Ω from a maximally extended manifold $\mathcal{M}$ so that $\mathcal{M} \setminus \Omega$ is not simply connected, and pass to the universal cover $\hat{\mathcal{M}}$; extend maximally the spacetime so obtained, if further needed. This provides many distinct maximal extensions. One is tempted to believe that such constructions can be used to classify all maximal analytic extensions, but this remains to be seen.

These examples exhibit the following undesirable feature: existence of maximally extended geodesics of finite affine length along which the spacetime is locally extendible in an intuitive sense. Indeed, if we choose $\mathcal{M}$ to be the Kruskal–Szekeres spacetime and Ω the bifurcation surface of the static Killing vector ∂_t , then the lifts to $\hat{\mathcal{M}}$ of those geodesics which meet Ω in $\mathcal{M}$ will be incomplete in $\hat{\mathcal{M}}$, ‘ceasing to exist’ at the values of the affine parameter at which they would have met Ω in $\mathcal{M}$. Examples which avoid this *local extendibility* feature can be constructed in spacetimes with certain symmetries, as follows.

Consider any spacetime $(\hat{\mathcal{M}}, \hat{g})$, and let $\mathcal{M}$ be a proper open subset of $\hat{\mathcal{M}}$ with the metric g obtained by restriction. Thus $(\hat{\mathcal{M}}, \hat{g})$ is an extension of $(\mathcal{M}, g)$.

Suppose that there exists a non-trivial isometry Ψ of $(\hat{\mathcal{M}}, \hat{g})$ satisfying:

- (a) Ψ has no fixed points;
 - (b) $\Psi(\mathcal{M}) \cap \mathcal{M} = \emptyset$; and
 - (c) Ψ^2 is the identity map.
- (4.4.1)

Then, by (a) and (c), $\hat{\mathcal{M}}/\Psi$ equipped with the obvious metric (still denoted by g) is a Lorentzian manifold. Furthermore, by (b), $\mathcal{M}$ embeds diffeomorphically into $\hat{\mathcal{M}}/\Psi$ in the obvious way. Therefore $\hat{\mathcal{M}}/\Psi$ provides an extension of $\mathcal{M}$ which is distinct from $(\hat{\mathcal{M}}, \hat{g})$.

It follows from Appendix C.4 that $(\hat{\mathcal{M}}/\Psi, g)$ is analytic if $(\mathcal{M}, g)$.

Keeping in mind that a spacetime must be time oriented by definition, $\hat{\mathcal{M}}/\Psi$ will be a spacetime if and only if Ψ preserves time-orientation. If $\hat{\mathcal{M}}$ is simply connected, then $\pi_1(\hat{\mathcal{M}}/\Psi) = \mathbb{Z}_2$.

EXAMPLE 4.4.1 As a definite example of this construction, denote by $(\hat{\mathcal{M}}, \hat{g})$ the Kruskal–Szekeres extension of the Schwarzschild spacetime $(\mathcal{M}, g)$; by the latter we mean a connected component of the set $\{r > 2m\}$ within $\hat{\mathcal{M}}$. Let (T, X) be the global coordinates on $\hat{\mathcal{M}}$ as defined in (4.2.32). Let $\hat{\Psi} : S^2 \rightarrow S^2$ be the antipodal map. For $p \in S^2$ consider the four isometries $\Psi_{\pm\pm}$ of the Kruskal–Szekeres spacetime defined as

$$\Psi_{\pm\pm}(T, X, p) = (\pm T, \pm X, \hat{\Psi}(p)).$$

Set $\mathcal{M}_{\pm\pm} := \hat{\mathcal{M}}/\Psi_{\pm\pm}$. Since Ψ_{++} is the identity, $\mathcal{M}_{++} = \hat{\mathcal{M}}$ is the Kruskal–Szekeres manifold, so nothing of interest here. Next, both manifolds $\mathcal{M}_{-\pm}$ are smooth maximal analytic Lorentzian extensions of $(\mathcal{M}, g)$, but are not spacetimes because the maps $\Psi_{-\pm}$ do not preserve time orientation. However, $\mathcal{M}_{+-}$ provides a maximal globally hyperbolic analytic extension of the Schwarzschild manifold distinct from $\hat{\mathcal{M}}$. This is the ‘ $\mathbb{RP}^3$ geon’ discussed in [194].

Theorem 4.3.23 shows that the above construction can be applied to any metric with a bifurcate horizon with constant surface gravity, e.g., to Kerr metrics of Section 4.6, p. 174, and to Emparan–Reall metrics of Section 5.3, p. 221. $\square$

4.4.2 Inextendibility

A *scalar invariant* is a function which can be calculated using the geometric objects at hand and which is invariant under coordinate transformations.

For instance, a function α_g which can be calculated in local coordinates from the metric g and its derivatives will be a scalar invariant if, for any local diffeomorphism ψ , we have

$$\alpha_g(p) = \alpha_{\psi^*g}(\psi^{-1}(p)). \quad (4.4.2)$$

In the case of the scalar invariant $g(X, X)$ calculated using a metric g and a Killing vector X , the invariance property (4.4.2) is replaced by

$$\alpha_{g,X}(p) = \alpha_{\psi^*g,(\psi^{-1})_*X}(\psi^{-1}(p)). \quad (4.4.3)$$

A scalar invariant f on $(\mathcal{M}, g)$ will be called a C^k -compatibility scalar if f satisfies the following property: for every C^k -extension $(\check{\mathcal{M}}, \check{g})$ of $(\mathcal{M}, g)$ and for any bounded timelike geodesic segment γ in $\mathcal{M}$ such that $\psi(\gamma)$ accumulates at the boundary $\partial(\psi(\mathcal{M}))$ (where ψ is the immersion map $\psi : \mathcal{M} \rightarrow \check{\mathcal{M}}$), the function f is bounded along γ .

Any constant function is a compatibility scalar in this terminology, albeit not very useful in practice. An example of a useful C^2 -compatibility scalar is the Kretschmann scalar $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$. Another example is provided by the norm $g(X, X)$ of a Killing vector X of g .⁹

THEOREM 4.4.2 *Let $(\check{\mathcal{M}}, \check{g})$ be an extension of $(\mathcal{M}, g)$. Let X be a Killing vector field on $(\mathcal{M}, g)$ and suppose that there exists a curve γ in $\mathcal{M}$ with end point $p \in \check{\mathcal{M}}$ on which $g(X, X)$ is unbounded. If γ is differentiable up to p , then the extended metric $\check{g}$ cannot be $C^{1,1}$ at p .*

Proof. Recall (cf. (4.3.22), p. 154) that a Killing vector field satisfies the set of equations

$$\nabla_\alpha \nabla_\beta X_\gamma = R_{\sigma\alpha\beta\gamma} X^\sigma. \quad (4.4.4)$$

This implies a linear system of equations

$$\begin{aligned} \frac{DX_\alpha}{ds} &= \dot{\gamma}^\mu \nabla_\mu X_\alpha =: \dot{\gamma}^\mu A_{\mu\alpha} \\ \frac{DA_{\alpha\beta}}{ds} &= \dot{\gamma}^\mu \nabla_\mu \nabla_\alpha X_\beta = R_{\gamma\mu\alpha\beta} \dot{\gamma}^\mu X^\gamma, \end{aligned} \quad (4.4.5)$$

satisfied by the Killing vector and its derivatives along γ .

Suppose that γ is differentiable up to its end point $p \in \check{\mathcal{M}}$. Equations (4.4.5) show that $X \circ \gamma$ extends by continuity to p . This implies that $g(X, X)$ remains bounded along γ , contradicting the hypothesis. $\square$

The next inextendibility criterion from [113] is often used.

PROPOSITION 4.4.3 *Let $k \geq 2$. Suppose that either every timelike geodesic in $(\mathcal{M}, g)$ is complete, or some C^k -compatibility scalar is unbounded on every timelike maximally extended geodesic. Then $(\mathcal{M}, g)$ is C^k -inextendible.*

Proof. Suppose there exists a C^k -extension $(\check{\mathcal{M}}, \check{g})$ of $(\mathcal{M}, g)$, with immersion $\psi : \mathcal{M} \rightarrow \check{\mathcal{M}}$. We identify $\mathcal{M}$ with its image $\psi(\mathcal{M})$ in $\check{\mathcal{M}}$.

Let $p \in \partial\mathcal{M}$ and let $\mathcal{O}$ be a globally hyperbolic neighbourhood of p . Let $q_n \in \mathcal{M}$ be a sequence of points approaching p ; thus, $q_n \in \mathcal{O}$ for n large enough.

Suppose, first, that there exists n such that $q_n \in I^+(p) \cup I^-(p)$. By global hyperbolicity of $\mathcal{O}$ there exists a timelike geodesic segment γ from q_n to p . Then the part of γ which lies within $\mathcal{M}$ is inextendible and has finite affine length.

⁹The inextendibility criterion based on a Killing vector seems to have first been used in [47] (see the second part of Proposition 5, p. 139 there).

Furthermore, every C^k -compatibility scalar is bounded on γ . But there are no such geodesics through q_n by hypothesis. We conclude that

$$(I^+(p) \cup I^-(p)) \cap \mathcal{M} = \emptyset. \quad (4.4.6)$$

Let $q \in (I^+(p) \cup I^-(p)) \cap \mathcal{O}$; thus $q \notin \mathcal{M}$ by (4.4.6). Since $I^+(q) \cup I^-(q)$ is open, and $p \in I^+(q) \cup I^-(q)$, we have $q_n \in I^+(q) \cup I^-(q)$ for all n sufficiently large, say $n \geq n_0$. Let γ be a timelike geodesic segment from q_{n_0} to q . Since q is not in $\mathcal{M}$, the part of γ that lies within $\mathcal{M}$ is, as before, inextendible within $\mathcal{M}$ and has finite affine length, with all C^k -compatibility scalars bounded. This is again incompatible with our hypotheses, and the result is established. $\square$

4.4.3 Uniqueness of a class of extensions

We have seen in Section 4.4.1 that distinct analytic extensions are possible in general. The aim of this section is to show the uniqueness of *simply connected analytic* extensions of analytic simply connected pseudo-Riemannian manifolds under a condition on geodesics. This is the contents of Corollary 4.4.7.

We start with some terminology. A maximally extended geodesic ray $\gamma : [0, s^+) \rightarrow \mathcal{M}$ will be called *s-complete* if $s_+ = \infty$ *unless* there exists some polynomial scalar invariant α such that

$$\limsup_{s \rightarrow s_+} |\alpha(\gamma(s))| = \infty.$$

A similar definition applies to maximally extended geodesics $\gamma : (s_-, s^+) \rightarrow \mathcal{M}$, with some polynomial scalar invariant (not necessarily the same) unbounded in the incomplete direction, if any. Here, by a *polynomial scalar invariant* we mean a scalar function which is a polynomial in the metric, its inverse, the Riemann tensor, and its derivatives.

It should be clear that this definition extends to include invariants such as the norm $g(X, X)$ of a Killing vector X , or of a Yano–Killing tensor, etc. But care should be taken not to take scalars such as $\ln(R_{ijkl}R^{ijkl})$ which are not defined for all regular geometries, and therefore could blow up along a curve even though the geometry remains regular; this is why we restricted attention to polynomials.

A Lorentzian manifold $(\mathcal{M}, g)$ will be said to be *s-complete* if every maximally extended geodesic is *s-complete*. The notions of *timelike s-completeness*, or *causal s-completeness*, are defined similarly, by specifying the causal type of the geodesics in the definition above.

We have the following version of [297, Theorem 6.3, p. 255] (compare also the Remark on p. 256 there), where geodesic completeness is weakened to *timelike s-completeness*.

THEOREM 4.4.4 *Let $(\mathcal{M}, g), (\mathcal{M}', g')$ be analytic Lorentzian manifolds of dimension $n+1$, $n \geq 1$, with $\mathcal{M}$ connected and simply connected, and $\mathcal{M}'$ timelike s-complete. Then every isometric immersion $f_U : U \subset \mathcal{M} \hookrightarrow \mathcal{M}'$, where U is an open subset of $\mathcal{M}$, extends uniquely to an isometric immersion $f : \mathcal{M} \hookrightarrow \mathcal{M}'$.*

We start by noting two preliminary lemmas, which are proved as in [297] by replacing ‘*affine mappings*’ there by ‘*isometric immersions*.’

LEMMA 4.4.5 ([297, Lemma 1, p. 252]) *Let $\mathcal{M}, \mathcal{M}'$ be analytic manifolds, with $\mathcal{M}$ connected. Let f, g be analytic mappings $\mathcal{M} \rightarrow \mathcal{M}'$. If f and g coincide on a non-empty open subset of $\mathcal{M}$, then they coincide everywhere.* $\square$

LEMMA 4.4.6 ([297, Lemma 4, p. 254]) *Let $(\mathcal{M}, g)$ and $(\mathcal{M}', g')$ be pseudo-Riemannian manifolds of same dimension, with $\mathcal{M}$ connected, and let f and g be isometric immersions of $\mathcal{M}$ into $\mathcal{M}'$. If there exists some point $x \in \mathcal{M}$ such that $f(x) = g(x)$ and $f_*(X) = g_*(X)$ for every vector X of $T_x\mathcal{M}$, then $f = g$ on $\mathcal{M}$.* $\square$

Before passing to the proof Theorem 4.4.4, we note a simple corollary.

COROLLARY 4.4.7 *Let $(\mathcal{M}, g)$, $(\mathcal{M}', g')$ be two connected, simply connected, s -complete analytic Lorentzian extensions of $(U, \dot{g})$. Then there exists an isometric diffeomorphism $f : \mathcal{M} \rightarrow \mathcal{M}'$.*

Proof of Corollary 4.4.7. Viewing U as a subset of $\mathcal{M}$, Theorem 4.4.4 provides an isometric immersion $f : \mathcal{M} \hookrightarrow \mathcal{M}'$ such that $f|_U = \text{id}_U$. Viewing U as a subset of $\mathcal{M}'$, Theorem 4.4.4 provides an isometric immersion $f' : \mathcal{M}' \hookrightarrow \mathcal{M}$ such that $f'|_U = \text{id}$. Then $f \circ f'$ is an isometry of $(\mathcal{M}', g')$ satisfying $(f \circ f')|_U = \text{id}_U$; hence $f \circ f' = \text{id}_{\mathcal{M}'}$ by Lemma 4.4.6. Similarly $f' \circ f = \text{id}_{\mathcal{M}}$, as desired. $\square$

We can turn our attention now to the proof of Theorem 4.4.4.

Proof of Theorem 4.4.4. Similarly to the proof of Theorem 6.1 in [297], we define an *analytic continuation of f_U along a continuous path $c : [0, 1] \rightarrow \mathcal{M}$* to be a set of mappings f_s , $0 \leq s \leq 1$, together with a family of open subsets U_s , $0 \leq s \leq 1$, satisfying the properties:

- $f_0 = f_U$ on $U_0 = U$;
- For every $s \in [0, 1]$, U_s is a neighbourhood of the point $c(s)$ of the path c , and f_s is an isometric immersion $f_s : U_s \subset \mathcal{M} \hookrightarrow \mathcal{M}'$; and
- For every $s \in [0, 1]$, there exists a number $\delta_s > 0$ such that for all $s' \in [0, 1]$, $(|s' - s| < \delta_s) \Rightarrow (c(s') \in U_s \text{ and } f_{s'} = f_s \text{ in a neighbourhood of } c(s'))$.

We need to prove that, under the hypothesis of s -completeness, such an analytic continuation does exist along any curve c . The argument is simplest for timelike curves, so let us first assume that c is timelike. To do so, we consider the set

$$A := \{s \in [0, 1] \mid \text{an analytic continuation exists along } c \text{ on } [0, s]\}. \quad (4.4.7)$$

A is non-empty, as it contains a neighbourhood of 0. Hence $\bar{s} := \sup A$ exists and is positive. We need to show that in fact, $\bar{s} = 1$ and can be reached. Assume that this is not the case. Let W be a normal convex neighbourhood of $c(\bar{s})$ such that every point x in W has a normal neighbourhood containing W . (Such a W exists from Theorem 8.7, Chapter III of [297].) We can choose $s_1 < \bar{s}$ such that $c(s_1) \in W$, and we let V be a normal neighbourhood of $c(s_1)$ containing W . Since $s_1 \in A$, f_{s_1} is well defined, and is an isometric immersion of a neighbourhood of $c(s_1)$ into $\mathcal{M}'$; we will extend it to $V \cap I^\pm(c(s_1))$. To do so, we know that $\exp : V^* \rightarrow V$ is a diffeomorphism, where V^* is a neighbourhood of 0 in $T_{c(s_1)}\mathcal{M}$; hence, in particular, for $y \in V \cap I^\pm(c(s_1))$, there exists a unique $X \in V^*$ such that $y = \exp X$. Define $X' := f_{s_1*}X$. Then X' is a vector tangent to $\mathcal{M}'$ at the point $f_{s_1}(c(s_1))$. Since y is in the timelike cone of $c(\bar{s})$, X is timelike, and so is X' , as f_{s_1} is isometric. We now need to prove the following.

LEMMA 4.4.8 *The geodesic $s \mapsto \exp(sX')$ of $\mathcal{M}'$ is well defined for $0 \leq s \leq 1$.*

Proof. Let

$$s^* := \sup\{s \in [0, 1] \mid \exp(s'X') \text{ exists } \forall s' \in [0, s]\}. \quad (4.4.8)$$

First, such a s^* exists, and is positive, and we note that if $s^* < 1$, then it is not reached. We wish to show that $s^* = 1$ and is reached. Hence, it suffices to show that ‘ s^* is not reached’ leads to a contradiction. Indeed, in such a case the timelike geodesic $s \mapsto \exp(sX')$ ends at a finite affine parameter; thus, there exists a scalar invariant φ such that $\varphi(\exp(sX'))$ is unbounded as $s \rightarrow s^*$. Now, for all $s < s^*$, we can define that $h(\exp(sX)) := \exp(sX')$, and this gives an extension h of f_{s_1} which is analytic (since it commutes with the exponential maps, which are analytic). By Lemma 4.4.6, h is in fact an isometric immersion. By definition of scalar invariants we have

$$\varphi(\exp(sX')) = \tilde{\varphi}(\exp(sX)),$$

$$g = -\left(1 - \frac{2m}{r} + \frac{Q^2}{r^2}\right)dt^2 + \frac{dr^2}{1 - \frac{2m}{r} + \frac{Q^2}{r^2}} + r^2 d\Omega^2, \quad (4.5.1)$$

where m is, as usual, the ADM mass of g and Q is the total electric charge. The electromagnetic potential takes the form

$$A = \frac{Q}{r} dt. \quad (4.5.2)$$

The equation $g(\partial_t, \partial_t) = 0$ has solutions $r = r_{\pm}$ provided that $|Q| \leq m$:

$$r_{\pm} = m \pm \sqrt{m^2 - Q^2}.$$

These hypersurfaces become Killing horizons, or bifurcate Killing horizons, in suitable extensions of the Reissner–Nordström metric.

Calculating as in Example 4.3.7, p. 152 one finds that the surface gravities of the Killing horizons $r = r_{\pm}$ of the Reissner–Nordström metric equal

$$\begin{aligned} \kappa_{\pm} &= -\frac{1}{2} \partial_r g_{tt}|_{r=r_{\pm}} = \frac{1}{2} \partial_r \left(1 - \frac{2m}{r} + \frac{Q^2}{r^2}\right) \Big|_{r=r_{\pm}} = \frac{mr_{\pm} - Q^2}{r_{\pm}^3} \\ &= \pm \frac{\sqrt{m^2 - Q^2}}{r_{\pm}^2}. \end{aligned} \quad (4.5.3)$$

For $r = r_+$ this is strictly positive unless $|Q| = m$; so we see that the Killing horizons in Reissner–Nordström black holes are non-degenerate for $|Q| < m$, and degenerate when $|Q| = m$.

The global structure of the standard maximal extensions of non-degenerate Reissner–Nordström spacetimes is presented in Example 6.3.1, p. 270, while that of degenerate solutions can be found in Example 6.3.6, p. 275.

INCIDENTALLY: Suppose that the metric (4.5.1) models an electron, for which

$$m_e \approx 9.11 \times 10^{-31} \text{ kg}, \quad Q_e \approx -1.60 \times 10^{-19} \text{ C}.$$

Our form of the metric requires units in which $G/c^2 = 1$ and $G/(4\pi\epsilon_0 c^4) = 1$. Using

$$\frac{1}{4\pi\epsilon_0} \equiv k_e \approx 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}, \quad G \approx 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2},$$

we find that

$$m_e \approx 6.75 \times 10^{-58} \text{ m} \times \frac{c^2}{G}, \quad Q_e \approx -1.38 \times 10^{-36} \text{ m} \times \sqrt{\frac{4\pi\epsilon_0 c^4}{G}},$$

leading to

$$\frac{|Q_e|}{m_e} \approx 2.04 \times 10^{21}.$$

We see that a point electron, if modelled by a RN metric, is described by a ‘nakedly singular’ metric, i.e. a RN metric without Killing horizons.

For a proton we have instead

$$m_p \approx 1.67 \times 10^{-27} \text{ kg},$$

with the charge $Q_p = -Q_e$, which gives

$$\frac{Q_p}{m_p} \approx 1.11 \times 10^{18},$$

leading again to a naked singularity when the RN metric is used. However, the proton is not a point particle, so the Reissner–Nordström metric applies, at best, only outside the charge radius of the proton $r_p \approx 0.85 \text{ fm}$. $\square$

In dimensions $n+1 \geq 5$ one has [371] the following counterpart of (4.5.1)–(4.5.2):

$$g = -\left(1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}\right) dt^2 + \frac{dr^2}{1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}} + r^2 d\Omega^2, \quad (4.5.4)$$

$$A = \frac{Q}{r^{n-2}} dr, \quad (4.5.5)$$

where m is related to the ADM mass, and Q to the total charge.

INCIDENTALLY: The RN metrics, when suitably extended, have the interesting property of being *timelike geodesically complete*, but not *null geodesically complete*. To see that, consider a timelike geodesic γ parameterized by proper time; thus, we have

$$-1 = -\left(1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}\right) \dot{t}^2 + \frac{\dot{r}^2}{1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}} + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2).$$

Conservation of ‘energy’, $g(\dot{\gamma}, \partial_t) = -E$, implies that

$$\left(1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}\right) \dot{t} = E;$$

hence

$$-1 = \frac{\dot{r}^2 - E^2}{1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}} + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2). \quad (4.5.6)$$

Equivalently,

$$\begin{aligned} r^2 &= E^2 - \left(1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}\right) \underbrace{\left(1 + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2)\right)}_{\geq 1} \\ &\leq E^2 - \left(1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}\right). \end{aligned} \quad (4.5.7)$$

As r approaches zero the right-hand side becomes negative, which is not possible. It follows that timelike geodesics cannot approach $r = 0$. It is then not too difficult to prove that maximally extended timelike geodesics are complete in the extensions of Figs. 6.3.1, p. 271, and 6.3.6, p. 275, and timelike geodesic completeness follows.

Obvious modifications of the above calculation similarly show that null geodesics with non-zero angular momentum, $g(\dot{\gamma}, \partial_\varphi) \neq 0$, cannot reach the singular boundary $\{r = 0\}$ and are complete.

On the other hand, radial null geodesics reach $r = 0$ in a finite affine parameter: for then we have zero at the left-hand side of (4.5.6), without an angular-momentum contribution, giving

$$\dot{r} = \pm E \implies r(s) - r_0 = \pm E(s - s_0).$$

Hence null radial geodesics reach $r = 0$ in finite affine time either to the future or to the past, showing null geodesic incompleteness. $\square$

4.6 The Kerr metric

The Kerr metric provides a ‘rotating generalization’ of the Schwarzschild metric. Its importance stems from the *black-hole uniqueness theorems*, which establish uniqueness of Kerr black holes under suitable global conditions (cf., e.g., [114] and references therein). It should, however, be kept in mind that the Schwarzschild metric describes not only spherically symmetric black holes, but also the *vacuum exterior region of any spherically symmetric* matter configuration. There is no such universality property for stationary axisymmetric configurations.

INCIDENTALLY: The construction of axisymmetric stationary stellar models is a rather complicated undertaking, we refer the reader to [345] for more information about the subject.

There have been many attempts to find a *Kerr fill-in*, i.e. a stationary solution with matter fields which are compactly supported in space and which has the Kerr metric in its exterior vacuum region. Such a fill-in satisfying the dominant energy condition, but not-known-to-be-related to some standard matter model (such as, e.g., a perfect fluid) has been constructed in [248] (compare [249]). $\square$

As such, the two parameter family of *Kerr metrics* in Boyer–Lindquist coordinates take the form

$$g = -dt^2 + \frac{2mr}{\Sigma}(dt - a \sin^2(\theta)d\varphi)^2 + (r^2 + a^2) \sin^2 \theta d\varphi^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2. \quad (4.6.1)$$

Here

$$\Sigma = r^2 + a^2 \cos^2(\theta), \quad \Delta = r^2 + a^2 - 2mr = (r - r_+)(r - r_-), \quad (4.6.2)$$

and $r_+ < r < \infty$, where

$$r_{\pm} = m \pm (m^2 - a^2)^{\frac{1}{2}}.$$

The metric satisfies the vacuum Einstein equations for any values of the parameters a and m , but we will mainly consider parameters in the range

$$0 < |a| \leq m.$$

The case $|a| = m$ will sometimes require separate consideration, as then Δ acquires one double root at $r = m$, instead of two simple ones. When $a = 0$, the Kerr metric reduces to the Schwarzschild metric, and therefore does not need to be discussed any further. The case $a < 0$ can be reduced to $a > 0$ by changing φ to $-\varphi$; this corresponds to a change of the direction of rotation. There is therefore no loss of generality to assume that $a > 0$, which will be done whenever the sign of a matters for the discussion at hand. The Kerr metrics with $|a| > m$ can be shown to be ‘nakedly singular’ (compare Eq. (4.6.5)), whence our disinterest for those solutions.

The metric (4.6.1) reduces clearly to the Schwarzschild one when $a = 0$. It turns out that the case $m = 0$ leads to Minkowski spacetime: for $a = 0$ this is obvious; for $a \neq 0$ the coordinate transformation (cf., e.g., [81, p. 102])

$$R^2 = r^2 + a^2 \sin^2(\theta), \quad R \cos(\Theta) = r \cos(\theta), \quad (4.6.3)$$

brings g to the Minkowski metric η in spherical coordinates:

$$\eta = -dt^2 + dR^2 + R^2 (d\Theta^2 + \sin^2(\Theta)d\varphi^2). \quad (4.6.4)$$

As $m = 0$ turns out to be Minkowski, and $a = 0$ Schwarzschild, it is customary to interpret m as a parameter related to mass, and a as a parameter related to rotation. This can be made precise by calculating the total mass and angular momentum of the solution using e.g. Hamiltonian methods. One then finds that m is indeed the total mass, while

$$J = ma$$

is the component of the total angular momentum in the direction of the axis of rotation $\sin(\theta) = 0$.

INCIDENTALLY: It might be of interest to put some numbers in. Consider, for instance, the Sun. As such, there are several ways of calculating the total angular momentum $J_\odot$ of our nearest stellar neighbour; see [270] for a discussion of the various estimates and their discrepancies. If we choose the averaged value [270]

$$J_\odot \approx 1.92 \times 10^{41} \text{ kg m}^2 \text{ s}^{-1}$$

for the angular momentum of the Sun, and keep in mind the estimate $M_\odot \approx 1.99 \times 10^{30} \text{ kg}$ for its mass (cf., e.g., <http://nssdc.gsfc.nasa.gov/planetary/factsheet/sunfact.html>), we find that

$$a_\odot = \frac{J_\odot}{M_\odot} \approx 0.96 \times 10^{10} \text{ m}^2 \text{ s}^{-1} \approx 322 \text{ m} \times c, \\ \frac{M_\odot G}{c^2} \approx 1.48 \text{ km}, \quad \frac{a_\odot}{M_\odot} \approx 0.22 \frac{G}{c}.$$

Keeping in mind that the units used in (4.6.1) are such that $G = c = 1$, we see that $|a| < m$ for the Sun, with both values of a and m being of similar order.

If we consider the Earth to be a rigidly rotating uniform sphere, the corresponding numbers are

$$J_\oplus \approx 7.10 \times 10^{33} \text{ kg m}^2 \text{ s}^{-1}, \quad M_\oplus \approx 5.98 \times 10^{24} \text{ kg}, \quad a_\oplus \approx 3.96 \text{ m} \times c, \\ \frac{M_\oplus G}{c^2} \approx 0.44 \text{ cm}, \quad \frac{a_\oplus c}{M_\oplus G} \approx 890.$$

We conclude that if the Earth collapsed to a Kerr metric without shedding angular-momentum, a naked singularity would result. $\square$

The metric (4.6.1) is not defined at points where Σ vanishes:

$$\Sigma = 0 \quad \Longleftrightarrow \quad r = 0, \quad \cos \theta = 0.$$

There is a ‘real singularity on Σ ’, in the sense that the metric cannot be extended across this set in a C^2 manner. The standard argument for this in the literature invokes the Kretschmann scalar (cf., e.g., [311])

$$R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = \frac{48m^2(r^2 - a^2 \cos^2 \theta)(\Sigma^2 - 16a^2 r^2 \cos^2(\theta))}{\Sigma^6}, \quad (4.6.5)$$

which is unbounded when the set $\{\Sigma = 0\}$ is approached from most directions.

Now, this does not quite settle the issue because $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = 0$ e.g. on all curves approaching $\{\Sigma = 0\}$ with $r^2 = a^2 \cos^2 \theta$. Similarly, $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ remains bounded on curves on which either $r^2 - a^2 \cos^2(\theta)$ or $\Sigma^2 - 16a^2 r^2 \cos^2(\theta)$ go to 0 sufficiently fast. So, one can imagine that spacetime could nevertheless be extended along some clever family of curves approaching Σ in a specific way.

Neither is this problem cured by considering the length of the Killing vector ∂_t ,

$$g(\partial_t, \partial_t) = -1 + \frac{2mr}{\Sigma},$$

which again tends to infinity as the set $\{\Sigma = 0\}$ is approached from most directions. This only implies inextendibility ‘along most directions’ at $\{\Sigma = 0\}$ by Theorem 4.4.2, p. 169.

The standard way of resolving this issue proceeds through a result of Carter [79, p. 1570] (compare [384, Proposition 4.5.1]), the proof of which requires considerable effort.

PROPOSITION 4.6.1 *Causal geodesics accumulating at $\{\Sigma = 0\}$ lie entirely in the equatorial plane $\{\cos \theta = 0\}$.* $\square$

INCIDENTALLY: An interesting study of geodesics of the Kerr metric, relevant for the observation of the ‘shadow’ of a Kerr black hole, can be found in [232]. $\square$

Now, on the equatorial plane we have

$$g(\partial_t, \partial_t)|_{\cos \theta=0} = -1 + \frac{2m}{r},$$

which is unbounded on any curve lying in this plane and approaching $\{\Sigma = 0\}$. We can therefore invoke Proposition 4.4.3, p. 169, to conclude that, indeed, no extensions are possible through $\{\Sigma = 0\}$.

A simpler resolution of this issue has been presented in [134], where it is observed that the following combination of curvature scalars,

$$(R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta})^2 + (\epsilon_{\alpha\beta\gamma\delta}R^{\alpha\beta}{}_{\mu\nu}R^{\mu\nu\gamma\delta})^2, \quad (4.6.6)$$

is unbounded on *any* curve accumulating at the set $\{\Sigma = 0\}$.

The Kerr–Schild coordinates presented in Section 4.6.9, p. 192, suggest strongly that the ‘singular set’ has a stringlike nature. The ‘ring’, as opposed to ‘string’, terminology has become standard in the literature. This might be related to the fact that the details of the blow-up of the curvature scalars depend upon the direction of approach to the string, as clearly seen in (4.6.5). Further support for the ‘ring singularity’ terminology can be found in [134].

The Kerr metric is stationary, with the Killing vector field $X = \partial_t$ generating asymptotic time translations, as well as axisymmetric, with the Killing vector field $Y = \partial_\varphi$ generating rotations.

The metric components $g_{\mu\nu}$ can be read off from the expanded version of (4.6.1):

$$\begin{aligned} g = & -\frac{\Delta - a^2 \sin^2(\theta)}{\Sigma} dt^2 - \frac{4amr \sin^2 \theta}{\Sigma} dt d\varphi + \\ & + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta d\varphi^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2. \end{aligned} \quad (4.6.7)$$

Because of the occurrence of the function Δ in the denominator of g_{rr} , the metric (4.6.7) is singular at $r = r_\pm$. Similarly to the Schwarzschild case, it turns out that the metric can be smoothly extended across both $r = r_+$ and $r = r_-$, with the sets

$$\mathcal{H}_\pm := \{r = r_\pm\}$$

being smooth null hypersurfaces in the extension.

INCIDENTALLY: Higher dimensional generalizations of the Kerr metric have been constructed by Myers and Perry [371].

We will give a detailed discussion of a family of maximal analytic extensions of the Kerr metric, and their global structure, in Section 7.3, p. 284. As a first step towards this we consider the extension obtained by replacing t with a new coordinate

$$v = t + \int \frac{r^2 + a^2}{\Delta} dr, \quad (4.6.8)$$

with a further replacement of φ by

$$\phi = \varphi + \int \frac{a}{\Delta} dr. \quad (4.6.9)$$

It is convenient to use the symbol $\hat{g}$ for the metric g in the new coordinate system, obtaining

$$\begin{aligned} \hat{g} = & -\left(1 - \frac{2mr}{\Sigma}\right) dv^2 + 2dr dv + \Sigma d\theta^2 - 2a \sin^2(\theta) d\phi dr \\ & + \frac{(r^2 + a^2)^2 - a^2 \sin^2(\theta) \Delta}{\Sigma} \sin^2(\theta) d\phi^2 - \frac{4amr \sin^2(\theta)}{\Sigma} d\phi dv. \end{aligned} \quad (4.6.10)$$

In order to see that (4.6.10) provides a smooth Lorentzian metric for $v \in \mathbb{R}$ and $r \in (0, \infty)$, note first that the coordinate transformation (4.6.8)–(4.6.9) has been

tailored to remove the $1/\Delta$ singularity in (4.6.7), so that all coefficients are now analytic functions on $\mathbb{R} \times (0, \infty) \times S^2$.

Next, a direct calculation of the determinant of $\hat{g}$ is somewhat painful; a simpler way is to proceed as follows: first, the calculation of the determinant of the metric (4.6.7) reduces to that of a two-by-two determinant in the (t, φ) variables, leading to

$$\det g = -\sin^2(\theta)\Sigma^2. \quad (4.6.11)$$

Now, it is very easy to check that the determinant of the Jacobi matrix

$$\partial(v, r, \theta, \phi)/\partial(t, r, \theta, \varphi)$$

equals one. It follows that $\det \hat{g} = -\sin^2(\theta)\Sigma^2$ for $r > r_+$. Analyticity implies that this equation holds globally, which (since Σ has no zeros) establishes the Lorentzian signature of $\hat{g}$ for all positive r .

Let us show that the region $r < r_+$ is a black-hole region, in the sense that

$$\text{observers, or signals, can enter this region, but can never leave it.} \quad (4.6.12)$$

For this, we start by noting that ∇r is a causal vector for $r_- \leq r \leq r_+$. A direct calculation using (4.6.10) is again somewhat lengthy; instead we use (4.6.7) in the region $r > r_+$ to obtain there

$$\hat{g}(\nabla r, \nabla r) = g(\nabla r, \nabla r) = g^{rr} = \frac{1}{g_{rr}} = \frac{\Delta}{\Sigma} = \frac{(r - r_+)(r - r_-)}{r^2 + a^2 \cos^2 \theta}. \quad (4.6.13)$$

But the left-hand side of this equation is an analytic function throughout the extended manifold $\mathbb{R} \times (0, \infty) \times S^2$, and uniqueness of analytic extensions implies that $\hat{g}(\nabla r, \nabla r)$ equals the expression at the extreme right of (4.6.13) throughout. (The intermediate equalities have only been assumed to be valid for $r > r_+$ in the calculation above, since g has only been defined for $r > r_+$.) Thus ∇r is spacelike if $r < r_-$ or $r > r_+$, null on the hypersurfaces $\{r = r_{\pm}\}$ (called ‘Killing horizons’, see Section 4.3.2), and timelike in the region $\{r_- < r < r_+\}$; note that this last region is empty when $|a| = m$.

We choose a time orientation so that ∇t is past pointing in the region $r > r_+$. Keeping in mind our signature of the metric, this means that t increases on future-directed causal curves in the region $r > r_+$.

Suppose, now, that $a^2 < m^2$, and consider a future-directed timelike curve $\gamma(s)$ that starts in the region $r > r_+$ and enters the region $r < r_+$. Since $\dot{\gamma}$ is timelike it meets the null hypersurface $\{r = r_+\}$ transversally, and thus r is decreasing along γ at least near the intersection point. As long as γ stays in the region $\{r_- < r < r_+\}$ the scalar product $g(\dot{\gamma}, \nabla r)$ has constant sign, since both $\dot{\gamma}$ and ∇r are timelike there. But

$$\frac{dr}{ds} = \dot{\gamma}^i \nabla_i r = g_{ij} \dot{\gamma}^i \nabla^j r = g(\dot{\gamma}, \nabla r), \quad (4.6.14)$$

and dr/ds is negative near the entrance point. We conclude that dr/ds is negative along such γ ’s on $\{r_- < r < r_+\}$. This implies that r is strictly decreasing along future-directed causal curves in the region $\{r_- < r < r_+\}$, so that such curves can only leave this region through the set $\{r = r_-\}$. In other words, no causal communication is possible from the region $\{r < r_+\}$ to the ‘exterior world’ $\{r > r_+\}$ in the extension that we constructed so far.

The Schwarzschild metric has the property that the set $g(X, X) = 0$, where X is the ‘static Killing vector’ ∂_t , coincides with the event horizon $r = 2m$. This is not the case any more for the Kerr metric, where we have

$$g(\partial_t, \partial_t) = \hat{g}(\partial_v, \partial_v) = \hat{g}_{vv} = -\left(1 - \frac{2mr}{r^2 + a^2 \cos^2 \theta}\right).$$

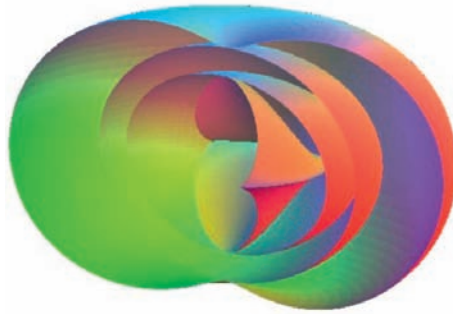


Fig. 4.6.1 A coordinate representation of the outer ergosphere $r = \hat{r}_+$, the event horizon $r = r_+$, the Cauchy horizon $r = r_-$, and the inner ergosphere $r = \hat{r}_-$ with the singular ring in Kerr spacetime; compare [394]. © Kayll Lake, reprinted with permission [311].

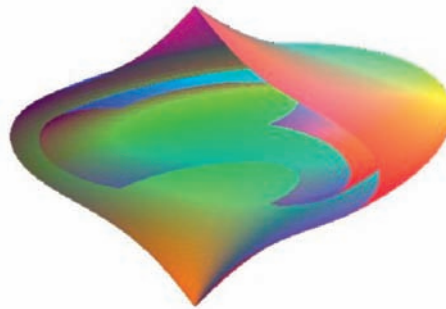


Fig. 4.6.2 Isometric embedding in Euclidean three space of the ergosphere (the outer hull), and part of the event horizon, for a rapidly rotating Kerr solution. The hole in the event horizon arises because there is no global isometric embedding for the event horizon when $a/m > \sqrt{3}/2$ [394]. Somewhat surprisingly, the embedding fails to represent accurately the fact that the cusps at the rotation axis are pointing inward, and not outward. © IOP Publishing. Reproduced with permission. All rights reserved.

The equation $\hat{g}(\partial_v, \partial_v) = 0$ defines instead a set called the *ergosphere*,

$$\hat{g}(\partial_v, \partial_v) = 0 \quad \Longleftrightarrow \quad \hat{r}_{\pm} = m \pm \sqrt{m^2 - a^2 \cos^2 \theta};$$

see Figs. 4.6.1 and 4.6.2. The ergosphere touches the horizons at the axes of symmetry $\cos \theta = \pm 1$. Note that $\partial \hat{r}_{\pm} / \partial \theta \neq 0$ at those axes, so the ergosphere has a cusp there. The region bounded by the outermost horizon $r = r_+$ and the outermost ergosphere $r = \hat{r}_+$ is called the *ergoregion*, with X spacelike in its interior.

It is important to realize that the ergospheres

$$\mathcal{E}_{\pm} := \{r = \hat{r}_{\pm}\}$$

are *not* Killing horizons for the Killing vector ∂_t . Recall that part of the definition of a Killing horizon $\mathcal{H}$ is the requirement that $\mathcal{H}$ is a *null* hypersurface. But this is not the case for $\mathcal{E}_{\pm}$: Indeed, note that the Killing vectors ∂_{φ} and ∂_t are both tangent to $\mathcal{E}_{\pm}$, and thus are all their linear combinations. Now, the character of the orbits of the isometry group $\mathbb{R} \times U(1)$ is determined by the sign of the determinant

$$\det \begin{pmatrix} g_{tt} & g_{t\varphi} \\ g_{t\varphi} & g_{\varphi\varphi} \end{pmatrix} = -\Delta \sin^2(\theta). \quad (4.6.15)$$

Therefore, when $\sin \theta = 0$ the orbits are either null or one-dimensional, while for $\theta \neq 0$ the orbits are timelike in the regions where $\Delta > 0$, spacelike where $\Delta < 0$ and null where $\Delta = 0$. Thus, at every point of $\mathcal{E}_\pm$ except at the intersection with the axis of rotation there exist linear combinations of ∂_t and ∂_φ which are timelike. This implies that these hypersurfaces are *not* null, as claimed.

We refer the reader to [79] and [384] for an exhaustive analysis of the geometry of the Kerr spacetime, and to [453] for a list of further useful references.

INCIDENTALLY: One of the most useful methods for analysing solutions of wave equations is the *energy method*. As an illustration, consider the wave equation

$$\square u = 0. \quad (4.6.16)$$

Let $\mathcal{S}_t$ be a foliation of $\mathcal{M}$ by spacelike hypersurfaces; the energy E_t of u on $\mathcal{S}_t$ associated to a vector field X is defined as

$$E(t) = \int_{\mathcal{S}_t} T^\mu{}_\nu X^\mu \eta_\nu,$$

where $T_{\mu\nu}$ is the usual energy–momentum tensor of a scalar field,

$$T_{\mu\nu} = \nabla_\mu u \nabla_\nu u - \frac{1}{2} \nabla^\alpha u \nabla_\alpha u g_{\mu\nu}.$$

The energy functional E has two important properties: (1) $E \geq 0$ if X is causal, and (2) $E(t)$ is conserved if X is a Killing vector field and, say, u has compact support on each of the $\mathcal{S}_t$.

Now, the existence of ergoregions where the Killing vector X becomes spacelike leads to an $E(t)$ which is not necessarily positive any more, and the energy stops being a useful tool for controlling the behaviour of the field. This is one of the obstacles to our understanding of both linear and non-linear solutions of wave equations on a Kerr background,¹⁰ not to mention the (still open, at the time of writing this book) question of non-linear stability of the Kerr black holes within the class of globally hyperbolic solutions of the vacuum Einstein equations; see Section 8.6, p. 334. $\square$

The hypersurfaces

$$\mathcal{H}_\pm := \{r = r_\pm\}$$

provide examples of *null acausal boundaries*. Because $g(\nabla r, \nabla r)$ vanishes at $\mathcal{H}_\pm$, the usual calculation (see Proposition 1.5.1, p. 11) shows that the integral curves of ∇r with $r = r_\pm$ are null geodesics. Such geodesics, tangent to a null hypersurface, are called *generators* of this hypersurface. A direct calculation of ∇r from (4.6.10) requires work which can be avoided as follows: in the coordinate system (t, r, θ, φ) of (4.6.7) one obtains immediately

$$\nabla r = g^{\mu\nu} \partial_\mu r \partial_\nu = \frac{\Delta}{\Sigma} \partial_r.$$

Now, under (4.6.8)–(4.6.9) the vector ∂_r transforms as

$$\partial_r \rightarrow \partial_r + \frac{a}{\Delta} \partial_\phi + \frac{r^2 + a^2}{\Delta} \partial_v.$$

More precisely, if we use the symbol $\hat{r}$ for the coordinate r in the coordinate system (v, r, θ, ϕ) , and retain the symbol r for the coordinate r in the coordinates (t, r, θ, φ) , we have

$$\partial_r = \frac{\partial \hat{r}}{\partial r} \partial_{\hat{r}} + \frac{\partial \phi}{\partial r} \partial_\phi + \frac{\partial v}{\partial r} \partial_v = \partial_{\hat{r}} + \frac{a}{\Delta} \partial_\phi + \frac{r^2 + a^2}{\Delta} \partial_v.$$

Forgetting the hat over r , we see that in the coordinates (v, r, θ, ϕ) we have

¹⁰See [53, 164] and references therein for further information on that subject.

$$\nabla r = \frac{1}{\Sigma} (\Delta \partial_r + a \partial_\phi + (r^2 + a^2) \partial_v).$$

Since Δ vanishes at $r = r_\pm$, and $r^2 + a^2$ equals $2mr_\pm$ there, we conclude that the ‘stationary-rotating’ Killing field $X + \Omega_+ Y$, where

$$X := \partial_t \equiv \partial_v, \quad Y := \partial_\phi \equiv \partial_\varphi, \quad \Omega_+ := \frac{a}{2mr_+} \equiv \frac{a}{a^2 + r_+^2}, \quad (4.6.17)$$

is proportional to ∇r on $\{r > r_+\}$:

$$X + \Omega_+ Y = \partial_v + \frac{a}{2mr_+} \partial_\phi = \frac{\Sigma}{a^2 + r_+^2} \nabla r \quad \text{on } \mathcal{H}_+.$$

It follows that $\partial_t + \Omega_+ \partial_\phi$ is null and *tangent* to the generators of the horizon $\mathcal{H}_+$. In other words, the generators of $\mathcal{H}_+$ are rotating with respect to the frame defined by the stationary Killing vector field X . This property is at the origin of the definition of Ω_+ as the *angular velocity* of the event horizon.

4.6.1 Komar integrals

An elegant way of associating global invariants with Killing vectors X is provided by Komar integrals, which for the Kerr metric are integrals of the form

$$\int_{r=R, t=T} \nabla^\alpha X^\beta dS_{\alpha\beta}, \quad (4.6.18)$$

where R and T are constants, and $dS_{\alpha\beta}$ form a basis of the space of two-forms defined as

$$dS_{\alpha\beta} = \frac{1}{2} \epsilon_{\alpha\beta\gamma\delta} dx^\gamma \wedge dx^\delta. \quad (4.6.19)$$

A key property of (4.6.18) is that in vacuum, and with zero cosmological constant, the integrals are independent of r and t . This follows from the divergence theorem together with the identity, which follows from (4.3.22), p. 154,

$$\nabla_\alpha \nabla^\alpha X^\beta = -R^\beta{}_\gamma X^\gamma.$$

In Kerr spacetime it is of interest to calculate (4.6.18) for both Killing vectors $X = \partial_t$ and $X = \partial_\phi$. In order to do the calculation for both vectors at once let us denote either ∂_t or ∂_ϕ by ∂_λ ; hence $X_\mu = g_{\mu\lambda}$. For the calculations we need the inverse metric, the components of which are

$$\begin{aligned} g^{tt} &= -1 - \frac{4mr(a^2 + r^2)}{(a^2 + r(r - 2m))(\cos(2\theta)a^2 + a^2 + 2r^2)}, \\ g^{rr} &= \frac{a^2 - 2mr + r^2}{a^2 \cos^2(\theta) + r^2}, \quad g^{\theta\theta} = \frac{1}{a^2 \cos^2(\theta) + r^2}, \\ g^{\varphi\varphi} &= \frac{\csc^2(\theta)(a^2 \cos(2\theta) + a^2 + 2r(r - 2m))}{(a^2 + r(r - 2m))(a^2 \cos(2\theta) + a^2 + 2r^2)}, \\ g^{t\varphi} &= -\frac{4amr}{(a^2 + r(r - 2m))(a^2 \cos(2\theta) + a^2 + 2r^2)}. \end{aligned} \quad (4.6.20)$$

Then

$$\begin{aligned} \int_{r=R, t=T} \nabla^\alpha X^\beta dS_{\alpha\beta} &= \int_{r=R, t=T} \nabla^{[\alpha} X^{\beta]} dS_{\alpha\beta} = \int_{r=R, t=T} \nabla_{[\mu} X_{\nu]} g^{\mu\alpha} g^{\nu\beta} dS_{\alpha\beta} \\ &= \int_{r=R, t=T} \partial_{[\mu} X_{\nu]} g^{\mu\alpha} g^{\nu\beta} dS_{\alpha\beta} = 2 \int_{r=R, t=T} \partial_{[\mu} X_{\nu]} g^{\mu t} g^{\nu r} dS_{tr} \end{aligned}$$

$$\begin{aligned}
&= 2 \int_{r=R, t=T} \partial_{[\mu} g_{\nu]\lambda} g^{\mu t} g^{\nu r} dS_{tr} = \int_{r=R, t=T} (\partial_{\mu} g_{r\lambda} - \partial_r g_{\mu\lambda}) g^{\mu t} g^{rr} dS_{tr} \\
&= - \int_{r=R, t=T} \partial_r g_{\mu\lambda} g^{\mu t} g^{rr} dS_{tr} = - \int_{r=R, t=T} (\partial_r g_{t\lambda} g^{tt} + \partial_r g_{\varphi\lambda} g^{\varphi t}) g^{rr} dS_{tr}.
\end{aligned}$$

As a by-product of T - and R -independence of (4.6.18), one can calculate the integrals by passing to the limit $R \rightarrow \infty$, which simplifies the calculations considerably. Thus

$$\int_{r=R, t=T} \nabla^{\alpha} X^{\beta} dS_{\alpha\beta} = - \lim_{R \rightarrow \infty} \int_{r=R, t=T} (\partial_r g_{t\lambda} g^{tt} + \partial_r g_{\varphi\lambda} g^{\varphi t}) g^{rr} r^2 \underbrace{\sin(\theta) d\theta d\varphi}_{=: d^2\mu}. \quad (4.6.21)$$

To finish the calculation we need the asymptotic behaviour of the metric functions for large r . We find that

$$\begin{aligned}
g &= - \left(1 - \frac{2m}{r} + O(r^{-2}) \right) dt^2 - \left(\frac{4am}{r} + O(r^{-2}) \right) \sin^2 \theta dt d\varphi \\
&\quad + (r^2 + O(1)) \sin^2 \theta d\varphi^2 + \left(1 + \frac{2m}{r} + O(r^{-2}) \right) dr^2 \\
&\quad + (r^2 + O(1)) d\theta^2.
\end{aligned} \quad (4.6.22)$$

This shows explicitly asymptotic flatness of the metric. For the inverse metric, one obtains

$$\begin{aligned}
g^{tt} &= -1 - \frac{2m}{r} + O(r^{-2}), \quad g^{rr} = 1 - \frac{2m}{r} + O(r^{-2}), \quad g^{\theta\theta} = \frac{1}{r^2} + O(r^{-3}), \\
g^{\varphi\varphi} &= \frac{1}{\sin^2(\theta)r^2} + O(r^{-3}), \quad g^{t\varphi} = -\frac{2am}{r^3} + O(r^{-4}).
\end{aligned} \quad (4.6.23)$$

We are ready to return to (4.6.21), which when $X = \partial_t$ becomes:

$$\begin{aligned}
\int_{r=R, t=T} \nabla^{\alpha} X^{\beta} dS_{\alpha\beta} &= \lim_{R \rightarrow \infty} \int_{r=R, t=T} (\partial_r g_{tt} - \partial_r g_{\varphi t} g^{\varphi t}) r^2 d^2\mu \\
&= \lim_{R \rightarrow \infty} \int_{r=R, t=T} \partial_r g_{tt} r^2 d^2\mu = -8\pi m.
\end{aligned} \quad (4.6.24)$$

When $X = \partial_{\varphi}$ we obtain instead

$$\begin{aligned}
\int_{r=R, t=T} \nabla^{\alpha} X^{\beta} dS_{\alpha\beta} &= \lim_{R \rightarrow \infty} \int_{r=R, t=T} (\partial_r g_{t\varphi} - \partial_r g_{\varphi\varphi} g^{\varphi t}) r^2 d^2\mu \\
&= 12\pi am \int_0^{\pi} \sin^3(\theta) d\theta = 16\pi am.
\end{aligned} \quad (4.6.25)$$

4.6.2 Non-degenerate solutions ($a^2 < m^2$): bifurcate horizons

The study of the global structure of Kerr is somewhat more involved than that of the spacetimes already encountered. An obvious extension of the coordinate system of (4.6.7) alternative to (4.6.8)–(4.6.9) is obtained when t is replaced by a new coordinate¹¹

$$u = t - \int_{r_+}^r \frac{r^2 + a^2}{\Delta} dr, \quad (4.6.26)$$

¹¹The discussion here is based on [62, 79]. I am grateful to Julien Cortier for useful discussions concerning this section.

with a further replacement of φ by

$$\psi = \varphi - \int_{r_+}^r \frac{a}{\Delta} dr. \quad (4.6.27)$$

If we use the symbol $\tilde{g}$ for the metric g in the new coordinate system, we obtain

$$\begin{aligned} \tilde{g} = & -\left(1 - \frac{2mr}{\Sigma}\right) du^2 - 2drdu + \Sigma d\theta^2 + 2a \sin^2(\theta) d\psi dr \\ & + \frac{(r^2 + a^2)^2 - a^2 \sin^2(\theta) \Delta}{\Sigma} \sin^2(\theta) d\psi^2 - \frac{4amr \sin^2(\theta)}{\Sigma} d\psi du. \end{aligned} \quad (4.6.28)$$

In the Schwarzschild case one replaces (t, r) with (u, v) , and with a little further work a well-behaved extension is obtained. It should be clear that this shouldn't be that simple for the Kerr metric, because the two extensions constructed so far involve incompatible redefinitions of the angular variable φ ; compare (4.6.9) and (4.6.27).

The calculations that follow are essentially a special case of the general construction of Rácz and Wald [408], presented in Theorem 4.3.23, p. 167, where the possibility of performing a simultaneous extension is traced back to the fact that the surface gravity is constant on the horizons $\{r = r_{\pm}\}$.

Now, recall that we have seen in Remark 4.2.10, p. 139, how to regularize two-dimensional Lorentzian metrics with a singularity structure as in (4.2.47). The first step of the calculation there gets rid of the zero in the denominator of g_{rr} *provided that there is a first-order zero in g_{tt} at that point*; then it is easy to remove the multiplicative first-order zero in g_{uv} by a logarithmic transformation to the variables $\hat{u}$ and $\hat{v}$ as in (4.2.49). Note that a first-order zero requires that

$$a^2 < m^2,$$

which we are going to assume in the remainder of this section; this is not an ad hoc restriction, as the geometry of the spacetime is essentially different in the *extreme case* $a^2 = m^2$; see the end of Section 4.6.9, p. 192.

However, under the current conditions the Kerr metric has a first-order pole in g_{rr} at $r = r_{\pm}$, but there is no zero in g_{tt} at those values of r . The trick is to change φ , near $r = r_{\pm}$, to a new angular variable

$$\varphi_{\pm} = \varphi - \frac{a}{2m\alpha_{\pm}} t, \quad (4.6.29)$$

choosing the free constants $\alpha_{\pm} \neq 0$ so that the new g_{tt} vanishes at $r_{\pm}$. Indeed, after tedious but otherwise straightforward algebra, in the coordinate system $(t, r, \theta, \varphi_{\pm})$ the metric (4.6.7) becomes

$$g = \frac{\Sigma}{\Delta} dr^2 + g_{tt} dt^2 + 2g_{\varphi_{\pm}t} d\varphi_{\pm} dt + \left(r^2 + a^2 + \frac{2mar}{\Sigma}\right) \sin^2(\theta) d\varphi_{\pm}^2 + \Sigma d\theta^2, \quad (4.6.30)$$

with

$$\begin{aligned} g_{tt} = & \left(\frac{a^2 \sin^2(\theta)}{(2m\alpha_{\pm})^2} - \frac{1}{\Sigma} + \frac{a^2 r \sin^2(\theta)}{2m\alpha_{\pm}^2 \Sigma} \right) \Delta + \frac{a^2 \sin^2(\theta)}{\alpha_{\pm}^2 \Sigma} (r - \alpha_{\pm})^2, \\ g_{\varphi_{\pm}t} = & \frac{a \sin^2(\theta)}{2m\alpha_{\pm} \Sigma} ((\Sigma + 2mr)\Delta + (2m)^2 r(r - \alpha_{\pm})); \end{aligned}$$

to avoid ambiguities, we emphasize that $g_{\varphi_{\pm}t} = g(\partial_{\varphi_{\pm}}, \partial_t)$. Recalling that $\Delta = (r - r_+)(r - r_-)$, one sees that the choice

$$\alpha_{\pm} = r_{\pm} \quad (4.6.31)$$

leads indeed to a zero of order one in g_{tt} , as desired. As a bonus one obtains a zero of order one in $g_{\varphi_{\pm}t}$, which will shortly be seen to be useful as well.

REMARK 4.6.2 It is of interest to check smoothness of the transition formulae from the coordinates $(u, v, \theta, \varphi_{\pm})$ to the coordinates (v, r, θ, ϕ) of (4.6.8)–(4.6.9), or to (u, r, θ, ψ) of (4.6.26)–(4.6.27). For example, near $r = r_+$ we have

$$\begin{aligned}\varphi_+ &= \varphi - \frac{a}{2mr_+}t = \psi + \int_{r_+}^r \frac{a}{\Delta} dr - \frac{a}{2mr_+} \left(u + \int_{r_+}^r \frac{r^2 + a^2}{\Delta} dr \right) \\ &= \psi - \frac{au}{2mr_+} + \frac{a}{2mr_+} \int_{r_+}^r \frac{2mr_+ - (r^2 + a^2)}{\Delta} dr.\end{aligned}\quad (4.6.32)$$

Now,

$$2mr_+ - (r^2 + a^2) = 2m(r_+ - r) - \Delta,$$

which vanishes at $r = r_+$. This shows that the integrand in (4.6.32) can be rewritten as a smooth function of r near $r = r_+$, and so the new angular coordinates φ_+ are smooth functions of (u, r, ψ) near $r = r_+$.

Similar calculations apply for φ_- near $r = r_-$, and for the coordinates (v, r, θ, ϕ) . $\square$

Keeping in mind (4.6.17), we see that (4.6.29) together with (4.6.31) is precisely what is needed for the Killing vectors $\partial_t + a(2mr_{\pm})^{-1}\partial_{\varphi}$, tangent to the generators of the horizons $\{r = r_{\pm}\}$, to annihilate $\varphi_{\pm}$:

$$(\partial_t + \frac{a}{2mr_{\pm}}\partial_{\varphi})\varphi_{\pm} = 0.$$

Thus $(\varphi_{\pm}, \theta)$ provide natural coordinates on the space of generators.

We can now get rid of the singularity in g_{rr} by introducing

$$u = t - f(r), \quad v = t + f(r), \quad f' = \frac{r^2 + a^2}{\Delta}, \quad (4.6.33)$$

so that, keeping in mind that $\Delta(r_{\pm}) = 0 \iff r_{\pm}^2 + a^2 = 2mr_{\pm}$,

$$f(r) = \frac{2mr_{\pm}}{r_{\pm} - r_{\mp}} \ln |r - r_{\pm}| + h_{\pm}(r),$$

where the functions $h_{\pm}$'s are smooth near $r = r_{\pm}$. This is somewhat similar to (4.2.48), but the function f has been chosen more carefully because of the θ -dependence of g_{rr} . One then has

$$dt = \frac{1}{2}(du + dv), \quad dr = \frac{\Delta}{2(r^2 + a^2)}(dv - du),$$

so that

$$\begin{aligned}g &= \frac{\Sigma\Delta}{4(r^2 + a^2)^2}(du - dv)^2 + \frac{g_{tt}}{4}(du + dv)^2 + g_{\varphi_{\pm}t}d\varphi_{\pm}(du + dv) \\ &\quad + \left(r^2 + a^2 + \frac{2mar}{\Sigma} \right) \sin^2(\theta)d\varphi_{\pm}^2 + \Sigma d\theta^2.\end{aligned}$$

There are no more unbounded terms in the metric, but one needs yet to get rid of a vanishing determinant: indeed, as seen in (4.6.11), the determinant of the metric in the (r, t, θ, φ) variables equals $-\sin^2(\theta)\Sigma$. Since the Jacobian of the map $(t, r, \varphi) \mapsto (t, r, \varphi_{\pm})$ is one, and that of the map $(t, r, \varphi_{\pm}) \mapsto (u, v, \varphi_{\pm})$ is $-2f' = -2(r^2 + a^2)/\Delta$, we find that the determinant of the metric in the $(u, v, \theta, \varphi_{\pm})$ coordinates equals

$$-\frac{\Sigma\Delta^2 \sin^2(\theta)}{4(r^2 + a^2)^2}, \quad (4.6.34)$$

which vanishes when Δ vanishes. To get rid of this problem we set, as in (4.2.49),

$$\hat{u} = -\exp(-cu), \quad \hat{v} = \exp(cv). \quad (4.6.35)$$

Now,

$$\begin{aligned} du &= -\frac{d\hat{u}}{c\hat{u}}, \quad dv = \frac{d\hat{v}}{c\hat{v}}, \\ \hat{u}\hat{v} &= -\exp(c(v-u)) = -|r-r_{\pm}|^{4cmr_{\pm}/(r_{\pm}-r_{\mp})} \exp(2ch_{\pm}(r)). \end{aligned}$$

As before, one chooses

$$c = \frac{r_{\pm} - r_{\mp}}{4mr_{\pm}},$$

so that, for $r > r_{\pm}$,

$$\hat{u}\hat{v} = -\exp(c(v-u)) = -(r-r_{\pm}) \exp\left(\frac{(r_{\pm} - r_{\mp})h_{\pm}(r)}{2mr_{\pm}}\right).$$

The functions

$$r \mapsto w := (r - r_{\pm}) \exp\left(\frac{(r_{\pm} - r_{\mp})h_{\pm}(r)}{2mr_{\pm}}\right) \quad (4.6.36)$$

have a non-vanishing derivative at $r = r_{\pm}$. Hence, by the analytic implicit function theorem, there exist near $w = 0$ analytic functions $r_{\pm}(w)$ inverting (4.6.36). So, near $r = r_{\pm}$ we can write

$$r - r_{\pm} = -\hat{u}\hat{v}H_{\pm}(-\hat{u}\hat{v}),$$

where the $H_{\pm}$'s are analytic near $\hat{u}\hat{v} = 0$, non-vanishing there, with a similar resulting formulae for Δ . Since g_{tt} and $g_{\varphi\pm t}$ both contain a multiplicative factor $r - r_{\pm} \sim \hat{u}\hat{v}$, we conclude that the coefficients $g_{\varphi\pm\hat{u}}$, $g_{\varphi\pm\hat{v}}$, as well as

$$g_{\hat{u}\hat{v}} = -\frac{1}{c^2\hat{u}\hat{v}}g_{uv}$$

can be analytically extended across $r = r_{\pm}$. This is somewhat less obvious for

$$g_{\hat{u}\hat{u}} = \frac{1}{(c\hat{u})^2}g_{uu}, \quad g_{\hat{v}\hat{v}} = \frac{1}{(c\hat{v})^2}g_{vv}.$$

However, with some work one obtains

$$\begin{aligned} g_{uu} = g_{vv} &= \frac{\Delta \sin^2(\theta)}{(4mr_{\pm})^2 \Sigma} \left\{ \frac{a^4 \sin^2(\theta)}{(r^2 + a^2)} \left[-(\Delta + 4mr)\Delta + (r_{\pm}^2 - r^2) \right] \right. \\ &\quad \left. + (\Delta + 6mr)\Delta + 2a^2(r^2 - r_{\pm}^2) \right\}. \end{aligned}$$

This shows that both g_{uu} and g_{vv} have a zero of order two at $r = r_{\pm}$, which is precisely what is needed to cancel the singularities arising from $\hat{u}^{-2}$ in $g_{\hat{u}\hat{u}}$ and from $\hat{v}^{-2}$ in $g_{\hat{v}\hat{v}}$.

The Jacobian of the map $(u, v) \mapsto (\hat{u}, \hat{v})$ equals $c^2\hat{u}\hat{v}$, and it follows from (4.6.34) that the metric has Lorentzian signature in the coordinates $(\hat{u}, \hat{v}, \theta, \varphi_{\pm})$.

The metric induced on the Boyer–Lindquist sections of the event horizons of the Kerr metric, as well as on the bifurcate Killing horizon, reads

$$ds^2 = (R^2 + a^2 \cos^2(\theta)) d\theta^2 + \frac{(R^2 + a^2)^2 \sin^2(\theta)}{R^2 + a^2 \cos^2(\theta)} d\varphi^2, \quad (4.6.37)$$

where $R = m \pm \sqrt{m^2 - a^2}$. We note that its Ricci scalar, which we denote by K , is [311]

$$K = \frac{(R^2 + a^2)(3a^2 \cos^2(\theta) - R^2)}{(R^2 + a^2 \cos^2(\theta))^3}.$$

4.6.3 Surface gravity, thermodynamical identities

Recall that the surface gravity κ_* of a Killing horizon $\mathcal{H}_*$ associated with a Killing vector X is defined through the formula

$$\partial_\mu (X^\alpha X_\alpha)|_{\mathcal{H}_*} = -2\kappa_* X_\mu. \quad (4.6.38)$$

The following provides a convenient procedure for calculating κ_* : let b be any one-form which extends smoothly across the horizon and such that $b(X) = 1$. Then κ_* can be obtained from the equation

$$-2\kappa_* = -2\kappa_* b(X) = b(\nabla(X^\alpha X_\alpha))|_{\mathcal{H}_*}.$$

Note that the leftmost side of the last equation is independent of the choice of b , and so is therefore the right-hand side.

In order to implement this for the Kerr metric, recall that the Killing vector

$$X_* := \partial_t + \frac{a}{2mr_*} \partial_\varphi \equiv \partial_t + \frac{a}{a^2 + r_*^2} \partial_\varphi =: \partial_t + \Omega_* \partial_\varphi \quad (4.6.39)$$

is null on the Killing horizon $\mathcal{H}_* = \{r = r_*\}$, where $r_* \in \{r_-, r_+\}$ is one of the roots of Δ . As already pointed out, the parameter Ω_* is called the *angular velocity* of the horizon. The equation $g(X_*, X_*)|_{r=r_*} = 0$ is most easily checked using the following rewriting of the metric:

$$\begin{aligned} g = & \Sigma \left(\frac{1}{\Delta} dr^2 + d\theta^2 \right) + \frac{\sin^2(\theta)}{\Sigma} (adt - (r^2 + a^2)d\varphi)^2 \\ & - \frac{\Delta}{\Sigma} (dt - a \sin^2(\theta) d\varphi)^2. \end{aligned} \quad (4.6.40)$$

Let us use the coordinates (4.6.8)–(4.6.9), so that

$$dv = dt + \frac{r^2 + a^2}{\Delta} dr, \quad (4.6.41)$$

$$d\phi = d\varphi - \frac{a}{\Delta} dr. \quad (4.6.42)$$

Setting

$$b = dt + \frac{r^2 + a^2}{\Delta} dr = dv,$$

we see that b extends smoothly across the Killing horizon $\mathcal{H}_*$ and satisfies $b(X_*) = 1$. Thus, using

$$\begin{aligned} g^{rr} &= \frac{\Delta}{r^2 + a^2 \cos^2 \theta} \\ g(X_*, X_*) &= -\frac{\Delta (r_*^2 + a^2 \cos^2 \theta)}{(a^2 + r_*^2)^2} + O((r - r_*)^2) \end{aligned} \quad (4.6.43)$$

we find that

$$\begin{aligned} \kappa_* &= -\frac{1}{2} b(\nabla(g(X_*, X_*))) = -\frac{1}{2} (dt + \frac{r^2 + a^2}{\Delta} dr) (g^{\nu\mu} \partial_\mu (g(X_*, X_*) \partial_\nu) \\ &= -\lim_{r \rightarrow r_*} \frac{(r^2 + a^2)}{2\Delta} g^{rr} \partial_r (g(X_*, X_*)) \\ &= \frac{\partial_r \Delta}{2(r^2 + a^2)} \Big|_{r=r_*} = \frac{r_* - m}{2mr_*} \\ &= \pm \frac{\sqrt{m^2 - a^2}}{2m(m \pm \sqrt{m^2 - a^2})}, \end{aligned} \quad (4.6.44)$$

where the plus sign applies to the *future event horizon* $\{r = r_+\}$, and the minus sign should be used for the *future Cauchy horizon* $\{r = r_-\}$.

In the extreme cases $m = \pm a$ only the plus sign is relevant. We see that κ vanishes then, but is not zero otherwise.

Let $J = ma$ be the ‘ z -axis component’ of the angular momentum vector, and let A_* be the area of the cross-sections of the event horizon: denoting by $(x^A) = (\theta, \varphi)$ we have, using the fact that the metric is t -independent,

$$\begin{aligned}
 A_* &= \int_{r=r_*, v=\text{const}} \sqrt{\det g_{AB}} d\theta d\varphi \\
 &= \lim_{r \rightarrow r_*} \int_{r=\text{const}', v=\text{const}} \sqrt{\det g_{AB}} d\theta d\varphi \\
 &= \lim_{r \rightarrow r_*} \int_{r=\text{const}', t=\text{const}+F(r)} \sqrt{\det g_{AB}} d\theta d\varphi \\
 &= \lim_{r \rightarrow r_*} \int_{r=\text{const}', t=\text{const}} \sqrt{\det g_{AB}} d\theta d\varphi \\
 &= 2\pi(r_*^2 + a^2) \int_0^\pi \sin(\theta) d\theta = 4\pi(r_*^2 + a^2). \tag{4.6.45}
 \end{aligned}$$

By a direct calculation, or by general considerations [29, 157, 275, 455], one has the ‘thermodynamical identity’

$$\delta m = \frac{\kappa_*}{8\pi} \delta A_* + \Omega_* \delta J. \tag{4.6.46}$$

(Some care must be taken with the overall sign in (4.6.44) when the identity (4.6.46) is considered, as that sign is related to various orientations involved. The positive sign for the horizon $r_* = r_+$ is clearly consistent in this context.)

4.6.4 Carter’s time machine

An intriguing feature of the Kerr metric in the region $\{r < 0\}$ is the existence of points at which

$$g_{\varphi\varphi} \equiv g(\partial_\varphi, \partial_\varphi) < 0. \tag{4.6.47}$$

In other words, there exists a non-empty region where the Killing vector ∂_φ is timelike. Indeed, we have

$$\begin{aligned}
 g_{\varphi\varphi} &= \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2(\theta)}{\Sigma} \sin^2(\theta) \\
 &= \sin^2(\theta) \left(\frac{2a^2 m r \sin^2(\theta)}{a^2 \cos^2(\theta) + r^2} + a^2 + r^2 \right) \\
 &= \frac{\sin^2(\theta) (a^4 + a^2 \cos(2\theta) \Delta + a^2 r(2m + 3r) + 2r^4)}{a^2 \cos(2\theta) + a^2 + 2r^2}. \tag{4.6.48}
 \end{aligned}$$

We are interested in the set where $g_{\varphi\varphi} < 0$. The second line above clearly shows that this never happens for $r \geq 0$, or for $|r|$ very large. Nevertheless, for all $m > 0$ the set

$$\begin{aligned}
 \mathcal{V} &:= \{g_{\varphi\varphi} < 0\} \\
 &= \{r < 0, \cos(2\theta) < \underbrace{-\frac{a^4 + 2a^2 m r + 3a^2 r^2 + 2r^4}{a^2 \Delta}}_{=: G(r)}, \\
 &\quad \Sigma \neq 0, \sin(\theta) \neq 0\} \tag{4.6.49}
 \end{aligned}$$

is not empty. In order to see this, note that $G(0) = -1$, and $G'(0) = -4m/a^2 < 0$. This implies that for small negative r we have $G(r) > -1$, and hence there exists

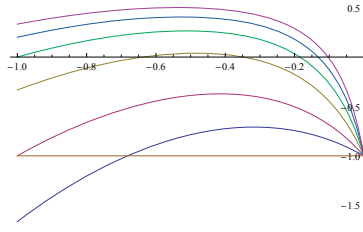


Fig. 4.6.3 The function $G(ax)$ of (4.6.49) with $m/a \in \{0.5, 1, 2, 3, 4, 5\}$, with the variable on the horizontal axis being $x = r/a$. There exist timelike orbits of ∂_φ in the region where $G(ax) > -1$.

a range of θ near $\theta = \pi/2$ for which the inequality defining $\mathcal{V}$ is satisfied. This is illustrated in Fig. 4.6.3.

It turns out that *any two points within $\mathcal{V}$ can be connected by a future-directed causal curve*. We show this in detail for points $p := (t, r, \theta, \varphi)$ and $p' := (t+T, r, \theta, \varphi)$ for any $T \in \mathbb{R}$: indeed, for n large consider the curve

$$[0, 2n\pi] \ni s \mapsto \gamma(s) = \left(t + \frac{T}{2n\pi}s, r, \theta, \varphi \pm s\right),$$

where the plus sign is chosen if ∂_φ is future directed in $\mathcal{V}$, while the negative sign is chosen otherwise. Then $\dot{\gamma} = \pm \partial_\varphi + \frac{T}{2n\pi} \partial_t$, which is timelike future directed for all n large enough. As the φ coordinate is 2π -periodic, the curve γ starts at p and ends at p' .

A similar argument applies for general pairs of points within $\mathcal{V}$.

In particular the choice $T = 0$ and $n = 1$ gives a closed timelike curve.

Interestingly enough, the non-empty region $\mathcal{V}$ can be used to *connect any two points p_1 and p_2 lying in the region $r < r_-$ by a future-directed timelike curve*. To see this, choose some future-directed timelike curve γ_1 from p_1 to some point $p \in \mathcal{V}$, and some future-directed timelike curve from some point $p' \in \mathcal{V}$ to p_2 . The existence of such curves γ_1 and γ_2 is easy to check, and follows e.g. by inspection of the projection diagram for the Kerr metric, Fig. 7.3.2, p. 288. We can then connect p_1 with p_2 by a future-directed causal curve by first following γ_1 from p_1 to p , then a future-directed causal curve γ from p to p' lying in $\mathcal{V}$, and then following γ_2 from p' to p_2 .

So, in fact, the region $\mathcal{V}$ provides a *time machine* for the region $r < r_-$, a property which seems to have been first observed by Carter [78, 79].

We have been assuming that $m > 0$ in our discussion of the time machine. It should be clear from the arguments given that the time-travel mechanism for Kerr metrics just described also works, now in the region $r > 0$, when $m < 0$.

4.6.5 Extreme case $a^2 = m^2$

The coordinate transformation leading to (4.6.10) can be used for $a = m$ as well, leading to

$$\begin{aligned} \hat{g} = & -\left(1 - \frac{2mr}{\Sigma}\right)dv^2 + 2drdv + \Sigma d\theta^2 - 2m \sin^2(\theta) d\phi dr \\ & + \frac{(r^2 + m^2)^2 - m^2 \sin^2(\theta) \Delta}{\Sigma} \sin^2(\theta) d\phi^2 - \frac{4m^2 r \sin^2(\theta)}{\Sigma} d\phi dv. \end{aligned} \quad (4.6.50)$$

As before, it holds that

$$\det g = -\sin^2(\theta) \Sigma^2, \quad (4.6.51)$$

which shows, using analyticity, that the metric is smooth and Lorentzian away from the set $\{\Sigma = 0\}$.

This provides an extension of the exterior region of the Kerr metric with $m = |a|$ across a degenerate Killing horizon.

A second similar extension is obtained by using instead the coordinates u and ψ of (4.6.26)–(4.6.27).

There is no construction analogous to that in Section 4.6.2 in the degenerate case.

The global structure of the standard maximal analytic extension obtained by repeatedly using the extensions just described is analysed in Section 7.3, and visualized by a projection diagram in Figure 7.3.2, p. 288.

Near-horizon geometry. The near-horizon metric of the extreme Kerr solution, which we will denote by g_{NHK} , can be obtained [30] by replacing first the coordinates (t, r, φ) of (4.6.7), p. 177, by new coordinates $(\hat{t}, \hat{r}, \hat{\phi})$ defined as

$$r = m + \epsilon \hat{r}, \quad t = \epsilon^{-1} \hat{t}, \quad \varphi = \hat{\phi} + \frac{\hat{t}}{2m\epsilon}, \quad (4.6.52)$$

and passing to the limit $\epsilon \rightarrow 0$; compare Section 4.3.6. Some algebra leads to

$$g_{\text{NHK}} = \frac{1 + \cos^2(\theta)}{2} \left[-\frac{\hat{r}^2}{r_0^2} d\hat{t}^2 + \frac{r_0^2}{\hat{r}^2} d\hat{r}^2 + r_0^2 d\theta^2 \right] + \frac{2r_0^2 \sin^2(\theta)}{1 + \cos^2(\theta)} \left(d\hat{\phi} + \frac{\hat{r}}{r_0^2} d\hat{t} \right)^2, \quad (4.6.53)$$

where $r_0 = \sqrt{2}m$. This metric is singular at $\hat{r} = 0$, but a second change of coordinates

$$v = \hat{t} - \frac{r_0^2}{\hat{r}}, \quad \tilde{\varphi} = \hat{\phi} - \log \left(\frac{\hat{r}}{r_0} \right), \quad (4.6.54)$$

leads to a manifestly regular form of the *near-horizon Kerr metric*:

$$g_{\text{NHK}} = \frac{1 + \cos^2(\theta)}{2} \left[-\frac{\hat{r}^2}{r_0^2} dv^2 + 2 dv d\hat{r} + r_0^2 d\theta^2 \right] + \frac{2r_0^2 \sin^2(\theta)}{1 + \cos^2(\theta)} \left(d\tilde{\varphi} + \frac{\hat{r}}{r_0^2} dv \right)^2. \quad (4.6.55)$$

This is again a vacuum solution of the Einstein equations with a degenerate horizon located at $\hat{r} = 0$, but with rather a different asymptotic behaviour as the radial variable $\hat{r}$ tends to infinity.

INCIDENTALLY: The reader might have noticed that the metric (4.6.55) is not directly in the form (4.3.46), p. 160, used to derive the near-horizon geometry equations (4.3.59). The coordinate transformation needed to bring the metric to the form (4.3.46), p. 160, has been derived as a power series in [323].

Cylindrical ends. It turns out that the degenerate Kerr spacetimes contain space-like hypersurfaces with constant mean curvature ('CMC slices') with *asymptotically conformally cylindrical ends*, in a sense which we make precise now: in Boyer–Lindquist coordinates the *extreme Kerr metrics*, with $a^2 = m^2$, take the form, changing φ to its negative if necessary,

$$g = -dt^2 + \frac{2mr}{r^2 + m^2 \cos^2(\theta)} (dt - m \sin^2(\theta) d\varphi)^2 + (r^2 + m^2) \sin^2(\theta) d\varphi^2 + \frac{r^2 + m^2 \cos^2(\theta)}{(r - m)^2} dr^2 + (r^2 + m^2 \cos^2(\theta)) d\theta^2. \quad (4.6.56)$$

The metric, say γ , induced on the slices $t = \text{const}$ reads, keeping in mind that $r > m$,

$$\begin{aligned}\gamma = & \frac{r^2 + m^2 \cos^2(\theta)}{(r - m)^2} dr^2 + (r^2 + m^2 \cos^2(\theta)) d\theta^2 \\ & + \frac{(r^2 + m^2)^2 - (r - m)^2 m^2 \sin^2(\theta)}{r^2 + m^2 \cos^2(\theta)} \sin^2(\theta) d\varphi^2.\end{aligned}\quad (4.6.57)$$

Introducing a new variable $x \in (-\infty, \infty)$ defined as

$$dx = -\frac{dr}{r - m} \iff x = -\ln\left(\frac{r}{m} - 1\right),$$

so that x tends to infinity as r approaches m from above, the metric (4.6.57) exponentially approaches

$$\begin{aligned}\gamma \rightarrow_{x \rightarrow \infty} & m^2(1 + \cos^2(\theta)) dx^2 + \dot{g} \\ = & m^2(1 + \cos^2(\theta)) \left(dx^2 + \underbrace{d\theta^2 + \frac{4 \sin^2(\theta)}{(1 + \cos^2(\theta))^2} d\varphi^2}_{=: \dot{h}} \right).\end{aligned}\quad (4.6.58)$$

We see that the space metric γ is, asymptotically, conformal to the product metric

$$dx^2 + \dot{h}$$

on the cylinder $\mathbb{R} \times S^2$.

INCIDENTALLY: The limiting metric, as one recedes to infinity along the cylindrical end of the extreme Kerr metric, can also be obtained from the metric (4.6.37) on the bifurcation surface of the event horizon by setting $a = m$ there,

$$ds^2|_{m=a} = m^2 \left((1 + \cos^2(\theta)) d\theta^2 + \frac{4 \sin^2(\theta)}{1 + \cos^2(\theta)} d\varphi^2 \right), \quad (4.6.59)$$

which indeed coincides with (4.6.58). $\square$

4.6.6 Maximal slices

Let us verify that the slices $t = \text{const}$ have constant vanishing mean curvature; we say that they are *maximal*. This follows from the fact that the unit normal n to these slices takes the form $n = n^t \partial_t + n^\varphi \partial_\varphi$, so that

$$\begin{aligned}\text{tr}_\gamma K &= \frac{1}{\sqrt{|\det g|}} \partial_\mu (\sqrt{|\det g|} n^\mu) \\ &= \frac{1}{\sqrt{|\det g|}} \left(\partial_t (\sqrt{|\det g|} n^t) + \partial_\varphi (\sqrt{|\det g|} n^\varphi) \right) = 0.\end{aligned}$$

It then follows from the scalar constraint equation that the scalar curvature R of these slices satisfies $R \geq 0$.

INCIDENTALLY: The *Lichnerowicz equation* provides a tool for constructing solutions of the scalar constraint equation. When studying the existence of solutions of this equation it is useful to control the sign of the scalar curvature of the metrics occurring in the problem at hand. Recall that the scalar curvature, say κ , of a metric of the form $d\theta^2 + e^{2f(\theta)} d\varphi^2$ equals

$$\kappa = -2(f'' + (f')^2).$$

Hence the sphere part $\dot{h}$ of the limiting conformal metric appearing in (4.6.58) has scalar curvature equal to

$$-\frac{4 \cos(2\theta)}{(\cos^2 \theta + 1)^2},$$

which is negative on the northern hemisphere and positive on the southern one. We also note that the metric (4.6.59) has scalar curvature

$$\kappa = \frac{2(3 \cos^2(\theta) - 1)}{m^2(1 + \cos^2(\theta))^3},$$

which changes sign as well. $\square$

4.6.7 The Ernst map for the Kerr metric

A key role for proving uniqueness of the Kerr black holes is a *harmonic map representation* of the field equations, which is obtained as follows: to every stationary axisymmetric solution of the vacuum Einstein field equations $(\mathcal{M}, g)$ one associates a pair of functions (f, ω) , where f is the norm of the axisymmetric Killing vector, say η ,

$$f = g(\eta, \eta),$$

while the function ω is called the *twist potential*. To define the latter one introduces, first, the *twist form* $\omega_\mu dx^\mu$ via the equation

$$\omega_\mu = \epsilon_{\mu\alpha\beta\gamma} \eta^\alpha \nabla^\beta \eta^\gamma.$$

It follows from the vacuum field equations that ω is closed; see (4.3.42), p. 158. So if, e.g., $\mathcal{M}$ is simply connected, there exists a function ω , called the *twist potential* such that

$$\omega_\mu = \partial_\mu \omega.$$

The complex valued function $f + i\omega$ is called the *Ernst potential*.

The Ernst potential defines a map from the two-dimensional manifold of orbits of the group $\mathbb{R} \times U(1)$, generated by time translations and rotations, to hyperbolic space. The map is harmonic when the four-dimensional metric satisfies the vacuum Einstein equations.

In Boyer–Lindquist coordinates of (4.6.7) the twist potential ω reads [168]

$$\omega = ma(\cos^3 \theta - 3 \cos \theta) - \frac{ma^3 \cos \theta \sin^4 \theta}{\Sigma}. \quad (4.6.60)$$

The Ernst potential $f + i\omega$ can now be obtained by reading $f = g_{\varphi\varphi}$ from (4.6.7).

4.6.8 The orbit-space metric

We have seen that the Kerr metric is asymptotically flat; we can therefore define its domain of outer communications $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$ using the prescription of Section 4.3.8. Let M denote the space of orbits of the isometry group in the d.o.c:

$$M := \langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle / (\mathbb{R} \times U(1)).$$

Note that, in Boyer–Lindquist coordinates, M can be viewed as the submanifold $\{t = 0 = \varphi\}$ of $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$, coordinatized by $\theta \in [0, \pi]$ and r , with $r_+ < r < \infty$.

Let $X_1 = \partial_t$, $X_2 = \eta = \partial_\varphi$, and let

$$\mathcal{A} := \{\eta = 0\} = \{\theta = 0\} \cup \{\theta = \pi\}$$

be the *axis of rotation*. We will use the same symbol $\mathcal{A}$ for the axis of rotation in $\mathcal{M}$, as well as for the corresponding set in M . The orbit-space metric, say q , on $M \setminus \mathcal{A}$ is defined as follows: for $Y, Z \in T(M \setminus \mathcal{A})$,

$$q(Y, Z) = g(Y, Z) - g^{ab} g(X_a, Y) g(X_b, Z), \quad (4.6.61)$$

where g^{ab} is the matrix inverse to $g(X_a, X_b)$. Note that $\det g(X_a, X_b) < 0$ on $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle \setminus \mathcal{A}$, which shows that q is well defined there.

Since $g_{\theta t} = g_{\theta\varphi} = g_{rt} = g_{r\varphi} = 0$, the two-dimensional metric q is obtained by simply forgetting the part of the metric involving dt and $d\varphi$:

$$q = (r^2 + a^2 \cos^2 \theta) \left(\frac{dr^2}{(r - r_+)(r - r_-)} + d\theta^2 \right). \quad (4.6.62)$$

So, $\{\theta = 0\}$ and $\{\theta = \pi\}$ are clearly smooth boundaries at finite distance for h , with h extending smoothly by continuity there. On the other hand, the nature of

the boundary $\{r = r_+\}$ (recall that $r > r_+$ on $\langle\langle \mathcal{M}_{\text{ext}} \rangle\rangle$, and hence on M) depends upon whether $r_+ = r_-$. In the subextreme case, where r_- and r_+ are distinct, the set $\{r = r_+\}$ is seen to be a totally geodesic boundary at finite distance after introducing a new coordinate x by the formula

$$\frac{dx}{dr} = \frac{1}{\sqrt{(r - r_+)(r - r_-)}}. \quad (4.6.63)$$

We then have

$$x_+ := \lim_{r \rightarrow r_+} x > -\infty, \quad (4.6.64)$$

as the right-hand side of (4.6.63) is integrable in r near $\{r = r_+\}$.

The same formula (4.6.63) in the extreme case $r_- = r_+$ gives a coordinate x which diverges logarithmically as r approaches r_- , leading to a *cylindrical end* for the metric q .

4.6.9 Kerr–Schild coordinates

In Kerr–Schild coordinates the Kerr metric takes the form (cf., e.g., [244])

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2m\tilde{r}^3}{\tilde{r}^4 + a^2z^2}\theta_\mu\theta_\nu =: \eta_{\mu\nu} + H\theta_\mu\theta_\nu, \quad \tilde{r}^2 + z^2 \neq 0,$$

where

$$\theta \equiv \theta_\mu dx^\mu := -\left(dt + \frac{1}{\tilde{r}^2 + a^2}[\tilde{r}(xdx + ydy) - a(xdy - ydx)] + \frac{z}{\tilde{r}}dz\right),$$

and where $\tilde{r}$ is defined implicitly as the solution of the equation

$$\tilde{r}^4 - \tilde{r}^2(x^2 + y^2 + z^2 - a^2) - a^2z^2 = 0. \quad (4.6.65)$$

This form of the metric makes manifest the asymptotic flatness of the metric, and turns out to be useful for understanding the singular set $\tilde{r} = 0 = z$.

The function $\tilde{r}$ above actually coincides with the Boyer–Lindquist coordinate previously denoted by r and, using the notation $(\tilde{t}, \tilde{r}, \theta, \phi)$ for the Boyer–Lindquist coordinates and (t, x, y, z) for the Kerr–Schild ones, the transformation between Boyer–Lindquist and Kerr–Schild coordinates reads (cf., e.g., [404])

$$x + iy = (\tilde{r} + ia)e^{i(\phi + r^\#)} \sin(\theta), \quad z = \tilde{r} \cos(\theta), \quad t = \tilde{t} + r^* - \tilde{r}, \quad (4.6.66)$$

where

$$r^* := \int \frac{\tilde{r}^2 + a^2}{\Delta} d\tilde{r}, \quad r^\# := \int \frac{a}{\Delta} d\tilde{r}. \quad (4.6.67)$$

Equation (4.6.66) shows that the Boyer–Lindquist singular set $\tilde{r} = 0 = \cos(\theta)$ becomes a coordinate cylinder in the coordinates (t, x, y, z) , or a ring in the coordinates (x, y, z) ,

$$x + iy = ae^{i\psi}, \quad (4.6.68)$$

with ψ in, say, $[0, 2\pi)$. Alternatively, one can first set z to 0 in (4.6.65) and divide by $\tilde{r}^2$ to obtain

$$\tilde{r}^2 - (x^2 + y^2 - a^2) = 0. \quad (4.6.69)$$

Setting $\tilde{r}$ to 0 reproduces a ring of radius a .

A detailed discussion of the geometry near the singular ring can be found in [134].

One checks that the covector field θ is null with respect both to the Minkowski metric and to the Kerr metric. This easily implies that

$$g^{\mu\nu} = \eta^{\mu\nu} - H\theta^\mu\theta^\nu, \quad (4.6.70)$$

where $\eta^{\mu\nu}$ is the inverse Minkowski metric. It follows from the lightlike character of θ that it does not matter whether the indices on θ have been raised with the Minkowski metric or with the Kerr metric in (4.6.70).

The lightlike character of θ also implies that

$$\det g_{\mu\nu} = -1. \quad (4.6.71)$$

This can be seen by noting that at every given point p we can find a Lorentz matrix which transforms θ_μ into $(h, h, 0, 0)$ for some number h . At p and in these coordinates $g_{\mu\nu}$ reads

$$\begin{pmatrix} -1 + Hh^2 & Hh^2 & 0 & 0 \\ Hh^2 & 1 + Hh^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (4.6.72)$$

and (4.6.71) immediately follows.

Equation (4.6.71) shows that the coordinates (t, x, y, z) provide a global coordinate system away from the singular ring, covering directly both Killing horizons.

4.6.10 Dain coordinates

Dain [168] has invented a system of coordinates which nicely exhibits the ‘Einstein–Rosen bridges’ of the Kerr metric. The aim is to write the space part of the Kerr metric in the form

$$g = e^{-2\tilde{U}+2\alpha} (d\rho^2 + dz^2) + \rho^2 e^{-2\tilde{U}} (d\varphi + \rho B_\rho d\rho + A_z dz)^2. \quad (4.6.73)$$

If $|a| \leq m$, let $r_+ = m + \sqrt{m^2 - a^2}$ be the largest root of Δ , and let $r_+ = 0$ otherwise. For

$$r > r_+,$$

so that $\Delta > 0$, define a new radial coordinate $\tilde{r}$ by

$$\tilde{r} = \frac{1}{2} (r - m + \sqrt{\Delta}). \quad (4.6.74)$$

After setting

$$\rho = \tilde{r} \sin \tilde{\theta}, \quad z = \tilde{r} \cos \tilde{\theta}, \quad (4.6.75)$$

one obtains (4.6.73).

Equation (4.6.74) is inverted as

$$r = \tilde{r} + m + \frac{m^2 - a^2}{4\tilde{r}}. \quad (4.6.76)$$

We emphasize that while those coordinates bring the metric to the form (4.6.73), familiar in the context of the reduction of the stationary axisymmetric vacuum Einstein equations to a harmonic map problem, the coordinate ρ in (4.6.75) is *not* the area coordinate needed for that reduction¹² *except* when $m = a$.

¹²The correct (ρ, z) coordinates for the harmonic map reduction are $\rho = \sqrt{\Delta} \sin(\theta)$, $z = (\tilde{r} - m) \cos \theta$. In the last coordinates the horizon lies on the axis $\rho = 0$, which is not the case for Dain’s coordinates except if $a = m$.

To analyse the behaviour of U near $\tilde{r} = 0$ we have to distinguish between the extreme and non-extreme cases. Let us first assume that $m^2 \neq a^2$. We can calculate $e^{\tilde{U}}$ from (4.6.7), and using (4.6.74) we then have

$$\tilde{U} = 2 \ln \left(\frac{2\tilde{r}}{m} \right) - \ln \left| 1 - \frac{a^2}{m^2} \right| + O(\tilde{r}). \quad (4.6.77)$$

With a little work it can now be seen that that $\tilde{r} = 0$ corresponds to another asymptotically flat region for the metric (4.6.73).

On the other hand, in the extreme case $m^2 = a^2$ one finds that

$$\tilde{U} = \ln \left(\frac{\tilde{r}}{2m} \right) + \frac{1}{2} \ln (1 + \cos^2(\theta)) + O(\tilde{r}). \quad (4.6.78)$$

This implies that the space geometry near $\tilde{r} = 0$ approaches that of an ‘asymptotically cylindrical end’, as discussed in general in Section 4.3.5.

4.6.11 Weyl coordinates

A useful ansatz for the analysis of stationary-axisymmetric solutions has been proposed by Weyl, namely

$$g = f^{-1} [h (d\rho^2 + d\zeta^2) + \rho^2 d\phi^2] - f (dt + ad\phi)^2. \quad (4.6.79)$$

The justification that suitably regular stationary-axisymmetric metrics on simply connected manifolds can be globally represented in this form can be found in [103].

Let us denote by $(t, \tilde{r}, \tilde{\theta}, \varphi)$ the Boyer–Lindquist coordinates for the Kerr metric, as in (4.6.1); we thus need to find coordinates for the two-dimensional Riemannian metric

$$\frac{\Sigma(\tilde{r})}{\Delta(\tilde{r})} d\tilde{r}^2 + \Sigma(\tilde{r}) d\tilde{\theta}^2,$$

which make it manifestly conformally flat. Such coordinates are called ‘isothermal coordinates’. Equation (4.6.79) can be obtained by setting (see, e.g., [345, (1.133), p. 27])

$$\rho := \sqrt{\tilde{r}^2 - 2m\tilde{r} + a^2} \sin(\tilde{\theta}) \equiv \sqrt{(\tilde{r} - r_-)(\tilde{r} - r_+)} \sin(\tilde{\theta}), \quad (4.6.80)$$

$$\zeta := (\tilde{r} - m) \cos(\tilde{\theta}), \quad (4.6.81)$$

with $\tilde{r} > r_+$.

To describe the inverse transformation, it is convenient to introduce a new variable z and two constants a_i by setting

$$a_2 - a_1 := 2\sqrt{m^2 - a^2}, \quad z = \zeta + \frac{a_1 + a_2}{2}.$$

Note that these quantities are defined only up to an overall shift $(a_1, a_2, z) \mapsto (a_1 + \lambda, a_2 + \lambda, z + \lambda)$, where λ is an arbitrary real constant. If one now replaces $\tilde{r}$ and $\cos \tilde{\theta}$ by new variables r_1 and r_2 defined as

$$\tilde{r} = m + \frac{r_1 + r_2}{2}, \quad \tilde{\theta} = \cos^{-1} \left(\frac{r_2 - r_1}{a_1 - a_2} \right),$$

then (4.6.80)–(4.6.81) can be solved for the r_i ’s in terms of ρ and z :

$$r_i := \sqrt{(z - a_i)^2 + \rho^2}.$$

The explicit form of the metric components in the coordinate system (t, ρ, φ, z) is not very enlightening, and will therefore not be given here.

4.7 Majumdar–Papapetrou multi-black holes

In all examples discussed so far the black-hole event horizon forms a connected hypersurface in spacetime. In fact [71, 102, 120], there are no *regular, static, vacuum* solutions with several black holes, consistently with the intuition that gravity is an attractive force. However, static multi-black holes become possible in presence of an electric field. Well-behaved examples are exhausted [148] by the *Majumdar–Papapetrou* black holes.

The metric g and the electromagnetic potential A take the form [335, 390]

$$g = -u^{-2}dt^2 + u^2(dx^2 + dy^2 + dz^2), \quad (4.7.1)$$

$$A = u^{-1}dt, \quad (4.7.2)$$

with a nowhere-vanishing function u . Einstein–Maxwell equations read then

$$\frac{\partial u}{\partial t} = 0, \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0. \quad (4.7.3)$$

The solutions will be called *standard MP black holes* if the coordinates x^μ of (4.7.1)–(4.7.2) cover the range $\mathbb{R} \times (\mathbb{R}^3 \setminus \{\vec{a}_i\})$ for a finite set of points $\vec{a}_i \in \mathbb{R}^3$, $i = 1, \dots, I$, and if the function u has the form

$$u = 1 + \sum_{i=1}^I \frac{\mu_i}{|\vec{x} - \vec{a}_i|}, \quad (4.7.4)$$

for some positive constants μ_i .

INCIDENTALLY: It has been proved in [136] (see also [141, 240]) that these are the only regular black holes within the MP class; the fact that all multi-component regular static black holes are in the MP class has been established in [148], building upon the work in [342, 419, 428]; a gap in [148] related to analyticity of the metric has been removed in [120].

When $I = \infty$, it is a standard fact in potential theory that if the series (4.7.4) converges at some point, then it converges to a smooth function everywhere away from the punctures $\vec{a}_i$. This case has been analysed in [100, Appendix B], where it was pointed out that the scalar $F_{\mu\nu}F^{\mu\nu}$ is unbounded whenever the $\vec{a}_i$'s have accumulation points. It follows from [136] that the case where $I = \infty$ and the $\vec{a}_i$'s do not have accumulation points cannot lead to regular asymptotically flat spacetimes.

Calculating the flux of the electric field on spheres $|\vec{x} - \vec{a}_i| = \epsilon \rightarrow 0$, one finds that μ_i is the electric charge carried by the puncture $\vec{x} = \vec{a}_i$: indeed, let $F = dA$ be the Maxwell tensor; we have

$$F = -u^{-2}du \wedge dt = -u^{-2}\partial_\ell u dx^\ell \wedge dt.$$

The flux of F through a two-dimensional hypersurface S is defined as

$$\int_S \star F,$$

where $\star$ is the Hodge dual; see Appendix D.1, p. 353. A convenient orthonormal basis of T^*M is given by the co-frame

$$\theta^0 = u^{-1}dt, \quad \theta^\ell = u dx^\ell,$$

in terms of which we have

$$F = -u^{-2}\partial_\ell u \theta^\ell \wedge \theta^0.$$

This gives

$$\begin{aligned}\star F &= -u^{-2} \partial_\ell u \star (\theta^\ell \wedge \theta^0) = \frac{1}{2} \sum_{\ell j k} u^{-2} \partial_\ell u \epsilon^{\ell j k} \theta^j \wedge \theta^k \\ &= \frac{1}{2} \sum_{ijk} \partial_\ell u \epsilon^{\ell j k} dx^j \wedge dx^k.\end{aligned}$$

Consider a sphere $S(\vec{a}_i, \epsilon)$ of radius ϵ centred at $\vec{a}_i$; shifting the coordinates by $\vec{a}_i$ we can assume that $\vec{a}_i = \vec{0}$. Then $\partial_\ell u$ approaches

$$-\mu_i \vec{x}/|\vec{x}|^3$$

on $S(\vec{a}_i, \epsilon)$ as ϵ tends to 0. Therefore

$$\lim_{\epsilon \rightarrow 0} \int_{S(\vec{a}_i, \epsilon)} \star F = -4\pi \mu_i.$$

In our conventions the right-hand side is -4π times the charge, which establishes the claim.

We will see shortly that punctures correspond to connected components of the event horizon, so μ_i can be thought of as the negative of the electric charge of the i th black hole.

Higher-dimensional generalizations of the MP solutions have been derived by Myers [370]. The metric and the electromagnetic potential take the form

$$g = -u^{-2} dt^2 + u^{\frac{2}{n-2}} \left((dx^1)^2 + \dots + (dx^n)^2 \right), \quad (4.7.5)$$

$$A = u^{-1} dt, \quad (4.7.6)$$

with u being time independent, and harmonic with respect to the flat metric $(dx^1)^2 + \dots + (dx^n)^2$. Then, a natural candidate potential u for solutions with black holes takes the form

$$u = 1 + \sum_{i=1}^N \frac{\mu_i}{|\vec{x} - \vec{a}_i|^{n-2}}, \quad (4.7.7)$$

for some $\vec{a}_i \in \mathbb{R}^n$. Here configurations with $N = \infty$ and which are periodic in some variables are also of interest, as they could lead to Kaluza–Klein-type four-dimensional solutions.

Let us point out some features of the geometries (4.7.5) with $N < \infty$. First, for large $|\vec{x}|$ we have

$$u = 1 + \frac{\sum_{i=1}^N \mu_i}{|\vec{x}|^{n-2}} + O(|\vec{x}|^{-(n-1)}),$$

so that the metric is asymptotically flat, with total ADM mass $\sum_{i=1}^N \mu_i$.

Next, choose any i and denote by $r := |\vec{x} - \vec{a}_i|$ a radial coordinate centred at $\vec{a}_i$. Then the space part, say g_s , of the metric (4.7.5) takes the form

$$\begin{aligned}g_s &= u^{\frac{2}{n-2}} \left((dx^1)^2 + \dots + (dx^n)^2 \right) = r^2 u^{\frac{2}{n-2}} \left(\frac{dr^2}{r^2} + h \right) \\ &= (r^{\frac{1}{n-2}} u)^{\frac{2}{n-2}} (d(\underbrace{\ln r}_{=:x})^2 + h) \\ &= (r^{\frac{1}{n-2}} u)^{\frac{2}{n-2}} (dx^2 + h),\end{aligned} \quad (4.7.8)$$

where h is the unit round metric on S^{n-1} . Now, the metric $dx^2 + h$ is the canonical, complete, product metric on the cylinder $\mathbb{R} \times S^{n-1}$. Further

$$r^{\frac{1}{n-2}} u \rightarrow_{\vec{x} \rightarrow \vec{a}_i} \mu_i > 0.$$

Therefore the space part of the Majumdar–Papapetrou metric approaches a multiple of the canonical metric on the cylinder $\mathbb{R} \times S^{n-1}$ as $\vec{x}$ approaches $\vec{a}_i$. We conclude that the space geometry is described by a complete metric which has one asymptotically flat region $|\vec{x}| \rightarrow \infty$ and N asymptotically cylindrical regions $\vec{x} \rightarrow \vec{a}_i$.

It has been shown by Hartle and Hawking [240] that, in dimension $n = 3$, every standard MP spacetime can be analytically extended to an electrovacuum spacetime with I black-hole regions. The calculation, which also provides some information in higher dimensions $n > 3$ but runs into difficulties there, proceeds as follows: let, as before, $r = |\vec{x} - \vec{a}_i|$; for r small we replace t by a new coordinate v defined as

$$v = t + f(r) \quad \Longrightarrow \quad dt = dv - f'(r)dr,$$

with a function f to be determined shortly. We obtain

$$\begin{aligned} g &= -u^{-2}(dv - f'dr)^2 + u^{\frac{2}{n-2}}(dr^2 + r^2h) \\ &= -u^{-2}dv^2 + 2u^{-2}f'dvdr + \left(u^{\frac{2}{n-2}} - u^{-2}(f')^2\right)dr^2 + u^{\frac{2}{n-2}}r^2h. \end{aligned} \quad (4.7.9)$$

We have already seen that the last term $u^{\frac{2}{n-2}}r^2h$ is well behaved for r small. Let us show that in some cases we can choose f to get rid of the singularity in g_{rr} . For this we Taylor expand the non-singular part of u near $\vec{a}_i$,

$$u = \underbrace{\frac{\mu_i}{r^{n-2}} + 1 + \sum_{j \neq i} \frac{\mu_j}{|\vec{a}_j - \vec{a}_i|^{n-2}}}_{=: \hat{u}} + r\hat{u} = \hat{u}\left(1 + O(r^{n-1})\right), \quad (4.7.10)$$

with $\hat{u}$ being an analytic function of r and of the angular variables, at least for small r . We choose f so that $\hat{u}^{\frac{2}{n-2}} - \hat{u}^{-2}(f')^2$ vanishes:

$$f' = \hat{u}^{\frac{n-1}{n-2}}.$$

This shows that the function

$$g_{rr} = u^{\frac{2}{n-2}} - u^{-2}(f')^2 = \underbrace{\hat{u}^{\frac{2}{n-2}}}_{\sim r^{-2}} \left[\left(\underbrace{\frac{u}{\hat{u}}}_{1+O(r^{n-1})} \right)^{\frac{2}{n-2}} - \left(\underbrace{\frac{\hat{u}}{u}}_{1+O(r^{n-1})} \right)^2 \right] = O(r^{n-3})$$

is an analytic function of r and of angular variables for small r .

The above works well when $n = 3$, in which case (4.7.9) reads

$$g = -\underbrace{u^{-2}}_{\sim r^2} dv^2 + 2\left(\underbrace{\frac{\hat{u}}{u}}_{=1+O(r^2)}\right)^2 dvdr + \underbrace{g_{rr}}_{=O(1)} dr^2 + \underbrace{u^2 r^2}_{=\mu_i^2 + O(r)} h.$$

At $r = 0$ the determinant of g equals $-\mu_i^4 \det h \neq 0$, which implies that $g_{\mu\nu}$ can be analytically extended across the null hypersurface

$$\mathcal{H}_i := \{r = 0\}$$

to a real-analytic Lorentzian metric defined in a neighbourhood of $\mathcal{H}_i$. By analyticity the extended metric is vacuum. Obviously $\mathcal{H}_i$ is a Killing horizon for the Killing vector $\partial_t = \partial_v$, since g_{vv} vanishes at $\mathcal{H}_i$.

We note that the differential of $g(\partial_v, \partial_v)$ vanishes at $r = 0$ as well, which shows that all horizons have vanishing surface gravity.

Let us return to general dimensions $n \geq 4$. The problem is that the determinant of the metric vanishes now at $r = 0$. One could hope that this can be repaired by a change of coordinates. For this, consider g_{rv} :

$$\begin{aligned}
g_{rv} dr dv &= u^{-2} f' dr dv = \left(\frac{\dot{u}}{u} \right)^2 \dot{u}^{\frac{3-n}{n-2}} dr dv = \left(1 + O(r^{n-2}) \right) \mu_i^{\frac{3-n}{n-2}} r^{n-3} dr dv \\
&= \left(1 + O(r^{n-2}) \right) \frac{\mu_i^{\frac{3-n}{n-2}}}{n-2} d(\underbrace{r^{n-2}}_{=: \rho}) dv.
\end{aligned}$$

We see that this term will give a non-vanishing contribution to the determinant if we introduce a new radial variable $\rho = r^{n-2}$. This, however, will wreak havoc in $g_{rr} dr^2$, as well as in various other terms because then $r = \rho^{\frac{1}{n-2}}$, which introduces fractional powers of the new coordinate ρ in the metric, leading to a continuous but non-manifestly differentiable extension.

Now, none of these problems occur if $N = 1$, in which case $u = \dot{u}$, hence $g_{rr} \equiv 0$. Furthermore,

$$g_{vv} = \dot{u}^{-2} = \left(1 + \frac{\mu_i}{\rho} \right)^{-2} = \frac{\rho^2}{(\mu_i + \rho)^2}, \quad (4.7.11)$$

$$u^{\frac{2}{n-2}} r^2 = (\dot{u} \rho)^{\frac{2}{n-2}} = (\mu_i + \rho)^{\frac{2}{n-2}}, \quad (4.7.12)$$

$$g_{rv} dr dv = \frac{\dot{u}^{\frac{3-n}{n-2}}}{(n-2)r^{n-3}} d\rho dv = \frac{(\dot{u} r^{n-2})^{\frac{3-n}{n-2}}}{(n-2)} d\rho dv = \frac{(\mu_i + \rho)^{\frac{3-n}{n-2}}}{(n-2)} d\rho dv, \quad (4.7.13)$$

which proves that the metric can be extended analytically across a Killing horizon $\{\rho = 0\}$, as desired. (The case $N = 1$ is of course spherically symmetric, so this calculation is actually a special case of Remark 4.2.10, p. 139.)

Equation (4.7.11) shows that the Killing vector $\partial_t = \partial_v$ is spacelike everywhere except at the horizon $\rho = 0$: $g(\partial_v, \partial_v) \geq 0$. In particular $g(\partial_v, \partial_v) \geq 0$ attains a minimum on the horizon; hence its derivative vanishes there. As before we conclude that the black hole is degenerate, $\kappa = 0$.

For $n \geq 4$ and $N > 1$ the above construction (or some slight variation thereof, with f not necessarily radial, chosen to obtain $g_{rr} = 0$) produces a metric which can at best be extended by continuity across a Killing horizon ‘located at $\vec{x} = \vec{a}_i$ ’, but the extensions so obtained do not appear to be differentiable. The optimal degree of differentiability that one can obtain does not seem to be known in general. As such, it has been shown in [458] that the metric cannot be extended smoothly when $n \geq 4$ and $N = 2$ or 3.

More can be said for axisymmetric solutions [75]: in dimension $n = 5$, C^2 extensions for multi-component axisymmetric configurations can be constructed, and it is argued there that generic such solutions do not possess C^3 extensions. Examples are constructed where smooth extensions are possible for one central component, or for an infinity string of components. In dimension $n \geq 5$, C^1 extensions for multi-component axisymmetric configurations can be constructed, and it is argued that generic such solutions do not possess C^2 extensions.

PROBLEM 4.7.1 Study, for $n \geq 4$, whether (4.7.7) can be corrected by a harmonic function to give a smooth event horizon. Alternatively, show that there are no regular static multi-component electrovacuum black holes in higher dimensions. $\square$

A visualization of the shadow left behind by the set of null geodesics which come in from infinity and are trapped by a MP black hole with two identical components can be found in Fig. 4.7.1, from [463].

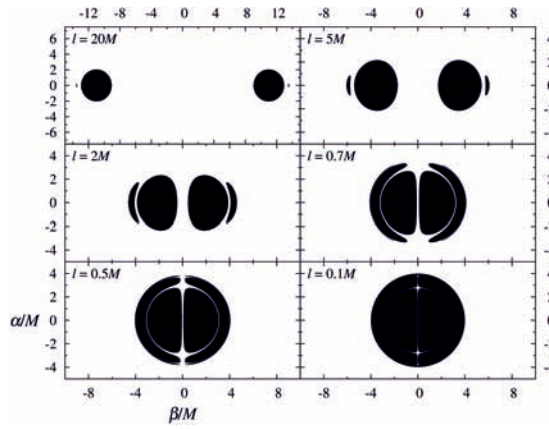


Fig. 4.7.1 The shadow of two MP black holes of identical mass for various distances l between the black holes. Reprinted with permission from [463]. © 2012 by the American Physical Society.

5

Further selected solutions

In previous chapters we presented the key notions associated with stationary black-hole spacetimes, as well as the minimal set of metrics needed to illustrate the basic features of the world of black-holes. In this chapter we present some further black holes, selected because of their physical and mathematical interest.

We start, in Section 5.1, with the Kerr–de Sitter/anti-de Sitter metrics, the cosmological counterparts of the Kerr metrics. Section 5.2 contains a description of the Kerr–Newman–de Sitter/anti-de Sitter metrics, which are the charged relatives of the metrics presented in Section 5.1. In Section 5.3 we analyse in detail the global structure of the Emparan–Reall ‘black rings’: these are five-dimensional black-hole spacetimes with $\mathbb{R} \times S^1 \times S^2$ -horizon topology. The Rasheed metrics of Section 5.4 provide an example of black holes arising in Kaluza–Klein theories. The Birmingham (which will also be referred to as Birmingham–Kottler, or Kottler) family of metrics, presented in Section 5.5, forms the most general class known of explicit static vacuum metrics with cosmological constant in all dimensions, with a wide range of horizon topologies.

5.1 The Kerr–de Sitter/anti-de Sitter metrics (with C. Ölz)

The Kerr–de Sitter (KdS) and the Kerr–anti-de Sitter (KAdS) metrics are solutions of the vacuum Einstein equations with a cosmological constant [80, 173]. They describe an axisymmetric stationary black hole solving the vacuum Einstein equations with a positive (KdS) or negative (KAdS) cosmological constant. We present here some of their global properties; see also [8, 81, 138]. Our presentation is based on unpublished joint work with Christa Ölz, with permission, cf. also [381]. The author of this book takes full responsibility for any mistakes here.

In *Boyer–Lindquist coordinates* the metric takes the form [80]¹

$$g = \Sigma \left(\frac{1}{\Delta_r} dr^2 + \frac{1}{\Delta_\theta} d\theta^2 \right) + \frac{\sin^2(\theta)}{\Sigma \Xi^2} \Delta_\theta (adt - (r^2 + a^2)d\varphi)^2 - \Delta_r \frac{1}{\Sigma \Xi^2} (dt - a \sin^2(\theta) d\varphi)^2, \quad (5.1.1)$$

where

$$\Sigma = r^2 + a^2 \cos^2(\theta), \quad (5.1.2)$$

$$\Delta_r = (r^2 + a^2) \left(1 - \frac{\Lambda}{3} r^2 \right) - 2mr, \quad (5.1.3)$$

¹The transformation between the coordinates used in [80] and the Boyer–Lindquist coordinates above is [81, p. 102]

$$\begin{aligned} \lambda = r, \quad \mu = a \cos(\theta), \quad \psi = \frac{1}{a\Xi} \varphi, \quad \chi + a^2 \psi = \frac{1}{\Xi} t, \\ p = a^2, \quad h = 1 - \frac{a^2 \Lambda}{3}, \quad e = 0, \quad q = 0. \end{aligned}$$

Note that Carter’s papers [80, 81] use the convention that *de Sitter* corresponds to $\Lambda < 0$; in other words, Carter’s Λ is the negative of ours. Carter explicitly assumes that $\Xi > 0$, but we will relax this assumptions in some of our considerations that follow.

$$\Delta_\theta = 1 + \frac{a^2 \Lambda}{3} \cos^2(\theta), \quad (5.1.4)$$

$$\Xi = 1 + \frac{a^2 \Lambda}{3}, \quad (5.1.5)$$

with $t \in \mathbb{R}$, $r \in \mathbb{R}$, and θ, φ being the standard coordinates on the sphere. Note that the metric functions are only well defined away from zeros of Σ and Δ_r , and the determinant vanishes at $\sin(\theta) = 0$.

When $m = 0$, g is the *de Sitter* ($\Lambda > 0$; ‘dS’) or the *anti-de Sitter* ($\Lambda < 0$; ‘AdS’) metric: Indeed, for $a = 0$, and after obvious renaming of coordinates, one obtains directly the standard form of the (A)dS metric in manifestly static coordinates (T, R, Θ, Φ) :

$$g_{(\Lambda)\text{dS}} = -(1 - \frac{\Lambda R^2}{3})dT^2 + \frac{1}{1 - \frac{\Lambda R^2}{3}}dR^2 + R^2(d\Theta^2 + \sin^2(\Theta)d\Phi^2). \quad (5.1.6)$$

For $a \neq 0 \neq \Xi$, an explicit coordinate transformation which brings the metric to the form (5.1.6) has been given in [81, p. 102]; see also [8, 247]:

$$\begin{aligned} T &= \frac{t}{\Xi}, \\ R^2 &= \frac{1}{\Xi} (r^2 \Delta_\theta + a^2 \sin^2(\theta)), \\ R \cos(\Theta) &= r \cos(\theta), \\ \Phi &= \varphi - a \frac{\Lambda}{3\Xi} t. \end{aligned} \quad (5.1.7)$$

If $a = 0 = \Lambda$ and $m \neq 0$ one obtains the Schwarzschild metric.

In the remainder of this section we will assume that $\Lambda \neq 0$.

When $m \neq 0$ but $a = 0$ one obtains the Schwarzschild–de Sitter or the Schwarzschild–anti-de Sitter metric:

$$g_{\text{S(A)dS}} = -(1 - \frac{\Lambda R^2}{3} - \frac{2m}{R})dT^2 + \frac{1}{1 - \frac{\Lambda R^2}{3} - \frac{2m}{R}}dR^2 + R^2(d\theta^2 + \sin^2(\theta)d\varphi^2). \quad (5.1.8)$$

The parameter m is related to the total mass; cf. (5.1.31).

From now on we assume $ma \neq 0$.

When $a < 0$, we can replace φ with $-\varphi$ to obtain a positive value of a , and therefore, to reduce the number of cases to be considered; we will assume that

$$a > 0. \quad (5.1.9)$$

With the same reasoning we require that

$$m > 0;$$

for $m < 0$ a new positive value for m is obtained by replacing r by $-r$.

The determinant of (5.1.1) is

$$\det(g) = -\frac{\Sigma^2}{\Xi^4} \sin^2(\theta) \quad (5.1.10)$$

and the metric is manifestly Lorentzian at $r = 0$, which shows that (5.1.1) defines a Lorentzian metric on any connected set on which the metric components remain bounded, i.e., away from zeros of Δ_r , the ‘ring singularity’ at $\Sigma = 0$, and the trivial spherical-coordinates singularity at $\theta \in \{0, \pi\}$; we will return to the last two issues shortly.

The inverse metric reads

$$\begin{aligned} g^{\mu\nu} \partial_\mu \partial_\nu = & -\frac{\Xi^2}{\Delta_r \Delta_\theta \Sigma} \left((a^2 + r^2)^2 \Delta_\theta - a^2 \sin^2(\theta) \Delta_r \right) \partial_t^2 \\ & - 2 \frac{\Xi^2}{\Delta_r \Delta_\theta \Sigma} a \left((a^2 + r^2) \Delta_\theta - \Delta_r \right) \partial_t \partial_\varphi + \frac{\Delta_r}{\Sigma} \partial_r^2 \\ & + \frac{\Xi^2}{\Delta_r \Delta_\theta \Sigma \sin^2(\theta)} \left(\Delta_r - a^2 \sin^2(\theta) \Delta_\theta \right) \partial_\varphi^2 + \frac{\Delta_\theta}{\Sigma} \partial_\theta^2. \end{aligned} \quad (5.1.11)$$

Note that

$$g^{tt} = \frac{g_{rr} g_{\theta\theta} g_{\varphi\varphi}}{\det(g)} = -\frac{\Xi^4}{\Delta_\theta} \times \frac{1}{\Delta_r} \times \frac{g_{\varphi\varphi}}{\sin^2(\theta)}$$

and

$$\text{sgn}(g^{rr}) = \text{sgn}(\Delta_r),$$

so either r or $-r$ is a time function when $\Delta_r < 0$, and t or $-t$ is a time function when $\Delta_r > 0$ and $g_{\varphi\varphi} > 0$. In the region where ∂_φ is timelike,

$$\{(a^2 + r^2) \Delta_\theta - a^2 \Delta_r \sin^2(\theta) < 0\},$$

which is non-empty for $\sin(\theta) \neq 0$, the orbits of the Killing vector ∂_φ form closed timelike curves.

The character of the principal orbits of the isometry group $\mathbb{R} \times U(1)$, i.e. those orbits which are two-dimensional submanifolds, is determined by the sign of the determinant

$$\det \begin{pmatrix} g_{tt} & g_{t\varphi} \\ g_{t\varphi} & g_{\varphi\varphi} \end{pmatrix} = -\frac{\Delta_r \Delta_\theta}{\Xi^4} \sin^2(\theta). \quad (5.1.12)$$

Therefore, for $\theta = 0$ the orbits are either null or one-dimensional, while for $\theta \neq 0$ the orbits are timelike in the regions where $\Delta_r > 0$, spacelike where $\Delta_r < 0$, and null where $\Delta_r = 0$. Strictly speaking, the set where Δ_r vanishes will only become part of the spacetime after an extension across the zeros of Δ_r has been constructed. This is done in Section 5.1.6, p. 208, with the connected components of the zero set of Δ_r becoming Killing horizons or bifurcate Killing horizons.

As discussed in the paragraph following (5.2.39), p. 221, existence of well-behaved spacelike hypersurfaces requires that

$$a^2 \Lambda > -3 \quad \Longleftrightarrow \quad \Xi > 0, \quad (5.1.13)$$

which will be assumed in what follows unless explicitly indicated otherwise.

The following identities are useful when studying the metrics (5.1.1):

$$g_{\varphi\varphi} + a \sin^2(\theta) g_{t\varphi} = \frac{\Delta_\theta (a^2 + r^2) \sin^2(\theta)}{\Xi^2}, \quad (5.1.14)$$

$$g_{t\varphi} + a \sin^2(\theta) g_{tt} = -\frac{a \Delta_\theta \sin^2(\theta)}{\Xi^2}, \quad (5.1.15)$$

$$a g_{\varphi\varphi} + (r^2 + a^2) g_{t\varphi} = \frac{a \Delta_r \sin^2(\theta)}{\Xi^2}, \quad (5.1.16)$$

$$a g_{t\varphi} + (r^2 + a^2) g_{tt} = -\frac{\Delta_r}{\Xi^2}, \quad (5.1.17)$$

and

$$g_{tt} g_{\varphi\varphi} - g_{t\varphi}^2 = -\frac{\Delta_r \Delta_\theta \sin^2(\theta)}{\Xi^4}. \quad (5.1.18)$$

5.1.1 Asymptotic behaviour

The K(A)dS metrics possess a boundary at infinity à la Penrose. We recall for the convenience of the reader (cf. Section 3.1) that a spacetime $(\mathcal{M}, g)$ admits a *conformal boundary at infinity* $\mathcal{I}$ if there exists a spacetime with non-empty boundary $(\tilde{\mathcal{M}}, \tilde{g})$ such that

1. $\mathcal{M}$ is the interior of $\tilde{\mathcal{M}}$ and $\mathcal{I} = \partial\tilde{\mathcal{M}}$, thus $\tilde{\mathcal{M}} = \mathcal{M} \cup \mathcal{I}$; and
2. There exists a function $\Omega \in C^\infty(\tilde{\mathcal{M}})$ such that (a) $\tilde{g} = \Omega^2 g$ on $\mathcal{M}$, (b) $\Omega > 0$ on $\mathcal{M}$, and (c) $\Omega = 0$ and $d\Omega \neq 0$ on $\mathcal{I}$.

This applies to the K(A)dS metric by choosing

$$\Omega = \sqrt{y^2}, \text{ where } y := \frac{1}{r}$$

(recall that we allow r to be negative). Then

$$\begin{aligned} \tilde{g} = \Omega^2 g = y^2 g = & -3 \frac{1 + a^2 y^2 \cos^2(\theta)}{-3a^2 y^4 + y^2(a^2 \Lambda - 3) + \Lambda + 6my^3} dy^2 \\ & + \frac{3a^2 \Delta_\theta y^4 \sin^2(\theta) - 3a^2 y^4 + y^2(a^2 \Lambda - 3) + \Lambda + 6my^3}{3\Xi^2(1 + a^2 y^2 \cos^2(\theta))} dt^2 \\ & - 2a \sin^2(\theta) \frac{a^2 \Lambda y^2(a^2 y^2 + 1) \cos^2(\theta) + a^2 \Lambda y^2 + \Lambda + 6my^3}{3\Xi^2(1 + a^2 y^2 \cos^2(\theta))} dt d\varphi \\ & + \sin^2(\theta) \left(\frac{a^4 \Lambda y^2 + a^2 y^2 \cos^2(\theta)(3\Xi(a^2 y^2 + 1) - 6my)}{3\Xi^2(1 + a^2 y^2 \cos^2(\theta))} \right. \\ & \left. + \frac{a^2(\Lambda + 6my^3 + 3y^2) + 3}{3\Xi^2(1 + a^2 y^2 \cos^2(\theta))} \right) d\varphi^2 + \frac{1 + a^2 y^2 \cos^2(\theta)}{\Delta_\theta} d\theta^2. \end{aligned} \quad (5.1.19)$$

All the metric coefficients can now be analytically extended across, and to a neighbourhood of, the set

$$\mathcal{I} := \{y = 0\}.$$

The limit $y \rightarrow 0$ gives

$$\lim_{y \rightarrow 0} y^2 g = -\frac{3}{\Lambda} dy^2 + \frac{\Delta_\theta \sin^2(\theta)}{\Xi^2} d\varphi^2 + \frac{\Lambda}{3\Xi^2} (dt - a \sin^2(\theta) d\varphi)^2 + \frac{1}{\Delta_\theta} d\theta^2, \quad (5.1.20)$$

which is manifestly Lorentzian at $\{y = 0\}$, and hence in neighbourhood of $\{y = 0\}$. Equation (5.1.20) shows that the metric $\tilde{g}$ induces on $\mathcal{I}$ a metric which is

$$\begin{cases} \text{Riemannian, for } \Lambda > 0; \\ \text{Lorentzian, for } \Lambda < 0, \end{cases} \text{ the conformal boundary } \mathcal{I} \text{ is } \begin{cases} \text{spacelike, if } \Lambda > 0; \\ \text{timelike, if } \Lambda < 0. \end{cases}$$

In [81, p. 102] it is emphasized that the K(A)dS metrics are *asymptotically* (A)dS, in the sense that the metrics approach the (A)dS metric as r goes to infinity. This can be immediately inferred from (5.1.20), where it is seen that the metric at $y = 0$ does not depend upon m , and hence coincides there with the corresponding conformal rescaling of the (A)dS metric. An explicit construction proceeds e.g. through the Kerr–Schild coordinates: Following [8, 226], we use the coordinate transformation

$$d\tau = \frac{1}{\Xi} dt + \frac{2mr}{(1 - \frac{r^2 \Lambda}{3}) \Delta_r} dr,$$

$$d\phi = d\varphi - \frac{a\Lambda}{3\Xi} dt + \frac{2mr}{(r^2 + a^2)\Delta_r} a dr \quad (5.1.21)$$

to obtain

$$g_{\text{K(A)dS}} = g_{\text{(A)dS}} + \frac{2mr}{\Sigma} (k_\mu dx^\mu)^2, \quad (5.1.22)$$

with

$$\begin{aligned} g_{\text{(A)dS}} = & -\frac{\left(1 - \frac{r^2\Lambda}{3}\right) \Delta_\theta}{\Xi} d\tau^2 + \frac{\Sigma}{\left(1 - \frac{r^2\Lambda}{3}\right) (r^2 + a^2)} dr^2 + \frac{\Sigma}{\Delta_\theta} d\theta^2 \\ & + \frac{(r^2 + a^2) \sin^2(\theta)}{\Xi} d\phi^2, \\ k_\mu dx^\mu = & \frac{\Delta_\theta}{\Xi} d\tau + \frac{\Sigma}{\left(1 - \frac{r^2\Lambda}{3}\right) (r^2 + a^2)} dr - \frac{a \sin^2(\theta)}{\Xi} d\phi. \end{aligned} \quad (5.1.23)$$

The vector field k^μ is null for both g and $g_{\text{(A)dS}}$, as seen by a direct calculation, and tangent to a null geodesic congruence, as noted in [226]. The key point here is that the $k_\mu dx^\mu$ contribution to the metric is subleading in (5.1.22) at large distances.

As the notation suggests, the metric $g_{\text{(A)dS}}$ as given by (5.1.23) is the (A)dS metric in unusual coordinates, which can be verified by using the coordinate transformation [81, 226]

$$R^2 = \frac{r^2 \Delta_\theta + a^2 \sin^2(\theta)}{\Xi}, \quad (5.1.24)$$

$$R^2 \sin^2(\Theta) = \frac{r^2 + a^2}{\Xi} \sin^2(\theta), \quad (5.1.25)$$

$$T = \tau, \quad (5.1.26)$$

$$\Phi = \phi. \quad (5.1.27)$$

This coordinate transformation brings (5.1.23) to the standard form (5.1.6) of the (A)dS metric. In order to invert (5.1.24)–(5.1.27) it is convenient to set

$$\ell := \sqrt{-\frac{3}{\Lambda}}, \quad \alpha := \frac{a}{\ell}, \quad (5.1.28)$$

keeping in mind (cf. (5.1.13)) that the existence of well-behaved spacelike hypersurfaces requires that $\alpha^2 < 1$. Equations (5.1.24)–(5.1.25) can be used to give

$$\sin(\theta) = \sqrt{\frac{(1 - \alpha^2) r^2 \sin^2(\Theta)}{-\alpha^2 (\ell^2 + r^2) \sin^2(\Theta) + \alpha^2 \ell^2 + r^2}}. \quad (5.1.29)$$

One can then solve (5.1.24) to obtain

$$\begin{aligned} 2r^2 = & R^2(1 - \alpha^2 \sin^2(\Theta)) - \alpha^2 \ell^2 \\ & + \sqrt{(\alpha^2 \ell^2 + \alpha^2 R^2 \sin^2(\Theta) - R^2)^2 + 4\alpha^2 \ell^2 R^2 \cos^2(\Theta)}, \end{aligned} \quad (5.1.30)$$

where we have chosen the solution for which $r \rightarrow_{R \rightarrow \infty} \infty$. The above formulae are handy when one wishes to calculate the Hamiltonian mass, say M , of the Kerr–AdS metric [247]

$$M = \frac{16\pi m}{(1 - \alpha^2)^2} = \frac{16\pi m \ell^2}{(\ell^2 - a^2)^2} = \frac{16\pi m}{\left(1 - \frac{\Lambda a^2}{3}\right)^2}, \quad (5.1.31)$$

which has the same sign as m .

5.1.2 The axis

The vanishing of the determinant (5.1.10) at $\sin(\theta) = 0$ shows that the metric in Boyer–Lindquist coordinates degenerates at the axis. This is the usual spherical-coordinates singularity, which can be cured by introducing new coordinates $(\bar{x}, \bar{y})$ defined as

$$\bar{x} = \sin(\theta) \cos(\varphi), \quad \bar{y} = \sin(\theta) \sin(\varphi). \quad (5.1.32)$$

(This will be seen to introduce a singularity at $\bar{x}^2 + \bar{y}^2 = 1$, which is irrelevant here as, at this stage, we are only interested in what is happening for small values of $\bar{x}^2 + \bar{y}^2$.)

We have

$$d\varphi = \frac{\bar{x}d\bar{y} - \bar{y}d\bar{x}}{\bar{x}^2 + \bar{y}^2},$$

$$d\theta^2 + \sin^2(\theta)d\varphi^2 = d\bar{x}^2 + d\bar{y}^2 + \frac{(\bar{x}d\bar{x} + \bar{y}d\bar{y})^2}{1 - \bar{x}^2 - \bar{y}^2},$$

which brings the metric (5.1.1) to the form

$$g = - \frac{\Delta_r - a^2(\bar{x}^2 + \bar{y}^2)\Delta_\theta}{\Xi^2\Sigma} dt^2 + 2a \frac{\Delta_r - (a^2 + r^2)\Delta_\theta}{\Xi^2\Sigma} (\bar{x}d\bar{y} - \bar{y}d\bar{x}) dt$$

$$+ a^2 \frac{6mr\Delta_\theta + (3 - \Lambda r^2)\Xi\Sigma}{3\Delta_\theta\Xi^2\Sigma} (\bar{x}d\bar{y} - \bar{y}d\bar{x})^2 + \frac{\Sigma}{\Delta_r} dr^2$$

$$+ \frac{\Sigma}{\Delta_\theta} \left(d\bar{x}^2 + d\bar{y}^2 + \frac{(\bar{x}d\bar{x} + \bar{y}d\bar{y})^2}{1 - \bar{x}^2 - \bar{y}^2} \right),$$

with

$$\Sigma = r^2 + a^2(1 - \bar{x}^2 - \bar{y}^2),$$

$$\Delta_\theta = 1 + \frac{a^2\Lambda}{3}(1 - \bar{x}^2 - \bar{y}^2).$$

The metric coefficients are smooth for $\bar{x}^2 + \bar{y}^2 < 1$ as long as one stays away from the ‘ring singularity’ $\Sigma = 0$ (which we are about to discuss), and the Killing horizons at $\Delta_r = 0$.

We have

$$\det(g) = - \frac{\Sigma^2}{(1 - \bar{x}^2 - \bar{y}^2)\Xi^4},$$

which makes clear the non-degeneracy of the metric tensor in the region of current interest.

5.1.3 The ‘singular ring’ $\Sigma = 0$

The set $\{\Sigma = 0\}$ corresponds to a geometric singularity in the metric. To see this, note that on the equatorial plane $\cos(\theta) = 0$ we have

$$g_{tt} = \frac{6m - r(3 - a^2\Lambda) + r^3\Lambda}{3r\Xi^2}.$$

It follows that the norm of the Killing vector ∂_t blows up as the set $\{\Sigma = 0\}$ is approached (where $r = 0$), along some directions when $m \neq 0$. This is not possible by Theorem 4.4.2 if the metric can be continued *throughout* this set in a C^2 manner.

To show nonexistence of any extensions, even *local*, across the singular set requires inspecting the curvature tensor. Indeed, a calculation shows that the scalar invariant

$$(R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta})^2 + (\epsilon_{\alpha\beta\gamma\delta}R^{\alpha\beta}{}_{\mu\nu}R^{\mu\nu\gamma\delta})^2 \quad (5.1.33)$$

is unbounded on *any* curve accumulating at the set $\{\Sigma = 0\}$ [134].

Whether the set $\{\Sigma = 0\}$ has the topology of a ring, and if so in which sense, is discussed in [134].

Suppose that Δ_r has no zeros; then the singular set

$$\{\Sigma = 0\} = \{r = \cos(\theta) = 0\}$$

is *not* shielded by horizons, and so the metric is ‘nakedly singular’. We will thus only be interested in those values of the parameters for which Δ_r has zeros. This leads to restrictions on m when $\Lambda \leq 0$, but does not exclude any values of m and a when $\Lambda > 0$.

5.1.4 Killing horizons

The metric coefficients in Boyer–Lindquist coordinates are singular at $\{\Delta_r = 0\}$. We will show in Section 5.1.6 how to extend the metric across these singularities. Zeros of Δ_r will give rise to Killing horizons in the extended spacetime.

In the remainder of this section, by *horizon* we will mean a *Killing horizon* $\mathcal{N}_\xi$ associated with a Killing vector ξ . Recall that this is defined as *an embedded null hypersurface which coincides with a connected component of the set*

$$\mathcal{N}_\xi := \{p : g(\xi, \xi)(p) = 0, \xi(p) \neq 0\}.$$

In order to find the location of the Killing horizons, we note that all Killing vector fields ξ of the K(A)dS metric are of the form

$$\xi = \bar{a}\partial_t + \bar{b}\partial_\varphi,$$

where $\bar{a}$ and $\bar{b}$ are constants. Now, since ∂_t and ∂_φ are commuting Killing vectors, we have

$$\mathcal{L}_{\partial_t}g = 0 = \mathcal{L}_{\partial_\varphi}g, \mathcal{L}_{\partial_t}\xi = 0 = \mathcal{L}_{\partial_\varphi}\xi \text{ and } \mathcal{L}_{\partial_t}g(\xi, \xi) = 0 = \mathcal{L}_{\partial_\varphi}g(\xi, \xi).$$

This implies that any Killing horizon $\mathcal{N}_\xi$ is invariant under time translations and rotations around the z -axis, with ∂_t and ∂_φ being tangent to $\mathcal{N}_\xi$. It follows that any linear combination

$$\bar{X}^\mu = \bar{\alpha}\partial_t + \bar{\beta}\partial_\varphi, \quad \bar{\alpha}, \bar{\beta} \in \mathbb{R},$$

is tangent to $\mathcal{N}_\xi$. A necessary and sufficient condition for the existence of such a linear combination is that the metric on the orbits of the Killing vectors is degenerate or, equivalently, the determinant (5.1.12) is zero. We conclude from this last equation that the set

$$\{g(\xi, \xi) = 0\}$$

will be part of a null hypersurface if and only if

$$\Delta_r = 0;$$

here we have anticipated the fact that the spacetime can be extended so that points with vanishing Δ_r are included in the extended manifold.

A calculation shows that the Killing vectors which are null on the set where Δ_r vanishes are of the form

$$\xi = c \left(\partial_t + \frac{a}{a^2 + r_h^2} \partial_\varphi \right),$$

where $c \in \mathbb{R}$. Following [8, 226] we choose $c = \Xi$, so that the Killing vector defining the Killing horizon at $\Delta_r = 0$ is

$$\xi = \Xi \partial_t + \Omega_h \partial_\varphi; \quad (5.1.34)$$

see also [131, Proposition 4.3] for a justification of this normalization based on Hamiltonian considerations.

The constant

$$\Omega_h = \frac{a\Xi}{a^2 + r_h^2}$$

can be thought of as the *angular velocity* of $\mathcal{N}_\xi$, though maybe Ω_h/Ξ would be a better candidate for the notion.

Recall that the surface gravity κ of $\mathcal{N}_\xi$ is defined as

$$\nabla^\mu (\xi^\nu \xi_\nu) |_{\mathcal{N}_\xi} = -2\kappa \xi^\mu.$$

With the choice (5.1.34), the surface gravity κ of the horizon at $r = r_h$ can be calculated as in (4.6.44), p. 186, and equals [8, 226]

$$\kappa = \frac{1 - r_h^2 \frac{\Lambda}{3}}{4mr_h} \Delta'_r(r_h). \quad (5.1.35)$$

We see that degenerate horizons, where $\kappa = 0$, occur at zeros of Δ_r of order two or higher.

5.1.5 The number and nature of Killing horizons

In order to understand the properties of the metric (5.1.1) one needs to understand the number and character of zeros of the metric function

$$\begin{aligned} \Delta_r &= (r^2 + a^2) \left(1 - \frac{\Lambda r^2}{3} \right) - 2mr \\ &= -\frac{\Lambda}{3} r^4 + \left(1 - \frac{a^2 \Lambda}{3} \right) r^2 - 2mr + a^2. \end{aligned} \quad (5.1.36)$$

Indeed, we have just seen that these zeros determine the location and character of the Killing horizons. This analysis is carried out in Sections 5.2.3 and 5.2.4 for the charged counterparts of the K(A)dS metrics; it suffices to set $q^2 \equiv |\Xi|e^2 = 0$ in the results there. We simply note that the key parameters which distinguish the various cases are

$$\beta = \left| \frac{3m}{\Lambda \sigma^3} \right| \geq 0, \quad \gamma = \frac{3a^2}{|\Lambda| \sigma^4} > 0, \quad (5.1.37)$$

with

$$\sigma := \sqrt{\left| \frac{1 - \frac{\Lambda}{3} a^2}{\Lambda} \right|}. \quad (5.1.38)$$

Figure 5.1.1 shows the dependence of the number and order of zeros of Δ_r with $\Lambda > 0$ on β and γ .

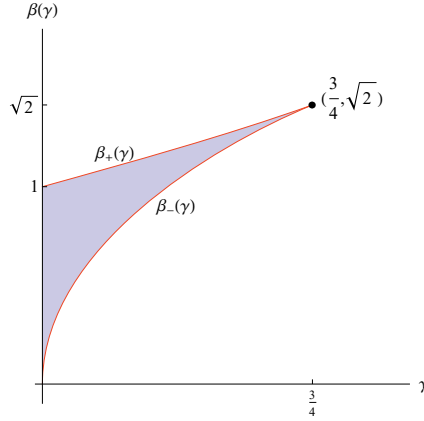


Fig. 5.1.1 The polynomial Δ_r has four simple zeros in the shaded region. Two of them merge to $x_h^{(1)} < x_h^{(2)} < x_h^{(3)} = x_h^{(4)}$ at β_+ and to $x_h^{(1)} < x_h^{(2)} = x_h^{(3)} < x_h^{(4)}$ at β_- ; all three positive zeros merge at the point $(3/4, \sqrt{2})$, which leads to one triple and one simple zero there. Here $x_h^{(i)} = r_h^{(i)}/\sigma$. In the unshaded region outside the graphs Δ_r has always two simple zeros. The parameters β , γ , and σ are defined in (5.1.37)–(5.1.38). Figure by Christa Ölz.

5.1.6 Extensions across Killing horizons

As already hinted at, the K(A)dS metrics can be analytically extended across the zeros of Δ_r , which become Killing horizons in the extended spacetime. The main aim of this section is to justify this. Here we concentrate on the local aspects of the problem; the question of the global structure of the extended spacetime will be addressed in Sections 7.4.1–7.4.3, pp. 293 and following.

Throughout this section both positive and negative values of the cosmological constant Λ are allowed as long as $\Xi > 0$.

Recall that the coefficient g_{rr} in the metric in Boyer–Lindquist coordinates (5.1.1) becomes infinite at zeros of $\Delta_r = 0$. In order to show that adjacent regions can be bridged over zeros of Δ_r , we need to find suitable Eddington–Finkelstein-type and Kruskal–Szekeres-type coordinate transformations.

In the remainder of this section we will ignore the term $\frac{\Sigma}{\Delta_\theta} d\theta^2$ of the metric in Eq. (5.1.1) since it is already regular. We therefore consider a metric, denoted by g_3 , of the form

$$g_3 = g_{tt} dt^2 + g_{rr} dr^2 + 2g_{t\varphi} dt d\varphi + g_{\varphi\varphi} d\varphi^2, \quad (5.1.39)$$

where

$$g_{tt} = -\frac{\Delta_r - \Delta_\theta a^2 \sin^2(\theta)}{\Sigma \Xi^2}, \quad (5.1.40)$$

$$g_{rr} = \frac{\Sigma}{\Delta_r}, \quad (5.1.41)$$

$$g_{t\varphi} = a \sin^2(\theta) \frac{\Delta_r - \Delta_\theta (r^2 + a^2)}{\Xi^2 \Sigma}, \quad (5.1.42)$$

$$g_{\varphi\varphi} = -\sin^2(\theta) \frac{a^2 \sin^2(\theta) \Delta_r - \Delta_\theta (r^2 + a^2)^2}{\Xi^2 \Sigma}. \quad (5.1.43)$$

Eddington–Finkelstein-type coordinates. We use an ansatz similar to the one for the Kerr metric,

$$dv = dt + \frac{n_1(r)}{\Delta_r} dr, \quad (5.1.44)$$

$$d\psi = d\varphi + \frac{n_2(r)}{\Delta_r} dr, \quad (5.1.45)$$

where the functions $n_1(r)$ and $n_2(r)$ will be chosen so that the coefficient of dr^2 vanishes. In these coordinates the metric g_3 of (5.1.39) takes the form

$$\begin{aligned} g_3 = & \frac{a^2 \sin^2(\theta) \Delta_\theta - \Delta_r}{\Xi^2 \Sigma} dv^2 - 2a \sin^2(\theta) \frac{(a^2 + r^2) \Delta_\theta - \Delta_r}{\Xi^2 \Sigma} dv d\psi \\ & + \sin^2(\theta) \frac{\Delta_\theta (a^2 + r^2)^2 - a^2 \sin^2(\theta) \Delta_r}{\Xi^2 \Sigma} d\psi^2 \\ & - 2 \frac{\Delta_\theta a \sin^2(\theta) \tilde{p}(r) - \Delta_r (n_1(r) - a n_2(r) \sin^2(\theta))}{\Delta_r \Xi^2 \Sigma} dr dv \\ & + 2 \sin^2(\theta) \frac{\Delta_\theta (a^2 + r^2) \tilde{p}(r) - \Delta_r a (n_1(r) - a n_2(r) \sin^2(\theta))}{\Delta_r \Xi^2 \Sigma} dr d\psi \\ & + \frac{\Delta_\theta \sin^2(\theta) \tilde{p}(r)^2 + \Delta_r \tilde{q}(r, \theta)}{\Delta_r^2 \Xi^2 \Sigma} dr^2, \end{aligned} \quad (5.1.46)$$

where

$$\tilde{p}(r) := a n_1(r) - (a^2 + r^2) n_2(r), \quad (5.1.47)$$

$$\tilde{q}(r, \theta) := \Xi^2 \Sigma^2 - (n_1(r) - a n_2(r) \sin^2(\theta))^2. \quad (5.1.48)$$

As is clear from (5.1.46), setting the coefficient of dr^2 zero by enforcing $\tilde{p}(r) = 0$ and $\tilde{q}(r, \theta) = 0$ leads to coefficients of $dr dv$ and $dv d\psi$, which are smooth near zeros of Δ_r . The desired vanishing of $\tilde{p}$ and q is achieved by choosing $n_1(r)$ and $n_2(r)$ as

$$n_1(r) = (a^2 + r^2) \Xi, \quad n_2(r) = a \Xi.$$

(Note that the pair $(-n_1(r), -n_2(r))$ leads to $\tilde{p}(r) = 0 = \tilde{q}(r, \theta)$ as well.) With this choice we obtain

$$\begin{aligned} g_3 = & \frac{-\Delta_r + a^2 \sin^2(\theta) \Delta_\theta}{\Xi^2 \Sigma} dv^2 + \frac{2}{\Xi} dr dv - 2a \sin^2(\theta) \frac{(a^2 + r^2) \Delta_\theta - \Delta_r}{\Xi^2 \Sigma} dv d\psi \\ & - \frac{2a \sin^2(\theta)}{\Xi} dr d\psi + \sin^2(\theta) \frac{(a^2 + r^2)^2 \Delta_\theta - a^2 \Delta_r \sin^2(\theta)}{\Xi^2 \Sigma} d\psi^2. \end{aligned}$$

The coefficients of the metric are all regular now, except at the ring singularity $\Sigma = 0$. The determinant of the full metric (5.1.1) in the (v, r, θ, ψ) -coordinates is

$$\det(g) = -\frac{\Sigma^2 \sin^2(\theta)}{\Xi^4}. \quad (5.1.49)$$

As already pointed out, we could have used

$$du = dt - \frac{n_1(r)}{\Delta_r} dr = dt - \frac{(a^2 + r^2) \Xi}{\Delta_r} dr, \quad (5.1.50)$$

$$d\tilde{\psi} = d\varphi - \frac{n_2(r)}{\Delta_r} dr = d\varphi - \frac{a \Xi}{\Delta_r} dr, \quad (5.1.51)$$

instead of (5.1.44)–(5.1.45). In these new coordinates, g_3 takes the form

$$g_3 = \frac{-\Delta_r + a^2 \sin^2(\theta) \Delta_\theta}{\Xi^2 \Sigma} du^2 - \frac{2}{\Xi} dr du - 2a \sin^2(\theta) \frac{(a^2 + r^2) \Delta_\theta - \Delta_r}{\Xi^2 \Sigma} du d\tilde{\psi} \\ + \frac{2a \sin^2(\theta)}{\Xi} dr d\tilde{\psi} + \sin^2(\theta) \frac{(a^2 + r^2)^2 \Delta_\theta - a^2 \Delta_r \sin^2(\theta)}{\Xi^2 \Sigma} d\tilde{\psi}^2,$$

with the determinant of g given again by (5.1.49).

The above applies regardless of whether $\Delta'(r_h)$ vanishes.

Kruskal–Szekeres-type coordinates. In order to include the bifurcation sphere of a non-degenerate horizon at a *first-order zero* $r = r_h$ of Δ_r we make the following ansatz for Kruskal–Szekeres-type coordinates,

$$u = t - f(r) \iff du = dt - f' dr, \quad (5.1.52)$$

$$v = t + f(r) \iff dv = dt + f' dr, \quad (5.1.53)$$

$$\psi = \varphi + \alpha t \iff d\psi = d\varphi + \alpha dt, \quad (5.1.54)$$

with a constant α to be determined. Equivalently,

$$dt = \frac{du + dv}{2}, \\ dr = \frac{dv - du}{2f'}, \\ d\varphi = d\psi - \frac{\alpha}{2} (du + dv).$$

Under this transformation, the metric (5.1.39) becomes

$$g_3 = \frac{-f'^2 \Delta_r^2 + \Xi^2 \Sigma^2 + f'^2 \Delta_r \sin^2(\theta) s(r, \theta)}{4f'^2 \Delta_r \Xi^2 \Sigma} (du^2 + dv^2) \\ + \frac{-f'^2 \Delta_r^2 - \Xi^2 \Sigma^2 + f'^2 \Delta_r \sin^2(\theta) s(r, \theta)}{2f'^2 \Delta_r \Xi^2 \Sigma} du dv \\ + \sin^2(\theta) \frac{a \Delta_r - (a^2 + r^2) \Delta_\theta b(r) + \alpha a^2 \Delta_r \sin^2(\theta)}{\Xi^2 \Sigma} (du d\psi + dv d\psi) \\ + \frac{(a^2 + r^2)^2 \Delta_\theta - a^2 \Delta_r \sin^2(\theta)}{\Xi^2 \Sigma} \sin^2(\theta) d\psi^2, \quad (5.1.55)$$

where

$$b(r) := a + (a^2 + r^2) \alpha \\ s(r, \theta) := -\alpha^2 a^2 \Delta_r \sin^2(\theta) - 2 \alpha a \Delta_r + b(r)^2 \Delta_\theta. \quad (5.1.56)$$

Choosing α and $f(r)$ appropriately will get rid of the first-order zero at $r = r_h$ in the denominator of the coefficients of du^2 , dv^2 , and $du dv$. Inspection of (5.1.55) shows that this can be achieved by choosing $f' \sim \Delta_r^{-1}$, and requiring $b(r)$ to vanish at the horizon. Moreover, and perhaps somewhat unexpectedly, we will gain an extra multiplicative factor of Δ_r in some coefficients, as needed for the final coordinate transformation.

Let us pass to the details of the above. We start with the coefficient of $du d\psi$ of g_3 , which is regular everywhere away from the ring singularity at $\Sigma = 0$. Choosing

$$\alpha = -\frac{a}{r_h^2 + a^2} \quad (5.1.57)$$

introduces a first-order zero at $r = r_h$ in the term $b(r) = a + \alpha(a^2 + r^2)$, so the coefficients of both $du d\psi$ and $dv d\psi$ have a first-order zero at $r = r_h$.

We continue by choosing

$$f'(r) = \frac{\chi(r)}{\Delta_r}, \quad (5.1.58)$$

where $\chi(r)$ is a smooth function of r which is bounded away from zero near $r = r_h$, leaving the coefficients of du^2 , dv^2 , and $du dv$ nonsingular at simple zeros of Δ_r . (We will see shortly that

$$\chi(r) = a^2 + r^2 \quad (5.1.59)$$

is fit for purpose, but other functions are possible as long as (5.1.68) holds.) The metric tensor g_3 becomes now

$$\begin{aligned} g_3 = & \frac{\Xi^2 \Sigma^2 \Delta_r - \chi^2 \Delta_r + \chi^2 a^2 \sin^2(\theta) \tilde{s}(r, \theta)}{4\chi^2 \Xi^2 \Sigma} (du^2 + dv^2) \\ & - \frac{\Xi^2 \Sigma^2 \Delta_r + \chi^2 \Delta_r - \chi^2 a^2 \sin^2(\theta) \tilde{s}(r, \theta)}{2\chi^2 \Xi^2 \Sigma} du dv \\ & + \frac{\left(1 - \frac{a^2 \sin^2(\theta)}{a^2 + r_h^2}\right) \Delta_r - (a^2 + r^2) \left(1 - \frac{a^2 + r^2}{a^2 + r_h^2}\right) \Delta_\theta}{\Xi^2 \Sigma} a \sin^2(\theta) (du d\psi + dv d\psi) \\ & + \frac{(a^2 + r^2)^2 \Delta_\theta - a^2 \sin^2(\theta) \Delta_r}{\Xi^2 \Sigma} \sin^2(\theta) d\psi^2, \end{aligned} \quad (5.1.60)$$

with

$$\tilde{s}(r, \theta) := \left(\frac{1}{a^2 + r_h^2} \left(2 - \frac{a^2 \sin^2(\theta)}{a^2 + r_h^2} \right) + \left(\frac{r_h^2 - r^2}{a^2 + r_h^2} \right)^2 \frac{\Delta_\theta}{\Delta_r} \right) \Delta_r. \quad (5.1.61)$$

(Note that, near $r = r_h$, this is a product of a smooth function with Δ_r .)

As such, the determinant of g in the (u, v, Θ, φ) coordinates vanishes at $r = r_h$. This can be seen by multiplying the determinant of g in the old coordinates, as given by (5.1.10), p. 201, by the inverse square of the determinant of the Jacobi matrix of the coordinate transformation. Since the determinant of the Jacobi matrix J equals

$$\det(J) = 2f'(r) = \frac{2\chi(r)}{\Delta_r},$$

the determinant, say $\det(\bar{g})$, of the metric (5.1.1) in the new coordinates is given by

$$\det(\bar{g}) = -\sin^2(\theta) \frac{\Sigma^2}{\Xi^4} \frac{\Delta_r^2}{4\chi(r)^2},$$

indeed vanishing at all zeros of Δ_r . Introducing, again near $r = r_h$, a new set of coordinates

$$\begin{aligned} \bar{u} = -\exp(-cu) & \iff du = \frac{\exp(cu)}{c} d\bar{u}, \\ \bar{v} = \exp(cv) & \iff dv = \frac{\exp(-cv)}{c} d\bar{v}, \end{aligned} \quad (5.1.62)$$

with an addition to the Jacobi determinant

$$\det(\bar{J}) = c^2 \exp(c(v - u)) = c^2 \exp(2cf(r)),$$

one obtains, still denoting by $\det(\bar{g})$ the determinant of the metric tensor in the new coordinates,

$$\det(\bar{g}) = -\sin^2(\theta) \frac{\Sigma^2}{\Xi^4} \frac{\Delta_r^2}{4\chi(r)^2} \frac{\exp(-4cf(r))}{c^4}.$$

Let us show that this will be nonzero near $r = r_h$ if c is chosen as

$$c = \lim_{r \rightarrow r_h} \frac{\Delta_r}{2\chi(r)|r - r_h|} =: \frac{1}{2A}. \quad (5.1.63)$$

Indeed, integrating the equation for f , for r near r_h one obtains

$$f(r) = A \ln|r - r_h| + h(r), \quad (5.1.64)$$

where h is smooth near r_h . This gives

$$\exp(-4cf(r)) = |r - r_h|^{-4cA} \exp(-4ch(r)) = (r - r_h)^{-2} \exp(-4ch(r)),$$

which is precisely what is needed to cancel the second-order zero at $r = r_h$ in the determinant of the metric in the new coordinates,

$$\det(\bar{g}) = -4A^4 \sin^2(\theta) \frac{\Sigma^2}{\Xi^4 \chi^2(r)} \frac{\Delta_r^2}{(r - r_h)^2} \exp\left(-\frac{2h(r)}{A}\right). \quad (5.1.65)$$

It remains to check that all metric functions are smooth in the new coordinates. Under (5.1.62) the metric becomes

$$\begin{aligned} g_3 = & A^2 \frac{\Xi^2 \Sigma^2 \Delta_r - \chi^2 \Delta_r + \tilde{s}(r, \theta) a^2 \chi^2 \sin^2(\theta)}{\chi^2 \Xi^2 \Sigma} \left(\exp\left(\frac{u}{A}\right) d\bar{u}^2 + \exp\left(-\frac{v}{A}\right) d\bar{v}^2 \right) \\ & - 2A^2 \exp\left(\frac{u-v}{2A}\right) \frac{\Xi^2 \Sigma^2 \Delta_r + \chi^2 \Delta_r - \tilde{s}(r, \theta) a^2 \chi^2 \sin^2(\theta)}{\chi^2 \Xi^2 \Sigma} d\bar{u} d\bar{v} \\ & + 2A \frac{\tilde{n}(r, \theta)}{\Xi^2 \Sigma} a \sin^2(\theta) \left(\exp\left(\frac{u}{2A}\right) d\bar{u} d\psi + \exp\left(-\frac{v}{2A}\right) d\bar{v} d\psi \right) \\ & + \frac{(a^2 + r^2)^2 \Delta_\theta - a^2 \sin^2(\theta) \Delta_r}{\Xi^2 \Sigma} \sin^2(\theta) d\psi^2, \end{aligned} \quad (5.1.66)$$

where A as in (5.1.63), $\tilde{s}$ as in (5.1.61), and

$$\tilde{n}(r, \theta) := \left(\left(1 - \frac{a^2 \sin^2(\theta)}{a^2 + r_h^2}\right) - \frac{(a^2 + r^2)(r_h^2 - r^2)}{(a^2 + r_h^2)\Delta_r} \Delta_\theta \right) \Delta_r, \quad (5.1.67)$$

The worrisome terms in (5.1.66) are those involving

$$e^{u/2A} = -\bar{u}^{-1}, \quad e^{-v/2A} = \bar{v}^{-1}, \quad e^{u/A} = -\bar{u}^{-2} \quad \text{and} \quad e^{-v/A} = \bar{v}^{-2}.$$

Let us show that some of those are innocuous. For this, suppose first that we are extending the metric from the region $r > r_h$; the extension in the other direction proceeds in a completely analogous manner. From the equations above we then have

$$\bar{u}\bar{v} = -\exp(c(v-u)) = -\exp(2cf(r)) = -(r - r_h) \exp(-2ch(r)).$$

Since $\tilde{n}$ factors out through $(r - r_h)$, smoothness of all metric functions except possibly those on the first line of (5.1.66) is apparent.

To finish the construction, a simple calculation shows that the choice

$$\chi(r_h) = (a^2 + r_h^2)\Xi \quad (5.1.68)$$

allows one to factor out $g_{uu} = g_{vv}$ through $(r - r_h)^2$. This cancels out the $\bar{u}^{-2}$ and $\bar{v}^{-2}$ terms in $g_{\bar{u}\bar{u}} = -\bar{u}^{-2}g_{uu}$ and in $g_{\bar{v}\bar{v}} = \bar{v}^{-2}g_{vv}$, leading to the promised smooth metric near $r = r_h$.

We emphasize that while the Eddington–Finkelstein-type coordinates always exist, the Kruskal–Szekeres-type ones exist only at those zeros of Δ_r which are of order one, i.e. $\Delta'(r_h) \neq 0$.

5.1.7 Principal null directions

As discussed in, e.g., [454, p. 179], the Weyl tensor C_{abcd} on a four-dimensional manifold

$$C_{abcd} = R_{abcd} - (g_{a[c} R_{d]b} - g_{b[c} R_{d]a}) + \frac{1}{3} R g_{a[c} g_{d]b},$$

possesses four, possibly coinciding, *principal null directions*. By definition, these are null vector fields k^a satisfying

$$k^b k^c k_{[e} C_{a]bc[d} k_{f]} = 0. \quad (5.1.69)$$

As proposed by Petrov and Pirani, one can use this fact to classify metrics according to the number of coinciding principal null directions. The metrics (5.1.1) are of Petrov type D [8, 173], which means that the Weyl tensor has two pairs of coinciding principal null directions. In this case one actually has

$$k^b k^c C_{abc[d} k_{e]} = 0 \quad (5.1.70)$$

(compare [454, Table 7.1]). The outgoing principal null vector field of the K(A)dS metric is [8]

$$l = \sqrt{\frac{\Delta_r}{2\Sigma}} \left(\partial_r + \frac{\Xi}{\Delta_r} V \right),$$

where the attribute *outgoing* refers to the fact that r is increasing along the integral curves of l . Here V is one of the *canonical vector fields*, as defined by analogy to the definitions in [384, p. 60]:

$$\begin{aligned} V &= a \partial_\varphi + (a^2 + r^2) \partial_t, \\ W &= \partial_\varphi + a \sin^2(\theta) \partial_t. \end{aligned}$$

These vector fields have particularly simple vector products:

$$\begin{aligned} g(V, V) &= -\frac{\Delta_r \Sigma}{\Xi^2}, \\ g(W, W) &= \frac{\Delta_\theta \Sigma \sin^2(\theta)}{\Xi^2}, \\ g(V, W) &= 0. \end{aligned}$$

A MATHEMATICA calculation shows the ingoing principal null vector field to be

$$n = \sqrt{\frac{\Delta_r}{2\Sigma}} \left(-\partial_r + \frac{\Xi}{\Delta_r} V \right).$$

Both l and n are defined up to a multiplicative factor, which we chose so that $l^\mu n_\mu = -1$.

The form of n can also be inferred from l by replacing (r, m) with $(-r, -m)$.

5.2 The Kerr–Newman–(anti-)de Sitter metrics (with C. Ölz)

The metrics we describe in this section are usually called *Kerr–Newman–de Sitter* when $\Lambda > 0$, and *Kerr–Newman–anti-de Sitter* when $\Lambda < 0$, though they have been

discovered (independently) by Carter [80] and by Demiański [173]. They can be written in a form [80, 436] formally identical to their vacuum counterparts (5.1.1),

$$g = \Sigma \left(\frac{1}{\Delta_r} dr^2 + \frac{1}{\Delta_\theta} d\theta^2 \right) + \frac{\sin^2 \theta}{\Sigma \Xi^2} \Delta_\theta (a dt - (r^2 + a^2) d\varphi)^2 - \Delta_r \frac{1}{\Sigma \Xi^2} (dt - a \sin^2 \theta d\varphi)^2, \quad (5.2.1)$$

where only the function Δ_r differs from the corresponding functions in vacuum (5.1.2)–(5.1.5):

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad (5.2.2)$$

$$\Delta_r = (r^2 + a^2) \left(1 - \frac{\Lambda}{3} r^2 \right) - 2mr + q^2, \quad (5.2.3)$$

$$\Delta_\theta = 1 + \frac{\Lambda}{3} a^2 \cos^2 \theta, \quad (5.2.4)$$

$$\Xi = 1 + \frac{\Lambda}{3} a^2. \quad (5.2.5)$$

We have again $t \in \mathbb{R}$, $r \in \mathbb{R}$ and we will assume that

$$\Lambda \neq 0, \quad a^2 + q^2 \neq 0.$$

We note that

$$\Xi > 0 \iff \Lambda \in \left(-\frac{3}{a^2}, \infty \right),$$

and since the case $\Xi = 0$ does not make sense in (5.2.1), the condition $\Xi \neq 0$ will also be assumed in the remainder of this section.

In [80] the electromagnetic potential is given by

$$A = q \frac{(a^2 + r^2) \sin(\alpha) \cos(\theta) - a r \cos(\alpha) \sin^2(\theta)}{\Xi \Sigma} d\varphi + q \frac{r \cos(\alpha) - a \sin(\alpha) \cos(\theta)}{\Xi \Sigma} dt, \quad (5.2.6)$$

where α is an arbitrary angle. The constant denoted by q here is denoted by e in [80], while the constant denoted by q in [80] should be set to 0. See footnote 1, p. 200 for the transformation between the coordinates used in [80] and the Boyer–Lindquist like coordinates used here.

It will sometimes be convenient to write

$$q^2 = |\Xi| e^2, \quad m = \Xi \mu, \quad (5.2.7)$$

where the rescaling of q^2 requires of course an absolute value for negative Ξ .

Clearly, all the properties of the K(A)dS metrics, presented in Section 5.1, which do not involve the explicit form of the metric function Δ_r remain unchanged.

5.2.1 The ‘ring singularity’

Similarly to the K(A)dS case, the set $\{\Sigma = 0\}$ corresponds to a geometric singularity in the metric. To see this, note that on the equatorial plane $\cos \theta = 0$ we have

$$g_{tt} = \frac{1}{3\Xi^2 r^2} (-3|\Xi| e^2 + 6\Xi \mu r + (\Lambda a^2 - 3) r^2 + \Lambda r^4). \quad (5.2.8)$$

It follows that the norm of the Killing vector ∂_t blows up as the set $\{\Sigma = 0\}$ is approached along some directions, which would not be possible if the metric could be continued across this set in a C^2 manner.

The fact that this set is singular regardless of the direction of approach is settled in [134], where it is established that the scalar function

$$(R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta})^2 + (\epsilon_{\alpha\beta\gamma\delta}R^{\alpha\beta}{}_{\mu\nu}R^{\mu\nu\gamma\delta})^2 \quad (5.2.9)$$

is unbounded on all curves accumulating at the set $\{\Sigma = 0\}$.

5.2.2 Extensions across Killing horizons

The construction of the extensions across Killing horizons is identical to that in Section 5.1.6.

5.2.3 The number and nature of horizons, $\Lambda > 0$

We can replace r by $-r$ if necessary to obtain a positive value of m and therefore, to reduce the number of cases that remain to be considered, we will assume

$$m \geq 0.$$

(When $\Lambda < 0$ one likes to think of $r \rightarrow \infty$ as corresponding to the physically relevant asymptotic region, in which case the replacement $r \mapsto -r$ is not relevant, and the sign of m will be chosen by different considerations.)

We wish to analyse the number and the character of zeros of the metric function

$$\begin{aligned} \Delta_r &= (r^2 + a^2) \left(1 - \frac{\Lambda r^2}{3} \right) - 2mr + q^2 \\ &= -\frac{\Lambda}{3}r^4 + \left(1 - \frac{\Lambda}{3}a^2 \right) r^2 - 2mr + a^2 + q^2. \end{aligned} \quad (5.2.10)$$

This will be needed to determine the location and character of the event horizons.

The case $\Lambda > 0$, $1 - \frac{\Lambda}{3}a^2 > 0$. By introducing a new variable x through the formula

$$r = \sigma x := \sqrt{\frac{1 - \frac{\Lambda}{3}a^2}{\Lambda}} x, \quad (5.2.11)$$

the equation $\Delta_r = 0$ becomes

$$P(x) := -x^4 + 3x^2 - 2\beta x + \gamma = 0, \quad (5.2.12)$$

where

$$\beta = \frac{3m}{\Lambda\sigma^3} \geq 0, \quad \gamma = \frac{3(a^2 + q^2)}{\Lambda\sigma^4}, \quad (5.2.13)$$

and note that $\gamma > 0$ under the current hypotheses. Thus, $P(x)$ is a polynomial of degree four with at most four distinct real roots, with $P(x) \rightarrow -\infty$ for $x \rightarrow \pm\infty$.

The analysis of (5.2.12) is particularly simple when $m = 0$, since then $\beta = 0$. The zeros of $P(x)$ then occur at

$$x_0^{(1,2,3,4)} = \pm \frac{1}{\sqrt{2}} \sqrt{3 \pm \sqrt{9 + 4\gamma}}. \quad (5.2.14)$$

A plot of P with $\gamma = 0$ can be found in Fig. 5.2.1. It is clear from the graph that, for $\gamma > 0$, the equation $P(x) = 0$ has always two first-order zeros.

In the remainder of this section we assume that

$$m > 0 \iff \beta > 0. \quad (5.2.15)$$

Since the function $x \mapsto -x^4 + 3x^2 = x^2(3 - x^2)$ is even, and βx changes sign at 0, the global maximum of P is attained at a negative value of x .

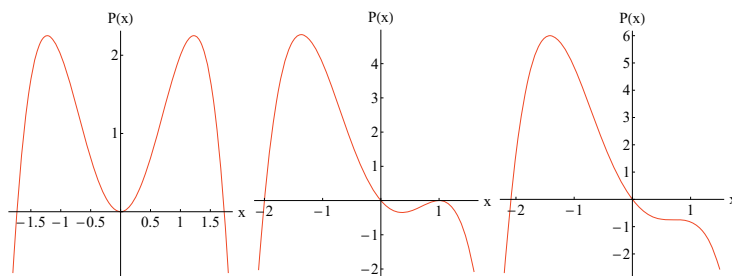


Fig. 5.2.1 The function $P(x)$ of (5.2.12) with $\gamma = 0$ when $\beta = 0$ (left), $\beta = 1$ (middle), and $\beta = \sqrt{2}$ (right).

$P(x)$ can have three extrema at most; we denote them x_1, x_2, x_3 with $x_1 < 0$. Since $P(0) = a^2 + q^2 > 0$ and $P(x) \rightarrow -\infty$ for $|x| \rightarrow \infty$ there always exists at least one strictly positive global maximum and hence at least two distinct real zeros of P .

In what follows we wish to analyse the character of the zeros of P . This requires finding joint zeros of P and one or more of its derivatives. A useful tool for this is provided by the *resultant* of two polynomials p and q , defined as follows: if

$$p(x) = \sum_{i=0}^n a_i x^{n-i}, \quad q(x) = \sum_{j=0}^m b_j x^{m-j},$$

are polynomials with nonzero coefficients a_0 and b_0 , and with zeros at α_i and β_j respectively, then the resultant of p and q is (see e.g. [406, pp. 20–2])

$$R(p, q) = a_0^m b_0^n \prod_{i=1}^n \prod_{j=1}^m (\alpha_i - \beta_j).$$

So the resultant vanishes if and only if p and q have common roots. The point is that the resultants can be calculated without a priori knowledge of the zeros of p and q , e.g. using MATHEMATICA.

Returning to the problem at hand, we consider first P with $\gamma = 0$:

$$P_\beta(x) := -x^4 + 3x^2 - 2\beta x. \quad (5.2.16)$$

Calculating the resultant of P_β and its derivative one finds that a double zero occurs only at $\beta = 0$ and $\beta = 1$. Similarly, one finds that P'_β and P''_β vanish simultaneously only if $\beta = \sqrt{2}$.

By inspection of the graph of P_0 we see that as β is increased from zero the negative maximum goes up; the zero at the origin splits into two, say $x_0^{(2)} = 0$, with $x_0^{(3)} = x_0^{(3)}(\beta)$ travelling to the right as β is increased. On the other hand, the zero at $\sqrt{3}$, say $x_0^{(4)}$, travels to the left as we increase β from 0. This happens until $x_0^{(3)}$ and $x_0^{(4)}$ merge, at $\beta = 1$. For $\beta > 1$ the polynomial P_β has only two simple real zeros. There is a further qualitative change which occurs at $\beta = \sqrt{2}$, since for $0 < \beta < \sqrt{2}$ there are precisely three extrema, while at $\sqrt{2}$ two rightmost extrema merge into an inflection point. For $\beta > \sqrt{2}$ the function P_β is concave and therefore has only one local extremum, which is also a global maximum.

Since $P = P_\beta + \gamma$ and $\gamma > 0$, the addition of γ amounts to shifting the graph up, which reduces the analysis of P to that of P_β . We proceed now to a systematic analysis of the general case, but in fact the results that follow can already be inferred from our discussion so far.

Using the Cardano algorithm to analyse the cubic equation $P'(x) = 0$, we define

$$d := \beta^2 - 2,$$

and find that there exist three distinct real solutions of the equation $P' = 0$ if and only if $d < 0$; three real solutions of which two or three could be identical for $d = 0$; and one real and two complex conjugate solutions for $d > 0$. Accordingly we will consider these cases separately, starting with $d > 0$.

The case $d > 0$. The condition for the existence of one real and two complex-conjugate solutions of the equation $P' = 0$ is $d > 0$, which translates to

$$m^2 > \frac{2}{3^5 \Lambda} (3 - a^2 \Lambda)^3. \quad (5.2.17)$$

The maximum of the function P must be at a real value; we denote it x_1 . Then

$$\begin{aligned} P'(x_1) &= -4x_1^3 + 6x_1 - 2\beta = 0, \\ P''(x_1) &= -12x_1^2 + 6 < 0, \end{aligned}$$

where the inequality is strict because the zero of P' at x_1 is necessarily simple. We conclude that $x_1^2 > \frac{1}{2}$. Since $P(0) > 0$, the zeros $x_0^{(1)}, x_0^{(2)}$ of $P(x)$ are ordered as $x_0^{(1)} < 0 < x_0^{(2)}$.

The case $d = 0$. In the case $d = 0$, m satisfies

$$m^2 = \frac{2}{3^5 \Lambda} (3 - a^2 \Lambda)^3 \iff \beta = \sqrt{2}. \quad (5.2.18)$$

$P(x)$ then has a maximum at $-\sqrt{2}$, and a saddle point at $x_2 = x_3 = 2^{-1/2}$. The graph of $P(x) - \gamma$ can be seen in Fig. 5.2.1. For $\gamma > 0$ and distinct from $3/4$ the function P has exactly two simple zeros. When

$$\gamma = \frac{3}{4} \iff q^2 = \frac{\Lambda \sigma^4}{4} - a^2$$

the function P has one simple zero $x_0^{(1)} = -3 \times 2^{-1/2}$ and one triple zero $x_0^{(2)} = 2^{-1/2}$. A triple zero of P occurs only for these values of the parameters; all the remaining zeros are simple or double.

Since $m^2 \geq 0$, a necessary condition for the case $d \leq 0$ is

$$\frac{a^2 \Lambda}{3} \leq 1. \quad (5.2.19)$$

The case $d < 0$. The condition $d < 0$ can be rewritten as

$$m^2 < \frac{2}{3^5 \Lambda} (3 - a^2 \Lambda)^3. \quad (5.2.20)$$

$P(x)$ has then three distinct real extremal points, one minimum at x_2 and two maxima at x_1 and x_3 . Therefore $P''(x_i)$ satisfies

$$P''(x_i) = -12x_i^2 + 6 < 0, \quad i = 1, 3, \quad P''(x_2) = -12x_2^2 + 6 > 0.$$

From this it follows that

$$1 < 2x_i^2, \quad i = 1, 3, \quad 2x_2^2 < 1, \quad (5.2.21)$$

and recall that we must also have

$$a < \sqrt{\frac{3}{\Lambda}}. \quad (5.2.22)$$

Rewriting $P'(x_i) = 0$ leads to

$$x_i(-4x_i^2 + 6) = 2\beta > 0, \quad (5.2.23)$$

so

$$\begin{aligned} x_i > 0 &\iff x_i^2 < \frac{3}{2} \\ x_i < 0 &\iff x_i^2 > \frac{3}{2}. \end{aligned}$$

Taking into account that the x_i 's also need to satisfy condition (5.2.21) we conclude that

$$x_1 < -\sqrt{\frac{3}{2}} < 0 < \sqrt{\frac{1}{2}} < x_2 < x_3 < \sqrt{\frac{3}{2}}. \quad (5.2.24)$$

Summarizing, P has exactly one extremum for negative x 's, which is a global maximum, and exactly one local minimum and maximum for positive x 's.

We turn our attention now to the sign of the zeros of $P(x)$, which we denote as $x_0^{(1)} < x_0^{(2)} \leq x_0^{(3)} \leq x_0^{(4)}$. Since P' has exactly one zero for $x < 0$, the function P has precisely one simple zero for $x \leq 0$. We thus have

$$x_0^{(1)} < 0 < x_0^{(2)} \leq x_0^{(3)} \leq x_0^{(4)}.$$

We have already seen that a zero of order three, $x_0^{(2)} = x_0^{(3)} = x_0^{(4)}$, cannot occur when $d > 0$. This leaves open the possibilities $x_0^{(2)} = x_0^{(3)} < x_0^{(4)}$ or $x_0^{(2)} < x_0^{(3)} = x_0^{(4)}$. Those will occur if and only if the resultant of P and P' vanishes. This happens when

$$\beta = \beta_{\pm} := \frac{1}{6} \sqrt{18(4\gamma + 1) \pm 2\sqrt{3}\sqrt{(3-4\gamma)^3}}. \quad (5.2.25)$$

One checks that both values are real for $0 \leq \gamma \leq 3/4$, and that $\beta_{\pm}$ are increasing functions of γ in that range, with

$$\beta_-(0) = 0, \quad \beta_+(0) = 1, \quad \beta_{\pm}(3/4) = \sqrt{2}. \quad (5.2.26)$$

We conclude that for $0 < \beta < \sqrt{2}$ there always exists a value of γ for which P has exactly one double zero, and that all zeros are simple for $\beta > \sqrt{2}$.

Some insight into (5.2.25)–(5.2.25) is provided by the following argument. Let $\beta \in (0, 1]$, and consider the graph of P as γ starts increasing from 0. For those values of β and for γ very small the function P has four roots, and as γ increases the two positive roots $x_0^{(2)}$ and $x_0^{(3)}$ merge at the value of $\gamma = \gamma_*$ for which $\beta = \beta_-(\gamma_*)$. For $\gamma > \gamma_*$ there are only two zeros of P .

Let next $\beta \in (1, \sqrt{2})$. Then P has only two zeros for small γ . As γ is increased the positive local maximum of P eventually meets the axis at the value γ_+ such that $\beta_+(\gamma_+) = \beta$. For $\gamma > \gamma_+$ and near to γ_+ the function P will have four zeros; this will remain true when increasing γ until a value γ_- is reached such that $\beta_-(\gamma_-) = \beta$; at this value the local minimum of P becomes a zero. For $\gamma > \gamma_-$ the polynomial P will have exactly two simple zeros.

Finally, for $\beta > \sqrt{2}$, the function P is concave so that there will only be two simple zeros of P , regardless of the value of γ .

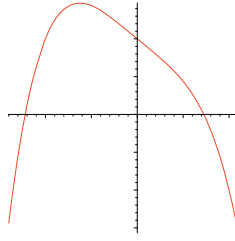


Fig. 5.2.2 The function $P(x)$ with $\Lambda a^2 = 3$, $m > 0$, and $\gamma = 1$.

The case $\Lambda > 0$, $1 - \frac{\Lambda}{3}a^2 = 0$. When

$$\Lambda a^2 = 3$$

the quadratic term in (5.2.10) vanishes, which simplifies the analysis considerably. Note that we necessarily have $a \neq 0$ in this section.

The simplest case $m = 0$ leads to two simple roots of Δ_r , since then

$$\begin{aligned} \Delta_r &= (a^2 + q^2 - a^{-2}r^4) = -a^{-2}(r^2 - a\sqrt{q^2 + a^2})(r^2 + a\sqrt{q^2 + a^2}) \\ &= -a^{-2}(r - r_+)(r - r_-)(r^2 + a\sqrt{q^2 + a^2}), \quad r_{\pm} = \pm\sqrt{a}(q^2 + a^2)^{\frac{1}{4}}. \end{aligned}$$

When $q = 0$ this is de Sitter spacetime in disguise: an explicit coordinate transformation which brings g to the de Sitter form can be found in [81, p. 102]; see also [8, 247, 457].

In the remainder of this section we assume that $m > 0$, as can be arranged by replacing r by its negative if necessary. By introducing a new variable x through the formula

$$r = \sigma x := (2ma^2)^{\frac{1}{3}} x, \quad (5.2.27)$$

the equation $\Delta_r = 0$ becomes

$$P(x) := -x^4 - x + \gamma = 0, \quad (5.2.28)$$

where

$$\gamma := \frac{(a^2 + q^2)a^2}{\sigma^4} \iff q^2 = \frac{\gamma\sigma^4 - a^4}{a^2}. \quad (5.2.29)$$

Since P is concave it has always two simple zeros; see Fig. 5.2.2.

We conclude that in the current case P has always two simple zeros, one positive and one negative.

The case $\Lambda > 0$, $1 - \frac{\Lambda}{3}a^2 < 0$. By introducing a new variable x through the formula

$$r = \sigma x := \sqrt{\frac{\frac{\Lambda}{3}a^2 - 1}{\Lambda}} x, \quad (5.2.30)$$

the equation $\Delta_r = 0$ becomes

$$P(x) := -x^4 - 3x^2 - 2\beta x + \gamma = 0, \quad (5.2.31)$$

where β and γ are again given by (5.2.13). Since P is concave now, and strictly positive at the origin, the function P has always precisely two simple zeros, one positive and one negative.

5.2.4 The number and nature of horizons, $\Lambda < 0$

When $\Lambda < 0$ the existence of zeros of Δ_r requires that

$$m > 0$$

and we will assume this in what follows; otherwise there are no horizons clothing the singular set $\{\Sigma = 0\}$.

The case $0 > \Lambda > -3a^{-2}$. We suppose now that $\Lambda < 0$ together with

$$1 + \frac{\Lambda}{3}a^2 > 0.$$

The change of variables (5.2.30) leads to the following canonical form of the equation $\Delta_r = 0$,

$$P(x) := x^4 + 3x^2 - 2|\beta|x + |\gamma| = 0, \quad (5.2.32)$$

with β and γ as in (5.2.13):

$$|\beta| = \frac{3m}{|\Lambda|\sigma^3} \geq 0, \quad |\gamma| = \frac{3(a^2 + q^2)}{|\Lambda|\sigma^4}. \quad (5.2.33)$$

Let $|\beta_*|$ be the value of $|\beta|$ at which P has a second-order zero; this can be found by calculating the resultant of P and P' :

$$|\beta_*| = \frac{\sqrt{-9 + 36|\gamma| + \sqrt{3(3 + 4|\gamma|)^3}}}{3\sqrt{2}}. \quad (5.2.34)$$

Since P is convex, strictly positive at the origin, for $|\beta| < |\beta_*|$ the function Δ_r has no zeros; when $|\beta| = |\beta_*|$ it has precisely one strictly positive second-order zero; for $|\beta| > |\beta_*|$ it has precisely two strictly positive first-order zeros.

The case $\Lambda = -3a^{-2}$. The value $\Lambda = -3a^{-2}$ does not make sense in the metric (5.2.1), as this leads to $\Xi = 0$, and thus to undefined metric coefficients. However, one can replace t by Ξt , φ by $\Xi\varphi$, and pass to the limit $\Xi \rightarrow 0$. This leads to the metric

$$g = \Sigma \left(\frac{1}{\Delta_r} dr^2 + \frac{1}{\sin^2(\theta)} d\theta^2 \right) + \frac{\sin^4(\theta)}{\Sigma} (a dt - (r^2 + a^2) d\varphi)^2 - \frac{\Delta_r}{\Sigma} (dt - a \sin^2(\theta) d\varphi)^2, \quad (5.2.35)$$

with Σ and Δ_r as in (5.2.2)–(5.2.4). Note that the ‘points’ $\sin \theta = 0$ are infinitely far away so that the question of possible conical singularities there does not arise, and hence there is no natural period for φ .

As before we assume that

$$a^2 + q^2 \neq 0, \quad \text{but now } \Lambda = -3a^{-2}.$$

The case $m = 0$ and $q \neq 0$ leads to a naked singularity and will not be considered any further.

The form (5.2.32) of P remains valid, and so does the associated discussion of its zeros.

The case $\Lambda < -3a^{-2}$. To restore some visual symmetry between θ and r , we start by introducing a coordinate

$$x := a \cos(\theta), \quad (5.2.36)$$

so that the metric becomes, after trivial rearrangements,

$$g = \frac{\Sigma}{\Delta_r} dr^2 - \Delta_r \frac{1}{a^2 \Sigma \Xi^2} (a dt - (a^2 - x^2) d\varphi)^2 \\ + \frac{\Sigma}{\Delta_x} dx^2 + \frac{1}{a^2 \Sigma \Xi^2} \Delta_x (a dt - (r^2 + a^2) d\varphi)^2, \quad (5.2.37)$$

where now

$$\Sigma = r^2 + x^2, \quad (5.2.38)$$

$$\Delta_x = \left(1 + \frac{\Lambda}{3} x^2\right) (a^2 - x^2), \quad (5.2.39)$$

with the remaining metric functions remaining unchanged. The first line of (5.2.37) has signature $(-, +)$ or $(+, -)$, depending upon the sign of Δ_r , assuming $\Delta_r \neq 0$. The second line has signature $(+, +)$ or $(-, -)$. It follows that the signature changes when crossing the singular set $\{\Delta_x = 0\}$, which implies that this singularity cannot be removed by a change of coordinates. We conclude that all solutions with $\Lambda < -3a^{-2}$ are nakedly singular. It is somewhat surprising that this singularity does not show in any curvature invariants that we have inspected.²

5.3 Emparan–Reall ‘black rings’

An interesting class of black-hole solutions of the $(4 + 1)$ -dimensional stationary vacuum Einstein equations has been found by Emparan and Reall [183] (see also [111, 182, 184] and references therein for further studies of the Emparan–Reall metrics). The metrics are asymptotically Minkowskian in spacelike directions, with an ergosurface and an event horizon having $S^1 \times S^2$ cross-sections. (The ‘ring’ terminology refers to the S^1 factor in $S^1 \times S^2$.) Our presentation is an expanded version of [183], with a somewhat different labelling of the constants appearing in the metric; furthermore, the gravitational coupling constant G from that reference has been set to one here.³

The Emparan–Reall metrics are the static members of the Pomeransky–Senkov family of stationary vacuum $(4 + 1)$ -dimensional metrics, presented in Section 7.6, p. 304.

INCIDENTALLY: While the mathematical interest of the black ring solutions is clear, their physical relevance is much less so, because of numerical evidence for their instability [177, 191, 253].

The starting point of the analysis is the metric

$$g = -\frac{F(x)}{F(y)} \left(dt + \sqrt{\frac{\nu}{\xi_F}} \frac{\xi_1 - y}{A} d\psi \right)^2 \\ + \frac{F(y)}{A^2 (x - y)^2} \left[-F(x) \left(\frac{dy^2}{G(y)} + \frac{G(y)}{F(y)} d\psi^2 \right) \right. \\ \left. + F(y) \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \right) \right], \quad (5.3.1)$$

²I am grateful to Kayll Lake and Sebastian Szybka for sharing the results of their calculations.

³I am grateful to R. Emparan and H. Reall for allowing me to reproduce their figures.

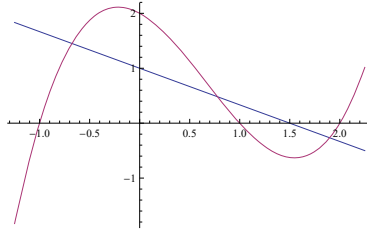


Fig. 5.3.1 Representative plots of F and G .

where $A > 0$, ν , and ξ_F are constants, and

$$F(\xi) = 1 - \frac{\xi}{\xi_F}, \quad (5.3.2)$$

$$G(\xi) = \nu\xi^3 - \xi^2 + 1 = \nu(\xi - \xi_1)(\xi - \xi_2)(\xi - \xi_3). \quad (5.3.3)$$

One can check, e.g. using the MATHEMATICA package xACT [340], that (5.3.1) solves the vacuum Einstein equations.⁴ The constant ν is chosen to satisfy

$$0 < \nu < \nu_* = \frac{2}{3\sqrt{3}}.$$

The upper bound is determined by the requirement that the three roots $\xi_1 < \xi_2 < \xi_3$ of G are distinct and real. Note that $G(0) = 1$ so that $\xi_1 < 0$. Further $G' = 3\nu\xi^2 - 2\xi > 0$ for $\xi < 0$, which implies that $\xi_2 > 0$. Hence,

$$\xi_1 < 0 < \xi_2 < \xi_3.$$

We will assume that⁵

$$\xi_2 < \xi_F < \xi_3.$$

A definite choice of ξ_F consistent with this hypothesis will be made shortly. See Fig. 5.3.1 for representative plots.

Requiring that

$$\xi_1 \leq x \leq \xi_2 \quad (5.3.4)$$

guarantees $G(x) \geq 0$ and $F(x) > 0$. On the other hand, both $G(y)$ and $F(y)$ will be allowed to change sign, as we will be working in the ranges

$$y \in (-\infty, \xi_1] \cup (\xi_F, \infty). \quad (5.3.5)$$

INCIDENTALLY: Explicit formulae for the roots of G , which are not particularly enlightening, can be found. For example, for $\nu \geq \nu_*$ one of the roots reads

$$\frac{\alpha}{6\nu} + \frac{2}{3\nu\alpha} + \frac{1}{3\nu}, \quad \text{where } \alpha = \sqrt[3]{-108\nu^2 + 8 + 12\sqrt{3}\sqrt{27\nu^2 - 4\nu}},$$

and a proper understanding of the various roots appearing in this equation also gives all solutions for $0 \leq \nu < \nu_*$. Alternatively, in this last range of ν the roots belong to the set $\{(z_k + \frac{1}{2})\frac{2}{3\nu}\}_{k=0}^2$, with

$$z_k = \cos\left(\frac{1}{3}\left[\arccos\left(1 - \frac{27\nu^2}{2}\right) + 2k\pi\right]\right).$$

⁴I thank Alfonso Garc  a-Parrado and Jos   Maria Mart  n-Garc  a for carrying out the xACT calculation.

⁵According to [183], the choice $\xi_F = \xi_2$ corresponds to the five-dimensional rotating black hole of [371], with one angular-momentum parameter set to 0.

Performing affine transformations of the coordinates, one can always achieve

$$\xi_1 = -1, \quad \xi_2 = 1,$$

but we will not impose these conditions.

Anticipating the analysis that follows, the variable ψ is chosen to be 2π periodic, which will guarantee asymptotic flatness. The variable φ is chosen to be periodic, with a period to be determined shortly so that the sets $x = \xi_1$ and $x = \xi_2$ are smooth rotation axes. The submanifolds with fixed (v, z, χ) will then become two-dimensional spheres.

5.3.1 The region $x \in \{\xi_1, \xi_2\}$

There is a potential singularity of the $G^{-1}(x)dx^2 + G(x)F^{-1}(x)d\varphi^2$ terms in the metric at $x = \xi_1$, which can be handled as follows: consider, first, a metric of the form

$$h = \frac{dx^2}{x - x_0} + (x - x_0)f(x)d\varphi^2, \quad f(x_0) > 0. \quad (5.3.6)$$

Introducing

$$\tilde{\rho} = 2\sqrt{x - x_0}, \quad \varphi = \lambda\tilde{\varphi}, \quad (5.3.7)$$

one obtains

$$h = d\tilde{\rho}^2 + \frac{\lambda^2 f\left(x_0 + \frac{\tilde{\rho}^2}{4}\right)}{4} \tilde{\rho}^2 d\tilde{\varphi}^2. \quad (5.3.8)$$

This defines a metric which smoothly extends through $\tilde{\rho} = 0$ (when f is smooth) if and only if $\tilde{\varphi}$ is periodically identified with period, say, 2π , and

$$\lambda = \frac{2}{\sqrt{f(x_0)}}. \quad (5.3.9)$$

REMARK 5.3.1 In order to show that (5.3.9) implies regularity, set $x^1 = \tilde{\rho} \cos \tilde{\varphi}$, $x^2 = \tilde{\rho} \sin \tilde{\varphi}$; we then have

$$\begin{aligned} h &= \underbrace{d\tilde{\rho}^2 + \tilde{\rho}^2 d\tilde{\varphi}^2}_{\delta_{ab} dx^a dx^b} + \frac{\lambda^2 \left(f\left(x_0 + \frac{\tilde{\rho}^2}{4}\right) - f(x_0) \right)}{4} \underbrace{\tilde{\rho}^2 d\tilde{\varphi}^2}_{\delta_{ab} dx^a dx^b - d\tilde{\rho}^2} \\ &= \delta_{ab} dx^a dx^b + \frac{\lambda^2 \left(f\left(x_0 + \frac{\tilde{\rho}^2}{4}\right) - f(x_0) \right)}{4} \left(\delta_{ab} dx^a dx^b - \tilde{\rho}^{-2} x^a x^b dx^a dx^b \right). \end{aligned}$$

As f is smooth, there exists a smooth function s such that

$$\frac{\lambda^2 \left(f\left(x_0 + \frac{\tilde{\rho}^2}{4}\right) - f(x_0) \right)}{4} = \tilde{\rho}^2 s(\tilde{\rho}^2),$$

so that

$$h = \left[\left(1 + s(\tilde{\rho}^2) \tilde{\rho}^2 \right) \delta_{ab} - s(\tilde{\rho}^2) x^a x^b \right] dx^a dx^b, \quad (5.3.10)$$

which is manifestly smooth. This shows the sufficiency of (5.3.9).

To show that (5.3.9) is necessary, note that from (5.3.8) we have $|D\tilde{\rho}|_h^2 = 1$. This implies that the integral curves of $D\tilde{\rho}$ are geodesics starting at $\{\tilde{\rho} = 0\}$. When $\{\tilde{\rho} = 0\}$ is a regular centre one can run backwards a calculation in the spirit of the one that led to (5.3.10), using normal coordinates centred at $\tilde{\rho} = 0$ as a starting point, to conclude that the unit vectors orthogonal to the vector $\partial_{\tilde{\rho}}$ take the form $\pm\chi(\tilde{\rho})\partial_{\tilde{\varphi}}$, where $\chi(\tilde{\rho})^2 \tilde{\rho}^2 \rightarrow_{\tilde{\rho} \rightarrow 0} 1$, and with $\partial_{\tilde{\varphi}}$ having periodic orbits with period 2π . Comparing with (5.3.8), (5.3.9) readily follows. $\square$

In order to apply (5.3.9) to the last line of (5.3.1) at $x_0 = \xi_1$ we have

$$\begin{aligned} & \frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \\ &= \frac{1}{\nu(x - \xi_2)(x - \xi_3)} \left(\frac{dx^2}{x - \xi_1} + \frac{\nu^2 \xi_F (x - \xi_1)(x - \xi_2)^2 (x - \xi_3)^2}{\xi_F - x} d\varphi^2 \right) \\ &= \frac{1}{\nu(x - \xi_2)(x - \xi_3)} \left(d\tilde{\rho}^2 + \frac{\lambda^2 \nu^2 \xi_F (x - \xi_2)^2 (x - \xi_3)^2}{4(\xi_F - x)} \tilde{\rho}^2 d\tilde{\varphi}^2 \right), \end{aligned} \quad (5.3.11)$$

so that (5.3.9) becomes

$$\lambda = \frac{2\sqrt{\xi_F - \xi_1}}{\nu\sqrt{\xi_F}(\xi_2 - \xi_1)(\xi_3 - \xi_1)}. \quad (5.3.12)$$

For further purposes it is convenient to rewrite (5.3.11) as

$$\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 = \frac{1}{H(x)} \left[d\tilde{\rho}^2 + \left(1 + s(\tilde{\rho}^2) \tilde{\rho}^2 \right) \tilde{\rho}^2 d\tilde{\varphi}^2 \right], \quad (5.3.13)$$

for a smooth function s with, of course,

$$H(\xi) = \nu(\xi - \xi_2)(\xi - \xi_3). \quad (5.3.14)$$

When $\xi_F > \xi_2$ one can repeat this analysis at $x = \xi_2$, obtaining instead

$$\lambda = \frac{2\sqrt{\xi_F - \xi_2}}{\nu\sqrt{\xi_F}(\xi_2 - \xi_1)(\xi_3 - \xi_2)}. \quad (5.3.15)$$

Since the left-hand sides of (5.3.12) and (5.3.15) are equal, so must be the right-hand sides; their equality determines ξ_F :

$$\xi_F = \frac{\xi_1 \xi_2 - \xi_3^2}{\xi_1 - 2\xi_3 + \xi_2}. \quad (5.3.16)$$

(Elementary algebra shows that $\xi_2 < \xi_F < \xi_3$, as desired.) It should be clear that with this choice of ξ_F , for $y \neq \xi_1$, the (x, φ) -part of the metric (5.3.1) is a smooth (in fact, analytic) metric on S^2 , with the coordinate x being the equivalent of the usual polar coordinate θ on S^2 , except possibly at those points where the overall conformal factor vanishes or acquires zeros, which will be analysed shortly. Anticipating, the set obtained by varying x and φ and keeping $y = \xi_1$ will be viewed as S^2 with the north pole $x = \xi_1$ removed.

5.3.2 Signature

The calculation of the determinant of (5.3.1) reduces to that of a two-by-two determinant in the (t, ψ) variables, which equals

$$\frac{F^2(x)G(y)}{A^2(x - y)^2 F(y)}, \quad (5.3.17)$$

leading to

$$\det g = -\frac{F^2(x)F^4(y)}{A^8(x - y)^8}, \quad (5.3.18)$$

so the signature is either $(-+++)$ or $(----)$, except perhaps at the singular points $x = y$, or $F(x) = 0$ (which does not happen when $\xi_F > \xi_2$, compare (5.3.4)), or $F(y) = 0$.

Now, $F(x) > 0$, $G(x) > 0$ (away from the axes $x \in \{\xi_1, \xi_2\}$); thus, by inspection of (5.3.1), the signature is

$$\left(\text{sign}(-F(y)), \text{sign}(-G(y)), \text{sign}(-F(y)G(y)), +, + \right). \quad (5.3.19)$$

An examination of the four possible cases shows that a Lorentzian signature is obtained except if $F(y) > 0$ and $G(y) > 0$, which occurs for $y \in (\xi_1, \xi_2)$. So y ’s in this last range will not be of interest to us.

We start by considering

$$y \leq \xi_1, \quad (5.3.20)$$

which leads to $F(y) > 0$ and $G(y) \leq 0$.

5.3.3 The rotation axis $y = \xi_1$

Note that $G(\xi_1)$ vanishes; however, it should be clear from what has been said that $-(\frac{dy^2}{G(y)} + \frac{G(y)}{F(y)}d\psi^2)$ is a smooth Riemannian metric if (ξ_1, ψ) are related to a new radial variable $\hat{\rho}$ and a new angular variable $\hat{\varphi}$ by

$$\hat{\rho} = 2\sqrt{\xi_1 - y} \in \mathbb{R}^+, \quad \psi = \lambda\hat{\varphi},$$

with λ given by (5.3.15) and $\hat{\varphi}$ being 2π -periodic. Analogously to (5.3.13), we thus have

$$-\left(\frac{dy^2}{G(y)} + \frac{G(y)}{F(y)}d\psi^2\right) = \frac{1}{H(y)} \left[d\hat{\rho}^2 + \left(1 + s(\hat{\rho}^2)\hat{\rho}^2\right)\hat{\rho}^2 d\hat{\varphi}^2 \right]. \quad (5.3.21)$$

Note that the remaining terms in (5.3.1) involving $d\psi$ are also well behaved: indeed, if we set $\hat{x}^1 = \hat{\rho} \cos \hat{\varphi}$, $\hat{x}^2 = \hat{\rho} \sin \hat{\varphi}$, then

$$(\xi_1 - y)d\psi = \frac{\lambda\hat{\rho}^2}{4}d\hat{\varphi} = \frac{\lambda}{4}(\hat{x}^1 d\hat{x}^2 - \hat{x}^2 d\hat{x}^1),$$

which is again manifestly smooth.

5.3.4 Asymptotic flatness

We turn our attention now to the singularity $x = y$. Given our ranges of coordinates, this only occurs for $x = y = \xi_1$. So, at this stage, the coordinate t parameterizes $\mathbb{R}$, the coordinates (y, ψ) are (related to polar) coordinates on $\mathbb{R}^2$, and the coordinates (x, φ) are coordinates on S^2 . If we think of $x = \xi_1$ as being the north pole of S^2 , and we denote it by N , then g is an analytic metric on

$$\underbrace{\mathbb{R}}_t \times \left(\left(\underbrace{\mathbb{R}^2}_{y, \psi \Leftrightarrow \hat{\rho}, \hat{\varphi}} \times \underbrace{S^2}_{x, \varphi \Leftrightarrow \hat{\rho}, \hat{\varphi}} \right) \setminus (\{0\} \times \{N\}) \right).$$

Before passing to a detailed analysis of the metric for x and y close to ξ_1 , it is useful to examine the leading order behaviour of the last two lines in (5.3.1). Recall that (5.3.7) with $x_0 = \xi_1$ gives $x = \xi_1 + \tilde{\rho}^2/4$, and using (5.3.13) we rewrite the last line of (5.3.1), for small $\tilde{\rho}$,

$$\frac{F(y)^2}{A^2(x-y)^2} \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)}d\varphi^2 \right) \approx \frac{F(\xi_1)^2}{A^2 H(\xi_1)(x-y)^2} (d\tilde{\rho}^2 + \tilde{\rho}^2 d\tilde{\varphi}^2).$$

Similarly, with $y = \xi_1 - \hat{\rho}^2/4$, and with $\hat{\rho}$ small, the second line of (5.3.1) reads, keeping in mind (5.3.21),

$$-\frac{F(x)F(y)}{A^2(x-y)^2} \left(\frac{dy^2}{G(y)} + \frac{G(y)}{F(y)}d\psi^2 \right) \approx \frac{F(\xi_1)^2}{A^2 H(\xi_1)(x-y)^2} (d\hat{\rho}^2 + \hat{\rho}^2 d\hat{\varphi}^2).$$

Since $x - y = (\tilde{\rho}^2 + \hat{\rho}^2)/4$, adding one obtains

$$\frac{16F(\xi_1)^2}{A^2H(\xi_1)} \times \frac{1}{(\tilde{\rho}^2 + \hat{\rho}^2)^2} (d\tilde{\rho}^2 + \tilde{\rho}^2 d\tilde{\varphi}^2 + d\hat{\rho}^2 + \hat{\rho}^2 d\hat{\varphi}^2) .$$

Up to an overall constant factor, this is a flat metric on $\mathbb{R}^4$, to which a *Kelvin inversion* $\vec{x} \mapsto \vec{x}/|\vec{x}|^2$ has been applied, rewritten using polar coordinates in two orthogonal planes.

We pass now to a complete analysis. Near the singular set $\mathbb{R} \times \{0\} \times \{N\}$, Emparan and Reall replace $(\tilde{\rho}, \hat{\rho})$ by new radial variables $(\tilde{r}, \hat{r})$ defined as

$$\tilde{r} = \frac{\tilde{\rho}}{B(\tilde{\rho}^2 + \hat{\rho}^2)}, \quad \hat{r} = \frac{\hat{\rho}}{B(\tilde{\rho}^2 + \hat{\rho}^2)}, \quad (5.3.22)$$

where B is a constant which will be determined shortly. This is inverted as

$$\tilde{\rho} = \frac{\tilde{r}}{B(\tilde{r}^2 + \hat{r}^2)}, \quad \hat{\rho} = \frac{\hat{r}}{B(\tilde{r}^2 + \hat{r}^2)}. \quad (5.3.23)$$

It is convenient to set

$$r = \sqrt{\tilde{r}^2 + \hat{r}^2}.$$

We note

$$x = \xi_1 + \frac{\tilde{\rho}^2}{4} = \xi_1 + \frac{\tilde{r}^2}{4B^2r^4}, \quad y = \xi_1 - \frac{\hat{\rho}^2}{4} = \xi_1 - \frac{\hat{r}^2}{4B^2r^4},$$

$$x - y = \frac{1}{4B^2r^2}.$$

This last equation shows that $x - y \rightarrow 0$ corresponds to $r \rightarrow \infty$.

Inserting (5.3.13) and (5.3.21) into (5.3.1) we obtain

$$g = -\frac{F(x)}{F(y)} \left(dt + \sqrt{\frac{\nu}{\xi_F}} \frac{\xi_1 - y}{A} d\psi \right)^2$$

$$+ \frac{F(y)}{A^2(x-y)^2 H(x)H(y)} \left[F(x)H(x) \left(d\hat{\rho}^2 + (1 + s(\hat{\rho}^2)\hat{\rho}^2)\hat{\rho}^2 d\hat{\varphi}^2 \right) \right.$$

$$\left. + F(y)H(y) \left(d\tilde{\rho}^2 + (1 + s(\tilde{\rho}^2)\tilde{\rho}^2)\tilde{\rho}^2 d\tilde{\varphi}^2 \right) \right]. \quad (5.3.24)$$

The simplest terms arise from the first line above,

$$-\frac{\xi_F - \xi_1 - \frac{\tilde{r}^2}{4B^2r^4}}{\xi_F - \xi_1 + \frac{\hat{r}^2}{4B^2r^4}} \left(dt + \lambda \sqrt{\frac{\nu}{\xi_F}} \frac{1}{4AB^2r^4} \hat{r}^2 d\hat{\varphi} \right)^2$$

$$= -\left(1 - \frac{1}{4(\xi_F - \xi_1)B^2r^2} + O(r^{-4}) \right) \left(dt + O(r^{-4})\hat{r}^2 d\hat{\varphi} \right)^2, \quad (5.3.25)$$

which has the right behaviour for asymptotic flatness.

In order to analyse the remaining terms, one needs to carefully keep track of all potentially singular terms in the metric: in particular, one needs to verify that the decay of the metric to the flat one is uniform with respect to directions, making sure that no problems arise near the rotation axes $\hat{r} = 0$ and $\tilde{r} = 0$. So we write the $\hat{\varphi}^2$ and the $\tilde{\varphi}^2$ terms from the last two lines of (5.3.24) as

$$\begin{aligned}
 g_{\hat{\varphi}\hat{\varphi}}d\hat{\varphi}^2 + g_{\tilde{\varphi}\tilde{\varphi}}d\tilde{\varphi}^2 &= \frac{F(y)}{A^2(x-y)^2H(x)H(y)} \left[F(x)H(x) \left(1 + s(\hat{\rho}^2)\hat{\rho}^2 \right) \hat{\rho}^2 d\hat{\varphi}^2 \right. \\
 &\quad \left. + F(y)H(y) \left(1 + s(\tilde{\rho}^2)\tilde{\rho}^2 \right) \tilde{\rho}^2 d\tilde{\varphi}^2 \right] \\
 &= \frac{16B^2F(y)}{A^2H(x)H(y)} \left[F(x)H(x) \left(1 + O(r^{-4})\hat{r}^2 \right) \hat{r}^2 d\hat{\varphi}^2 \right. \\
 &\quad \left. + F(y)H(y) \left(1 + O(r^{-4})\tilde{r}^2 \right) \tilde{r}^2 d\tilde{\varphi}^2 \right]. \tag{5.3.26}
 \end{aligned}$$

From

$$d\tilde{\rho} = \frac{1}{Br^4} \left((\hat{r}^2 - \tilde{r}^2)d\tilde{r} - 2\tilde{r}\hat{r}d\hat{r} \right), \quad d\hat{\rho} = \frac{1}{Br^4} \left((\tilde{r}^2 - \hat{r}^2)d\hat{r} - 2\hat{r}\tilde{r}d\tilde{r} \right),$$

one finds that

$$\begin{aligned}
 g_{\hat{r}\hat{r}} &= \frac{(4B)^2F(y)}{A^2H(x)H(y)r^4} \left(F(x)H(x)(\hat{r}^2 - \tilde{r}^2)^2 + 4F(y)H(y)\hat{r}^2\tilde{r}^2 \right) \\
 &= \frac{(4B)^2F(y)}{A^2H(x)H(y)} \left(F(x)H(x) + 4(F(y)H(y) - F(x)H(x))\frac{\hat{r}^2\tilde{r}^2}{r^4} \right) \\
 &= \frac{(4B)^2F(y)}{A^2H(x)H(y)} \left(F(x)H(x) + O(r^{-4})\hat{r}^2 \right), \tag{5.3.27}
 \end{aligned}$$

$$\begin{aligned}
 g_{\tilde{r}\tilde{r}} &= \frac{(4B)^2F(y)}{A^2H(x)H(y)r^4} \left(F(y)H(y)(\hat{r}^2 - \tilde{r}^2)^2 + 4F(x)H(x)\hat{r}^2\tilde{r}^2 \right) \\
 &= \frac{(4B)^2F(y)}{A^2H(x)H(y)} \left(F(y)H(y) + O(r^{-4})\tilde{r}^2 \right), \tag{5.3.28}
 \end{aligned}$$

$$\begin{aligned}
 g_{\hat{r}\tilde{r}} &= \frac{2(4B)^2F(y)}{A^2H(x)H(y)r^4} \hat{r}\tilde{r}(\tilde{r}^2 - \hat{r}^2)(F(y)H(y) - F(x)H(x)) \\
 &= O(r^{-4})\hat{r}\tilde{r}. \tag{5.3.29}
 \end{aligned}$$

It is clearly convenient to choose B so that

$$\frac{(4B)^2F^2(\xi_1)}{A^2H(\xi_1)} = 1,$$

and with this choice (5.3.25)–(5.3.29) give

$$\begin{aligned}
 g &= - \left(1 + O(r^{-2}) \right) \left(dt + O(r^{-4})\hat{r}^2d\hat{\varphi} \right)^2 + O(r^{-4})\tilde{r}d\tilde{r}\hat{r}d\hat{r} \\
 &\quad + \left(1 + O(r^{-2}) \right) \left(d\hat{r}^2 + \hat{r}^2d\hat{\varphi}^2 \right) + O(r^{-4})\hat{r}^4d\hat{\varphi}^2 \\
 &\quad + \left(1 + O(r^{-2}) \right) \left(d\tilde{r}^2 + \tilde{r}^2d\tilde{\varphi}^2 \right) + O(r^{-4})\tilde{r}^4d\tilde{\varphi}^2. \tag{5.3.30}
 \end{aligned}$$

To obtain a manifestly asymptotically flat form one sets

$$\hat{y}^1 = \hat{r} \cos \hat{\varphi}, \quad \hat{y}^2 = \hat{r} \sin \hat{\varphi}, \quad \tilde{y}^1 = \tilde{r} \cos \tilde{\varphi}, \quad \tilde{y}^2 = \tilde{r} \sin \tilde{\varphi},$$

then

$$\begin{aligned}
 \hat{r}d\hat{r} &= \hat{y}^1d\hat{y}^1 + \hat{y}^2d\hat{y}^2, & \hat{r}^2d\hat{\varphi} &= \hat{y}^1d\hat{y}^2 - \hat{y}^2d\hat{y}^1, \\
 \tilde{r}d\tilde{r} &= \tilde{y}^1d\tilde{y}^1 + \tilde{y}^2d\tilde{y}^2, & \tilde{r}^2d\tilde{\varphi} &= \tilde{y}^1d\tilde{y}^2 - \tilde{y}^2d\tilde{y}^1.
 \end{aligned}$$

Introducing $(x^\mu) = (t, \hat{y}^1, \hat{y}^2, \tilde{y}^1, \tilde{y}^2)$, (5.3.30) gives indeed an asymptotically flat metric:

$$g = \left(\eta_{\mu\nu} + O(r^{-2}) \right) dx^\mu dx^\nu.$$

5.3.5 The limits $y \rightarrow \pm\infty$

In order to understand the geometry when $y \rightarrow -\infty$, one replaces y by

$$Y = -1/y.$$

Surprisingly, the metric can be analytically extended across $\{Y = 0\}$ to negative Y : indeed, we have

$$\begin{aligned} g = & -F(x) \left[\frac{dt^2}{F(y)} + 2\sqrt{\frac{\nu}{\xi_F}} \frac{\xi_1 - y}{AF(y)} dt d\psi \right. \\ & + \frac{1}{A^2} \left(\frac{\nu(\xi_1 - y)^2}{\xi_F - y} + \frac{G(y)}{(x - y)^2} \right) d\psi^2 + \frac{F(y)y^4}{A^2(x - y)^2 G(y)} dY^2 \Big] \\ & + \frac{F^2(y)}{A^2(x - y)^2} \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \right) \\ & \xrightarrow{y \rightarrow -\infty} \\ & -F(x) \left[2\frac{\sqrt{\nu\xi_F}}{A} dt d\psi - \frac{2\nu\xi_1 + 2\nu x - 1 - \nu\xi_F}{A^2} d\psi^2 + \frac{1}{A^2\nu\xi_F} dY^2 \right] \\ & + \frac{1}{A^2\xi_F^2} \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \right) \Big]. \end{aligned} \quad (5.3.31)$$

Calculating directly, or using (5.3.18) and the transformation law for $\det g$, one has

$$\det g = -\frac{F^2(x)F^4(y)y^4}{A^8(x - y)^8} \xrightarrow{y \rightarrow -\infty} -\frac{F^2(x)}{A^8\xi_F^4}, \quad (5.3.32)$$

which shows that the metric remains non-degenerate up to $\{Y = 0\}$. Further, one checks that all functions in (5.3.31) extend analytically to small negative Y ; e.g.,

$$g(\partial_t, \partial_t) = g_{tt} = -\frac{F(x)}{F(y)} = -\frac{\xi_F - x}{\xi_F - y} = -\frac{(\xi_F - x)Y}{Y\xi_F + 1}, \quad (5.3.33)$$

etc.

To take advantage of the work done so far, in the region $Y < 0$ we replace Y with a new coordinate

$$z = -Y^{-1} > 0,$$

obtaining a metric which has the same form as (5.3.1):

$$\begin{aligned} g = & -\frac{F(x)}{F(z)} \left(dt + \sqrt{\frac{\nu}{\xi_F}} \frac{\xi_1 - z}{A} d\psi \right)^2 \\ & + \frac{F(z)}{A^2(x - z)^2} \left[-F(x) \left(\frac{dz^2}{G(z)} + \frac{G(z)}{F(z)} d\psi^2 \right) \right. \\ & \left. + F(z) \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \right) \right]. \end{aligned} \quad (5.3.34)$$

By continuity, or by (5.3.19), the signature remains Lorentzian, and (taking into account our previous analysis of the zeros of $G(x)$) the metric is manifestly regular in the range

$$\xi_3 < z < \infty. \quad (5.3.35)$$

5.3.6 Ergoregion

Note that the ‘stationary’ Killing vector

$$X := \partial_t,$$

which was timelike in the region $Y > 0$, is now spacelike in view of (5.3.33). In analogy with the Kerr solution, the part of the region where X is spacelike which lies outside of the black-hole is called an *ergoregion*. (The fact that the region $\{\xi_3 < z\}$ lies outside of the black hole will be justified shortly.)

Since $g(X, X) = 0$ on the hypersurface $\{Y = 0\}$, this hypersurface is part of the boundary of the ergoregion, and the question arises whether this is a Killing horizon. Recall that, by definition, a Killing vector X is *normal* to its Killing horizon; in other words, it is orthogonal to every vector tangent to the Killing horizon (compare Appendix E). But, from (5.3.34), we find that

$$\begin{aligned} g(\partial_t, \partial_\psi) &= -\sqrt{\frac{\nu}{\xi_F}} \frac{F(x)(\xi_1 - z)}{AF(z)} \\ &= -\sqrt{\frac{\nu}{\xi_F}} \frac{F(x)(\xi_1 - z)\xi_F}{A(\xi_F - z)} \\ &= -\sqrt{\frac{\nu}{\xi_F}} \frac{F(x)(\xi_1 + Y^{-1})\xi_F}{A(\xi_F + Y^{-1})} \\ &= -\sqrt{\frac{\nu}{\xi_F}} \frac{F(x)(\xi_1 Y + 1)\xi_F}{A(\xi_F Y + 1)} \xrightarrow{Y \rightarrow 0} -\sqrt{\frac{\nu}{\xi_F}} \frac{F(x)\xi_F}{A}. \end{aligned}$$

Since ∂_ψ is tangent to $\{Y = 0\}$, and since this last expression is not identically zero, we conclude that ∂_t is *not* normal to $\{Y = 0\}$. Hence $\{Y = 0\}$ is *not* a Killing horizon. Now, the part of the boundary of an ergoregion which lies outside the black hole is called an *ergosurface*. In the current case its topology is $S^1 \times S^2$: the factor S^1 corresponds to the rotations generated by ψ , and the factor S^2 to the spheres coordinatized by x and φ . Note that in the Kerr solution the ergosurface ‘touches’ the event horizon at the axis of rotation, while here the event horizon and the ergosurface are separated by an open set.

5.3.7 Black ring

The metric (5.3.34) has a problem at $z = \xi_3$ because $G(\xi_3) = 0$. We have already shown how to solve that in regions where F was positive, but now $F(z) < 0$ so the previous analysis does not apply. Instead we replace ψ with a new (periodic) coordinate χ defined as

$$d\chi = d\psi + \frac{\sqrt{-F(z)}}{G(z)} dz. \quad (5.3.36)$$

However, this coordinate transformation wreaks havoc in the first line of (5.3.40). This is fixed if we replace t with a new coordinate v :

$$dv = dt + \sqrt{\frac{\nu}{\xi_F}} (z - \xi_1) \frac{\sqrt{-F(z)}}{AG(z)} dz. \quad (5.3.37)$$

INCIDENTALLY: The integrals above can be evaluated explicitly; for example, in (5.3.37) we have

$$\int \frac{\sqrt{z - \xi_F}}{(z - \xi_2)(z - \xi_3)} dz = \frac{\sqrt{\xi_3 - \xi_F}}{\xi_3 - \xi_2} \ln(z - \xi_3) + H(z), \quad (5.3.38)$$

where H is an analytic function defined in (ξ_F, ∞) :

$$H(z) = \frac{2}{\xi_3 - \xi_2} \left[\sqrt{\xi_F - \xi_2} \arctan \left(\sqrt{\frac{z - \xi_F}{\xi_F - \xi_2}} \right) - \sqrt{\xi_3 - \xi_F} \ln \left(\sqrt{z - \xi_F} + \sqrt{\xi_3 - \xi_F} \right) \right]. \quad (5.3.39)$$

□

In the (v, x, z, χ, φ) -coordinates the metric takes the form

$$\begin{aligned} ds^2 = & -\frac{F(x)}{F(z)} \left(dv - \sqrt{\frac{\nu}{\xi_F}} \frac{z - \xi_1}{A} d\chi \right)^2 \\ & + \frac{1}{A^2(x - z)^2} \left[F(x) \left(-G(z) d\chi^2 + 2\sqrt{-F(z)} d\chi dz \right) \right. \\ & \left. + F(z)^2 \left(\frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\varphi^2 \right) \right]. \end{aligned} \quad (5.3.40)$$

This is regular at

$$\mathcal{E} := \{z = \xi_3\},$$

and the metric can be analytically continued into the region $\xi_F < z \leq \xi_3$. One can check directly from (5.3.40) that $g(\nabla z, \nabla z)$ vanishes at $\mathcal{E}$. However, it is simplest to use (5.3.34) to obtain

$$g(\nabla z, \nabla z) = g^{zz} = -\frac{A^2(x - z)^2 G(z)}{F(x)F(z)} \quad (5.3.41)$$

in the region $\{z > \xi_3\}$, and to invoke analyticity to conclude that this equation remains valid on $\{z > \xi_F\}$. Equation (5.3.41) shows that $\mathcal{E}$ is a null hypersurface, with z being a time function on $\{z < \xi_3\}$, which is contained in a black-hole region by the usual arguments (compare the paragraph around (4.6.14), p. 178).

Anticipating, we will show in Section 5.3.10, p. 237, that a subset of $\mathcal{E}$ in a natural extension forms the event horizon, with topology $\mathbb{R} \times S^1 \times S^2$: this is a ‘black ring’.

5.3.8 Some further properties

It follows from (5.3.40) that the Killing vector field

$$\zeta = \frac{\partial}{\partial v} + \frac{A\sqrt{\xi_F}}{\sqrt{\nu}(\xi_3 - \xi_1)} \frac{\partial}{\partial \chi} = \frac{\partial}{\partial t} + \frac{A\sqrt{\xi_F}}{\sqrt{\nu}(\xi_3 - \xi_1)} \frac{\partial}{\partial \psi} \quad (5.3.42)$$

is light-like at $\mathcal{E}$, which is therefore a *Killing horizon*. Equation (5.3.42) shows that the horizon is ‘rotating’, with angular velocity

$$\Omega_{\mathcal{E}} = \frac{A\sqrt{\xi_F}}{\lambda(\xi_3 - \xi_1)\sqrt{\nu}} = \frac{A\sqrt{\nu}\xi_F(\xi_2 - \xi_1)}{2\sqrt{\xi_F - \xi_1}}; \quad (5.3.43)$$

recall that λ has been defined in (5.3.15). More precisely, in the coordinate system (v, χ, z, x, φ) the generators of the horizon are the curves

$$s \mapsto (v + s, \chi + \lambda\Omega_{\mathcal{E}}s, \xi_3, x, \varphi).$$

We wish, next, to calculate the surface gravity of the Killing horizon $\mathcal{E}$. For this we start by noting that

$$\begin{aligned} \zeta^\flat &:= g_{\mu\nu} \zeta^\mu dx^\nu = g_{v\mu} dx^\mu + \lambda\Omega_{\mathcal{E}} g_{\chi\nu} dx^\nu \\ &= -\frac{F(x)}{F(z)} \left(1 - \lambda\Omega_{\mathcal{E}} \sqrt{\frac{\nu}{\xi_F}} \frac{z - \xi_1}{A} \right) \left(dv - \sqrt{\frac{\nu}{\xi_F}} \frac{z - \xi_1}{A} d\chi \right) \end{aligned}$$

$$\begin{aligned}
 & + \frac{1}{A^2(x-z)^2} \lambda \Omega_{\mathcal{E}} F(x) \left(-G(z) d\chi + \sqrt{-F(z)} dz \right) \\
 & = - \frac{F(x)(\xi_3 - z)}{F(z)(\xi_3 - \xi_1)} \left(dv - \sqrt{\frac{\nu}{\xi_F}} \frac{z - \xi_1}{A} d\chi \right) \\
 & + \frac{1}{A^2(x-z)^2} \lambda \Omega_{\mathcal{E}} F(x) \left(-G(z) d\chi + \sqrt{-F(z)} dz \right) \\
 & = \Big|_{z=\xi_3} \frac{\lambda \Omega_{\mathcal{E}} F(x) \sqrt{-F(\xi_3)}}{A^2(x - \xi_3)^2} dz, \tag{5.3.44}
 \end{aligned}$$

$$\begin{aligned}
 g(\zeta, \zeta) & = - \frac{F(x)}{F(z)} \left(1 - \lambda \Omega_{\mathcal{E}} \sqrt{\frac{\nu}{\xi_F}} \frac{z - \xi_1}{A} \right)^2 - \frac{\lambda^2 \Omega_{\mathcal{E}}^2 F(x) G(z)}{A^2(x-z)^2} \\
 & = - \frac{F(x)(\xi_3 - z)^2}{F(z)(\xi_3 - \xi_1)^2} - \frac{\lambda^2 \Omega_{\mathcal{E}}^2 F(x) G(z)}{A^2(x-z)^2}, \\
 d(g(\zeta, \zeta))|_{z=\xi_3} & = - \frac{\lambda^2 \Omega_{\mathcal{E}}^2 F(x)}{A^2(x - \xi_3)^2} G'(\xi_3) dz = -2\kappa \xi^b. \tag{5.3.45}
 \end{aligned}$$

Comparing (5.3.44) with (5.3.45) we conclude that

$$\kappa = \frac{\lambda \Omega_{\mathcal{E}} G'(\xi_3)}{2\sqrt{-F(\xi_3)}} = \frac{A\sqrt{\nu}}{2} \frac{\xi_F(\xi_3 - \xi_2)}{\sqrt{\xi_3 - \xi_F}}. \tag{5.3.46}$$

Since $\kappa \neq 0$, one can further extend the spacetime obtained so far to one which contains a bifurcate Killing horizon, and a white-hole region; we present the construction in Section 5.3.9. The global structure of the resulting spacetime resembles somewhat that of the Kruskal–Szekeres extension of the Schwarzschild solution.

The plot of Ω_H and κ (as well as some other quantities of geometric interest) in terms of ν can be found in Fig. 5.3.2.

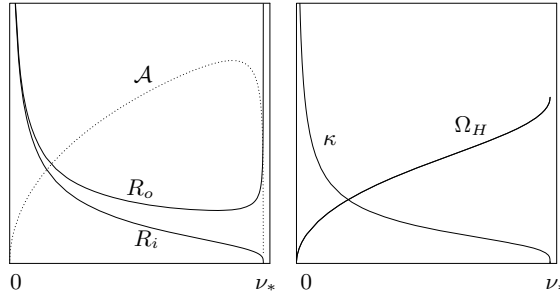


Fig. 5.3.2 Plots, as functions of ν at fixed total mass m , of the radius of curvature R_i at $x = \xi_2$ of the S^1 factor of the horizon, the curvature radius R_o at $x = \xi_1$, total area $\mathcal{A}$ of the ring, surface gravity κ , and angular velocity at the horizon Ω_H . All quantities are rendered dimensionless by dividing by an appropriate power of m . Reprinted with permission from [183]. © Copyright 2002 by the American Physical Society.

It is essential to understand the nature of the orbits of the isometry group, e.g. to make sure that the domain of outer communications does not contain any closed timelike curves. We have the following:

- The Killing vector ∂_t is timelike iff

$$F(y) > 0 \iff y < \xi_F.$$

- The Killing vector ∂_φ is always spacelike.
- From (5.3.1) we have

$$g(\partial_\psi, \partial_\psi) = \frac{\nu F(x)(\xi_1 - y)}{A^2(x - y)^2(\xi_F - y)} \times \underbrace{\left((\xi_F - y)(\xi_2 - y)(\xi_3 - y) - (\xi_1 - y)(x - y)^2 \right)}_{(*)}. \quad (5.3.47)$$

For $y < \xi_1$ we can write

$$\underbrace{(\xi_F - y)}_{\geq (x-y)} \underbrace{(\xi_2 - y)}_{> (\xi_1 - y)} \underbrace{(\xi_3 - y)}_{> (x-y)} > (\xi_1 - y)(x - y)^2,$$

which leads to $g_{\psi\psi} \geq 0$. Similarly, for $y > \xi_3$,

$$\underbrace{(y - \xi_F)}_{\leq (y-x)} \underbrace{(y - \xi_2)}_{< (y-\xi_1)} \underbrace{(y - \xi_3)}_{< (y-x)} < -(\xi_1 - y)(x - y)^2,$$

so that ∂_ψ is spacelike or vanishing throughout the domain of outer communications.

- The metric induced on the level sets of t has the form

$$g_{yy}dy^2 + g_{\psi\psi}d\psi^2 + g_{xx}dx^2 + g_{\varphi\varphi}d\varphi^2. \quad (5.3.48)$$

We have just seen that $g_{\psi\psi}$ is non-negative, and g_{xx} and $g_{\varphi\varphi}$ also are for x in the range (5.3.4). Further

$$g_{yy} = -\frac{F(x)F(y)}{A^2(x - y)^2G(y)} = \frac{F(x)}{A^2(x - y)^2\xi_F\nu} \times \frac{(y - \xi_F)}{(y - \xi_1)(y - \xi_2)(y - \xi_3)},$$

an expression which is again non-negative in the ranges of interest. It follows that the hypersurfaces $\{t = \text{const}\}$ are spacelike.

The main topological features of the manifold $\mathcal{M}$ constructed so far are summarized in Fig. 5.3.3, see also Fig. 5.3.4. One thus finds that

$$\mathcal{M} = \mathbb{R} \times \left[\left(\mathbb{R}^2 \times S^2 \right) \setminus \underbrace{(\vec{0}, N)}_{=: i^o} \right],$$

where $\vec{0}$ is the origin of $\mathbb{R}^2$, and N is the north pole of S^2 , with the first $\mathbb{R}$ factor corresponding to time. The point i^o which has been removed from the $\mathbb{R}^2 \times S^2$ factor can be thought of as representing ‘spatial infinity’.

INCIDENTALLY: The metric, say h , induced on the sections of the horizon $\{v = \text{const}, z = \xi_3\}$ can be obtained from (5.3.48) by first neglecting the dy^2 terms, and then passing to the limit $y \rightarrow \xi_3$. (By general arguments, or by a direct calculation starting from (5.3.40), this coincides with the metric of the sections $\{v = \text{const}\}$ of the event horizon $\mathcal{E}$.) One finds that

$$h = \frac{\lambda^2\nu(\xi_F - x)(\xi_3 - \xi_1)^2}{\xi_F A^2(\xi_3 - \xi_F)} d\varphi^2 + \frac{F^2(\xi_3)}{A^2(x - \xi_3)^2} \left(\frac{dx^2}{G(x)} + \frac{\lambda^2 G(x)}{F(x)} d\tilde{\varphi}^2 \right),$$

so that (recall (5.3.15))

$$\sqrt{\det h} = \frac{\lambda^2\nu^{1/2}(\xi_3 - \xi_F)^{3/2}(\xi_3 - \xi_1)}{\xi_F^3 A^3(x - \xi_3)^2} = \frac{4(\xi_3 - \xi_F)^{3/2}(\xi_F - \xi_1)}{A^3\nu^{3/2}\xi_F^3(\xi_3 - \xi_1)^2(\xi_2 - \xi_1)^2} \times \frac{1}{(x - \xi_3)^2}.$$

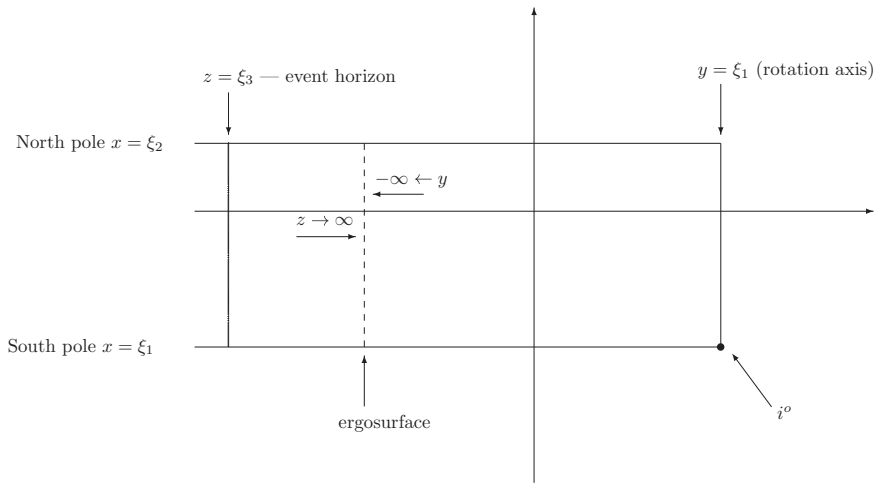


Fig. 5.3.3 Space sections of the Emparan–Reall black holes, with the angular variables φ and ψ suppressed. The x variable runs along the vertical axis, the y variable runs along the horizontal axis to the right of the ergosurface, while the z coordinate is used horizontally to the left of the ergosurface. i^o is the point at infinity.

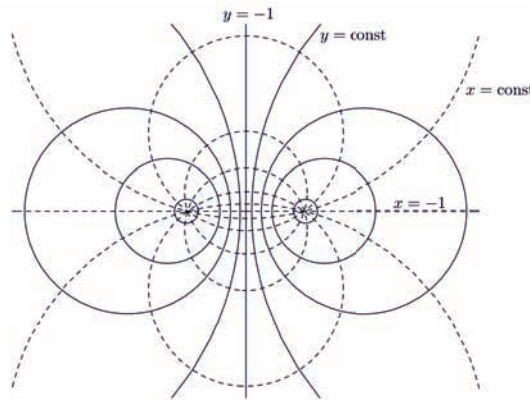


Fig. 5.3.4 Ring coordinates for flat four-dimensional space, on a section at constant ϕ and ψ (and $\phi + \pi$, $\psi + \pi$). Dashed circles correspond to spheres at constant $|x| \in [0, 1]$, solid circles to spheres at constant $y \in [-\infty, -1]$. Spheres at constant y collapse to zero size at the location of the ring of radius R , $y = -\infty$. The disk bounded by the ring is an axis for ∂_ϕ at $x = +1$. © IOP Publishing. Reproduced with permission. All rights reserved.

By integration in $x \in (\xi_1, \xi_2)$ and in the angular variables $\tilde{\varphi}, \hat{\varphi} \in (0, 2\pi)$ one obtains the area of the sections of the horizon:

$$\mathcal{A} = \frac{16\pi^2}{A^3\nu^{3/2}} \frac{(\xi_3 - \xi_F)^{3/2}(\xi_F - \xi_1)}{\xi_F^3(\xi_3 - \xi_2)(\xi_2 - \xi_1)(\xi_3 - \xi_1)^2}. \quad (5.3.49)$$

□

If $\nu = \nu_*$ then the black ring and the black hole degenerate to the same solution with $\xi_2 = \xi_F = \xi_3$. This is the $\mu = a^2$ limit of the five-dimensional rotating black hole, for which the horizon disappears, and is replaced by a naked singularity.

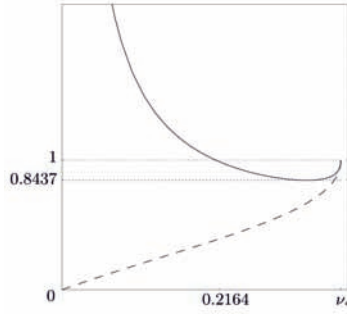


Fig. 5.3.5 $(27\pi/32)J^2/m^3$ as a function of the parameter ν . The solid line corresponds to the Emparan–Reall solutions, the dashed line to the Myers–Perry black holes. The two dotted lines delimit the values for which both solutions with the same mass and spin exist. Reprinted with permission from [183]. © Copyright 2002 by the American Physical Society.

The mass m and the angular momentum J can be calculated using Komar integrals:

$$m = \frac{3\pi}{2A^2} \frac{\xi_F - \xi_1}{\nu \xi_1^2 (\xi_2 - \xi_1)(\xi_3 - \xi_1)}, \quad (5.3.50)$$

$$J = \frac{2\pi}{A^3} \frac{(\xi_F - \xi_1)^{5/2}}{\nu^{3/2} \xi_F^3 (\xi_2 - \xi_1)^2 (\xi_3 - \xi_1)^2}. \quad (5.3.51)$$

Thus, m and J are rather complicated functions of the independent parameters A and ν in view of (5.3.16).

An important family of black-hole metrics, not discussed in this work, are the Myers–Perry five-dimensional black holes with spin bounded from above [371]:

$$\frac{J^2}{m^3} < \frac{32}{27\pi}. \quad (5.3.52)$$

The corresponding ratio for the solutions here is

$$\frac{J^2}{m^3} = \frac{32}{27\pi} \frac{(\xi_3 - \xi_1)^3}{(2\xi_3 - \xi_1 - \xi_2)^2 (\xi_2 - \xi_1)}. \quad (5.3.53)$$

These ratios are plotted as a function of ν in Fig. 5.3.5. Rather surprisingly, this ratio is bounded *from below*:

$$\frac{J^2}{m^3} > 0.8437 \frac{32}{27\pi}. \quad (5.3.54)$$

For $0.2164 < \nu < \nu_*$, there are *two* black ring solutions with the same values of m and J (but different A). Moreover, these satisfy the bound (5.3.52) so there is also a black hole with the same values of m and J . This shows that the uniqueness theorems valid in four dimensions, which classify stationary well-behaved vacuum black holes in terms of mass and angular momentum, do not have a simple generalization to five dimensions; compare [255].

Some algebra shows that the quantities m, J, Ω_H, κ and A satisfy a *Smarr relation*

$$m = \frac{3}{2} \left(\frac{\kappa A}{8\pi} + \Omega_H J \right). \quad (5.3.55)$$

5.3.9 A Kruskal–Szekeres type extension

So far we have seen how to construct an Eddington–Finkelstein-type extension of the ER metric (5.3.1) across a Killing horizon. We will denote by $(\mathcal{M}_{I \cup II}, g)$ that extension. We will denote by $(\mathcal{M}_I, g)$ the subset of $(\mathcal{M}_{I \cup II}, g)$ exterior to the Killing horizon; as in Fig. 5.3.6, p. 238.

To construct a Kruskal–Szekeres-type extension we follow [113] and consider, first, the form of the metric (5.3.34) with the coordinate z in the range $z \in (\xi_3, \infty)$. There we define new coordinates w, v by the formulae

$$dv = dt + \frac{bdz}{(z - \xi_3)(z - \xi_0)}, \quad dw = dt - \frac{bdz}{(z - \xi_3)(z - \xi_0)}, \quad (5.3.56)$$

where b and ξ_0 are constants to be chosen shortly. Similarly to the construction of the extension of the Kerr metric in [62, 79], we define a new angular coordinate $\hat{\psi}$ by

$$d\hat{\psi} = d\psi - a dt, \quad (5.3.57)$$

where a is a constant to be chosen later. Let

$$\sigma := \frac{1}{A} \sqrt{\frac{\nu}{\xi_F}}. \quad (5.3.58)$$

Using (5.3.56)–(5.3.57), we obtain

$$dt = \frac{1}{2}(dv + dw), \quad (5.3.59)$$

$$dz = \underbrace{\frac{(z - \xi_3)(z - \xi_0)}{2b}}_{=: H(z)/2} (dv - dw), \quad d\psi = d\hat{\psi} + \frac{a}{2}(dv + dw), \quad (5.3.60)$$

which leads to

$$g_{vv} = g_{ww} = -\frac{F(x)}{4F(z)} \left(1 + a\sigma(\xi_1 - z)\right)^2 - \frac{F(x)F(z)}{4A^2(x - z)^2} \left(\frac{a^2 G(z)}{F(z)} + \frac{H^2(z)}{G(z)}\right), \quad (5.3.61)$$

$$g_{vw} = -\frac{F(x)}{4F(z)} \left(1 + a\sigma(\xi_1 - z)\right)^2 - \frac{F(x)F(z)}{4A^2(x - z)^2} \left(\frac{a^2 G(z)}{F(z)} - \frac{H^2(z)}{G(z)}\right), \quad (5.3.62)$$

$$g_{v\hat{\psi}} = g_{w\hat{\psi}} = -\frac{F(x)}{2F(z)} \sigma(\xi_1 - z) \left(1 + a\sigma(\xi_1 - z)\right) - \frac{F(x)G(z)a}{2A^2(x - z)^2}, \quad (5.3.63)$$

$$g_{\hat{\psi}\hat{\psi}} = -\frac{F(x)}{F(z)} \sigma^2(\xi_1 - z)^2 - \frac{F(x)G(z)}{A^2(x - z)^2}. \quad (5.3.64)$$

The Jacobian of the coordinate transformation is

$$\frac{\partial(w, v, \hat{\psi}, x, \varphi)}{\partial(t, z, \psi, x, \varphi)} = 2 \frac{\partial v}{\partial z} = \frac{2b}{(z - \xi_2)(z - \xi_3)}.$$

In the original coordinates (t, z, ψ, x, φ) the determinant of g was

$$\det(g_{(t, z, \psi, x, \varphi)}) = -\frac{F^2(x)F^4(z)}{A^8(x - z)^8}, \quad (5.3.65)$$

so that in the new coordinates it reads

$$\det(g_{(w, v, \hat{\psi}, x, \varphi)}) = -\frac{F^2(x)F^4(z)(z - \xi_2)^2(z - \xi_3)^2}{4A^8 b^2(x - z)^8}. \quad (5.3.66)$$

This last expression is negative on $(\xi_F, \infty) \setminus \{\xi_3\}$, and has a second-order zero at $z = \xi_3$. In order to remove this degeneracy one introduces

$$\hat{v} = \exp(cv), \quad \hat{w} = -\exp(-cw), \quad (5.3.67)$$

where c is some constant to be chosen. Hence we have

$$d\hat{v} = c\hat{v}dv, \quad d\hat{w} = -c\hat{w}dw, \quad (5.3.68)$$

and the determinant in the coordinates $(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)$ reads

$$\det(g_{(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)}) = -\frac{F^2(x)F^4(z)(z - \xi_2)^2(z - \xi_3)^2}{4A^8b^2(x - z)^8c^4\hat{v}^2\hat{w}^2}. \quad (5.3.69)$$

But one has $\hat{v}^2\hat{w}^2 = \exp(2c(v - w))$, so that

$$\begin{aligned} \hat{v}^2\hat{w}^2 &= \exp\left(4cb \int \frac{1}{(z - \xi_2)(z - \xi_3)} dz\right) \\ &= \exp\left(\frac{4cb}{(\xi_3 - \xi_2)}(\ln(z - \xi_3) - \ln(z - \xi_2))\right). \end{aligned} \quad (5.3.70)$$

Taking into account (5.3.70), and the determinant (5.3.69), we choose the constant c to satisfy

$$\frac{2cb}{(\xi_3 - \xi_2)} = 1. \quad (5.3.71)$$

We obtain

$$\hat{v}\hat{w} = -\frac{z - \xi_3}{z - \xi_2}, \quad (5.3.72)$$

and

$$\det(g_{(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)}) = -\frac{F^2(x)F^4(z)(z - \xi_2)^4}{4A^8b^2(x - z)^8c^4}. \quad (5.3.73)$$

With this choice, the determinant of g in the $(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)$ coordinates extends analytically to a negative function on $\{z \in (\xi_F, \infty)\}$. In fact, z is an analytic function of $\hat{v}\hat{w}$ on $\{\hat{v}\hat{w} \neq -1\}$ (that last set corresponds to $z = \infty \Leftrightarrow Y = 0$; we will return to this shortly):

$$z = \frac{\xi_3 + \xi_2\hat{v}\hat{w}}{1 + \hat{v}\hat{w}}. \quad (5.3.74)$$

In the $(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)$ coordinates, one obtains the coefficients of the metric from (5.3.68) using

$$\begin{aligned} g_{\hat{v}\hat{v}} &= \frac{1}{c^2\hat{v}^2}g_{vv}, & g_{\hat{w}\hat{w}} &= \frac{1}{c^2\hat{w}^2}g_{ww}, \\ g_{\hat{v}\hat{w}} &= -\frac{1}{c^2\hat{v}\hat{w}}g_{vw}, & g_{\hat{v}\hat{\psi}} &= \frac{1}{c\hat{v}}g_{v\psi}, & g_{\hat{w}\hat{\psi}} &= -\frac{1}{c\hat{w}}g_{w\psi}. \end{aligned} \quad (5.3.75)$$

In order to show that the coefficients of the metric are analytic on the set

$$\left\{\hat{w}, \hat{v} \mid z(\hat{v}\hat{w}) > \xi_F\right\} = \left\{\hat{w}, \hat{v} \mid -1 < \hat{v}\hat{w} < \frac{\xi_3 - \xi_F}{\xi_F - \xi_2}\right\} \quad (5.3.76)$$

it is convenient to write

$$\begin{aligned} g_{\hat{v}\hat{v}} &= \frac{1}{c^2\hat{v}^2\hat{w}^2}\hat{w}^2g_{vv}, & g_{\hat{w}\hat{w}} &= \frac{1}{c^2\hat{v}^2\hat{w}^2}\hat{v}^2g_{ww}, \\ g_{\hat{v}\hat{w}} &= -\frac{1}{c^2\hat{v}\hat{w}}g_{vw}, & g_{\hat{v}\hat{\psi}} &= \frac{1}{c\hat{v}\hat{w}}\hat{w}g_{v\psi}, & g_{\hat{w}\hat{\psi}} &= -\frac{1}{c\hat{v}\hat{w}}\hat{v}g_{w\psi}. \end{aligned} \quad (5.3.77)$$

Hence, to make sure that all the coefficients of metric are well behaved at $\{\hat{w}, \hat{v} \in \mathbb{R} \mid z = \xi_3\}$ (i.e. $\hat{v} = 0$ or $\hat{w} = 0$), it suffices to check that there is a multiplicative

factor $(z - \xi_3)^2$ in $g_{vv} = g_{ww}$, as well as a multiplicative factor $(z - \xi_3)$ in g_{vw} and in $g_{v\hat{\psi}} = g_{w\hat{\psi}}$. In view of (5.3.61)–(5.3.64), one can see that this will be the case if, first, a is chosen so that $1 + a\sigma(\xi_1 - z) = a\sigma(\xi_3 - z)$, that is

$$a = \frac{1}{\sigma(\xi_3 - \xi_1)}, \quad (5.3.78)$$

and then, if ξ_0 and b are chosen, such that

$$0 = -\frac{a^2\nu\xi_F(\xi_3 - \xi_1)(\xi_3 - \xi_2)}{\xi_3 - \xi_F} + \frac{(\xi_3 - \xi_0)^2}{\nu b^2(\xi_3 - \xi_1)(\xi_3 - \xi_2)}. \quad (5.3.79)$$

With the choice $\xi_0 = \xi_2$, (5.3.79) will hold if we set

$$b^2 = \frac{(\xi_3 - \xi_F)}{\nu^2 a^2 \xi_F (\xi_3 - \xi_1)^2}. \quad (5.3.80)$$

So far we have been focusing on the region $z \in (\xi_F, \infty)$, which overlaps only with part of the manifold ‘ $\{z \in (\xi_3, \infty] \cup [-\infty, \xi_1]\}$ ’. A well-behaved coordinate on that last region is $Y = -1/z$. This allows one to go smoothly through $Y = 0$ in (5.3.72):

$$\hat{v}\hat{w} = -\frac{1 + \xi_3 Y}{1 + \xi_2 Y} \iff Y = -\frac{1 + \hat{v}\hat{w}}{\xi_3 + \xi_2 \hat{v}\hat{w}}. \quad (5.3.81)$$

In other words, $\hat{v}\hat{w}$ extends analytically to the region of interest, $0 \leq Y \leq -1/\xi_1$ (and in fact beyond, but this is irrelevant to us). Similarly, the determinant

$$\det(g_{(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)})$$

extends analytically across $Y = 0$, being the ratio of two polynomials of order eight in z (equivalently, in Y), with limit

$$\det(g_{(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi)}) \rightarrow_{z \rightarrow \infty} -\frac{F^2(x)}{4A^8 b^2 c^4 \xi_F^4}. \quad (5.3.82)$$

We conclude that the construction so far produces an analytic Lorentzian metric on the set

$$\hat{\Omega} := \left\{ \hat{w}, \hat{v} \mid -\frac{\xi_3 - \xi_1}{\xi_2 - \xi_1} \leq \hat{v}\hat{w} < \frac{\xi_3 - \xi_F}{\xi_F - \xi_2} \right\} \times S_{\hat{\psi}}^1 \times S_{(x, \varphi)}^2, \quad (5.3.83)$$

where a subscript on S^k points to the names of the corresponding local variables.

The map

$$(\hat{w}, \hat{v}, \hat{\psi}, x, \varphi) \mapsto (-\hat{w}, -\hat{v}, -\hat{\psi}, x, -\varphi) \quad (5.3.84)$$

is an orientation-preserving analytic isometry of the analytically extended metric on $\hat{\Omega}$. It follows that the manifold

$$\hat{\mathcal{M}}$$

obtained by gluing together $\hat{\Omega}$ and two isometric copies of $(\mathcal{M}_I, g)$ can be equipped with the obvious Lorentzian metric, denoted by $\hat{g}$, which is furthermore analytic. The second copy of $(\mathcal{M}_I, g)$ will be denoted by $(\mathcal{M}_{III}, g)$; compare Fig. 5.3.6. The reader should keep in mind the polar character of the coordinates around the relevant axes of rotation, and the special character of the ‘point at infinity’ $z = \xi_1 = x$.

5.3.10 Global structure

Our discussion of the global structure of $(\hat{\mathcal{M}}, \hat{g})$ follows closely [113]. The reader will also find there a (rather involved) proof of global hyperbolicity of $\hat{\mathcal{M}}$.

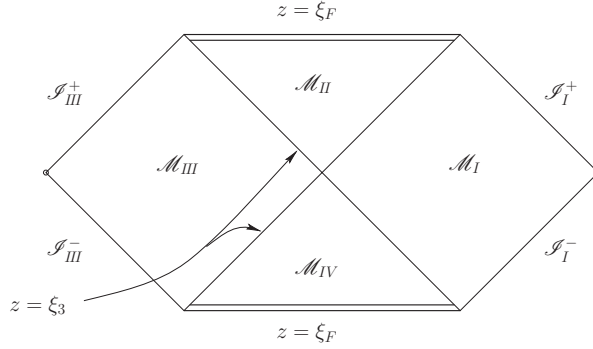


Fig. 5.3.6 $\hat{\mathcal{M}}$ with its various subsets. For example, $\mathcal{M}_{I \cup II}$ is the union of $\mathcal{M}_I$ and of $\mathcal{M}_{II}$ and of that part of $\{z = \xi_3\}$ which lies in the intersection of their closures; this is the manifold constructed in [183]. Very roughly speaking, the various $\mathcal{J}$'s correspond to $x = z = \xi_1$. It should be stressed that this is neither a *conformal* diagram, nor is the spacetime a product of the figure times $S^2 \times S^1$: $\mathcal{M}_I$ cannot be the product of the depicted diamond with $S^2 \times S^1$, as this product is not simply connected, while $\mathcal{M}_I$ is. But the diagram represents accurately the causal relations between the various $\mathcal{M}_N$'s, as well as the geometry near the bifurcate horizon $z = \xi_F$, as the manifold *does have* a product structure there. Reprinted from [113], with permission.

The event horizon has $S^2 \times S^1 \times \mathbb{R}$ topology. The analysis in Section 5.3.9 shows that the set

$$\mathcal{E} := \{z = \xi_3\}$$

is a bifurcate Killing horizon in $\hat{\mathcal{M}}$. In this section we wish to show that a subset of $\mathcal{E}$ is actually a black-hole event horizon, with $S^2 \times S^1 \times \mathbb{R}$ topology. For this, note first that

$$g(\nabla z, \nabla z) = g^{zz} = -\frac{A^2(x-z)^2 G(z)}{F(x)F(z)} \quad (5.3.85)$$

in the region $\{z > \xi_3\}$, and by analyticity this equation remains valid on $\{z > \xi_F\}$. Equation (5.3.85) shows that $\mathcal{E}$ is a null hypersurface, with z being a time function on $\{\xi_F < z < \xi_3\}$. The usual choice of time orientation implies that z is strictly decreasing along future-directed causal curves in the region $\{\hat{v} > 0, \hat{w} > 0\}$, and strictly increasing along such curves in the region $\{\hat{v} < 0, \hat{w} < 0\}$. In particular no causal future-directed curve can leave the region $\{\hat{v} > 0, \hat{w} > 0\}$. Hence the spacetime contains a black-hole region.

However, it is not clear that $\mathcal{E}$ is the *event horizon* within the Emparan–Reall spacetime $(\mathcal{M}_{I \cup II}, g)$, because the actual event horizon could be enclosing the region $z < \xi_3$. To show that this is not the case, consider the ‘area function’, defined as the determinant, say W , of the matrix

$$g(K_i, K_j),$$

where the K_i 's, $i = 1, 2, 3$, are the Killing vectors ∂_t , ∂_ψ , and ∂_φ in the asymptotically flat region. In the coordinates of (5.3.34) this equals

$$\frac{F(x)G(x)F(z)G(z)}{A^4(x-z)^4}. \quad (5.3.86)$$

Analyticity implies that this formula is valid throughout $\mathcal{M}_{I \cup II}$, as well as $\hat{\mathcal{M}}$. Now,

$$F(z)G(z) = \frac{\nu}{\xi_F} (\xi_F - z)(z - \xi_1)(z - \xi_2)(z - \xi_3),$$

and, in view of the range of the variable x , the sign of (5.3.86) depends only upon the values of z . Since $F(z)G(z)$ behaves as $-\nu z^4/\xi_F$ for large z , W is negative both for $z < \xi_1$ and for $z > \xi_3$. Hence, at each point p of those two regions the set of vectors in $T_p\mathcal{M}$ spanned by the Killing vectors is timelike. So, suppose for contradiction, that the event horizon $\mathcal{H}$ intersects the region $\{z \in (\xi_3, \infty]\} \cup \{z \in [-\infty, \xi_1]\}$; here ‘ $z = \pm\infty$ ’ should be understood as $Y = 0$, as already mentioned in the introduction. Since $\mathcal{H}$ is a null hypersurface invariant under isometries, every Killing vector is tangent to $\mathcal{H}$. However, at each point at which W is negative there exists a linear combination of the Killing vectors which is timelike. This gives a contradiction because no timelike vectors are tangent to a null hypersurface.

We conclude that $\{z = \xi_3\}$ forms indeed the event horizon in the spacetime $(\mathcal{M}_{I \cup II}, g)$, with topology $\mathbb{R} \times S^1 \times S^2$.

The argument just given also shows that the domain of outer communications within $(\mathcal{M}_I, g)$ coincides with $(\mathcal{M}_I, g)$.

Similarly, one finds that the domain of outer communications within $(\widehat{\mathcal{M}}, \widehat{g})$, or that within $(\mathcal{M}_{I \cup II}, g)$, associated with an asymptotic region lying in $(\mathcal{M}_I, g)$, is $(\mathcal{M}_I, g)$. The boundary of the d.o.c. in $(\widehat{\mathcal{M}}, \widehat{g})$ is a subset of the set $\{z = \xi_3\}$, which can be found by inspection of Fig. 5.3.6.

Inextendibility at $z = \xi_F$, maximality. The obvious place where $(\widehat{\mathcal{M}}, \widehat{g})$ could be enlarged is at $z = \xi_F$. To show that no extension is possible there, note that the Lorentzian length of the Killing vector ∂_t satisfies

$$g(\partial_t, \partial_t) = -\frac{F(x)}{F(z)} \rightarrow_{\xi_F < z \rightarrow \xi_F} \infty \quad (\text{recall that } F(x) \geq 1 - \frac{\xi_2}{\xi_F} > 0). \quad (5.3.87)$$

Inextendibility of the spacetime across the boundary $\{z = \xi_F\}$ follows from this and from Theorem 4.4.2, p. 169.

An alternative way, demanding somewhat more work, of proving that the ER metrics are C^2 -inextendible across $\{z = \xi_F\}$, is to note that $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ is unbounded along any curve along which z approaches ξ_F . This has been pointed out to us by Harvey Reall (private communication), and can be verified using the symbolic algebra package xACT [340]:⁶

$$R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = \frac{12A^4\xi_F^4 G(\xi_F)^2(x-z)^4(1+O(z-\xi_F))}{(\xi_F-x)^2(z-\xi_F)^6}. \quad (5.3.88)$$

(Note that the factor $(x-z)^4$ is strictly bounded away from 0 for $z \rightarrow \xi_F$.)

The following result is established in [113].

THEOREM 5.3.2 *All maximally extended causal geodesics in $(\widehat{\mathcal{M}}, \widehat{g})$ either are complete or reach a singular boundary $\{z = \xi_F\}$ in finite affine time.*

This, together with Proposition 4.4.3, p. 169 gives the following.

THEOREM 5.3.3 *$(\widehat{\mathcal{M}}, \widehat{g})$ is maximal within the class of C^2 Lorentzian manifolds.*

Conformal infinity $\mathcal{I}$. In this section we address the question of existence of conformal completions at null infinity à la Penrose, as defined in Section 3.1, p. 85, for a class of higher dimensional stationary spacetimes that includes the Emparan–Reall metrics; see the Appendix of [169] and [165] for the $(3+1)$ -dimensional case.

We start by noting that any stationary asymptotically flat spacetime which is vacuum, or electrovacuum, outside of a spatially compact set is necessarily *asymptotically Schwarzschildian*, in the sense that there exists a coordinate system in

⁶I am grateful to Alfonso Garcia-Parrado and José María Martín García for carrying out the calculation.

which the leading order terms of the metric have the Schwarzschild form, with the error terms falling off one power of r faster,

$$g = g_m + O(r^{-(n-1)}) \quad (5.3.89)$$

in spacetime dimension $n + 1$, where g_m is the Schwarzschild metric of mass m , and the size of the decay of the error terms in (5.3.89) is measured in a manifestly asymptotically Minkowskian coordinate system. The proof of this fact is outlined briefly in [41, Section 2]. In that last reference it is also shown that the remainder term has a full asymptotic expansion in terms of inverse powers of r in dimension $2k + 1$, $k \geq 3$, or in dimension $4 + 1$ for static metrics. Otherwise, the remainder is known to have an asymptotic expansion in terms of inverse powers of r and of $\ln r$, and whether there will be non-trivial logarithmic terms in the expansion is not known in general.

In higher dimensions, the question of existence of a conformal completion at null infinity is straightforward: we start by writing the $(n + 1)$ -dimensional Minkowski metric as

$$\eta = -dt^2 + dr^2 + r^2 h, \quad (5.3.90)$$

where h is the round unit metric on an $(n - 2)$ -dimensional sphere. Replacing t by the standard retarded time $u = t - r$, one is led to the following form of the metric g ,

$$g = -du^2 - 2du dr + r^2 h + O(r^{-(n-2)}) dx^\mu dx^\nu, \quad (5.3.91)$$

where the dx^μ 's are the manifestly Minkowskian coordinates $(t, x^1, \dots, x^n)$ for η . Setting $x = 1/r$ in (5.3.91) one obtains

$$g = \frac{1}{x^2} \left(-x^2 du^2 + 2du dx + h + O(x^{n-4}) dy^\alpha dy^\beta \right), \quad (5.3.92)$$

with correction terms in (5.3.92) which will extend smoothly to $x = 0$ in the coordinate system $(y^\mu) = (u, x, v^A)$, where the v^A 's are local coordinates on S^{n-2} . For example, a term $O(r^{-2}) dx^i dx^j$ in g will contribute a term

$$O(r^{-2}) dr^2 = O(r^{-2}) x^{-4} dx^2 = x^{-2} (O(1) dx^2),$$

which is bounded up to $x = 0$ after a rescaling by x^2 . The remaining terms in (5.3.92) are analysed similarly.

In dimension $4 + 1$, care has to be taken to make sure that the correction terms do not affect the signature of the metric so extended; in higher dimension this is already apparent from (5.3.92).

So, to construct a conformal completion at null infinity for the Emparan–Reall metric it suffices to verify that the determinant of the conformally rescaled metric, when expressed in the coordinates described earlier, does not vanish at $x = 0$. This is indeed the case, and can be seen by calculating the Jacobian of the map

$$(t, z, \psi, x, \varphi) \mapsto (u, x, v^A);$$

the result can then be used to calculate the determinant of the metric in the new coordinates, making use of the formula for the determinant of the metric in the original coordinates.

For a general stationary, vacuum $(4+1)$ -dimensional metric one can always transform to the coordinates, alluded to earlier, in which the metric takes a manifestly Schwarzschildian form in leading order. Instead of using $(u = t - r, x = 1/r)$ one can use coordinates $(u_m, x = 1/r)$, where u_m is the corresponding null coordinate

u for the $(4+1)$ -dimensional Schwarzschild metric. This will lead to a conformally rescaled metric with the correct signature on the conformal boundary. Note, however, that this transformation might introduce log terms in the metric, even if there were none to start with; this is why we did not use this earlier.

In summary, whenever a stationary, vacuum, $(n+1)$ -dimensional, $4 \neq n \geq 3$, asymptotically flat metric has an asymptotic expansion in terms of inverse powers of r , one is led to a smooth $\mathcal{I}$. This is the case for any such metric in dimensions $3+1$ or $2k+1$, $k \geq 3$. In the remaining dimensions one always has a polyhomogeneous conformal completion at null infinity, with a conformally rescaled metric which is C^{n-4} up to boundary. For the Emparan–Reall metric there exists a completion which has no logarithmic terms, and is thus C^∞ up to boundary.

Uniqueness and non-uniqueness of extensions. In Section 4.4.1 we have seen several constructions of distinct maximal analytic extensions of an analytic spacetime, which all apply here.

In particular the construction around (4.4.1), p. 168, applies, leading to a space-time analogous to the $\mathbb{RP}^3$ geon of Example 4.4.1. Indeed, let $(\hat{\mathcal{M}}, \hat{g})$ be our extension, as constructed earlier, of the domain of outer communication $(\mathcal{M}_I, g)$ within the Emparan–Reall spacetime $(\mathcal{M}_{I \cup II}, g)$, and let $\Psi : \hat{\mathcal{M}} \rightarrow \hat{\mathcal{M}}$ be defined as

$$\Psi(\hat{v}, \hat{w}, \hat{\psi}, x, \varphi) = (\hat{w}, \hat{v}, \hat{\psi} + \pi, x, -\varphi). \quad (5.3.93)$$

By inspection of (5.3.61)–(5.3.63) and (5.3.75), the map Ψ is an isometry, and clearly satisfies the three conditions (a), (b), and (c) spelled out in (4.4.1). Then $\hat{\mathcal{M}}/\Psi$ is a maximal, orientable, time-orientable, analytic extension of $\mathcal{M}_I$ distinct from $\hat{\mathcal{M}}$. This extension remains asymptotically flat in the usual sense but the topology of sections of the event horizon is now a non-trivial quotient of $S^1 \times S^2$, resulting from the identifications

$$(\hat{\psi}, \varphi) \sim (\hat{\psi} + \pi, -\varphi). \quad (5.3.94)$$

Further similar examples can be constructed using isometries which do not preserve orientation and/or time orientation.

On the positive side, we have the following uniqueness result for our extension $(\hat{\mathcal{M}}, \hat{g})$ of the Emparan–Reall spacetime $(\mathcal{M}_I, g)$, which follows immediately from Corollary 4.4.7, p. 171, and from the properties of causal geodesics of $(\hat{\mathcal{M}}, \hat{g})$ spelled out in Theorem 5.3.2.

THEOREM 5.3.4 *$(\hat{\mathcal{M}}, \hat{g})$ is unique within the class of simply connected analytic extensions of $(\mathcal{M}_I, g)$ which have the property that all maximally extended causal geodesics on which $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ is bounded are complete.*

As usual, uniqueness here is understood up to isometry.

The examples presented in Section 4.4.1 show that the hypotheses of Theorem 5.3.4 are optimal.

INCIDENTALLY: It is natural to raise another uniqueness question, namely of the singling-out features of the Emparan–Reall metric amongst all five-dimensional vacuum stationary black-hole spacetimes. Partial answers to this can be found in the work of Hollands and Yazadjiev [256].

5.3.11 Other coordinate systems

An alternative convenient form of the Emparan–Reall metric has been given in [184],

$$g = \frac{R^2 F(x)}{(x-y)^2} \left(\frac{dx^2}{G(x)} - \frac{dy^2}{G(y)} + \frac{G(x)}{F(x)} d\varphi^2 - \frac{G(y)}{F(y)} d\psi^2 \right)$$

$$-\frac{F(y)}{F(x)} \left(dt - \frac{CR(1+y)}{F(y)} d\psi \right)^2, \quad (5.3.95)$$

where

$$F(z) = 1 + \lambda z, \quad G(z) = (1 - z^2)(1 + \nu z), \quad C = \sqrt{\frac{\lambda(1+\lambda)(\lambda-\nu)}{1-\lambda}},$$

with

$$\lambda = \frac{2\nu}{1+\nu^2}, \quad 0 < \nu < 1.$$

If we denote by $\{\hat{t}, \hat{x}, \hat{y}, \hat{\psi}, \hat{\varphi}\}$ the original coordinates of (5.3.1), then we have the relation

$$t = \hat{t}, \quad x = \frac{\hat{\lambda} - \hat{x}}{-1 + \hat{\lambda}\hat{x}}, \quad y = \frac{\hat{\lambda} - \hat{y}}{-1 + \hat{\lambda}\hat{y}}, \quad \varphi = \frac{1 - \hat{\lambda}\hat{\nu}}{\sqrt{1 - \hat{\lambda}^2}}\hat{\varphi}, \quad \psi = \frac{1 - \hat{\lambda}\hat{\nu}}{\sqrt{1 - \hat{\lambda}^2}}\hat{\psi}, \quad (5.3.96)$$

where

$$\hat{\nu} = \frac{\nu - \lambda}{\lambda\nu - 1}, \quad \hat{\lambda} = \lambda, \quad \nu = \frac{\hat{\nu} - \hat{\lambda}}{\hat{\lambda}\hat{\nu} - 1}.$$

The transformation (5.3.96) brings the metric (5.3.95) into the form

$$g = -\frac{\hat{F}(\hat{x})}{\hat{F}(\hat{y})}(d\hat{t} + \hat{A}\sqrt{\hat{\lambda}\hat{\nu}}(1 + \hat{y})d\hat{\psi})^2 \\ + \frac{\hat{A}^2}{(\hat{x} - \hat{y})^2} \left[\hat{F}(\hat{y})^2 \left(\frac{d\hat{x}^2}{\hat{G}(\hat{x})} + \frac{\hat{G}(\hat{x})}{\hat{F}(\hat{x})} d\hat{\varphi}^2 \right) - \hat{F}(\hat{x}) \left(\frac{\hat{F}(\hat{y})}{\hat{G}(\hat{y})} d\hat{y}^2 + \hat{G}(\hat{y}) d\hat{\psi}^2 \right) \right], \quad (5.3.97)$$

where

$$\hat{F}(z) = 1 - \hat{\lambda}z, \quad \hat{G}(z) = (1 - z^2)(1 - \hat{\nu}z), \quad \hat{A} = -R\sqrt{\frac{(1 - \hat{\lambda}\hat{\nu})}{1 - \hat{\lambda}^2}}.$$

Simple rescalings and redefinitions of constants bring (5.3.97) to the form (5.3.1).

5.4 Rasheed's metrics

Kaluza–Klein spacetimes have attracted a lot of attention as providers of interesting models in theoretical physics. In vacuum, and with a vanishing cosmological constant, the simplest such solutions are obtained by taking the product of a known vacuum solution with a flat torus. A more sophisticated class of black-hole solutions of this kind has been discovered by Rasheed in [409]. The aim of this section is to analyse their geometry.⁷

The Rasheed metrics take the form

$$g = \frac{B}{A} (dx^4 + 2A_\mu dx^\mu)^2 + \sqrt{\frac{A}{B}} g_{(3)}, \quad (5.4.1)$$

where a , M , P , Q , and Σ are real numbers satisfying

$$\frac{Q^2}{\Sigma + M\sqrt{3}} + \frac{P^2}{\Sigma - M\sqrt{3}} = \frac{2\Sigma}{3}, \quad (5.4.2)$$

⁷This section is reprinted (up to minor modifications) with permission from [33, Appendix A].
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$$M^2 + \Sigma^2 - P^2 - Q^2 \neq 0, \quad (M + \Sigma/\sqrt{3})^2 - Q^2 \neq 0, \quad (M - \Sigma/\sqrt{3})^2 - P^2 \neq 0, \quad (5.4.3)$$

$$M \pm \frac{\Sigma}{\sqrt{3}} \neq 0, \quad F^2 := \frac{[(M + \Sigma/\sqrt{3})^2 - Q^2][(M - \Sigma/\sqrt{3})^2 - P^2]}{M^2 + \Sigma^2 - P^2 - Q^2} > 0, \quad (5.4.4)$$

and where

$$g_{(3)} = -\frac{G}{\sqrt{AB}} (dt + \omega^0_\phi d\phi)^2 + \frac{\sqrt{AB}}{\Delta} dr^2 + \sqrt{AB} d\theta^2 + \frac{\Delta\sqrt{AB}}{G} \sin^2(\theta) d\phi^2, \quad (5.4.5)$$

with

$$A = \left(r - \Sigma/\sqrt{3}\right)^2 - \frac{2P^2\Sigma}{\Sigma - M\sqrt{3}} + a^2 \cos^2(\theta) + \frac{2JPQ \cos(\theta)}{(M + \Sigma/\sqrt{3})^2 - Q^2},$$

$$B = \left(r + \Sigma/\sqrt{3}\right)^2 - \frac{2Q^2\Sigma}{\Sigma + M\sqrt{3}} + a^2 \cos^2(\theta) - \frac{2JPQ \cos(\theta)}{(M - \Sigma/\sqrt{3})^2 - P^2},$$

$$G = r^2 - 2Mr + P^2 + Q^2 - \Sigma^2 + a^2 \cos^2(\theta),$$

$$\Delta = r^2 - 2Mr + P^2 + Q^2 - \Sigma^2 + a^2,$$

$$\omega^0_\phi = \frac{2J \sin^2(\theta)}{G} [r + E],$$

$$J^2 = a^2 F^2, \quad (5.4.6)$$

whereas E is given by

$$E = -M + \frac{(M^2 + \Sigma^2 - P^2 - Q^2)(M + \Sigma/\sqrt{3})}{(M + \Sigma/\sqrt{3})^2 - Q^2}. \quad (5.4.7)$$

In Kaluza–Klein theories, vacuum metrics on the Kaluza–Klein bundle lead to solutions of the Einstein–Maxwell–dilaton field equations. In the Rasheed case the physical-space Maxwell potential is given by

$$2A_\mu dx^\mu = \frac{C}{B} dt + \left(\omega^5_\phi + \frac{C}{B} \omega^0_\phi\right) d\phi, \quad (5.4.8)$$

where

$$C = 2Q \left(r - \Sigma/\sqrt{3}\right) - \frac{2PJ \cos(\theta)(M + \Sigma/\sqrt{3})}{(M - \Sigma/\sqrt{3})^2 - P^2}, \quad (5.4.9)$$

$$\omega^5_\phi = \frac{H}{G}, \quad (5.4.10)$$

and

$$H := 2P\Delta \cos(\theta) - \frac{2QJ \sin^2(\theta) [r(M - \Sigma/\sqrt{3}) + M\Sigma/\sqrt{3} + \Sigma^2 - P^2 - Q^2]}{[(M + \Sigma/\sqrt{3})^2 - Q^2]}. \quad (5.4.11)$$

The Rasheed metrics (5.4.1) have been found by applying a solution-generating technique ([409], compare [216]) to the Kerr metrics. This guarantees that these

metrics solve the five-dimensional vacuum Einstein equations when the constraint (5.4.3) is satisfied.

Let us address the question of the global structure of the metrics above. We have

$$\det g = -A^2 \sin^2(\theta),$$

which shows that the metrics are smooth and Lorentzian except possibly at the zeros of A , B , G , Δ , and $\sin(\theta)$.

After a suitable periodicity of ϕ as in Section 5.4.3 has been imposed, regularity at the axes of rotation away from the zeros of denominators follows from the factorizations

$$\left(\frac{\Delta}{G} - 1\right) = \frac{a^2 \sin^2(\theta)}{a^2 \cos^2(\theta) - 2Mr + P^2 + Q^2 + r^2 - \Sigma^2}, \quad (5.4.12)$$

$$2A_\phi - 2P \frac{\Delta}{G} \cos(\theta) = \frac{\sin^2(\theta)}{G} \left(K + \frac{2JC}{B} [r + E] \right), \quad (5.4.13)$$

where

$$K := -\frac{2QJ [r(M - \Sigma/\sqrt{3}) + M\Sigma/\sqrt{3} + \Sigma^2 - P^2 - Q^2]}{[(M + \Sigma/\sqrt{3})^2 - Q^2]}. \quad (5.4.14)$$

It will be seen later that, after restricting the parameter ranges as in (5.4.20)–(5.4.21) and (5.4.23)–(5.4.24), the location of Killing horizons is determined by the zeros of

$$\begin{vmatrix} g_{tt} & g_{t\phi} & g_{t4} \\ g_{\phi t} & g_{\phi\phi} & g_{\phi 4} \\ g_{4t} & g_{4\phi} & g_{44} \end{vmatrix} = -\Delta \sin^2(\theta), \quad (5.4.15)$$

and thus by the real roots $r_+ \geq r_-$ of Δ , if any:

$$r_\pm = M \pm \sqrt{M^2 + \Sigma^2 - P^2 - Q^2 - a^2}. \quad (5.4.16)$$

5.4.1 Zeros of the denominators

The norms

$$g_{tt} = \frac{W}{AB} \text{ and } g_{44} = \frac{B}{A}$$

of the Killing vectors ∂_t and ∂_4 are geometric invariants, where $W = -GA + C^2$. So zeros of A and of AB correspond to singularities in the five-dimensional geometry except if

1. a zero of A is a joint zero of A , B and W , or if
2. a zero of B which is not a zero of A is also a zero of W .

Setting

$$\mathcal{A} := \frac{2JPQ}{a^2 \left((M + \Sigma/\sqrt{3})^2 - Q^2 \right)}, \quad (5.4.17)$$

one checks that if

$$\begin{cases} \frac{2P^2\Sigma}{\Sigma - M\sqrt{3}} - a^2(1 - |\mathcal{A}|) = 0, & \text{when } |\mathcal{A}| > 2 \text{ or} \\ \frac{2P^2\Sigma}{\Sigma - M\sqrt{3}} + \frac{a^2\mathcal{A}^2}{4} = 0, & \text{when } |\mathcal{A}| \leq 2, \end{cases} \quad (5.4.18)$$

then A vanishes exactly at one point. Otherwise the set of zeros of A forms a curve in the (r, θ) plane. Let $\theta \mapsto r_A^+(\theta)$ denote the curve, say γ , corresponding to the set of largest zeros of A .

Note that W and A are polynomials in r , with A of second order. If W/A is smooth, the remainder of the polynomial division of W by $r - r_A^+$ must vanish on the part of γ that lies outside the horizon. One can calculate this remainder with MATHEMATICA, obtaining a function of θ which vanishes at most at isolated points, if at all. It follows that the division of W by A is singular on the closure of the domain of outer communications (d.o.c.), i.e. the region $\{r \geq r_+\}$, if A has zeros there, except perhaps when (5.4.18) holds.

One can likewise exclude a joint zero of W and B in the closure of the d.o.c. without a zero of A , except possibly for the case where this zero is isolated for B as well, which happens if

$$\begin{cases} \frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} - a^2(1-|\mathcal{B}|) = 0, & \text{if } |\mathcal{B}| > 2 \text{ or} \\ \frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} + \frac{a^2\mathcal{B}^2}{4} = 0, & \text{if } |\mathcal{B}| \leq 2. \end{cases} \quad (5.4.19)$$

See [261] for a more detailed analysis of the borderline cases.

Summarizing, a necessary condition for a black hole without obvious singularities in the closure of the domain of outer communications is that all zeros of A lie under the outermost Killing horizon $r = r_+$. One finds that this will be the case if and only if $|\mathcal{A}| > 2$ and

$$\begin{cases} \frac{2P^2\Sigma}{\Sigma-M\sqrt{3}} - a^2(1-|\mathcal{A}|) < 0, & \text{or} \\ M + \sqrt{M^2 + \Sigma^2 - P^2 - Q^2 - a^2} > \frac{\Sigma}{3} + \sqrt{\frac{2P^2\Sigma}{\Sigma-M\sqrt{3}} - a^2(1-|\mathcal{A}|)}, \end{cases} \quad (5.4.20)$$

or $|\mathcal{A}| \leq 2$ and

$$\begin{cases} \frac{2P^2\Sigma}{\Sigma-M\sqrt{3}} + \frac{a^2\mathcal{A}^2}{4} < 0, & \text{or} \\ M + \sqrt{M^2 + \Sigma^2 - P^2 - Q^2 - a^2} > \frac{\Sigma}{3} + \sqrt{\frac{2P^2\Sigma}{\Sigma-M\sqrt{3}} + \frac{a^2\mathcal{A}^2}{4}}, \end{cases} \quad (5.4.21)$$

except perhaps when (5.4.18) holds.

An identical argument applies to the zeros of B , with the zeros of B lying on a curve unless (5.4.19) holds. Ignoring this last case, the zeros of B need similarly be hidden behind the outermost Killing horizon. Setting

$$\mathcal{B} := -\frac{2JPQ}{a^2 \left((M - \Sigma/\sqrt{3})^2 - P^2 \right)}, \quad (5.4.22)$$

one finds that this will be the case if and only if $|\mathcal{B}| > 2$ and

$$\begin{cases} \frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} - a^2(1-|\mathcal{B}|) < 0, & \text{or} \\ M + \sqrt{M^2 + \Sigma^2 - P^2 - Q^2 - a^2} > -\frac{\Sigma}{3} + \sqrt{\frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} - a^2(1-|\mathcal{B}|)}, \end{cases} \quad (5.4.23)$$

or $|\mathcal{B}| \leq 2$ and

$$\begin{cases} \frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} + \frac{a^2\mathcal{B}^2}{4} < 0, & \text{or} \\ M + \sqrt{M^2 + \Sigma^2 - P^2 - Q^2 - a^2} > -\frac{\Sigma}{3} + \sqrt{\frac{2Q^2\Sigma}{\Sigma+M\sqrt{3}} + \frac{a^2\mathcal{B}^2}{4}}, \end{cases} \quad (5.4.24)$$

except perhaps when (5.4.19) holds.

While the above guarantees a lack of obvious singularities in the *domain of outer communications* $\{r > r_+\}$ (d.o.c.), there could still be causality violations there. Ideally the d.o.c. should be globally hyperbolic, a question which we have not attempted to address. Barring global hyperbolicity, a decent d.o.c. should at least

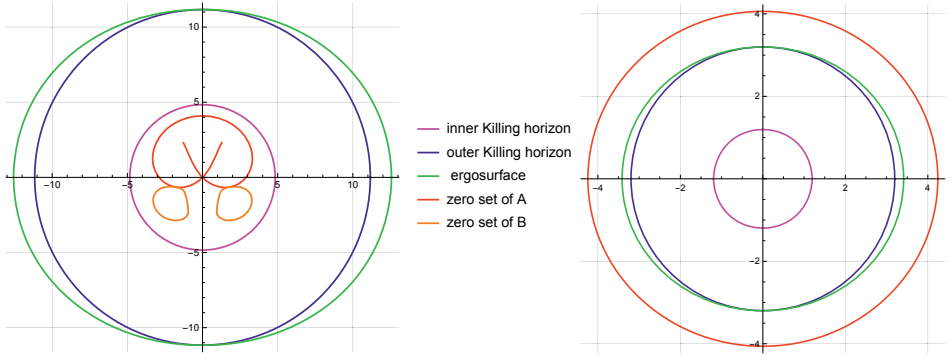


Fig. 5.4.1 Two sample plots for the location of the ergosurface (zeros of G), the outer and inner Killing horizons (zeros of Δ), and the zeros of A, B . (Left) $M = 8$, $a = \frac{33}{10}$, $Q = \frac{8}{5}$, $\Sigma = -\frac{23}{5}$, $P = -\frac{1}{5}\sqrt{\frac{2(4105960\sqrt{3}+2770943)}{12813}} \approx -7.86$, with zeros of A and B under both horizons, consistently with (5.4.20)–(5.4.21) and (5.4.23)–(5.4.24). (Right) $M = 1$, $a = 1$, $Q = 0$, $\Sigma = \sqrt{6}$, $P = \sqrt{4 - 2\sqrt{2}} \approx 1.08$; here (5.4.20)–(5.4.21) are violated, while the zeros of B occur at negative r .

admit a time function, and the function t provides an obvious candidate. In order to study the issue we note the identity

$$g^{00} = \frac{4J^2[r + E]^2 \sin^2(\theta) - AB\Delta}{A\Delta G}. \quad (5.4.25)$$

A MATHEMATICA calculation shows that the numerator factorizes through G , so that g^{00} extends smoothly through the ergosphere. When $P = 0$ one can verify that g^{00} is negative on the d.o.c. For $P \neq 0$ one can find open sets of parameters which guarantee that g^{00} is strictly negative for $r > r_+$ when A and B have no zeros there. An example is given by the condition

$$r_+ \geq \frac{EM + q}{M + E}, \quad (5.4.26)$$

which is sufficient but not necessary, where $q := P^2 + Q^2 - \Sigma^2 + a^2$.

In Fig. 5.4.1 we show the locations of the zeros of A and B for some specific sets of parameters satisfying, or violating, the conditions above.

Another potential source of singularities of the metric (5.4.1) could be the zeros of G . It turns out that they do not matter, which can be seen as follows: the relevant metric coefficient is $g_{\phi\phi}$, which reads

$$g_{\phi\phi} = \frac{B}{A} \left(\omega^5_{\phi} + \frac{C}{B} \omega^0_{\phi} \right)^2 + \sqrt{\frac{A}{B}} \left(-\frac{G}{\sqrt{AB}} (\omega^0_{\phi})^2 + \frac{\Delta\sqrt{AB}}{G} \sin^2 \theta \right). \quad (5.4.27)$$

Taking into account a G^{-1} factor in ω^0_{ϕ} , it follows that $g_{\phi\phi}$ can be written as a fraction $(\dots)/ABG^2$. A MATHEMATICA calculation shows that the denominator $(\dots)$ factorizes through AG^2 , which shows indeed that the zeros of G are innocuous for the problem at hand.

Let us write $ds_{(4)}^2$ as ${}^{(4)}g_{ab}dx^a dx^b$. The factorization just described works for $g_{\phi\phi}$ but does *not* work for ${}^{(4)}g_{\phi\phi}$. From what has been said we see that the quotient metric ${}^{(4)}g_{ab}dx^a dx^b$ is always singular in the d.o.c.

5.4.2 Regularity at the outer Killing horizon $\mathcal{H}^+$

The location of the outer Killing horizon $\mathcal{H}^+$ of the Killing field

$$k := \partial_t + \Omega_\phi \partial_\phi + \Omega_4 \partial_{x^4} \quad (5.4.28)$$

is given by the larger root r_+ of Δ , cf. (5.4.16). The condition that $\mathcal{H}_+$ is a Killing horizon for k is that the pullback of $g_{\mu\nu}k^\nu$ to $\mathcal{H}^+$ vanishes. This, together with

$$\Delta|_{\mathcal{H}^+} = 0, \quad G|_{\mathcal{H}^+} = -a^2 \sin^2(\theta), \quad (5.4.29)$$

yields

$$\begin{aligned} \Omega_\phi &= -\frac{1}{\omega_\phi^0} \Big|_{\mathcal{H}^+} = \frac{a^2}{2J} (r_+ + E)^{-1}, \\ \Omega_4 &= -\frac{2(A_t \omega_\phi^0 - A_\phi)}{\omega_\phi^0} \Big|_{\mathcal{H}^+} \\ &= \frac{Q(-3Mr_+ - \sqrt{3}M\Sigma + 3P^2 + 3Q^2 + \sqrt{3}r\Sigma - 3\Sigma^2)}{(E + r_+)(3M^2 + 2\sqrt{3}M\Sigma - 3Q^2 + \Sigma^2)}. \end{aligned} \quad (5.4.30)$$

After the coordinate transformation

$$\bar{\phi} = \phi - \Omega_\phi dt, \quad \bar{x}^4 = x^4 - \Omega_4 dt, \quad (5.4.31)$$

the metric (5.4.1) becomes

$$g = g_S + \frac{dr^2}{\Delta} + \Delta U dt^2, \quad (5.4.32)$$

where g_S is a smooth $(0,2)$ -tensor, with $U := g_{tt}/\Delta$ extending smoothly across $\Delta = 0$. Introducing a new time coordinate by

$$\tau = t - \sigma \ln(r - r_+) \Rightarrow d\tau = dt - \frac{\sigma}{r - r_+} dr, \quad (5.4.33)$$

where σ is a constant to be determined, (5.4.32) takes the form

$$\begin{aligned} g &= g_S + \Delta U \left(d\tau + \frac{\sigma}{r - r_+} dr \right)^2 + \frac{dr^2}{\Delta} \\ &= g_S + \Delta U d\tau^2 + \frac{2\Delta U \sigma}{r - r_+} d\tau dr + \left(\frac{1}{\Delta} + \frac{\Delta U \sigma^2}{(r - r_+)^2} \right) dr^2 \\ &= g_S + \Delta U d\tau^2 + \frac{2\Delta U \sigma}{r - r_+} d\tau dr + \underbrace{\frac{(r - r_+)^2 + \Delta^2 \sigma^2 U}{\Delta (r - r_+)^2}}_V dr^2. \end{aligned} \quad (5.4.34)$$

In order to obtain a smooth metric in the domain of outer communication the constant σ must be chosen so that the numerator of V has a triple-zero at $r = r_+$. A MATHEMATICA computation gives an explicit formula for the desired constant σ , which is too lengthy to be explicitly presented here. This establishes smooth extendibility of the metric in suitable coordinates across $r = r_+$.

5.4.3 Asymptotic behaviour

When $P = 0$ the Rasheed metrics satisfy the KK-asymptotic flatness conditions. This can be seen by introducing manifestly asymptotically flat coordinates (t, x, y, z) in the usual way. With some work one finds that the metric takes the form

$$\begin{pmatrix} \frac{2M}{r} + \frac{2\Sigma}{\sqrt{3}r} - 1 & 0 & 0 & 0 & \frac{2Q}{r} \\ 0 & \frac{2Mx^2}{r^3} - \frac{2\Sigma}{\sqrt{3}r} + 1 & \frac{2Mxy}{r^3} & \frac{2Mxz}{r^3} & 0 \\ 0 & \frac{2Mxy}{r^3} & \frac{2My^2}{r^3} - \frac{2\Sigma}{\sqrt{3}r} + 1 & \frac{2Myz}{r^3} & 0 \\ 0 & \frac{2Mxz}{r^3} & \frac{2Myz}{r^3} & \frac{2Mz^2}{r^3} - \frac{2\Sigma}{\sqrt{3}r} + 1 & 0 \\ \frac{2Q}{r} & 0 & 0 & 0 & \frac{4\Sigma}{\sqrt{3}r} + 1 \end{pmatrix} + O(r^{-2}). \quad (5.4.35)$$

However, when $P \neq 0$, the Rasheed metrics do *not* satisfy the KK-asymptotic flatness requirements anymore: instead, the space of Rasheed metrics decomposes into sectors, labelled by $P \in \mathbb{R}$, in which the metrics g asymptote to the background metric

$$\bar{g} := (dx^4 + 2P \cos(\theta)d\varphi)^2 - dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2(\theta)d\varphi^2. \quad (5.4.36)$$

The metrics (5.4.1) and (5.4.36) are singular at $\sin(\theta) = 0$. This can be resolved by replacing x^4 with $\bar{x}^4$, respectively with $\tilde{x}^4$, on the coordinate patches

$$\begin{cases} \bar{x}^4 := x^4 + 2P\varphi, & \theta \in [0, \pi), \\ \tilde{x}^4 := x^4 - 2P\varphi, & \theta \in (0, \pi]. \end{cases} \quad (5.4.37)$$

Indeed, the one-form

$$dx^4 + 2P \cos(\theta)d\varphi = d\bar{x}^4 + 2P(\cos(\theta) - 1)d\varphi = d\bar{x}^4 - \frac{2P}{r(r+z)}(xdy - ydx)$$

is smooth for $r > 0$ on $\{\theta \in [0, \pi)\}$. Similarly the one-form

$$dx^4 + 2P \cos(\theta)d\varphi = d\tilde{x}^4 + 2P(\cos(\theta) + 1)d\varphi = d\tilde{x}^4 + \frac{2P}{r(r-z)}(xdy - ydx)$$

is smooth on $\{\theta \in (0, \pi], r > 0\}$. Smoothness of both g and of the background metric $\bar{g}$ outside of the event horizons readily follows.

We note the relation

$$\bar{x}^4 = \tilde{x}^4 + 4P\varphi, \quad (5.4.38)$$

which implies a smooth geometry with periodic coordinates $\bar{x}^4$ and $\tilde{x}^4$ if and only if

$$\text{both } \bar{x}^4 \text{ and } \tilde{x}^4 \text{ are periodic with period } 8P\pi. \quad (5.4.39)$$

From this perspective x^4 is *not a coordinate* anymore: instead the basic coordinates are $\bar{x}^4$ for $\theta \in [0, \pi)$ and $\tilde{x}^4$ for $\theta \in (0, \pi]$, with dx^4 (but *not* x^4) well defined away from the axes of rotation $\{\sin(\theta) = 0\}$ as

$$dx^4 = \begin{cases} d\bar{x}^4 - 2Pd\varphi, & \theta \in [0, \pi), \\ d\tilde{x}^4 + 2Pd\varphi, & \theta \in (0, \pi]. \end{cases} \quad (5.4.40)$$

Curvature of the asymptotic background. We continue with a calculation of the curvature tensor of the asymptotic background. It is convenient to work in the coframe

$$\bar{\Theta}^0 = dt, \quad \bar{\Theta}^1 = dx, \quad \bar{\Theta}^2 = dy, \quad \bar{\Theta}^3 = dz, \quad \bar{\Theta}^4 = dx^4 + 2P \cos(\theta)d\varphi, \quad (5.4.41)$$

which is manifestly smooth after replacing dx^4 as in (5.4.40). Using

$$d\bar{\Theta}^4 = -2P \sin(\theta) d\theta \wedge d\varphi = -2P \frac{x^i}{r^3} \partial_i (dx \wedge dy \wedge dz) = -\frac{P}{r^3} \epsilon_{ij\hat{k}} x^{\hat{i}} dx^{\hat{j}} \wedge dx^{\hat{k}}, \quad (5.4.42)$$

where $\epsilon_{ij\hat{k}} \in \{0, \pm 1\}$ denotes the usual epsilon symbol, one finds the non-vanishing connection coefficients

$$\bar{\omega}^4_{\hat{i}} = \frac{P}{r^3} \epsilon_{ij\hat{k}} x^{\hat{j}} \bar{\Theta}^{\hat{k}}, \quad \bar{\omega}^{\hat{i}}_{\hat{j}} = \frac{P}{r^3} \epsilon_{ij\hat{k}} x^{\hat{k}} \bar{\Theta}^{\hat{i}}, \quad (5.4.43)$$

where $x^{\hat{i}} \equiv x^i$. This leads to the curvature forms

$$\begin{aligned} \bar{\Omega}^{\hat{i}}_{\hat{j}} &= \frac{P}{r^3} \epsilon_{ij\hat{k}} \left(-\frac{3}{r^2} x^{\hat{k}} x^{\hat{\ell}} + \delta^{\hat{k}}_{\hat{\ell}} \right) \bar{\Theta}^{\hat{\ell}} \wedge \bar{\Theta}^{\hat{i}} - \frac{2P^2}{r^6} \epsilon_{i\hat{m}(\hat{k}} \epsilon_{\hat{j})\hat{n}\hat{\ell}} x^{\hat{m}} x^{\hat{n}} \bar{\Theta}^{\hat{k}} \wedge \bar{\Theta}^{\hat{\ell}}, \\ \bar{\Omega}^4_{\hat{i}} &= \frac{P}{r^3} \epsilon_{ij\hat{k}} \left(-\frac{3}{r^2} x^{\hat{j}} x^{\hat{\ell}} + \delta^{\hat{j}}_{\hat{\ell}} \right) \bar{\Theta}^{\hat{\ell}} \wedge \bar{\Theta}^{\hat{k}} + \frac{P^2}{r^6} \epsilon_{\hat{k}\hat{m}\hat{j}} \epsilon_{\hat{k}\hat{i}\hat{\ell}} x^{\hat{m}} x^{\hat{\ell}} \bar{\Theta}^{\hat{j}} \wedge \bar{\Theta}^{\hat{i}}; \end{aligned} \quad (5.4.44)$$

hence the non-vanishing curvature tensor components

$$\begin{aligned} \bar{\mathbf{R}}^{\hat{i}}_{\hat{j}\hat{k}\hat{4}} &= \frac{P}{r^3} \epsilon_{ij\hat{\ell}} \left(-\frac{3}{r^2} x^{\hat{\ell}} x^{\hat{k}} + \delta^{\hat{\ell}}_{\hat{k}} \right), \\ \bar{\mathbf{R}}^4_{\hat{i}\hat{j}\hat{4}} &= \frac{P^2}{r^6} \epsilon_{\hat{k}\hat{m}\hat{j}} \epsilon_{\hat{k}\hat{i}\hat{\ell}} x^{\hat{m}} x^{\hat{\ell}}, \\ \bar{\mathbf{R}}_{ij\hat{k}\hat{\ell}} &= -\frac{2P^2}{r^6} (\epsilon_{ij\hat{n}} \epsilon_{\hat{k}\hat{\ell}\hat{m}} + \epsilon_{i\hat{m}[\hat{k}} \epsilon_{\hat{\ell}]\hat{j}\hat{n}}) x^{\hat{m}} x^{\hat{n}}. \end{aligned}$$

The non-vanishing components of the Ricci tensor read

$$\bar{\mathbf{R}}_{ij\hat{j}} = -\frac{2P^2}{r^6} \epsilon_{\hat{k}\hat{m}\hat{i}} \epsilon_{\hat{k}\hat{n}\hat{j}} x^{\hat{m}} x^{\hat{n}}, \quad \bar{\mathbf{R}}_{44} = -\frac{P^2}{r^6} \epsilon_{\hat{k}\hat{m}\hat{i}} \epsilon_{\hat{k}\hat{i}\hat{\ell}} x^{\hat{m}} x^{\hat{\ell}}.$$

Subsequently the Ricci scalar is $\bar{\mathbf{R}} = -2P^2/r^4$.

5.4.4 Global charges

Let p_μ be the Hamiltonian momentum of the level sets of t , and let $p_{\mu,\text{ADM}}$ be the ADM four-momentum of the space metric $g_{ij} dx^i dx^j$. Then

$$\begin{aligned} p_{i,\text{ADM}} &= p_i = 0, \quad p_{0,\text{ADM}} = M - \frac{\Sigma}{\sqrt{3}}, \\ p_0 &= \begin{cases} 2\pi M, & P = 0; \\ 4\pi P M, & P \neq 0, \end{cases} \quad p_4 = \begin{cases} 2\pi Q, & P = 0; \\ 8\pi P Q, & P \neq 0. \end{cases} \end{aligned}$$

The Komar integrals associated with $X = \partial_t$ are

$$\frac{1}{8\pi} \lim_{R \rightarrow \infty} \int_{S(R)} \int_{S^1} X^{\alpha;\beta} dS_{\alpha\beta} = \begin{cases} 2\pi (M + \frac{\Sigma}{\sqrt{3}}), & P = 0; \\ 8\pi P (M + \frac{\Sigma}{\sqrt{3}}), & P \neq 0. \end{cases}$$

The Komar integrals associated with $X = \partial_4$ are

$$\frac{1}{8\pi} \lim_{R \rightarrow \infty} \int_{S(R)} \int_{S^1} X^{\alpha;\beta} dS_{\alpha\beta} = \begin{cases} 4\pi Q, & P = 0; \\ 16\pi P Q, & P \neq 0. \end{cases}$$

5.5 Birmingham metrics

The *Birmingham metrics* [55], also known as *generalized Kottler metrics* [299], are higher-dimensional counterparts of the Schwarzschild metric with a non-zero cosmological constant thrown in. These are $(n+1)$ -dimensional metrics, $n \geq 3$, of the form

$$g = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 \underbrace{\check{h}_{AB}(x^C)dx^A dx^B}_{=: \check{h}}, \quad (5.5.1)$$

where $\check{h}$ is a *Riemannian Einstein metric* on a compact $(n-1)$ -dimensional manifold $\check{N}$, and where we denote by x^A the local coordinates on $\check{N}$. As first pointed out by Birmingham in [55], for any $m \in \mathbb{R}$ and

$$\ell \in \mathbb{R}^* \cup \sqrt{-1}\mathbb{R}^*$$

the function

$$f = \frac{\check{R}}{(n-1)(n-2)} - \frac{2m}{r^{n-2}} - \frac{r^2}{\ell^2}, \quad (5.5.2)$$

where $\check{R}$ is the (constant) scalar curvature of $\check{h}$, leads to a vacuum metric,

$$R_{\mu\nu} = \frac{n}{\ell^2} g_{\mu\nu}, \quad (5.5.3)$$

and where ℓ is a constant related to the cosmological constant $\Lambda \in \mathbb{R}$ as

$$\frac{1}{\ell^2} = \frac{2\Lambda}{n(n-1)}. \quad (5.5.4)$$

A comment about negative Λ , and thus purely complex ℓ 's, is in order. When considering a negative cosmological constant, (5.5.4) requires $\ell \in \sqrt{-1}\mathbb{R}$, which is awkward to work with. So when $\Lambda < 0$ it is convenient to change r^2/ℓ^2 in f to $-r^2/\ell^2$, change the sign in (5.5.4), and use a real ℓ . We will often do this without further ado.

Clearly, n cannot be 2 in (5.5.2), and we therefore exclude this dimension in what follows.

INCIDENTALLY: Actually the formulae above without the $\check{R}$ -term apply when $n = 2$. Indeed, and perhaps surprisingly, when $\Lambda < 0$ one obtains a one-parameter family of static vacuum black holes in spacetime dimension three.⁸ These solutions have been discovered by Bañados, Teitelboim, and Zanelli [25]. The static, circularly symmetric, vacuum solutions read explicitly

$$ds^2 = -\left(\frac{r^2}{\ell^2} - m\right)dt^2 + \frac{dr^2}{\frac{r^2}{\ell^2} - m} + r^2 d\phi^2, \quad (5.5.5)$$

where m is related to the total mass and $\ell^2 = -1/\Lambda$. For $m > 0$, this can be extended, as in (4.2.13) with $V^2 = r^2/\ell^2 - m$, to a black-hole spacetime with event horizon located at $r_H = \ell\sqrt{m}$.

There exist rotating counterparts of (5.5.5), discussed in [11, 24, 25, 64, 435].

The multiplicative factor two in front of m is convenient in dimension three when $\check{h}$ is a unit round metric on S^2 , and we will keep this form regardless of topology and dimension of $\check{N}$.

⁸There are no such vacuum black holes with $\Lambda > 0$, or with $\Lambda = 0$ and degenerate horizons [269].

There is a rescaling of the coordinate $r = b\bar{r}$, with $b \in \mathbb{R}^*$, which leaves (5.5.1)–(5.5.2) unchanged if moreover

$$\bar{\check{h}} = b^2 \check{h}, \quad \bar{m} = b^{-n} m, \quad \bar{t} = bt. \quad (5.5.6)$$

We can use this to achieve

$$\beta := \frac{\check{R}}{(n-1)(n-2)} \in \{0, \pm 1\}. \quad (5.5.7)$$

This scaling will be used from now on. The set $\{r = 0\}$ corresponds to a singularity when $m \neq 0$. Except in the case $m = 0$ and $\beta = -1$, by an appropriate choice of the sign of b we can always achieve $r > 0$ in the regions of interest. This will also be assumed from now on.

The global structure of the standard extensions of the Birmingham metrics depends upon the number of zeros of f , and their order. The various cases are discussed in Sections 6.2.5, 6.3, and 6.6 (cf. in particular Examples 6.3.2–6.3.4).

5.5.1 ‘Thermodynamics’

In this section we derive an identity, Eq. (5.5.13), relating variations of the mass, within the family of Birmingham metrics, with that of the area of the Killing horizons. Such identities are part of a direction of studies known as the ‘thermodynamics of black holes’. A mathematical fallout of such considerations is the proof, in [254], of linear instability of a class of black string solutions. A closely related observation, in [177], is that of a correspondence between linear instabilities and ‘local Penrose inequalities’.

Since the Schwarzschild–(anti-)de Sitter metrics are part of the Birmingham family, the identity (5.5.13) applies in particular to the S(A)dS metrics.

The locations of Killing horizons of the Birmingham metrics are defined, in space-dimension n , by the condition

$$f(r_0) = \beta - \frac{2m}{r_0^{n-1}} - \frac{r_0^2}{\ell^2} = 0.$$

Thus, variations of the metric on the horizons satisfy

$$0 = \delta f|_{r=r_0} = \left[(\partial_r f) \delta r - \frac{2}{r^{n-2}} \delta m \right] \Big|_{r=r_0}; \quad (5.5.8)$$

equivalently

$$\delta m = \frac{1}{2(n-1)} (\partial_r f) \delta(r^{n-1}) = \frac{1}{(n-1)\sigma_{n-1}} \frac{(\partial_r f)}{2} \Big|_{r=r_0} \delta A, \quad (5.5.9)$$

where $r^{n-1}\sigma_{n-1}$ is the ‘area’ of the cross-section of the horizon (area if $n = 3$, volume if $n = 4$, hypervolume otherwise).

The multiplicative coefficient above,

$$\kappa := \frac{(\partial_r f)}{2} \Big|_{r=r_0}, \quad (5.5.10)$$

coincides with the surface gravity of the horizon, defined through the usual formula

$$\nabla_K K = -\kappa K, \quad (5.5.11)$$

where K is the Killing vector field which is null on the horizon. To see this, we rewrite the spacetime metric (5.5.1) as usual as

$$g = -f du^2 - 2 du dr + r^2 \check{h},$$

where $du = dt - \frac{1}{f} dr$. The Killing field $K = \partial_u = \partial_t$ is indeed tangent to the horizon and null on it. Formula (5.5.11) implies that

$$\kappa = -\Gamma_{uu}^u = -\frac{1}{2}g^{u\lambda}(2g_{\lambda u,u} - g_{uu,\lambda}). \quad (5.5.12)$$

The inverse metric equals

$$g^\sharp = -2 \frac{\partial}{\partial u} \frac{\partial}{\partial r} + f \left(\frac{\partial}{\partial r} \right)^2 + r^{-2} \check{h}^\sharp,$$

whence $g^{u\lambda} = -\delta_r^\lambda$, and

$$\kappa = -\frac{1}{2}g_{uu,r} = \left. \frac{(\partial_r f)}{2} \right|_{r=r_0},$$

as claimed.

We conclude that on Killing horizons it holds

$$\delta m = \frac{1}{(n-1)\sigma_{n-1}} \kappa \delta A. \quad (5.5.13)$$

Equation (5.5.13) is often referred to as the *first law of black-hole dynamics*.

5.5.2 Curvature

In this section we calculate the curvature of metrics of the form

$$g = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 \underbrace{\check{h}_{AB}(x^C)dx^A dx^B}_{=: \check{h}}, \quad (5.5.14)$$

which includes all Birmingham metrics, and hence also the Schwarzschild–anti-de Sitter metrics.

INCIDENTALLY: We recall that the curvature tensor of the space part of the metric (5.5.14) has been calculated in Example 1.6.7, p. 17. $\square$

A corollary of the calculation to follow is the proof that such metrics are singular at $r = 0$. For black holes this will typically (but not always) happen in the region where $f < 0$, and therefore we will start by assuming the negativity of f in our calculations. (We will return to the case $f > 0$ at the end of the section.) For visual clarity it is convenient to make the following replacements and redefinitions,

$$r \rightarrow \tau, \quad t \rightarrow x, \quad f \rightarrow -e^{2\chi}, \quad (5.5.15)$$

which bring g to the form

$$g = -e^{-2\chi(\tau)} d\tau^2 + e^{2\chi(\tau)} dx^2 + \tau^2 \check{h}. \quad (5.5.16)$$

To calculate the Riemann tensor we use the moving-frames formalism. For $A = 1, \dots, n$ let $\check{\theta}^A$ be an ON-coframe for $\check{h}$,

$$\check{h} = \sum_{A=1}^{n-1} \check{\theta}^A \otimes \check{\theta}^A,$$

and let $\check{\omega}_{AB}$ and $\check{\Omega}_{AB}$ be the associated connection and curvature forms. It holds that

$$0 = d\check{\theta}^A + \check{\omega}^A{}_B \wedge \check{\theta}^B,$$

$$\check{\Omega}^A{}_B = d\check{\omega}^A{}_B + \check{\omega}^A{}_C \wedge \check{\omega}^C{}_B.$$

Let θ^μ be the g -ON-coframe

$$\theta^0 = e^{-\chi} d\tau, \quad \theta^A = \tau \check{\theta}^A, \quad \theta^n = e^\chi dx.$$

The vanishing of torsion gives

$$\begin{aligned} 0 &= d\theta^0 + \omega^0{}_\mu \wedge \theta^\mu = \omega^0{}_n \wedge \theta^n + \omega^0{}_A \wedge \theta^A, \\ 0 &= d\theta^A + \omega^A{}_\mu \wedge \theta^\mu = d\tau \wedge \check{\theta}^A + \tau d\check{\theta}^A + \omega^A{}_\mu \wedge \theta^\mu \\ &= d\tau \wedge \check{\theta}^A + \omega^A{}_0 \wedge \theta^0 + \omega^A{}_n \wedge \theta^n + (\omega^A{}_B - \check{\omega}^A{}_B) \wedge \theta^B, \\ 0 &= d\theta^n + \omega^n{}_\mu \wedge \theta^\mu = d(e^\chi) \wedge dx + \omega^n{}_\mu \wedge \theta^\mu \\ &= e^\chi \dot{\chi} \theta^0 \wedge \theta^n + \omega^n{}_0 \wedge \theta^0 + \omega^n{}_A \wedge \theta^A. \end{aligned}$$

This is solved by setting

$$\begin{aligned} \omega^n{}_A &= 0, \\ \omega^n{}_0 &= e^\chi \dot{\chi} \theta^n = \frac{1}{2} (e^{2\chi}) dx, \\ \omega^A{}_0 &= e^\chi \check{\theta}^A, \\ \omega^A{}_B &= \check{\omega}^A{}_B. \end{aligned}$$

The curvature two-forms are thus

$$\begin{aligned} \Omega^0{}_n &= d\omega^0{}_n + \omega^0{}_\mu \wedge \omega^\mu{}_n = d\omega^0{}_n = \frac{1}{2} \frac{d^2(e^{2\chi})}{d\tau^2} d\tau \wedge dx \\ &= \frac{1}{2} \frac{d^2(e^{2\chi})}{d\tau^2} \theta^0 \wedge \theta^n = \frac{1}{2} \frac{d^2(e^{2\chi})}{d\tau^2} \delta^0_{[\mu} g_{\nu]n} \theta^\mu \wedge \theta^\nu, \end{aligned} \quad (5.5.17)$$

$$\begin{aligned} \Omega^0{}_A &= d\omega^0{}_A + \omega^0{}_\mu \wedge \omega^\mu{}_A = \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \theta^0 \wedge \check{\theta}_A + e^\chi d\theta_A + e^\chi \theta_B \wedge \omega^B{}_A \\ &= \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \theta^0 \wedge \theta_A = \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \delta^0_{[\mu} g_{\nu]A} \theta^\mu \wedge \theta^\nu, \end{aligned} \quad (5.5.18)$$

$$\begin{aligned} \Omega^n{}_A &= d\omega^n{}_A + \omega^n{}_\mu \wedge \omega^\mu{}_A = \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \theta^n \wedge \theta_A \\ &= \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \delta^n_{[\mu} g_{\nu]A} \theta^\mu \wedge \theta^\nu, \end{aligned} \quad (5.5.19)$$

$$\begin{aligned} \Omega^A{}_B &= d\omega^A{}_B + \omega^A{}_\mu \wedge \omega^\mu{}_B = \check{\Omega}^A{}_B + e^{2\chi} \tau^{-2} \theta^A \wedge \theta_B \\ &= \frac{1}{2} \check{\Omega}^A{}_{BCD} \check{\theta}^C \wedge \check{\theta}^D + e^{2\chi} \tau^{-2} \theta^A \wedge \theta_B \\ &= \frac{1}{2} \tau^{-2} (\check{\Omega}^A{}_{BCD} + 2e^{2\chi} \delta^A_{[C} \delta_{D]B}) \theta^C \wedge \theta^D. \end{aligned} \quad (5.5.20)$$

Using

$$\Omega^\mu{}_\nu = \frac{1}{2} R^\mu{}_{\nu\alpha\beta} \theta^\alpha \wedge \theta^\beta, \quad (5.5.21)$$

we conclude that, up to symmetries, the non-zero frame components of the Riemann tensor are

$$R^0{}_{n\mu\nu} = \frac{d^2(e^{2\chi})}{d\tau^2} \delta^0_{[\mu} g_{\nu]n}, \quad (5.5.22)$$

$$R^0{}_{A\mu\nu} = \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \delta^0_{[\mu} g_{\nu]A}, \quad (5.5.23)$$

$$R^n{}_{A\mu\nu} = \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \delta_{[\mu}^n g_{\nu]A}, \quad (5.5.24)$$

$$R^A{}_{BCD} = \tau^{-2} (\check{\Omega}^A{}_{BCD} + 2e^{2\chi} \delta_{[C}^A \delta_{D]B}). \quad (5.5.25)$$

Hence the non-vanishing components of the Ricci tensor are

$$R_{00} = -\frac{1}{2} \left(\frac{d^2(e^{2\chi})}{d\tau^2} + (n-1) \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \right) = -R_{nn}, \quad (5.5.26)$$

$$R_{AB} = \tau^{-2} \check{R}_{AB} + \left((n-2) \tau^{-2} e^{2\chi} + \frac{d(e^{2\chi})}{d\tau} \tau^{-1} \right) g_{AB}. \quad (5.5.27)$$

If $\check{h}$ is Einstein, $\check{R}_{AB} = (\check{R}/(n-1))\check{h}_{AB}$, the last equation becomes

$$R_{AB} = \tau^{-2} \left(\frac{\check{R}}{n-1} + (n-2)e^{2\chi} + \tau \frac{d(e^{2\chi})}{d\tau} \right) g_{AB}. \quad (5.5.28)$$

It is now straightforward to check that for any $m \in \mathbb{R}$, $\ell \in \mathbb{R}^*$ and $n \geq 3$ the function

$$e^{2\chi} = -\frac{\check{R}}{(n-1)(n-2)} + \frac{2m}{\tau^{n-2}} + \frac{\tau^2}{\ell^2} \quad (5.5.29)$$

(compare with (5.5.2) and (5.5.14)–(5.5.15)) leads to a vacuum metric:

$$R_{\mu\nu} = \frac{n}{\ell^2} g_{\mu\nu}. \quad (5.5.30)$$

For further reference we note that the Ricci scalar R equals, quite generally,

$$R = \frac{d^2(e^{2\chi})}{d\tau^2} + (n-1) \left(2 \frac{d(e^{2\chi})}{d\tau} \tau^{-1} + (n-2) \tau^{-2} e^{2\chi} \right) + \tau^{-2} \check{R}. \quad (5.5.31)$$

Suppose that g is a Birmingham metric with $m = 0$; thus

$$e^{2\chi} = -\beta + \frac{\tau^2}{\ell^2}$$

for a constant β . Then

$$\frac{1}{2} \frac{d^2(e^{2\chi})}{d\tau^2} = \frac{1}{2} \frac{d(e^{2\chi})}{d\tau} \tau^{-1} = \tau^{-2} (e^{2\chi} + \beta) = \frac{1}{\ell^2}.$$

If $\check{h}$ is a space-form, with

$$\check{\Omega}^A{}_{BCD} = 2\beta \delta_{[C}^A \delta_{D]B},$$

consistently with (5.5.7), we obtain

$$R_{\mu\nu\rho\sigma} = \frac{2}{\ell^2} g_{\mu[\rho} g_{\sigma]\nu}.$$

If, however, $\check{h}$ is *not* a space-form, we have

$$\check{\Omega}^A{}_{BCD} = 2\beta \delta_{[C}^A \delta_{D]B} + r^A{}_{BCD},$$

for some non-identically vanishing tensor $r^A{}_{BCD}$, with all traces zero. Hence we obtain

$$R_{\mu\nu\rho\sigma} = \frac{2}{\ell^2} g_{\mu[\rho} g_{\sigma]\nu} + \tau^{-2} r_{\mu\nu\rho\sigma},$$

where the functions $r_{\mu\nu\rho\sigma}$ are τ -independent in the current frame, and vanish whenever one of the indices is 0 or n . This gives

$$R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = \frac{2n(n+1)}{\ell^4} + r^{\mu\nu\rho\sigma} r_{\mu\nu\rho\sigma}$$

$$= \frac{2n(n+1)}{\ell^4} + \tau^{-4} \sum_{A,B,C,D=1}^{n-1} (r_{ABCD})^2,$$

which is singular at $\tau = 0$.

Recall, now, that the calculations so far also apply to

$$g = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2\check{h} \quad (5.5.32)$$

with the following replacements and redefinitions:

$$r \rightarrow \tau, \quad t \rightarrow x, \quad f \rightarrow -e^{2\chi}. \quad (5.5.33)$$

Strictly speaking, f should be negative when using (5.5.33), but the final formulae hold regardless of the sign of f . (Alternatively, we could replace χ by $\chi + \pi\sqrt{-1}/2$ in all equations above.) For convenience of cross-referencing we rewrite the formulae obtained so far in this notation:

$$R^r{}_{t\mu\nu} = -f''\delta_{[\mu}^r g_{\nu]t}, \quad (5.5.34)$$

$$R^r{}_{A\mu\nu} = -f'r^{-1}\delta_{[\mu}^r g_{\nu]A}, \quad (5.5.35)$$

$$R^t{}_{A\mu\nu} = -f'r^{-1}\delta_{[\mu}^t g_{\nu]A}, \quad (5.5.36)$$

$$R^A{}_{BCD} = r^{-2}(\check{\Omega}^A{}_{BCD} - 2f\delta_{[C}^A\delta_{D]B}), \quad (5.5.37)$$

$$R_{rt} = 0 = R_{tA} = R_{rA}, \quad (5.5.38)$$

$$R_{rr} = \frac{1}{2}(f'' + (n-1)f'r^{-1}) = -R_{tt}, \quad (5.5.39)$$

$$R_{AB} = r^{-2}\check{R}_{AB} - ((n-2)r^{-2}f + f'r^{-1})g_{AB}. \quad (5.5.40)$$

$$R = -f'' - (n-1)(2f'r^{-1} + (n-2)r^{-2}f) + r^{-2}\check{R}. \quad (5.5.41)$$

If $\check{h}$ is Einstein

$$\check{R}_{AB} = \frac{\check{R}}{n-1}\check{h}_{AB}, \quad (5.5.42)$$

the last equation becomes

$$R_{AB} = r^{-2}\left(\frac{\check{R}}{n-1} - (n-2)f - rf'\right)g_{AB}. \quad (5.5.43)$$

As before, for any $m \in \mathbb{R}$ and $\ell \in \mathbb{R}^*$ the function

$$f = \frac{\check{R}}{(n-1)(n-2)} - \frac{2m}{r^{n-2}} - \varepsilon \frac{r^2}{\ell^2}, \quad \varepsilon \in \{0, \pm 1\} \quad (5.5.44)$$

leads then to an Einstein metric:

$$R_{\mu\nu} = \varepsilon \frac{n}{\ell^2} g_{\mu\nu}. \quad (5.5.45)$$

5.5.3 Euclidean Birmingham (Schwarzschild–(anti-)de Sitter) metrics

An important role in Euclidean quantum gravity [224] is played by solutions of the field equations with Riemannian signature. Example of such metrics are provided by the Euclidean equivalent of the Birmingham metrics, also known as *Euclidean*

Schwarzschild-de Sitter metrics or *Euclidean Schwarzschild-anti-de Sitter metrics* and which, in $(n+1)$ -dimensions, take the form

$$g = \left(\underbrace{\frac{r^2}{\ell^2} + \kappa - \frac{2m}{r^{n-2}}}_{=:F(r)} \right) dt^2 + \frac{dr^2}{\frac{r^2}{\ell^2} + \kappa - \frac{2m}{r^{n-2}}} + r^2 h_\kappa, \quad (5.5.46)$$

where $\ell > 0$ and m are real constants, $\kappa \in \{0, \pm 1\}$, and $(\check{N}, h_\kappa)$ is an $(n-1)$ -dimensional Einstein manifold with Ricci tensor equal to $(n-2)\kappa h_\kappa$.

The metrics (5.5.46) are obtained from the Birmingham metrics by replacing dt^2 with $-dt^2$. One can think of this as the result of a ‘complex coordinate transformation’, where t is replaced with $\sqrt{-1}t$; this is called a *Wick rotation* in the physics literature. Such a substitution preserves the condition that the Ricci tensor is proportional to the metric, which can be seen as follows.

Quite generally, let g be a metric such that $\partial_t g_{\mu\nu} = 0$ in a suitable coordinate system. Consider the tensor field, say $g(a)$, where every occurrence of dt in g is replaced by adt , where $a \in \mathbb{C}$. Let $R_{\mu\nu}(a)$ denote the Ricci tensor of $g(a)$, calculated using the usual formulae, which can be also used for complex valued tensor fields. Then $R_{\mu\nu}(a) - \lambda g_{\mu\nu}(a)$ is a holomorphic function of a away from the set where $\det g(a)_{\mu\nu}$ vanishes. When $a \in \mathbb{R}^+$ the metric $g(a)$ can be obtained from g by a bona fide coordinate transformation $t \mapsto at$; hence $R_{\mu\nu}(a) - \lambda g_{\mu\nu}(a)$ with $a \in \mathbb{R}$ vanishes if

$$R_{\mu\nu}(1) - \lambda g_{\mu\nu}(1) = R_{\mu\nu} - \lambda g_{\mu\nu}$$

did. Since a holomorphic function vanishing on the real positive axis vanishes everywhere, we conclude that $R_{\mu\nu}(a) - \lambda g(a)_{\mu\nu} = 0$ for all a on the connected component of $\mathbb{C}$ containing 1 on which $\det g(a)_{\mu\nu} \neq 0$. Using the notation of Appendix H.1 for the metric $g(1)_{\mu\nu} \equiv g_{\mu\nu}$ we have (see (H.3), p. 364)

$$\det g(a)_{\mu\nu} = -a^2 N^2 \det g_{ij}, \quad (5.5.47)$$

which shows that the argument applies to all $a \in \mathbb{C}^*$; in particular this applies to $a = \sqrt{-1}$, leading to the desired replacement of dt^2 by $-dt^2$ in the metric.

We conclude that g given by (5.5.46) is indeed an Einstein metric.

Let $r_* > 0$ be any *first-order* zero of g_{tt} ,

$$\frac{r_*^2}{\ell^2} + \kappa - \frac{2m}{r_*^{n-2}} = 0.$$

After introducing a new coordinate ρ by the formula

$$\rho(r) = \int_{r_*}^r \frac{1}{\sqrt{\frac{s^2}{\ell^2} + \kappa - \frac{2m}{s^{n-2}}}} ds, \quad (5.5.48)$$

one can rewrite the metric (5.5.46) as

$$g = d\rho^2 + \rho^2 H(\rho) dt^2 + r^2 h_\kappa, \quad (5.5.49)$$

where H is obtained by dividing g_{tt} by ρ^2 . Elementary analysis, using the fact that r_* is a simple zero of F , shows that

$$H(0) = \frac{F'(r_*)^2}{4}.$$

This implies that a periodic identification of t with period

$$T := \frac{4\pi}{F'(r_*)}$$

guarantees that $d\rho^2 + \rho^2 H(\rho) dt^2$ is a smooth metric on $\mathbb{R}^2$ with a rotation axis at $\rho = 0$. As a result, (5.5.49) defines a smooth Riemannian metric on

$$M := \mathbb{R}^2 \times \check{N}.$$

The metric (5.5.46) can be smoothly conformally compactified by introducing, for large r , a coordinate $x := 1/r$ and rescaling:

$$x^2 g = \left(\frac{1}{\ell^2} + \kappa - 2mx^n \right) dt^2 + \frac{dx^2}{\frac{1}{\ell^2} + \kappa x^2 - 2mx^n} + h_\kappa. \quad (5.5.50)$$

Hence, M admits a conformal boundary $\partial M := \{x = 0\}$ diffeomorphic to $S^1 \times \check{N}$, with conformal metric

$$\frac{dt^2}{\ell^2} + h_\kappa. \quad (5.5.51)$$

5.5.4 Horowitz–Myers-type metrics

Consider an $(n+1)$ -dimensional metric, $n \geq 3$, of the Birmingham form

$$g = f(r) d\psi^2 + \frac{dr^2}{f(r)} + r^2 \underbrace{\check{h}_{AB}(x^C) dx^A dx^B}_{=: \check{h}}, \quad (5.5.52)$$

except that now $\check{h}$ is a Riemannian or pseudo-Riemannian Einstein metric on an $(n-1)$ -dimensional manifold $\check{N}$ with constant scalar curvature $\check{R}$ and, similarly to the last section, the x^A 's are local coordinates on $\check{N}$. This metric can be formally obtained from (5.5.1) by changing t to $i\psi$. It therefore follows that for $m \in \mathbb{R}$ and $\ell \in \mathbb{R}^*$ the function

$$f = \beta - \frac{2m}{r^{n-2}} - \varepsilon \frac{r^2}{\ell^2}, \quad \varepsilon \in \{0, \pm 1\}, \quad \beta = \frac{\check{R}}{(n-1)(n-2)} \in \{0, \pm 1\}, \quad (5.5.53)$$

leads to a metric satisfying (5.5.3),

$$R_{\mu\nu} = \frac{\varepsilon n}{\ell^2} g_{\mu\nu}, \quad \varepsilon \in \{0, \pm 1\}, \quad (5.5.54)$$

where ℓ is a constant related to the cosmological constant as in (5.5.4). The treatment in the last section of zeros of f applies, leading to axes of rotation. Similarly the analysis there of conformal boundaries applies, leading to a timelike null infinity when $\check{h}$ has Lorentzian signature, with global structure reminiscent of that of anti-de Sitter spacetime, cf. Fig. 6.2.8, p. 269.

$\beta = 0$, $n = 3$. In [260] Horowitz and Myers consider the case $n+1 = 4$, $\varepsilon = -1$,⁹ and choose $\check{h} = -\ell^{-2} dt^2 + d\varphi^2$, with φ being a 2π -periodic coordinate on S^1 . Thus

$$g = -\frac{r^2}{\ell^2} dt^2 + f(r) \ell^2 \lambda^2 d\alpha^2 + \frac{dr^2}{f(r)} + r^2 d\varphi^2. \quad (5.5.55)$$

Replacing r with $1/x$ one finds that the timelike infinity $\mathcal{I} \approx \mathbb{R} \times S^1 \times S^1$ is conformally flat:

⁹The case $\beta = 0$ and $\varepsilon = 1$ leads to a signature $(+---)$ for large r ; our signature $(-+++)$ is recovered by multiplying the metric by -1 , but then one is back in the case $\varepsilon = -1$ after renaming m to $-m$.

$$x^2 g \rightarrow_{r \rightarrow \infty} -dt^2 + \ell^2(\lambda^2 d\alpha^2 + dx^2 + d\varphi^2). \quad (5.5.56)$$

Some comments about factors of ℓ are in order: if we think of r as having a dimension of length, then ℓ , t , and ψ also have dimension of length, m has dimension length^{n-1} , while f , x , and the x^A 's (and thus φ) are dimensionless.

A striking feature of the Horowitz–Myers metrics is the negativity of its total mass, when measured with respect to the massless Kottler metrics [34, 152, 260].

INCIDENTALLY: Uniqueness theorems for the metrics (5.5.55) have been established in [211, 213, 460]. $\square$

$\beta = \pm 1$, $n = 3$. We consider the metric (5.5.52) with $\varepsilon = -1$ (compare footnote 9 on p. 257) and $\check{h}$ of the form

$$\check{h} = \begin{cases} d\theta^2 + \sin^2(\theta) d\varphi^2, & \beta = 1; \\ d\theta^2 + \sinh^2(\theta) d\varphi^2, & \beta = -1. \end{cases} \quad (5.5.57)$$

In regions where f is positive, one obtains a Lorentzian metric after a ‘double Wick rotation’

$$\theta = i\ell^{-1}t, \quad \varphi = i\phi,$$

resulting in

$$g = -\frac{r^2}{\ell^2} dt^2 + \frac{dr^2}{f(r)} + f(r)\ell^2 \lambda^2 d\alpha^2 + r^2 \begin{cases} \sinh^2(\ell^{-1}t) d\phi^2, & \beta = 1; \\ \sin^2(\ell^{-1}t) d\phi^2, & \beta = -1. \end{cases} \quad (5.5.58)$$

Taking α and ϕ periodic one obtains again a conformal infinity diffeomorphic to $\mathbb{R} \times \mathbb{T}^2$. Note that the conformal metric at the conformal boundary is *not* conformally stationary anymore, as opposed to the metrics (5.5.56). We have not attempted to study the nature of the singularities of g at $\sinh(\ell^{-1}t) = 0$ or at $\sin(\ell^{-1}t) = 0$.

6

Extensions, conformal diagrams

In this chapter we present a systematic approach to extensions for a class of spacetimes, as needed to construct black-hole spacetimes. On the way we introduce the conformal diagrams, which are a useful tool for visualizing the geometry of the extensions.

We focus on metrics containing a two-by-two stationary Lorentzian block. We discuss causality for such metrics in Section 6.1; the possible building blocks are described in Section 6.2; these building blocks are put together in Section 6.3. The general rules governing the construction are explained in Section 6.4, with the causal aspects of the construction highlighted in the short Section 6.5. The method is applied to those Birmingham metrics which have not been analysed previously in Section 6.6.

6.1 Causality for a class of block-diagonal metrics

We start with a construction of extensions for metrics of the form

$$g = -Fdt^2 + F^{-1}dr^2 + \underbrace{h_{AB}dx^A dx^B}_{=:h}, \quad F = F(r), \quad (6.1.1)$$

where $h := h_{AB}(t, r, x^C)dx^A dx^B$ is a family of *Riemannian* metrics on an $(n-1)$ -dimensional manifold N^{n-1} , possibly depending upon t and r . Our analysis is based upon that of Walker [456].

There is a long list of important examples:

1. $F = 1 - \frac{2m}{r}$, $h = r^2 d\Omega^2$; this is the usual $(3+1)$ -dimensional Schwarzschild solution. For $m \neq 0$ the function F has a simple zero at $r = 2m$.
2. $F = 1 - \frac{2m}{r^{n-1}}$, $h = r^2 \check{h}$, where $\check{h}$ is the unit round metric on an $(n-1)$ -dimensional sphere S^{n-1} ; this is the $(n+1)$ -dimensional Schwarzschild–Tangherlini solution. Here F has again one simple positive zero for $m > 0$.
3. $F = 1 - \frac{2m}{r} + \frac{Q^2}{r^2}$, $h = r^2 d\Omega^2$; this is the $(3+1)$ -dimensional Reissner–Nordström metric, solution of the Einstein–Maxwell equations, with total electric charge Q , and with associated Maxwell potential $A = Q/r$ (compare Section 4.5, p. 172). If we assume that $|Q| < m$, then F has two distinct positive zeros

$$r_{\pm} = m \pm \sqrt{m^2 - |Q|^2},$$

and a single zero when $|Q| = m$. This last case is referred to as *extreme*, or *degenerate*.

4. $F = 1 - \frac{2m}{r^{n-2}} + \frac{Q^2}{r^{2(n-2)}}$, $h = r^2 d\Omega^2$; this is an $(n+1)$ -dimensional generalization of the Reissner–Nordström metric, solution of the Einstein–Maxwell equations, with total electric charge Q , and with associated Maxwell potential $A = Q/r$.
5. $F = -\frac{\Lambda}{3}r^2 + \kappa - \frac{2m}{r}$, where Λ is the cosmological constant, and $h = r^2 \check{h}_{\kappa}$, with $\check{h}_{\kappa}$ having constant Gauss curvature κ :

$$\check{h}_{\kappa} = \begin{cases} d\theta^2 + \sin^2 \theta d\varphi^2, & \kappa > 0; \\ d\theta^2 + d\varphi^2, & \kappa = 0; \\ d\theta^2 + \sinh^2 \theta d\varphi^2, & \kappa < 0; \end{cases}$$

these are the (3+1)-dimensional Kottler metrics, also known as the Schwarzschild–(anti-)de Sitter metrics.

6. $F(r) = 1 - \frac{2m}{r^{n-2}} - \frac{r^2}{\ell^2}$, where $\ell > 0$ is related to the cosmological constant Λ by the formula $2\Lambda = n(n-1)/\ell^2$; these are solutions of the vacuum Einstein equations if $h = r^2\check{h}$, provided that $\check{h}$ is an *Einstein metric on an $(n-1)$ -dimensional manifold N^{n-1} with scalar curvature $(n-1)(n-2)$* , already encountered in Section 5.5, p. 250. Note that $m = 0$ requires $(N^{n-1}, \check{h})$ to be the unit round metric if one wants to avoid a singularity at finite distance, at $r = 0$, along the level sets of t .

7. We note finally the metrics

$$g = -(\lambda - \Lambda r^2)dt^2 + (\lambda - \Lambda r^2)^{-1}dr^2 + |\Lambda|^{-1}\check{h}_k, \quad (6.1.2)$$

with $k = \pm 1$, $k\Lambda > 0$, $\lambda \in \mathbb{R}$. The case $k = 1$ has been discovered by Nariai [374].

REMARK 6.1.1 It is worth pointing out that the study of the conformal structure for (more general) metrics of the form

$$^{(2)}g = -F(r)H_1(r)dt^2 + F^{-1}(r)H_2(r)dr^2, \quad (6.1.3)$$

where H_1 and H_2 are strictly positive in the range of r of interest, can be reduced to that for the metric (6.1.1) by writing

$$^{(2)}g = \sqrt{H_1 H_2} \left(-\hat{F}dt^2 + \hat{F}^{-1}dr^2 \right), \text{ where } \hat{F} = \sqrt{\frac{H_1}{H_2}}F. \quad (6.1.4)$$

6.1.1 Riemannian aspects

We will be mainly interested in functions F which change sign: thus, we assume that there exists a real number r_0 such that

$$F(r_0) = 0.$$

Not unexpectedly, the global properties of g will depend upon the nature of the zero of F . For example, for the Schwarzschild metric with $m > 0$ we have a first-order zero of F at $2m$. This implies that the distance, on the level sets of t , along radial curves to the set $\{r = 2m\}$ is finite: indeed, letting γ denote any of the curves $r \mapsto (r, \theta, \varphi)$, the length of γ in the region $\{r > 2m\}$ is

$$\begin{aligned} \int \sqrt{g(\dot{\gamma}, \dot{\gamma})}dr &= \int \frac{dr}{\sqrt{1 - \frac{2m}{r}}} \\ &= \sqrt{r^2 - 2mr} + m \ln \left(r - m + \sqrt{r^2 - 2mr} \right) + C, \end{aligned}$$

where C depends upon the starting point. This has a finite limit as r approaches $2m$. More generally, if F has a first-order zero, then F behaves as some constant C_1 times $(r - r_0)$ near r_0 , giving a radial distance

$$\int \sqrt{g(\dot{\gamma}, \dot{\gamma})}dr = \int \frac{1}{\sqrt{F(r)}} \approx \int \frac{1}{\sqrt{C_1(r - r_0)}} < \infty,$$

since $x^{-1/2}$ is integrable near $x = 0$.

On the other hand, for the *extreme* Reissner–Nordström metric we have, by definition,

$$F(r) = \left(1 - \frac{m}{r}\right)^2,$$

leading to a radial distance

$$\begin{aligned} \int \sqrt{g(\dot{\gamma}, \dot{\gamma})} dr &= \int \frac{1}{1 - \frac{m}{r}} dr = \int \frac{r}{r - m} dr = \int \frac{r - m + m}{r - m} dr \\ &= r + \ln(r - m), \end{aligned} \quad (6.1.5)$$

which diverges as $r \rightarrow m$. Quite generally, if F has a zero of order $k \geq 2$ at r_0 , then F behaves as some constant C_2 times $(r - r_0)^k$ near r_0 , giving a radial distance

$$\int \sqrt{g(\dot{\gamma}, \dot{\gamma})} dr = \int \frac{1}{\sqrt{F(r)}} \approx \int \frac{1}{(C_2(r - r_0))^{k/2}} \rightarrow_{r \rightarrow r_0} \infty,$$

since the integral of $x^{-k/2}$ near $x = 0$ diverges for $k \geq 2$.

The above considerations are closely related to the embedding diagrams, already seen in Section 4.2.6, p. 143, for the Schwarzschild metric. In the Schwarzschild case those led to a hypersurface in Euclidean $\mathbb{R}^{n+1}$ which could be smoothly continued across its boundary. It is not too difficult to verify that this will be the case for any function F which has a first-order zero at $r_0 > 0$.

One can likewise attempt to embed in four-dimensional Euclidean space the $t = \text{const}$ slice of the extreme Reissner–Nordström metric. The embedding equation arising from (4.2.71), p. 143, now reads

$$\left(\frac{dz}{dr}\right)^2 + 1 = \frac{1}{F(r)}. \quad (6.1.6)$$

For r close to and larger than m we obtain

$$\frac{dz}{dr} \approx \frac{1}{1 - \frac{m}{r}}.$$

Integrating as in (6.1.5), one obtains a logarithmic divergence of the graphing function z near $r = m$:

$$z(r) \approx m \ln(r - m).$$

This behaviour can also be inferred from the exact formula

$$z = 2\sqrt{m(2r - m)} + m \ln \left(\frac{\sqrt{m(2r - m)} + m}{\sqrt{m(2r - m)} - m} \right).$$

The embedding, near $r = m$, is depicted in Fig. 6.1.1.

Quite generally, the behaviour near r_0 of the embedding function z , solution of (6.1.6), depends only upon the order of the zero of F at r_0 . For $r_0 > 0$, and for all orders of that zero larger than or equal to 2, if the ‘angular part’ h of the metric (6.1.1) is of the form $h = r^2 \check{h}$, where $\check{h}$ does not depend upon r , then the geometry of the slices $t = \text{const}$ resembles more and more that of a ‘cylinder’ $dx^2 + r_0^2 \check{h}$ as r_0 is approached.

6.1.2 Causality

To understand causality for metrics of the form (6.1.1), the guiding principles for the analysis that follows will be:

1. The $t - r$ part of g plays a key role;

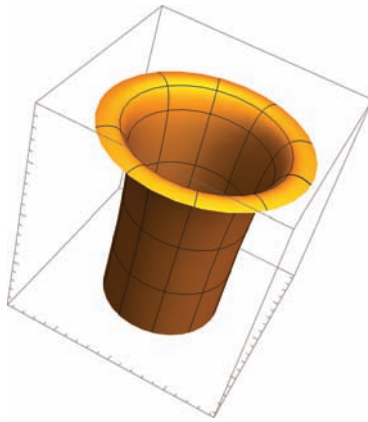


Fig. 6.1.1 An isometric embedding in four-dimensional Euclidean space (one dimension suppressed; circles should be S^2 's) of a slice of constant time in extreme Reissner–Nordström spacetime near $r = m$. I am grateful to Maciek Maliborski for providing the figure.

2. Multiplying the metric by a nowhere-vanishing function does not matter;
3. The geometry of bounded sets is easier to visualize than that of unbounded ones.

Concerning point 1, consider a timelike curve $\gamma(s) := (t(s), r(s), x^A(s))$ for the metric (6.1.1). We have

$$\begin{aligned}
 0 > g(\dot{\gamma}, \dot{\gamma}) &= -F \left(\frac{dt}{ds} \right)^2 + \frac{1}{F} \left(\frac{dr}{ds} \right)^2 + h_{AB} \left(\frac{dx^A}{ds} \right)^2 \left(\frac{dx^B}{ds} \right)^2 \\
 \implies 0 > g(\dot{\gamma}, \dot{\gamma}) &= -F \left(\frac{dt}{ds} \right)^2 + \frac{1}{F} \left(\frac{dr}{ds} \right)^2.
 \end{aligned} \tag{6.1.7}$$

Thus, curves which are timelike for g project to curves $(t(s), r(s))$ which are timelike for the metric

$${}^{(2)}g := -F dt^2 + \frac{1}{F} dr^2. \tag{6.1.8}$$

Similarly, (6.1.7) shows that, for any set of constants x_0^A , a curve $(t(s), r(s))$ which is timelike for the metric ${}^{(2)}g$ lifts to a curve $(t(s), r(s), x_0^A)$ which is timelike for g .

Identical statements hold for causal curves.

Concerning point 2 we note that, for any positive function Ω , the sign of $g(\dot{\gamma}, \dot{\gamma})$ is the same as that of $\Omega^2 g(\dot{\gamma}, \dot{\gamma})$. Hence, causality for a metric g is, in many respects, identical to that of the ‘conformally rescaled’ metric $\Omega^2 g$. This justifies the terminology *conformal diagrams* for the diagrams we are about to construct. These diagrams are sometimes referred to as *Penrose diagrams*, or *Carter diagrams*, or a mixture of the names with varying order.

Concerning point 3, it is best to proceed via examples, presented in the next section.

6.2 The building blocks

We proceed to gather a collection of building blocks that will be used to depict the global structure of spacetimes of interest. We start with the following.

6.2.1 Two-dimensional Minkowski spacetime

Let $g^{(2)}$ be the two-dimensional Minkowski metric

$$g^{(2)} = -dt^2 + dx^2, \quad (t, x) \in \mathbb{R}^2.$$

In order to map *conformally* $\mathbb{R}^2$ to a bounded set, we first introduce two null coordinates u and v ,

$$u = t - x, \quad v = t + x \quad \Longleftrightarrow \quad t = \frac{u + v}{2}, \quad x = \frac{v - u}{2}, \quad (6.2.1)$$

with $g^{(2)}$ taking the form

$$g^{(2)} = -du \, dv.$$

We have $(t, x) \in \mathbb{R}^2$ if and only if $(u, v) \in \mathbb{R}^2$. We bring the last $\mathbb{R}^2$ to a bounded set by introducing

$$U = \arctan(u), \quad V = \arctan(v), \quad (6.2.2)$$

thus

$$(U, V) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Using

$$\frac{du}{dU} = \frac{1}{\cos^2 U}, \quad \frac{dv}{dV} = \frac{1}{\cos^2 V},$$

the metric becomes

$$g^{(2)} = -\frac{1}{\cos^2 U \cos^2 V} dU dV. \quad (6.2.3)$$

This looks somewhat more familiar if we make one last change of coordinates similar to that in (6.2.1),

$$U = T - X, \quad V = T + X \quad \Longleftrightarrow \quad T = \frac{U + V}{2}, \quad X = \frac{V - U}{2}, \quad (6.2.4)$$

leading to

$$g^{(2)} = \frac{1}{\cos^2(T - X) \cos^2(T + X)} (-dT^2 + dX^2).$$

We conclude that the Minkowski metric on $\mathbb{R}^2$ is conformal to the Minkowski metric on a diamond

$$\{-\pi/2 < T - X < \pi/2, \quad -\pi/2 < T + X < \pi/2\};$$

see Fig. 6.2.1.

REMARK 6.2.1 Equation (6.2.3) shows that $g^{uu} = g^{vv} = 0$. This implies (see Proposition 1.5.1, p. 11) that the curves $s \mapsto (u, v = s)$ are future-directed null geodesics along which V approaches $\pi/2$ to the future, and $-\pi/2$ to the past. A similar observation applies to the null geodesics $s \mapsto (u = s, v)$. Thus, the union of the boundary intervals

$$\mathscr{I}^- := \{U \in (-\pi/2, \pi/2), \quad V = -\pi/2\} \cup \{V \in (-\pi/2, \pi/2), \quad U = -\pi/2\}$$

can be thought of as describing initial points of null future-directed geodesics; this set is usually denoted by $\mathscr{I}^-$, and is called *past null infinity*. Similarly, the set $\mathscr{I}^+$, called *future null infinity*, defined as

$$\mathscr{I}^+ := \{U \in (-\pi/2, \pi/2), \quad V = \pi/2\} \cup \{V \in (-\pi/2, \pi/2), \quad U = \pi/2\},$$

is the set of end points of future-directed null geodesics. Compare Section 3.1, p. 85.

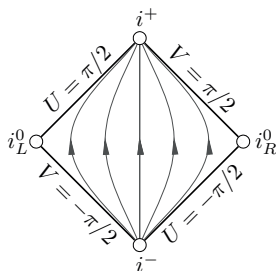


Fig. 6.2.1 The conformal diagram for $(1+1)$ -dimensional Minkowski spacetime; see also Remark 6.2.1. I am very grateful to Christa Ölz (see [382]) and Michał Eckstein for providing the figures in this section.

Next, every *timelike* future-directed geodesic acquires an end point at

$$i^+ := (V = \pi/2, U = \pi/2),$$

called *future timelike infinity*, and an initial point at

$$i^- := (V = -\pi/2, U = -\pi/2),$$

called *past timelike infinity*. Finally, all spacelike geodesics accumulate at both i_R^0 and i_L^0 . $\square$

6.2.2 Higher dimensional Minkowski spacetime

We write the $(n+1)$ -dimensional Minkowski metric, $n \geq 2$, using spherical coordinates,

$$\eta = -dt^2 + dr^2 + r^2 d\Omega^2,$$

where the symbol $d\Omega^2$ denotes the unit round metric on an $(n-1)$ -dimensional sphere. (We hope that the reader will not confuse $d\Omega^2$ with the square of the differential of the conformal factor Ω used in the definition of conformal completions.) In view of our principle that ‘for causality only the $t-r$ part of the metric matters’, to understand global causality it suffices to consider the two-dimensional metric

$$\stackrel{(2)}{g} = -dt^2 + dr^2.$$

But this is the two-dimensional Minkowski metric, so the calculations done in two dimensions apply, with x in (6.2.1) replaced by r . However, one must keep in mind the following:

1. First, $r \equiv x \geq 0$, as opposed to $x \in \mathbb{R}$ previously. In the notation of (6.2.1)–(6.2.4) this leads to

$$\begin{aligned} x \geq 0 &\iff v \geq u \\ &\iff \tan V \geq \tan U \\ &\iff V \geq U \\ &\iff X \geq 0. \end{aligned}$$

So instead of Fig. 6.2.1 one obtains Fig. 6.2.2a.

2. Next, Fig. 6.2.2a suggests that $r = 0$ is a boundary of the spacetime, which is not the case; instead it is an axis of rotation where the spheres of constant coordinates t and r degenerate to points. A more faithful representation in dimension $2+1$ is provided by Fig. 6.2.2b, which also gives an idea how Fig. 6.2.2a should be understood in higher dimensions.

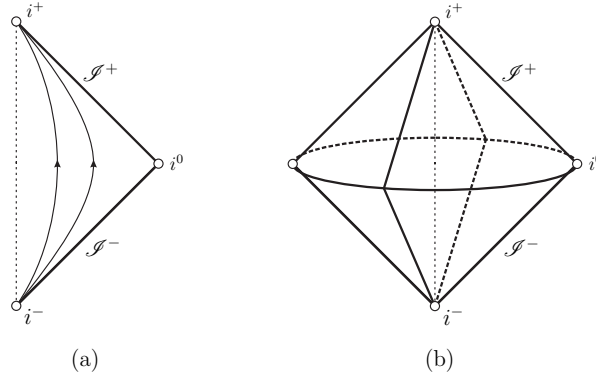


Fig. 6.2.2 Conformal structure of $(1+n)$ -dimensional Minkowski spacetime, $n \geq 2$. See also Fig. 6.2.3.

3. Finally, the conformal nature of the point i^0 of Fig. 6.2.2 needs a more careful investigation: for this, let us write $R \geq 0$ for X , and return to the equations

$$\eta = -dt^2 + dr^2 + r^2 d\Omega^2 = \frac{1}{\cos^2(T-R)\cos^2(T+R)}(-dT^2 + dR^2) + r^2 d\Omega^2,$$

$$T+R = \arctan(t+r), \quad T-R = \arctan(t-r).$$

Now,

$$r = \frac{1}{2}(\tan(R-T) + \tan(R+T)) = \frac{\sin(2R)}{\cos(2R) + \cos(2T)},$$

$$\cos(T-R)\cos(R+T) = \frac{1}{2}(\cos(2R) + \cos(2T)),$$

leading to

$$\eta = \frac{1}{(\cos(2R) + \cos(2T))^2} \underbrace{(4(-dT^2 + dR^2) + \sin^2(2R)d\Omega^2)}_{=: \hat{g}_E}. \quad (6.2.5)$$

The metric

$$\hat{g}_{S^3} := 4dR^2 + \sin^2(2R)d\Omega^2 = d\psi^2 + \sin^2(\psi)d\Omega^2, \text{ where } \psi := 2R,$$

is readily recognized to be the unit round metric on S^n , with $R = 0$ being the north pole, and $2R = \pi$ being the south pole. Hence

$$\hat{g}_E = -d\tau^2 + \hat{g}_{S^3}, \text{ where } \tau := 2T,$$

is the product metric on the *Einstein cylinder* $\mathbb{R} \times S^n$. Now, for $\tau \in (-\pi, \pi)$ and $\psi \in (0, \pi)$ the condition of positivity of the conformal factor,

$$\cos(\tau) + \cos(\psi) > 0,$$

is equivalent to

$$-\pi + \psi < \tau < \pi - \psi. \quad (6.2.6)$$

Thus, we have the following.

PROPOSITION 6.2.2 *For $n \geq 2$ the Minkowski metric is conformal to the metric on the open subset (6.2.6) of the Einstein cylinder $\mathbb{R} \times S^n$, cf. Fig. 6.2.3. $\square$*

From what has been said, it should be clear that i^0 is actually a single point in the conformally rescaled spacetime. Future null infinity $\mathcal{J}^+$ is the future light cone of i^0 in the Einstein cylinder, and reconverges at i^+ to the past light cone of i^+ . Similarly $\mathcal{J}^-$ is the past light cone of i^0 , and reconverges at i^- to the future light cone of i^- .

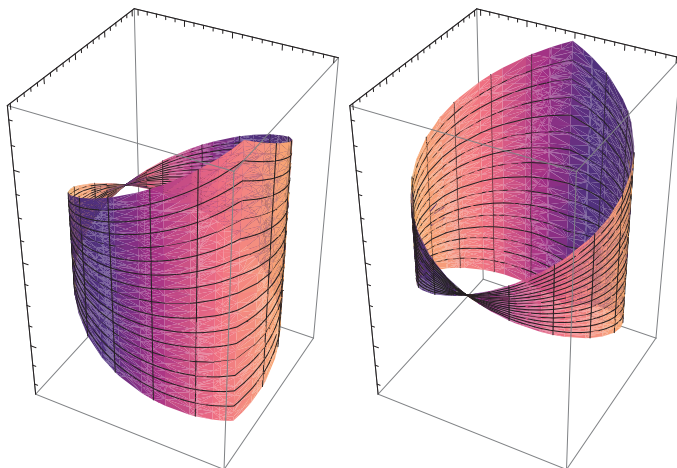


Fig. 6.2.3 The embedding of Minkowski spacetime into the Einstein cylinder $\mathbb{R} \times S^3$ viewed from two different perspectives, with two space dimensions suppressed. The lowest point is i^- , the highest i^+ , the touching point is i^0 .

6.2.3 $\int F^{-1}$ diverging at both ends

We return now to a metric $\overset{(2)}{g}$ of the form (6.1.8) with non-constant F , and consider an open interval $I = (r_1, r_2)$, with $r_1 \in \mathbb{R} \cup \{-\infty\}$, $r_2 \in \mathbb{R} \cup \{\infty\}$, such that F has constant sign on I . We choose some $r_* \in I$, and we assume that

$$\lim_{r \rightarrow r_1} \int_r^{r_*} \frac{ds}{|F(s)|} = \infty, \quad \lim_{r \rightarrow r_2} \int_{r_*}^r \frac{ds}{|F(s)|} = \infty. \quad (6.2.7)$$

Equation (6.2.7) will hold in the following cases of interest:

1. At the event horizons of all classical black holes: Schwarzschild with or without cosmological constant, Kerr–Newman with or without cosmological constant, etc. More generally, (6.2.7) will hold if $r_1 \in \mathbb{R}$ and if F extends differentiably across r_1 , with $F(r_1) = 0$; note that the left integral will then diverge regardless of the order of the zero of F at r_1 . A similar statement holds for r_2 .
2. In the asymptotically flat regions of asymptotically flat spacetimes. Quite generally, (6.2.7) will hold if $r_2 = \infty$ and if F is bounded away from zero near r_2 , as is the case for asymptotically flat regions where $F(r) \rightarrow 1$ as $r \rightarrow \infty$.

Note that the sign of F determines the causal character of the Killing vector $X := \partial_t$: X will be timelike if $F > 0$ and spacelike otherwise. Alternatively, t or $-t$ will be a time function if $F > 0$, while r or $-r$ will be a time function in regions where F is negative.

We introduce a new coordinate x defined as

$$x(r) = \int_{r_*}^r \frac{ds}{F(s)} \quad \implies \quad dx = \frac{dr}{F(r)}. \quad (6.2.8)$$

This gives

$$\overset{(2)}{g} = -F dt^2 + \frac{1}{F} (\underbrace{dr}_{F dx})^2 = F(-dt^2 + dx^2). \quad (6.2.9)$$

In view of (6.2.7) the coordinate x ranges over $\mathbb{R}$. So, if $F > 0$, then $\overset{(2)}{g}$ is conformal to the two-dimensional Minkowski metric, and thus the causal structure is that in

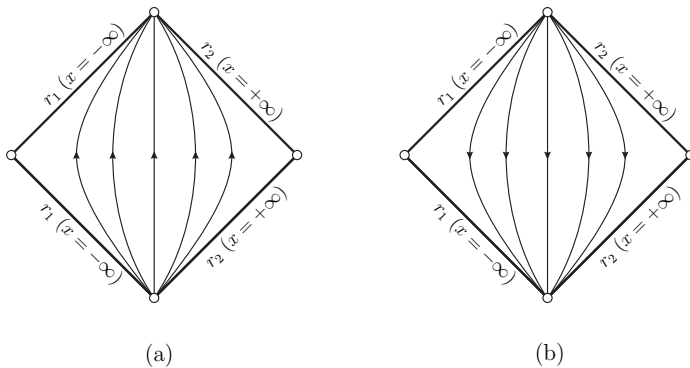


Fig. 6.2.4 The conformal structure for $F > 0$ according to whether the Killing vector $X = \partial_t$ is future pointing (left) or past pointing (right). Note that in some cases r might have a wrong orientation (this occurs e.g. in region *III* in the Kruskal–Szekeres spacetime of Fig. 4.2.7), in which case one also needs to consider the mirror reflections of the above diagrams with respect to the vertical axis; compare Fig. 6.2.6.

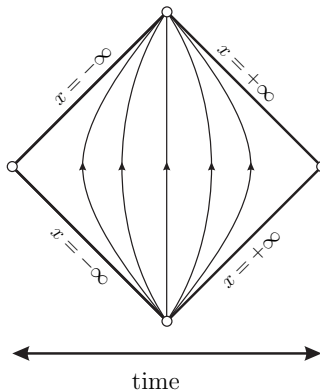


Fig. 6.2.5 The conformal structure for $F < 0$.

Fig. 6.2.4. Otherwise, for *negative* F , we obtain a Minkowski metric in which x corresponds to time and t corresponds to space, leading to a causal structure as in Fig. 6.2.5. Rotating Fig. 6.2.5 so that time flows to the future along the vertical positively oriented axis we obtain the four possible diagrams of Fig. 6.2.6.

6.2.4 $\int F^{-1}$ diverging at one end only

We consider again a general $g^{(2)}$ of the form (6.1.8), with F defined on an open interval $I = (r_1, r_2)$, with $r_1 \in \mathbb{R} \cup \{-\infty\}$, $r_2 \in \mathbb{R} \cup \{\infty\}$, such that F has constant sign on I . We choose some $r_* \in I$ and, instead of (6.2.7), we assume that

$$\lim_{r \rightarrow r_1} \int_r^{r_*} \frac{ds}{|F(s)|} < \infty, \quad \lim_{r \rightarrow r_2} \int_{r_*}^r \frac{ds}{|F(s)|} = \infty. \quad (6.2.10)$$

The case of r_1 interchanged with r_2 in (6.2.10) is analysed by replacing r by $-r$ in what follows. Note that this introduces the need of applying a mirror symmetry to the diagrams that follow.

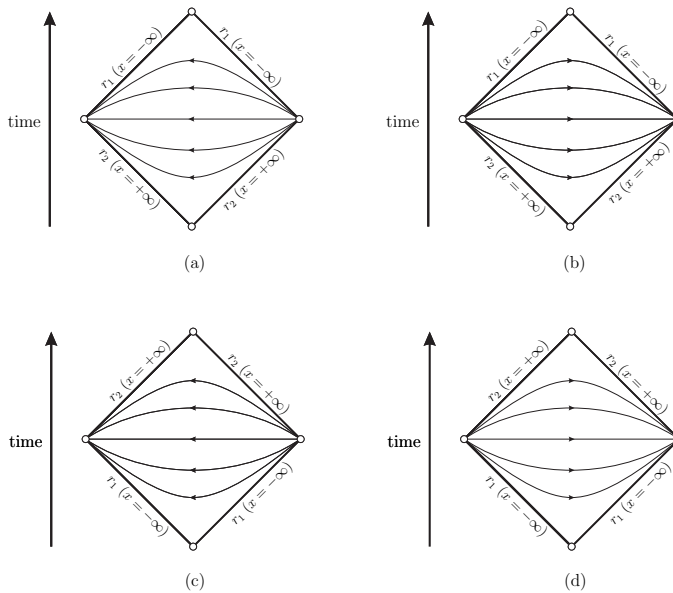


Fig. 6.2.6 Figure 6.2.5 rotated so that time flows in the positive vertical direction. Four different diagrams are possible, according to whether the Killing vector $X = \partial_t$ is pointing left or right, and whether ∇r is future- or past pointing.

The conditions in (6.2.10) arise in the following cases of interest:

1. We have $r_1 \in \mathbb{R}$, with the set $\{r = r_1\}$ corresponding either to a singularity or to an axis of rotation. We encountered the latter possibility when analysing $(n + 1)$ -dimensional Minkowski spacetime. The former situation occurs e.g. in Schwarzschild spacetime under the horizons, with $r_1 = 0$ and $r_2 = 2m$.
2. An example with $r_1 = -\infty$ is provided by anti-de Sitter spacetime, where F behaves as r^2 for large r . The variable r here should be *the negative of* the usual radial coordinate in anti-de Sitter. Yet another example of this kind occurs in the de Sitter metric, where F behaves as $-r^2$ for large $|r|$, so that r is a time function there.

Instead of (6.2.8) we introduce a new coordinate x defined as

$$x(r) = \int_{r_1}^r \frac{ds}{F(s)}. \quad (6.2.11)$$

Equation (6.2.9) remains unchanged, but now the coordinate x ranges over $[0, \infty)$. This has already been analysed in the context of higher dimensional Minkowski spacetime, resulting in the conformal diagrams of Fig. 6.2.7.

6.2.5 Birmingham metrics with $\Lambda < 0$ and $m = 0$

We consider a positive function $F : [0, \infty) \rightarrow \mathbb{R}$ which has no zeros, with

$$x_\infty := \int_0^\infty \frac{dr}{F(r)} < \infty. \quad (6.2.12)$$

This will be e.g. the case for the generalized Kottler metrics (as described in Section 5.5, p. 250), with negative Λ , with $\check{R} > 0$, and with vanishing mass $m = 0$ so that

$$F(r) = \frac{r^2}{\ell^2} + 1.$$

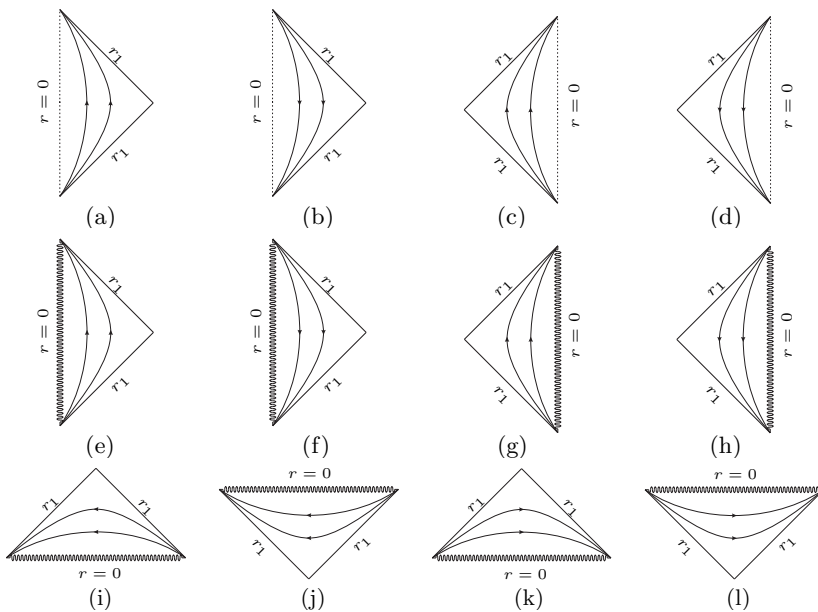


Fig. 6.2.7 Some possible diagrams for (6.2.10). Time always flows forwards along the vertical axis. In (a)–(d) the set $\{r = 0\}$ corresponds to an axis of rotation; in (e)–(h) it is a singularity. There should be four more such figures where $\{r = 0\}$ should be replaced by $\{r = \infty\}$, corresponding to an asymptotic region. Similarly there should be four more figures similar to (i)–(l), where a singularity $\{r = 0\}$ is replaced by an asymptotic region $\{r = \infty\}$.

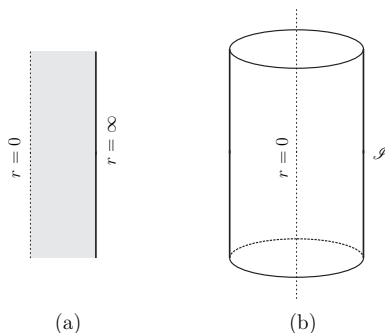


Fig. 6.2.8 The conformal structure of anti-de Sitter spacetime. The two-dimensional projection is the shaded strip $0 \leq r < \infty$ of (a). Since $\{r = 0\}$ is a centre of rotation, a more faithful representation is provided by the solid cylinder of (b).

In this case, we use (6.2.8) with $r_* = 0$, then $x \in [0, x_\infty)$; we obtain that $g^{(2)}$ is conformal to a Minkowski metric on a strip $\mathbb{R}_{\text{time}} \times [0, x_\infty)$, as in Fig. 6.2.8a. If the ‘internal space’ $\tilde{N}$ is a sphere S^{n-1} , then $\{r = 0\} \equiv \{x = 0\}$ is a rotation axis, so a more adequate representation of the resulting spacetime is provided by Fig. 6.2.8b.

6.3 Putting things together

We have now at our disposal a variety of building blocks and a natural question arises, whether more interesting spacetimes can be constructed using those. We start

by noting that no C^2 -extensions are possible across a boundary near which $|F|$ approaches infinity: indeed, $F = -g(X, X)$, where $X = \partial_t$ is a Killing vector. Now, Theorem 4.4.2, p. 169, shows that for any Killing vector the scalar function $g(X, X)$ is bounded on compact sets, which justifies the claim. It follows that boundaries at which F becomes unbounded correspond either to a spacetime singularity, as is the case in the Schwarzschild metric at $r = 0$, or to a ‘boundary at infinity’ representing ‘points lying infinitely far away’.

So it remains to consider boundaries at which F tends to a finite value.

6.3.1 Four-blocks gluing

We have seen in Remark 4.2.10, p. 139, how to glue four blocks together, assuming a *first-order zero* of F . This allows us to reproduce immediately the Penrose diagram of the Schwarzschild spacetime, by gluing together across $r_1 = 2m$ two copies of block (a) from Fig. 6.2.4 (one for which r increases from left to right, corresponding to the usual $r > 2m$ Schwarzschild region, with a mirror image thereof where r decreases from left to right), as well as blocks (i) and (j) from Fig. 6.2.7.

Some further significant examples are as follows:

EXAMPLE 6.3.1 (The conformal structure of non-extreme Reissner–Nordström black holes) Let us consider a C^k function

$$F : [0, \infty) \rightarrow \mathbb{R},$$

for some $k \geq 1$, such that F has precisely two first-order zeros at $0 < r_1 < r_2 < \infty$, and assume that

$$\lim_{r \rightarrow \infty} F(r) = 1, \quad \int_0^{r_1} \frac{dr}{F(r)} < \infty.$$

We further assume that the set $\{r = 0\}$ corresponds to a spacetime singularity. This is the behaviour exhibited by the electrovacuum Reissner–Nordström black holes with $|Q| < m$; compare Section 4.5, p. 172.

One possible construction of a (maximal, analytic) extension of the region $\{r > r_2\}$ proceeds as follows: we start by noting that this region corresponds to the block of Fig. 6.2.4a; this is block *I* in Fig. 6.3.1. We can perform a four-block gluing by joining together the left–right mirror image of block (a) from Fig. 6.2.4, corresponding to the region $\{r > r_2\}$ where now r *decreases* from left to right (this is block *III* in Fig. 6.3.1), as well as blocks (b) and (d) from Fig. 6.2.6, corresponding to two regions $\{r_1 < r < r_2\}$. This results in the spacetime consisting of the union of blocks *I* to *IV* in Fig. 6.3.1. This spacetime can be further extended to the future, via a four-blocks gluing, by adding two triangles (c) and (g) from Fig. 6.2.7, and yet another region $\{r_1 < r < r_2\}$ from Fig. 6.2.6. This leads to a spacetime consisting of the union of blocks *I* to *VII* in Fig. 6.3.1. One can now continue periodically in time, both to the future and to the past, obtaining the infinite sequence of blocks of Fig. 6.3.1.

Note that identifying periodically in time, with distinct periods, provides a countable infinity of distinct alternative extensions. The resulting spacetimes contain closed timelike curves, and no black-hole region.

Further maximal analytic distinct extensions can be obtained by removing a certain number of bifurcation spheres from the spacetime depicted in Fig. 6.3.1, and passing to the universal cover of the resulting spacetime. There are then no causality violations, as opposed to the examples of the previous paragraph. On the other hand, the current construction leads to spacetimes containing incomplete geodesics on which e.g. the norm of the Killing vector ∂_t remains bounded, while no such geodesics exist in the spacetimes of the previous paragraph.

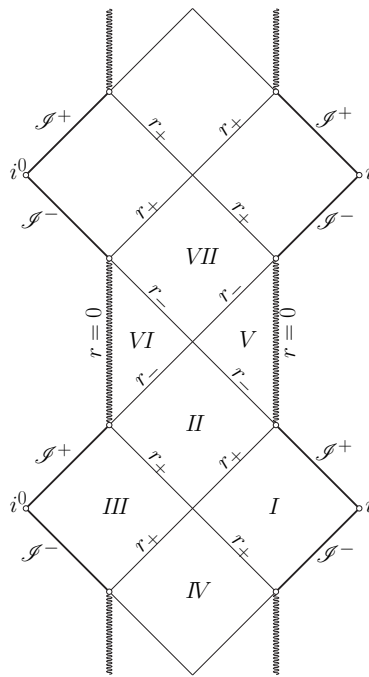


Fig. 6.3.1 A maximal analytic extension for the Reissner–Nordström metric with $|Q| < m$.

EXAMPLE 6.3.2 (Birmingham/‘Schwarzschild–de Sitter’ metrics) We consider a function $F \in C^k([0, \infty))$, $k \geq 1$, which has precisely two first-order zeros at $0 < r_1 < r_2 < \infty$, which is negative for large r , and which satisfies

$$\int_{2r_2}^{\infty} \frac{dr}{|F(r)|} < \infty.$$

We again assume that the set $\{r = 0\}$ corresponds to a spacetime singularity. This is the behaviour of the Birmingham metrics of Section 5.5, p. 250) with cosmological constant $\Lambda > 0$ under suitable restrictions on m . Recall that the metric takes the form

$$ds^2 = -F(r)dt^2 + \frac{dr^2}{F(r)} + r^2\check{h}, \quad \text{where} \quad F(r) = 1 - \frac{2m}{r^{n-2}} - \frac{r^2}{\ell^2}, \quad (6.3.1)$$

where $\ell > 0$ is related to the cosmological constant Λ by the formula $2\Lambda = n(n-1)/\ell^2$, while $\check{h}$ denotes an Einstein metric with constant scalar curvature $(n-1)(n-2)$ on an $(n-1)$ -dimensional manifold $\check{N}$. Representative graphs of the function F are shown in Fig. 6.3.2. We will only consider the case $m > 0$ and

$$\left(\frac{2}{(n-1)(n-2)} \right)^{n-2} \Lambda^{n-2} m^2 n^2 < 1, \quad (6.3.2)$$

which are sufficient and necessary conditions for exactly two distinct positive first-order zeros of F . When $n = 3$, the condition (6.3.2) reads $9m^2\Lambda < 1$, and the case of equality is referred to as the extreme Kottler–Schwarzschild–de Sitter spacetime. In the limit where Λ tends to 0 with m held constant, the spacetime metric approaches the Schwarzschild metric with mass m , and in the limit where m goes to 0 with Λ held constant the metric tends to that of the de Sitter spacetime with cosmological constant Λ .

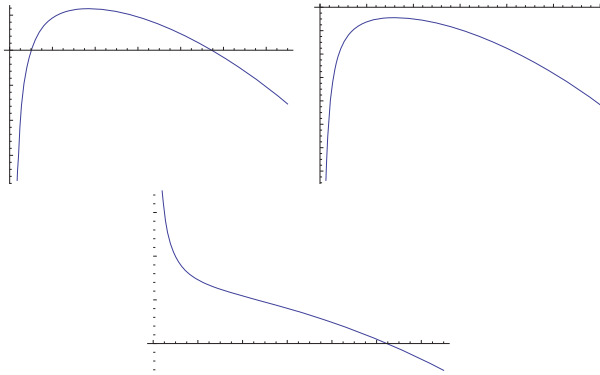


Fig. 6.3.2 The function F when (from left to right) (a) m is positive but smaller than the threshold of (6.3.2), (b) m is positive and larger than the threshold, and (c) m is negative.

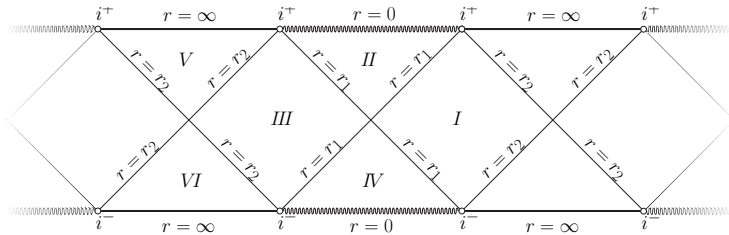


Fig. 6.3.3 A maximal extension of the class (6.3.2) of generalized Kottler (Schwarzschild-de Sitter) metrics with positive cosmological constant and mass.

To obtain a maximal extension, as $F(r)$ is positive for $r \in (r_1, r_2)$ we can choose, say, block (a) of Fig. 6.2.4 as the starting point of the construction; this is block I of Fig. 6.3.3. A four-block gluing can then be done using further the mirror reflection of block (b) of Fig. 6.2.4 as well as blocks (j) and (k) of Fig. 6.2.7 to extend I to the spacetime consisting of blocks I – IV of Fig. 6.3.3. Continuing similarly across $r = r_2$, etc., one obtains the infinite sequence of blocks of Fig. 6.3.3.

Let us denote by $(\mathcal{M}, g)$ the spacetime constructed as in Fig. 6.3.3, then $\mathcal{M}$ is diffeomorphic to $\mathbb{R}_{\text{time}} \times \mathbb{R}_{\text{space}} \times \check{N}$. Note that Fig. 6.3.3 remains unchanged when shifted by two blocks to the left or right. This leads to a discrete isometry of the associated spacetime $(\mathcal{M}, g)$; let's call it ψ . Given $k \in \mathbb{N}$, one can then consider the quotient manifold $\mathcal{M}/\psi^k$, with the obvious metric. This is the same as introducing periodic identifications in Fig. 6.3.3, identifying a block with its image obtained by shifting by a multiple of $2k$ blocks to the left or to the right. The resulting spacetime will have topology $\mathbb{R} \times S^1 \times \check{N}$; in particular it will contain compact spacelike hypersurfaces, with topology $S^1 \times \check{N}$. For distinct k 's the resulting spacetimes will be diffeomorphic, but not isometric.

EXAMPLE 6.3.3 (Kottler-de Sitter metrics with positive cosmological constant, and vanishing mass parameter m) We consider the metrics (6.3.1) with $m = 0$; thus $F(r) = 1 - r^2/\ell^2$. Then F has one simple zero for positive r . By arguments already given, one is led to the conformal diagram of Fig. 6.3.4.

EXAMPLE 6.3.4 (Kottler-anti-de Sitter metrics with negative cosmological constant) The reader should have no difficulties showing that the metrics (6.3.1) with $\Lambda < 0$ and $m \neq 0$ can be extended to a spacetime as in Fig. 7.4.7, p. 301, without,

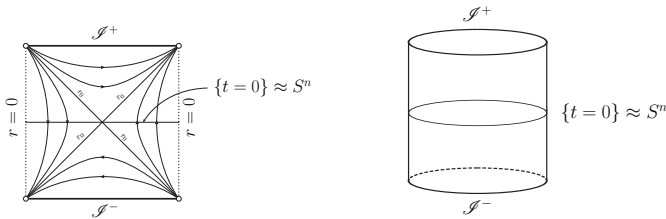


Fig. 6.3.4 The generalized Kottler (de Sitter) metrics with positive cosmological constant and vanishing mass parameter m . (a) In this a conformal diagram, the lines $\{r = 0\}$ are centres of rotation. (b) makes it clearer that the Cauchy surface $\{t = 0\}$, as well as $\mathcal{I}^+$ and $\mathcal{I}^-$, has spherical topology.

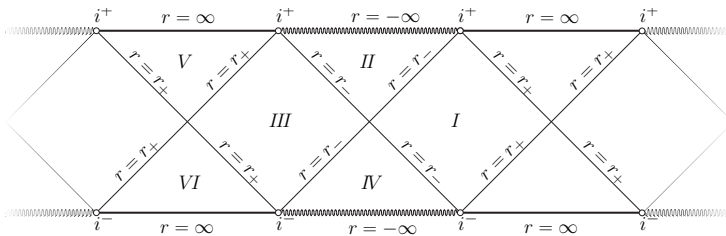


Fig. 6.3.5 A maximal extension of Nariai metrics with $\lambda > 0$ and $\Lambda > 0$.

however, the shaded region there as there are no time machines in the solution when the angular-momentum parameter a vanishes.

EXAMPLE 6.3.5 (Nariai metrics with $\lambda\Lambda > 0$) The Nariai metrics can be written in the form

$$g = -(\lambda - \Lambda r^2)dt^2 + (\lambda - \Lambda r^2)^{-1}dr^2 + |\Lambda|^{-1}\check{h}_k, \quad (6.3.3)$$

with constants satisfying $k = \pm 1$, $k\Lambda > 0$, $\lambda \in \mathbb{R}$. The metric g will satisfy the Lorentzian $(n + 1)$ -dimensional vacuum Einstein equations with cosmological constant proportional to Λ (equal Λ in spacetime dimension four) if and only if $\check{h}_k$ is a Riemannian Einstein metric on a $(n - 1)$ -dimensional manifold $\check{N}$ with scalar curvature equal to a suitable, dimension-dependent constant, whose sign coincides with that of k .

The lapse function g_{tt} has two first-order zeros $r_- < r_+$ if and only if $\lambda\Lambda > 0$, with the Killing vector ∂_t timelike between r_- and r_+ ; in all remaining cases $\lambda\Lambda \leq 0$ we obtain directly an inextendible spacetime, without Killing horizons, and thus a somewhat dull product structure. When $r \rightarrow \pm\infty$ we have $|g_{tt}| \rightarrow \infty$, and since the norm of a Killing vector is a geometric invariant, no extension is possible there. One can then obtain a global extension shown in Fig. 6.3.5.

The case $\lambda\Lambda > 0$ but $\Lambda < 0$ leads to a global structure described by rotating Fig. 6.3.5 by 90° .

For further reference we note alternative forms of g . When $\lambda > 0$ and $\Lambda > 0$, a constant rescaling of t and r leads to

$$g = \Lambda^{-1} \left(-(1 - r^2)dt^2 + \frac{dr^2}{1 - r^2} + \check{h}_k \right). \quad (6.3.4)$$

In the region $r^2 < 1$ (regions I and III in Fig. 6.3.5) we can set $r = \cos(x)$, so that

$$g = \Lambda^{-1} \left(-\sin^2(x)dt^2 + dx^2 + \check{h}_k \right). \quad (6.3.5)$$

In the region $r^2 > 1$ (regions *II*, *IV*, *V*, and *VI* in Fig. 6.3.5) we can set $r = \cosh(\tau)$ and $y = t$ in (6.3.4), which results in

$$g = \Lambda^{-1} \left(-d\tau^2 + \sinh^2(\tau) dy^2 + \check{h}_k \right). \quad (6.3.6)$$

In either case, the space part of the metric has cylindrical structure, with a product metric on $\mathbb{R} \times \check{N}$.

Amusingly, the metric (6.3.6) can be obtained from (6.3.5) by replacing $x \mapsto i\tau$ and $t \mapsto y$. Further complex substitutions in (6.3.6), namely $\tau \mapsto \tau + i\pi/2$ and $y \mapsto iy$, lead to the metric

$$g = \Lambda^{-1} \left(-d\tau^2 + \cosh^2(\tau) dy^2 + \check{h}_k \right), \quad (6.3.7)$$

with cylindrical spatial slices and boring global structure.

When λ and Λ are both negative, a constant rescaling of t and r leads instead to

$$g = |\Lambda|^{-1} \left(-(r^2 - 1) dt^2 + \frac{dr^2}{r^2 - 1} + \check{h}_k \right), \quad (6.3.8)$$

subsequently leading to obvious sign changes in (6.3.5)–(6.3.7).

6.3.2 Two-blocks gluing

The four-blocks gluing construction requires a first-order zero of F , but there exist metrics of interest where F has zeros of order two. Examples are provided by the extreme Reissner–Nordström metrics, with $|Q| = m$, or by the extreme generalized Kottler metrics, for which the inequality in (6.3.2) is an equality. In such cases a two-block gluing applies, which works regardless of the order of the zero of F , and which proceeds as follows: consider a function F defined on an interval I , which might or might not change sign on I , with one single zero there. As in (4.2.48), p. 140, we introduce functions u and v defined as

$$u = t - f(r), \quad v = t + f(r), \quad f' = \frac{1}{F}. \quad (6.3.9)$$

But now one does *not* use u and v simultaneously; instead one considers, first, a coordinate system (u, r) , so that

$$\stackrel{(2)}{g} = -F \left(\underbrace{dt}_{du + \frac{1}{F} dr} \right)^2 + \frac{1}{F} dr^2 = -F du^2 - 2 du dr.$$

Since $\det \stackrel{(2)}{g} = -1$, the resulting metric extends smoothly as a Lorentzian metric to the whole interval of definition, say I , of F . If we further replace u by a coordinate $U = \arctan u$, as in (6.2.2), each level set of U is extended from its initial range $r \in (r_1, r_2)$ to the whole range I . In terms of the blocks of, say, Fig. 6.2.4, this provides a way of extending across the lower-left interval $r = r_1$ and/or across the upper-right interval $r = r_2$; similarly for the remaining blocks. Equivalently, the (u, r) -coordinates allow one to attach another block from our collection at the boundaries $V = -\pi/2$ and $V = \pi/2$, with $V = \arctan v$ as in (6.2.2).

Next, using (v, r) as coordinates one obtains

$$\stackrel{(2)}{g} = -F \left(\underbrace{dt}_{dv - \frac{1}{F} dr} \right)^2 + \frac{1}{F} dr^2 = -F dv^2 + 2 dv dr.$$

The (V, r) -coordinates provide a way of extending across the boundary intervals $U = \pm\pi/2$; in Fig. 6.2.4 these are the (open) lower-right or upper-left boundary intervals.

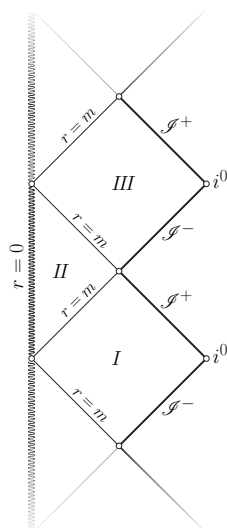


Fig. 6.3.6 A maximally extended extreme Reissner–Nordström spacetime.

EXAMPLE 6.3.6 (The global structure of extreme Reissner–Nordström black holes $|Q| = m$) For extreme Reissner–Nordström metrics the function F equals

$$F(r) = \left(1 - \frac{m}{r}\right)^2.$$

Although this is not needed for our purposes here, we note the explicit form of the function f in (6.3.9):

$$f(r) = r - \frac{m^2}{r - m} + 2m \ln |r - m|.$$

To construct a maximal extension of the exterior region $r \in (m, \infty)$, we start with block (a) from Fig. 6.2.4 with $r \in (m, \infty)$; this is region *I* in Fig. 6.3.6. This can be extended across the upper-left interval with block (e) from Fig. 6.2.7, providing region *II* in Fig. 6.3.6. That last block can be extended by another block identical to *I*. Continuing in this way leads to the infinite sequence of blocks of Fig. 6.3.6.

6.4 General rules

For definiteness we will assume in this section that the full spacetime metric takes the form

$$-F(r)dt^2 + \frac{dr^2}{F(r)} + \underbrace{r^2 \check{h}_{AB}(x^C) dx^A dx^B}_{=:h}, \quad (6.4.1)$$

which holds for many metrics of interest. The reader should have no difficulties adapting the discussion here to more general metrics, with a positive-definite h depending possibly on all variables. The function r in (6.4.1) will be referred to as the *radius function*.

Our constructions so far lead to the following picture: consider any two blocks from the collection provided so far, with corresponding functions F_1 and F_2 , and Killing vectors X_1 and X_2 generating translations in the coordinate t of (6.4.1). Then we have the following rules:

1. Before any gluing all two-dimensional coordinate domains should be viewed as open subsets of the plane, without their boundaries.

2. Two such blocks can be attached together across an *open* boundary interval to obtain a metric of class C^k if the corresponding radius functions take the same finite value at the boundary, and if the function F_2 extends F_1 in a C^k way across the boundary. One might sometimes have to change the space orientation $x \rightarrow -x$ of one of the blocks to achieve this, and perhaps also the time orientation so that the (extended) Killing vector X_1 matches X_2 at the relevant boundary interval.
3. The calculation in Example 4.3.7, p. 152, shows that the surface gravity of a horizon $r = r_*$, where $F(r_*) = 0$, for metrics of the form (6.4.1) equals $F'(r_*)/2$. Hence, in view of what has just been said, a necessary and sufficient condition for a C^1 gluing of the metric across an open boundary interval is equality of the *radius functions* r together with equality of the *surface gravities* of the Killing vectors ∂_t at the boundary in question.
4. Two-block gluings only attach the common *open* boundary interval to the existing structure, so that the result is again an open subset of $\mathbb{R}^2$. In particular two-block gluings never attach the corners of the blocks to the spacetime.
5. Four-block gluings can only be done across a first-order zero of F at which F is differentiable. A function F which is of C^k -differentiability class leads then to a metric which is of C^{k-1} -differentiability class.
6. A four-block gluing attaches to the spacetime the common corner of the four-blocks, as well as the four open intervals accumulating at the corner. (The result is of course again an open subset of $\mathbb{R}^2$.)

6.5 Black holes / white holes

One of the points of the conformal diagrams above is that one can, by *visual inspection*, decide whether a spacetime, constructed by the prescription just given, contains a black-hole region. The key observation is that each boundary which is represented by a line of a 45° slope corresponds to a null hypersurface in spacetime. If the spacetime is faithfully represented by a collection of blocks on the plane, the corresponding hypersurfaces are two-sided in spacetime, and can therefore

only be crossed by future-directed causal curves from one side to the other.

So consider a spacetime which contains a block with a boundary which has a slope of either 45° or -45° . Let Γ denote a straight line in the plane which contains that boundary. Assuming the usual time orientation, it should be clear that no future-directed causal curve with initial point in that part of the plane which lies above Γ will ever reach that part of the plane which lies under Γ . In other words, the region above Γ is inaccessible to any observer that remains entirely under Γ .

We conclude that if a *physically preferred* block lies under Γ , then anything above Γ will belong to a *black-hole region*, as defined relatively to that block.

One can similarly talk about *white-hole regions* by reversing time orientation.

As an example consider block I of Fig. 6.3.1, p. 271, describing one connected component of the infinitely many ‘exterior’, $r > r_+$, regions of a maximally extended non-degenerate Reissner–Nordström solution. Everything lying above the line of a 45° slope bounding this block to the future belongs to a black-hole region, as defined with respect to this block. Everything lying below the line of a -45° slope bounding this block to the past belongs to a white-hole region, as defined with respect to block I .

Note that this argument might fail if the spacetime is *not* faithfully represented by a subset of the plane, for example if some identifications between various blocks are made, as already mentioned at the end of Example 6.3.1, p. 270.

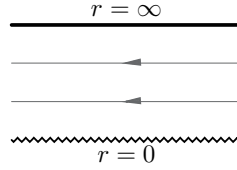


Fig. 6.6.1 The (t, r) -conformal diagram when the function f in (6.6.1) is negative.

6.6 Birmingham metrics

Further examples of interesting conformal diagrams can be constructed for the *Birmingham metrics* of Section 5.5, p. 250. Recall that these are $(n+1)$ -dimensional metrics, $n \geq 3$, of the form

$$g = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 \underbrace{\check{h}_{AB}(x^C)dx^A dx^B}_{=: \check{h}}, \quad (6.6.1)$$

where $\check{h}$ is a *Riemannian metric* on a compact $(n-1)$ -dimensional manifold $\check{N}$. As already pointed out, the global structure of g is determined by the number and nature of the zeros of f , regardless of the details of the function f .

6.6.1 Cylindrical solutions

Consider, first, the case where f has no zeros. Since f is negative for large $|r|$, f is negative everywhere. It therefore makes sense to rename r to $\tau > 0$, t to x , and $-f$ to $F > 0$, leading to the metric

$$g = -\frac{d\tau^2}{F(\tau)} + F(\tau)dx^2 + \tau^2 \check{h}. \quad (6.6.2)$$

The non-zero level sets of the time coordinate τ are infinite cylinders with topology $\mathbb{R} \times \check{N}$, with a product metric. Note that the extrinsic curvature of those level sets is never zero because of the τ^2 term in front of $\check{h}$, except possibly for the $\{\tau = 0\}$ -slice in the case $\beta = -1$ and $m = 0$.

Assuming that $m \neq 0$, the region $\tau \in (0, \infty)$ is a ‘big-bang–big freeze’ spacetime with cylindrical spatial sections. The corresponding conformal diagram is an infinite horizontal strip with a singular spacelike boundary at $\tau = 0$, and a smooth conformal spacelike boundary at $\tau = \infty$; see Fig. 6.6.1.

In the case $m = 0$ and $\beta = 0$ the spatial sections are again cylindrical, with the boundary $\{\tau = 0\}$ being now at infinite temporal distance: Indeed, setting $T = \ln \tau$, in this case we can write

$$\begin{aligned} g &= -\ell^2 \frac{d\tau^2}{\tau^2} + \frac{\tau^2}{\ell^2} dx^2 + \tau^2 \check{h} \\ &= -\ell^2 dT^2 + e^{2T} \left(\frac{dx^2}{\ell^2} + \check{h} \right). \end{aligned}$$

When $\check{h}$ is a flat torus, this is one of the forms of the de Sitter metric [244, p. 125].

The next case which we consider is $f \leq 0$, with f vanishing precisely at one positive value $r = r_0$. This occurs if and only if $\beta = 1$ and

$$r_0 = \sqrt{\frac{n}{n-2}} \ell, \quad m = \frac{r_0^n}{(n-2)\ell^2}. \quad (6.6.3)$$

A $(r \equiv \tau, t \equiv x)$ -conformal diagram can be found in Fig. 6.6.2.

From now on we assume that f has positive zeros.

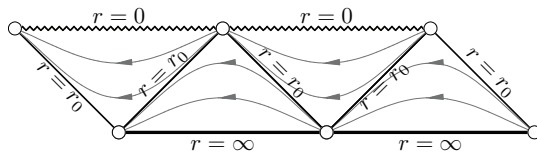


Fig. 6.6.2 The conformal diagram for the Birmingham metrics with positive cosmological constant and $f \leq 0$, vanishing precisely at r_0 .

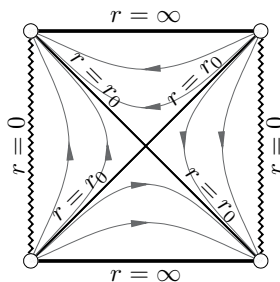


Fig. 6.6.3 The conformal diagram for the Birmingham metrics with positive cosmological constant and $m < 0$, $\beta \in \mathbb{R}$, or $m = 0$ and $\beta = 1$, with r_0 defined by the condition $f(r_0) = 0$. The set $\{r = 0\}$ is a singularity unless the metric is the de Sitter metric ($\check{N} = S^{n-1}$ and $m = 0$), or a suitable quotient thereof so that $\{r = 0\}$ corresponds to a centre of (possibly local) rotational symmetry.

6.6.2 Naked singularities

Assuming that $m = 0$ but $\beta \neq 0$, we must have $\beta = 1$ in view of our hypothesis that f has positive zeros. For $r \geq 0$ the function f has exactly one zero, $r = \ell$. The boundaries $\{r = 0\}$ and $\{r = \ell\}$ of the set $\{r \in [0, \ell]\}$ correspond either to regular centres of symmetry, in which case the level sets of t are S^n 's or their quotients, or to conical singularities. See Fig. 6.6.3.

If $m < 0$ the function $f : (0, \infty) \rightarrow \mathbb{R}$ is monotonically decreasing, tending to minus infinity as r tends to 0, where a naked singularity occurs, and to minus infinity when r tends to ∞ ; hence f has then precisely one zero. The conformal diagram can be seen in Fig. 6.6.3.

6.6.3 Spatially periodic time-symmetric initial data

We continue with the remaining cases, that is, f having zeros and $m > 0$. The function $f : (0, \infty) \rightarrow \mathbb{R}$ is then concave and thus has precisely two first-order zeros, except for the case already discussed in (6.6.3). A conformal diagram for a maximal extension of the spacetime, for the two first-order-zeros cases, is provided by Fig. 6.6.4. The level sets of t within each of the diamonds in that figure can be smoothly continued across the bifurcation surfaces of the Killing horizons to smooth spatially periodic Cauchy surfaces.

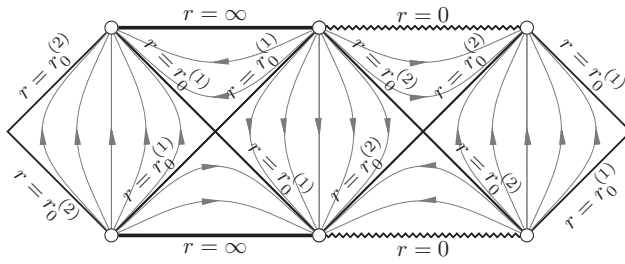


Fig. 6.6.4 The conformal diagram for Birmingham metrics with $\Lambda > 0$ and exactly two first-order zeros of f .

7

Projection diagrams

We have just seen that a very useful tool for visualizing the geometry of two-dimensional Lorentzian manifolds is that of *conformal Carter–Penrose diagrams*. For spherically symmetric geometries, or more generally for metrics in block-diagonal form, the two-dimensional conformal diagrams provide useful information about the four-dimensional geometry as well, since many essential aspects of the spacetime geometry are described by the $t - r$ sector of the metric.

The question then arises as to whether a similar device can be used for metrics which are not in block-diagonal form. In the following sections we show¹ that one can usefully represent classes of non-spherically symmetric geometries in terms of two-dimensional diagrams, called *projection diagrams*, using an auxiliary two-dimensional metric constructed out of the spacetime metric. Whenever such a construction can be carried out, the issues such as stable causality, global hyperbolicity, the existence of an event or Cauchy horizon, the causal nature of boundaries, and the existence of conformally smooth infinities become evident by inspection of the diagrams, in a way completely analogous to the block-diagonal case.

The reader will notice that the conformal diagrams of the previous chapter are a special case of the projection diagrams, which we are about to present.

We present the definition of a projection diagram in Section 7.1.

In Section 7.7 we show that projection diagrams can also be constructed in some non-black-hole spacetimes with Cauchy horizons. The simplest examples are presented in Section 7.2. The projection diagrams for the Kerr metrics are constructed in Section 7.3. This is extended to cover the Kerr–Newman metrics in Section 7.3, the Kerr–de Sitter metrics in Section 7.4.1, the Kerr–Newman–de Sitter metrics in Section 7.4.2, the Kerr–Newman–anti-de Sitter metrics in Section 7.4.3. The diagrams are constructed for the Pomeransky–Senkov metrics in Section 7.6.

7.1 The definition

Let $(\mathcal{M}, g)$ be a smooth spacetime, and let $\mathbb{R}^{1,n}$ denote the $(n + 1)$ -dimensional Minkowski spacetime. We wish to construct a map π from $(\mathcal{M}, g)$ to $\mathbb{R}^{1,1}$ which allows one to obtain information about the causality properties of $(\mathcal{M}, g)$. Ideally, π should be defined and differentiable throughout $\mathcal{M}$. However, already the example of Minkowski spacetime, discussed in Section 6.2.2, p. 264, shows that such a requirement is too restrictive: the map used there is not differentiable at the axis of rotation. So, while we will require that π is defined everywhere, it will be convenient to require that π be differentiable, and a submersion, on a subset of $\mathcal{M}$ which we will denote by $\mathcal{U}$. (Recall that π is a submersion if the tangent map π_* is surjective at every point.) This allows us to talk about ‘the projection diagram of Minkowski spacetime’, or ‘the projection diagram of Kerr spacetime’, rather than of ‘the projection diagram of the subset $\mathcal{U}$ of Minkowski spacetime’, etc. Note that the latter terminology would be more precise, and will sometimes be used, but appears to be an overkill in most cases.

¹The material in this section, including figures, draws extensively on [138], with permission. © 2012 by the American Physical Society.

Now, to preserve causality it appears a good idea to map timelike vectors to timelike vectors. This will be part of our definition: π will be required to have this property on $\mathcal{U}$. But note that a necessary condition for existence of a map from $\mathcal{M}$ to $\mathbb{R}^{1,1}$ which maps timelike vectors to timelike vectors is *stable causality* of $\mathcal{U}$: indeed, if t is a time function on $\mathbb{R}^{1,1}$, then $t \circ \pi$ will be a time function on $\mathcal{U}$ for such maps; but the existence of a time function on $\mathcal{U}$ is precisely the definition of stable causality. So causality violations provide an obvious obstruction for the construction of π .

Having accepted that $\mathcal{U}$ might not be the whole of $\mathcal{M}$, a possible requirement could be that $\mathcal{U}$ is dense in $\mathcal{M}$, as is the case for Minkowski spacetime. Keeping in mind that the Kerr spacetime contains causality-violating regions, which obviously have to be excluded from the domain where π has good causality properties, we see that the density requirement cannot be imposed in general. Clearly, one would like $\mathcal{U}$ to be as large as possible: the larger $\mathcal{U}$, the more information we will get about $\mathcal{M}$. We leave it as an open question as to whether there is an optimal largeness condition which can be imposed on $\mathcal{U}$. We simply use $\mathcal{U}$ as part of the input data of the definition, hoping secretly that it is as large as can be.

As already mentioned, we will require timelike vectors to be mapped to timelike vectors. Note that if some timelike vectors in the image of π within Minkowski spacetime will not arise as projections of timelike vectors, then there will be Minkowskian timelike curves in the image of π which will have nothing to do with causal curves in $\mathcal{M}$. But then no much insight into the causality of $\mathcal{M}$ will be gained by inspecting causal curves in $\mathbb{R}^{1,1}$. In order to avoid this, one is finally led to the following definition.

DEFINITION 7.1.1 *Let $(\mathcal{M}, g)$ be a Lorentzian manifold. A projection diagram is a pair $(\pi, \mathcal{U})$, where*

$$\mathcal{U} \subset \mathcal{M},$$

is open and non-empty, and where

$$\pi : \mathcal{M} \rightarrow \mathcal{W} := \pi(\mathcal{M}) \subset \mathbb{R}^{1,1}$$

is a continuous map, differentiable on an open dense subset of $\mathcal{M}$, such that $\pi|_{\mathcal{U}}$ is a smooth submersion. Moreover:

1. *for every smooth timelike curve $\sigma \subset \pi(\mathcal{U})$ there exists a smooth timelike curve γ in $(\mathcal{U}, g)$ such that σ is the projection of γ : $\sigma = \pi \circ \gamma$;*
2. *the image $\pi \circ \gamma$ of every smooth timelike curve $\gamma \subset \mathcal{U}$ is a timelike curve in $\mathbb{R}^{1,1}$.*

Some further comments are in order:

First, we assumed for simplicity that $(\mathcal{M}, g)$, $\pi|_{\mathcal{U}}$, and the causal curves in the definition are smooth, though assuming that π is C^1 on $\mathcal{U}$ would suffice for most purposes.

Next, as already mentioned, the requirement that timelike curves in $\pi(\mathcal{U})$ arise as projections of timelike curves in $\mathcal{M}$ ensures that causal relations on $\pi(\mathcal{U})$, which can be seen by inspection of $\pi(\mathcal{U})$, reflect causal relations on $\mathcal{M}$. Conditions 1 and 2 taken together ensure that causality on $\pi(\mathcal{U})$ represents as accurately as possible causality on $\mathcal{U}$.

The second condition of the definition is of course equivalent to the requirement that the images by π_* of timelike vectors in $T\mathcal{U}$ are timelike. This implies further that the images by π_* of causal vectors in $T\mathcal{U}$ are causal. But it should be kept in mind that projections lose information, so that the images by π_* of many null vectors in $T\mathcal{U}$ will be timelike. And, of course, many spacelike vectors will be mapped to causal vectors under π_* .

The curve-equivalent of the last remarks is that images of causal curves in $\mathcal{U}$ are causal in $\pi(\mathcal{U})$; that many spacelike curves in $\mathcal{U}$ will be mapped to causal curves in $\pi(\mathcal{U})$; and that many null curves in $\mathcal{U}$ will be mapped to timelike ones in $\pi(\mathcal{U})$.

The requirement that π is a submersion guarantees that open sets are mapped to open sets. This, in turn, ensures that projection diagrams with the same set $\mathcal{U}$ are locally unique, up to a local conformal isometry of two-dimensional Minkowski spacetime. We do not know whether two surjective projection diagrams $\pi_i : \mathcal{U} \rightarrow \mathcal{W}_i$, $i = 1, 2$, with identical domain of definition $\mathcal{U}$ are (globally) unique, up to a conformal isometry of $\mathcal{W}_1$ and $\mathcal{W}_2$. It would be of interest to settle this question.

In many examples of interest the set $\mathcal{U}$ will *not* be connected; we will see that this happens in the Kerr spacetime.

Recall that a map is proper if inverse images of compact sets are compact. In the definition we could further have required π to be proper; indeed, many projection diagrams in the following sections have this property. This is actually useful, as then the inverse images of globally hyperbolic subsets of $\mathcal{W}$ are globally hyperbolic, and so global hyperbolicity, or lack thereof, can be established by visual inspection of $\mathcal{W}$. It appears, however, more convenient to talk about *proper projection diagrams* whenever π is proper, allowing for non-properness in general.

As such, we have assumed for simplicity that π maps $\mathcal{M}$ into a subset of Minkowski spacetime. In some applications it might be natural to consider more general two-dimensional manifolds as the target of π ; this requires only a trivial modification of the definition. An example is provided by the Gowdy metrics on a torus, discussed at the end of Section 7.2, where the natural image manifold for π is $(-\infty, 0) \times S^1$, equipped with a flat product metric. Similarly, maximal extensions of the class of Kerr–Newman–de Sitter metrics of Figure 7.4.3, p. 297, require the image of π to be a suitable Riemann surface.

Given a spacetime $(\mathcal{M}, g)$, our construction of a projection diagram will proceed through the construction of a tensor field of signature $(-, +, 0, 0)$, which will be denoted by γ and which will be called the *projection metric*. The tensor field γ will be locally conformal to the pull-back $\pi^*\eta$ to $\mathcal{M}$ of the two-dimensional Minkowski metric η on $\mathcal{W}$, encoding thus the causality properties of the map π of Definition 7.1.1.

7.2 Simplest examples

The simplest examples of projection diagrams have already been constructed for metrics of the form

$$g = e^f (-F dt^2 + F^{-1} dr^2) + \underbrace{h_{AB} dx^A dx^B}_{=: h}, \quad F = F(r), \quad (7.2.1)$$

where $h = h_{AB}(t, r, x^C) dx^A dx^B$ is a family of *Riemannian* metrics on an $(n-1)$ -dimensional manifold N^{n-1} , possibly depending upon t and r , and f is a function which is allowed to depend upon all variables. It should be clear that any manifestly conformally flat representation of any extension, defined on $\mathcal{W} \subset \mathbb{R}^{1,1}$, of the two-dimensional metric

$$\gamma := -F dt^2 + F^{-1} dr^2,$$

discussed in Section 6.3, provides a projection diagram for $(\mathcal{W} \times N^{n-1}, g)$.

In particular, introducing spherical coordinates (t, r, x^A) on

$$\mathcal{U} := \{(t, \vec{x}) \in \mathbb{R}^{n+1}, |\vec{x}| \neq 0\} \subset \mathbb{R}^{1,n} \quad (7.2.2)$$

and forgetting about the $(n-1)$ -sphere part of the metric leads to a projection diagram for Minkowski spacetime which coincides with the usual conformal diagram

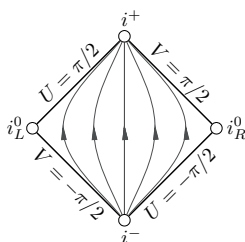


Fig. 7.2.1 The conformal diagram for $(1+1)$ -dimensional Minkowski spacetime provides also a possible projection diagram for $(n+1)$ -dimensional Minkowski spacetime, $n \geq 2$.

of the fixed-angles subsets of Minkowski spacetime (see the left image in Fig. 6.2.2, p. 265). The set $\mathcal{U}$ defined in (7.2.2) cannot be extended to include the world-line passing through the origin of $\mathbb{R}^n$ since the map π fails to be differentiable there. This diagram is proper, but fails to represent correctly the nature of the spacetime near the set $|\vec{x}| = 0$.

On the other hand, a globally defined projection diagram for Minkowski spacetime (thus, $(\mathcal{U}, g) = \mathbb{R}^{1,n}$) can be obtained by writing $\mathbb{R}^{1,n}$ as a product $\mathbb{R}^{1,1} \times \mathbb{R}^{n-1}$, and forgetting about the second factor. This leads to a projection diagram of Fig. 7.2.1; compare Fig. 6.2.1, p. 264.² This diagram, which is *not* proper, fails to represent correctly the connectedness of $\mathcal{I}^+$ and $\mathcal{I}^-$ when $n > 1$.

It will be seen in Section 7.5 that yet another choice of π and of the set $(\mathcal{U}, g) \subset \mathbb{R}^{1,n}$ leads to a third projection diagram for Minkowski spacetime.

A further example of non-uniqueness is provided by the projection diagrams for Taub–NUT metrics, discussed in Section 7.7.2.

These examples show that there is no uniqueness in the projection diagrams, and that various such diagrams might carry different information about the causal structure. It is clear that for spacetimes with intricate causal structure, some information will be lost when projecting to two dimensions. This raises the interesting question as to whether there exists a notion of optimal projection diagram for specific spacetimes. In any case, the examples we give in what follows appear to depict the essential causal properties of the associated spacetime, except perhaps for the black ring diagrams of Sections 7.5–7.6.

Non-trivial examples of metrics of the form (7.2.1) are provided by the Gowdy metrics on a torus [230]. These are vacuum $U(1) \times U(1)$ -symmetric metrics which can globally be written in the form [95, 230]

$$g = e^f (-dt^2 + d\theta^2) + |t| \left(e^P (dx^1 + Q dx^2)^2 + e^{-P} (dx^2)^2 \right), \quad (7.2.3)$$

with $t \in (-\infty, 0)$ and $(\theta, x^1, x^2) \in S^1 \times S^1 \times S^1$. Unwrapping θ from S^1 to $\mathbb{R}$ and projecting away the x^1 and x^2 coordinates, one obtains a projection diagram, the image of which is the half-space $t < 0$ in Minkowski spacetime. This can be further compactified as in Section 6.2.4, keeping in mind that the asymptotic behaviour of the metric for large negative values of t [415] is not compatible with the existence of a smooth conformal completion of the full spacetime metric across past null infinity. Note that this projection diagram fails to represent properly the existence of Cauchy horizons for non-generic [417] Gowdy metrics.

Similarly, generic Gowdy metrics on $S^1 \times S^2$, S^3 , or $L(p, q)$ can be written in the form [95, 230]

²All figures in this section are reprinted from [138] with permission. © 2012 by the American Physical Society.

$$g = e^f (-dt^2 + d\theta^2) + R_0 \sin(t) \sin(\theta) \left(e^P (dx^1 + Q dx^2)^2 + e^{-P} (dx^2)^2 \right), \quad (7.2.4)$$

with $(t, \theta) \in (0, \pi) \times [0, \pi]$, leading to the Gowdy square as the projection diagram for the spacetime. (This is the diagram of Fig. 7.5.1, p. 303, where the lower boundary corresponds to $t = 0$, the upper boundary corresponds to $t = \pi$, the left boundary corresponds to the axis of rotation $\theta = 0$, and the right boundary is the projection of the axis of rotation $\theta = \pi$. The diagonals, denoted as $y = y_h$ in Fig. 7.5.1, correspond in the Gowdy case to the projection of the set where the gradient of the area $R = R_0 \sin(t) \sin(\theta)$ of the orbits of the isometry group $U(1) \times U(1)$ becomes null or vanishes, and do not have any further geometric significance. The lines with the arrows in Fig. 7.5.1 are irrelevant for the Gowdy metrics, as the orbits of the isometry group of the spacetime metric, which are spacelike throughout the Gowdy square, have been projected away.)

Let us now pass to the construction of projection diagrams for families of metrics of interest which are not of the simple form (7.2.1).

7.3 The Kerr metrics

Consider the Kerr metric in Boyer–Lindquist coordinates,

$$g = -\frac{\Delta - a^2 \sin^2(\theta)}{\Sigma} dt^2 - \frac{2a \sin^2(\theta) (r^2 + a^2 - \Delta)}{\Sigma} dt d\varphi \\ + \frac{\sin^2(\theta) \left((r^2 + a^2)^2 - a^2 \sin^2(\theta) \Delta \right)}{\Sigma} d\varphi^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2. \quad (7.3.1)$$

Here

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 + a^2 - 2mr = (r - r_+)(r - r_-), \quad (7.3.2)$$

for some real parameters a and m , with

$$r_{\pm} = m \pm (m^2 - a^2)^{\frac{1}{2}}, \quad \text{and we assume that } 0 < |a| \leq m.$$

Recall that in the region $r \leq 0$ there exists a non-empty domain on which the Killing vector ∂_φ becomes timelike:

$$\mathcal{V} = \{g_{\varphi\varphi} < 0\} \\ = \left\{ r < 0, \cos(2\theta) < -\frac{a^4 + 2a^2mr + 3a^2r^2 + 2r^4}{a^2\Delta} \right. \\ \left. \Sigma \neq 0, \sin(\theta) \neq 0 \right\} \quad (7.3.3)$$

(see (4.6.49), p. 187). Since the orbits of ∂_φ are periodic, this leads to causality violations, as described in detail in Section 4.6.4. But, as pointed out earlier, the existence of a projection diagram implies stable causality of the spacetime. It is thus clear that we will need to remove the region where ∂_φ is timelike to construct the diagram. It is, however, not clear a priori whether removing $\mathcal{V}$ from $\mathcal{M}$ suffices; it turns out that this will be the case.

We start by recalling (4.6.48)

$$g_{\varphi\varphi} = \sin^2(\theta) \left(\frac{2a^2mr \sin^2(\theta)}{a^2 \cos^2(\theta) + r^2} + a^2 + r^2 \right) \\ = \frac{\sin^2(\theta) (a^4 + a^2 \cos(2\theta)\Delta + a^2r(2m + 3r) + 2r^4)}{a^2 \cos(2\theta) + a^2 + 2r^2}, \quad (7.3.4)$$

where the first line makes clear the non-negativity of $g_{\varphi\varphi}$ for $r \geq 0$.

To fulfill the requirements of our definition, we will be projecting out the θ and φ variables. We thus need to find a two-dimensional metric γ with the property that g -timelike vectors $X^t\partial_t + X^r\partial_r + X^\theta\partial_\theta + X^\varphi\partial_\varphi$ project to γ -timelike vectors $X^t\partial_t + X^r\partial_r$. For this, in the region where ∂_φ is spacelike (which thus includes $\{r > 0\}$) it turns out to be convenient to rewrite the $t - \varphi$ part of the metric as

$$\begin{aligned} g_{tt}dt^2 + 2g_{t\varphi}dtd\varphi + g_{\varphi\varphi}d\varphi^2 \\ = g_{\varphi\varphi}\left(d\varphi + \frac{g_{t\varphi}}{g_{\varphi\varphi}}dt\right)^2 + \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}\right)dt^2, \end{aligned} \quad (7.3.5)$$

with

$$g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} = -\frac{2\Delta\Sigma}{a^4 + a^2\Delta\cos(2\theta) + a^2r(2m+3r) + 2r^4}.$$

For $r > 0$ and $\Delta > 0$ it holds that

$$\frac{\Delta\Sigma}{(a^2 + r^2)^2} \leq \left|g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}\right| \leq \frac{\Delta\Sigma}{r(a^2(2m+r) + r^3)}, \quad (7.3.6)$$

with the infimum attained at $\theta \in \{0, \pi\}$ and maximum at $\theta = \pi/2$. One of the key facts for us is that $g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}$ has constant sign, and is in fact negative in this region.

In the region $r > 0$, $\Delta > 0$ consider any vector

$$X = X^t\partial_t + X^r\partial_r + X^\theta\partial_\theta + X^\varphi\partial_\varphi$$

which is causal for the metric g . Let $\Omega(r, \theta)$ be any positive function. Since both $g_{\theta\theta}$ and the first term in (7.3.5) are positive, while the coefficient of dt^2 in (7.3.5) is negative, we have

$$\begin{aligned} 0 &\geq \Omega^2 g(X, X) = \Omega^2 g_{\mu\nu} X^\mu X^\nu \geq \Omega^2 \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}\right) (X^t)^2 + \Omega^2 g_{rr} (X^r)^2 \\ &\geq \inf_\theta \left(\Omega^2 \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}\right) (X^t)^2 + \Omega^2 g_{rr} (X^r)^2 \right) \\ &\geq -\sup_\theta \left(\Omega^2 \left|g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}}\right| (X^t)^2 + \inf_\theta (\Omega^2 g_{rr}) (X^r)^2 \right). \end{aligned} \quad (7.3.7)$$

Thus, regardless of the choice of Ω , g -causality of X enforces a sign on the expression given in the last line of (7.3.7). We will therefore use this expression, with a suitable choice of Ω , to define the desired projection metric γ . It is simplest to choose Ω so that both extrema in (7.3.7) are attained at the same value of θ , say θ_* , while keeping those features of the coefficients which are essential for the problem at hand. It is convenient, but not essential, to have θ_* independent of r . We will therefore make the choice

$$\Omega^2 = \frac{r^2 + a^2}{\Sigma}, \quad (7.3.8)$$

but other choices are possible, and might be more convenient for some purposes. Here the Σ factor has been included to get rid of the angular dependence in

$$\Omega^2 g_{rr} = \Omega^2 \frac{\Sigma}{\Delta}$$

while the numerator $r^2 + a^2$ has been added to ensure that the metric coefficient γ_{rr} in (7.3.10) tends to 1 as r recedes to infinity. With this choice of Ω , Eq. (7.3.7) is equivalent to the statement that

$$\pi_*(X) := X^t \partial_t + X^r \partial_r \quad (7.3.9)$$

is a causal vector in the two-dimensional Lorentzian metric

$$\gamma := -\frac{\Delta(r^2 + a^2)}{r(a^2(2m+r) + r^3)} dt^2 + \frac{(r^2 + a^2)}{\Delta} dr^2. \quad (7.3.10)$$

Using the methods of Walker [456] reviewed in Section 6.3, in the region $r_+ < r < \infty$ the metric γ is conformal to a flat metric on the interior of a diamond, with the conformal factor extending smoothly across that part of its boundary at which $r \rightarrow r_+$ when $|a| < m$. This remains true when $|a| = m$ except at the leftmost corner i_L^0 of Fig. 7.2.1.

To make things clear, the map π of the definition of a projection diagram is the projection $(t, r, \theta, \varphi) \mapsto (t, r)$. The fact that g -causal curves are mapped to γ -causal curves follows from the construction of γ . In order to prove the lifting property, let $\sigma(s) = (t(s), r(s))$ be a γ -causal curve; then the curve

$$(t(s), r(s), \pi/2, \varphi(s)) ,$$

where $\varphi(s)$ satisfies

$$\frac{d\varphi}{ds} = -\frac{g_{t\varphi}}{g_{\varphi\varphi}} \frac{dt}{ds}$$

is a g -causal curve which projects to σ .

For causal vectors in the region $r > 0$, $\Delta < 0$, we have instead

$$\begin{aligned} 0 &\geq \Omega^2 g(X, X) \geq \Omega^2 \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right) (X^t)^2 + \Omega^2 g_{rr} (X^r)^2 \\ &\geq \inf_{\theta} \left(\Omega^2 \left| g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right| \right) (X^t)^2 - \sup_{\theta} (\Omega^2 |g_{rr}|) (X^r)^2. \end{aligned} \quad (7.3.11)$$

Since the inequalities in (7.3.6) are reversed when $\Delta < 0$, choosing the same factor Ω one concludes again that $X^t \partial_t + X^r \partial_r$ is γ -causal in the metric (7.3.10) whenever it is in the metric g . Using again the results of Section 6.3, in the region $r_- < r < r_+$, such a metric is conformal to a flat two-dimensional metric on the interior of a diamond, with the conformal factor extending smoothly across those parts of its boundary where $r \rightarrow r_+$ or $r \rightarrow r_-$.

When $|a| < m$ the metric coefficients in γ extend analytically from the $(r > r_+)$ range to the $(r_- < r < r_+)$ range. As described in Section 6.3.1, one can then smoothly glue together four diamonds as above to a single diamond on which $r_- < r < \infty$.

The singularity of γ at $r = 0$ reflects the fact that the metric g is singular at $\Sigma = 0$. This singularity persists even if $m = 0$, which might at first seem surprising since then there is no geometric singularity at $\Sigma = 0$ anymore. However, this singularity of γ reflects the singularity of the associated coordinates on Minkowski spacetime (compare (4.6.3), p. 175), with the set $r = 0$ in the projection metric corresponding to a boundary of the projection diagram.

For $r < 0$ we have $\Delta > 0$, and the inequality (7.3.7) still applies in the region where ∂_φ is spacelike. Setting

$$\mathcal{U} := \mathcal{M} \setminus \overline{\mathcal{V}},$$

where $\mathcal{V}$ is given by (7.3.3), throughout $\mathcal{U}$ we have

$$\frac{a^4 + 2a^2mr + 3a^2r^2 + 2r^4}{a^2(a^2 - 2mr + r^2)} > 1 \iff r(a^2(2m+r) + r^3) > 0. \quad (7.3.12)$$

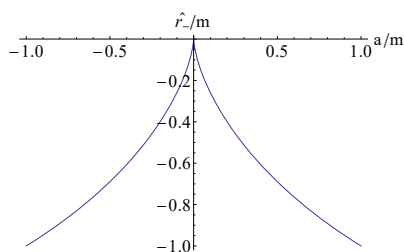


Fig. 7.3.1 The radius $\hat{r}_-$, scaled by m , of the ‘left boundary’ of the time-machine region as a function of a/m .

Equivalently,

$$r < \hat{r}_- := \frac{\sqrt[3]{\sqrt{3}\sqrt{a^6 + 27a^4m^2} - 9a^2m}}{3^{2/3}} - \frac{a^2}{\sqrt[3]{3}\sqrt[3]{\sqrt{3}\sqrt{a^6 + 27a^4m^2} - 9a^2m}} < 0; \quad (7.3.13)$$

see Fig. 7.3.1. In the region $r < \hat{r}_-$ the inequalities (7.3.6) hold again, and so the projected vector $\pi_*(X)$ as defined by (7.3.9) is causal, for g -causal X , in the metric γ given by (7.3.10). One concludes that the four-dimensional region $\{-\infty < r < r_-\}$ has the causal structure which projects to those diamonds of, e.g., Fig. 7.3.2 with $\hat{r}_+ = 0$ which contain a shaded region. Those shaded regions, which correspond to the projection of both the singularity $r = 0$, $\theta = \pi/2$ and the time-machine region $\mathcal{V}$ of (7.3.3), belong to $\mathcal{W} = \pi(\mathcal{M})$ but *not* to $\pi(\mathcal{U})$. Causality within the shaded region is *not* represented in any useful way by a flat two-dimensional metric there, as causal curves can exit this region earlier, in Minkowskian time on the diagram, than they entered it. This results in causality violations throughout the enclosing diamond unless the shaded region is removed.

The projection diagrams for the usual maximal extensions of the Kerr–Newman metrics can be found in Fig. 7.3.2.

REMARK 7.3.1 Some general remarks concerning projection diagrams for the Kerr family of metrics are in order. Anticipating, the remarks here apply also to projection diagrams of Kerr–Newman metrics, with or without a cosmological constant and regardless of its sign, to be discussed in the sections to follow.

The shaded regions in figures such as Fig. 7.3.2 contain the singularity $\Sigma = 0$ and the time-machine set $\{g_{\varphi\varphi} < 0\}$; they belong to the set $\mathcal{W} = \pi(\mathcal{M})$ but do *not* belong to the set $\pi(\mathcal{U})$, on which causality properties of two-dimensional Minkowski spacetime reflect those of $\mathcal{U} \subset \mathcal{M}$. We emphasize that there are closed timelike curves through every point in the preimage under π of the entire diamonds containing the shaded areas; this is discussed in detail for the Kerr metric in Section 4.6.4, and applies *as is* to all Kerr-type metrics. On the other hand, if the preimages of the shaded region are removed from $\mathcal{M}$, the causality relations in the resulting spacetimes are accurately represented by the diagrams, which are then proper.

The parameters $\hat{r}_\pm$ are determined by the mass and the charge parameters (see (7.4.26)), with $\hat{r}_+ = 0$ when the charge e vanishes, and $\hat{r}_+$ positive otherwise. The boundaries $r = \pm\infty$ correspond to smooth conformal boundaries at infinity, with causal character determined by Λ . The arrows indicate the spatial or timelike character of the orbits of the isometry group.

Loosely speaking, ‘maximal diagrams’ are obtained when continuing the diagrams shown in all allowed directions. It should be kept in mind that the resulting subsets of $\mathbb{R}^2$ are not simply connected in some cases, which implies that many alternative non-isometric maximal extensions of the spacetime can be obtained by taking

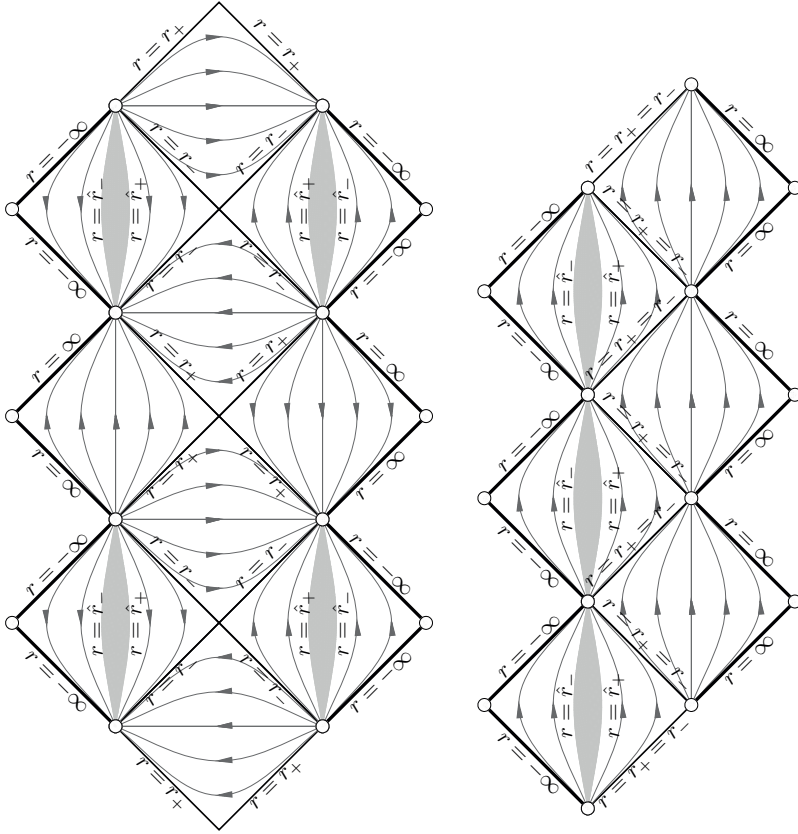


Fig. 7.3.2 A projection diagram for the Kerr–Newman metrics with two distinct zeros of Δ (left diagram) and one double zero (right diagram); see Remark 7.3.1. In the Kerr case $Q = 0$ we have $\hat{r}_+ = 0$, with $\hat{r}_-$ given by (7.3.12).

various coverings of the planar diagram. One can also make use of the symmetries of the diagram to produce distinct quotients. $\square$

7.3.1 Uniqueness of extensions

Let us denote by $(\mathcal{M}_{\text{Kerr}}, g_{\text{Kerr}})$ the spacetime with projection diagram visualized in Fig. 7.3.2, continued indefinitely to the future and the past in the obvious way, including the preimages of the shaded regions except for the singular set $\{\Sigma = 0\}$. Note that $(\mathcal{M}_{\text{Kerr}}, g_{\text{Kerr}})$ is *not* simply connected because loops circling around $\{\Sigma = 0\}$ cannot be homotoped to a point. Let us denote by $(\hat{\mathcal{M}}_{\text{Kerr}}, \hat{g}_{\text{Kerr}})$ the universal covering space of $(\mathcal{M}_{\text{Kerr}}, g_{\text{Kerr}})$ with the pull-back metric.

The question then arises of the uniqueness of the extension so obtained. To address this, we start by recalling an-already mentioned result of Carter [79] (compare [384, Theorem 4.3.1, p. 189]):

PROPOSITION 7.3.2 *The Kretschmann scalar $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ is unbounded on all maximally extended incomplete causal geodesics in $(\mathcal{M}_{\text{Kerr}}, g_{\text{Kerr}})$.*

Proposition 7.3.2 together with Corollary 4.4.7, p. 171, implies the following.

THEOREM 7.3.3 *Let $(\mathcal{M}, g)$ denote the region $\{r > r_+\}$ of a Kerr metric with $|a| \leq m$. Then $(\hat{\mathcal{M}}_{\text{Kerr}}, \hat{g}_{\text{Kerr}})$ is the unique simply connected analytic extension of $(\mathcal{M}, g)$ such that all maximally extended causal geodesics along which $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ are bounded are complete.* $\square$

Theorem 7.3.3 makes it clear in which sense $(\widehat{\mathcal{M}}_{\text{Kerr}}, \widehat{g}_{\text{Kerr}})$ is unique. However, the extension $(\mathcal{M}_{\text{Kerr}}, g_{\text{Kerr}})$ appears to be more economical, if not more natural. It would be of interest to find a natural condition which singles it out.

7.3.2 Two-dimensional submanifolds of Kerr spacetime

One can find e.g. in [81, 244] conformal diagrams for the symmetry axes in the maximally extended Kerr spacetime. These diagrams are identical with those of Fig. 7.3.2, except for the absence of shading. (The authors of [81, 244] seem to indicate that the subset $r = 0$ plays a special role in their diagrams, which is not the case as the singularity $r = \cos \theta = 0$ does not intersect the symmetry axes.) Now, the symmetry axes are totally geodesic submanifolds, being the collection of fixed points of the isometry group generated by the rotational Killing vector field. They can be thought of as the submanifolds $\theta = 0$ and $\theta = \pi$ (with the remaining angular coordinate irrelevant then) of the extended Kerr spacetime. As such, another totally geodesic two-dimensional submanifold in Kerr is the equatorial plane $\theta = \pi/2$, which is the set of fixed points of the isometry $\theta \mapsto \pi - \theta$. This leads one to enquire about the global structure of this submanifold or, more generally, of various families of two-dimensional submanifolds on which θ is kept fixed. The discussion that follows illustrates clearly the distinction between projection diagrams, in which one projects out the θ and φ variables, and conformal diagrams, for submanifolds where θ and either φ or the angular variable $\tilde{\varphi}$ of (7.3.16) are fixed.

An obvious family of two-dimensional Lorentzian submanifolds to consider is that of submanifolds, which we denote as $N_{\theta, \varphi}$, which are obtained by keeping θ and φ fixed. The metric, say $g(\theta)$, induced by the Kerr metric on $N_{\theta, \varphi}$ reads

$$g(\theta) = -\frac{\Delta - a^2 \sin^2(\theta)}{\Sigma} dt^2 + \frac{\Sigma}{\Delta} dr^2 =: -F_1(r) dt^2 + \frac{dr^2}{F_2(r)}. \quad (7.3.14)$$

For $m^2 - a^2 \cos^2(\theta) > 0$ the function F_1 has two first-order zeros at the intersection of $N_{\theta, \varphi}$ with the boundary of the ergoregion $\{g(\partial_t, \partial_t) > 0\}$:

$$r_{\theta, \pm} = m \pm \sqrt{m^2 - a^2 \cos^2(\theta)}. \quad (7.3.15)$$

The key point is that these zeros are distinct from those of F_2 if $\cos^2 \theta \neq 1$, which we assume in the remainder of this section. Since $r_{\theta, +}$ is larger than the largest zero of F_2 , the metric $g(\theta)$ is a priori only defined for $r > r_{\theta, +}$. One checks that its Ricci scalar diverges as $(r - r_{\theta, +})^{-2}$ when $r_{\theta, +}$ is approached; therefore those submanifolds do not extend smoothly across the ergosphere, and will thus be of no further interest to us.

We consider, next, the two-dimensional submanifolds, say $\tilde{N}_{\theta, \tilde{\varphi}}$, of the Kerr spacetime obtained by keeping θ and $\tilde{\varphi}$ fixed, where $\tilde{\varphi}$ is a new angular coordinate defined as

$$d\tilde{\varphi} = d\varphi + \frac{a}{\Delta} dr. \quad (7.3.16)$$

Using further the coordinate v defined as

$$dv = dt + \frac{(a^2 + r^2)}{\Delta} dr, \quad (7.3.17)$$

the metric, say $\tilde{g}(\theta)$, induced on $\tilde{N}_{\theta, \tilde{\varphi}}$ takes the form

$$\tilde{g}(\theta) = -\frac{\tilde{F}(r)}{\Sigma} dv^2 + 2dvdr$$

$$= -\frac{\tilde{F}(r)}{\Sigma} dv \left(dv - 2\frac{\Sigma}{\tilde{F}(r)} dr \right), \quad (7.3.18)$$

where $\tilde{F}(r) := r^2 + a^2 \cos^2(\theta) - 2mr$. The zeros of $\tilde{F}(r)$ are again given by (7.3.15). Setting

$$du = dv - 2\frac{\Sigma}{\tilde{F}(r)} dr \quad (7.3.19)$$

brings (7.3.18) to the form

$$\tilde{g}(\theta) = -\frac{\tilde{F}(r)}{\Sigma} du dv.$$

The usual Kruskal–Szekeres-type analysis applies to this metric, leading to a conformal diagram as in the left image of Fig. 7.3.2 *with no shadings*, and with $r_{\pm}$ there replaced by $r_{\theta,\pm}$, as long as $\tilde{F}$ has two distinct zeros.

Several comments are in order.

First, the event horizons *within* $\tilde{N}_{\theta,\varphi}$ do not coincide with the intersection of the event horizons of the Kerr spacetime with $\tilde{N}_{\theta,\varphi}$. This is not difficult to understand by noting that the class of causal curves that lie within $\tilde{N}_{\theta,\varphi}$ is smaller than the class of causal curves in spacetime, and there is therefore no a priori reason to expect that the associated horizons will be the same. In fact, it should be clear that the event horizons within $\tilde{N}_{\theta,\varphi}$ should be located on the boundary of the ergoregion, since in two spacetime dimensions the boundary of an ergoregion is necessarily a null hypersurface. This illustrates the fact that conformal diagrams for submanifolds might fail to represent correctly the location of horizons. The reason that the conformal diagrams for the symmetry axes correctly reflect the global structure of the spacetime is an accident related to the fact that the ergosphere touches the event horizon there.

This last issue acquires a dramatic dimension for extreme Kerr black holes, for which $|a| = m$, where for $\theta \in (0, \pi)$ the global structure of maximally extended $\tilde{N}_{\theta,\varphi}$'s is represented by an unshaded version of the left image in Fig. 7.3.2, while the conformal diagrams for the axisymmetry axes are given by the unshaded version of the right image in Fig. 7.3.2.

Next, another dramatic change arises in the global structure of the $\tilde{N}_{\theta,\varphi}$'s with $\theta = \pi/2$. Indeed, in this case we have $r_{\theta,+} = 2m$, as in Schwarzschild spacetime, and $r_{\theta,-} = 0$, *regardless of whether the metric is underspinning* ($a^2 < m^2$), *extreme* ($a^2 = m^2$), *or overspinning*. Since $r_{\theta,-}$ coincides now with the location of the singularity, $\tilde{N}_{\theta,\varphi}$ acquires two connected components, one where $r > 0$ and a second one with $r < 0$. The conformal diagram of the first one is identical to that of the Schwarzschild spacetime with positive mass, while the second is identical to that of Schwarzschild with negative mass; see Fig. 7.3.3. We thus obtain the unexpected conclusion, that *the singularity* $r = \cos(\theta) = 0$ *has a spacelike character when approached with positive* r *within the equatorial plane, and a timelike one when approached with negative* r *within that plane*. This is rather obvious in retrospect, since the metric induced by Kerr on $\tilde{N}_{\pi/2,\varphi}$ coincides, when $m > 0$, with the one induced by the Schwarzschild metric with positive mass in the region $r > 0$ and with the Schwarzschild metric with negative mass $-m$ in the region $r < 0$.

Note finally that, surprisingly enough, even for overspinning Kerr metrics there will be a range of angles θ near $\pi/2$ so that $\tilde{F}$ will have two distinct first-order zeros. This implies that, for such θ , the global structure of maximally extended $\tilde{N}_{\theta,\varphi}$'s will be similar to that of the corresponding submanifolds of the underspinning Kerr solutions. This should be compared with the projection diagram for overspinning Kerr spacetimes in Fig. 7.3.4.

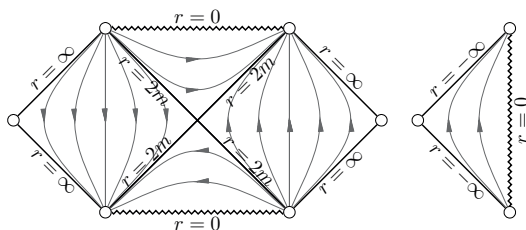


Fig. 7.3.3 The conformal diagram for a maximal analytic extension of the metric induced by the Kerr metric, with arbitrary $a \in \mathbb{R}$, on the submanifolds of constant angle $\tilde{\varphi}$ within the equatorial plane $\theta = \pi/2$, with $r > 0$ (left) and $r < 0$ (right).

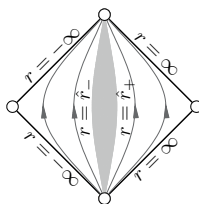


Fig. 7.3.4 A projection diagram for overspinning Kerr–Newman spacetimes.

7.3.3 The orbit-space metric on $\mathcal{M}/\mathbf{U}(1)$

Let h denote the tensor field obtained by quotienting out in the Kerr metric g the $\eta := \partial_{\varphi}$ direction,

$$h(X, Y) = g(X, Y) - \frac{g(X, \eta)g(Y, \eta)}{g(\eta, \eta)}. \quad (7.3.20)$$

(This is *not* the same metric as in Section 4.6.8, p. 191, where the whole group $\mathbb{R} \times \mathbf{U}(1)$ has been quotiented out instead.) The tensor field h projects to the natural quotient metric on the manifold part of $\mathcal{M}/\mathbf{U}(1)$. In the region where η is spacelike, the quotient space $\mathcal{M}/\mathbf{U}(1)$ has the structure of a manifold with boundary, where the boundary is the image, under the quotient map, of the axis of rotation

$$\mathcal{A} := \{\eta = 0\}.$$

Using t, r, θ as coordinates on the quotient space we find a diagonal metric

$$h = h_{tt}dt^2 + \frac{\Sigma}{\Delta}dr^2 + \Sigma d\theta^2, \quad (7.3.21)$$

where

$$h_{tt} = g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}},$$

as in (7.3.5). Thus, the metric γ of (7.3.10) is directly constructed out of the (t, r) -part of the quotient-space metric h .

INCIDENTALLY: We note that a Penrose diagram for the quotient-space metric has been constructed in [225]. $\square$

7.4 The Kerr–Newman metrics

The analysis of the Kerr–Newman metrics is essentially identical to that of the Kerr metric: the metric takes the same general form (7.3.1), except that now

$$\Delta = r^2 + a^2 + e^2 - 2mr =: (r - r_+)(r - r_-),$$

and we assume that $e^2 + a^2 \leq m$ so that the roots are real. We have

$$g_{\varphi\varphi} = \frac{\sin^2(\theta) \left((r^2 + a^2)^2 - a^2 \Delta \sin^2(\theta) \right)}{\Sigma}, \quad (7.4.1)$$

$$g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} = -\frac{\Delta \Sigma}{(r^2 + a^2)^2 - a^2 \Delta \sin^2(\theta)}, \quad (7.4.2)$$

and note that the sign of the denominator in (7.4.2) coincides with the sign of $g_{\varphi\varphi}$. Hence

$$\text{sign} \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right) = -\text{sign}(\Delta) \text{sign}(g_{\varphi\varphi}).$$

For $g_{\varphi\varphi} > 0$, which is the main region of interest, we conclude that the minimum of $(g_{tt} - g_{t\varphi}^2/g_{\varphi\varphi}) \Sigma^{-1} \Delta^{-1}$ is assumed at $\theta = \frac{\pi}{2}$ and the maximum at $\theta = 0, \pi$, so for all r for which $g_{\varphi\varphi} > 0$ we have

$$-\frac{\Delta \Sigma}{(r^2 + a^2)^2 - a^2 \Delta} \leq g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \leq -\frac{\Delta \Sigma}{(r^2 + a^2)^2}. \quad (7.4.3)$$

Choosing the conformal factor as

$$\Omega^2 = \frac{r^2 + a^2}{\Sigma}$$

we obtain, for g -causal vectors X ,

$$\begin{aligned} 0 &\geq \Omega^2 g(X, X) = \Omega^2 g_{\mu\nu} X^\mu X^\nu \geq \Omega^2 \left(g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right) (X^t)^2 + \Omega^2 g_{rr} (X^r)^2 \\ &\geq -\frac{\Delta (r^2 + a^2)}{(r^2 + a^2)^2 - a^2 \Delta} (X^t)^2 + \frac{(r^2 + a^2)}{\Delta} (X^r)^2. \end{aligned} \quad (7.4.4)$$

This leads to the projection metric

$$\begin{aligned} \gamma &:= -\frac{\Delta (r^2 + a^2)}{(r^2 + a^2)^2 - a^2 \Delta} dt^2 + \frac{(r^2 + a^2)}{\Delta} dr^2 \\ &= -\frac{\Delta (r^2 + a^2)}{a^2 (r(2m + r) - e^2) + r^4} dt^2 + \frac{(r^2 + a^2)}{\Delta} dr^2, \end{aligned} \quad (7.4.5)$$

which is Lorentzian if and only if r is such that $g_{\varphi\varphi} > 0$ for all $\theta \in [0, \pi]$.

Now, it follows from (7.4.1) that $g_{\varphi\varphi}$ will have the *wrong* sign if

$$0 > (r^2 + a^2)^2 - a^2 \Delta \sin^2(\theta). \quad (7.4.6)$$

This does not happen when $\Delta \leq 0$, and hence in a neighbourhood of both horizons. On the other hand, for $\Delta > 0$, a necessary condition for (7.4.6) is

$$0 > (r^2 + a^2)^2 - a^2 \Delta = r^4 + r^2 a^2 + 2 m r a^2 - a^2 e^2 =: f(r). \quad (7.4.7)$$

The second derivative of f is positive; hence f' has exactly one real zero. Note that f is strictly smaller than the corresponding function for the Kerr metric, where $e = 0$; thus the interval where f is negative encloses the corresponding interval for Kerr. We conclude that f is negative on an interval $(\hat{r}_-, \hat{r}_+)$, with $\hat{r}_- < 0 < \hat{r}_+ < r_-$.

The corresponding projection diagrams are identical to those of the Kerr space-time, see Fig. 7.3.2, with the minor modification that the region to be excised from the diagram is $\{r \in (\hat{r}_-, \hat{r}_+)\}$, with now $\hat{r}_+ > 0$, while we had $\hat{r}_+ = 0$ in the uncharged case.

7.4.1 The Kerr–de Sitter metrics

The Kerr–(anti-)de Sitter family of metrics has been discussed at length in Section 5.1, p. 200; for the convenience of the reader we repeat some of the formulae there, as needed for the problem at hand here.

The Kerr–de Sitter metric in Boyer–Lindquist-like coordinates reads

$$g = \frac{\Sigma}{\Delta_r} dr^2 + \frac{\Sigma}{\Delta_\theta} d\theta^2 + \frac{\sin^2(\theta)}{\Xi^2 \Sigma} \Delta_\theta (adt - (r^2 + a^2)d\varphi)^2 - \frac{1}{\Xi^2 \Sigma} \Delta_r (dt - a \sin^2(\theta) d\varphi)^2, \quad (7.4.8)$$

where

$$\Sigma = r^2 + a^2 \cos^2(\theta), \quad \Delta_r = (r^2 + a^2) \left(1 - \frac{\Lambda}{3} r^2\right) - 2m\Xi r, \quad (7.4.9)$$

and

$$\Delta_\theta = 1 + \frac{\Lambda}{3} a^2 \cos^2(\theta), \quad \Xi = 1 + \frac{\Lambda}{3} a^2, \quad (7.4.10)$$

for some real parameters a and m , where Λ is the cosmological constant. In this section we assume that

$$\Lambda > 0, m > 0 \text{ and } a > 0.$$

Recall that

$$g^{tt} = \frac{g_{rr} g_{\theta\theta} g_{\varphi\varphi}}{\det(g)} = -\frac{\Xi^4}{\Delta_\theta} \times \frac{1}{\Delta_r} \times \frac{g_{\varphi\varphi}}{\sin^2 \theta}, \quad (7.4.11)$$

which shows that either t or its negative is a time function whenever Δ_r and $g_{\varphi\varphi}/\sin^2 \theta$ are positive. (As usual, chronology is violated on the set where $g_{\varphi\varphi}$ is negative; we will return to this shortly.) One also has

$$g(\nabla r, \nabla r) \equiv g^{rr} = \frac{\Delta_r}{\Sigma}, \quad (7.4.12)$$

which shows that r or its negative is a time function in the region where $\Delta_r < 0$.

From what has been said in Section 5.2.3, p. 215 (compare Section 5.1.5, p. 207), the function Δ_r has exactly *two distinct first-order real* zeros, one of them strictly negative and the other strictly positive, when

$$m^2 > \frac{2}{3^5 \Xi^2 \Lambda} (3 - a^2 \Lambda)^3. \quad (7.4.13)$$

It has *at least two, and up to four, possibly but not necessarily distinct*, real roots when

$$a^2 \Lambda \leq 3, \quad m^2 \leq \frac{2}{3^5 \Xi^2 \Lambda} (3 - a^2 \Lambda)^3. \quad (7.4.14)$$

Under the current assumptions the smallest root, say r_1 , is always simple and strictly negative; the remaining ones are positive. We can thus order the roots as

$$r_1 < 0 < r_2 \leq r_3 \leq r_4, \quad (7.4.15)$$

when there are four real ones, and we set $r_3 \equiv r_4 := r_2$ when there are only two real roots $r_1 < r_2$. The function Δ_r is strictly positive for $r \in (r_1, r_2)$, and for $r \in (r_3, r_4)$ whenever the last interval is not empty; Δ_r is negative or vanishing otherwise.

It holds that

$$g_{\varphi\varphi} = \frac{\sin^2(\theta) \left(\Delta_\theta (r^2 + a^2)^2 - a^2 \Delta_r \sin^2(\theta) \right)}{\Xi^2 \Sigma} \quad (7.4.16)$$

$$= \frac{\sin^2(\theta)}{\Xi} \left(\frac{2a^2 m r \sin^2(\theta)}{a^2 \cos^2(\theta) + r^2} + a^2 + r^2 \right). \quad (7.4.17)$$

The second line is manifestly non-negative for $r \geq 0$, and positive there away from the axis $\sin(\theta) = 0$. The first line is manifestly non-negative for $\Delta_r \leq 0$, and hence also in a neighbourhood of this set.

Next

$$\begin{aligned} g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} &= - \frac{\Delta_\theta \Delta_r \Sigma}{\Xi^2 \left(\Delta_\theta (r^2 + a^2)^2 - \Delta_r a^2 \sin^2(\theta) \right)} \\ &= - \frac{\Delta_\theta \Delta_r \Sigma}{\Xi^2 (A(r) + B(r) \cos(2\theta))}, \end{aligned} \quad (7.4.18)$$

with

$$A(r) = \frac{\Xi}{2} (a^4 + 3a^2 r^2 + 2r^4 + 2a^2 m r), \quad (7.4.19)$$

$$B(r) = \frac{a^2}{2} \Xi (a^2 + r^2 - 2m r). \quad (7.4.20)$$

We have

$$\begin{aligned} A(r) + B(r) &= \Xi (a^2 + r^2)^2, \\ A(r) - B(r) &= r^2 \Xi \left(a^2 + r^2 + 2 \frac{a^2 m}{r} \right), \end{aligned} \quad (7.4.21)$$

which confirms that for $r > 0$, or for large negative r , we have $A > |B| > 0$, as needed for $g_{\varphi\varphi} \geq 0$. The function

$$f(r, \theta) := \frac{(A(r) + B(r) \cos(2\theta))}{\Delta_\theta} \equiv \frac{(A(r) + B(r) \cos(2\theta))}{1 + \frac{\Lambda}{3} a^2 \cos^2(\theta)}$$

satisfies

$$\frac{\partial f}{\partial \theta} = - \frac{a^2 \Xi}{\Delta_\theta^2} \Delta_r \sin(2\theta), \quad (7.4.22)$$

which has the same sign as $-\Delta_r \sin(2\theta)$. In any case, its extrema are achieved at $\theta = 0, \pi/2$, and π . Accordingly, this is where the extrema of the right-hand side of (7.4.18) are achieved as well. In particular, for $\Delta_r > 0$, we find that

$$\frac{\Delta_r \Sigma}{(a^2 + r^2)^2} \leq \Xi^2 \left| g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right| \leq \frac{\Sigma \Delta_r}{\Xi r (a^2 (2m + r) + r^3)}, \quad (7.4.23)$$

with the minimum attained at $\theta = 0$ and the maximum attained at $\theta = \pi/2$.

To obtain the projection diagram, we can now repeat word for word the analysis carried out for the Kerr metrics on the set $\{g_{\varphi\varphi} > 0\}$. Choosing a conformal factor Ω^2 equal to

$$\Omega^2 = \frac{r^2 + a^2}{\Sigma}, \quad (7.4.24)$$

one is led to a projection metric

$$\gamma := -\frac{(r^2 + a^2)\Delta_r}{\Xi^3 r (a^2(2m + r) + r^3)} dt^2 + \frac{r^2 + a^2}{\Delta_r} dr^2. \quad (7.4.25)$$

It remains to understand the set

$$\mathcal{V} := \{g_{\varphi\varphi} < 0\}$$

where $g_{\varphi\varphi}$ is negative. To avoid repetitiveness, we will do it simultaneously for both the charged and the uncharged cases, where (7.4.16) still applies (but not (7.4.17) for $e \neq 0$) with Δ_r given by (7.4.26); the Kerr–de Sitter case is obtained by setting $e = 0$ in what follows. A calculation shows that $g_{\varphi\varphi}$ is the product of a non-negative function with

$$\chi := 2a^2 m r - a^2 e^2 + r^2 a^2 + r^4 + (r^2 a^2 - 2a^2 m r + a^2 e^2 + a^4) \cos^2(\theta).$$

This is clearly positive for all r and all $\theta \neq \pi/2$ when $m = e = 0$, which shows that $\mathcal{V} = \emptyset$ in this case.

Next, the function χ is sandwiched between the two following functions of r , obtained by setting $\cos(\theta) = 0$ or $\cos^2(\theta) = 1$ in χ :

$$\begin{aligned} \chi_0 &:= r^4 + r^2 a^2 + 2a^2 m r - a^2 e^2, \\ \chi_1 &:= (r^2 + a^2)^2. \end{aligned}$$

Hence, χ is positive for all r when $\cos^2(\theta) = 1$. Next, for $m > 0$ the function χ_0 is negative for negative r near zero. Further, χ_0 is convex. We conclude that, for $m > 0$, the set on which χ_0 is non-positive is a non-empty interval $[\hat{r}_-, \hat{r}_+]$ containing the origin. We have already seen that $g_{\varphi\varphi}$ is non-negative wherever $\Delta_r \leq 0$, and since $r_2 > 0$ we must have

$$r_1 < \hat{r}_- \leq \hat{r}_+ < r_2.$$

In fact, when $e = 0$ the value of $\hat{r}_-$ is given by (7.3.13) with m there replaced by m , with $\hat{r}_- = 0$ if and only if $m = 0$.

We conclude that if $m = e = 0$ the time-machine set is empty, while if $|m| + e^2 > 0$ there are always causality violations ‘produced’ in the non-empty region $\{\hat{r}_- \leq r \leq \hat{r}_+\}$.

The projection diagrams for the Kerr–Newman–de Sitter family of metrics depend upon the number of zeros of Δ_r and their nature, and can be found in Figs. 7.4.1–7.4.4.

7.4.2 The Kerr–Newman–de Sitter metrics

In the standard Boyer–Lindquist coordinates the Kerr–Newman–de Sitter metric takes the form (7.4.8), with all the functions as in (7.4.9)–(7.4.10) except for Δ_r , which instead takes the form

$$\Delta_r = \left(1 - \frac{1}{3}\Lambda r^2\right)(r^2 + a^2) - 2\Xi m r + \Xi e^2, \quad (7.4.26)$$

where $\sqrt{\Xi}e$ is the electric charge of the spacetime. In this section we assume that

$$\Lambda > 0, \quad m \geq 0, \quad a > 0, \quad e \neq 0.$$

The calculations of the previous section, and the analysis of zeros of Δ_r , remain identical except for the following equations: first, we have now

$$g_{\varphi\varphi} = \frac{\sin^2(\theta)}{\Xi} \left(\frac{a^2(2mr - e^2)\sin^2(\theta)}{a^2 \cos^2(\theta) + r^2} + a^2 + r^2 \right), \quad (7.4.27)$$

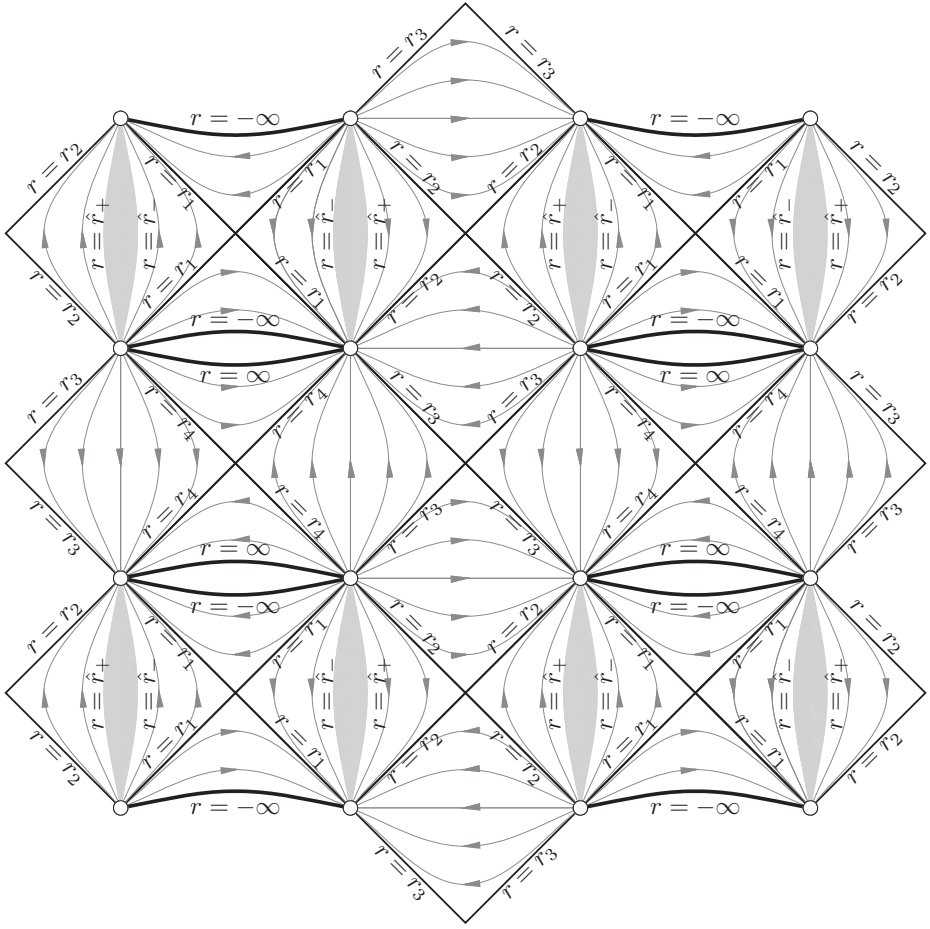


Fig. 7.4.1 A projection diagram for the Kerr–Newman–de Sitter metric with four distinct zeros of Δ_r ; see Remark 7.3.1.

and the sign of $g_{\varphi\varphi}$ requires further analysis; we will return to this shortly. Next, we still have

$$\begin{aligned} g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} &= - \frac{\Delta_\theta \Delta_r \Sigma}{\Xi^2 \left(\Delta_\theta (r^2 + a^2)^2 - \Delta_r a^2 \sin^2(\theta) \right)} \\ &= - \frac{\Delta_\theta \Delta_r \Sigma}{\Xi^2 (A(r) + B(r) \cos(2\theta))}, \end{aligned} \quad (7.4.28)$$

but now

$$A(r) = \frac{\Xi}{2} (a^4 + 3a^2 r^2 + 2r^4 + 2a^2 m r - a^2 e^2), \quad (7.4.29)$$

$$B(r) = \frac{a^2}{2} \Xi (a^2 + r^2 - 2m r + e^2), \quad (7.4.30)$$

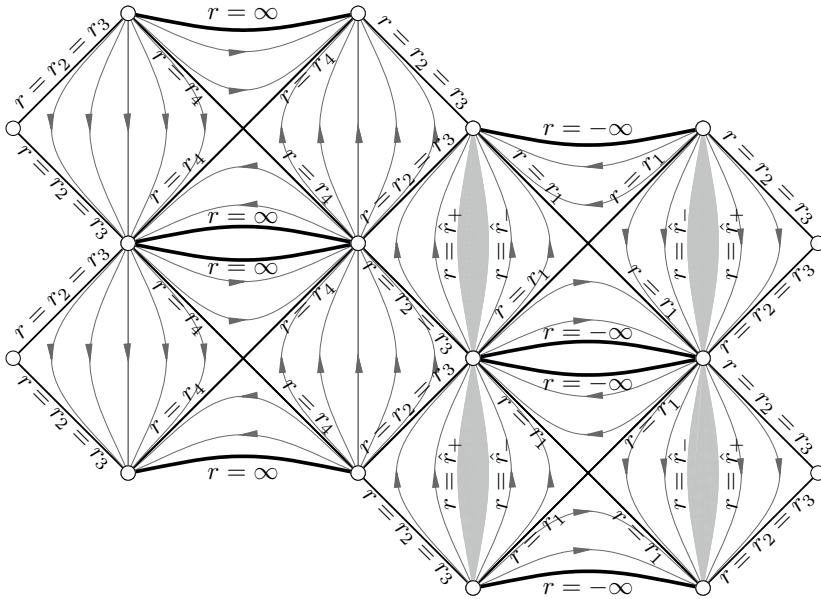


Fig. 7.4.2 A projection diagram for the Kerr–Newman–de Sitter metrics with three distinct zeros of Δ_r , $r_1 < 0 < r_2 = r_3 < r_4$; see Remark 7.3.1.

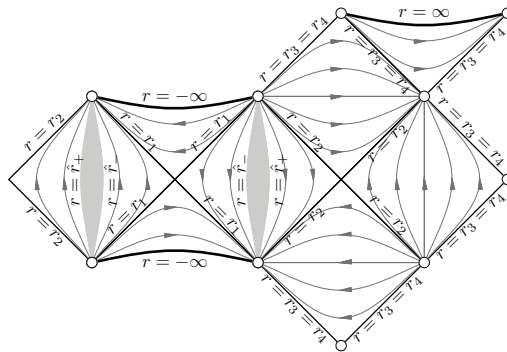


Fig. 7.4.3 A projection diagram for the Kerr–Newman–de Sitter metrics with three distinct zeros of Δ_r , $r_1 < 0 < r_2 < r_3 = r_4$; see Remark 7.3.1. Note that one cannot continue the diagram simultaneously across all boundaries $r = r_3$ on $\mathbb{R}^2$, but this can be done on an appropriate Riemann surface.

with

$$\begin{aligned} A(r) + B(r) &= \Xi (a^2 + r^2)^2, \\ A(r) - B(r) &= r^2 \Xi \left(a^2 + r^2 + 2 \frac{a^2 m}{r} - \frac{a^2 e^2}{r^2} \right). \end{aligned} \quad (7.4.31)$$

Equation (7.4.22) remains unchanged, and for $\Delta_r > 0$, we find that

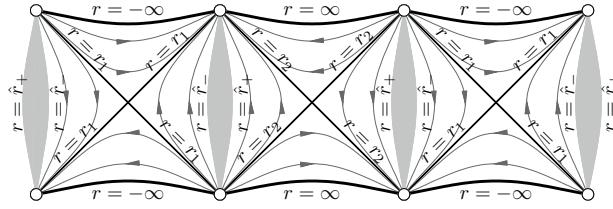


Fig. 7.4.4 A projection diagram for the Kerr–Newman–de Sitter metrics with two distinct first-order zeros of Δ_r , $r_1 < 0 < r_2$ and $m > 0$; see Remark 7.3.1. The diagram for a first-order zero at r_1 and third-order zero at $r_2 = r_3 = r_4$ would be identical except for the bifurcation surface of the bifurcate Killing horizon at the intersection of the lines $r = r_2$, which does not exist in the third-order case and has therefore to be removed from the diagram.

$$\frac{\Delta_r \Sigma}{(a^2 + r^2)^2} \leq \Xi^2 \left| g_{tt} - \frac{g_{t\varphi}^2}{g_{\varphi\varphi}} \right| \leq \frac{\Sigma \Delta_r}{\Xi (a^2(2mr - e^2 + r^2) + r^4)}, \quad (7.4.32)$$

with the minimum attained at $\theta = 0$ and the maximum attained at $\theta = \pi/2$. This leads to the projection metric

$$\gamma := -\frac{\Delta_r}{\Xi^3 (a^2(2mr - e^2 + r^2) + r^4)} dt^2 + \frac{1}{\Delta_r} dr^2. \quad (7.4.33)$$

We recall that the analysis of the time-machine set $\{g_{\varphi\varphi} < 0\}$ has already been carried out at the end of Section 7.4.1, where it was shown that for $e \neq 0$ causality violations always exist, and arise from the non-empty region $\{\hat{r}_- \leq r \leq \hat{r}_+\}$.

The projection diagrams for the Kerr–Newman–de Sitter family of metrics can be found in Figures 7.4.1–7.4.4.

7.4.3 The Kerr–Newman–anti-de Sitter metrics

We consider again the metric (7.4.8)–(7.4.10), with Δ_r given now by (7.4.26), assuming that

$$a^2 + e^2 > 0, \quad \Lambda < 0.$$

While the local calculations carried out in Section 7.4.1 remain unchanged, one needs to re-examine the occurrence of zeros of Δ_r .

We start by noting that the requirement that $\Xi \neq 0$ imposes that

$$1 + \frac{\Lambda}{3} a^2 \neq 0.$$

Next, a negative Ξ would lead to a function Δ_θ which changes sign. By inspection, one finds that the signature changes from $(-+++)$ to $(+---)$ across these zeros, which implies nonexistence of a coordinate system in which the metric could be smoothly continued there.³ From now on we thus assume that

$$\Xi \equiv 1 + \frac{\Lambda}{3} a^2 > 0. \quad (7.4.34)$$

As such, those metrics for which Δ_r has no zeros are nakedly singular whenever

$$e^2 + |m| > 0. \quad (7.4.35)$$

³Somewhat surprisingly, no curvature invariants for the overspinning metrics that we calculated were singular at $\Delta_\theta = 0$.

This can be easily seen from the following formula for g_{tt} on the equatorial plane:

$$g_{tt} = \frac{1}{3\Xi^2 r^2} (-3\Xi e^2 + 6\Xi m r + (\Lambda a^2 - 3) r^2 + \Lambda r^4). \quad (7.4.36)$$

So, under (7.4.35) the norm of the Killing vector ∂_t is unbounded and the metric cannot be C^2 -continued across $\{\Sigma = 0\}$ by usual arguments.

Turning our attention, first, to the region where $r > 0$, the occurrence of zeros of Δ_r requires that

$$m \geq m_c(a, e, \Lambda) > 0.$$

Hence, there is a positive threshold for the mass of a black hole at given a and e . The solution with $m = m_c$ has the property that Δ_r and its r -derivative have a joint zero, and can thus be found by equating to zero the resultant of these two polynomials in r . An explicit formula for $M_c = \Xi m_c$ can be given, which takes a relatively simple form when expressed in terms of suitably renormalized parameters. We set

$$\begin{aligned} \alpha &= \sqrt{\frac{|\Lambda|}{3}} a \iff a = \alpha \sqrt{\frac{3}{|\Lambda|}}, \\ \gamma &= 9 \frac{\alpha^2 + \frac{|\Lambda|}{3} q^2}{(1 + \alpha^2)^2} \iff q^2 := \Xi e^2 = \frac{3}{|\Lambda|} \left(\left(\frac{1 + \alpha^2}{3} \right)^2 \gamma - \alpha^2 \right), \\ \beta &= \frac{3\sqrt{|\Lambda|}}{(1 + \alpha^2)^{3/2}} m \Xi \iff M := \Xi m = \frac{(1 + \alpha^2)^{3/2}}{3\sqrt{|\Lambda|}} \beta. \end{aligned}$$

Letting β_c be the value of β corresponding to m_c , one finds that

$$\begin{aligned} \beta_c &= \frac{\sqrt{-9 + 36\gamma + \sqrt{3}\sqrt{(3 + 4\gamma)^3}}}{3\sqrt{2}} \\ \iff M_c^2 &:= m_c^2 \Xi^2 = \frac{(1 + \alpha^2)^3 \left(-9 + 36\gamma + \sqrt{3}\sqrt{(3 + 4\gamma)^3} \right)}{162|\Lambda|}. \end{aligned} \quad (7.4.37)$$

The graph of β_c as a function of α can be found in Fig. 7.4.5 with $q = 0$, and Fig. 7.4.6 as a function of a and q .

Note that if $q = 0$, then γ can be used as a replacement for a ; otherwise, γ is a substitute for q at fixed a .

When $e = 0$ we have $M_c = a + O(a^3)$ for small a , and $M_c \rightarrow \frac{8}{3\sqrt{|\Lambda|}}$ as $|a| \nearrow \sqrt{3/|\Lambda|}$.

According to [247], the physically relevant mass of the solution is M and *not* m .

We have $d^2\Delta_r/dr^2 > 0$, so that the set $\{\Delta_r \leq 0\}$ is an interval (r_-, r_+) , with $0 < r_- < r_+$.

It follows from (7.4.16) that $g_{\varphi\varphi}/\sin^2(\theta)$ is positive for $r > 0$, and the analysis of the time-machine set is identical to the case $\Lambda > 0$ as long as $\Xi > 0$, which is assumed. We note that stable causality of each region on which Δ_r has constant sign follows from (7.4.11) or (7.4.12).

The projection metric is formally identical to that derived in Section 7.4.1, with projection diagrams as in Fig. 7.4.7.

7.5 The Emparan–Reall metrics

We consider the Emparan–Reall black-ring metric of Section 5.3, p. 221,

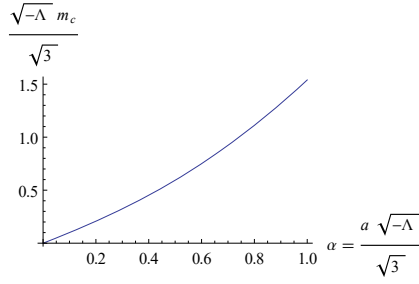


Fig. 7.4.5 The critical mass parameter $M_c \sqrt{|\Lambda/3|} := \Xi m_c \sqrt{3/|\Lambda|}$ as a function of $|a| \sqrt{|\Lambda/3|}$ when $q = 0$.

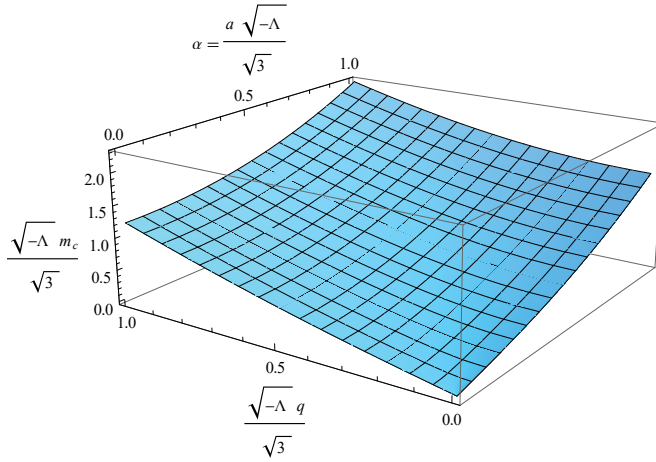


Fig. 7.4.6 The critical mass parameter $M_c \sqrt{\frac{|\Lambda|}{3}}$ as a function of $\alpha = a \sqrt{\frac{|\Lambda|}{3}}$ and $q \sqrt{\frac{|\Lambda|}{3}}$.

$$ds^2 = -\frac{F(y)}{F(x)} \left(dt - C R \frac{1+y}{F(y)} d\psi \right)^2 + \frac{R^2}{(x-y)^2} F(x) \left[-\frac{G(y)}{F(y)} d\psi^2 - \frac{dy^2}{G(y)} + \frac{dx^2}{G(x)} + \frac{G(x)}{F(x)} d\phi^2 \right], \quad (7.5.1)$$

where

$$F(\xi) = 1 + \lambda \xi, \quad G(\xi) = (1 - \xi^2)(1 + \nu \xi), \quad (7.5.2)$$

and

$$C = \sqrt{\lambda(\lambda - \nu) \frac{1 + \lambda}{1 - \lambda}}. \quad (7.5.3)$$

The parameter λ is chosen to be

$$\lambda = \frac{2\nu}{1 + \nu^2}, \quad (7.5.4)$$

with the parameter ν lying in $(0, 1)$, so that

$$0 < \nu < \lambda < 1. \quad (7.5.5)$$

The coordinates x, y lie in the ranges $-\infty \leq y \leq -1$, $-1 \leq x \leq 1$, assuming further that $(x, y) \neq (-1, -1)$. The event horizons are located at $y = y_h = -1/\nu$ and the

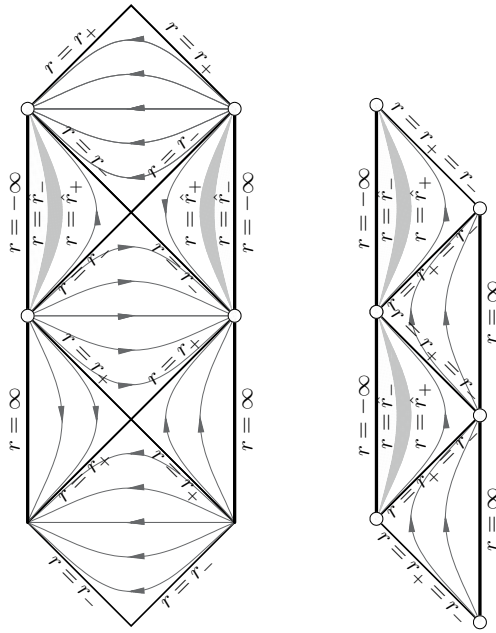


Fig. 7.4.7 The projection diagrams for the Kerr–Newman–anti-de Sitter metrics with two distinct zeros of Δ_r (left diagram) and one double zero (right diagram); see Remark 7.3.1.

ergosurface is at $y = y_e = -1/\lambda$. The ∂_ψ -axis is at $y = -1$ and the ∂_ϕ -axis is split into two parts $x = \pm 1$. Spatial infinity i^0 corresponds to $x = y = -1$. The metric becomes singular as $y \rightarrow -\infty$.

Although this is not immediately apparent from the current form of the metric, we have seen that ∂_ψ is spacelike or vanishing in the region of interest, with $g_{\psi\psi} > 0$ away from the rotation axis $y = -1$. Now, the metric (7.5.1) may be rewritten in the form

$$g = \left(g_{tt} - \frac{g_{t\psi}^2}{g_{\psi\psi}} \right) dt^2 - \frac{R^2}{(x-y)^2} \frac{F(x)}{G(y)} dy^2 + \underbrace{g_{\psi\psi} \left(d\psi + \frac{g_{t\psi}}{g_{\psi\psi}} dt \right)^2 + g_{xx} dx^2 + g_{\phi\phi} d\phi^2}_{\geq 0}. \quad (7.5.6)$$

We have

$$g_{tt} - \frac{g_{t\psi}^2}{g_{\psi\psi}} = - \frac{G(y)F(y)F(x)}{F(x)^2G(y) + C^2(1+y)^2(x-y)^2}. \quad (7.5.7)$$

It turns out that there is a non-obvious factorization of the denominator as

$$F(x)^2G(y) + C^2(1+y)^2(x-y)^2 = -F(y)I(x, y),$$

where I is a second-order polynomial in x and y with coefficients depending upon ν , sufficiently complicated so that it cannot be usefully displayed here. The polynomial I turns out to be non-negative, which can be seen as follows: one introduces new, non-negative, variables and parameters (X, Y, σ) via the equations

$$x = X - 1, \quad y = -Y - 1, \quad \nu = \frac{1}{1 + \sigma}, \quad (7.5.8)$$

with $0 \leq X \leq 2$, $0 \leq Y < +\infty$, $0 < \sigma < +\infty$. A MATHEMATICA calculation shows that in this parameterization the function I is a rational function of the new variables, with a simple denominator which is explicitly non-negative, while the numerator is a complicated polynomial in X , Y , σ with, however, *all coefficients positive*.

Let $\Omega = (x - y)/\sqrt{F(x)}$; then the function

$$\kappa(x, y) := \Omega^2 \left(g_{tt} - \frac{g_{t\psi}^2}{g_{\psi\psi}} \right) = -\frac{G(y)F(y)}{\frac{F(x)^2}{(x-y)^2}G(y) + C^2(1+y)^2} \quad (7.5.9)$$

has extrema in x only for $x = y = -1$ and $x = -1/\lambda < -1$. This may be seen from its derivative with respect to x , which is explicitly non-positive in the ranges of variables of interest:

$$\frac{\partial \kappa}{\partial x} = -\frac{2G(y)^2 F(y)^2 F(x)(x-y)}{(F(x)^2 G(y) + C^2(1+y)^2(x-y)^2)^2} = -\frac{2G(y)^2 F(x)(x-y)}{I(x, y)^2}. \quad (7.5.10)$$

Therefore,

$$\frac{(1+y)^2 G(y)}{I(-1, y)} = \kappa(-1, y) \geq \kappa(x, y) \geq \kappa(1, y) = \frac{(1-y)^2 G(y)}{I(1, y)}.$$

Since both $I(-1, y)$ and $I(1, y)$ are positive, in the domain of outer communications $\{-1/\nu < y \leq -1\}$ where $G(y)$ is negative we obtain

$$\frac{-G(y)(1+y)^2}{I(-1, y)} \leq \left| \Omega^2 \left(g_{tt} - \frac{g_{t\psi}^2}{g_{\psi\psi}} \right) \right| \leq \frac{-G(y)(1-y)^2}{I(1, y)}. \quad (7.5.11)$$

One finds that

$$I(1, y) = \frac{1+\lambda}{1-\lambda}(-1+y^2)(1-y(\lambda-\nu)-\lambda\nu),$$

which leads to the projection metric

$$\gamma := \chi(y) \frac{G(y)}{(-1-y)} dt^2 - \frac{R^2}{G(y)} dy^2, \quad (7.5.12)$$

where, using the variables (7.5.8) to make manifest the positivity of χ in the range of variables of interest,

$$\begin{aligned} \chi(y) &= \frac{(1-y)(1-\lambda)}{(1+\lambda)(1-y(\lambda-\nu)-\lambda\nu)} \\ &= \frac{(2+Y)\sigma(1+\sigma)(2+2\sigma+\sigma^2)}{(2+\sigma)^3(2+Y+\sigma)} > 0. \end{aligned}$$

The calculation of (6.1.4) leads to the conformal metric

$$\stackrel{(2)}{g} = R \sqrt{\frac{\chi}{|1+y|}} \left(-\hat{F} dt^2 + \hat{F}^{-1} dr^2 \right), \text{ where } \hat{F} = -\frac{1}{R} \sqrt{\frac{\chi}{|1+y|}} G. \quad (7.5.13)$$

Since the integral of $\hat{F}^{-1}$ diverges at the event horizon, and is finite at $y = -1$ (which corresponds to *both* an axis of rotation and the asymptotic region at infinity), the analysis in Sections 6.2.4 and 6.3 shows that the corresponding projection diagram is as in Fig. 7.5.1.

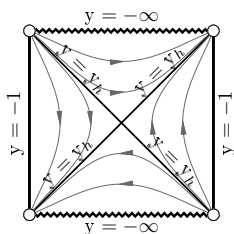


Fig. 7.5.1 The projection diagram for the Emparan–Reall black rings. The arrows indicate the causal character of the orbits of the isometry group. The boundary $y = -1$ is covered, via the projection map, by the axis of rotation and by spatial infinity i^0 . Curves approaching the conformal null infinities $\mathcal{J}^\pm$ asymptote to the missing corners in the diagram.

It is instructive to compare this to the projection diagram for five-dimensional Minkowski spacetime

$$(t, \hat{r} \cos \phi, \hat{r} \sin \phi, \tilde{r} \cos \psi, \tilde{r} \sin \psi) \equiv (t, \hat{x}, \hat{y}, \tilde{x}, \tilde{y}) \in \mathbb{R}^5$$

parameterized by ring-type coordinates:

$$y = -\frac{\hat{r}^2}{(\hat{r}^2 + \tilde{r}^2)^2} - 1, \quad x = \frac{\tilde{r}^2}{(\hat{r}^2 + \tilde{r}^2)^2} - 1, \quad \hat{r} = \sqrt{\hat{x}^2 + \hat{y}^2}, \quad \tilde{r} = \sqrt{\tilde{x}^2 + \tilde{y}^2}.$$

For fixed $x \neq 0, y \neq 0$ we obtain a torus as φ and ψ vary over S^1 . The image of the resulting map is the set $x \geq -1, y \leq -1, (x, y) \neq (-1, -1)$. Since

$$x - y = \frac{1}{\hat{r}^2 + \tilde{r}^2},$$

the spheres $\hat{r}^2 + \tilde{r}^2 =: r^2 = \text{const}$ are mapped to subsets of the lines $x = y + 1/r^2$, and the limit $r \rightarrow \infty$ corresponds to $0 \leq x - y \rightarrow 0$ (hence $x \rightarrow -1$ and $y \rightarrow -1$). The inverse transformation reads

$$\hat{r} = \frac{\sqrt{-y-1}}{x-y}, \quad \tilde{r} = \frac{\sqrt{x+1}}{x-y}.$$

The Minkowski metric takes the form

$$\begin{aligned} \eta &= -dt^2 + d\hat{x}^2 + d\hat{y}^2 + d\tilde{x}^2 + d\tilde{y}^2 \\ &= -dt^2 + d\hat{r}^2 + \hat{r}^2 d\varphi^2 + d\tilde{r}^2 + \tilde{r}^2 d\psi^2 \\ &= -dt^2 + \frac{dy^2}{4(-y-1)(x-y)^2} + \underbrace{\frac{dx^2}{4(x+1)(x-y)^2} + \hat{r}^2 d\varphi^2 + \tilde{r}^2 d\psi^2}_{\geq 0}. \end{aligned}$$

Thus, for any η -causal vector X ,

$$\eta(X, X) \geq -(X^t)^2 + \frac{(X^y)^2}{4(-y-1)(x-y)^2}.$$

There is a problem with the right-hand side since, at fixed y , x is allowed to go to infinity, and so there is no positive lower bound on the coefficient of $(X^y)^2$. However, if we restrict attention to the set

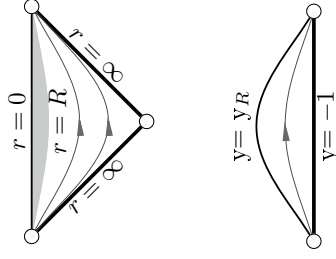


Fig. 7.5.2 The projection diagram for the complement of a world-tube $\mathbb{R} \times B(R)$ in five-dimensional Minkowski spacetime using spherical coordinates (left, where the shaded region has to be removed), or using ring coordinates (right). In the right image the right boundary $y = -1$ is covered, via the projection map, both by the axis of rotation and by spatial infinity, while null infinity projects to the missing points at the top and at the bottom of the diagram.

$$r = \sqrt{\hat{r}^2 + \tilde{r}^2} \geq R$$

for some $R > 0$, we obtain

$$\eta(X, X) \geq -(X^t)^2 + \frac{R^4 (X^y)^2}{4(-y-1)}.$$

This leads to the conformal projection metric, for $-1 - \frac{1}{R^2} =: y_R \leq y \leq -1$,

$$\begin{aligned} \gamma &:= -dt^2 + \frac{R^4 dy^2}{4|y+1|} \\ &= -dt^2 + \left(d \left(R^2 \sqrt{|y+1|} \right) \right)^2 \\ &= \frac{R^2}{2\sqrt{|y+1|}} \left(-\frac{2\sqrt{|y+1|}}{R^2} dt^2 + \frac{R^2}{2\sqrt{|y+1|}} dy^2 \right). \end{aligned} \quad (7.5.14)$$

Introducing a new coordinate $y' = -R^2 \sqrt{-y-1}$ we have

$$\gamma = -dt^2 + dy'^2,$$

where $-1 \leq y' \leq 0$. Therefore, the projection diagram corresponds to a subset of the standard diagram for a two-dimensional Minkowski spacetime; see Fig. 7.5.2.

7.6 The Pomeransky–Senkov metrics

We consider the *Pomeransky–Senkov metric* [405],

$$\begin{aligned} g &= \frac{2H(x, y)k^2}{(1-\nu)^2(x-y)^2} \left(\frac{dx^2}{G(x)} - \frac{dy^2}{G(y)} \right) - 2 \frac{J(x, y)}{H(y, x)} d\varphi d\psi \\ &\quad - \frac{H(y, x)}{H(x, y)} (dt + \Omega)^2 - \frac{F(x, y)}{H(y, x)} d\psi^2 + \frac{F(y, x)}{H(y, x)} d\varphi^2, \end{aligned} \quad (7.6.1)$$

where Ω is a 1-form given by

$$\Omega = M(x, y)d\psi + P(x, y)d\varphi.$$

The somewhat lengthy formulae for the metric functions may be found in [112, 405].⁴ We only note the function

⁴We use (ψ, φ) where Pomeransky and Senkov use (φ, ψ) .

$$G(y) = (1 - y^2)(\nu x^2 + \lambda y + 1).$$

The zeros of its second factor determine the location of the Killing horizons.

The metric depends on three constants: k , ν , λ , where k is assumed to be in $\mathbb{R}^*$, while the parameters λ and ν are restricted to the set⁵

$$\{(\nu, \lambda) : \nu \in (0, 1), 2\sqrt{\nu} \leq \lambda < 1 + \nu\}. \quad (7.6.2)$$

The coordinates x , y , φ , ψ , and t vary within the ranges $-1 \leq x \leq 1$, $-\infty < y < -1$, $0 \leq \varphi \leq 2\pi$, $0 \leq \psi \leq 2\pi$, and $-\infty < t < \infty$.

A Cauchy horizon is located at

$$y_c := -\frac{\lambda + \sqrt{\lambda^2 - 4\nu}}{2\nu},$$

and the event horizon corresponds to

$$y_h := -\frac{\lambda - \sqrt{\lambda^2 - 4\nu}}{2\nu}.$$

Using an appropriate Gauss diagonalization, the metric may be rewritten in the form

$$g = \overbrace{\frac{g_{t\psi}^2 g_{\varphi\varphi} - 2g_{t\varphi} g_{t\psi} g_{\psi\varphi} + g_{t\varphi}^2 g_{\psi\psi} + g_{tt}(g_{\psi\varphi}^2 - g_{\varphi\varphi} g_{\psi\psi})}{g_{\psi\varphi}^2 - g_{\varphi\varphi} g_{\psi\psi}}}^{(*)} dt^2 + g_{yy} dy^2 + g_{xx} dx^2 + \underbrace{\left(g_{\varphi\varphi} - \frac{g_{\psi\varphi}^2}{g_{\psi\psi}} \right) \left(d\varphi + \frac{g_{t\varphi} - \frac{g_{t\psi} g_{\psi\varphi}}{g_{\psi\psi}}}{g_{\varphi\varphi} - \frac{g_{\psi\varphi}^2}{g_{\psi\psi}}} dt \right)^2 + \frac{(g_{t\psi} dt + g_{\psi\varphi} d\varphi + g_{\psi\psi} d\psi)^2}{g_{\psi\psi}}}_{(**)}. \quad (7.6.3)$$

The positive-definiteness of $(**)$ for $y > y_c$ has been established in [112, 145]. Note that $g_{\psi\psi} < 0$ would give a timelike Killing vector ∂_ψ , and that $g_{\varphi\varphi} g_{\psi\psi} - g_{\psi\varphi}^2 < 0$ would lead to some combination of the periodic Killing vectors ∂_φ and ∂_ψ being timelike, so the term $(**)$ in (7.6.3) is non-negative on any region where there are no obvious causality violations.

The coefficient $(*)$ in front of dt^2 is negative for $y > y_h$ and positive for $y < y_h$, vanishing at $y = y_h$. This may be seen in the reparameterized form of the Pomeransky–Senkov solution that was introduced in [145]: indeed, let us introduce new coordinates a , b replacing x and y , respectively, and let us parameterize ν , λ by c , d , with c , d non-negative, and where the coordinates a , b are non-negative in the region $\{y > y_c\}$:

$$\begin{aligned} x &= -1 + \frac{2}{1+a}, \\ y &= -1 - \frac{d(4+c+2d)}{(1+b)(2+c)}, \\ \nu &= \frac{1}{(1+d)^2}, \end{aligned}$$

⁵ $\nu = 0$ corresponds to the Emparan–Reall metric which has been already analysed in Section 7.5.

$$\lambda = 2 \frac{2d^2 + 2(2+c)d + (2+c)^2}{(2+c)(1+d)(2+c+2d)}. \quad (7.6.4)$$

Set

$$\kappa := (*) \Omega^2, \quad (7.6.5)$$

$$\Omega^2 := \frac{(x-y)^2(1-\nu)^2}{2k^2 H(x,y)}. \quad (7.6.6)$$

Using MATHEMATICA one finds that κ takes the form

$$\kappa = -\Omega^2(y - y_h)Q,$$

where $Q = Q(a, b, c, d)$ is a huge rational function in (a, b, c, d) with *all coefficients positive*. To obtain the corresponding projection metric γ one would have, e.g., to find sharp lower and upper bounds for Q , at fixed y , which would lead to

$$\gamma := -(y - y_h) \sup_{y \text{ fixed}} |Q| dt^2 - \frac{1}{G(y)} dy^2.$$

This requires analysing a complicated rational function, which has not been done in the literature so far.

We expect the corresponding projection diagram to look like that for Kerr–anti-de Sitter spacetime of Fig. 7.4.7, with $r = \infty$ there replaced by $y = -1$, $r = -\infty$ replaced by $y = 1$ with an appropriate analytic continuation of the metric to positive y 's (compare [112]), r_+ replaced by y_h , and r_- replaced by y_c . The shaded regions in the negative region there might be non-connected for some values of parameters, and always extend to the boundary at infinity in the relevant diamond [112].

Recall that a substantial part of the work in [112] was to show that the function $H(x, y)$ had no zeros for $y > y_c$. We note that the reparameterization

$$y \rightarrow -1 - \frac{cd}{(1+b)(2+c+2d)}$$

of [145] (with the remaining formulae (7.6.4) remaining the same) gives

$$H(x, y) = \frac{P(a, b, c, d)}{(1+a)^2(1+b)^2(2+c)^2(1+d)^6(2+c+2d)^4},$$

where P is a huge polynomial with *all coefficients positive* for $y > y_h$. This establishes immediately the positivity of $H(x, y)$ in the domain of outer communications. However, the positivity of $H(x, y)$ in the whole range $y > y_c$ has only been established so far using the rather more involved analysis in [112].

7.7 $U(1) \times U(1)$ symmetry with compact Cauchy horizons

The Kerr–Newman–(A)dS family of metrics possesses regions where both Killing vectors are timelike. Periodically identifying the orbits of the Killing vector ∂_t there provides explicit examples of maximal, four-dimensional, $U(1) \times U(1)$ symmetric, electrovacuum or vacuum models, with or without cosmological constant, containing a spatially compact partial Cauchy surface. Similarly, five-dimensional, $U(1) \times U(1) \times U(1)$ symmetric, spatially compact vacuum models with spatially compact partial Cauchy surfaces can be constructed using the Emparan–Reall or Pomeransky–Senkov metrics. In this section we will show how the projection diagrams constructed so far can be used to understand maximal (non-globally hyperbolic) extensions of the maximal globally hyperbolic regions in such models, and for the Taub–NUT metrics.

7.7.1 Building blocks and periodic identifications

The diamonds and triangles which have been used so far to construct our diagrams will be referred to as *blocks*. Here a triangle is understood *up to diffeomorphism*; thus planar sets with three corners, connected by smooth curves intersecting only at the corners which are *not* necessarily straight lines, are also considered to be triangles.

In the interior of each block one can periodically identify points lying along the orbits of the action of the $\mathbb{R}$ factor of the isometry group. Here we are only interested in the connected component of the identity of the group, which is $\mathbb{R} \times \mathrm{U}(1)$ in the four-dimensional case, and $\mathbb{R} \times \mathrm{U}(1) \times \mathrm{U}(1)$ in the five-dimensional case.

Note that isometries of spacetime extend smoothly across all block boundaries. For example, in the coordinates $(v, r, \theta, \tilde{\varphi})$ discussed in the paragraph around (7.3.16), p. 289, translations in t become translations in v ; similarly for the $(u, r, \theta, \tilde{\varphi})$ coordinates. Using the coordinates $(\hat{u}, \hat{v}, \theta, \tilde{\varphi})$ of (4.2.49), p. 140 near the intersection of two Killing horizons, translations in t become boosts in the (U, V) plane.

Consider one of the blocks, out of any of the diagrams constructed earlier, in which the orbits of the isometry group are spacelike. (Note that no such diamond or triangle has a shaded area which needs to be excised in the examples seen so far, as the shadings occur only within those building blocks where the isometry orbits are timelike.) It can be seen that the periodic identifications result then in a spatially compact maximal globally hyperbolic spacetime with $S^1 \times S^2$ spatial topology, respectively with $S^1 \times S^1 \times S^2$ topology.

Now, each diamond in our diagrams has four null boundaries which naturally split into pairs, as follows: in each block in which the isometry orbits are spacelike, we will say that two boundaries are *orbit-adjacent* if both boundaries lie to the future of the block, or both to the past. In a block where the isometry orbits are timelike, boundaries will be said *orbit-adjacent* if they are both to the left or both to the right.

One out of each pair of orbit-adjacent null boundaries of a block with spacelike isometry-orbits corresponds, in the periodically identified spacetime, to a compact Cauchy horizon across which the spacetime can be continued to a periodically identified adjacent block. Which of the two adjacent boundaries will become a Cauchy horizon is a matter of choice; once such a choice has been made, the other boundary *cannot* be attached anymore: those geodesics which, in the unidentified spacetime, would have been crossing the second boundary become, in the periodically identified spacetime, incomplete inextendible geodesics. This behaviour is well known from Taub–NUT spacetimes [127, 358, 443], and is easily seen as follows.

Consider a sequence of points $p_i := (t_i, r_i)$ such that p_i converges to a point p on a horizon in a projection diagram in which no periodic identifications have been made. Let $T > 0$ be the period with which the points are identified along the isometry orbits; thus for every $n \in \mathbb{Z}$ points (t, r) and $(t + nT, r)$ represent the same point of the quotient manifold. It should be clear from the form of the Eddington–Finkelstein-type coordinates u and v used to perform the two distinct extensions (see the paragraph around (7.3.16), p. 289) that there exists a sequence $n_i \in \mathbb{Z}$ such that, passing to a subsequence if necessary, the sequence $q_i = (t_i + n_i T, r_i)$ converges to some point q in the companion orbit-adjacent boundary; see Fig. 7.7.1.

Denote by $[p]$ the class of p under the equivalence relation $(t, r) \sim (t + nT, r)$, where $n \in \mathbb{Z}$ and T is the period. Suppose that one could construct simultaneously an extension of the quotient manifold across both orbit-adjacent boundaries. Then the sequence of points $[q_i] = [p_i]$ would have two distinct points $[p]$ and $[q]$ as limit points, which is not possible. This establishes our claim.

Returning to our main line of thought, note that a periodically identified building block in which the isometry orbits are timelike will have obvious causality violations

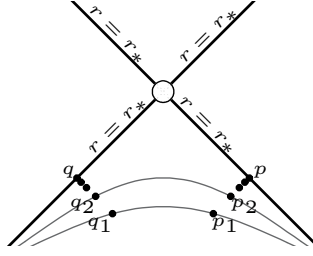


Fig. 7.7.1 The sequences q_i and p_i . Rotating the figure by integer multiples of 90° shows that the problem of non-unique limits arises on any pair of orbit-adjacent boundaries.

throughout, as a linear combination of the periodic Killing vectors becomes timelike there.

The branching construction, where one out of the pair of orbit-adjacent boundaries is chosen to perform the extension, can be continued at each block in which the isometry orbits are spacelike. This shows that maximal extensions are obtained from any connected union of blocks such that in each block an extension is carried out across precisely one out of each pair of orbit-adjacent boundaries. Some such subsets of the plane might only comprise a finite number of blocks, as seen trivially in Fig. 7.4.4. Clearly an infinite number of distinct finite, semi-infinite, or infinite sequences of blocks can be constructed in the diagram of Fig. 7.4.1. Two sequences of blocks which are not related by one of the discrete isometries of the diagram will lead to non-isometric maximal extensions of the maximal globally hyperbolic initial region.

7.7.2 Taub–NUT metrics

We have seen at the end of Section 7.2 how to construct a projection diagram for Gowdy cosmological models. Those models all contain $U(1) \times U(1)$ as part of their isometry group. The corresponding projection diagrams constructed in Section 7.2 were obtained by projecting out the isometry orbits. This is rather different from the remaining projection diagrams constructed in this work, where only one of the coordinates along the Killing orbits was projected out.

It is instructive to carry out explicitly both procedures for the Taub–NUT metrics, which belong to the Gowdy class. Using Euler angles (ζ, θ, φ) to parameterize S^3 , the Taub–NUT metrics [375, 443] take the form

$$g = -U^{-1}dt^2 + (2\ell)^2 U(d\zeta + \cos(\theta) d\varphi)^2 + (t^2 + \ell^2)(d\theta^2 + \sin^2(\theta) d\varphi^2). \quad (7.7.1)$$

Here

$$U(t) = -1 + \frac{2(mt + \ell^2)}{t^2 + \ell^2} = \frac{(t_+ - t)(t - t_-)}{t^2 + \ell^2},$$

with

$$t_{\pm} := m \pm \sqrt{m^2 + \ell^2}.$$

Further, ℓ and m are real numbers with $\ell > 0$. The region $\{t \in (t_-, t_+)\}$ will be referred to as the *Taub spacetime*.

The metric induced on the sections $\theta = \text{const}$, $\varphi = \text{const}'$, of the Taub spacetime reads

$$\gamma_0 := -U^{-1}dt^2 + (2\ell)^2 U d\zeta^2. \quad (7.7.2)$$

As discussed by Hawking and Ellis [244], this is a metric to which the methods of Section 6.1, p. 259, apply *provided that the 4π -periodic identifications in ζ are relaxed*. Since U has two simple zeros, and no singularities, the conformal diagram for

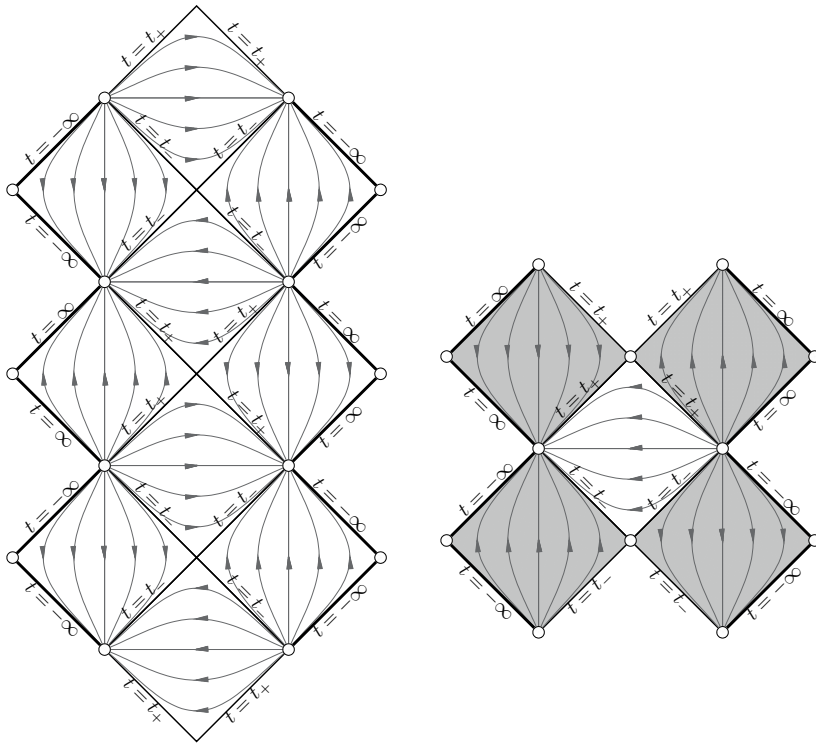


Fig. 7.7.2 The left diagram is the conformal diagram for an extension of the universal covering space of the sections $\theta = \text{const}$, $\varphi = \text{const}'$, of the Taub spacetime. The right diagram represents simultaneously the four possible diagrams for the maximal extensions, within the Taub–NUT class, with compact Cauchy horizons, of the Taub space–time. After invoking the left–right symmetry of the diagram, which lifts to an isometry of the extended spacetime, the four diagrams lead to two non-isometric spacetimes.

the corresponding maximally extended two-dimensional spacetime equipped with the metric γ_0 coincides with the left diagram in Fig. 7.7.2; compare [244, Fig. 33]. The discussion in the last paragraph of the previous section applies and, together with the left diagram in Fig. 7.7.2, provides a family of simply connected maximal extensions of the sections $\theta = \text{const}$, $\varphi = \text{const}'$, of the Taub spacetime.

However, it is not clear how to relate the above to extensions of the four-dimensional spacetime. Note that projecting out the ζ and φ variables in the region where $U > 0$, using the projection map $\pi_1(t, \zeta, \theta, \varphi) := (t, \theta)$, one is left with the two-dimensional metric

$$\gamma_1 := -U^{-1}dt^2 + (t^2 + \ell^2)d\theta^2, \quad (7.7.3)$$

which leads to the flat metric on the Gowdy square as the projection metric. (The coordinate t here is not the same as the Gowdy t coordinate, but the projection diagram remains a square.) And one is left wondering how this fits with the previous picture.

Now, one can attempt instead to project out the θ and φ variables, with the projection map

$$\pi_2(t, \zeta, \theta, \varphi) := (t, \zeta). \quad (7.7.4)$$

For this we note the trivial identity

$$g_{\zeta\zeta}d\zeta^2 + 2g_{\varphi\zeta}d\varphi d\zeta + g_{\varphi\varphi}d\varphi^2 = \left(g_{\zeta\zeta} - \frac{g_{\varphi\zeta}^2}{g_{\varphi\varphi}}\right)d\zeta^2 + \underbrace{g_{\varphi\varphi}\left(d\varphi + \frac{g_{\varphi\zeta}}{g_{\varphi\varphi}}d\zeta\right)^2}_{(*)}. \quad (7.7.5)$$

Since the left-hand side is positive-definite on Taub space, where $U > 0$, both $g_{\zeta\zeta} - \frac{g_{\varphi\zeta}^2}{g_{\varphi\varphi}}$ and $g_{\varphi\varphi}$ are non-negative there. Indeed,

$$g_{\varphi\varphi} = (\ell^2 + t^2) \sin^2(\theta) + 4\ell^2 U \cos^2(\theta), \quad (7.7.6)$$

$$g_{\zeta\zeta} - \frac{g_{\varphi\zeta}^2}{g_{\varphi\varphi}} = (2\ell)^2 \left(1 - \frac{(2\ell)^2 U \cos^2(\theta)}{g_{\varphi\varphi}}\right) U$$

$$= \underbrace{\frac{4\ell^2 (\ell^2 + t^2) \sin^2(\theta)}{(\ell^2 + t^2) \sin^2(\theta) + 4\ell^2 U \cos^2(\theta)}}_{(**)} U. \quad (7.7.7)$$

However, perhaps not unsurprisingly given the character of the coordinates involved, the function $(**)$ in (7.7.7) does not have a positive lower bound independent of $\theta \in [0, 2\pi]$, which is unfortunate for our purposes. To sidestep this drawback we choose a number $0 < \epsilon < 1$ and restrict ourselves to the range $\theta \in [\theta_\epsilon, \pi - \theta_\epsilon]$, where $\theta_\epsilon \in [0, \pi/2]$ is defined by

$$\sin^2(\theta_\epsilon) = \epsilon.$$

Now, $g_{\varphi\varphi}$ is positive for large t , independently of θ . Next, $g_{\varphi\varphi}$ equals $4\ell^2 U$ at the axes of rotation $\sin(\theta) = 0$, and equals $\ell^2 + t^2$ at $\theta = \pi/2$. Hence, keeping in mind that U is monotonic away from (t_-, t_+) , for ϵ small enough there will exist values

$$\hat{t}_\pm(\epsilon), \text{ with } \hat{t}_-(\epsilon) < t_- < 0 < t_+ < \hat{t}_+(\epsilon)$$

such that $g_{\varphi\varphi}$ will be negative somewhere in the region $(\hat{t}_-(\epsilon), t_-) \cup (t_+, \hat{t}_+(\epsilon))$, and will be positive outside of this region. We choose those numbers to be optimal with respect to those properties.

On the other hand, for ϵ close enough to 1 the metric coefficient $g_{\varphi\varphi}$ will be positive for all $\theta \in [\theta_\epsilon, \pi - \theta_\epsilon]$ and $t < t_-$. In this case we set $\hat{t}_-(\epsilon) = t_-$, so that the interval $(\hat{t}_-(\epsilon), t_-)$ is empty. Similarly, there will exist a range of ϵ for which $\hat{t}_+(\epsilon) = t_+$, and $(t_+, \hat{t}_+(\epsilon)) = \emptyset$. The relevant ranges of ϵ will coincide only if $m = 0$.

We note that

$$\partial_\theta \left(g_{\zeta\zeta} - \frac{g_{\varphi\zeta}^2}{g_{\varphi\varphi}} \right) = \frac{16\ell^4 U^2 (\ell^2 + t^2) \sin(2\theta)}{((\ell^2 + t^2) \sin^2(\theta) + 4\ell^2 U \cos^2(\theta))^2},$$

which shows that, for

$$t \notin (\hat{t}_-(\epsilon), t_-) \cup (t_+, \hat{t}_+(\epsilon)) \text{ and } \theta \in (\theta_\epsilon, \pi - \theta_\epsilon), \quad (7.7.8)$$

the multiplicative coefficient $(**)$ of U in (7.7.7) will satisfy

$$(**) \geq \frac{4\ell^2 (\ell^2 + t^2) \sin^2(\theta_\epsilon)}{(\ell^2 + t^2) \sin^2(\theta_\epsilon) + 4\ell^2 U \cos^2(\theta_\epsilon)} =: f_\epsilon(t). \quad (7.7.9)$$

We are ready now to construct the projection metric in the region (7.7.8). Removing from the metric tensor (7.7.1) the terms $(*)$ appearing in (7.7.5), as well as the $d\theta^2$ terms, and using (7.7.9) one finds, for g -causal vectors X ,

$$g(X, X) \geq \gamma_2((\pi_2)_*X, (\pi_2)_*X),$$

with π_2 as in (7.7.4), and where

$$\gamma_2 := -U^{-1}dt^2 + f_\epsilon U d\zeta^2. \quad (7.7.10)$$

Since U has exactly two simple zeros and is finite everywhere, and for ϵ such that $g_{\varphi\varphi}$ is positive on the region $\theta \in [\theta_\epsilon, \pi - \theta_\epsilon]$, the projection diagram for that region, *in a spacetime in which no periodic identifications in ζ are made*, is given by the left diagram of Fig. 7.7.2. The reader should have no difficulties finding the corresponding diagrams for the remaining values of ϵ .

However, we are in fact interested in those spacetimes where ζ is 4π periodic. This has two consequences: (a) there are closed timelike Killing orbits in all the regions where U is negative, and (b) no simultaneous extensions are possible across two orbit-adjacent boundaries. It then follows (see the right diagram of Fig. 7.7.2) that there are, within the Taub–NUT class, only two non-isometric, maximal, vacuum extensions across compact Cauchy horizons of the Taub spacetime. (Compare [143, Proposition 4.5 and Theorem 1.2] for the local uniqueness of extensions, and [105] for a discussion of extensions with non-compact Killing horizons.)

8

Dynamical black holes

All black-hole spacetimes seen so far have been stationary. This is mainly due to the fact that explicit non-stationary metrics are hard to come by. This chapter provides an excursion into the world of dynamical solutions.

The closest one can come to explicit time-dependent black-hole metrics is provided by the Robinson–Trautman family metrics, which are explicit except for one function satisfying a parabolic evolution equation. In Section 8.1 we present some properties of those metrics.

One way of constructing dynamical black holes is by solving a Cauchy problem: one prescribes initial data containing regions which are likely to be contained in a black-hole region after evolving the initial data. Some aspects of this, relevant for black holes, are briefly reviewed in Section 8.2.

In stationary spacetimes it is most convenient to define the black-hole region using the flow of the Killing vector field, as presented in Section 4.3.8, p. 166. In non-stationary spacetimes there is no such device, and a different approach is needed. The standard way to do this invokes conformal completions, which we critically review in Section 8.3, where we also discuss alternative possibilities.

8.1 Robinson–Trautman spacetimes

The Robinson–Trautman (RT) metrics are vacuum metrics which can be viewed as evolving from data prescribed on a single null hypersurface.

From a physical point of view, the RT metrics provide examples of isolated gravitationally radiating systems. In fact, these metrics were hailed to be the first exact nonlinear solutions describing such a situation. Their discovery [418] was a breakthrough in the conceptual understanding of gravitational radiation in Einstein’s theory.

INCIDENTALLY: The RT metrics were the only example of vacuum dynamical black holes without any symmetries and with exhaustively described global structure until the construction, in 2013 [160], of a large class of such spacetimes using ‘scattering data’ at the horizon and at future null infinity. See Section 8.6.

A class of dynamical, vacuum, multi-black holes with a positive cosmological constant has been constructed in [283]. $\square$

There are several interesting features exhibited by the RT metrics: First, and rather unexpectedly, in this class of metrics the Einstein equations reduce to a single parabolic fourth-order equation. Next, the evolution is unique within the class, in spite of a ‘naked singularity’ at $r = 0$. Last but not least, they possess remarkable extendibility properties.

By definition, the Robinson–Trautman spacetimes can be foliated by a *null, hypersurface-orthogonal, shear-free, expanding* geodesic congruence. It has been shown by Robinson and Trautman [412] that in such a spacetime there always exists a coordinate system in which the metric takes the form

$$g = -\Phi du^2 - 2du dr + r^2 e^{2\lambda} \underbrace{\dot{h}_{ab}(x^c) dx^a dx^b}_{=: \dot{h}}, \quad \lambda = \lambda(u, x^a), \quad (8.1.1)$$

$$\Phi = \frac{R}{2} + \frac{r}{12m} \Delta_h R - \frac{2m}{r}, \quad R = R(h_{ab}) \equiv R(e^{2\lambda} \dot{h}_{ab}), \quad (8.1.2)$$

where the x^a 's are local coordinates on the two-dimensional smooth Riemannian manifold $(^2M, \dot{h})$, $m \neq 0$ is a constant which is related to the total Trautman–Bondi mass of the metric, and R is the Ricci scalar of the metric

$$h := e^{2\lambda} \dot{h}.$$

In writing (8.1.1)–(8.1.2) we have ignored those spacetimes which admit a congruence as above and where the parameter m vanishes.

The Einstein equations for a metric of the form (8.1.1) reduce to a single equation

$$\partial_u h_{ab} = \frac{1}{12m} \Delta_h R h_{ab} \iff \partial_u \lambda = \frac{1}{24m} e^{-2\lambda} \Delta_h (e^{-2\lambda} (\dot{R} - 2\Delta_h \lambda)), \quad (8.1.3)$$

where Δ_h is the Laplace operator of the two-dimensional metric $\dot{h} = \dot{h}_{ab} dx^a dx^b$, and $\dot{R}$ is the Ricci scalar of the metric $\dot{h}$.

Equation (8.1.3) will be referred to as the RT equation. It is first order in the ‘time’ u , is fourth order in the space variables x^a , and belongs to the family of parabolic equations. The Cauchy data for (8.1.3) consist of a function $\lambda_0(x^a) \equiv \lambda(u = u_0, x^a)$, which is equivalent to prescribing the metric $g_{\mu\nu}$ of the form (8.1.1) on a null hypersurface $\{u = u_0, r \in (0, \infty)\} \times ^2M$. Without loss of generality, translating u if necessary, we can assume that $u_0 = 0$.

Note that the initial data hypersurface asymptotes to a curvature singularity at $r = 0$, with the scalar $R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta}$ diverging as r^{-6} when $r = 0$ is approached. This is a ‘white-hole singularity’, familiar to all known stationary black-hole spacetimes.

INCIDENTALLY: The RT equation (8.1.3) has been considered in a completely different context by Calabi [74]. $\square$

The function $\lambda \equiv 0$ solves (8.1.3) when $\dot{h}$ is the unit round metric on the sphere. The metric (8.1.1) is then the Schwarzschild metric in retarded Eddington–Finkelstein coordinates.

It follows from the theory of parabolic equations that for $m < 0$ the evolution problem for (8.1.3) is locally well posed backwards in u , while for $m > 0$ the RT equation can be locally solved forwards in u . Redefining u to $-u$ transforms (8.1.3) with $m < 0$, $u \leq 0$ to the same equation with a new mass parameter $-m > 0$ and with $u \geq 0$. Thus, when discussing (8.1.3) it suffices to assume $m > 0$. On the other hand, the global properties of the associated spacetimes will be different, and will need separate discussion.

Note that solutions of typical parabolic equations, including (8.1.3), immediately become analytic. This implies that for smooth but *not analytic* initial functions λ_0 , the equation will *not* be solvable backwards in u when $m > 0$, or forwards in u when $m < 0$.

In [97, 98, 144] the following has been proved.

1. When $m > 0$ solutions of (8.1.3) with, say smooth, initial data at $u = 0$ exist for all $u \geq 0$. The proof consists in showing that all Sobolev norms of the solution remain finite during the evolution. The first key to this is the monotonicity of the Trautman–Bondi mass, which for RT metrics equals [430]

$$m_{\text{TB}} = \frac{m}{4\pi} \int_{S^2} e^{3\lambda} d\mu_{\dot{h}}. \quad (8.1.4)$$

The second is the monotonicity property of

$$\int_{^2M} (R - \mathring{R})^2 \quad (8.1.5)$$

discovered by Calabi [74] and, independently, by Lukács, Perjés, Porter, and Sebestyén [330].

2. Let $m > 0$. There exists a strictly increasing sequence of real numbers $\lambda_i > 0$, integers n_i with $n_1 = 0$, and functions $\varphi_{i,j} \in C^\infty(^2M)$, $0 \leq j \leq n_i$, such that, possibly after performing a conformal transformation of $\mathring{h}$, solutions of (8.1.3) have a full asymptotic expansion of the form

$$\lambda(u, x^a) = \sum_{i \geq 1, 0 \leq j \leq n_i} \varphi_{i,j}(x^a) u^j e^{-\lambda_i u/m}, \quad (8.1.6)$$

when u tends to infinity. The result is obtained by a delicate asymptotic analysis of solutions of the RT equation.

The decay exponents λ_i and the n_i 's are determined by the spectrum of $\Delta_{\mathring{h}}$. For example, if $(^2M, \mathring{h})$ is a round two sphere, we have [98]

$$\lambda_i = 2i, \quad i \in \mathbb{N}, \quad \text{with } n_1 = \dots = n_{14} = 0, \quad n_{15} = 1. \quad (8.1.7)$$

REMARK 8.1.1 The first global existence result for the RT equation has been obtained by Rendall [413] for a restricted class of near-Schwarzschildian initial data. Global existence and convergence to a round metric for all smooth initial data has been established in [97]. There the uniformization theorem for compact two-dimensional manifolds has been assumed. An alternative proof of global existence, which establishes the uniformization theorem as a by-product, has been given by Struwe [437]. $\square$

The RT metrics all possess a smooth conformal boundary *à la Penrose* at ' $r = \infty$ '. To see this, one can replace r by a new coordinate $x = 1/r$, which brings the metric (8.1.1) to the form

$$g = x^{-2} \left(- \left(\frac{Rx^2}{2} + \frac{x\Delta_h R}{12m} - 2mx^3 \right) du^2 + 2du dx + e^{2\lambda\mathring{h}} \right), \quad (8.1.8)$$

so that the spacetime metric g multiplied by a conformal factor x^2 smoothly extends to $\{x = 0\}$.

In what follows we shall take $(^2M, \mathring{h})$ to be a two-dimensional sphere equipped with the unit round metric. See [98] for a discussion of other topologies.

8.1.1 $m > 0$

Let us assume that $m > 0$. Following an observation of Schmidt reported in [447], the hypersurface ' $u = \infty$ ' can be attached to the manifold $\{r \in (0, \infty), u \in [0, \infty)\} \times ^2M$ as a null boundary by introducing Kruskal–Szekeres-type coordinates $(\hat{u}, \hat{v})$, defined in a way identical to those for the Schwarzschild metric:

$$\hat{u} = -\exp\left(-\frac{u}{4m}\right), \quad \hat{v} = \exp\left(\frac{u+2r}{4m} + \ln\left(\frac{r}{2m} - 1\right)\right). \quad (8.1.9)$$

This brings the metric to the form

$$g = -\frac{32m^3 \exp\left(-\frac{r}{2m}\right)}{r} d\hat{u} d\hat{v} + r^2 e^{2\lambda\mathring{h}} \\ - 16m^2 \exp\left(\frac{u}{2m}\right) \left(\frac{R}{2} - 1 + \frac{r\Delta_h R}{12m}\right) d\hat{u}^2. \quad (8.1.10)$$

Note that $g_{\hat{u}\hat{u}}$ vanishes when $\lambda \equiv 0$, and one recovers the Schwarzschild metric in Kruskal–Szekeres coordinates. Equations (8.1.6)–(8.1.7) imply that $g_{\hat{u}\hat{u}}$ decays as

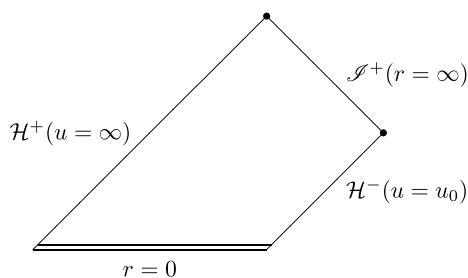


Fig. 8.1.1 A projection diagram for RT metrics with $m > 0$.

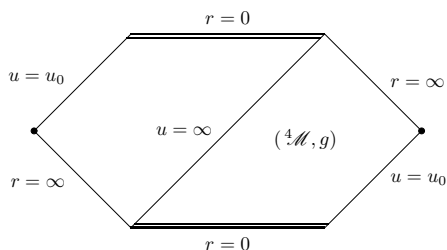


Fig. 8.1.2 Vacuum RT extensions beyond $\mathcal{H}^+ = \{u = \infty\}$. Any two RT metrics with the same mass parameter m can be glued across the null hypersurface $\mathcal{H}^+$, leading to a metric of C^5 -differentiability class.

$e^{u/2m} \times e^{-2u/m} = \hat{u}^6$. Hence g approaches the Schwarzschild metric as $O(\hat{u}^6)$ when the null hypersurface

$$\mathcal{H}^+ := \{\hat{u} = 0\}$$

is approached. A projection diagram, as defined in Chapter 7, with the 2M factor projected out, can be found in Fig. 8.1.1.

In terms of $\hat{u}$ the expansion (8.1.6) becomes

$$\lambda(\hat{u}, x^a) = \sum_{i \geq 1, 0 \leq j \leq n_i} \varphi_{i,j}(x^a) (-4m \log(|\hat{u}|))^j \hat{u}^{8i}, \quad (8.1.11)$$

which can be extended to $\hat{u} > 0$ as an even function of $\hat{u}$. This expansion carries over to similar expansions of R and $\Delta_h R$, and results in an asymptotic expansion of the form

$$g_{\hat{u}\hat{u}}(\hat{u}, x^a) = \sum_{i \geq 1, 0 \leq j \leq n_i} \psi_{i,j}(x^a) (\log |\hat{u}|)^j \hat{u}^{8i-2}, \quad (8.1.12)$$

for some functions $\psi_{i,j}$. It follows from (8.1.2) that the even extension of $g_{\hat{u}\hat{u}}$ will be of C^{117} -differentiability class.

In fact, any two such even functions $g_{\hat{u}\hat{u}}$ can be continued into each other across $u = 0$ to a function of C^5 -differentiability class. It follows that:

1. Any two RT metrics can be joined together as in Fig. 8.1.2 to obtain a spacetime with a metric of C^5 -differentiability class. In particular g can be glued to a Schwarzschild metric beyond $\mathcal{H}$, resulting in a C^5 metric.
2. It follows from (8.1.2) that g can be glued to itself in the C^{117} -differentiability class.

The vanishing, or not, of the expansion functions $\varphi_{i,j}$ in (8.1.6) with $j \geq 1$ turns out to play a key role for the smoothness of the metric at $\mathcal{H}$. Indeed, the first non-vanishing function $\varphi_{i,j}$ with $j \geq 1$ will lead to a $\psi_{i,j} (\ln |\hat{u}|)^j \hat{u}^{8i-2}$ term in the

asymptotic expansion of $g_{\hat{u}\hat{u}}$. As a result, $g_{\hat{u}\hat{u}}$ will be extendable to an even function of $\hat{u}$ of C^{8i-3} -differentiability class, but not better. It is shown in [144] that:

1. A generic function $\lambda(0, x^a)$ close to zero leads to a solution with $\psi_{15,1} \neq 0$, resulting in metrics which are extendible across $\mathcal{H}$ in the C^{117} -differentiability class, but not C^{118} , in the coordinate system above.
2. There exists an infinite-dimensional family of non-generic initial-data functions $\lambda(0, x^a)$ for which $\psi_{15,1} \equiv 0$. An even extension of $g_{\hat{u}\hat{u}}$ across $\mathcal{H}$ results in a metric of C^{557} -differentiability class, but not C^{558} , in the coordinate system above.

The question arises as to whether the above differentiability issues are related to a poor choice of coordinates. By analysing the behaviour of the derivatives of the Riemann tensor on geodesics approaching $\mathcal{H}$, one can show [144] that the metrics of point 1 cannot be extended across $\mathcal{H}$ in the class of spacetimes with metrics of C^{123} -differentiability class. Similarly the metrics of point 2 cannot be extended across $\mathcal{H}$ in the class of spacetimes with metrics of C^{564} -differentiability class. One expects that the differentiability mismatches are not a real effect, but result from a non-optimal inextendibility criterion used.

Summarizing, we have the following.

THEOREM 8.1.2 *Let $m > 0$. For any $\lambda_0 \in C^\infty(S^2)$ there exists a Robinson–Trautman spacetime $({}^4\mathcal{M}, g)$ with a ‘half-complete’ $\mathcal{I}^+$, the global structure of which is shown in Fig. 8.1.1. Moreover:*

1. *$({}^4\mathcal{M}, g)$ is smoothly extendible to the past through $\mathcal{H}^-$. If, however, λ_0 is not analytic, then no vacuum Robinson–Trautman extensions through $\mathcal{H}^-$ exist.*
2. *There exist infinitely many non-isometric vacuum Robinson–Trautman C^5 extensions¹ of $({}^4\mathcal{M}, g)$ through $\mathcal{H}^+$, which are obtained by gluing to $({}^4\mathcal{M}, g)$ any other positive mass Robinson–Trautman spacetime, as shown in Fig. 8.1.2.*
3. *There exist infinitely many C^{117} vacuum RT extensions of $({}^4\mathcal{M}, g)$ through $\mathcal{H}^+$. One such extension is obtained by gluing a copy of $({}^4\mathcal{M}, g)$ to itself, as shown in Fig. 8.1.2.*
4. *For any $6 \leq k \leq \infty$ there exists an open set $\mathcal{O}_k$ of Robinson–Trautman spacetimes, in a $C^k(S^2)$ topology on the set of the initial-data functions λ_0 , for which no C^{123} extensions beyond $\mathcal{H}^+$ exist, vacuum or otherwise. For any u_0 there exists an open ball $\mathcal{B}_k \subset C^k(S^2)$ around the initial data for the Schwarzschild metric, $\lambda_0 \equiv 0$, such that $\mathcal{O}_k \cap \mathcal{B}_k$ is dense in $\mathcal{B}_k$.*

The picture that emerges from Theorem 8.1.2 is the following: generic initial data lead to a spacetime which has no RT vacuum extension to the past of the initial surface, even though the metric can be smoothly extended (in the non-vacuum class); and generic data sufficiently close² to Schwarzschildian ones lead to a spacetime for which no smooth vacuum RT extensions exist beyond $\mathcal{H}^+$. This shows that considering smooth extensions across $\mathcal{H}^+$ leads to non-existence, while giving up the requirement of smoothness of extensions beyond $\mathcal{H}^+$ leads to non-uniqueness. It follows that global well-posedness of the general relativistic initial value problem completely fails in the class of positive mass Robinson–Trautman metrics.

REMARK 8.1.3 There are two striking differences between the global structure seen in Fig. 8.1.2 and the usual Penrose diagram for Schwarzschild spacetime. The first is

¹By this we mean that the metric can be C^5 extended beyond $\mathcal{H}^+$; the extension can actually be chosen to be of $C^{5,\alpha}$ -differentiability class, for any $\alpha < 1$.

²It is rather clear from the results of [144] that generic RT spacetimes will not be smoothly extendible across $\mathcal{H}^+$, without any restrictions on the ‘size’ of the initial data; but no rigorous proof is available.

the lack of past null infinity, which we have seen to be unavoidable in the RT case. The second is the lack of the past event horizon, sections of which can be technically described as a *marginally past-outer-trapped surface*. The existence of such surfaces in RT spacetimes is a non-trivial property which has been established in [447]. $\square$

8.1.2 $m < 0$

Unsurprisingly, and as already mentioned, the global structure of RT spacetimes turns out to be different when $m < 0$, which we assume now. As already noted, in this case we should take $u \leq 0$, in which case the expansion (8.1.6) again applies with $u \rightarrow -\infty$.

The existence of future null infinity as in (8.1.8) applies without further due, except that now the coordinate u belongs to $(-\infty, 0]$.

The new aspect is the possibility of attaching a conformal boundary at past null infinity, $\mathcal{I}^-$, which is carried out by first replacing u with a new coordinate v defined as [423]

$$v = u + 2r + 4m \ln \left(\left| \frac{r}{2m} - 1 \right| \right). \quad (8.1.13)$$

In the coordinate system (v, r, x^a) the metric becomes

$$g = - \left(1 - \frac{2m}{r} \right) dv^2 + 2dv dr + r^2 e^{2\lambda} \mathring{h} + \left(\frac{R}{2} - 1 + \frac{r}{12m} \Delta_h R \right) \left(dv - \frac{2dr}{1 - \frac{2m}{r}} \right)^2. \quad (8.1.14)$$

The last step is the usual replacement of r by $x = 1/r$:

$$g = x^{-2} \left[-x^2(1 - 2mx)dv^2 - 2dv dx + e^{2\lambda} \mathring{h} + \left(\frac{R - 2}{2x^2} + \frac{12m\Delta_h R}{x^3} \right) \left(x^2 dv + \frac{2dx}{1 - 2mx} \right)^2 \right]. \quad (8.1.15)$$

One notices that all terms in the conformally rescaled metric $x^2 g$ extend smoothly to smooth functions of (v, x^a) at the conformal boundary $\{x = 0\}$ except possibly for

$$\left(\frac{R - 2}{2x^2} + \frac{12m\Delta_h R}{x^3} \right) \times \left(\frac{4dx^2}{(1 - 2mx)^2} + \frac{4x^2 dv dx}{1 - 2mx} \right). \quad (8.1.16)$$

Now, from the definition of v we have

$$\begin{aligned} \exp \left(-\frac{2u}{m} \right) &= \left(\frac{r}{2m} - 1 \right)^8 \exp \left(\frac{2v - 4r}{|m|} \right) \\ &= \left(\frac{1 - 2mx}{2mx} \right)^8 \exp \left(\frac{2v}{|m|} \right) \exp \left(-\frac{4}{|m|x} \right). \end{aligned}$$

Using the fact that $\lambda = O(\exp(-2u/m))$, similarly for all angular derivatives of λ , we see that all three functions λ , $R - 2$, and $\Delta_h R$ decay to zero, as x approaches zero, faster than any negative power of x . In fact, the offending terms (8.1.16) extend smoothly by zero across $\{x = 0\}$. We conclude that the conformally rescaled metric $x^2 g$ smoothly extends to $\mathcal{I}^- := \{x = 0\}$.

Summarizing, we have the following.

THEOREM 8.1.4 *Let $m < 0$. For any $\lambda_0 \in C^\infty({}^2M)$ there exists a unique RT spacetime $({}^4\mathcal{M}, g)$ with a complete i^0 in the sense of [21], a complete $\mathcal{I}^-$, and ‘a piece of $\mathcal{I}^+$ ’, as shown in Fig. 8.1.3. Moreover:*

1. $({}^4\mathcal{M}, g)$ is smoothly extendible through $\mathcal{H}^+$, but

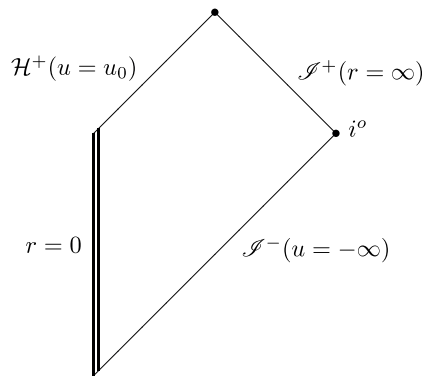


Fig. 8.1.3 A projection diagram for RT metrics with $m < 0$.

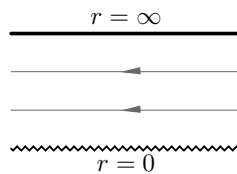


Fig. 8.1.4 The projection diagram for a Robinson–Trautman metric when $R = \mathring{R}$ is constant and the function $\mathring{\Phi}$ is negative.

2. if λ_0 is not analytic, there exist no vacuum RT extensions through $\mathcal{H}^+$.

The generic non-extendibility of the metric through $\mathcal{H}^+$ in the vacuum RT class is rather surprising, and seems to be related to a similar non-extendibility result for compact non-analytic Cauchy horizons in the polarized Gowdy class, cf. [128]. Since it may well be possible that there exist vacuum extensions which are not in the RT class, this result does not unambiguously demonstrate a failure of Einstein equations to propagate generic data forwards in u in such a situation; however, it certainly shows that the forward evolution of the metric via Einstein equations breaks down in the class of RT metrics with $m < 0$.

8.1.3 $\Lambda \neq 0$

So far we have assumed a vanishing cosmological constant. It turns out that there exists a straightforward generalization of RT metrics to $\Lambda \neq 0$. The metric retains its form (8.1.1), with the function Φ of (8.1.2) taking instead the form

$$\Phi = \frac{R}{2} + \frac{r}{12m} \Delta_h R - \frac{2m}{r} - \frac{\Lambda}{3} r^2. \quad (8.1.17)$$

We continue to assume that $m \neq 0$.

It turns out that the key equation (8.1.3) remains the same; thus λ tends to zero and Φ tends to the function

$$\mathring{\Phi} = \frac{\mathring{R}}{2} - \frac{2m}{r} - \frac{\Lambda}{3} r^2 \quad (8.1.18)$$

as u approaches infinity, for a constant $\mathring{R}$. Now, it follows from the generalized Birkhoff Theorem 4.2.3, p. 126 that a RT metric for which $\Phi \equiv \mathring{\Phi}$ as given by the last equation is a *Birmingham* metric, presented in Section 5.5, p. 250. We conclude that a RT metric with $\Lambda \neq 0$ approaches, asymptotically in u , a Birmingham metric. The projection diagrams for these last metrics can be found in Section 5.5; for the convenience of the reader we repeat them in Figs. 8.1.4–8.1.7.

The global structure of the spacetimes with $\Lambda \neq 0$ and $\lambda \neq 0$ should be clear from the analysis of the case $\Lambda = 0$: one needs to cut one of the building blocks

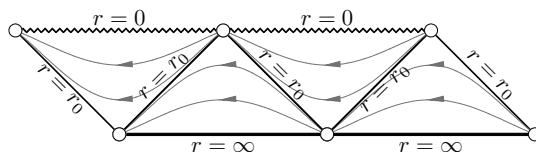


Fig. 8.1.5 The projection diagram for RT metrics with $\Lambda > 0$, constant $R \equiv \mathring{R}$, and $\dot{\Phi} \leq 0$, with $\dot{\Phi}$ vanishing precisely at r_0 .

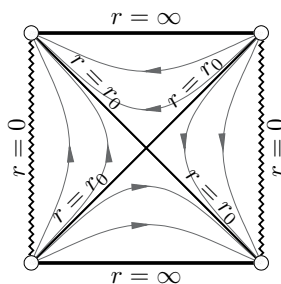


Fig. 8.1.6 The projection diagram for Robinson–Trautman metrics with $m < 0$, $\Lambda > 0$, constant $R \equiv \mathring{R}$, or $m = 0$ and $\mathring{R} = 1$, with r_0 defined by the condition $\dot{\Phi}(r_0) = 0$. The set $\{r = 0\}$ is a singularity unless the metric is the de Sitter metric (${}^2M = S^2$ and $m = 0$), or a suitable quotient thereof so that $\{r = 0\}$ corresponds to a centre of (possibly local) rotational symmetry.

of Figs. 8.1.4–8.1.7 with a line with a $\pm 45^\circ$ slope, corresponding to the initial data hypersurface $u_0 = 0$. This hypersurface should *not* coincide with one of the Killing horizons there, where $\dot{\Phi}$ vanishes. The Killing horizons with the opposite slope in the diagrams should be ignored. Depending upon the sign of m , one can evolve to the future or to the past of the associated spacetime hypersurface until a conformal boundary at infinity or a Killing horizon $\dot{\Phi}(r_0) = 0$ with the same slope is reached.

The metric will always be smoothly conformally extendable through the conformal boundaries at infinity.

As discussed in [57], the extendibility properties across the horizons which are approached as $m \times u$ tends to plus infinity will depend upon the surface gravity of the horizon and the spectrum of $\mathring{h}$. For simplicity we assume that

$${}^2M = S^2 \iff \mathring{R} > 0;$$

a similar analysis can be carried out for other topologies.

Consider, first, a zero $r = r_0$ of $\dot{\Phi}$ such that

$$c = \frac{\dot{\Phi}'(r_0)}{2} > 0.$$

Similarly to (8.1.9), introduce Kruskal–Szekeres-type coordinates $(\hat{u}, \hat{v})$ defined as

$$\hat{u} = -e^{-cu}, \quad \hat{v} = e^{c(u+2F(r))}, \quad (8.1.19)$$

where

$$F' = \frac{1}{\dot{\Phi}}. \quad (8.1.20)$$

This brings the metric to the form

$$g = -\frac{e^{-2cF(r)}\dot{\Phi}}{c^2} d\hat{u} d\hat{v} + r^2 e^{2\lambda} \mathring{h} - \frac{e^{2cu}}{c^2} \left(\underbrace{\frac{R - \mathring{R}}{2} + \frac{r\Delta_h R}{12m}}_{O(\exp(-2u/m))} \right) d\hat{u}^2. \quad (8.1.21)$$

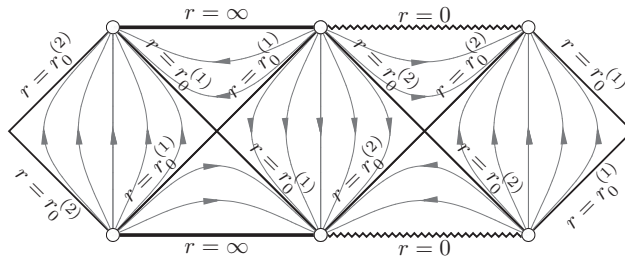


Fig. 8.1.7 The projection diagram for Robinson–Trautman metrics with $\Lambda > 0$, constant $R \equiv \mathring{R}$ and exactly two first-order zeros of $\mathring{\Phi}$.

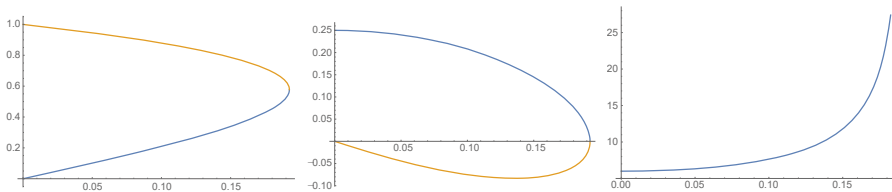


Fig. 8.1.8 The value of the real positive zero of $\mathring{\Phi}$ (left), the product $m \times c(m, \Lambda)$ (middle), and the function $2/(m \times c(m, \Lambda)) - 2$ which determines the differentiability class of the extension through the black-hole event horizon (right), as functions of m , with $\mathring{R} = 2$ and $\Lambda = 3$.

It is elementary to show that $g_{\hat{u}\hat{v}}$ extends smoothly across $\{r = r_0\}$. Next we have

$$g_{\hat{u}\hat{u}} = O\left(e^{2(c-\frac{1}{m})u}\right) = O\left(\hat{u}^{2(\frac{1}{mc}-1)}\right) \quad (8.1.22)$$

which will extend continuously across a horizon $\{\hat{u} = 0\}$ provided that

$$\frac{1}{mc} > 1 \quad \Longleftrightarrow \quad mF'(r_0) < 2. \quad (8.1.23)$$

In fact when (8.1.23) holds, then for any $\epsilon > 0$ the extension to any other RT solution will be of $C^{[2(\frac{1}{mc}-1)]-\epsilon}$ -differentiability class.

When $\Lambda > 0$, the parameter $c = c(m, \Lambda)$ can be made as small as desired by making m approach from below the critical value

$$m_c = \frac{1}{3\sqrt{\Lambda}},$$

for which c vanishes. For $m > m_c$ the function $\mathring{\Phi}$ has no (real) zeros, and for $0 < m < m_c$ all zeros are simple.

It follows from Fig. 8.1.8 that the extension through the black-hole event horizon is at least of C^6 -differentiability class, and becomes as differentiable as desired when the critical mass is approached.

The calculation above breaks down for degenerate horizons, where $m = m_c$, for which $c = 0$. In this case an extension across a degenerate horizon can be obtained by replacing u by a coordinate v defined as

$$v = u + 2F(r), \text{ with again } \frac{dF}{dr} = \frac{1}{\mathring{\Phi}}. \quad (8.1.24)$$

An explicit formula for F can be found, which is not very enlightening. Since $\mathring{\Phi}$ has now a quadratic zero, we find that for r approaching r_0 we have, after choosing an integration constant appropriately,

$$u \approx v + \frac{1}{3(r-r_0)} \implies \mathring{\Phi} \sim u^{-2} \text{ and } e^{-\frac{2u}{m}} \sim e^{-\frac{2}{3m(r-r_0)}}, \quad (8.1.25)$$

where $f_1 \sim f_2$ is used to indicate that $|f_1/f_2|$ is bounded by a positive constant from both above and below over compact intervals of v .

Using $du = dv - 2dr/\mathring{\Phi}$ we find that

$$g = -\Phi dv^2 + 2 \underbrace{\frac{2\Phi - \mathring{\Phi}}{\mathring{\Phi}}}_{1+O(\exp(-2u/m))} dv dr - 4 \underbrace{\frac{\Phi - \mathring{\Phi}}{\mathring{\Phi}^2}}_{O(u^4 \exp(-2u/m))} dr^2 + r^2 e^{2\lambda} \mathring{h}. \quad (8.1.26)$$

It easily follows that g_{vr} can be *smoothly* extended by a constant function across $r = r_0$, and that g_{rr} can be *again smoothly* extended by the constant function 0. We conclude that any RT metric with a degenerate horizon can be smoothly continued across the horizon to a Schwarzschild–de Sitter or Schwarzschild–anti-de Sitter metric with the same mass parameter m , as first observed in [57].

INCIDENTALLY: Some results on higher-dimensional generalizations of RT metrics can be found in [403].

8.2 Initial data sets with trapped surfaces

It follows from Theorem 3.3.18, p. 101, that, under appropriate global conditions, existence of a future-outer-trapped or marginally trapped surface implies that of a non-empty black-hole region. So, one strategy in constructing black-hole spacetimes is to solve the Cauchy problem with initial data which contain trapped, or marginally trapped, surfaces [10, 43, 166, 167, 228, 320, 343].

It should be emphasized that the existence of disconnected apparent horizons within an initial data set does not guarantee the existence of an event horizon with disconnected components in the associated spacetime, because our understanding of the long time behaviour of solutions of Einstein equations is way too poor. Some very partial results concerning such questions can be found in [135].

8.2.1 Brill–Lindquist initial data

Probably the simplest example of initial data containing trapped surfaces is provided by the time-symmetric initial data of Brill and Lindquist [65]. Here the space metric at time $t = 0$ takes the form

$$g = \psi^{4/(n-2)} \left((dx^1)^2 + \dots + (dx^n)^2 \right), \quad (8.2.1)$$

with

$$\psi = 1 + \sum_{i=1}^I \frac{m_i}{2|\vec{x} - \vec{x}_i|^{n-2}}.$$

The positions of the poles $\vec{x}_i \in \mathbb{R}^n$ and the values of the mass parameters $m_i \in \mathbb{R}$ are arbitrary. If all the m_i are positive and sufficiently small, then for each i there exists a small minimal surface with the topology of a sphere which encloses $\vec{x}_i$ [135]. From [289], in dimension $3+1$ the associated maximal globally hyperbolic development possesses a $\mathcal{I}^+$ which is complete to the past. However, $\mathcal{I}^+$ cannot be smooth [452], and it is not known how large it is to the future. One expects that the intersection of the event horizon with the initial data surface will have more than one connected component for sufficiently small values of $m_i/|\vec{x}_k - \vec{x}_j|$, but no proof of this exists in the literature.

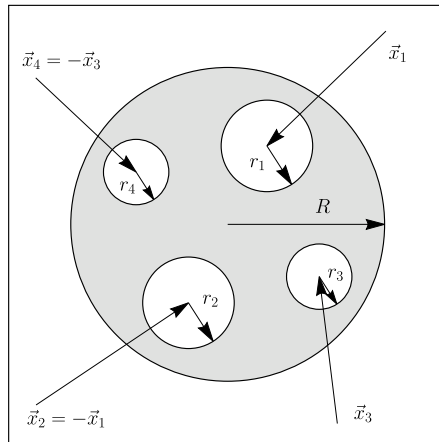


Fig. 8.2.1 ‘Many Schwarzschild’ initial data with four black holes. The initial data are *exactly* Schwarzschild within the four innermost circles and outside the outermost one. The free parameters are R , $(\vec{x}_1, r_1, m_1)$, and $(\vec{x}_3, r_3, m_3)$, with sufficiently small m_a ’s. One requires $m_2 = m_1$, $r_2 = r_1$, $m_4 = m_3$ and $r_4 = r_3$. Figure by M.Maliborski, reproduced with permission.

8.2.2 The ‘many Schwarzschild’ initial data

There is a well-known special case of (8.2.1), which is the space part of the Schwarzschild metric centred at $\vec{x}_0$ with mass m in isotropic coordinates,

$$g = \left(1 + \frac{m}{2|\vec{x} - \vec{x}_0|^{n-2}}\right)^{4/(n-2)} \delta, \quad (8.2.2)$$

where δ is the Euclidean metric. Abusing terminology, we call (8.2.2) the space Schwarzschild metric, or simply the Schwarzschild metric. The sphere $|\vec{x} - \vec{x}_0| = m/2$ is minimal, and the region $|\vec{x} - \vec{x}_0| < m/2$ corresponds to the second asymptotic region. This feature of the geometry, as connecting two asymptotic regions, is sometimes referred to as the *Einstein–Rosen bridge*; see Fig. 4.2.4, p. 145.

Now fix the radii $0 \leq 4R_1 < R_2 < \infty$. Denoting by $B(\vec{a}, R)$ the open coordinate ball centred at $\vec{a}$ with radius R , choose points

$$\vec{x}_i \in \Gamma_0(4R_1, R_2) := \begin{cases} B(0, R_2) \setminus \overline{B(0, 4R_1)}, & R_1 > 0 \\ B(0, R_2), & R_1 = 0, \end{cases}$$

and radii r_i , $i = 1, \dots, 2N$, so that the closed balls $\overline{B(\vec{x}_i, 4r_i)}$ are all contained in $\Gamma_0(4R_1, R_2)$ and are pairwise disjoint. Set

$$\Omega := \Gamma_0(R_1, R_2) \setminus \left(\cup_i \overline{B(\vec{x}_i, r_i)}\right). \quad (8.2.3)$$

We assume that the $\vec{x}_i$ and r_i are chosen so that Ω is invariant with respect to the reflection $\vec{x} \rightarrow -\vec{x}$. Consider a collection of nonnegative mass parameters, arranged into a vector as

$$\vec{M} = (m, m_0, m_1, \dots, m_{2N}),$$

where $0 < 2m_i < r_i$, $i \geq 1$, and in addition with $2m_0 < R_1$ if $R_1 > 0$ but $m_0 = 0$ if $R_1 = 0$. We assume that the mass parameters associated to the points $\vec{x}_i$ and $-\vec{x}_i$ are the same. The remaining entry m is explained below.

Using the gluing constructions developed in [115, 116, 154] one can show that there exists a number $\delta > 0$ such that if

$$\sum_{i=0}^{2N} |m_i| \leq \delta, \quad (8.2.4)$$

then there exists a number

$$m = \sum_{i=0}^{2N} m_i + O(\delta^2)$$

and a C^∞ metric $\hat{g}_{\vec{M}}$ which is a solution of the time-symmetric vacuum constraint equation

$$R(\hat{g}_{\vec{M}}) = 0,$$

such that:

1. on the punctured balls $B(\vec{x}_i, 2r_i) \setminus \{\vec{x}_i\}$, $i \geq 1$, $\hat{g}_{\vec{M}}$ is the Schwarzschild metric, centred at $\vec{x}_i$, with mass m_i ;
2. on $\mathbb{R}^n \setminus \overline{B(0, 2R_2)}$, $\hat{g}_{\vec{M}}$ agrees with the Schwarzschild metric centred at 0, with mass m ; and
3. if $R_1 > 0$, then $\hat{g}_{\vec{M}}$ agrees on $B(0, 2R_1) \setminus \{0\}$ with the Schwarzschild metric centred at 0, with mass m_0 .

By point 1 each of the spheres $|\vec{x} - \vec{x}_i| = m_i/2$ is an apparent horizon.

A key feature of those initial data is that we have complete control of the *spacetime metric* within the domains of dependence of $B(\vec{x}_i, 2r_i) \setminus \{\vec{x}_i\}$ and of $\mathbb{R}^n \setminus \overline{B(0, 2R_2)}$, where the spacetime metric is a Schwarzschild metric.

Because of the high symmetry, one expects that ‘all black holes will eventually merge’, so that the event horizon will be a connected hypersurface in spacetime.

8.2.3 Black holes and conformal gluing methods

A recent alternate technique for gluing initial data sets is given in Refs. [271, 272, 282]. In this approach, general initial data sets on compact manifolds or with asymptotically Euclidean or hyperboloidal ends are glued together to produce solutions of the constraint equations on the connected sum manifolds. Only very mild restrictions on the original initial data are needed. The connecting ‘neck’ regions produced by this construction are again of Schwarzschild type. The overall strategy of the construction is similar to that used in many previous gluing constructions in geometry. Namely, one takes a family of approximate solutions to the constraint equations and then attempts to perturb the members of this family to exact solutions. There is a parameter η which measures the size of the neck, or gluing region; the main difficulty is caused by the tension between the competing demands that the approximate solutions become more nearly exact as $\eta \rightarrow 0$ while the underlying geometry and analysis become more singular. In this approach, the conformal method of solving the constraints is used, and the solution involves a conformal factor which is exponentially close to one (as a function of η) away from the neck region. It has been shown [129] that the deformation can actually be localized near the neck in generic [42] situations.

Consider, now, an asymptotically flat time-symmetric initial data set, to which several other time-symmetric initial data sets have been glued by this method. If the gluing regions are made small enough, the existence of a non-trivial minimal surface, hence of an apparent horizon, follows by standard results. This implies the existence of a black-hole region in the maximal globally hyperbolic development of the data.

It is shown in [135] that the intersection of the event horizon with the initial data hypersurface will have more than one connected component for several families of glued initial data sets.

8.3 Black holes without Scri

8.3.1 The shortcomings of the conformal approach

The standard way of defining *non-stationary* black-hole regions is by using the conformal completions of Section 3.1, see (3.1.9), p. 88; compare Section 4.3.8, p. 166. Thus the black-hole region $\mathcal{B}$ is defined as

$$\mathcal{B} := \tilde{\mathcal{M}} \setminus J^-(\mathcal{I}^+). \quad (8.3.1)$$

This raises various questions. Our presentation in this section follows closely [94].

- **Non-equivalent Scris:** Conformal completions at null infinity are not unique in general; an example is provided by the Taub–NUT metrics [94]. This raises the question, *how many different conformal completions can a spacetime have?* Building upon an approach proposed by Geroch [220], one can prove existence and uniqueness of *maximal* strongly causal conformal completions in the smooth category, *provided there exists a non-trivial one* [105, Theorem 5.3] (compare [220, Theorem 2, p. 14]). But the existence of a non-empty strongly causal completion seems to be difficult to establish in general situations. While globally hyperbolic conformal completions (in the sense of manifolds with boundary) are necessarily strongly causal, it should be borne in mind that good causal properties of a spacetime might fail to survive the process of adding a conformal boundary.
- **Poorly differentiable Scris:** In standard treatments [220, 244, 454] it is assumed that both the conformal completion $\tilde{\mathcal{M}} = \mathcal{M} \cup \mathcal{I}$ and the extended metric $\tilde{g}$ are smooth, or have a high degree of differentiability [397]. This is a restriction which excludes most spacetimes which are asymptotically Minkowskian in lightlike directions; see [18, 92, 140, 387, 451] and references therein. In this context a breakthrough result is that of Hintz and Vasy [251], who prove that initial data which are polyhomogeneous at spatial infinity lead to spacetimes with a polyhomogeneous Scri. Here polyhomogeneity means that the fields have expansions in terms of inverse powers of a radial coordinate r and powers of $\ln r$ (thus, $r^{-i} \ln^j r$) as r goes to infinity; this should be contrasted with expansions purely in terms of inverse powers of r , which hold in spacetimes with a smooth conformal boundary at infinity. Further results concerning the existence of spacetimes with a poorly differentiable Scri can be found in [147, 318]. Low differentiability properties of $\mathcal{I}$ change the ‘peeling properties’ of the gravitational field [132]. However, essential properties of black holes should be unaffected by conformal completions with, e.g., polyhomogeneous differentiability as considered in [19, 132]. It should nevertheless be borne in mind that smoothness has been assumed in many treatments, so that in a complete theory the validity of various claims needs to be re-examined.
- **The structure of i^+ :** The current theory of black holes is entirely based on intuitions originating in the Kerr and Schwarzschild geometries. In those spacetimes we have a family of preferred ‘stationary’ observers which follow the orbits of the Killing vector field ∂_t in the asymptotic region, and their past coincides with that of $\mathcal{I}^+$. It is customary to denote by i^+ the set consisting of the points $t = \infty$, where t is the Killing time parameter for those observers. Now, the usual conformal diagrams for those spacetimes leave the highly misleading impression that i^+ is a regular point in the conformally rescaled manifold,

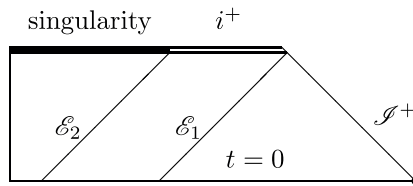


Fig. 8.3.1 An asymptotically flat spacetime with an unusual i^+ . Reprinted with permission from [94].

which is not the case. In dynamical cases the situation is likely to become worse. For example, one can imagine black-hole spacetimes with a conformal diagram which, to the future of a Cauchy hypersurface $t = 0$, looks as in Fig. 8.3.1. In that diagram the set i^+ should be thought of as the addition to the spacetime manifold $\mathcal{M}$ of a set of points $\{t = \infty, q \in \mathcal{O}\}$, where $t \in [0, \infty)$ is the proper time for a family of observers $\mathcal{O}$. The part of the boundary of $\tilde{\mathcal{M}}$ corresponding to i^+ is a singularity of the conformally rescaled metric, but we assume that it does *not* correspond to singular behaviour in the physical spacetime. In this spacetime there is the usual event horizon $\mathcal{E}_1$ corresponding to the boundary of the past of $\mathcal{S}^+$, which is completely irrelevant for the family of observers $\mathcal{O}$, and an event horizon $\mathcal{E}_2$ which is the boundary of the *true* black-hole region for the family $\mathcal{O}$; i.e., the region that is not accessible to observations for the family $\mathcal{O}$. Clearly the usual black-hole definition conveys a biased picture in this setting.

- **Causal regularity of Scri:** As already pointed out, in order to be able to prove interesting results the definition (8.3.1) should be complemented by further conditions on $\tilde{\mathcal{M}}$: see the discussion after Example 3.5.2, p. 110, or Section 3.5.1. Now, it appears reasonable to impose causal regularity conditions on a spacetime, but the physical properties of a black hole should not depend upon the causal regularity—or lack thereof—of some artificial boundary which is being attached to the spacetime. Physically motivated restrictions are relevant when dealing with physical objects; they are not when non-physical constructs are considered.
- **Inadequacy for numerical purposes:** Most numerical studies of black holes are performed on numerical grids which cover finite spacetime regions. Clearly, it would be convenient to have a setup which is more compatible in spirit with such calculations than the Scri one.

We will present, in Sections 8.3.3 and 8.4.1, two approaches in which the above listed problems are avoided. Before doing this, let us expand on the last point raised above.

8.3.2 Numerical black holes

There has been considerable progress in the numerical analysis of black-hole solutions of Einstein's equations. Here one of the objectives is to write a stable code which would solve the full four-dimensional Einstein equations, with initial data containing a non-connected black-hole region that eventually merges into a connected one. One wishes to be able to consider initial data which do not possess any symmetries, and which have various parameters—such as the masses of the individual black holes, their angular momenta, and the distances between them—which can be varied in significant ranges. Finally one wishes the code to run until the solution settles to a state close to equilibrium. The challenge then is to calculate the gravitational wave forms for each set of parameters, which could then be used in the gravitational wave observatories to determine the parameters of the collapsing black holes out of the

measured signals. This program has been undertaken for years by several groups of researchers, with steady progress being made [1, 27, 56, 59, 63, 72, 178, 229, 245, 407].

One of the challenges, which deserves further theoretical investigations, is to associate in a meaningful way global parameters such as mass, angular momentum, and the amount of energy radiated to a numerical solutions of the equations. The current solution to this problem appears to have a largely ad-hoc character.

There is a fundamental difficulty, in the context of numerical simulations, of deciding whether one is dealing with the desired black-hole initial data: the definition (8.3.1) of a black hole requires a conformal boundary $\mathcal{S}$ satisfying some—if not all—properties discussed e.g., in Section 3.5.1. Clearly there is no way of ensuring those requirements in a calculation performed on a finite spacetime grid.³

In practice what one does is to set up initial data on a finite grid so that the region near the boundary is close to flat (in the conformal approach the whole asymptotically flat region is covered by the numerical grid, and does not need to be near the boundary of the numerical grid; this distinction does not affect the discussion here). Then one evolves the initial data as long as the code allows. The gravitational waves emitted by the system are then extracted out of the metric near the boundary of the grid. Now, our understanding of energy emitted by gravitational radiation is essentially based on an analysis of the metric in an asymptotic region where g is nearly flat. In order to recover useful information out of the numerical data it is thus necessary for the metric near the boundary of the grid to remain close to a flat one. If we want to be sure that the information extracted contains all the essential dynamical information about the system, the metric near the boundary of the grid should quiet down to an almost stationary state as time evolves. Now, it is straightforward to setup a mathematical framework to describe such situations without having to invoke conformal completions; this is done in the next section.

8.3.3 Naive black holes

Consider a *globally hyperbolic* spacetime $\mathcal{M}$ which contains a region covered by coordinates (t, x^i) with ranges

$$r := \sqrt{\sum_i (x^i)^2} \geq R_0, \quad T_0 - R_0 + r \leq t < \infty, \quad (8.3.2)$$

such that the metric g satisfies there

$$|g_{\mu\nu} - \eta_{\mu\nu}| \leq C_1 r^{-\alpha} \leq C_2, \quad \alpha > 0, \quad (8.3.3)$$

for some positive constants C_1, C_2, α ; clearly, C_2 can be chosen to be less than or equal to $C_1 R_0^{-\alpha}$. Making R_0 larger one can thus make C_2 as small as needed to guarantee that objects algebraically constructed out of g (such as $g^{\mu\nu}, \sqrt{\det g}$) are well controlled. To be able to prove theorems about such spacetimes one would need to impose some further, perhaps not necessarily uniform, decay conditions on a finite number of derivatives of g ; there are various possibilities here, but we shall ignore this for the moment. Then one can define the *exterior region* $\mathcal{M}_{\text{ext}}$, the *black-hole region* $\mathcal{B}$, and the *future event horizon* $\mathcal{E}$ as

$$\mathcal{M}_{\text{ext}} := \cup_{\tau \geq T_0} J^-(\mathcal{S}_{\tau, R_0}) = J^-(\cup_{\tau \geq T_0} \mathcal{S}_{\tau, R_0}), \quad (8.3.4)$$

³The conformal approach developed by Friedrich (c.f., e.g., [197, 198] and references therein) provides an ideal numerical framework for studying gravitational radiation in situations where the extended spacetime is smoothly conformally compactifiable across i^+ , since then one can hope that the code will be able to ‘calculate $\mathcal{S}_{\text{cri}}$ ’ globally to the future of the initial hyperboloidal hypersurface. It is, however, not clear whether a conformal approach could provide more information than the non-conformal ones when i^+ is itself a singularity of the conformally rescaled equations, as is the case for black holes.

$$\mathcal{B} := \mathcal{M} \setminus \mathcal{M}_{\text{ext}}, \quad \mathcal{E} := \partial \mathcal{B}, \quad (8.3.5)$$

where

$$\mathcal{S}_{\tau, R_0} := \{t = \tau, r = R_0\}. \quad (8.3.6)$$

We will refer to the definition (8.3.2)–(8.3.6) as that of a *naive* black hole.

In the setup of Eqs. (8.3.2)–(8.3.6) an arbitrarily chosen R_0 has been used; for this definition to make sense $\mathcal{B}$ so defined should *not* depend upon this choice. This is indeed the case, as can be seen as follows.

PROPOSITION 8.3.1 *Let $\mathcal{O}_a \subset \mathbb{R}^3 \setminus B(0, R_0)$, $a = 1, 2$, and let $\mathcal{U}_a \subset \mathcal{M}$ be of the form $\{(t \geq T_0 - R_0 + r(\vec{x}), \vec{x}) \mid \vec{x} \in \mathcal{O}_a\}$ in the coordinate system of (8.3.3). Then*

$$I^-(\mathcal{U}_1) = J^-(\mathcal{U}_1) = I^-(\mathcal{U}_2) = J^-(\mathcal{U}_2).$$

Proof. If Γ is a future-directed causal path from $p \in \mathcal{M}$ to $q = (t, \vec{x}) \in \mathcal{U}_1$, then the path obtained by concatenating Γ with the path $[0, 1] \ni s \rightarrow (t(s) := t + s, \vec{x}(s) := \vec{x})$ is a causal path which is not a null geodesic; hence, it can be deformed to a timelike path from p to $(t + 1, \vec{x}) \in \mathcal{U}_1$. It follows that $I^-(\mathcal{U}_1) = J^-(\mathcal{U}_1)$; clearly the same holds for $\mathcal{U}_2$. Next, let $\vec{x}_a \in \mathcal{O}_a$, and let $\gamma : [0, 1] \rightarrow \mathbb{R}^3 \setminus B(0, R_0)$ be any differentiable path such that $\gamma(0) = \vec{x}_1$ and $\gamma(1) = \vec{x}_2$. Then for any $t_0 \geq T_0 - R_0 + r(\vec{x}_1)$ the causal curve $[0, 1] \ni s \rightarrow \Gamma(s) = (t := Cs + t_0, \vec{x}(s) := \gamma(s))$ will be causal for the metric g by (8.3.3) if the constant C is chosen large enough, with a similar result holding when $\vec{x}_1$ is interchanged with $\vec{x}_2$. The equality $I^-(\mathcal{U}_1) = I^-(\mathcal{U}_2)$ easily follows from this observation. $\square$

Summarizing, Proposition 8.3.1 shows that there are many possible equivalent definitions of $\mathcal{M}_{\text{ext}}$: in (8.3.4) one can replace $J^-(\mathcal{S}_{\tau, R_0})$ with $J^-(\mathcal{S}_{\tau, R_1})$ for any $R_1 \geq R_0$, but also simply with

$$\cup_{\tau \geq 0} J^-(t + \tau, q) = J^-(\cup_{\tau \geq 0} (t + \tau, q)),$$

for any $p = (t, q) \in \mathcal{M}$ and which belongs to the region covered by the coordinate system (t, x^i) .

The following remarks concerning (8.3.4)–(8.3.5) are in order.

- For vacuum, stationary, asymptotically flat spacetimes the definition is equivalent to the usual one with $\mathcal{I}$, as in (3.1.9), p. 88. This follows from the fact that in such a spacetime one can introduce, at a sufficiently large distance, suitably controlled coordinates satisfying (8.3.3) for all times [165]. The result then follows from [169] by constructing Bondi coordinates (u, r, θ, φ) in the asymptotic region and using $\Omega = 1/r$ as the compactifying factor. However, one does not expect the existence of a smooth $\mathcal{I}^+$ to follow from (8.3.2)–(8.3.3) in general.
- Suppose that $\mathcal{M}$ admits a conformal completion as defined in Section 3.1, and that $\mathcal{I}$ is *semi-complete to the future*, in the sense that the Geroch–Horowitz condition, presented after Example 3.5.2, p. 110, holds with $\mathcal{I} \approx \mathbb{R} \times S^2$ there replaced by $\mathcal{I} \supset \mathbb{R}^+ \times S^2$. Then for any finite interval $[T_0, T_1]$ there exists $R_0(T_0, T_1)$ and a coordinate system satisfying (8.3.3) and covering a set $r \geq R_0(T_0, T_1)$, $T_0 - R_0 \leq t \leq T_1 - R_0$. This follows from the Tamburino–Winicour construction of Bondi coordinates (u, r, θ, φ) near $\mathcal{I}^+$ [439], followed by the introduction of the usual Minkowskian coordinates $t = u + r$, $x = r \sin \theta \cos \varphi$, etc. The problem is that $R(T_1, T_2)$ could shrink to zero as T_2 goes to infinity. Thus, when $\mathcal{I}^+$ exists, conditions (8.3.2)–(8.3.3) are *uniformity* conditions on $\mathcal{I}^+$ to the future: the metric remains uniformly controlled on a uniform neighbourhood of $\mathcal{I}^+$ as the retarded time goes to infinity.
- It is likely that the future geodesically complete spacetimes of Friedrich [195, 199], Christodoulou and Klainerman [93], Lindblad and Rodnianski [325, 326]

(see also [284, 329]), Hintz and Vasy [251], or the small data black holes presented in Section 8.6, as well as the Robinson–Trautman black holes discussed in Section 8.1, p. 312, admit coordinate systems satisfying (8.3.2)–(8.3.3).

It is not clear whether asymptotically flat spacetimes in which no such control is available exist at all; in fact, it is tempting to formulate the following version of the (*weak*) *cosmic censorship* conjecture:

The maximal globally hyperbolic development of generic,⁴ asymptotically flat, vacuum initial data contains a region with coordinates satisfying (8.3.2)–(8.3.3).

Whatever the status of this conjecture, one can hardly envisage numerical simulations leading to the calculation of an essential fraction of the total energy radiated away in spacetimes in which some uniformity conditions do not hold.

8.4 Apparent horizons

In spite of their name, apparent horizons are *not* horizons. In [244] they are defined on spacelike hypersurfaces $\mathcal{S} \subset \mathcal{M}$, as follows: let $\Omega \subset \mathcal{M}$ be the set covered by (*future*) *trapped surfaces*; recall that (cf. Section 3.3.2, p. 90) these are defined as compact, boundaryless, smooth surfaces $S \subset \mathcal{S}$ with the property that

$$\theta(S) := \lambda - (g^{ij} - n^i n^j) K_{ij} < 0, \quad (8.4.1)$$

where λ is the (outward) mean extrinsic curvature of S in $\mathcal{S}$, n^i is the outward pointing unit normal to S in $\mathcal{S}$, and K_{ij} is the extrinsic curvature of $\mathcal{S}$ in $\mathcal{M}$. There is no reason for Ω to be non-empty in general; on the other hand, in appropriately censored spacetimes, a non-empty Ω implies the existence of a black-hole region. Hawking and Ellis define the apparent horizon $\mathcal{A}$ as

$$\mathcal{A} := \partial\Omega, \quad (8.4.2)$$

and argue that

$$\theta(\mathcal{A}) = 0. \quad (8.4.3)$$

However, the arguments in [244] establish (8.4.3) only if one assumes that $\mathcal{A}$ is C^2 . The problem is that $\mathcal{A}$ could be a priori a very rough set, with θ undefined in the classical sense (compare Section 2.10.3, in which case the arguments of [244] do not suffice to conclude).

INCIDENTALLY: If the extrinsic curvature K_{ij} of the initial data hypersurface $\mathcal{S}$ vanishes, then (8.4.2) is the equation for a minimal surface. Now, in the course of their proof of the Penrose inequality, G. Huisken and T. Ilmanen [266] prove that the outermost minimal (in the sense of calculus of variations) surface is always smooth, which supports the validity of (8.4.3) at least for the outermost component of $\mathcal{A}$. However, they provide examples which show that this outermost minimal surface might not be the boundary of Ω in general. $\square$

Codimension-two manifolds $\mathcal{A}$ satisfying (8.4.3) are also called *marginally outer trapped surfaces* (MOTS). We refer the reader to [123, Section 7] for a review of MOTS and their properties.

The existence of a C^2 compact, boundaryless MOTS implies the existence of a black hole in a Scri framework, assuming appropriate causal properties of $\mathcal{S}^+$; see Theorem 3.3.18, p. 101.

INCIDENTALLY: Some partial results concerning the differentiability properties of $\mathcal{A}$ have been obtained by M. Kriele and S. Hayward in [301]. R. Howard and J. Fu have shown [264] that $\partial\Omega$ satisfies (8.4.2) in a *viscosity sense*—this is defined as follows: let

⁴The examples constructed by Christodoulou [90] with spherically symmetric gravitating scalar fields suggest that the genericity condition is unavoidable, though no corresponding vacuum examples are known.

D be an open set, and let $p \in \partial D$. Then U is an inner (outer) support domain if U is an open subset of D (of $\mathbb{C}D$), ∂U is a C^2 hypersurface, and $p \in \partial U$. One says that $\theta(\partial D) \leq 0$ (respectively $\theta(\partial D) \geq 0$) in the viscosity sense if for any point $p \in \partial D$ and for any inner support (outer support) domain we have

$$\theta(\partial U)(p) \leq 0 \quad (\text{respectively } \theta(\partial U)(p) \geq 0).$$

This coincides with the usual inequality $\theta \leq 0$ ($\theta \geq 0$) on any subset of ∂D which is C^2 . Finally $\theta(\partial D) = 0$ in the viscosity sense if for any point $p \in \partial D$ we have both $\theta(\partial D) \leq 0$ in the viscosity sense and $\theta(\partial D) \geq 0$ in the viscosity sense. $\square$

A MOTS $\Sigma \subset \mathcal{S}$ is said to be outermost if Σ separates $\mathcal{S}$, and if there are no weakly trapped hypersurfaces homologous to Σ in the outer region. Building upon a closely related result due to him and Schoen [210], in [207] Galloway proves the following, which can be thought of as a higher-dimensional generalization of the Topology Theorem 3.3.1.

THEOREM 8.4.1 (Higher-dimensional horizon topology theorem [207]) *Let Σ be an outermost MOTS in a spacelike hypersurface $\mathcal{S}$, and assume that the dominant energy condition (cf. (4.3.27), p. 156) holds in a spacetime neighbourhood of Σ . Then Σ is of positive Yamabe type.*

As an immediate corollary, we obtain the following.

COROLLARY 8.4.2 *Compact cross-sections of event horizons in regular stationary black-hole spacetimes obeying the dominant energy condition are of positive Yamabe type.*

In particular, there can be no toroidal horizons in spacetimes as in the corollary.

We refer the reader to [14] for further results about MOTS, and to [15] for results concerning their evolution. Numerical treatments of apparent horizons can be found in [180, 446] and references therein.

8.4.1 Quasi-local black holes

As already argued, the naive approach of the previous section should be more convenient for numerical simulations of black-hole spacetimes, as compared to the usual one based on Scri. It appears to be even more convenient to have a framework in which all the issues are localized in space; we describe one such framework here. When numerically modelling an asymptotically flat spacetime, whether in a conformal or a direct approach, a typical numerical grid will contain large spheres $S(R)$ on which the metric is nearly flat, so that an inequality such as (8.3.3) will hold in a neighbourhood of $S(R)$. On slices $t = \text{const}$ the condition (8.3.3) is usually complemented with a fall-off condition on the derivatives of the metric

$$|\partial_\sigma g_{\mu\nu}| \leq Cr^{-\alpha-1}. \quad (8.4.4)$$

However, condition (8.4.4) is inadequate in the radiation regime, where retarded time derivatives of the metric are not expected to falloff faster than r^{-1} . It turns out that there is a condition on derivatives of the metric in null directions which is fulfilled at large distance both in spacelike and in null directions: Let K_a , $a = 1, 2$, be null future-pointing vector fields on $S(R)$ orthogonal to $S(R)$, with K_1 inward pointing and K_2 —outward pointing; these vector fields are unique up to scaling. Let χ_a denote the associated null second fundamental forms defined as

$$\forall X, Y \in TS(R) \quad \chi_a(X, Y) := g(\nabla_X K_a, Y). \quad (8.4.5)$$

It can be checked, e.g. using the asymptotic expansions for the connection coefficients near $\mathcal{S}^+$ from [130, Appendix C], that χ_1 is negative definite and χ_2 is positive definite for Bondi spheres $S(R)$ sufficiently close to $\mathcal{S}^+$; similarly for $\mathcal{S}^-$.

An alternative calculation proceeds along the lines of (3.3.27), p. 103. This property is not affected by the rescaling freedom at hand. Following Galloway [204], a two-dimensional spacelike submanifold of a four-dimensional spacetime will be called *weakly null convex* if χ_1 is semi-positive definite, with the trace of χ_2 negative.⁵ The null convexity condition is easily verified for sufficiently large spheres in a region asymptotically flat in the sense of (8.4.4). It does also hold for large spheres in a large class of spacetimes with a negative cosmological constant. The null convexity condition is then the condition which we propose as a starting point to defining ‘quasi-local’ black holes and horizons. The point is that several of the usual properties of black holes carry over to the weakly null convex setting. In retrospect, the situation can be summarized as follows: the usual theory of Scri-based black holes exploits the existence of conjugate points on appropriate null geodesics whenever those are complete to the future; this completeness is guaranteed by the fact that the conformal factor goes to zero at the conformal boundary at an appropriate rate. Galloway’s discovery in [204] is that weak null convexity of large spheres near Scri provides a second, in principle completely independent, mechanism to produce the needed focusing behaviour.

In what follows we will consider a globally hyperbolic spacetime $(\mathcal{M}, g)$ equipped with a time function t . Let $\mathcal{T} \subset \mathcal{M}$ be a finite union of connected timelike hypersurfaces $\mathcal{T}_\alpha$ in $\mathcal{M}$. We set

$$\mathcal{S}_\tau := \{t = \tau\}, \quad \mathcal{T}(\tau) := \mathcal{T} \cap \mathcal{S}_\tau, \quad \mathcal{T}_\alpha(\tau) := \mathcal{T}_\alpha \cap \mathcal{S}_\tau. \quad (8.4.6)$$

For further purposes anything that happens on the exterior side of $\mathcal{T}$ is completely irrelevant, so it is convenient to think of $\mathcal{T}$ as a boundary of $\mathcal{M}$; global hyperbolicity should then be understood in the sense that $(\overline{\mathcal{M}} := \mathcal{M} \cup \mathcal{T}, g)$ is strongly causal, and that $J^+(p; \overline{\mathcal{M}}) \cap J^-(q; \overline{\mathcal{M}})$ is compact in $\mathcal{M}$ for all $p, q \in \mathcal{M}$.

One can also think of each $\mathcal{T}_\alpha$ as a family of observers.

Recall that the null convergence condition is the requirement that

$$\text{Ric}(X, X) \geq 0 \quad \text{for all null vectors } X \in TM. \quad (8.4.7)$$

The following *topological censorship* theorem of Galloway [204] is a special case of Theorem 3.3.4, p. 92.

THEOREM 8.4.3 *Suppose that a globally hyperbolic spacetime $(\overline{\mathcal{M}}, g)$ satisfying the null convergence condition (8.4.7) has a timelike boundary $\mathcal{T} = \cup_{\alpha=1}^I \mathcal{T}_\alpha$ and a time function t such that the level sets of t are Cauchy surfaces, with each section $\mathcal{T}(\tau)$ of $\mathcal{T}$ being null convex. Then distinct $\mathcal{T}_\alpha$ ’s cannot communicate with each other:*

$$\alpha \neq \beta \quad J^+(\mathcal{T}_\alpha) \cap J^-(\mathcal{T}_\beta) = \emptyset.$$

Theorem 8.4.3 and arguments familiar from Section 3.3 lead to the following.

THEOREM 8.4.4 (Galloway [204]) *Under the hypotheses of Theorem 8.4.3 suppose further that the cross-sections $\mathcal{T}_\alpha(\tau)$ of $\mathcal{T}_\alpha$ have spherical topology.⁶ Then the α -domain of outer communication*

$$\langle\langle \mathcal{T}_\alpha \rangle\rangle := J^+(\mathcal{T}_\alpha) \cap J^-(\mathcal{T}_\alpha) \quad (8.4.8)$$

is simply connected.

⁵Galloway defines *null convexity* through the requirement of positive definiteness of χ_1 and negative definiteness of χ_2 . However, he points out himself [204, p. 1472] that the weak null convexity as defined above suffices for his arguments to go through.

⁶The reader is referred to [209] and references therein for results without the hypothesis of spherical topology. The results there, presented in a Scri setting, generalize immediately to the weakly null convex one.

It follows in particular from Theorem 8.4.4 that $\overline{\mathcal{M}}$ can be replaced by a subset thereof such that $\mathcal{T}$ is connected in the new spacetime, with all essential properties relevant for the discussion in the remainder of this section being unaffected by that replacement. We shall not do that, to avoid a lengthy discussion of which properties are relevant and which are not, but the reader should keep in mind that the hypothesis of connectedness of $\mathcal{T}$ can indeed be done without any loss of generality for most purposes.

We define the *quasi-local black-hole region* $\mathcal{B}_{\mathcal{T}_\alpha}$ and the *quasi-local event horizon* $\mathcal{E}_{\mathcal{T}_\alpha}$ associated with the hypersurface $\mathcal{T}_\alpha$ by

$$\mathcal{B}_{\mathcal{T}_\alpha} := \mathcal{M} \setminus J^-(\mathcal{T}_\alpha), \quad \mathcal{E}_{\mathcal{T}_\alpha} := \partial \mathcal{B}_{\mathcal{T}_\alpha}. \quad (8.4.9)$$

If $\mathcal{T}$ is the hypersurface $\cup_{\tau \geq T_0} \mathcal{S}_{\tau, R_0}$ of Section 8.3.3, then the resulting black-hole region coincides with that defined in (8.3.5); hence, it does not depend upon the choice of R_0 by Proposition 8.3.1. However, $\mathcal{B}_{\mathcal{T}_\alpha}$ might depend upon the chosen family of observers $\mathcal{T}_\alpha$ in general. It is certainly necessary to impose some further conditions on $\mathcal{T}$ to reduce this dependence. A possible condition, suggested by the geometry of the large coordinate spheres considered in the previous section, could be that the light cones of the induced metric on $\mathcal{T}$ are uniformly controlled from both outside and inside by those of two static, future complete reference metrics on $\mathcal{T}$. However, neither the results above nor the results that follow require that condition.

The Scri equivalents of Theorem 8.4.4 [67, 104, 124, 149, 203, 209, 212, 277] allow one to control the topology of ‘good’ sections of the horizon, and for the standard stationary black holes this does lead to the usual $S^2 \times \mathbb{R}$ topology of the horizon, cf. Theorem 3.3.1, p. 90. In particular, in stationary, asymptotically flat, appropriately regular spacetimes the intersection of a partial Cauchy hypersurface with an event horizon will necessarily be a finite union of spheres. In general spacetimes such intersections do not even need to be manifolds: for example, in the usual spherically symmetric collapsing star the intersection of the event horizon with level sets of a time function will be a point at the time of appearance of the event horizon. We refer the reader to [117, Section 3] for other such examples, including one in which the topology of sections of horizon changes type from toroidal to spherical as time evolves. This behaviour can be traced back to the existence of past end points of the generators of the horizon. Nevertheless, some sections of the horizon have controlled topology. For instance, we have the following version of Theorem 3.3.1.

THEOREM 8.4.5 *Under the hypotheses of Theorem 8.4.3, consider a connected component $\mathcal{T}_\alpha$ of $\mathcal{T}$ such that $\mathcal{E}_{\mathcal{T}_\alpha} \neq \emptyset$. Let*

$$\mathcal{C}_\alpha(\tau) := \partial J^+(\mathcal{T}_\alpha(\tau)).$$

If $\mathcal{C}_\alpha(\tau) \cap \mathcal{E}_{\mathcal{T}_\alpha}$ is a topological manifold, then each connected component thereof has spherical topology.

Proof. Consider the open subset $\mathcal{M}_\tau$ of $\overline{\mathcal{M}}$ defined as

$$\mathcal{M}_\tau := I^+(\mathcal{C}_\alpha(\tau); \overline{\mathcal{M}}) \cap I^-(\mathcal{T}_\alpha; \overline{\mathcal{M}}) \subset \langle \langle \mathcal{T}_\alpha \rangle \rangle.$$

We claim that $(\mathcal{M}_\tau, g|_{\mathcal{M}_\tau})$ is globally hyperbolic: indeed, let $p, q \in \mathcal{M}_\tau$; global hyperbolicity of $\overline{\mathcal{M}}$ shows that $J^-(p; \overline{\mathcal{M}}) \cap J^+(q; \overline{\mathcal{M}})$ is a compact subset of $\overline{\mathcal{M}}$, which is easily seen to be included in $\mathcal{M}_\tau$. It follows that $J^-(p; \mathcal{M}_\tau) \cap J^+(q; \mathcal{M}_\tau)$ is compact, as desired. By the usual decomposition we thus have

$$\mathcal{M}_\tau \approx \mathbb{R} \times \mathcal{S},$$

where $\mathcal{S}$ is a Cauchy hypersurface for $\mathcal{M}_\tau$. Applying Theorem 8.4.4 to the globally hyperbolic spacetime $\mathcal{M}_\tau$ (which has a weakly null convex boundary $\mathcal{T}_\alpha \cap \{t > \tau\}$)

one finds that $\mathcal{M}_\tau$ is simply connected, and thus so is $\mathcal{S}$. Since $\mathcal{C}_\alpha(\tau)$ and $\mathcal{E}_\alpha$ are null hypersurfaces in $\mathcal{M}$, it is easily seen that the closure in $\overline{\mathcal{M}}$ of the Cauchy surface $\{0\} \times \mathcal{S}$ intersects $\mathcal{E}_\alpha$ precisely at $\mathcal{C}_\alpha(\tau) \cap \mathcal{E}_{\mathcal{T}_\alpha}$. It follows that $\mathcal{S}$ is a compact, simply connected, three-dimensional topological manifold with boundary, and a classical result [246, Lemma 4.9] shows that each connected component of $\partial\mathcal{S}$ is a sphere. The result follows now from $\partial\mathcal{S} \approx \mathcal{C}_\alpha(\tau) \cap \mathcal{E}_{\mathcal{T}_\alpha}$. $\square$

Yet another class of ‘good sections’ of $\mathcal{E}_{\mathcal{T}}$ can be characterized⁷ as follows: suppose that $\langle\langle\mathcal{T}_\alpha\rangle\rangle \cap \mathcal{S}_\tau$ is a submanifold with boundary of $\mathcal{M}$ which is, moreover, a retract of $\langle\langle\mathcal{T}_\alpha\rangle\rangle$. Then $\langle\langle\mathcal{T}_\alpha\rangle\rangle \cap \mathcal{S}_\tau$ is simply connected by Theorem 8.4.4, and spherical topology of all boundary components of $\langle\langle\mathcal{T}_\alpha\rangle\rangle \cap \mathcal{S}_\tau$ follows again from [246, Lemma 4.9]. It is not clear whether there always exist time functions t such that the retract condition is satisfied; similarly it is not clear that there always exist τ ’s for which the conditions of Theorem 8.4.5 are met for metrics which are not stationary (one would actually want ‘a lot of τ ’s’). It would be of interest to understand this better.

We have an area theorem for $\mathcal{E}_{\mathcal{T}}$.

THEOREM 8.4.6 *Under the hypotheses of Theorem 8.4.3, suppose further that $\mathcal{E}_{\mathcal{T}} \neq \emptyset$. Let $\mathcal{S}_a$, $a = 1, 2$ be two achronal spacelike embedded hypersurfaces of C^2 differentiability class; set $S_a = \mathcal{S}_a \cap \mathcal{E}_{\mathcal{T}}$. Then we have the following:*

1. *The area of S_a is well defined.*
2. *If*

$$S_1 \subset J^-(S_2) ,$$

then the area of S_2 is larger than or equal to that of S_1 . (Moreover, this is true even if the area of S_1 is counted with multiplicity⁸ of generators provided that $S_1 \cap S_2 = \emptyset$.)

Sketch of proof. The result is obtained by a mixture of methods of [117] and of [204], and proceeds by contradiction: assume that the Alexandrov divergence $\theta_{\mathcal{A}l}$ of $\mathcal{E}_{\mathcal{T}}$ is negative, and consider the $S_{\epsilon,\eta,\delta}$ deformation of the horizon as constructed in Proposition 4.1 of [117], with parameters chosen so that $\theta_{\epsilon,\eta,\delta} < 0$. Global hyperbolicity implies the existence of an achronal null geodesic from $S_{\epsilon,\eta,\delta}$ to some cut $\mathcal{T}(\tau)$ of $\mathcal{T}$. The geodesic can further be chosen to be ‘extremal’, in the sense that it meets $\mathcal{T}(t)$ for the smallest possible value of t among all generators of the boundary of $J^+(S_{\epsilon,\eta,\delta})$ meeting $\mathcal{T}$. The argument of the proof of Theorem 1 of [204] shows that this is incompatible with the null energy condition and with the weak null convexity of $\mathcal{T}(\tau)$. It follows that $\theta_{\mathcal{A}l} \geq 0$, and the result follows from [117, Proposition 3.3 and Theorem 6.1]. $\square$

It immediately follows from the proof above that, under the hypotheses of Theorem 8.4.3, the occurrence of twice-differentiable future-trapped (compact) surfaces implies the presence of a black-hole region. The same result holds for semi-convex compact surfaces trapped in an Alexandrov sense (that is, (8.4.1) holds with λ there defined in a way which should be clear from the discussion following Theorem 2.10.24). The proof that the existence of marginally trapped surfaces, defined in a classical sense, signals the occurrence of black holes, requires considerably more work; see Theorem 3.3.18, p. 101.

In summary, we have shown that the quasi-local black holes, defined using weakly null convex timelike hypersurfaces, or boundaries, possess several properties usually

⁷I am grateful to G. Galloway for useful discussions concerning this question, as well as many other points presented in this section.

⁸See [117] for details.

associated with the Scri-based black holes, without the associated problems. We believe they provide a reasonable alternative, well suited for numerical calculations.

8.5 Christodoulou's trapped surfaces

In Minkowski spacetime, outgoing light fronts emanating from a round sphere increase in area, while ingoing ones decrease. Clearly, something similar will hold for weak gravitational fields. A trapped surface, as defined in Section 3.3.2, p. 90, will have radically different properties: roughly speaking, it is a surface from which both ingoing and outgoing light fronts decrease in area.

Penrose has shown (see Theorem 3.4.2, p. 109) that, under natural hypotheses, the existence of trapped surfaces leads to geodesic incompleteness; hence, the interest in understanding the formation of such surfaces. Furthermore, the existence of a trapped surface will sometimes signal the presence of a black hole, though this requires various supplementary conditions; see Theorem 3.3.18.

The monumental book of Christodoulou [89] is devoted to the proof of the theorem that sufficiently focused 'short pulse' initial data on a light cone lead to the formation of a trapped surface within their Cauchy development.

The approach is based on the characteristic Cauchy problem, where the initial data are prescribed on a light cone of a point. Christodoulou considers a family of initial data governed by a small parameter δ , which can be thought of as the duration of a burst of intense gravitational waves. The initial data are taken to be Minkowskian near the tip, which sidesteps difficulties occurring there [108]. The intensity of the waves increases, in a specific way, as δ goes to 0. This setup is referred to as the 'short pulse method'. Most of the book consists in a formidable argument showing that a certain set of inequalities can be 'bootstrapped' when the vacuum Einstein equations hold. It follows that any solution of the vacuum Einstein equations with initial data satisfying the bootstrap conditions will also satisfy the bootstrap inequalities. This eventually leads to the existence of trapped surfaces in the region where the inequalities hold. The transition from initial data on a light cone in spacetime to initial data on past null infinity is obtained by receding to past timelike infinity with the tip of the light cone.

All this results in the following.

THEOREM 8.5.1 (Christodoulou) *Let k, ℓ be positive constants, $k > 1$, $\ell < 1$. Let us be given smooth asymptotic initial data at past null infinity which is trivial at early times. Suppose that the incoming energy per unit solid angle in each direction in the advanced time interval $[0, \delta]$ is not less than $k/8\pi$. Then if δ is suitably small, the maximal development of the data contains a closed trapped surface S with area larger than or equal to $4\pi\ell^2$.*

A more detailed overview of Christodoulou's work can be found in [158].

The proof of Theorem 8.5.1 is truly visionary, of terrifying complexity, opening new perspectives in the analysis of the Einstein equations.

The theorem has been followed up by studies reducing the number of derivatives needed, or proposing alternative trapping conditions [12, 288, 292, 293]. The result has been generalized to Einstein–Maxwell equations in [462].

From the perspective of this book, a key addition to the theorem is due to Li and Mei [322], who show how to adapt Christodoulou's construction to obtain black-hole spacetimes containing an asymptotically flat Cauchy surface and a complete future null infinity. This improves upon previous related work of Li and Yu [324], where black-hole spacetimes with a past-complete piece of future null infinity were constructed.

8.6 Small perturbations of the Schwarzschild metric

Since the pioneering work of Christodoulou and Klainerman on the stability of Minkowski spacetime, many researchers have been looking into ways to address the question of stability of Kerr black holes. The first naive guess would be to study the stability of Schwarzschild black holes, but those cannot be stable since a generic small perturbation will introduce angular momentum. The strategy to solve the problem is to study, as a first step, linear wave equations on black-hole backgrounds, with the hope that sufficiently robust linear decay estimates can be ‘bootstrapped’ to produce the proof of nonlinear stability.

INCIDENTALLY: A sample of papers on the linear problem includes [17, 60, 61, 162–164, 179, 267, 281, 341, 441, 442]; see also [192] and references therein. $\square$

The first nonlinear result in a related spirit was obtained by Dafermos, Holzegel, and Rodnianski. In [160] the authors establish the existence of a large class of dynamical vacuum black-hole spacetimes whose exterior geometry asymptotes in time to a fixed Schwarzschild or Kerr metric. The spacetimes are constructed by solving a backwards scattering problem for the vacuum Einstein equations with characteristic data prescribed on the event horizon and at null infinity. The solutions are parameterized by scattering data, and the class admits the full functional degrees of freedom to specify data for the Einstein equations. The resulting spacetime $(\mathcal{M}, g)$ contains a hyperboloidal hypersurface extending to the conformal boundary at infinity; one expects that $(\mathcal{M}, g)$ can be continued backwards so that it contains an asymptotically flat initial data hypersurface, but this has not been established so far. There is another catch, that the scattering data considered decay exponentially fast, which is not expected for general such solutions, where polynomial decay is anticipated along $\mathcal{I}^+$. As a result, the solutions converge to stationarity exponentially fast, with their rate of decay being related to the surface gravity of the event horizon.

Whatever the minor shortcomings, this was until 2017 the only construction of a family of black-hole solutions parameterized by the expected number of functions, without imposing an ansatz on the form of the metric, as in the case of the Robinson–Trautman metrics of Section 8.1 or as in [58, 159], or assuming symmetries as, e.g., in [91].

The two next breakthrough results concern the stability of the characteristic Cauchy problem for the Schwarzschild metric. The idea is to construct spacetimes covered by two transverse foliations by null surfaces, leading to a metric of the form

$$g = -4\Omega^2 du dv + g_{AB}(d\theta^A - b^A dv)(d\theta^B - b^B dv). \quad (8.6.1)$$

For example, the Schwarzschild metric can be written in this form,

$$g = -\left(1 - \frac{2m}{r}\right) du dv + r^2 d\Omega^2, \quad (8.6.2)$$

as seen in (4.2.25), p. 133. Metrics in the *doubly null gauge* of (8.6.1) can be constructed by prescribing, e.g., g_{AB} on $\{u = 0\}$ as well as on $\{v = 0\}$, together with Ω and the curl of the one-form $b_A dx^A$ on the sphere $\{u = v = 0\}$; see, e.g., [73, 139].

One has the following result, reported in [161].

THEOREM 8.6.1 (Dafermos, Holzegel, Rodnianski and Taylor) *There exists a submanifold of codimension-three within the moduli space of vacuum general relativistic characteristic initial data such that the following holds: for all initial data in this submanifold which are sufficiently close to the Schwarzschild ones, the maximal globally hyperbolic development $(\mathcal{M}, g)$ contains a region $\mathcal{R}$ covered by a doubly null foliation such that:*

1. $\mathcal{M}$ admits a conformal completion at infinity $\mathcal{I}^+$, with $\mathcal{I}^+$ complete to the future;
2. such that $\mathcal{R} \subset J^-(\mathcal{I}^+)$, with a smooth, affine complete, event horizon $J^-(\mathcal{I}^+) \subset \overline{\mathcal{R}}$; and
3. with the metric asymptoting, inverse polynomially, to a nearby Schwarzschild metric on $\mathcal{R}$.

The submanifold, alluded to above, consists of those data for which the total Arnowitt–Deser–Misner (ADM) angular momentum vector tends to zero as one recedes to infinity in u . The codimension of this manifold is thus determined by the three components of the ADM angular momentum vector. The precise property is that every nearby three-dimensional plane intersects a data set evolving to Schwarzschild. Solutions with initial data which are *not* in the submanifold of the theorem do *not* converge to a Schwarzschild solution. Note that the ADM angular momentum is not necessarily vanishing for elements of this subset as nontrivial angular momentum is in general radiated to null infinity. Thus, the vanishing of the ADM angular momentum is neither necessary nor sufficient to be in the subset.

The next breakthrough result, of Klainerman and Szeftel, established by different techniques, is concerned with a class of perturbations of the Schwarzschild initial data which guarantee a priori that the metric will asymptote to a Schwarzschild metric. Namely, in [295] one considers axisymmetric perturbations satisfying a polarization condition. Axisymmetry guarantees that the total angular momentum is conserved, so that if the solution tends to a Kerr one, and if the initial data have vanishing angular momentum, then the end solution must be the Schwarzschild solution. The polarization condition is the requirement of the existence of a cylindrical-type coordinate system (t, ρ, φ, z) in which the metric can be written as

$$g = g_{\varphi\varphi}d\varphi^2 + g_{ab}dx^a dx^b, \quad \partial_\varphi g_{\mu\nu} = 0, \quad (8.6.3)$$

where $(x^a) = (t, \rho, z)$. In other words, $g_{\varphi\mu} = 0$ for $\mu \neq \varphi$, with the Killing vector field being ∂_φ .

The properties of the space of solutions are very similar to those of Theorem 8.6.1.

THEOREM 8.6.2 (Klainerman and Szeftel [295]) *The maximal globally hyperbolic development of an axially symmetric, polarized, vacuum, characteristic initial data sufficiently close to Schwarzschild data with positive mass admits a conformal completion $\mathcal{I}^+$ which is complete to the future, with the metric converging on $J^-(\mathcal{I}^+)$ to a nearby Schwarzschild solution.*

The reader is referred to [295] for the precise definition of closedness. A detailed description of the rates of approach of the metric to the final end state can be found in [295, Section 3.3.2].

Both theorems above concern the characteristic initial value problem. The drawback is that such data can only be evolved to the future. There is an elegant procedure, essentially due to Corvino and Schoen [154, 156], which allows one to associate in a simple way many such initial data to initial data which are asymptotically flat in spacelike directions; see also [115, 116, 155, 171, 172]. In these references it is shown how to construct large families of asymptotically flat vacuum initial data which are exactly stationary outside of a compact set. The procedure includes initial data sets which are arbitrarily small perturbations of the Schwarzschild metric or of the Kerr metric. The maximal globally hyperbolic development of such data will contain characteristic initial data surfaces with non-trivial data, as needed for Theorems 8.6.1 and 8.6.2.

INCIDENTALLY: It is expected, in the fall of 2019, that the above results will be generalized in a near future to show that small perturbations of Kerr metrics lead to spacetimes in which the metric asymptotes to a nearby Kerr one. $\square$

The question, what happens under general perturbations of the Kerr metrics, appears to be completely out of reach of the existing mathematical techniques.

Appendices

In this set of appendices we present auxiliary material useful in the book. Appendices A and B shortly review the notions of Lie and covariant derivatives. Appendix C contains proofs of the basic identities satisfied by the curvature tensor. A short overview of differential forms is provided in Appendix D. Appendix E discusses the algebraic properties of null hyperplanes, as needed for a proper understanding of the geometry of null hypersurfaces, which in turn is presented in Appendix F. Appendix G lists the basic facts concerning the general relativistic Cauchy problem. Appendix H contains a collection of identities often used in explicit calculations.

A The Lie derivative

We give a short overview of the definition of the Lie derivative and its properties. An extended introduction to the notion can be found in e.g. [110, Section 6.1].

Here, and elsewhere, vector fields are identified with *linear, homogeneous, first-order partial differential operators on functions*. In local coordinates, a vector field X is represented by the partial differential operator $X^i \partial_i$, acting on functions in the obvious way:

$$X(f) := X^i \frac{\partial f}{\partial x^i}, \quad (\text{A.1})$$

which we often write as $X^i \partial_i f$.

Given two vector fields X, Y and a function f , their *commutator* is defined as

$$[X, Y](f) := X(Y(f)) - Y(X(f)). \quad (\text{A.2})$$

We have the *Jacobi identity*: for any triple of vector fields X, Y, Z it holds that

$$[Z, [Y, X]] + [X, [Z, Y]] + [Y, [X, Z]] = 0. \quad (\text{A.3})$$

Given a vector field X , the *Lie derivative* $\mathcal{L}_X$ is an operation on tensor fields, defined as follows.

For a function f , one sets

$$\mathcal{L}_X f := X(f). \quad (\text{A.4})$$

For a vector field Y , the Lie derivative coincides with the Lie bracket:

$$\mathcal{L}_X Y := [X, Y]. \quad (\text{A.5})$$

For a one-form α , $\mathcal{L}_X \alpha$ is defined by imposing the Leibniz rule written the wrong-way round:

$$(\mathcal{L}_X \alpha)(Y) := \mathcal{L}_X(\alpha(Y)) - \alpha(\mathcal{L}_X Y). \quad (\text{A.6})$$

For tensor products, the Lie derivative is defined by imposing linearity under addition together with the Leibniz rule:

$$\mathcal{L}_X(\alpha \otimes \beta) = (\mathcal{L}_X \alpha) \otimes \beta + \alpha \otimes \mathcal{L}_X \beta.$$

Since a general tensor A is a sum of tensor products,

$$A = A^{a_1 \dots a_p}_{b_1 \dots b_q} \partial_{a_1} \otimes \dots \otimes \partial_{a_p} \otimes dx^{b_1} \otimes \dots \otimes dx^{b_q},$$

the above requirements provide a definition of the Lie derivative for any tensor.

A useful property of Lie derivatives is

$$\mathcal{L}_{[X,Y]} = [\mathcal{L}_X, \mathcal{L}_Y], \quad (\text{A.7})$$

where, for a tensor T , the commutator $[\mathcal{L}_X, \mathcal{L}_Y]T$ is defined in the usual way:

$$[\mathcal{L}_X, \mathcal{L}_Y]T := \mathcal{L}_X(\mathcal{L}_Y T) - \mathcal{L}_Y(\mathcal{L}_X T). \quad (\text{A.8})$$

B Covariant derivatives

We review briefly the notion of a covariant derivative.

A *covariant derivative* is a map which to a vector field X and a tensor field T assigns a tensor field of the same type as T , denoted by $\nabla_X T$, with the following properties:

1. $\nabla_X T$ is linear with respect to addition with respect to both X and T :

$$\nabla_{X+Y} T = \nabla_X T + \nabla_Y T, \quad \nabla_X(T+Y) = \nabla_X T + \nabla_X Y; \quad (\text{B.1})$$

2. $\nabla_X T$ is linear with respect to multiplication of X by functions f ,

$$\nabla_{fX} T = f \nabla_X T; \quad (\text{B.2})$$

3. and, finally, $\nabla_X T$ satisfies the *Leibniz rule* under multiplication of T by a differentiable function f :

$$\nabla_X(fT) = f \nabla_X T + X(f)T. \quad (\text{B.3})$$

By definition, if T is a tensor field of rank (p, q) , i.e., p -contravariant and q -covariant, then for any vector field X the field $\nabla_X T$ is again a tensor of rank (p, q) . Since $\nabla_X T$ is linear in X , the field ∇T can be viewed as a tensor field of rank $(p, q+1)$.

It is natural to ask whether covariant derivatives do exist at all in general and, if so, how many of them can there be. First, it immediately follows from the axioms above that if D and ∇ are two covariant derivatives, then

$$\Delta(X, T) := D_X T - \nabla_X T$$

is multi-linear with respect to both addition and multiplication by functions, since the non-homogeneous terms $X(f)T$ in (B.3) cancel. So Δ is a tensor field. Reciprocally, if ∇ is a covariant derivative and $\Delta(X, T)$ is bilinear with respect to addition and multiplication by functions, then

$$D_X T := \nabla_X T + \Delta(X, T) \quad (\text{B.4})$$

is a new covariant derivative. So, at least locally, there are as many covariant derivatives on tensors of valence (r, s) as tensors of valence $(r+s, r+s+1)$.

We note that the sum of two covariant derivatives is *not* a covariant derivative. However, *convex* combinations of covariant derivatives, with coefficients which may vary from point to point, are again covariant derivatives. This remark allows one to construct covariant derivatives using partitions of unity: Let, indeed, $\{\mathcal{O}_i\}_{i \in \mathbb{N}}$ be

an open covering of M by coordinate patches and let φ_i be an associated partition of unity. In each of those coordinate patches we can decompose a tensor field T as

$$T = T^{i_1 \dots i_k}_{j_1 \dots j_\ell} \partial_{i_1} \otimes \dots \otimes \partial_{i_k} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_\ell}, \quad (\text{B.5})$$

and define

$$D_X T := \sum_i \varphi_i X^j \partial_j (T^{i_1 \dots i_k}_{j_1 \dots j_\ell}) \partial_{i_1} \otimes \dots \otimes \partial_{i_k} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_\ell}. \quad (\text{B.6})$$

This procedure, which depends upon the choice of the coordinate patches and the choice of the partition of unity, defines *one* covariant derivative; all other covariant derivatives are then obtained from D using (B.4).

The *canonical covariant derivative on functions* is defined as

$$\nabla_X f := X(f),$$

and we will always use the above.

Let ∇ be a covariant derivative defined for vector fields; the *torsion tensor* T is defined by the formula

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y], \quad (\text{B.7})$$

where $[X, Y]$ is the Lie bracket.

Suppose that we are given a covariant derivative on vector fields. There is then a natural way of inducing a covariant derivative on one-forms by imposing the condition that *the duality operation be compatible with the Leibniz rule*: given two vector fields X and Y together with a field of one-forms α , one sets

$$(\nabla_X \alpha)(Y) := X(\alpha(Y)) - \alpha(\nabla_X Y). \quad (\text{B.8})$$

One extends ∇ to tensors of arbitrary valence as follows: if T is r covariant and s contravariant one defines

$$\begin{aligned} (\nabla_X T)(X_1, \dots, X_r, \alpha_1, \dots, \alpha_s) &:= X(T(X_1, \dots, X_r, \alpha_1, \dots, \alpha_s)) \\ &\quad - T(\nabla_X X_1, \dots, X_r, \alpha_1, \dots, \alpha_s) - \dots - T(X_1, \dots, \nabla_X X_r, \alpha_1, \dots, \alpha_s) \\ &\quad - T(X_1, \dots, X_r, \nabla_X \alpha_1, \dots, \alpha_s) - \dots - T(X_1, \dots, X_r, \alpha_1, \dots, \nabla_X \alpha_s). \end{aligned} \quad (\text{B.9})$$

One of the fundamental results in pseudo-Riemannian geometry is the existence of a torsion-free connection which preserves the metric, called the *Levi-Civita connection*.

THEOREM B.1 *Let g be a two-covariant symmetric non-degenerate tensor field on a manifold M . Then there exists a unique connection ∇ such that*

1. $\nabla g = 0$, and
2. *the torsion tensor T of ∇ vanishes.* □

Unless explicitly indicated otherwise, the Levi-Civita connection is used throughout this book, in particular in what follows.

The following result is useful for calculations.

PROPOSITION B.2 *Let g be a Lorentzian metric.*

1. *For every $p \in M$ there exists a neighbourhood thereof with a coordinate system such that $g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, 1, \dots, 1)$ at p .*
2. *If g is differentiable, then the coordinates can be further chosen so that*

$$\partial_\sigma g_{\alpha\beta} = 0 \quad (\text{B.10})$$

at p . Equivalently, the Christoffel symbols associated with the Levi-Civita connection vanish at p . □

Normal coordinates centred at a point p , which are obtained by shooting geodesics from p , provide an example of coordinates as above. The following gives necessary and sufficient condition for a coordinate system to be normal.

PROPOSITION B.3 (Thomas [445]) *Let $\{x^\mu\}$ be a local coordinate system defined on a star-shaped domain containing the origin. The following conditions are equivalent:*

1. For every $a^\mu \in \mathbb{R}^n$ the rays $s \rightarrow sa^\mu$ are geodesics;
2. $\Gamma^\mu_{\alpha\beta}(x)x^\alpha x^\beta = 0$;
3. $\frac{\partial g_{\gamma\alpha}}{\partial x^\beta}(x)x^\alpha x^\beta = 0$;
4. $g_{\alpha\beta}(x)x^\beta = g_{\alpha\beta}(0)x^\beta$.

Proof. $1 \iff 2$: The rays $\gamma^\mu(s) = sa^\mu$ are geodesics if and only if

$$0 = \underbrace{\frac{d^2 \gamma^\mu}{ds^2}}_{=0} + \Gamma^\mu_{\alpha\beta}(sa^\sigma) \frac{d\gamma^\alpha}{ds} \frac{d\gamma^\beta}{ds} = \Gamma^\mu_{\alpha\beta}(sa^\sigma) a^\alpha a^\beta.$$

Multiplying by s^2 and setting $x^\mu = sa^\mu$, the result follows.

$3 \iff 4$:

$$\begin{aligned} g_{\mu\alpha}(x^\sigma)x^\alpha &= g_{\mu\alpha}(0)x^\alpha \iff g_{\mu\alpha}(sa^\sigma)a^\alpha = g_{\mu\alpha}(0)a^\alpha \\ &\iff \frac{d}{ds}(g_{\mu\alpha}(sa^\sigma)a^\alpha) = 0 \\ &\iff \frac{\partial g_{\mu\alpha}(x^\sigma)}{\partial x^\beta} x^\alpha x^\beta = 0. \end{aligned}$$

$2 \implies 4$: From the formula for the Christoffel symbols in terms of the metric we have

$$\Gamma^\mu_{\alpha\beta}(x)x^\alpha x^\beta = 0 \iff \left(2 \frac{\partial g_{\mu\alpha}}{\partial x^\beta} - \frac{\partial g_{\alpha\beta}}{\partial x^\mu}\right) x^\alpha x^\beta = 0. \quad (\text{B.11})$$

Multiplying by x^μ we obtain

$$\begin{aligned} \frac{\partial g_{\mu\alpha}(x^\sigma)}{\partial x^\beta} x^\alpha x^\beta x^\mu &= 0 \iff \frac{\partial g_{\mu\alpha}(sa^\sigma)}{\partial x^\beta} a^\alpha a^\beta a^\mu = 0 \\ &\iff \frac{d}{ds}(g_{\mu\alpha}(sa^\sigma)a^\alpha a^\mu) = 0 \\ &\iff g_{\mu\alpha}(sa^\sigma)a^\alpha a^\mu = g_{\mu\alpha}(0)a^\alpha a^\mu \\ &\iff g_{\mu\alpha}(x^\sigma)x^\alpha x^\mu = g_{\mu\alpha}(0)x^\alpha x^\mu. \end{aligned}$$

Differentiating, it follows that

$$\frac{\partial g_{\mu\alpha}(x^\sigma)}{\partial x^\gamma} x^\alpha x^\mu + 2g_{\gamma\alpha}(x^\sigma)x^\alpha = 2g_{\gamma\alpha}(0)x^\alpha.$$

Substituting this into the last term in (B.11) one obtains

$$\frac{\partial g_{\mu\alpha}}{\partial x^\beta}(x^\sigma)x^\alpha x^\beta + g_{\mu\alpha}(x^\sigma)x^\alpha - g_{\mu\alpha}(0)x^\alpha = 0. \quad (\text{B.12})$$

This implies that

$$\frac{d}{ds}(g_{\mu\alpha}(sa^\mu)sa^\alpha - g_{\mu\alpha}(0)sa^\alpha) = 0,$$

and the result follows by integration.

3&4 $\implies$ 2: Point 4 implies that

$$g_{\alpha\beta}(x^\gamma)x^\alpha x^\beta = g_{\alpha\beta}(0)x^\alpha x^\beta.$$

Differentiating one obtains

$$\frac{\partial g_{\alpha\beta}(x^\gamma)}{\partial x^\mu} x^\alpha x^\beta + 2g_{\alpha\mu}(x^\gamma)x^\alpha = 2g_{\alpha\mu}(0)x^\alpha.$$

The last two terms are equal by point 4, so that

$$\frac{\partial g_{\alpha\beta}(x^\gamma)}{\partial x^\mu} x^\alpha x^\beta = 0.$$

This shows that the last term in (B.11) vanishes, so does the next-to-last by point 3, and the proof is complete. $\square$

C Curvature

In this section we will give complete proofs of the standard identities satisfied by the curvature tensor, as the proofs are often skipped in the literature.

Let ∇ be a covariant derivative defined for vector fields. Its curvature tensor is defined by the formula

$$\boxed{R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z}, \quad (\text{C.1})$$

where, as elsewhere, $[X, Y]$ is the Lie bracket defined in (A.2). We note the obvious anti-symmetry

$$R(X, Y)Z = -R(Y, X)Z. \quad (\text{C.2})$$

Let us check that (C.1) defines a tensor field. Multi-linearity with respect to addition is obvious, but multiplication by functions require more work.

First, we have

$$\begin{aligned} R(fX, Y)Z &= \nabla_{fX} \nabla_Y Z - \nabla_Y \nabla_{fX} Z - \nabla_{[fX, Y]} Z \\ &= f \nabla_X \nabla_Y Z - \nabla_Y (f \nabla_X Z) - \underbrace{\nabla_{f[X, Y] - Y(f)X} Z}_{= f \nabla_{[X, Y]} Z - Y(f) \nabla_X Z} \\ &= f R(X, Y)Z. \end{aligned}$$

INCIDENTALLY: The simplest proof of linearity with respect to multiplication by functions in the last slot proceeds via an index calculation in adapted coordinates; so while we will do the more elegant, index-free version shortly, let us do the ugly one first. We use the coordinate system of Proposition B.2, p. 339, in which the first derivatives of the metric vanish at the prescribed point p :

$$\begin{aligned} \nabla_i \nabla_j Z^k &= \partial_i (\partial_j Z^k - \Gamma_{\ell j}^k Z^\ell) + \underbrace{0 \times \nabla Z}_{\text{at } p} \\ &= \partial_i \partial_j Z^k - \partial_i \Gamma_{\ell j}^k Z^\ell \quad \text{at } p. \end{aligned} \quad (\text{C.3})$$

Antisymmetrizing in i and j , the terms involving the second derivatives of Z drop out, so the result is indeed linear in Z . So $\nabla_i \nabla_j Z^k - \nabla_j \nabla_i Z^k$ is a tensor field linear in Z , and therefore can be written as $R_{\ell ij}^k Z^\ell$.

Note that $\nabla_i \nabla_j Z^k$ is, by definition, the tensor field describing the collection of first covariant derivatives of the tensor field $\nabla_j Z^k$, while (C.1) involves covariant derivatives of vector fields only, so the equivalence of both approaches requires a further argument. This is provided in the calculation below leading to (C.5). $\square$

We continue with

$$\begin{aligned}
 R(X, Y)(fZ) &= \nabla_X \nabla_Y (fZ) - \nabla_Y \nabla_X (fZ) - \nabla_{[X, Y]}(fZ) \\
 &= \left\{ \nabla_X \left(Y(f)Z + f \nabla_Y Z \right) \right\} - \left\{ \cdots \right\}_{X \leftrightarrow Y} \\
 &\quad - [X, Y](f)Z - f \nabla_{[X, Y]} Z \\
 &= \left\{ \underbrace{X(Y(f))Z}_a + \underbrace{Y(f) \nabla_X Z + X(f) \nabla_Y Z + f \nabla_X \nabla_Y Z}_b \right\} - \left\{ \cdots \right\}_{X \leftrightarrow Y} \\
 &\quad - \underbrace{[X, Y](f)Z - f \nabla_{[X, Y]} Z}_c.
 \end{aligned}$$

Now, a together with its counterpart with X and Y interchanged cancel out with c , while b is symmetric with respect to X and Y and therefore cancels out with its counterpart with X and Y interchanged, leading to the desired equality

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

In a coordinate basis $\{e_a\} = \{\partial_\mu\}$ we find that⁹ (recall that $[\partial_\mu, \partial_\nu] = 0$)

$$\begin{aligned}
 R^\alpha_{\beta\gamma\delta} &:= \langle dx^\alpha, R(\partial_\gamma, \partial_\delta)\partial_\beta \rangle \\
 &= \langle dx^\alpha, \nabla_\gamma \nabla_\delta \partial_\beta \rangle - \langle \cdots \rangle_{\delta \leftrightarrow \gamma} \\
 &= \langle dx^\alpha, \nabla_\gamma (\Gamma^\sigma_{\beta\delta} \partial_\sigma) \rangle - \langle \cdots \rangle_{\delta \leftrightarrow \gamma} \\
 &= \langle dx^\alpha, \partial_\gamma (\Gamma^\sigma_{\beta\delta}) \partial_\sigma + \Gamma^\rho_{\sigma\gamma} \Gamma^\sigma_{\beta\delta} \partial_\rho \rangle - \langle \cdots \rangle_{\delta \leftrightarrow \gamma} \\
 &= \{ \partial_\gamma \Gamma^\alpha_{\beta\delta} + \Gamma^\alpha_{\sigma\gamma} \Gamma^\sigma_{\beta\delta} \} - \{ \cdots \}_{\delta \leftrightarrow \gamma},
 \end{aligned}$$

leading finally to

$$R^\alpha_{\beta\gamma\delta} = \partial_\gamma \Gamma^\alpha_{\beta\delta} - \partial_\delta \Gamma^\alpha_{\beta\gamma} + \Gamma^\alpha_{\sigma\gamma} \Gamma^\sigma_{\beta\delta} - \Gamma^\alpha_{\sigma\delta} \Gamma^\sigma_{\beta\gamma}. \quad (\text{C.4})$$

In a general frame some supplementary commutator terms will appear in the formula for $R^\alpha_{\beta\gamma\delta}$.

We note the following.

THEOREM C.1 *There exists a coordinate system in which the metric tensor field has vanishing second derivatives at p if and only if its Riemann tensor vanishes at p . Furthermore, there exists a coordinate system in which the metric tensor field has constant entries near p if and only if the Riemann tensor vanishes near p .*

Proof. The condition is necessary, since the Riemann tensor is a tensor. The proof of sufficiency can be found in e.g. [314, Theorem 7.3]. $\square$

The calculation of the curvature tensor may be a very traumatic experience. There is one obvious case where things are painless, when all $g_{\mu\nu}$'s are constants: in this case the Christoffels vanish, and so does the curvature tensor. Metrics with the last property are called *flat*.

For more general metrics, one way out is to use symbolic computer algebra. This can, e.g., be done online on <http://grtensor.phy.queensu.ca/NewDemo>. MATHEMATICA packages to do this can be downloaded from this URL, or <https://sites.math.washington.edu/~lee/Ricci/> or <http://www.xact.es/xTensor/>. This last package is the least user-friendly at the time of writing this book, but is the most flexible, especially for more involved computations.

⁹The reader is warned that certain authors use other sign conventions either for $R(X, Y)Z$ or for $R^\alpha_{\beta\gamma\delta}$, or both. A useful table that lists the sign conventions for a series of standard GR references can be found on the backside of the front cover of [360].

INCIDENTALLY: We also note an algorithm of Benenti [46] to calculate the curvature tensor, starting from the variational principle for geodesics, which avoids writing out explicitly all the Christoffel coefficients. $\square$

Equation (C.1) is most frequently used ‘upside-down’, not as a definition of the Riemann tensor, but as a tool for calculating what happens when one changes the order of covariant derivatives. Recall that for partial derivatives we have

$$\partial_\mu \partial_\nu Z^\sigma = \partial_\nu \partial_\mu Z^\sigma ,$$

but this is not true in general if partial derivatives are replaced by covariant ones:

$$\nabla_\mu \nabla_\nu Z^\sigma \neq \nabla_\nu \nabla_\mu Z^\sigma .$$

To find the correct formula let us consider the tensor field S defined as

$$Y \longrightarrow S(Y) := \nabla_Y Z .$$

In local coordinates, S takes the form

$$S = \nabla_\mu Z^\nu dx^\mu \otimes \partial_\nu .$$

It follows from the Leibniz rule—or, equivalently, from the definitions in Appendix B—that we have

$$\begin{aligned} (\nabla_X S)(Y) &= \nabla_X(S(Y)) - S(\nabla_X Y) \\ &= \nabla_X \nabla_Y Z - \nabla_{\nabla_X Y} Z . \end{aligned}$$

The commutator of the derivatives can then be calculated as

$$\begin{aligned} (\nabla_X S)(Y) - (\nabla_Y S)(X) &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{\nabla_X Y} Z + \nabla_{\nabla_Y X} Z \\ &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \\ &\quad + \nabla_{[X,Y]} Z - \nabla_{\nabla_X Y} Z + \nabla_{\nabla_Y X} Z \\ &= R(X, Y)Z - \nabla_{T(X,Y)} Z . \end{aligned} \tag{C.5}$$

Writing ∇S in the usual form

$$\nabla S = \nabla_\sigma S_\mu{}^\nu dx^\sigma \otimes dx^\mu \otimes \partial_\nu = \nabla_\sigma \nabla_\mu Z^\nu dx^\sigma \otimes dx^\mu \otimes \partial_\nu ,$$

we are thus led to

$$\nabla_\mu \nabla_\nu Z^\alpha - \nabla_\nu \nabla_\mu Z^\alpha = R^\alpha{}_{\sigma\mu\nu} Z^\sigma - T^\sigma{}_{\mu\nu} \nabla_\sigma Z^\alpha . \tag{C.6}$$

In the important case of vanishing torsion, the coordinate-component equivalent of (C.1) is thus

$$\nabla_\mu \nabla_\nu X^\alpha - \nabla_\nu \nabla_\mu X^\alpha = R^\alpha{}_{\sigma\mu\nu} X^\sigma . \tag{C.7}$$

An identical calculation gives, still for torsionless connections,

$$\nabla_\mu \nabla_\nu a_\alpha - \nabla_\nu \nabla_\mu a_\alpha = -R^\sigma{}_{\alpha\mu\nu} a_\sigma . \tag{C.8}$$

For a general tensor t and torsion-free connection, each tensor index comes with a corresponding Riemann tensor term:

$$\begin{aligned} \nabla_\mu \nabla_\nu t_{\alpha_1 \dots \alpha_r}{}^{\beta_1 \dots \beta_s} - \nabla_\nu \nabla_\mu t_{\alpha_1 \dots \alpha_r}{}^{\beta_1 \dots \beta_s} = \\ -R^\sigma{}_{\alpha_1 \mu \nu} t_{\sigma \dots \alpha_r}{}^{\beta_1 \dots \beta_s} - \dots - R^\sigma{}_{\alpha_r \mu \nu} t_{\alpha_1 \dots \sigma}{}^{\beta_1 \dots \beta_s} \\ + R^{\beta_1}{}_{\sigma \mu \nu} t_{\alpha_1 \dots \alpha_r}{}^{\sigma \dots \beta_s} + \dots + R^{\beta_s}{}_{\sigma \mu \nu} t_{\alpha_1 \dots \alpha_r}{}^{\beta_1 \dots \sigma} . \end{aligned} \tag{C.9}$$

C.1 Bianchi identities

In this section we derive the Bianchi identities. A derivation using moving frames can be found in Section 1.6, p. 12.

We have already seen the anti-symmetry property of the Riemann tensor, which in the index notation corresponds to the equation

$$R^\alpha{}_{\beta\gamma\delta} = -R^\alpha{}_{\beta\delta\gamma}. \quad (\text{C.10})$$

There are a few other identities satisfied by the Riemann tensor; we start with the *first Bianchi identity*. Let $A(X, Y, Z)$ be any expression depending upon three vector fields X, Y, Z which is antisymmetric in X and Y ; we set

$$\sum_{[XYZ]} A(X, Y, Z) := A(X, Y, Z) + A(Y, Z, X) + A(Z, X, Y). \quad (\text{C.11})$$

Thus $\sum_{[XYZ]}$ is a sum over cyclic permutations of the vectors X, Y, Z . Clearly,

$$\sum_{[XYZ]} A(X, Y, Z) = \sum_{[XYZ]} A(Y, Z, X) = \sum_{[XYZ]} A(Z, X, Y). \quad (\text{C.12})$$

Suppose, first, that X, Y , and Z commute. Using (C.12) together with the definition (B.7) of the torsion tensor T we calculate

$$\begin{aligned} \sum_{[XYZ]} R(X, Y)Z &= \sum_{[XYZ]} (\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z) \\ &= \sum_{[XYZ]} \left(\nabla_X \nabla_Y Z - \nabla_Y \underbrace{(\nabla_Z X + T(X, Z))}_{\text{we have used } [X, Z]=0, \text{ see (B.7)}} \right) \\ &= \underbrace{\sum_{[XYZ]} \nabla_X \nabla_Y Z - \sum_{[XYZ]} \nabla_Y \nabla_Z X}_{=0 \text{ (see (C.12))}} - \sum_{[XYZ]} \nabla_Y \underbrace{(T(X, Z))}_{=-T(Z, X)} \\ &= \sum_{[XYZ]} \nabla_X (T(Y, Z)), \end{aligned}$$

and in the last step we have again used (C.12). This can be somewhat rearranged by using the definition of the covariant derivative of a higher-order tensor (compare (B.9))—equivalently, using the Leibniz rule rewritten upside-down:

$$(\nabla_X T)(Y, Z) = \nabla_X (T(Y, Z)) - T(\nabla_X Y, Z) - T(Y, \nabla_X Z).$$

This leads to

$$\begin{aligned} \sum_{[XYZ]} \nabla_X (T(Y, Z)) &= \sum_{[XYZ]} \left((\nabla_X T)(Y, Z) + T(\nabla_X Y, Z) + T(Y, \underbrace{\nabla_X Z}_{=T(X, Z) + \nabla_Z X}) \right) \\ &= \sum_{[XYZ]} \left((\nabla_X T)(Y, Z) - \underbrace{T(T(X, Z), Y)}_{=-T(Z, X)} \right) \\ &\quad + \underbrace{\sum_{[XYZ]} T(\nabla_X Y, Z) + \sum_{[XYZ]} \underbrace{T(Y, \nabla_Z X)}_{=-T(\nabla_Z X, Y)}}_{=0 \text{ (see (C.12))}} \end{aligned}$$

$$= \sum_{[XYZ]} \left((\nabla_X T)(Y, Z) + T(T(X, Y), Z) \right).$$

Summarizing, we have obtained the first Bianchi identity,

$$\sum_{[XYZ]} R(X, Y)Z = \sum_{[XYZ]} \left((\nabla_X T)(Y, Z) + T(T(X, Y), Z) \right), \quad (\text{C.13})$$

under the hypothesis that X , Y , and Z commute. However, both sides of this equation are tensorial with respect to X , Y , and Z , so that they remain correct without the commutation hypothesis.

We are mostly interested in connections with vanishing torsion, in which case (C.13) can be rewritten as

$$R^\alpha{}_{\beta\gamma\delta} + R^\alpha{}_{\gamma\delta\beta} + R^\alpha{}_{\delta\beta\gamma} = 0. \quad (\text{C.14})$$

Equivalently,

$$R^\alpha{}_{[\beta\gamma\delta]} = 0, \quad (\text{C.15})$$

where brackets over indices denote complete antisymmetrization, e.g.

$$\begin{aligned} A_{[\alpha\beta]} &= \frac{1}{2}(A_{\alpha\beta} - A_{\beta\alpha}), \\ A_{[\alpha\beta\gamma]} &= \frac{1}{6}(A_{\alpha\beta\gamma} - A_{\beta\alpha\gamma} + A_{\gamma\alpha\beta} - A_{\gamma\beta\alpha} + A_{\alpha\gamma\beta} - A_{\beta\gamma\alpha}), \end{aligned}$$

etc.

Our next goal is the *second Bianchi identity*. We consider four vector fields X , Y , Z , and W , and we assume again that everybody commutes with everybody else. We calculate

$$\begin{aligned} \sum_{[XYZ]} \nabla_X (R(Y, Z)W) &= \sum_{[XYZ]} \left(\underbrace{\nabla_X \nabla_Y \nabla_Z W}_{=R(X, Y)\nabla_Z W + \nabla_Y \nabla_X \nabla_Z W} - \nabla_X \nabla_Z \nabla_Y W \right) \\ &= \sum_{[XYZ]} R(X, Y)\nabla_Z W \\ &\quad + \underbrace{\sum_{[XYZ]} \nabla_Y \nabla_X \nabla_Z W - \sum_{[XYZ]} \nabla_X \nabla_Z \nabla_Y W}_{=0}. \end{aligned}$$

Next,

$$\begin{aligned} \sum_{[XYZ]} (\nabla_X R)(Y, Z)W &= \sum_{[XYZ]} \left(\nabla_X (R(Y, Z)W) \right. \\ &\quad \left. - R(\nabla_X Y, Z)W - R(Y, \underbrace{\nabla_X Z}_{=\nabla_Z X + T(X, Z)})W - R(Y, Z)\nabla_X W \right) \\ &= \sum_{[XYZ]} \nabla_X (R(Y, Z)W) \\ &\quad - \underbrace{\sum_{[XYZ]} R(\nabla_X Y, Z)W - \sum_{[XYZ]} \underbrace{R(Y, \nabla_Z X)W}_{=-R(\nabla_Z X, Y)W}}_{=0} \\ &\quad - \sum_{[XYZ]} \left(R(Y, T(X, Z))W + R(Y, Z)\nabla_X W \right) \end{aligned}$$

$$= \sum_{[XYZ]} \left(\nabla_X (R(Y, Z)W) - R(T(X, Y), Z)W - R(Y, Z)\nabla_X W \right).$$

It follows now from (C.16) that the first term cancels out the third one, leading to

$$\sum_{[XYZ]} (\nabla_X R)(Y, Z)W = - \sum_{[XYZ]} R(T(X, Y), Z)W, \quad (\text{C.16})$$

which is the desired second Bianchi identity for commuting vector fields. As before, because both sides are multi-linear with respect to addition and multiplication by functions, the result remains valid for arbitrary vector fields.

For torsionless connections, the components-equivalent of (C.16) reads

$$R^\alpha{}_{\mu\beta\gamma;\delta} + R^\alpha{}_{\mu\gamma\delta;\beta} + R^\alpha{}_{\mu\delta\beta;\gamma} = 0. \quad (\text{C.17})$$

INCIDENTALLY: In the case of the Levi-Civita connection, the proof of the second Bianchi identity is simplest in coordinates in which the derivatives of the metric vanish at p : indeed, a calculation very similar to that leading to (C.22) gives

$$\begin{aligned} \nabla_\delta R_{\alpha\mu\beta\gamma}(0) &= \partial_\delta R_{\alpha\mu\beta\gamma}(0) \\ &= \frac{1}{2} \left\{ \partial_\delta \partial_\beta \partial_\mu g_{\alpha\gamma} - \partial_\delta \partial_\beta \partial_\alpha g_{\mu\gamma} - \partial_\delta \partial_\gamma \partial_\mu g_{\alpha\beta} + \partial_\delta \partial_\gamma \partial_\alpha g_{\mu\beta} \right\} (0). \end{aligned} \quad (\text{C.18})$$

and (C.17) follows by inspection $\square$

C.2 Pair interchange symmetry

There is one more identity satisfied by the curvature tensor which is specific to the curvature tensor associated with the Levi-Civita connection, namely

$$g(X, R(Y, Z)W) = g(Y, R(X, W)Z). \quad (\text{C.19})$$

If one sets

$$R_{\alpha\beta\gamma\delta} := g_{\alpha\mu} R^\mu{}_{\beta\gamma\delta}, \quad (\text{C.20})$$

then (C.19) is equivalent to

$$R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta}. \quad (\text{C.21})$$

We will present two proofs of (C.19). The first is direct, but not very elegant. The second is prettier, but less insightful.

For the ugly proof, we suppose that the metric is twice-differentiable. By point 2 of Proposition B.2, in a neighbourhood of any point $p \in M$ there exists a coordinate system in which the connection coefficients $\Gamma^\alpha{}_{\beta\gamma}$ vanish at p . Equation (C.4) evaluated at p therefore reads

$$\begin{aligned} R^\alpha{}_{\beta\gamma\delta} &= \partial_\gamma \Gamma^\alpha{}_{\beta\delta} - \partial_\delta \Gamma^\alpha{}_{\beta\gamma} \\ &= \frac{1}{2} \left\{ g^{\alpha\sigma} \partial_\gamma (\partial_\delta g_{\sigma\beta} + \partial_\beta g_{\sigma\delta} - \partial_\sigma g_{\beta\delta}) \right. \\ &\quad \left. - g^{\alpha\sigma} \partial_\delta (\partial_\gamma g_{\sigma\beta} + \partial_\beta g_{\sigma\gamma} - \partial_\sigma g_{\beta\gamma}) \right\} \\ &= \frac{1}{2} g^{\alpha\sigma} \left\{ \partial_\gamma \partial_\beta g_{\sigma\delta} - \partial_\gamma \partial_\sigma g_{\beta\delta} - \partial_\delta \partial_\beta g_{\sigma\gamma} + \partial_\delta \partial_\sigma g_{\beta\gamma} \right\}. \end{aligned}$$

Equivalently,

$$R_{\sigma\beta\gamma\delta}(0) = \frac{1}{2} \left\{ \partial_\gamma \partial_\beta g_{\sigma\delta} - \partial_\gamma \partial_\sigma g_{\beta\delta} - \partial_\delta \partial_\beta g_{\sigma\gamma} + \partial_\delta \partial_\sigma g_{\beta\gamma} \right\} (0). \quad (\text{C.22})$$

This last expression is obviously symmetric under the exchange of $\sigma\beta$ with $\gamma\delta$, leading to (C.21).

The above calculation traces back the pair-interchange symmetry to the definition of the Levi-Civita connection in terms of the metric tensor. As already mentioned, there exists a more elegant proof, where the origin of the symmetry is perhaps somewhat less apparent, which proceeds as follows: we start by noting that

$$0 = \nabla_a \nabla_b g_{cd} - \nabla_b \nabla_a g_{cd} = -R^e{}_{cab} g_{ed} - R^e{}_{dab} g_{ce}, \quad (\text{C.23})$$

leading to anti-symmetry in the first two indices:

$$R_{abcd} = -R_{bacd}. \quad (\text{C.24})$$

Next, using the cyclic symmetry for a torsion-free connection, we have

$$\begin{aligned} R_{abcd} + R_{cabd} + R_{bcad} &= 0, \\ R_{bcda} + R_{dbca} + R_{cdba} &= 0, \\ R_{cdab} + R_{acdb} + R_{dacb} &= 0, \\ R_{dabc} + R_{bdac} + R_{abdc} &= 0. \end{aligned} \quad (\text{C.25})$$

The desired equation (C.21) follows now by adding the first two and subtracting the last two equations, using (C.24).

A somewhat longer index-free proof of (C.24) proceeds as follows: one starts by noting that

$$g(R(X, Y)Z, Z) = 0. \quad (\text{C.26})$$

To prove this, one writes that

$$\begin{aligned} g(\nabla_X \nabla_Y Z, Z) &= X(g(\nabla_Y Z, Z)) - g(\nabla_Y Z, \nabla_X Z) \\ &= \frac{1}{2} X(Y(g(Z, Z))) - g(\nabla_Y Z, \nabla_X Z). \end{aligned}$$

It follows that

$$g(\nabla_X \nabla_Y Z, Z) - \frac{1}{2} X(Y(g(Z, Z)))$$

is symmetric in X and Y , and (C.26) follows. We can do now the usual ‘polarization formula’ calculation to obtain

$$\begin{aligned} 0 &= g(\nabla_X \nabla_Y (W + Z), W + Z) \\ &= g(\nabla_X \nabla_Y W, W) + g(\nabla_X \nabla_Y W, Z) + g(\nabla_X \nabla_Y Z, W) + g(\nabla_X \nabla_Y Z, Z) \\ &= g(\nabla_X \nabla_Y W, Z) + g(\nabla_X \nabla_Y Z, W), \end{aligned}$$

which is equivalent to (C.24).

REMARK C.2 In dimension two, the pair-interchange symmetry and the anti-symmetry in the last two indices immediately imply that the only non-zero components of the Riemann tensor are

$$R_{1212} = -R_{2112} = R_{2121} = -R_{2112}.$$

This is equivalent to the formula

$$R_{abcd} = \frac{R}{2} (g_{ac} g_{bd} - g_{ad} g_{bc}),$$

as easily checked at a point p in a coordinate system where g_{ab} is diagonal at p .

In dimension three, a similar argument gives

$$R_{abcd} = (P_{ac} g_{bd} - P_{ad} g_{bc} + g_{ac} P_{bd} - g_{ad} P_{bc}), \quad (\text{C.27})$$

where

$$P_{ab} := R_{ab} - \frac{R}{2} g_{ab}.$$

INCIDENTALLY: It is natural to enquire about the number of independent components of a tensor with the symmetries of a metric Riemann tensor in dimension n ; the calculation proceeds as follows: as R_{abcd} is symmetric under the exchange of ab with cd , and anti-symmetric in each of these pairs, we can view it as a symmetric map from the space of anti-symmetric tensor with two indices. Now, the space of anti-symmetric tensors is $N = n(n-1)/2$ -dimensional, while the space of symmetric maps in dimension N is $N(N+1)/2$ -dimensional, so we obtain at most

$$\frac{n(n-1)(n^2-n+2)}{8}$$

free parameters. However, we need to take into account the cyclic identity:

$$R_{dabc} + R_{dbca} + R_{dcab} = 0. \quad (\text{C.28})$$

If $a = b$ this reads

$$R_{daac} + R_{daca} + R_{dcaa} = 0,$$

which has already been accounted for. Similarly if $a = d$ we obtain

$$R_{abca} + R_{bc aa} + R_{caba} = 0,$$

which holds in view of the previous identities. We conclude that the only new identities which could possibly arise are those where $abcd$ are all distinct. (Another way to see this is to note the identity

$$R_{a[bcd]} = R_{[abcd]}, \quad (\text{C.29})$$

which holds for any tensor satisfying

$$R_{abcd} = R_{[ab]cd} = R_{ab[cd]} = R_{cdab}, \quad (\text{C.30})$$

and which can be proved by writing explicitly all the terms in $R_{[abcd]}$; this is the same as adding the left-hand sides of the first and third equations in (C.25), and removing those of the second and fourth.)

It should be clear that for each distinct set of four indices the Bianchi identity provides a constraint which is independent of the symmetries listed in (C.30). In dimension four (C.28) provides thus four candidate equations for another constraint, labelled by d , but it is easily checked that they all coincide either directly or using (C.29). This leads to 20 free parameters at each space point. (Strictly speaking, to prove this one would still need to show that there are no further algebraic identities satisfied by the Riemann tensor, which is indeed the case).

Note that (C.29) shows that in dimension $n \geq 4$ the Bianchi identity introduces $\binom{n}{4}$ new constraints, leading to

$$\frac{n(n-1)(n^2-n+2)}{8} - \frac{n(n-1)(n-2)(n-3)}{12} = \frac{n^2(n^2-1)}{12} \quad (\text{C.31})$$

independent components at each point. $\square$

C.3 Curvature of product metrics

Let (M, g) and (N, h) be two pseudo-Riemannian manifolds; on the product manifold $M \times N$ we define a metric $g \oplus h$ as follows: every element of $T(M \times N)$ can be uniquely written as $X \oplus Y$ for some $X \in TM$ and $Y \in TN$. We set

$$(g \oplus h)(X \oplus Y, \hat{X} \oplus \hat{Y}) = g(X, \hat{X}) + h(Y, \hat{Y}).$$

Let ∇ be the Levi-Civita connection associated with g , D that associated with h , and $\mathscr{D}$ the one associated with $g \oplus h$. To understand the structure of ∇ , we note that sections of $T(M \times N)$ are linear combinations, with coefficients in $C^\infty(M \times N)$, of elements of the form $X \oplus Y$, where $X \in \Gamma(TM)$ and $Y \in \Gamma(TN)$. (Thus, X does

not depend upon $q \in N$ and Y does not depend upon $p \in M$.) We claim that for such fields $X \oplus Y$ and $W \oplus Z$ we have

$$\mathcal{D}_{X \oplus Y}(W \oplus Z) = \nabla_X W \oplus D_Y Z. \quad (\text{C.32})$$

(If true, (C.32) together with the Leibniz rule characterizes $\mathcal{D}$ uniquely.) To verify (C.32), we check first that $\mathcal{D}$ has no torsion:

$$\begin{aligned} \mathcal{D}_{X \oplus Y}(W \oplus Z) - \mathcal{D}_{W \oplus Z}(X \oplus Y) &= \nabla_X W \oplus D_Y Z - \nabla_W X \oplus D_Z Y \\ &= (\nabla_X W - \nabla_W X) \oplus (D_Y Z - D_Z Y) \\ &= [X, W] \oplus [Y, Z] \\ &= [X \oplus Y, W \oplus Z]. \end{aligned}$$

(In the last step we have used $[X \oplus 0, 0 \oplus Z] = [0 \oplus Y, W \oplus 0] = 0$.) Next, we check metric compatibility:

$$\begin{aligned} &X \oplus Y \left((g \oplus h)(W \oplus Z, \hat{W} \oplus \hat{Z}) \right) \\ &= X \oplus Y \left(g(W, \hat{W}) + h(Z, \hat{Z}) \right) \\ &= \underbrace{X(g(W, \hat{W}))}_{g(\nabla_X W, \hat{W}) + g(W, \nabla_X \hat{W})} + \underbrace{Y(h(Z, \hat{Z}))}_{h(D_Y Z, \hat{Z}) + h(Z, D_Y \hat{Z})} \\ &= \underbrace{g(\nabla_X W, \hat{W}) + h(D_Y Z, \hat{Z})}_{(g \oplus h)(\nabla_X W \oplus D_Y Z, \hat{W} \oplus \hat{Z})} + \underbrace{g(W, \nabla_X \hat{W}) + h(Z, D_Y \hat{Z})}_{(g \oplus h)(W \oplus Z, \nabla_X \hat{W} \oplus D_Y \hat{Z})} \\ &= (g \oplus h)(\mathcal{D}_{X \oplus Y} W \oplus Z, \hat{W} \oplus \hat{Z}) + (g \oplus h)(W \oplus Z, \mathcal{D}_{X \oplus Y} \hat{W} \oplus \hat{Z}). \end{aligned}$$

The uniqueness of Levi-Civita connections proves (C.32).

Let $\text{Riem}(k)$ denote the Riemann tensor of the metric k . It should be clear from (C.32) that the Riemann tensor of $g \oplus h$ has a sum structure,

$$\text{Riem}(g \oplus h) = \text{Riem}(g) \oplus \text{Riem}(h). \quad (\text{C.33})$$

More precisely,

$$\text{Riem}(g \oplus h)(X \oplus Y, \hat{X} \oplus \hat{Y})W \oplus Z = \text{Riem}(g)(X, \hat{X})W \oplus \text{Riem}(h)(Y, \hat{Y})Z. \quad (\text{C.34})$$

This implies that

$$\text{Ric}(g \oplus h) = \text{Ric}(g) \oplus \text{Ric}(h), \quad (\text{C.35})$$

in the sense that

$$\text{Ric}(g \oplus h)(X \oplus Y, \hat{X} \oplus \hat{Y}) = \text{Ric}(g)(X, \hat{X}) \oplus \text{Ric}(h)(Y, \hat{Y}), \quad (\text{C.36})$$

and

$$\text{tr}_{g \oplus h} \text{Ric}(g \oplus h) = \text{tr}_g \text{Ric}(g) + \text{tr}_h \text{Ric}(h). \quad (\text{C.37})$$

C.4 Analyticity of isometries

We give the proof of an unpublished result of Nomizu on the analyticity of maps preserving analytic affine connections. The analyticity of isometries of analytic manifolds is a direct consequence of this.

THEOREM C.3 *Suppose M and M' are real analytic manifolds each provided with a real analytic affine connection. Then a (smooth) diffeomorphism f from M onto M' which preserves the affine connections is real analytic.*

Proof. It follows from the analytic implicit function theorem that for M with an analytic affine connection ∇ , the exponential mapping $\exp_p : T_p M \rightarrow M$ is real analytic on a neighbourhood of 0 in $T_p M$, say U (so that $\exp_p(U)$ is a normal neighbourhood of p). We have for each $p \in M$

$$f(\exp_p(X)) = \exp_{f(p)}(f_*(p)X) \quad \text{for every } X \in V,$$

where V is an open neighbourhood of 0 in $T_p M$. The analyticity of f follows immediately from this equation. $\square$

D Exterior algebra

A preferred class of tensors is provided by those that are totally antisymmetric in all indices. Such k -covariant tensors are called k -forms. They are of special interest because they can naturally be used for integration. Furthermore, on such tensors one can introduce a differentiation operation, called an *exterior derivative*, which does not require a connection.

INCIDENTALLY: An extensive treatment of differential forms can be found in [235].

By definition, functions are *zero-forms*, and covectors are *one-forms*.

Let α_i , $i = 1, \dots, k$, be a collection of one-forms; the *exterior product* of the α_i 's is a k -form defined as

$$(\alpha_1 \wedge \dots \wedge \alpha_k)(X_1, \dots, X_k) = \det(\alpha_i(X_j)), \quad (\text{D.1})$$

where $\det(\alpha_i(X_j))$ denotes the determinant of the matrix obtained by applying all the α_i 's to all the vectors X_j . For example,

$$(dx^a \wedge dx^b)(X, Y) = X^a Y^b - Y^a X^b.$$

Note that

$$dx^a \wedge dx^b = dx^a \otimes dx^b - dx^b \otimes dx^a,$$

which is twice the antisymmetrization $dx^{[a} \otimes dx^{b]}$.

Quite generally, if α is a totally anti-symmetric k -covariant tensor with coordinate coefficients $\alpha_{a_1 \dots a_k}$, then

$$\begin{aligned} \alpha &= \alpha_{a_1 \dots a_k} dx^{a_1} \otimes \dots \otimes dx^{a_k} \\ &= \alpha_{a_1 \dots a_k} dx^{[a_1} \otimes \dots \otimes dx^{a_k]} \\ &= \frac{1}{k!} \alpha_{a_1 \dots a_k} dx^{a_1} \wedge \dots \wedge dx^{a_k} \\ &= \sum_{a_1 < \dots < a_k} \alpha_{a_1 \dots a_k} dx^{a_1} \wedge \dots \wedge dx^{a_k}. \end{aligned} \quad (\text{D.2})$$

The middle formulae exhibits the factorial coefficients needed to go from tensor components to the components with respect to the basis $\{dx^{a_1} \wedge \dots \wedge dx^{a_k}\}$.

Equation (D.2) makes it clear that, in dimension n , for any non-trivial k -form we have $k \leq n$. It also shows that the dimension of the space of k -forms, with $0 \leq k \leq n$, equals

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

A differential form is defined as a linear combination of k -forms, with k possibly taking different values for different summands.

Let Y be a vector and α a k -form. The *contraction* $Y \rfloor \alpha$, also called the *interior product* of Y and α , is a $(k-1)$ -form defined as

$$(Y \rfloor \alpha)(X_1, \dots, X_{k-1}) := \alpha(Y, X_1, \dots, X_{k-1}). \quad (\text{D.3})$$

The operation $Y \rfloor$ is often denoted by i_Y .

Let α be a k -form and β an ℓ -form; the exterior product $\alpha \wedge \beta$ of α and β , also called a *wedge product*, is defined using bilinearity:

$$\begin{aligned}
 \alpha \wedge \beta &\equiv \left(\sum_{a_1 < \dots < a_k} \alpha_{a_1 \dots a_k} dx^{a_1} \wedge \dots \wedge dx^{a_k} \right) \wedge \left(\sum_{b_1 < \dots < b_\ell} \beta_{b_1 \dots b_\ell} dx^{b_1} \wedge \dots \wedge dx^{b_\ell} \right) \\
 &:= \sum_{a_1 < \dots < a_k, b_1 < \dots < b_\ell} \alpha_{a_1 \dots a_k} \beta_{b_1 \dots b_\ell} \\
 &\quad \times dx^{a_1} \wedge \dots \wedge dx^{a_k} \wedge dx^{b_1} \wedge \dots \wedge dx^{b_\ell}. \tag{D.4}
 \end{aligned}$$

The product so defined is associative:

$$\alpha \wedge (\beta \wedge \gamma) = (\alpha \wedge \beta) \wedge \gamma =: \alpha \wedge \beta \wedge \gamma. \tag{D.5}$$

INCIDENTALLY: In order to establish (D.5), we start by rewriting the definition of the wedge product of a k -form α and ℓ -form β as

$$(\alpha \wedge \beta)(X_1, \dots, X_{k+l}) := \frac{1}{k!} \frac{1}{\ell!} \sum_{\pi \in S_{k+l}} \text{sgn}(\pi) (\alpha \otimes \beta)(X_{\pi(1)}, \dots, X_{\pi(k+l)}), \tag{D.6}$$

where $X_i \in \Gamma(TM)$ for $i = 1, \dots, k+l$. Here S_p denotes the group of permutations of p elements and $\text{sgn}(\pi)$ the signature of the permutation π .

Let S_p denote the group of permutations of p elements and let $\Omega^\ell(M)$ denote the space of ℓ -forms. For $\alpha \in \Omega^k(M)$, $\beta \in \Omega^\ell(M)$, and $\gamma \in \Omega^m(M)$ we have

$$\begin{aligned}
 &((\alpha \wedge \beta) \wedge \gamma)(X_1, \dots, X_{k+l+m}) \\
 &= \frac{1}{(k+l)!m!} \sum_{\pi \in S_{k+l+m}} \text{sgn}(\pi) ((\alpha \wedge \beta) \otimes \gamma)(X_{\pi(1)}, \dots, X_{\pi(k+l+m)}) \\
 &= \frac{1}{(k+l)!m!} \sum_{\pi \in S_{k+l+m}} \text{sgn}(\pi) (\alpha \wedge \beta)(X_{\pi(1)}, \dots, X_{\pi(k+l)}) \cdot \gamma(X_{\pi(k+l+1)}, \dots, X_{\pi(k+l+m)}) \\
 &= \frac{1}{(k+l)!k!l!m!} \sum_{\pi \in S_{k+l+m}} \text{sgn}(\pi) \sum_{\pi' \in S_{k+l}} \text{sgn}(\pi') (\alpha \otimes \beta)(X_{\pi'(\pi(1))}, \dots, X_{\pi'(\pi(k+l))}) \\
 &\quad \cdot \gamma(X_{\pi(k+l+1)}, \dots, X_{\pi(k+l+m)}). \tag{D.7}
 \end{aligned}$$

We introduce a new permutation $\pi'' \in S_{k+l+m}$ such that

$$\pi''(\pi(i)) = \begin{cases} \pi'(\pi(i)) & \text{for } 1 \leq i \leq k+l, \\ \pi(i) & \text{for } i > k+l, \end{cases}$$

which implies that $\text{sgn}(\pi'') = \text{sgn}(\pi')$. One then obtains

$$\begin{aligned}
 &((\alpha \wedge \beta) \wedge \gamma)(X_1, \dots, X_{k+l+m}) \\
 &= \frac{1}{(k+l)!k!l!m!} \sum_{\pi' \in S_{k+l}} \text{sgn}(\pi') \sum_{\pi \in S_{k+l+m}} \text{sgn}(\pi) ((\alpha \otimes \beta) \otimes \gamma)(X_{\pi''(\pi(1))}, \dots, X_{\pi''(\pi(k+l+m))}).
 \end{aligned}$$

Set $\sigma := \pi'' \circ \pi$. Then $\text{sgn}(\sigma) = \text{sgn}(\pi'')\text{sgn}(\pi)$; thus

$$\text{sgn}(\pi) = \text{sgn}(\sigma)\text{sgn}(\pi'') = \text{sgn}(\sigma)\text{sgn}(\pi')$$

and we get

$$\begin{aligned}
 &((\alpha \wedge \beta) \wedge \gamma)(X_1, \dots, X_{k+l+m}) \\
 &= \frac{1}{(k+l)!k!l!m!} \underbrace{\sum_{\pi' \in S_{k+l}} (\text{sgn}(\pi'))^2}_{=(k+l)!}
 \end{aligned}$$

$$\begin{aligned}
& \sum_{\sigma \in S_{k+l+m}} \operatorname{sgn}(\sigma) ((\alpha \otimes \beta) \otimes \gamma)(X_{\sigma(1)}, \dots, X_{\sigma(k+l+m)}) \\
&= \frac{1}{k!l!m!} \sum_{\sigma \in S_{k+l+m}} \operatorname{sgn}(\sigma) ((\alpha \otimes \beta) \otimes \gamma)(X_{\sigma(1)}, \dots, X_{\sigma(k+l+m)}). \quad (\text{D.8})
\end{aligned}$$

A similar calculation gives

$$\begin{aligned}
& (\alpha \wedge (\beta \wedge \gamma))(X_1, \dots, X_{k+l+m}) \\
&= \frac{1}{k!l!m!} \sum_{\sigma \in S_{k+l+m}} \operatorname{sgn}(\sigma) (\alpha \otimes (\beta \otimes \gamma))(X_{\sigma(1)}, \dots, X_{\sigma(k+l+m)}), \quad (\text{D.9})
\end{aligned}$$

and the associativity of the wedge product follows.

The above calculations lead to the following form of the wedge product of n forms, where associativity is hidden in the notation:

$$\begin{aligned}
& (\alpha_1 \wedge \dots \wedge \alpha_n)(X_1, \dots, X_{k_1+\dots+k_n}) \\
&= \frac{1}{k_1! \dots k_n!} \sum_{\pi \in S_{k_1+\dots+k_n}} \operatorname{sgn}(\pi) (\alpha_1 \otimes \dots \otimes \alpha_n)(X_{\pi(1)}, \dots, X_{\pi(k_1+\dots+k_n)}), \quad (\text{D.10})
\end{aligned}$$

where $\alpha_i \in \Omega^{k_i}(M)$ for $i = 1, \dots, n$ and $X_j \in \Gamma(TM)$ for $j = 1, \dots, k_1 + \dots + k_n$.

Let us apply the last formula to one-forms: if $\alpha_i \in \Omega^1(M)$ we have

$$\begin{aligned}
(\alpha_1 \wedge \dots \wedge \alpha_n)(X_1, \dots, X_n) &= \sum_{\pi \in S_n} \operatorname{sgn}(\pi) (\alpha_1 \otimes \dots \otimes \alpha_n)(X_{\pi(1)}, \dots, X_{\pi(n)}) \\
&= \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_{i=1}^n \alpha_i(X_{\pi(i)}) \\
&= \det(\alpha_i(X_j)), \quad (\text{D.11})
\end{aligned}$$

where we have used the Leibniz formula for determinants. $\square$

The *exterior derivative* of a differential form is defined as follows:

1. For a zero form f , the exterior derivative of f is its *usual differential* df .
2. For a k -form α , its *exterior derivative* $d\alpha$ is a $(k+1)$ -form defined as

$$d\alpha \equiv d \left(\frac{1}{k!} \alpha_{\mu_1 \dots \mu_k} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k} \right) := \frac{1}{k!} d\alpha_{\mu_1 \dots \mu_k} \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}. \quad (\text{D.12})$$

Equivalently

$$\begin{aligned}
d\alpha &= \frac{1}{k!} \partial_\beta \alpha_{\mu_1 \dots \mu_k} dx^\beta \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k} \\
&= \frac{1}{k!} \partial_{[\beta} \alpha_{\mu_1 \dots \mu_k]} dx^\beta \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k} \\
&= \frac{k+1}{(k+1)!} \partial_{[\beta} \alpha_{\mu_1 \dots \mu_k]} dx^\beta \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}, \quad (\text{D.13})
\end{aligned}$$

which can also be written as

$$(d\alpha)_{\mu_1 \dots \mu_{k+1}} = (k+1) \partial_{[\mu_1} \alpha_{\mu_2 \dots \mu_{k+1}]} . \quad (\text{D.14})$$

INCIDENTALLY: In this context it is clearly more convenient *not* to use the summation convention when dealing with differential forms, using instead explicit ordered sums: if α is a k -form we have

$$\begin{aligned}\alpha &= \frac{1}{k!} \alpha_{\mu_1 \dots \mu_k} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k} \\ &= \sum_{\mu_1 < \mu_2 < \dots < \mu_k} \alpha_{\mu_1 \dots \mu_k} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}.\end{aligned}\quad (\text{D.15})$$

Then (D.12) can be rewritten as

$$d\alpha = \sum_{\mu_1 < \mu_2 < \dots < \mu_k} d\alpha_{\mu_1 \dots \mu_k} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}.\quad (\text{D.16})$$

□

One easily checks, using $\partial_\alpha \partial_\beta y^\gamma = \partial_\beta \partial_\alpha y^\gamma$, that the exterior derivative behaves as a tensor under coordinate transformations. An ‘active way’ of saying this is

$$d(\phi^* \alpha) = \phi^*(d\alpha),\quad (\text{D.17})$$

for any differentiable map ϕ . The tensorial character of d is also made clear by noting that for any torsion-free connection ∇ we have

$$\partial_{[\mu_1} \alpha_{\mu_2 \dots \mu_{k+1}]} = \nabla_{[\mu_1} \alpha_{\mu_2 \dots \mu_{k+1}]}. \quad (\text{D.18})$$

Again by symmetry of second derivatives, it immediately follows from (D.12) that $d(df) = 0$ for any function, and subsequently also for any differential form:

$$d^2 \alpha := d(d\alpha) = 0. \quad (\text{D.19})$$

A coordinate-free definition of $d\alpha$ is

$$\begin{aligned}d\alpha(X_0, X_1, \dots, X_k) &= \sum_{0 \leq j \leq k} (-1)^j X_j \left(\alpha(X_0, \dots, \widehat{X_j}, \dots, X_k) \right) \\ &+ \sum_{0 \leq i < j \leq k} (-1)^{i+j} \alpha([X_i, X_j], X_0, \dots, \widehat{X_j}, \dots, X_k),\end{aligned}\quad (\text{D.20})$$

where $\widehat{X_\ell}$ denotes the omission of the vector X_ℓ .

It is not too difficult to prove that if $\overset{k}{\alpha}$ is a k -form and $\overset{\ell}{\beta}$ is an ℓ -form, then the following version of the Leibniz rule holds:

$$d(\overset{k}{\alpha} \wedge \overset{\ell}{\beta}) = (d \overset{k}{\alpha}) \wedge \overset{\ell}{\beta} + (-1)^k \overset{k}{\alpha} \wedge (d \overset{\ell}{\beta}). \quad (\text{D.21})$$

D.1 Hodge duality

In dimension n , let $\sigma \in \{\pm 1\}$ denote the parity of a permutation; set

$$\epsilon_{\mu_1 \dots \mu_n} = \begin{cases} \sqrt{|\det g_{\alpha\beta}|} \sigma(\mu_1 \dots \mu_n) & \text{if } (\mu_1 \dots \mu_n) \text{ is a permutation of } (1 \dots n); \\ 0 & \text{otherwise.} \end{cases}$$

The *Hodge dual* $\star \alpha$ of a k -form $\alpha = \alpha_{\mu_1 \dots \mu_k} dx^{\mu_1} \otimes \dots \otimes dx^{\mu_k}$ is a $(n - k)$ -form defined as

$$\star \alpha = \frac{1}{k!(n - k)!} \epsilon_{\mu_1 \dots \mu_k \mu_{k+1} \dots \mu_n} \alpha^{\mu_1 \dots \mu_k} dx^{\mu_{k+1}} \otimes \dots \otimes dx^{\mu_n}. \quad (\text{D.22})$$

Equivalently,

$$\star \alpha_{\mu_{k+1} \dots \mu_n} = \frac{1}{k!(n - k)!} \epsilon_{\mu_1 \dots \mu_k \mu_{k+1} \dots \mu_n} \alpha^{\mu_1 \dots \mu_k}. \quad (\text{D.23})$$

For example, in Euclidean three-dimensional space,

$$\star 1 = dx \wedge dy \wedge dz, \quad \star dx = dy \wedge dz, \quad \star(dy \wedge dz) = dx, \quad \star(dx \wedge dy \wedge dz) = 1,$$

etc. In Minkowski four-dimensional spacetime we have, e.g.,

$$\begin{aligned}\star dt &= -dx \wedge dy \wedge dz, & \star dx &= -dy \wedge dz \wedge dt, \\ \star(dt \wedge dx) &= -dy \wedge dz, & \star(dx \wedge dy) &= -dz \wedge dt, & \star(dx \wedge dy \wedge dz) &= -dt.\end{aligned}$$

E Null hyperplanes

Some of the objects that occur in Lorentzian geometry, and which possess rather disturbing properties, are *null hyperplanes* and *null hypersurfaces*, and it appears useful to include a short discussion of those. Perhaps the most unusual feature of such objects is that the direction normal is actually tangential as well. Furthermore, because the normal has no natural normalization in this case, there is no natural measure induced on a null hypersurface by the ambient metric.

In this section we present some algebraic preliminaries concerning null hyperplanes; null hypersurfaces will be discussed in Appendix F.

Let W be a real vector space, and recall that its dual W^* is defined as the set of all linear maps from W to $\mathbb{R}$ (in this work only vector spaces over the reals are relevant, but the field makes no difference for the discussion that follows). To avoid unnecessary complications we assume that W is finite dimensional. Then W^* has the same dimension as W : indeed, if $\{e_i\}_{i=1}^N$ is a basis of W , then the collection of linear maps $\{\theta^i\}_{i=1}^N$, defined by the condition $\theta^j(e_i) = \delta_i^j$, forms a basis of W^* , called the *dual basis*.

We suppose that W is equipped with a (a) *bilinear*, (b) *symmetric*, and (c) *non-degenerate form* q . Thus,

$$q : W \times W \rightarrow \mathbb{R}$$

satisfies

$$(a) \quad q(\lambda X + \mu Y, Z) = \lambda q(X, Z) + \mu q(Y, Z), \quad (b) \quad q(X, Y) = q(Y, X),$$

and we also have the implication

$$(c) \quad \forall Y \in W \quad q(X, Y) = 0 \implies X = 0. \quad (E.1)$$

(Strictly speaking, we should have indicated linearity with respect to the second variable in (a) as well, but this property follows from (a) and (b) above.) By an abuse of terminology, we will call q a *scalar product*; note that standard algebra textbooks often add the condition of positive-definiteness to the definition of a scalar product, which we do not include here.

Let $V \subset W$ be a vector subspace of W . The *annihilator* V^0 of W is defined as the set of linear forms on W which vanish on V :

$$V^0 := \{\alpha \in W^* : \forall Y \in V \quad \alpha(Y) = 0\} \subset W^*.$$

V^0 is obviously a linear subspace of W^* .

Because q is non-degenerate, it defines a linear isomorphism, denoted by $\flat$, between W and W^* by the formula

$$X^\flat(Y) = q(X, Y).$$

Indeed, the map $X \mapsto X^\flat$ is clearly linear. Next, it has no kernel by (E.1). Since the dimensions of W and W^* are the same, it must be an isomorphism. The inverse map is denoted by $\sharp$. Thus, by definition we have

$$q(\alpha^\sharp, Y) = \alpha(Y).$$

The map $\flat$ is nothing but ‘the lowering of the index on a vector using the metric q ’, while $\sharp$ is the ‘raising of the index on a one-form using the inverse metric’.

For further purposes it is useful to recall the following

PROPOSITION E.1

$$\dim V + \dim V^0 = \dim W.$$

Proof. Let $\{e_i\}_{i=1,\dots,\dim V}$ be any basis of V ; we can complete $\{e_i\}$ to a basis $\{e_i, f_a\}$, with $a = 1, \dots, \dim W - \dim V$, of W . Let $\{e_i^*, f_a^*\}$ be the dual basis of W^* . It is straightforward to check that V^0 is spanned by $\{f_a^*\}$, which gives the result. $\square$

The quadratic form q defines the notion of orthogonality:

$$V^\perp := \{Y \in W : \forall X \in V \ q(X, Y) = 0\}.$$

A chase through the definitions above shows that

$$V^\perp = (V^0)^\#.$$

Proposition E.1 implies the following.

PROPOSITION E.2

$$\dim V + \dim V^\perp = \dim W.$$

This implies, again regardless of signature, the following.

PROPOSITION E.3

$$(V^\perp)^\perp = V.$$

Proof. The inclusion $(V^\perp)^\perp \supset V$ is obvious from the definitions. The equality follows now because both spaces have the same dimension, as a consequence of Proposition (E.2). $\square$

Now,

$$X \in V \cap V^\perp \implies q(X, X) = 0, \quad (\text{E.2})$$

so that X vanishes if q is positive- or negative-definite, leading to $\dim V \cap \dim V^\perp = \{0\}$ in those cases. However, this does not have to be the case anymore for non-definite scalar products q .

A vector subspace V of W is called a *hyperplane* if

$$\dim V = \dim W - 1.$$

Proposition E.2 implies then

$$\dim V^\perp = 1,$$

regardless of the signature of q . Thus, given a hyperplane V there exists a vector w such that

$$V^\perp = \mathbb{R}w.$$

If q is Lorentzian, we say that

$$V \text{ is } \begin{cases} \text{spacelike} & \text{if } w \text{ is timelike;} \\ \text{timelike} & \text{if } w \text{ is spacelike;} \\ \text{null} & \text{if } w \text{ is null.} \end{cases}$$

An argument based e.g. on Gram–Schmidt orthonormalization shows that if V is spacelike, then the scalar product defined on V by restriction is positive-definite; similarly, if V is timelike, then the resulting scalar product is Lorentzian. The last case, of a null V , leads to a degenerate-induced scalar product. In fact, we claim that

$$V \text{ is null if and only if } V \text{ contains its normal.} \quad (\text{E.3})$$

To see (E.3), suppose that $V^\perp = \mathbb{R}w$, with w null. Since $q(w, w) = 0$ we have $w \in (\mathbb{R}w)^\perp$, and from Proposition E.3

$$w \in (\mathbb{R}w)^\perp = (V^\perp)^\perp = V.$$

Since V does not contain its normal in the remaining cases, the equivalence is established.

As discussed in more detail in the next section, a hypersurface $\mathcal{N} \subset \mathcal{M}$ is called *null* if at every $p \in \mathcal{N}$ the scalar product restricted to $T_p\mathcal{N}$ is degenerate. Equivalently, the tangent space $T_p\mathcal{N}$ is a null subspace of $T_p\mathcal{M}$. So (E.2) shows that vectors normal to a null hypersurface $\mathcal{N}$ are also tangent to $\mathcal{N}$.

F The geometry of null hypersurfaces

In this appendix we review some aspects of the geometry of null hypersurfaces. We follow the exposition in [117], which in turn is based on [205].

A C^k *null hypersurface* in a spacetime $(\mathcal{M}, g)$, $k \geq 1$, is a C^k embedded submanifold $\mathcal{N}$ of $\mathcal{M}$ of codimension one such that the pullback of the metric g to $\mathcal{N}$ is degenerate. As explained in Appendix E, each such hypersurface $\mathcal{N}$ admits a C^{k-1} non-vanishing *future-directed* null vector field $L \in \Gamma T\mathcal{N}$ such that the normal space of L at a point $p \in \mathcal{N}$ coincides with the tangent space of $\mathcal{N}$ at p , i.e., $L_p^\perp = T_p\mathcal{N}$ for all $p \in \mathcal{N}$. In particular, tangent vectors to $\mathcal{N}$ not parallel to L are spacelike. We note that the vector field L is unique up to a positive scale factor.

The integral curves of L , when suitably parameterized, are null geodesics, called the *null geodesic generators* of $\mathcal{N}$: indeed, since $g(L, L) = 0$ on $\mathcal{N}$, we have that $X(g(L, L)) = 0$ for all vectors X tangent to $\mathcal{N}$. This implies that $d(g(L, L))$ annihilates $T\mathcal{N}$; hence, it is conormal to $\mathcal{N}$. Now, $g(L, \cdot)$ annihilates $T\mathcal{N}$ as well, so we conclude that

$$\nabla(g(L, L)) \sim L.$$

Let u be any defining function for $\mathcal{N}$, i.e., $\mathcal{N} = \{u = 0\}$, with du nowhere vanishing on $\mathcal{N}$. Since du annihilates $T\mathcal{N}$ as well, L must be proportional to ∇u , so let us first assume that $L = \nabla u$. Then

$$\begin{aligned} \nabla_\mu(g(L, L)) &= \nabla_\mu(L^\alpha L_\alpha) = 2L^\alpha \nabla_\mu L_\alpha = 2L^\alpha \nabla_\mu \nabla_\alpha u = 2L^\alpha \nabla_\alpha \nabla_\mu u \\ &= 2L^\alpha \nabla_\alpha L_\mu \sim L_\mu. \end{aligned} \quad (\text{F.1})$$

We have thus shown that the integral curves of ∇u are null geodesics, though perhaps not affinely parameterized. Now, multiplying ∇u by a function will not change its integral curves, but only their parameterization, so the result remains true for all L proportional to ∇u .

Since L is orthogonal to $\mathcal{N}$ we can introduce the *null Weingarten map* and *null second fundamental form* of $\mathcal{N}$ with respect to L in a manner roughly analogous to what is done for spacelike hypersurfaces or hypersurfaces in a Riemannian manifold, as follows. We start by introducing an equivalence relation on tangent vectors: for $X, X' \in T_p\mathcal{N}$, we say that $X' = X \bmod L$ if and only if $X' - X = \lambda L$ for some $\lambda \in \mathbb{R}$. Let $\bar{X}$ denote the equivalence class of X . Now, if $X' = X \bmod L$ and $Y' = Y \bmod L$, then obviously

$$g(X', Y') = g(X, Y), \quad (\text{F.2})$$

since $g(L, Z)$ vanishes for all $Z \in T\mathcal{N}$. Similarly, writing $X' = X + aL$ for some function a ,

$$g(\nabla_{X'} L, Y') = g(\nabla_{X+aL} L, Y') = g(\nabla_X L + \underbrace{a \nabla_L L}_{\sim L}, Y') = g(\nabla_X L, Y).$$

Hence, for various quantities of interest, components along L are irrelevant. For this reason one works with the tangent space of $\mathcal{N}$ modded out by L , i.e.,

$$(T_p\mathcal{N})/L = \{\bar{X} \mid X \in T_p\mathcal{N}\} \text{ and } (T\mathcal{N})/L = \cup_{p \in \mathcal{N}} (T_p\mathcal{N})/L.$$

In spacetimes of dimension $n + 1$, the bundle $(T\mathcal{N})/L$ is a vector bundle over $\mathcal{N}$ of dimension $(n - 1)$, and does not depend on the particular choice of null vector field L .

There is a natural positive definite metric h in $(T\mathcal{N})/L$ induced from g : For each $p \in \mathcal{N}$, define $h: (T_p\mathcal{N})/L \times (T_p\mathcal{N})/L \rightarrow \mathbb{R}$ by

$$h(\overline{X}, \overline{Y}) = g(X, Y). \quad (\text{F.3})$$

Equation (F.2) shows that h is well defined.

The *null Weingarten map* $b = b_L$ of $\mathcal{N}$ with respect to L is, for each point $p \in \mathcal{N}$, a linear map

$$b: (T_p\mathcal{N})/L \rightarrow (T_p\mathcal{N})/L$$

defined by

$$b(\overline{X}) = \overline{\nabla_X L}. \quad (\text{F.4})$$

To see that b is well defined, let $X' = X + \lambda L$, then

$$\nabla_{X'} L = \nabla_{X+\lambda L} L = \nabla_X L + \underbrace{\lambda \nabla_X L}_{\sim L},$$

which shows that the equivalence classes $\overline{\nabla_{X'} L}$ and $\overline{\nabla_X L}$ coincide, as needed.

If $\tilde{L} = fL$, $f \in C^1(\mathcal{N})$ is any other future-directed null vector field tangent to $\mathcal{N}$, then

$$\nabla_X \tilde{L} = f \nabla_X L \bmod L.$$

Thus,

$$b_f L = f b_L. \quad (\text{F.5})$$

It follows that the Weingarten map b of $\mathcal{N}$ is defined *only up to a positive scale factor*.

The *null second fundamental form* $B = B_L$ of $\mathcal{N}$ with respect to L is the bilinear form associated to b via h : for each $p \in \mathcal{N}$, the map

$$B: (T_p\mathcal{N})/L \times (T_p\mathcal{N})/L \rightarrow \mathbb{R}$$

is defined by

$$B(\overline{X}, \overline{Y}) := h(b(\overline{X}), \overline{Y}) = g(\nabla_X L, Y). \quad (\text{F.6})$$

Now,

$$h(b(\overline{X}), \overline{Y}) = g(\nabla_X L, Y) = g(X, \nabla_Y L) = h(\overline{X}, b(\overline{Y})). \quad (\text{F.7})$$

This shows that b is self-adjoint with respect to h , and that B is symmetric.

INCIDENTALLY: In a manner analogous to the second fundamental form for spacelike hypersurfaces, a null hypersurface is totally geodesic if and only if B vanishes identically [310, Theorem 30]. $\square$

The *null mean curvature* of $\mathcal{N}$ with respect to L is the continuous scalar field $\theta \in C^0(\mathcal{N})$ defined by

$$\theta = \text{tr}_h b, \quad (\text{F.8})$$

also called *divergence*, or *expansion*, of the null hypersurface. Let $e_1, e_2, \dots, e_{n-1}$ be $n-1$ orthonormal spacelike vectors (with respect to g) tangent to $\mathcal{N}$ at p . Then $\{\bar{e}_1, \bar{e}_2, \dots, \bar{e}_{n-1}\}$ is an orthonormal basis (with respect to h) of $(T_p\mathcal{N})/L$. Hence at p ,

$$\begin{aligned}
\theta &= \operatorname{tr} b = \sum_{i=1}^{n-1} h(b(\bar{e}_i), \bar{e}_i) \\
&= \sum_{i=1}^{n-1} g(\nabla_{e_i} L, e_i).
\end{aligned} \tag{F.9}$$

Let Σ be the intersection, transverse to L , of a hypersurface in $\mathcal{M}$ with $\mathcal{N}$. Then Σ is a C^2 $(n-1)$ -dimensional spacelike submanifold of $\mathcal{M}$ contained in $\mathcal{N}$ which meets L orthogonally. From (F.9), by definition of the divergence along a submanifold,

$$\theta|_{\Sigma} = \operatorname{div}_{\Sigma} L. \tag{F.10}$$

Hence, the null mean curvature gives a measure of the divergence of the null generators of $\mathcal{N}$.

INCIDENTALLY: Given Σ and L as above, let $\underline{L}$ be the second future-directed null vector normal to Σ normalized so that

$$g(\underline{L}, L) = -2. \tag{F.11}$$

Set

$$h_{\mu\nu} := g_{\mu\nu} + \frac{1}{2}(\underline{L}_{\mu} L_{\nu} + \underline{L}_{\nu} L_{\mu}). \tag{F.12}$$

Then $h^{\mu}_{\nu} := g^{\mu\alpha} h_{\alpha\nu}$ is the projector on $T\Sigma$: indeed, $X \in T\Sigma$ if and only if $g(X, \underline{L}) = g(X, L) = 0$, but then $h^{\mu}_{\nu} X^{\nu} = X^{\mu}$ readily follows. On the other hand, $h^{\mu}_{\nu} L^{\nu} = h^{\mu}_{\nu} \underline{L}^{\nu} = 0$ holds by definition of h .

Along Σ , elements of $T\mathcal{N}/L$ can be represented by vectors in $T\Sigma$. Then h in (F.12) coincides with h of (F.3), and the former can be thought of a spacetime equivalent of the latter. Note, however, that the definition (F.3) does not require any supplementary structures, while (F.12) requires Σ , or at least $\underline{L}$.

From the definition of θ we have

$$\begin{aligned}
\theta &= h^{\mu\nu} \nabla_{\mu} L_{\nu} \\
&= g^{\mu\nu} \nabla_{\mu} L_{\nu} + \frac{1}{2}(\underline{L}^{\mu} L^{\nu} \nabla_{\mu} L_{\nu} + \underline{L}^{\nu} L^{\mu} \nabla_{\mu} L_{\nu}).
\end{aligned} \tag{F.13}$$

In some situations L arises from a null vector field defined in a neighbourhood of $\mathcal{N}$, and is autoparallel on $\mathcal{N}$. Then $g(L, L) = 0$ near $\mathcal{N}$, which implies that the middle term in (F.13) drops out. Further, $\nabla_L L = 0$ on $\mathcal{N}$, which shows that the last terms drops out. One then finds a convenient formula:

$$\text{If } g(L, L) = 0 \text{ near } \mathcal{N} \text{ and if } \nabla_L L = 0 \text{ on } \mathcal{N} \text{ then } \theta = \nabla_{\mu} L^{\mu}. \tag{F.14}$$

(This should be compared with the formula $H = \nabla_{\mu} n^{\mu}$ for the mean curvature H of a non-characteristic hypersurface $\mathcal{S}$, where n is a unit normal to $\mathcal{S}$, without the need of any further hypotheses.) $\square$

EXAMPLE F.1 Let $\mathcal{N}$ be a null hypersurface $u := x - t = 0$ in Minkowski spacetime; we choose $-t + x$ instead of $-x + t$ because of the convention that $L := \nabla u$ in the calculations above should be future pointing. Then L is null everywhere, and autoparallel everywhere; thus,

$$\theta = \nabla_{\alpha} \nabla^{\alpha} u = \square g u = 0.$$

Next, still in Minkowski spacetime, let $\mathcal{N}$ be a future null cone $\mathcal{N} = \dot{I}^+(p) - \{p\}$, $u := -t + r = 0$. Then ∇u is null, geodesic, and future pointing, so

$$\theta = \square u = \partial_{\mu}(\eta^{\mu\nu} \partial_{\nu} u) = \partial_{\mu}(\eta^{\mu\nu} \partial_{\nu} r) = \partial_i(\eta^{ij} \partial_j r) = \partial_i \left(\frac{x^i}{r} \right) = \frac{n-1}{r}. \tag{F.15}$$

Hence $\dot{I}^+(p) - \{p\}$ has positive null mean curvature. Changing time orientation, we obtain that a past null cone $\mathcal{N} = \dot{I}^-(p) - \{p\}$ in Minkowski spacetime has negative null mean curvature, $\theta < 0$.

Finally, in an arbitrary smooth spacetime, consider a future null cone $\mathcal{N} = \dot{I}^+(p) - \{p\}$ near its tip. In geodesic coordinates $\dot{I}^+(p)$ is given again by the equation $-t + r = 0$.

In those coordinates the metric coincides with the Minkowskian metric up to terms quadratic in the coordinates, and the derivatives of the metric vanish at p . With a little work one now checks that (F.15) holds near p up to terms which are $O(1)$. $\square$

EXAMPLE F.2 Let $\mathcal{S}$ be a spacelike hypersurface in $\mathcal{N}$ with induced metric γ and extrinsic curvature K , and let Σ be the intersection of $\mathcal{N}$ with $\mathcal{S}$; this is a smooth submanifold of $\mathcal{S}$, of codimension one in $\mathcal{S}$ since the intersection is transverse. Let n be the field of unit normals to Σ within $\mathcal{S}$; we can choose both L and the direction of n so that

$$L = T + n,$$

where T is the field of unit normals to $\mathcal{S}$ in $\mathcal{M}$. Let h denote the metric induced by g on Σ . Representing, as before, elements of $T\mathcal{M}/L$ by vectors tangents to Σ , one finds that, in local coordinates x^A on $\mathcal{S}$,

$$b_{AB} = K_{AB} + \lambda_{AB}, \quad \theta = h^{AB}K_{AB} + H, \quad (\text{F.16})$$

where λ_{AB} is the extrinsic curvature (second fundamental form) of Σ within $(\mathcal{S}, \gamma)$, and H is the mean curvature of Σ within $(\mathcal{S}, \gamma)$. $\square$

Equation (F.5) shows that if $\tilde{L} = fL$, then $\tilde{\theta} = f\theta$. Thus, the null mean curvature inequalities $\theta \geq 0$ and $\theta \leq 0$, are invariant under positive rescaling of L .

The null second fundamental form of a null hypersurface obeys a well-defined comparison theory roughly similar to the comparison theory satisfied by the second fundamental forms of a family of parallel spacelike hypersurfaces (cf. Eschenburg [186]).

Let $\eta: (a, b) \rightarrow \mathcal{M}$, $s \rightarrow \eta(s)$, be a future-directed affinely parameterized null geodesic generator of $\mathcal{N}$. For each $s \in (a, b)$, let

$$b(s) := b_{\eta'(s)} : (T_{\eta(s)}\mathcal{N})/\eta'(s) \rightarrow (T_{\eta(s)}\mathcal{N})/\eta'(s)$$

be the Weingarten map based at $\eta(s)$ with respect to the null vector $L = \eta'(s)$.

To continue, we need to define how a derivative $b(\bar{X})'$ of an equivalence class of vectors $b(\bar{X})$ can be calculated. Let $s \rightarrow V(s)$ be a $T\mathcal{N}/L$ -vector field along η ; i.e., for each s , $V(s)$ is an element of $T_{\eta(s)}\mathcal{N}/L$. Say $s \rightarrow V(s)$ is smooth if (at least locally) there is a smooth—in the usual sense—vector field $s \rightarrow Y(s)$ along η such that $V(s) = \overline{Y(s)}$ for each s . Then define the covariant derivative of $s \rightarrow V(s)$ along η by $V'(s) = \overline{Y'(s)}$, where Y' is the usual covariant derivative $\nabla_{\dot{\eta}}Y$. Then V' so defined is independent of the choice of Y : indeed, if $Y'(s) = Y(s) + \alpha(s)\dot{\eta}(s)$ for some smooth function α , then

$$\overline{\nabla_{\dot{\eta}}Y'} = \overline{\nabla_{\dot{\eta}}Y} + \overline{\frac{d\alpha}{ds}\dot{\eta} + \underbrace{\alpha \nabla_{\dot{\eta}}\dot{\eta}}_{\sim \dot{\eta}}} = \overline{\nabla_{\dot{\eta}}Y}.$$

We are ready now to derive the equation satisfied by the null Weingarten map b of a smooth null hypersurface $\mathcal{N}$ (cf. [38, p. 431]):

$$b' + b^2 + R = 0. \quad (\text{F.17})$$

Here $'$ denotes covariant differentiation in the direction $\eta'(s)$, in the sense just defined for vectors, and extended to tensors in the usual way: if $X = X(s)$ is a vector field along η tangent to $\mathcal{N}$, then

$$b'(\bar{X}) := (b(\bar{X}))' - b(\bar{X}'). \quad (\text{F.18})$$

Finally, $R: T_{\eta(s)}\mathcal{N}/\eta'(s) \rightarrow T_{\eta(s)}\mathcal{N}/\eta'(s)$ is the curvature endomorphism defined by

$$R(\bar{X}) = \overline{R(X, \eta'(s))\eta'(s)},$$

where $(X, Y, Z) \rightarrow R(X, Y)Z$ is the Riemann curvature tensor of $\mathcal{M}$ (recall that $R(X, Y)Z = \nabla_X\nabla_YZ - \nabla_Y\nabla_XZ - \nabla_{[X, Y]}Z$).

In order to prove (F.17), fix a point $p = \eta(s_0)$, $s_0 \in (a, b)$, on η . On a neighbourhood U of p in $\mathcal{H}$ we can scale the null vector field L so that L is a geodesic vector field, $\nabla_L L = 0$, and so that L , restricted to η , is the velocity vector field to η , i.e., for each s near s_0 , $L_{\eta(s)} = \eta'(s)$. Let $X \in T_p \mathcal{N}$. Shrinking U if necessary, we can extend X to a smooth vector field on U so that $[X, L] = \nabla_X L - \nabla_L X = 0$. Then

$$R(X, L)L = \nabla_X \nabla_L L - \nabla_L \nabla_X L - \nabla_{[X, L]} L = -\nabla_L \nabla_L X.$$

Hence along η we have

$$X'' = -R(X, \eta')\eta'.$$

Thus, from Eq. (F.18), at the point p we have

$$\begin{aligned} b'(\bar{X}) &= \overline{\nabla_X L'} - b(\overline{\nabla_L X}) = \overline{\nabla_L X'} - b(\overline{\nabla_X L}) \\ &= \overline{X''} - b(b(\bar{X})) = -\overline{R(X, \eta')\eta'} - b^2(\bar{X}) \\ &= -R(\bar{X}) - b^2(\bar{X}), \end{aligned} \tag{F.19}$$

which establishes Eq. (F.17).

Equation (F.17) leads to the well-known *Raychaudhuri equation* for an *irrotational* null geodesic congruence in general relativity: By taking the trace of (F.17) we obtain the following formula for the derivative of the null mean curvature $\theta = \theta(s)$ along η , in spacetime dimension $n + 1$,

$$\theta' = -\text{Ric}(\eta', \eta') - \sigma^2 - \frac{1}{n-1}\theta^2, \tag{F.20}$$

where σ is the trace-free part of b . This equation shows how the Ricci curvature of spacetime influences the null mean curvature of a null hypersurface.

INCIDENTALLY: In the calculations above we have assumed smoothness of all objects involved. If we suppose that $\mathcal{N}$ is only C^2 , then a priori b is only a C^0 tensor field so that there is no reason that the derivative b' should exist. The main point of the following result, proved in [117], is that b' does exist for C^2 hypersurfaces and satisfies the expected differential equation. As the function $s \mapsto R_{\eta(s)}$ is C^∞ , then the Riccati equation implies that actually the dependence of $b_{\eta(s)}$ on s is C^∞ .

PROPOSITION F.3 Let $\mathcal{N}$ be a C^2 null hypersurface in the $(n+1)$ -dimensional spacetime $(\mathcal{M}, g)$ and let b be the one-parameter family of Weingarten maps along an affine parameterized null generator η . Then the covariant derivative b' defined by (F.18) exists and satisfies (F.17). $\square$

In the previous result, the hypersurface $\mathcal{N}$ had to be of at least C^2 differentiability class. Now, in various applications one has to consider hypersurfaces $\mathcal{N}$ obtained as a collection of null geodesics normal to a C^2 surface. A naive inspection of the problem at hand shows that such $\mathcal{N}$'s could in principle be of C^1 differentiability only. It is shown in [117] that one does nevertheless have C^2 differentiability of the resulting hypersurface in such situations.

PROPOSITION F.4 Consider a C^{k+1} spacelike submanifold $N \subset \mathcal{M}$ of co-dimension two in an $(n+1)$ -dimensional spacetime $(\mathcal{M}, g)$, with $k \geq 1$. Let $\mathbf{k}$ be a non-vanishing C^k null vector field along N , and let $\mathcal{U} \subseteq \mathbb{R} \times N \rightarrow \mathcal{M}$ be the set of points where the function

$$f(t, p) := \exp_p(t\mathbf{k}(p))$$

is defined. If $f_{(t_0, p_0)*}$ is injective, then there is an open neighbourhood $\mathcal{O}$ of (t_0, p_0) so that the image $f[\mathcal{O}]$ is a C^{k+1} embedded hypersurface in $\mathcal{M}$. $\square$

The result is somewhat surprising as the function $p \mapsto \mathbf{k}(p)$ used in the definition of f is only C^k . Indeed, f will not be of C^{k+1} differentiability class in general, which can be seen as follows: let $t \rightarrow r(t)$ be a C^{k+1} curve in the x - y plane of Minkowski 3-space which is *not* of C^{k+2} differentiability class. Let $t \rightarrow \mathbf{n}(t)$ be the spacelike unit normal field along the curve in the x - y plane; then $t \rightarrow \mathbf{n}(t)$ is C^k and is *not* C^{k+1} . Let

$T = (0, 0, 1)$ be the unit normal to the x - y plane. Then $L(t) = \mathbf{n}(t) + T$ is a C^k normal null field along $t \rightarrow r(t)$. The normal exponential map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ in the direction L is given by $f(s, t) = r(t) + s[\mathbf{n}(t) + T]$, and hence $df/dt = r'(t) + sn'(t)$, showing explicitly that the regularity of f can be no greater than the regularity of $\mathbf{n}(t)$, and hence no greater than the regularity of $r'(t)$. $\square$

We continue with a calculation tying the divergence θ of a null hypersurface with Jacobians; this, together with the Raychaudhuri equation, is the key to the *black-hole area theorem* of Section 3.5, p. 109. Let us start by recalling the definition of the Jacobian, as needed in the context. Let $\phi: M \rightarrow N$ be a C^1 map between Riemannian manifolds, with $\dim M \leq \dim N$. Let¹⁰ $n = \dim M$ and let $e_1, \dots, e_n$ be an orthonormal basis of $T_p M$; then the Jacobian¹¹ of ϕ at p is

$$J(\phi)(p) = \|\phi_{*p}e_1 \wedge \phi_{*p}e_2 \wedge \cdots \wedge \phi_{*p}e_n\|.$$

Here $\|\cdot\|$ denotes the norm, defined by the metric, on the space of forms. When $\dim M = \dim N$ and both M and N are oriented with ω_M being the volume form on M , and ω_N being the volume form on N , then $J(\phi)$ can also be described as the positive scalar satisfying $\phi^*(\omega_N) = \pm J(\phi)\omega_M$.

Let S be a C^2 codimension-two acausal spacelike submanifold of a smooth space-time M , and let L be a past-directed C^1 null vector field along S . Consider the normal exponential map in the direction L , $\Phi: \mathbb{R} \times S \rightarrow M$, defined by $\Phi(s, x) = \exp_x sL$. (Φ need not be defined on all of $\mathbb{R} \times S$.) Suppose the null geodesic $\eta: s \rightarrow \Phi(s, p)$ meets a given acausal spacelike hypersurface Σ at $\eta(1)$. Then there is a neighbourhood W of p in S such that each geodesic $s \rightarrow \Phi(s, x)$, $x \in W$ meets Σ , and so determines a C^1 map $\phi: W \rightarrow \Sigma$, which is the projection into Σ along these geodesics. Let $J(\phi)$ denote the Jacobian determinant of ϕ at p . $J(\phi)$ may be computed as follows. Let $\{X_1, X_2, \dots, X_k\}$ be an orthonormal basis for the tangent space $T_p S$. Similarly, we have

$$J(\phi)(p) = \|\phi_{*p}X_1 \wedge \phi_{*p}X_2 \wedge \cdots \wedge \phi_{*p}X_k\|.$$

Suppose there are no focal points to S along $\eta|_{[0,1]}$. Then by shrinking W and rescaling L if necessary, $\Phi: [0, 1] \times W \rightarrow M$ is a C^1 embedded null hypersurface N such that $\Phi(\{1\} \times W) \subset \Sigma$. Extend L to be the C^1 past-directed null vector field, $L = \Phi_*(\frac{\partial}{\partial s})$ on N . Let $\theta = \theta(s)$ be the null mean curvature of N with respect to $-L$ along η . We then have the following.

PROPOSITION F.5 *With $\theta = \theta(s)$ as described above:*

1. *If there are no focal points to S along $\eta|_{[0,1]}$, then*

$$J(\phi) = \exp \left(- \int_0^1 \theta(s) ds \right). \quad (\text{F.21})$$

2. *If $\eta(1)$ is the first focal point to S along $\eta|_{[0,1]}$, then*

$$J(\phi) = 0.$$

REMARK F.6 In particular, if N has nonnegative null mean curvature with respect to the future-pointing null normal, i.e., if $\theta \geq 0$, we obtain that $J(\phi) \leq 1$. $\square$

¹⁰Note that this is different from our usual convention elsewhere that the dimension of spacetime is $n + 1$.

¹¹The Jacobian $J(\phi)$ here is *not* the usual Jacobian which occurs in the change-of-variables theorem for Lebesgue measure on $\mathbb{R}^n$, but contains the appropriate $\sqrt{\det g_{ij}}$ factors occurring in the definition of the measure associated with a Riemannian metric, see [189, Def. 2.10, p. 423]. A clear exposition of the Jacobians that occur in the area and co-area formulas can be found in [187, Section 3.2] in the flat $\mathbb{R}^n$ case. See also [263, Appendix, p. 66] for the smooth Riemannian case.

REMARK F.7 Recall that θ was only defined when a normalization of L has been chosen. We stress that in (F.21) L is tangent to an affinely parameterized geodesic, with s being an affine distance along η , and with p corresponding to $s = 0$ and $\phi(p)$ corresponding to $s = 1$. $\square$

Proof. 1. To relate $J(\phi)$ to the null mean curvature of N , extend the orthonormal basis $\{X_1, X_2, \dots, X_k\}$ to Lie parallel vector fields $s \rightarrow X_i(s)$, $i = 1, \dots, k$, along η , $\mathcal{L}_L X_i = 0$ along η . Then

$$\begin{aligned} J(\phi) &= \|\phi_{*p} X_1 \wedge \phi_{*p} X_2 \wedge \dots \wedge \phi_{*p} X_k\| \\ &= \|X_1(1) \wedge X_2(1) \wedge \dots \wedge X_k(1)\| \\ &= \sqrt{g} \Big|_{s=1}, \end{aligned}$$

where $g = \det[g_{ij}]$, and $g_{ij} = g_{ij}(s) = \langle X_i(s), X_j(s) \rangle$. We claim that along η ,

$$\theta = -\frac{1}{\sqrt{g}} \frac{d}{ds} \sqrt{g}.$$

To see this, set $b_{ij} = B(\bar{X}_i, \bar{X}_j)$, where B is the null second fundamental form of N with respect to $-L$, $h_{ij} = h(\bar{X}_i, \bar{X}_j) = g_{ij}$, and let g^{ij} be the i, j -th entry of the inverse matrix $[g_{ij}]^{-1}$. Then $\theta = g^{ij} b_{ij}$. Differentiating g_{ij} along η we obtain

$$\begin{aligned} \frac{d}{ds} g_{ij} &= L \langle X_i, X_j \rangle = \langle \nabla_L X_i, X_j \rangle + \langle X_i, \nabla_L X_j \rangle \\ &= \langle \nabla_{X_i} L, X_j \rangle + \langle X_i, \nabla_{X_j} L \rangle \\ &= -(b_{ij} + b_{ji}) = -2b_{ij}. \end{aligned}$$

Thus,

$$\theta = g^{ij} b_{ij} = -\frac{1}{2} g^{ij} \frac{d}{ds} g_{ij} = -\frac{1}{2} \frac{1}{g} \frac{dg}{ds} = -\frac{1}{\sqrt{g}} \frac{d}{ds} \sqrt{g},$$

as claimed. Integrating along η from $s = 0$ to $s = 1$,

$$J(\phi) = \sqrt{g} \Big|_{s=1} = \sqrt{g} \Big|_{s=0} \cdot \exp \left(- \int_0^1 \theta ds \right) = \exp \left(- \int_0^1 \theta ds \right).$$

2. Suppose now that $\eta(1)$ is a focal point to S along η , but that there are no focal points to S along η prior to that. Then we can still construct the C^1 map $\Phi: [0, 1] \times W \rightarrow M$, with $\Phi(\{1\} \times W) \subset \Sigma$, such that Φ is an embedding when restricted to a sufficiently small open set in $[0, 1] \times W$ containing $[0, 1] \times \{p\}$. The vector fields $s \rightarrow X_i(s)$, $s \in [0, 1]$, $i = 1, \dots, k$, may be constructed as above, and are Jacobi fields along $\eta|_{[0,1]}$, which extend smoothly to $\eta(1)$. Since $\eta(1)$ is a focal point, the vectors $\phi_* X_1 = X_1(1)$, $\dots$, $\phi_* X_k = X_k(1)$ must be linearly dependent, which implies that $J(\phi) = 0$. $\square$

INCIDENTALLY: It has been pointed out in [280] that *every null hypersurface carries a canonical $(n-1)$ -form Λ , annihilating L* . The construction proceeds as follows: Let $p \in \mathcal{N}$ and denote by

$$P: T_p \mathcal{N} \rightarrow (T_p \mathcal{N})/L$$

the canonical projection. Let μ_h denote the canonical volume form associated with the metric h on $(T_p \mathcal{N})/L$; then the pull-back

$$\Lambda := P^* \mu_h \tag{F.22}$$

defines the required field of $(n-1)$ -forms on $\mathcal{S}$.

Since $PL = 0$ we have

$$\Lambda(L, \dots) = \mu_h(PL, \dots) = 0;$$

thus Λ vanishes on L .

Had we a metric on $\mathcal{N}$, then the associated Hodge dual of Λ would provide a canonical vector field on $\mathcal{N}$. However, $\mathcal{N}$ is degenerate, and there is no natural metric on $\mathcal{N}$. $\square$

G The general relativistic Cauchy problem

A systematic way of constructing solutions of Einstein equations, possibly with cosmological constant Λ and matter fields, proceeds by solving a Cauchy problem. In this appendix we introduce some associated notations and terminology; the reader is referred to [32, 416] and references therein for more information about the topic.

A *general relativistic initial data set* $(\mathcal{S}, h, K)$ is a triple where $\mathcal{S}$ is an n -dimensional manifold, h is a Riemannian metric on $\mathcal{S}$, and K is a symmetric two-covariant tensor field on $\mathcal{S}$. The tensor field $h = h_{ij}dx^i dx^j$ corresponds to the metric induced on the hypersurface $\mathcal{S}$ by the spacetime metric in the associated spacetime, and $K = K_{ij}dx^i dx^j$ is the extrinsic curvature tensor of $\mathcal{S}$. In the notation of Section H.1 below, $h = h_{ij}dx^i dx^j$ is the tensor field $g_{ij}dx^i dx^j$ of (H.1), and K is the tensor field (H.4).

The equation

$$R(h) = 16\pi\rho + 2\Lambda + |K|_h^2 - (\text{tr}_h K)^2, \quad (\text{G.1})$$

where $R(h)$ is the Ricci scalar of the metric h and ρ is the energy density of matter fields, is called the *scalar constraint equation*. The equation

$$D_j K^j_k - D_k K^j_j = 8\pi J_k, \quad (\text{G.2})$$

where J_i is the momentum vector of matter fields, is called the *vector constraint equation*.

Equations (G.1)–(G.2) taken together are known as the *general relativistic constraint equations*. They provide a necessary condition for $(\mathcal{S}, h, K)$ to arise from a hypersurface in a spacetime satisfying the Einstein equations.

The set $(\mathcal{S}, h, K)$ is called a *vacuum general relativistic initial data set* if (h, K) satisfy the scalar and vector constraint equations with $\rho \equiv 0 \equiv J_k$.

A development of a vacuum initial data set $(\mathcal{S}, h, K)$ is a vacuum spacetime $(\mathcal{M}, g)$ together with an embedding $i: \mathcal{S} \rightarrow \mathcal{M}$ such that h is the pull-back to $\mathcal{S}$ of the metric induced by g on $\mathcal{S}$, i.e. $h = i^*g$, and K is the pull-back to $\mathcal{S}$ of the second fundamental form of $i(\mathcal{S})$ in $(\mathcal{M}, g)$. The tensor field K is usually called the *extrinsic curvature tensor* in the physics literature.

Let us for simplicity assume that $(\mathcal{S}, h, K)$ is smooth. The notion of global hyperbolicity is described in detail in Section 2.8, p. 56. We note the following reformulation of the key Theorem 3.6.2, p. 114, of Choquet-Bruhat and Geroch.

THEOREM G.1 *Let $(\mathcal{S}, h, K)$ be a vacuum initial data set. There exists a smooth globally hyperbolic development $(\mathcal{M}, g, i)$ of $(\mathcal{S}, h, K)$ such that for every smooth globally hyperbolic development $(\tilde{\mathcal{M}}, \tilde{g}, \tilde{i})$ of $(\mathcal{S}, h, K)$ there exists an isometric embedding $\Psi: \tilde{\mathcal{M}} \rightarrow \mathcal{M}$ satisfying*

$$\Psi \circ \tilde{i} = i.$$

A development $(\mathcal{M}, g)$ with the properties stated in Theorem G.1 is called a *maximal globally hyperbolic development* (MGHD). It should be clear from Theorem G.1 that maximal globally hyperbolic developments are unique up to isometry,

and inextendible in the class of smooth globally hyperbolic vacuum spacetimes. Extensions of the MGHD, if they exist, occur across Cauchy horizons.

Examples of extendible MGHD are known. This is considered to be a problem because solutions of Einstein equations stop being unique in general past Cauchy horizons. The *strong cosmic censorship conjecture*, essentially due to Penrose [396] (see also [96]) is the expectation that *the MGHD of generic vacuum initial data $(\mathcal{S}, h, K)$, with $\mathcal{S}$ compact or with $(\mathcal{S}, h, K)$ asymptotically flat, is inextendible.*

The *weak cosmic censorship conjecture* is a weaker version thereof, where one hopes that *violations of strong cosmic censorship in asymptotically flat spacetimes are clothed by event horizons.* A possible precise version thereof is the conjecture that *MGHDs of generic, vacuum, asymptotically flat initial data admit a differentiable conformal boundary with generators which are complete to the future in the shear-free gauge.*

H A collection of identities

We include here a collection of useful identities, mostly compiled by Erwann Delay. I am grateful to Erwann for allowing me to include his set here.

H.1 ADM notation

Given a coordinate system (t, x^i) with ∇t timelike, every Lorentzian metric g can be parameterized as

$$g = -N^2 dt^2 + g_{ij}(dx^i + N^i dt)(dx^j + N^j dt). \quad (\text{H.1})$$

This is known as the Arnowitt–Deser–Misner (ADM) notation. Letting $\tilde{g}^{ij}$ denote the inverse matrix to g_{ij} , we then have

$$g^{kl} = \tilde{g}^{kl} - \frac{N^l N^k}{N^2}, \quad g_{0k} = N_k, \quad g^{0k} = \frac{N^k}{N^2}, \quad N^2 = -\frac{1}{g^{00}}, \quad g_{00} = N^k N_k - N^2, \quad (\text{H.2})$$

where $N^k := \tilde{g}^{kl} N_l$. It also holds that

$$\det g_{\mu\nu} = -N^2 \det g_{ij}. \quad (\text{H.3})$$

One sets

$$K_{kl} = -N \Gamma_{kl}^0 = \frac{1}{2N} (D_l N_k + D_k N_l - \partial_0 g_{kl}). \quad (\text{H.4})$$

It holds that

$$\begin{aligned} \Gamma_{k0}^0 &= \partial_k \log N - \frac{N^l}{N} K_{lk}, & \Gamma_{00}^0 &= \partial_0 \log N + N^k \partial_k \log N - \frac{N^l N^k}{N} K_{lk}, \\ \Gamma_{ij}^k &= \tilde{\Gamma}_{ij}^k + \frac{N^k}{N} K_{ij}, & \Gamma_{0j}^k &= D_j N^k - N K^k_j + \frac{N^k}{N} (N^l K_{lj} - D_j N). \end{aligned}$$

H.2 Some commutators

Here are some formulae for the commutation of derivatives:

$$\begin{aligned} \nabla_m \nabla_l t_{ik} - \nabla_l \nabla_m t_{ik} &= R^p{}_{klm} t_{ip} + R^p{}_{ilm} t_{kp}, \\ \nabla_i \nabla_j V^l - \nabla_j \nabla_i V^l &= R^l{}_{kij} V^k, \\ \nabla^k \nabla_k |df|^2 &= 2(\nabla^l f \nabla_l \nabla^k \nabla_k f + \text{Ric}(\nabla f, \nabla f) + |\nabla \nabla f|^2), \\ \nabla^k \nabla_k \nabla_i \nabla_j f - \nabla_i \nabla_j \nabla^k \nabla_k f &- R_{kj} \nabla^k \nabla_i f - R_{ki} \nabla^k \nabla_j f + 2R_{qjli} \nabla^q \nabla^l f \\ &= (\nabla_i R_{kj} + \nabla_j R_{ki} - \nabla_k R_{ij}) \nabla^k f. \end{aligned}$$

H.3 Bianchi identities

The Bianchi identities for a Levi-Civita connection:

$$\begin{aligned} R^i{}_{jkl} + R^i{}_{ljk} + R^i{}_{klj} &= 0, \\ \nabla_l R^\ell{}_{ijk} + \nabla_k R^\ell{}_{ilj} + \nabla_j R^\ell{}_{ikl} &= 0, \\ \nabla_\ell R^\ell{}_{ijk} + \nabla_k R_{ij} - \nabla_j R_{ik} &= 0, \\ \nabla^k R_{ik} - \frac{1}{2} \nabla_k R &= 0. \end{aligned}$$

H.4 Linearizations

Linearizations for various objects of interest:

$$\begin{aligned} D_g \Gamma_{ij}^k(g)h &= \frac{1}{2}(\nabla_i h^k{}_j + \nabla_j h^k{}_i - \nabla^k h_{ij}), \\ 2[D_g \text{Riem}(g)h]_{sklm} &= \nabla_l \nabla_k h_{sm} - \nabla_l \nabla_s h_{km} + \nabla_m \nabla_s h_{kl} \\ &\quad - \nabla_m \nabla_k h_{sl} + R^p{}_{klm} h_{ps} + R^p{}_{sml} h_{pk}, \\ 2[D_g \text{Riem}(g)h]^i{}_{klm} &= \nabla_l \nabla_k h^i{}_m - \nabla_l \nabla^i h_{km} + \nabla_m \nabla^i h_{kl} \\ &\quad - \nabla_m \nabla_k h^i{}_l + g^{is} R^p{}_{sml} h_{pk} - R^p{}_{klm} h^i{}_p, \\ D_g \text{Ric}(g)h &= \frac{1}{2} \Delta_L h - \text{div}^* \text{div}(Gh), \\ \Delta_L h_{ij} &= -\nabla^k \nabla_k h_{ij} + R_{ik} h^k{}_j + R_{jk} h^k{}_i - 2R_{ikjl} h^{kl}, \\ Gh &= h - \frac{1}{2} \text{tr} h g, \quad (\text{div} h)_i = -\nabla^k h_{ik}, \quad \text{div}^* w = \frac{1}{2}(\nabla_i w_j + \nabla_j w_i), \\ D_g R(g)h &= -\nabla^k \nabla_k (\text{tr} h) + \nabla^k \nabla^l h_{kl} - R^{kl} h_{kl}, \\ [D_g R(g)]^* f &= -\nabla^k \nabla_k f g + \nabla \nabla f - f \text{Ric}(g). \end{aligned}$$

H.5 Warped products

Let (M, g) , $\nabla := \nabla_g$, $f : M \rightarrow \mathbb{R}$ and

$$(\mathcal{M} = M \times_f I, \tilde{g} = -f^2 dt^2 + g),$$

then for X, Y tangent to M and V, W tangent to I , we have

$$\begin{aligned} \text{Ric}(\tilde{g})(X, Y) &= \text{Ric}(g)(X, Y) - f^{-1} \nabla \nabla f(X, Y), \\ \text{Ric}(\tilde{g})(X, V) &= 0 = \tilde{g}(X, V), \\ \text{Ric}(\tilde{g})(V, W) &= -f^{-1} \nabla^k \nabla_k f \tilde{g}(V, W). \end{aligned}$$

Let (M, g) , $\nabla := \nabla_g$, $f : M \rightarrow \mathbb{R}$ and let $(\mathcal{M} = M \times_f I, \tilde{g} = \epsilon f^2 dt^2 + g)$, $\epsilon = \pm 1$, $x^a = (x^0 = t, x^i = (x^1, \dots, x^n))$,

$$\begin{aligned} \tilde{\Gamma}_{00}^0 &= \tilde{\Gamma}_{ij}^0 = \tilde{\Gamma}_{i0}^k = 0, \quad \tilde{\Gamma}_{i0}^0 = f^{-1} \partial_i f, \quad \tilde{\Gamma}_{00}^k = -\epsilon f \nabla^k f, \quad \tilde{\Gamma}_{ij}^k = \Gamma_{ij}^k, \\ \tilde{R}^l{}_{ijk} &= R^l{}_{ijk}, \quad \tilde{R}^l{}_{0j0} = -\epsilon f \nabla_j \nabla^l f, \quad \tilde{R}^0{}_{ij0} = f^{-1} \nabla_j \nabla_i f, \\ \tilde{R}^0{}_{ijk} &= \tilde{R}^l{}_{ij0} = \tilde{R}^l{}_{0jk} = \tilde{R}^0{}_{0jk} = \tilde{R}^0{}_{0j0} = 0, \\ \tilde{R}_{mijk} &= R_{mijk}, \quad \tilde{R}_{0ijk} = 0, \quad \tilde{R}_{0ij0} = \epsilon f \nabla_j \nabla_i f, \\ \tilde{R}_{ik} &= R_{ik} - f^{-1} \nabla_k \nabla_i f, \quad \tilde{R}_{0k} = 0, \quad \tilde{R}_{00} = -\epsilon f \nabla^i \nabla_i f, \\ \tilde{R} &= R - 2f^{-1} \nabla^i \nabla_i f. \end{aligned}$$

H.6 Conformal transformations

The Weyl tensor:

$$W_{ijkl} = R_{ijkl} - \frac{1}{n-2}(R_{ik}g_{jl} - R_{il}g_{jk} + R_{jl}g_{ik} - R_{jk}g_{il}) \\ + \frac{R}{(n-1)(n-2)}(g_{jl}g_{ik} - g_{jk}g_{il}).$$

If $g' = e^f g$ for some twice-differentiable function f , we have

$$W_i{}^j{}_{kl}(e^f g) = W_i{}^j{}_{kl}(g).$$

The Schouten tensor:

$$S_{ij} = \frac{1}{n-2}[2R_{ij} - \frac{R}{n-1}g_{ij}].$$

Under a conformal transformation $g' = e^f g$, we have

$$\Gamma_{ij}'^k - \Gamma_{ij}^k = \frac{1}{2}(\delta_j^k \partial_i f + \delta_i^k \partial_j f - g_{ij} \nabla^k f). \\ R_{ij}' = R_{ij} - \frac{n-2}{2} \nabla_i \nabla_j f + \frac{n-2}{4} \nabla_i f \nabla_j f - \frac{1}{2}(\nabla^k \nabla_k f + \frac{n-2}{2} |df|^2) g_{ij} \\ R' = e^{-f} [R - (n-1) \nabla^i \nabla_i f - \frac{(n-1)(n-2)}{4} \nabla^i f \nabla_i f].$$

Specializing to $g' = e^{\frac{2}{n-2}u} g$,

$$R_{ij}' = R_{ij} - \nabla_i \nabla_j u + \frac{1}{n-2} \nabla_i u \nabla_j u - \frac{1}{n-2} (\nabla^k \nabla_k u + |du|^2) g_{ij}.$$

In the notation $g' = v^{\frac{2}{n-2}} g$,

$$R_{ij}' = R_{ij} - v^{-1} \nabla_i \nabla_j v + \frac{n-1}{n-2} v^{-2} \nabla_i v \nabla_j v - \frac{1}{n-2} v^{-1} (\nabla^k \nabla_k v) g_{ij}.$$

If we write instead $g' = \phi^{4/(n-2)} g$, then

$$R_{ij}' = R_{ij} - 2\phi^{-1} \nabla_i \nabla_j \phi + \frac{2n}{n-2} \phi^{-2} \nabla_i \phi \nabla_j \phi - \frac{2}{n-2} \phi^{-1} (\nabla^k \nabla_k \phi + \phi^{-1} |d\phi|^2) g_{ij}, \\ R' \phi^{(n+2)/(n-2)} = -\frac{4(n-1)}{n-2} \nabla^k \nabla_k \phi + R\phi.$$

When we have two metrics g and g' at our disposal, then

$$T_{ij}^k := \Gamma_{ij}'^k - \Gamma_{ij}^k = \frac{1}{2} g'^{kl} (\nabla_i g'_{lj} + \nabla_j g'_{li} - \nabla_l g'_{ij}).$$

$$\text{Riem}^i{}_{klm} - \text{Riem}^i{}_{klm} = \nabla_l T_{km}^i - \nabla_m T_{kl}^i + T_{jl}^j T_{km}^i - T_{jm}^i T_{kl}^j.$$

Under $g' = e^f g$, the Laplacian acting on functions transforms as

$$\nabla'^k \nabla'_k v = e^{-f} (\nabla^k \nabla_k v + \frac{n-2}{2} \nabla^k f \nabla_k v).$$

For symmetric tensors we have instead

$$\nabla'^k \nabla'_k u_{ij} = e^{-f} \left[\nabla^k \nabla_k u_{ij} + \frac{n-6}{2} \nabla^k f \nabla_k u_{ij} - (\nabla_i f \nabla^k u_{kj} + \nabla_j f \nabla^k u_{ki}) \right. \\ \left. + (\nabla^k f \nabla_i u_{kj} + \nabla^k f \nabla_j u_{ki}) + (\frac{3-n}{2} \nabla^k f \nabla_k f - \nabla^k \nabla_k f) u_{ij} \right. \\ \left. - \frac{n}{4} (\nabla_i f \nabla^k f u_{kj} + \nabla_j f \nabla^k f u_{ki}) + \frac{1}{2} \nabla_i f \nabla_j f u_k^k + \frac{1}{2} g_{ij} u_{kl} \nabla^k f \nabla^l f \right].$$

H.7 Laplacians on tensors

For symmetric u 's and arbitrary T 's let

$$(Du)_{kij} := \frac{1}{\sqrt{2}}(\nabla_k u_{ij} - \nabla_j u_{ik}),$$

then

$$(D^*T)_{ij} = \frac{1}{2\sqrt{2}}(-\nabla^k T_{kij} - \nabla^k T_{kji} + \nabla^k T_{ijk} + \nabla^k T_{jik}).$$

Further

$$D^*Du_{ij} = -\nabla^k \nabla_k u_{ij} + \frac{1}{2}(\nabla^k \nabla_i u_{jk} + \nabla^k \nabla_j u_{ik}),$$

and

$$\operatorname{div}^* \operatorname{div} u = -\frac{1}{2}(\nabla_i \nabla^k u_{jk} + \nabla_j \nabla^k u_{ik});$$

thus,

$$(D^*D + \operatorname{div}^* \operatorname{div})u_{ij} = -\nabla^k \nabla_k u_{ij} + \frac{1}{2}(R_{kj}u_i^k + R_{ki}u_j^k - 2R_{qjli}u^{ql}).$$

H.8 Stationary metrics

Let (M, γ) be a Riemannian or pseudo-Riemannian three-dimensional manifold, define $\lambda : M \rightarrow \mathbb{R}$, $\xi : M \rightarrow T^*M$, $(N = I \times M, g)$ by the formulae

$$g(t, x) = \begin{pmatrix} \lambda & {}^t\xi \\ \xi & \lambda^{-1}({}^t\xi \xi - \gamma) \end{pmatrix} = \lambda(dt + \lambda^{-1}\xi_i dx^i)^2 - \lambda^{-1}\gamma_{ij}dx^i dx^j.$$

Let $w = -\lambda^2 *_\gamma d(\lambda^{-1}\xi)$. $\nabla = \nabla_g$, $E^i = \gamma^{is}E_s$. Then

$$\operatorname{Ric}(\gamma)_{ij} = \frac{1}{2}\lambda^{-1}(\nabla_i \lambda \nabla_j \lambda + w_i w_j) + \lambda^{-2}(\operatorname{Ric}(g)_{ij} - \operatorname{Ric}(g)_{cd} \xi^c \xi^d \gamma_{ij}),$$

$$\nabla^i \nabla_i \lambda = \lambda^{-1}(|d\lambda|^2 - |w|^2) - 2\lambda^{-1} \operatorname{Ric}(g)_{ab} \xi^a \xi^b,$$

$$\nabla^i (\lambda^{-2} w_i) = 0,$$

$$\lambda(*_\gamma dw)^i = -2\lambda^{-1} T(g)_c^i \xi^c, \quad \operatorname{Ric}(g) = G(T(g)).$$

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